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Class 12 Physics Chapterwise PYQs

2026 – 2003 | All CBSE Board Papers

Chapterwise previous year questions, sorted by marks and year

Chapter 3: Current Electricity

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HOW TO USE THIS BOOKLET

1. **Attempt a section** on your own. Questions run newest year first.
2. **Check the Answer Key,** then follow a question number to its solution.
3. **Study the solution.** Note each *Common Mistake,* then return via "Back to Q n".

SECTION A

1-Mark Questions

Multiple choice · 126 questions

> *Attempt these first, then check the Answer Key and open any solution.*

1. Two heaters rated as (P_1, V) and (P_2, V) are connected in series across a dc source of V volt. The power consumed by the combination will be:

(A) $(P_1 + P_2)$

(B) $\dfrac{P_1 + P_2}{2}$

(C) $\dfrac{P_1 P_2}{2(P_1 + P_2)}$

(D) $\dfrac{P_1 P_2}{4(P_1 + P_2)}$

Check Solution → **[2026 • Set 55-1-1]**

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2. A copper wire is stretched to increase its length by 1%. Then the change in its resistance is close to:

(A) 1%

(B) 4%

(C) −4%

(D) 2%

Check Solution → **[2026 • Set 55-1-2]**

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3. A current of 1.5 A is maintained in a copper wire of length 1 m with area of cross-section 1.7×10^{-7} m^2. The magnitude of electric field in the wire is [ρ_{Cu} = 1.7×10^{-7} $\Omega\, m$]:

(A) 0.15 V/m

(B) 0.30 V/m

(C) 1.5 V/m

(D) 3.0 V/m

Check Solution → [2026 • Set 55-1-3]

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4. **Assertion (A):** Two electric heaters of power P_1 and P_2 ($>P_1$) are joined in series across a dc source of voltage V. The power consumed by the combination will be less than that consumed by P_1 when connected across the same source.
Reason (R): The power consumed by an electric device when connected to a dc source of voltage V is proportional to its resistance.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false and Reason (R) is also false.

Check Solution → [2026 • Set 55-2-1]

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5. **Assertion (A):** In a Wheatstone bridge circuit, if we interchange the position of the cell and the galvanometer, the balance condition $\frac{P}{Q} = \frac{R}{S}$ remains unchanged.
Reason (R): $\frac{P}{Q} = \frac{R}{S} \Rightarrow \frac{Q}{S} = \frac{P}{R}$ so balance condition remains same.
Select the correct answer from the codes (A), (B), (C) and (D) given below:

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → [2026 • Set 55-3-1]

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6. Two wires of same length and same area of cross-section but made of different material of resistivities ρ_1 and ρ_2 are connected in series. The equivalent resistivity of the combination is:

(A) $\frac{1}{2}(\rho_1 + \rho_2)$

(B) $\rho_1 + \rho_2$

(C) $\sqrt{\rho_1 \rho_2}$

(D) $2(\rho_1 + \rho_2)$

Check Solution → [2026 • Set 55-4-1]

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7. Directions: Questions are Assertion (A) and Reason (R) type. Two statements are given - one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (A), (B), (C) and (D).

Assertion (A): The work done, in taking a unit charge around a closed loop of an electric circuit involving cells and resistors in the loop, is zero.

Reason (R): The potential at a point depends on the location of the point in the loop. After completing one round, the charge comes back to the point of start.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

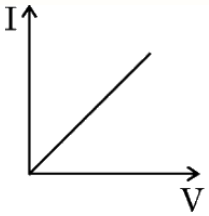
(C) Assertion (A) is true, but Reason (R) is false.

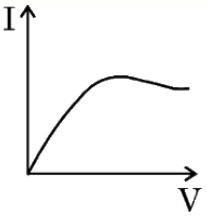
(D) Both Assertion (A) and Reason (R) are false.

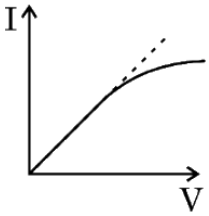
Check Solution → [2026 • Set 55-4-1]

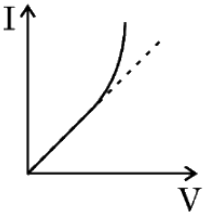
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8. If resistor X were made of manganin and readings for V and I are taken without switching off the circuit, the graph between V and I will be as:

(A) 

(B) 

(C) 

(D) 

(A) Graph (A)

(B) Graph (B)

(C) Graph (C)

(D) Graph (D)

Check Solution → [2026 • Set 55-4-1]

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9. Error in the value of X obtained from different sets of voltmeter and ammeter readings, is:

(A) due to error in voltmeter reading only.

(B) due to error in ammeter reading only.

(C) equal to the sum of error in voltmeter reading and error in ammeter reading.

(D) equal to error in voltmeter reading divided by the error in ammeter reading.

Check Solution → [2026 • Set 55-4-1]

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10. If the movable end of rheostat is moved towards P, then:

(A) reading in ammeter decreases and reading in voltmeter increases.

(B) readings in both voltmeter and ammeter increase.

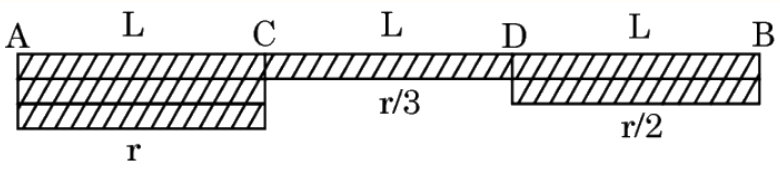
(C) reading in ammeter increases and reading in voltmeter decreases.

(D) readings in both voltmeter and ammeter decrease.

Check Solution → [2026 • Set 55-4-1]

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11. Suppose the unknown resistance X is replaced by a wire made of the same metal. This wire consists of three parts, of the same length L but has radii r, $r/3$ and $r/2$ as shown in the figure.



For a particular setting of the rheostat, let v_1, v_2 and v_3 be the value of drift velocities in parts AC, CD and DB. Then:

(A) $v_1 > v_2 > v_3$

(B) $v_2 > v_3 > v_1$

(C) $v_3 > v_2 > v_1$

(D) $v_1 = v_2 = v_3$

Check Solution → **[2026 • Set 55-4-1]**

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12. Consider the same wire, as shown in figure in question (iv) (a) connected in place of X. For a particular setting of rheostat, let E_1, E_2 and E_3 be the value of electric fields in part AC, CD and DB. Then:

(A) $E_1 = E_2 = E_3$

(B) $E_1 > E_2 > E_3$

(C) $E_2 > E_3 > E_1$

(D) $E_2 > E_1 > E_3$

Check Solution → **[2026 • Set 55-4-1]**

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13. A wire is gradually stretched to make it 100% longer. Its resistance will increase by:

(A) 100%

(B) 200%

(C) 300%

(D) 400%

Check Solution → **[2026 • Set 55-4-2]**

---

14. Which of the following characteristics of electrons determines the current in the conductor?

(A) Drift velocity only

(B) Thermal velocity only

(C) Both drift velocity and thermal velocity

(D) Neither drift nor thermal velocity

Check Solution → [2026 • Set 55-4-2]

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15. Which of the following statements is true about mobility of charge carriers in a metal?

(A) Mobility increases with increase in applied electric field.

(B) Mobility decreases with increase in temperature.

(C) Mobility is independent of the mass of the charge carrier.

(D) Mobility increases with increase in temperature.

Check Solution → [2026 • Set 55-4-3]

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16. The resistivity ρ of a metal increases with rise in temperature because:

(A) only relaxation time τ of electrons decreases with temperature.

(B) only number of electrons per unit volume 'n' increases appreciably.

(C) τ decreases with temperature but 'n' does not change appreciably.

(D) τ decreases with temperature and 'n' increases.

Check Solution → [2026 • Set 55-4-3]

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17. Two wires P and Q are made of the same material. The wire Q has twice the diameter and half the length as that of wire P. If the resistance of wire P is R, the resistance of the wire Q will be:

(A) R

(B) $\frac{R}{2}$

(C) $\frac{R}{8}$

(D) 2R

Check Solution → [2025 • Set 55-1-1]

18. The resistance of a wire of length L and radius r is R. Which one of the following would provide a wire of the same material of resistance $\frac{R}{2}$?

(A) Using a wire of same radius and twice the length

(B) Using a wire of same radius and half length

(C) Using a wire of same length and twice the radius

(D) Using a wire of same length and half the radius

Check Solution → [2025 • Set 55-1-2]

19. Two conductors A and B of the same material have their lengths in the ratio 1:2 and radii in the ratio 2:3. If they are connected in parallel across a battery, the ratio $\frac{v_A}{v_B}$ of the drift velocities of electrons in them will be:

(A) 2

(B) $\frac{1}{2}$

(C) $\frac{8}{9}$

(D) $\frac{9}{8}$

Check Solution → [2025 • Set 55-1-3]

20. A student has three resistors, each of resistance R. To obtain a resistance of $\frac{2}{3}$R, she should connect

(A) all the three resistors in series.

(B) all the three resistors in parallel.

(C) two resistors in series and then this combination in parallel with the third resistor.

(D) two resistors in parallel and then this combination in series with the third resistor.

Check Solution → [2025 • Set 55-2-1]

21. A battery of e.m.f. 12 V and internal resistance 0.5 Ω is connected to a 9.5 Ω resistor through a key. The ratio of potential difference between the two terminals of the battery, when the key is open to that when the key is closed, is

(A) 1.05

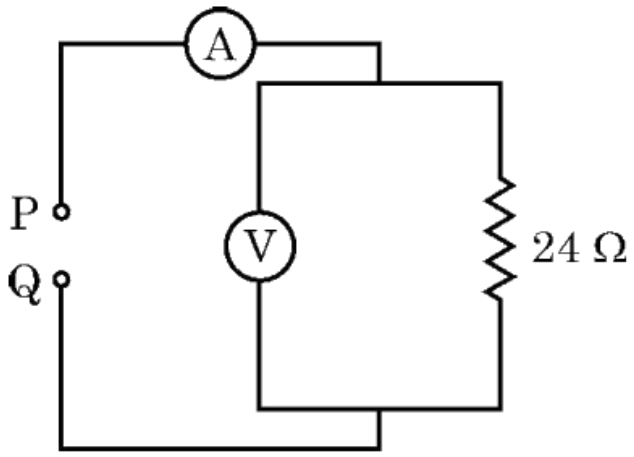
(B) 1

(C) 0.95

(D) 1.1

Check Solution → [2025 • Set 55-2-2]

22. Which pair of readings of ideal voltmeter and ideal ammeter in the given circuit is possible when a suitable power source of 3 Ω internal resistance is connected between P and Q?



(A) 12.0 V, 2.0 A

(B) 2.0 V, 0.5 A

(C) 6.0 V, 2.0 A

(D) 12 V, 0.5 A

Check Solution → [2025 • Set 55-2-3]

23. A current flows through a cylindrical conductor of radius R. The current density at a point in the conductor is $j = \alpha r$ (along its axis), here α is a constant and r is

distance from the axis of the conductor. The current flowing through the portion of the conductor from $r = 0$ to $r = \frac{R}{2}$ is proportional to :

(A) R

(B) R^2

(C) R^3

(D) R^4

Check Solution → [2025 • Set 55-4-1]

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24. When the switch of the circuit is turned on, the filament of the bulb glows instantaneously because :

(A) the electrons coming from the power source move fast through the initially empty filament.

(B) the filament may be old having low resistance.

(C) electric field is established instantaneously across the filament which pushes the electrons.

(D) free electrons in the filament travel with the speed of light.

Check Solution → [2025 • Set 55-4-2]

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25. Three wires A, B and C of the same material have lengths and area of cross-sections as $(2l, \frac{A}{2})$, (l, A) and $(\frac{l}{2}, 2A)$, respectively. If the resistances of these wires are R_A, R_B and R_C respectively, then :

(A) $R_A > R_B > R_C$

(B) $R_B > R_C > R_A$

(C) $R_B > R_A > R_C$

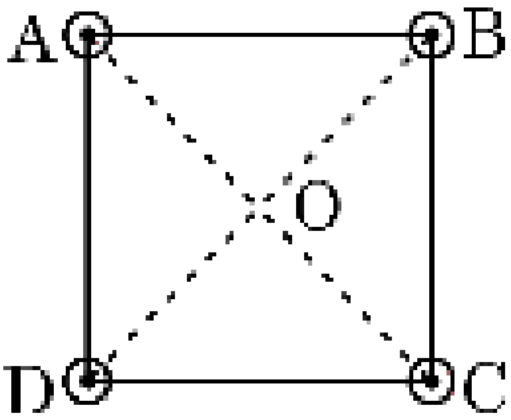
(D) $R_A > R_C > R_B$

Check Solution → [2025 • Set 55-4-3]

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26. Four resistors, each of resistance R and a key K are connected as shown in the figure. The equivalent resistance between points A and B when key K is open,

will be:



(A) 4R

(B) ∞

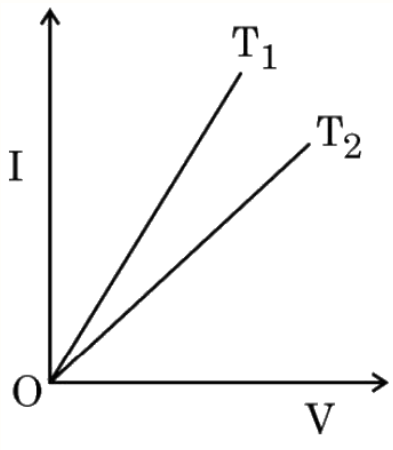
(C) $\frac{R}{4}$

(D) $\frac{4R}{3}$

Check Solution → [2025 • Set 55-5-1]

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27. The figure shows the voltage (V) versus the current (I) graphs for a wire at two temperatures T_1 and T_2. One can conclude that:



(A) $T_1 > T_2$

(B) $T_1 < T_2$

(C) $T_1 = T_2/3$

(D) $T_1 < T_2/3$

Check Solution → [2025 • Set 55-6-1]

28. If R_s and R_p are the equivalent resistances of n resistors, each of value R, in series and parallel combinations respectively, then the value of $(R_s - R_p)$ is:

(A) $\frac{n^2-1}{n} R$

(B) $\frac{n}{n^2-1} R$

(C) $(\frac{n-1}{n})^2 R$

(D) $\frac{(n+1)^2}{n^2} R$

Check Solution → [2025 • Set 55-6-1]

29. A wire of resistance R, connected to an ideal battery consumes a power P. If the wire is gradually stretched to double its initial length, and connected across the same battery, the power consumed will be:

(A) $\frac{P}{2}$

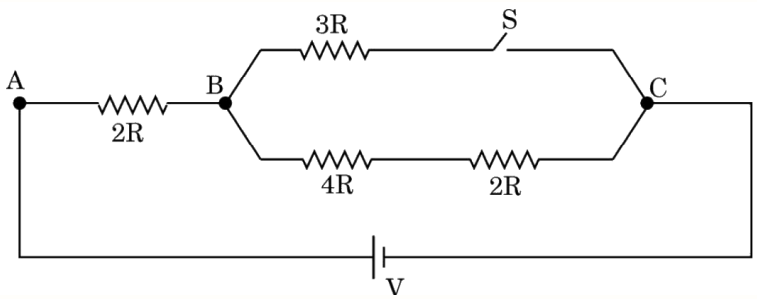
(B) $\frac{P}{4}$

(C) P

(D) 2P

Check Solution → [2025 • Set 55-6-2]

30. The ratio of potential difference across AB in the circuit shown for the case (i) when switch S is closed and (ii) when S is open is:



(A) $\frac{1}{4}$

(B) $\frac{1}{2}$

(C) 1

(D) 2

Check Solution → [2025 • Set 55-6-3]

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31. The effective resistance between points A and B in the given circuit is:

(A) 6 Ω

(B) 5 Ω

(C) $\frac{6}{5}$ Ω

(D) 2 Ω

Check Solution → [2025 • Set 55-7-1]

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32. **Case Study:** In a metallic conductor, an electron, moving due to thermal motion, suffers collisions with the heavy fixed ions but after collision, it will emerge out with the same speed but in random directions. If we consider all the electrons, their average velocity will be zero. When an electric field is applied, electrons move with an average velocity, known as drift velocity (v_d). The average time between successive collisions is known as relaxation time (τ). The magnitude of drift velocity per unit electric field is called mobility (μ). An expression for current through the conductor can be obtained in terms of drift velocity, number of electrons per unit volume (n), electronic charge ($-e$), and the cross-sectional area (A) of the conductor. This expression leads to an expression between current density ($\vec{J}$) and the electric field ($\vec{E}$). Hence, an expression for resistivity (ρ) of a metal is obtained. This expression helps us to understand increase in resistivity of a metal with increase in its temperature, in terms of change in the relaxation time (τ) and change in the number density of electrons (n).

(i) Consider two cylindrical conductors A and B, made of the same metal

connected in series to a battery. The length and the radius of B are twice that of A. If μ_A and μ_B are the mobility of electrons in A and B respectively, then $\frac{\mu_A}{\mu_B}$ is:

(A) $\frac{1}{2}$

(B) $\frac{1}{4}$

(C) 2

(D) 1

Check Solution → [2025 • Set 55-7-1]

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33. **(ii)** A wire of length 0.5 m and cross-sectional area 1.0×10^{-7} m^2 is connected to a battery of 2 V that maintains a current of 1.5 A in it. The conductivity of the material of the wire (in Ω^{-1} m^{-1}) is:

(A) 2.5×10^4

(B) 3.0×10^5

(C) 3.75×10^6

(D) 5.0×10^7

Check Solution → [2025 • Set 55-7-1]

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34. **(iii)** The temperature coefficient of resistance of nichrome is 1.70×10^{-4} $°C^{-1}$. In order to increase resistance of a nichrome wire by 8.5%, the temperature of the wire should be increased by:

(A) 250°C

(B) 500°C

(C) 850°C

(D) 1000°C

Check Solution → [2025 • Set 55-7-1]

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35. **(iv) (a)** Consider the contribution of the following two factors I and II in resistivity of a metal:

I. Relaxation time of electrons

II. Number of electrons per unit volume

The resistivity of a metal increases with increase in its temperature because:

OR

(b) A steady current flows in a copper wire of non-uniform cross-section.

Consider the following three physical quantities:

I. Electric field

II. Current density

III. Drift speed

Then at the different points along the wire:

(A) (a) I decreases and II increases. OR (b) II and III change, but I is constant.

(B) (a) I increases and II is almost constant. OR (b) I and II change, but III is constant.

(C) (a) Both I and II increase. OR (b) I and III change, but II is constant.

(D) (a) I decreases and II is almost constant. OR (b) All I, II and III change.

Check Solution → **[2025 • Set 55-7-1]**

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36. A steady current I is passed through a conductor at room temperature for time t. It is observed that its temperature rises by $0.5^{\circ}C$. If 2I current is passed through the conductor (at room temperature) for the same duration, the rise in its temperature will be approximately:

(A) $1.0^{\circ}C$

(B) $1.5^{\circ}C$

(C) $2.0^{\circ}C$

(D) $4.0^{\circ}C$

Check Solution → **[2025 • Set 55-7-3]**

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37. Consider the circuit shown in the figure. The potential difference between points A and B is:

(A) $6\ V$

(B) $8\ V$

(C) $9\ V$

(D) $12\ V$

Check Solution → [2024 • Set 55-1-1]

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38. A student is asked to connect four cells, each of emf E and internal resistance r, in series. But she/he connects one cell wrongly in series with the other cells. The equivalent emf and the equivalent internal resistance of the combination will be:

(A) 4E and 2r

(B) 4E and 3r

(C) 3E and 4r

(D) 2E and 4r

Check Solution → [2024 • Set 55-1-2]

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39. In a uniform straight wire, conduction electrons move along $+x$ direction. Let $\vec{E}$ and $\vec{j}$ be the electric field and current density in the wire, respectively. Then:

(A) $\vec{E}$ and $\vec{j}$ both are along $+x$ direction.

(B) $\vec{E}$ and $\vec{j}$ both are along $-x$ direction.

(C) $\vec{E}$ is along $+x$ direction, but $\vec{j}$ is along $-x$ direction.

(D) $\vec{E}$ is along $-x$ direction, but $\vec{j}$ is along $+x$ direction.

Check Solution → [2024 • Set 55-1-3]

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40. Electrons drift with speed v_d in a conductor with potential difference V across its ends. If V is reduced to $(\frac{V}{2})$, their drift speed will become:

(A) $\frac{v_d}{2}$

(B) v_d

(C) $2v_d$

(D) $4v_d$

Check Solution → [2024 • Set 55-2-1]

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41. When the terminals of a cell are connected to a conductor of resistance R, an electric current flows through the circuit. The electrolyte of the cell also offers some resistance in the path of the current, like the conductor. This resistance offered by the electrolyte is called internal resistance of the cell (r). It depends upon the nature of the electrolyte, the area of the electrodes immersed in the electrolyte and the temperature. Due to internal resistance, a part of the energy supplied by the cell is wasted in the form of heat.
When no current is drawn from the cell, the potential difference between the two electrodes is known as emf of the cell (ε). With a current drawn from the cell, the potential difference between the two electrodes is termed as terminal potential difference (V).
Choose the incorrect statement:

(A) The potential difference (V) between the two terminals of a cell in a closed circuit is always less than its emf (ε), during discharge of the cell.

(B) The internal resistance of a cell decreases with the decrease in temperature of the electrolyte.

(C) When current is drawn from the cell then $V=\varepsilon-Ir$.

(D) The graph between potential difference between the two terminals of the cell (V) and the current (I) through it is a straight line with a negative slope.

Check Solution → [2024 • Set 55-2-1]

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42. Two cells of emfs 2.0 V and 6.0 V and internal resistances 0.1 Ω and 0.4 Ω respectively, are connected in parallel. The equivalent emf of the combination will be:

(A) 2.0 V

(B) 2.8 V

(C) 6.0 V

(D) 8.0 V

Check Solution → [2024 • Set 55-2-1]

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43. Dipped in the solution, the electrode exchanges charges with the electrolyte. The positive electrode develops a potential V_+ ($V_+>0$), and the negative electrode develops a potential $-(V_-)$ ($V_->0$), relative to the electrolyte adjacent to it. When no current is drawn from the cell then:

(A) $\varepsilon=V_++V_->0$

(B) $\varepsilon=V_+-V_->0$

(C) $\varepsilon=V_++V_-<0$

(D) $\varepsilon=V_++V_-=0$

Check Solution → [2024 • Set 55-2-1]

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44. (a) Five identical cells, each of emf 2 V and internal resistance 0.1 Ω are connected in parallel. This combination in turn is connected to an external resistor of 9.98 Ω. The current flowing through the resistor is:

OR

(b) Potential difference across a cell in the open circuit is 6 V. It becomes 4 V when a current of 2 A is drawn from it. The internal resistance of the cell is:

(A) (a) 0.05 A (b) 1.0 Ω

(B) (a) 0.1 A (b) 1.5 Ω

(C) (a) 0.15 A (b) 2.0 Ω

(D) (a) 0.2 A (b) 2.5 Ω

Check Solution → [2024 • Set 55-2-1]

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45. **Assertion (A):** When electrons drift in a conductor, it does not mean that all free electrons in the conductor are moving in the same direction.

Reason (R): The drift velocity is superposed over large random velocities of electrons.

(A) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) If both Assertion (A) and Reason (R) are true and Reason (R) is not the correct explanation of Assertion (A).

(C) If Assertion (A) is true and Reason (R) is false.

(D) If both Assertion (A) and Reason (R) are false.

Check Solution → **[2024 • Set 55-4-1]**

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46. A battery supplies 0.9 A current through a 2 Ω resistor and 0.3 A current through a 7 Ω resistor when connected one by one. The internal resistance of the battery is:

(A) 2 Ω

(B) 1.2 Ω

(C) 1 Ω

(D) 0.5 Ω

Check Solution → **[2024 • Set 55-5-1]**

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47. An ammeter and a voltmeter are connected in series to a battery. Their readings are noted as 'A' and 'V' respectively. If a resistor is connected in parallel with the voltmeter, then:

(A) A will increase, V will decrease.

(B) A will decrease, V will increase.

(C) Both A and V will decrease.

(D) Both A and V will increase.

Check Solution → **[2024 • Set 55-5-2]**

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48. A heater coil rated as (P, V) is cut into two equal parts. One of the parts is then connected to a battery of V volt. The power consumed by it will be:

(A) P

(B) $\frac{P}{2}$

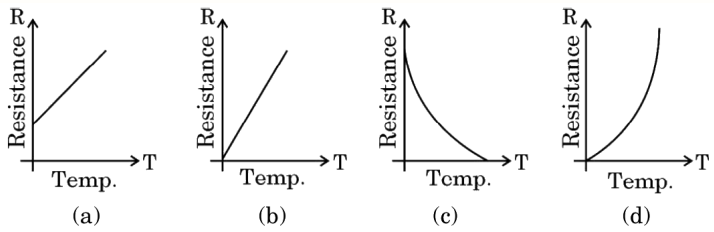
(C) $\frac{P}{4}$

(D) 2P

Check Solution → [2024 • Set 55-5-3]

---

49. For a metallic conductor, the correct representation of variation of resistance R with temperature T is:



(A) Graph (a)

(B) Graph (b)

(C) Graph (c)

(D) Graph (d)

Check Solution → [2023 • Set 55-1-1]

---

50. The potential difference across a cell in an open circuit is 8 V. It falls to 4 V when a current of 4 A is drawn from it. The internal resistance of the cell is:

(A) 4 Ω

(B) 8 Ω

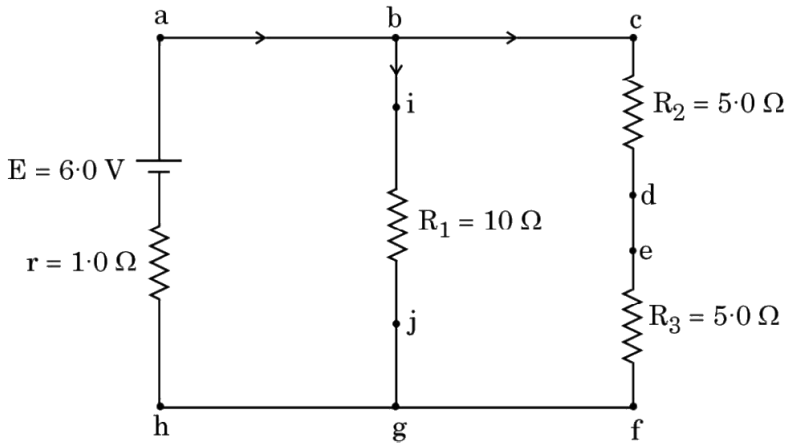
(C) 2 Ω

(D) 1 Ω

Check Solution → [2023 • Set 55-1-1]

---

51. The following figure shows a circuit diagram. We can find the currents through and potential differences across different resistors using Kirchhoff's rules.

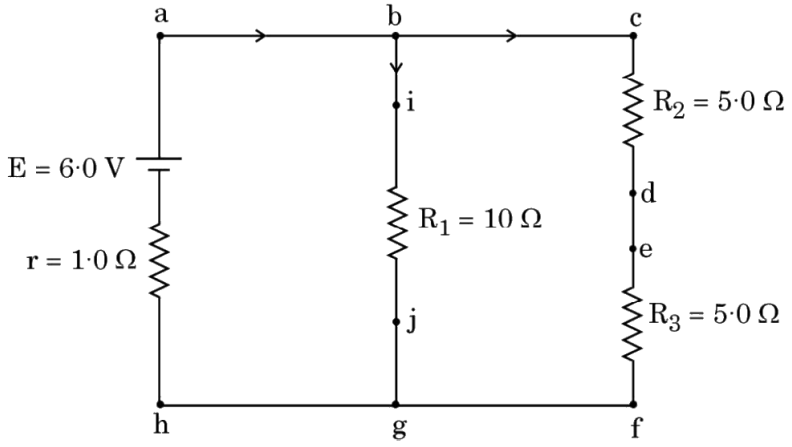


Answer the following questions based on the above:

Which points are at the same potential in the circuit?

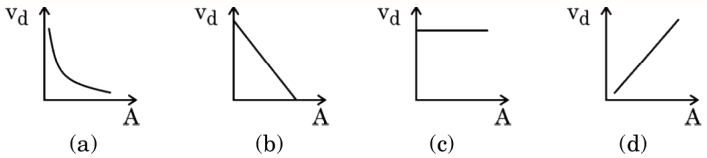
Check Solution → [2023 • Set 55-1-1]

52. Refer to the circuit diagram. What is the current through arm bg?



Check Solution → [2023 • Set 55-1-1]

53. A steady current flows through a metallic wire whose area of cross-section (A) increases continuously from one end of the wire to the other. The magnitude of drift velocity (v_d) of the free electrons as a function of 'A' can be shown by:



(A) Graph (a)

(B) Graph (b)

(C) Graph (c)

(D) Graph (d)

Check Solution → [2023 • Set 55-1-1]

---

54. The masses of two cylindrical wires of copper are in the ratio of 1:3 and their lengths are in the ratio of 5:3. The ratio of their resistances will be:

(A) 1:8

(B) 2:5

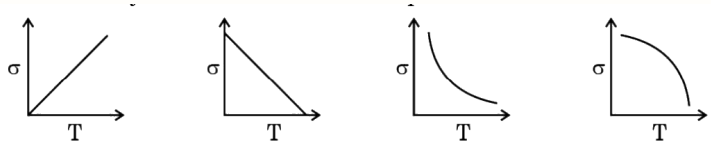
(C) 2:3

(D) 3:5

Check Solution → [2023 • Set 55-1-2]

---

55. Which one of the following is the correct representation of variation of conductivity of a conductor with temperature?



(a) (b) (c) (d)

(A) Graph (a)

(B) Graph (b)

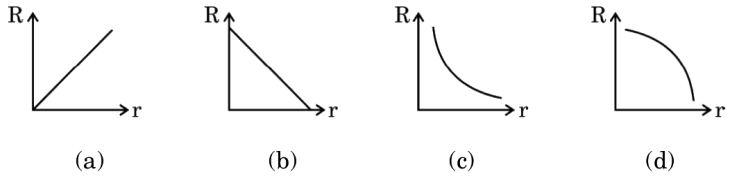
(C) Graph (c)

(D) Graph (d)

Check Solution → [2023 • Set 55-1-2]

---

56. The correct graph showing the variation of the resistance (R) of a cylindrical metal wire as a function of its radius (r), keeping its length and temperature constant, is:



(A) Graph (a)

(B) Graph (b)

(C) Graph (c)

(D) Graph (d)

Check Solution → [2023 • Set 55-1-3]

---

57. The number of electrons flowing through a conductor per second is 3.3×10^{19}. The current flowing through the conductor is:

(A) 2.0 A

(B) 3.3 A

(C) 4.8 A

(D) 5.3 A

Check Solution → [2023 • Set 55-1-3]

---

58. A current of 0.8 A flows in a conductor of 40 Ω for 1 minute. The heat produced in the conductor will be

(A) 1445 J

(B) 1536 J

(C) 1569 J

(D) 1640 J

Check Solution → [2023 • Set 55-2-1]

---

59. A cell of emf E is connected across an external resistance R. When current I is drawn from the cell, the potential difference across the electrodes of the cell drops to V. The internal resistance 'r' of the cell is

(A) $(\frac{E-V}{V})R$

(B) $(\frac{E-V}{E})R$

(C) $\frac{E-V}{I}$

(D) $(\frac{E-V}{I})r$

Check Solution → [2023 • Set 55-2-1]

---

60. A steady current of 8 mA flows through a wire. The number of electrons passing through a cross-section of the wire in 10 s is

(A) 4.0×10^{16}

(B) 5.0×10^{17}

(C) 1.6×10^{16}

(D) 1.0×10^{17}

Check Solution → [2023 • Set 55-2-2]

---

61. A conductor of 10 Ω is connected across a 6 V ideal source. The power supplied by the source to the conductor is

(A) 1.8 W

(B) 2.4 W

(C) 3.6 W

(D) 7.2 W

Check Solution → [2023 • Set 55-2-2]

---

62. The current in a device varies with time t as I = 6t, where I is in mA and t is in s. The amount of charge that passes through the device during t = 0 s to t = 3 s is

(A) 10 mC

(B) 18 mC

(C) 27 mC

(D) 54 mC

Check Solution → [2023 • Set 55-2-3]

---

63. **Assertion (A):** The internal resistance of a cell is constant.
Reason (R): Ionic concentration of the electrolyte remains same during use of a cell.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false and Reason (R) is also false.

Check Solution → [2023 • Set 55-3-1]

---

64. The current density due to drift of electrons in a conductor is given by (symbols have their usual meanings):

(A) $neAv_d$

(B) $\frac{neA}{v_d}$

(C) $\frac{ne}{v_d}$

(D) nev_d

Check Solution → [2023 • Set 55-3-1]

---

65. The emf and internal resistance of a cell are E and r respectively. It is connected across an external resistance $R = 2r$. The potential drop across the terminals of the cell will be:

(A) $\frac{E}{3}$

(B) $\frac{E}{2}$

(C) $\frac{2E}{3}$

(D) E

Check Solution → [2023 • Set 55-3-2]

---

66. The SI unit of mobility of charge carriers is:

(A) $\Omega\ s^{-1}$

(B) $m^2V^{-1}s^{-1}$

(C) $m\ s^{-1}V^{-1}$

(D) $\Omega\ m$

Check Solution → [2023 • Set 55-3-2]

---

67. A battery of emf E and internal resistance r is connected to an external circuit. The potential drop within the battery is proportional to:

(A) current in the circuit

(B) total resistance of the circuit

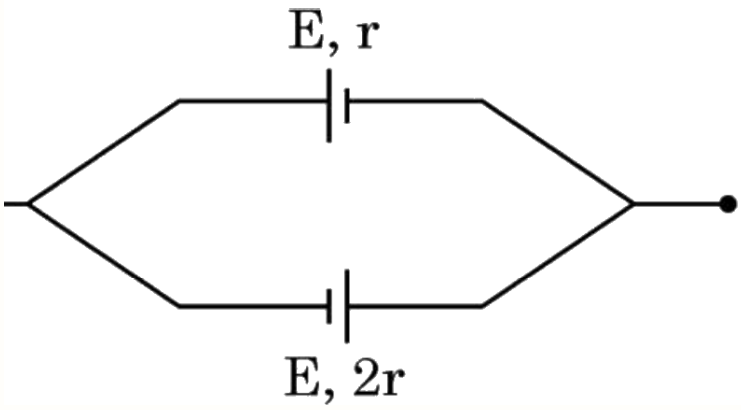
(C) emf of the battery

(D) power dissipated in the circuit

Check Solution → [2023 • Set 55-3-3]

---

68. Two cells of emf E each and internal resistances r and 2r are connected in parallel as shown in the figure. The equivalent emf of the combination is:



(A) zero

(B) $\frac{E}{2}$

(C) $\frac{3E}{2}$

(D) E

Check Solution → [2023 • Set 55-3-3]

---

69. Two wires of equal length and radii r and 2r are connected in series. Their resistivities are 2ρ and ρ respectively. For the same current passing through the combination, the potential drop across their ends will be in the ratio of:

(A) 2 : 1

(B) 1 : 2

(C) 4 : 1

(D) 8 : 1

Check Solution → [2023 • Set 55-3-3]

---

70. **Assertion (A):** When three electric bulbs of power 200 W, 100 W and 50 W are connected in series to a source, the power consumed by the 50 W bulb is maximum.

Reason (R): In a series circuit, current is the same through each bulb, but the potential difference across each bulb is different.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

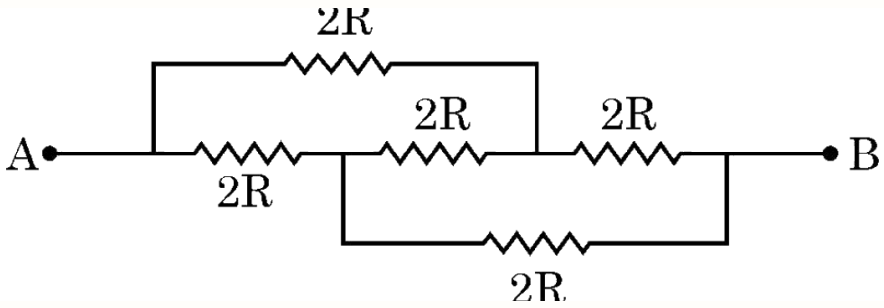
(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false and Reason (R) is also false.

Check Solution → [2023 • Set 55-4-1]

---

71. **Assertion (A):** The equivalent resistance between points A and B in the given network is 2R.
Reason (R): All the resistors are connected in parallel.



(A) Both Assertion (A) and Reason (R) are true and (R) is the correct explanation of (A).

(B) Both Assertion (A) and Reason (R) are true and (R) is NOT the correct explanation of (A).

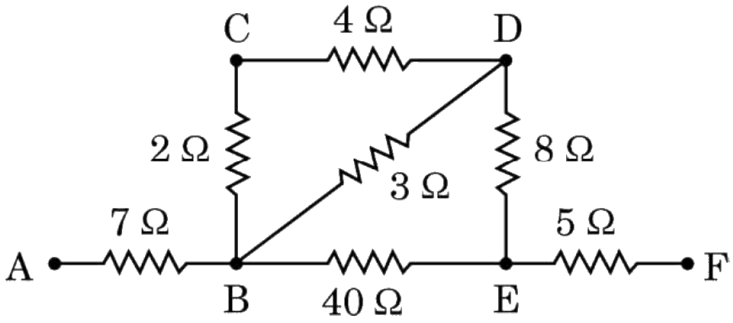
(C) Assertion (A) is true and Reason (R) is false.

(D) Assertion (A) is false and Reason (R) is also false.

Check Solution → [2023 • Set 55-5-1]

---

72. **Assertion (A):** The given figure does not show a balanced Wheatstone bridge.
Reason (R): For a balanced bridge small current should flow through the galvanometer.



(A) Both Assertion (A) and Reason (R) are true and (R) is the correct explanation of (A).

(B) Both Assertion (A) and Reason (R) are true and (R) is NOT the correct explanation of (A).

(C) Assertion (A) is true and Reason (R) is false.

(D) Assertion (A) is false and Reason (R) is also false.

Check Solution → **[2023 • Set 55-5-3]**

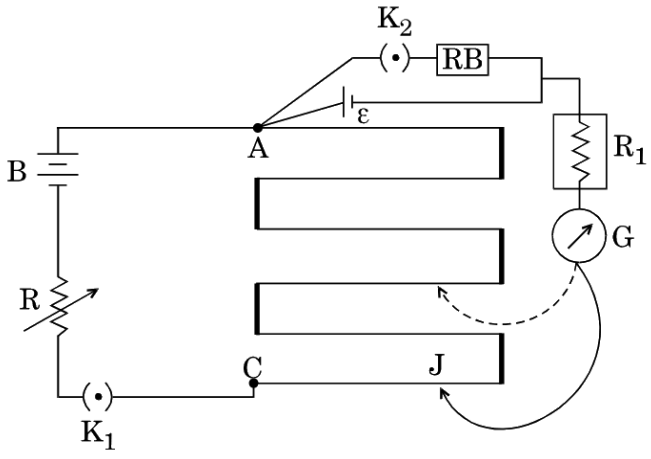
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73. Two wires X and Y of the same material and of equal lengths having area of cross-section A and 2A respectively, are connected in parallel across an ideal battery of emf E. What is the ratio of current density (J_X / J_Y) in them?

Check Solution → **[2021]**

---

74. A potentiometer draws a steady current from the battery (B) of emf E in the circuit as shown in the figure. The balancing length for the cell of emf ε when key K_2 is kept open, is $_1$. If key K_2 is closed and a resistance S is introduced in the resistance box (RB), the balancing length becomes $_2$.



The balancing length $_1 > _2$ because

(A) $E > \varepsilon$.

(B) positive terminal of the battery is connected to positive terminal of the cell.

(C) the voltage drop across the cell is greater than that across the battery.

(D) terminal potential difference of the cell is less than its emf.

Check Solution → [2021]

---

75. The high resistance R_1 in the circuit is used to

(A) keep the balance point near the mid-point of the potentiometer wire AC.

(B) reduce the current supplied by the battery (B) to zero at balance point.

(C) protect the galvanometer from damage due to large current.

(D) make the galvanometer more sensitive.

Check Solution → [2021]

---

76. If the key K_2 is kept open and resistance R_1 is increased, the balancing length $_1$ of the wire will

(A) increase.

(B) decrease.

(C) remain unaffected.

(D) become zero.

Check Solution → [2021]

---

77. Two potentiometers P and Q use the same wire of lengths 5 m and 10 m respectively, between A and C. They are connected to battery (B) in the circuit separately. The potentiometer which is preferred to compare the emfs of two cells is

(A) P, because potential gradient is less.

(B) Q, because potential gradient is less.

(C) P, because current is more.

(D) Q, because resistance of the potentiometer is less.

Check Solution → [2021]

---

78. The reason for preferring manganin wire in a potentiometer is because

(A) of high value of its temperature coefficient of resistivity.

(B) of its low resistivity.

(C) of low value of its temperature coefficient of resistivity.

(D) its resistance decreases with increase in temperature.

Check Solution → [2021]

---

79. A potentiometer can measure emf of a cell because

(A) the sensitivity of potentiometer is large.

(B) no current is drawn from the cell at balance.

(C) no current flows in the wire of potentiometer at balance.

(D) internal resistance of cell is neglected.

Check Solution → [2020 • Set 55-1-1]

---

80. Two resistors R_1 and R_2 of 4 Ω and 6 Ω are connected in parallel across a battery. The ratio of power dissipated in them, $P_1 : P_2$ will be

(A) 4 : 9

(B) 3 : 2

(C) 9 : 4

(D) 2 : 3

Check Solution → [2020 • Set 55-1-1]

---

81. Kirchhoff's first rule at a junction in an electrical network, deals with conservation of

(A) energy

(B) charge

(C) momentum

(D) both energy and charge

Check Solution → [2020 • Set 55-1-2]

---

82. The resistance of a metal wire increases with increasing temperature on account of

(A) decrease in free electron density.

(B) decrease in relaxation time.

(C) increase in mean free path.

(D) increase in the mass of electron.

Check Solution → [2020 • Set 55-1-2]

---

83. Two unequal resistors are connected in series across a battery. Then the

(A) potential difference across each resistor is the same.

(B) current in the smaller resistor is larger.

(C) potential difference across the bigger resistor is greater.

(D) power dissipated in both resistors is the same.

Check Solution → [2020 • Set 55-1-3]

84. A cell of internal resistance r connected across an external resistance R can supply maximum current when

(A) $R = r$

(B) $R \gg r$

(C) $R = \frac{r}{2}$

(D) $R = 0$

Check Solution → [2020 • Set 55-2-1]

85. In a current carrying conductor, the ratio of the electric field and the current density at a point is called

(A) Resistivity

(B) Conductivity

(C) Resistance

(D) Mobility

Check Solution → [2020 • Set 55-2-1]

86. Resistivity of a given conductor depends upon

(A) temperature.

(B) length of conductor.

(C) area of cross-section.

(D) shape of the conductor.

Check Solution → [2020 • Set 55-2-2]

87. The ratio of current density and electric field is called

(A) Resistivity

(B) Conductivity

(C) Drift velocity

(D) Mobility

Check Solution → [2020 • Set 55-2-2]

---

88. For a fixed potential difference applied across a conductor, the drift speed of free electrons does not depend upon

(A) free electron density in the conductor.

(B) mass of the electrons.

(C) length of the conductor.

(D) temperature of the conductor.

Check Solution → [2020 • Set 55-2-3]

---

89. The electrical resistance of a conductor

(A) varies directly proportional to its area of cross-section.

(B) decreases with increase in its temperature.

(C) decreases with increase in its conductivity.

(D) is independent of its shape but depends only on its volume.

Check Solution → [2020 • Set 55-3-1]

---

90. $m^2\ V^{-1}\ s^{-1}$ is the SI unit of which of the following?

(A) Drift velocity

(B) Mobility

(C) Resistivity

(D) Potential gradient

Check Solution → [2020 • Set 55-3-1]

---

91. The element of a heater is rated (P, V). If it is connected across a source of voltage $\frac{V}{2}$, then the power consumed by it will be

(A) P

(B) 2P

(C) $\frac{P}{2}$

(D) $\frac{P}{4}$

Check Solution → [2020 • Set 55-3-1]

---

92. n resistors, each of resistance R are connected (a) in series, and (b) in parallel. Each combination is then connected to a source of emf E. The ratio of heat produced per second in the two cases will be

(A) n:1

(B) 1:n

(C) n^2:1

(D) 1:n^2

Check Solution → [2020 • Set 55-3-2]

---

93. A uniform wire of 16 Ω resistance is bent round in the form of a circle. When connected in a circuit between its two diametrically opposite points, its effective resistance will be

(A) 8 Ω

(B) 16 Ω

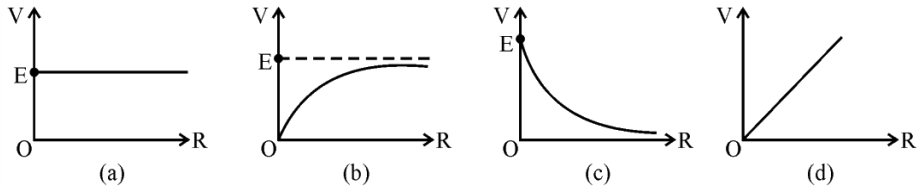
(C) 4 Ω

(D) 2 Ω

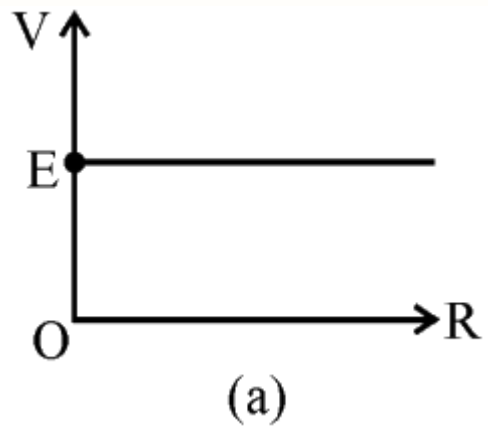
Check Solution → [2020 • Set 55-3-3]

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94. A cell of emf (E) and internal resistance r is connected across a variable external resistance R. The graph of terminal potential difference V as a function of R is —

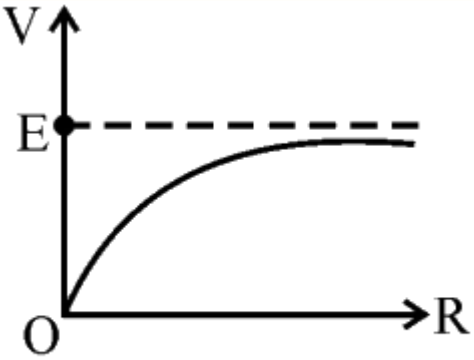

V
E
O
R
(a)
V
E
O
R
(b)
V
E
O
R
(c)
V
O
R
(d)

(A)



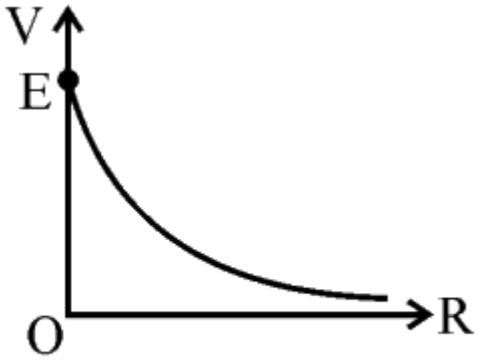
(a)

(B)



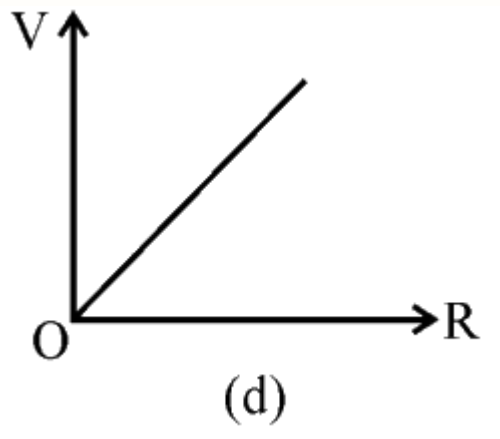
(b)

(C)



(c)

(D)



Check Solution → [2020 • Set 55-4-1]

---

95. Fill in the blank with appropriate answer.

To minimize the percentage error in the determination of unknown resistance of a conductor in meter bridge experiment, the balance point is adjusted near _____ of the wire.

OR

In potentiometer, a long uniform wire is used to _____ potential gradient along the wire.

Check Solution → [2020 • Set 55-4-1]

---

96. A copper wire of non-uniform area of cross-section is connected to a d.c. battery. The physical quantity which remains constant along the wire is _____ .

Check Solution → [2020 • Set 55-5-1]

---

97. How does the mobility of electrons in a conductor change, if the potential difference applied across the conductor is doubled, keeping the length and temperature of the conductor constant?

Check Solution → [2019 • Set 55-1-1]

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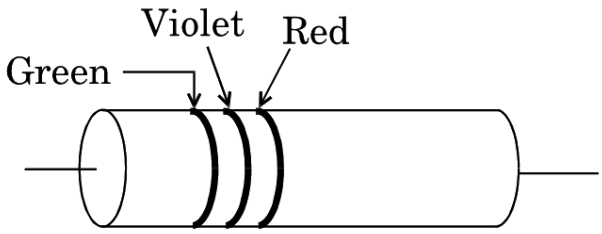
98. When a potential difference is applied across the ends of a conductor, how is the drift velocity of the electrons related to the relaxation time?

Check Solution → [2019 • Set 55-1-2]

99. How is the drift velocity in a conductor affected with the rise in temperature?

Check Solution → [2019 • Set 55-1-3]

100. A carbon resistor is shown in the figure. Using colour code, write the value of the resistance.



Check Solution → [2019 • Set 55-3-1]

101. Draw a plot of resistivity of copper as a function of temperature.

Check Solution → [2019 • Set 55-3-3]

102. Under what condition will the current in a wire be the same when connected in series and in parallel of n identical cells each having internal resistance r and external resistance R?

Check Solution → [2019 • Set 55-4-1]

103. Why is the terminal voltage of a cell generally less than the emf of the cell?

Check Solution → [2019 • Set 55-4-2]

104. Name the charge carriers for the flow of current in a (i) conductor and (ii) electrolyte.

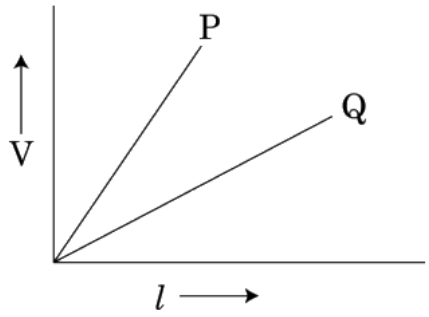
Check Solution → [2019 • Set 55-4-3]

105. Nichrome and copper wires of same length and same radius are connected in series. Current I is passed through them. Which wire gets heated up more? Justify your answer.

Check Solution → [2017]

---

106. The variation of potential difference V with length l in case of two potentiometer wires P and Q is as shown. Which one of these will you prefer for comparing emfs of two primary cells and why?



Check Solution → [2016]

---

107. Define the term conductivity of a conductor. On what factors does it depend?

Check Solution → [2016]

---

108. Define mobility of a charge carrier. What is its relation with relaxation time?

Check Solution → [2016]

---

109. Two wires one of copper and other of manganin have same resistance and equal length. Which wire is thicker and why?

Check Solution → [2016]

---

110. Define the term 'relaxation time' in a conductor.

Check Solution → [2016]

111. Distinguish between emf and terminal voltage of a cell.

Check Solution → [2015]

112. Define the term 'Mobility' of charge carriers in a conductor. Write its S.I. unit.

Check Solution → [2014]

113. Show variation of resistivity of copper as a function of temperature in a graph.

Check Solution → [2014]

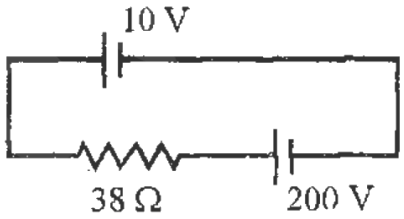
114. Define the term 'electrical conductivity' of a metallic wire. Write its S.I. unit.

Check Solution → [2014]

115. Define the term 'drift velocity' of charge carriers in a conductor and write its relationship with the current flowing through it.

Check Solution → [2014]

116. A 10 V battery of negligible internal resistance is connected across a 200 V battery and a resistance of 38 Ω as shown in the figure. Find the value of the current in circuit.



Check Solution → [2013]

117. The emf of a cell is always greater than its terminal voltage. Why? Give reason.

Check Solution → [2013]

---

118. Two conducting wires X and Y of same diameter but different materials are joined in series across a battery. If the number density of electrons in X is twice that in Y, find the ratio of drift velocity of electrons in the two wires.

Check Solution → [2012]

---

119. Two wires of equal length, one of copper and the other of manganin have the same resistance. Which wire is thicker?

Check Solution → [2012]

---

120. When electrons drift in a metal from lower to higher potential, does it mean that all the free electrons of the metal are moving in the same direction?

Check Solution → [2012]

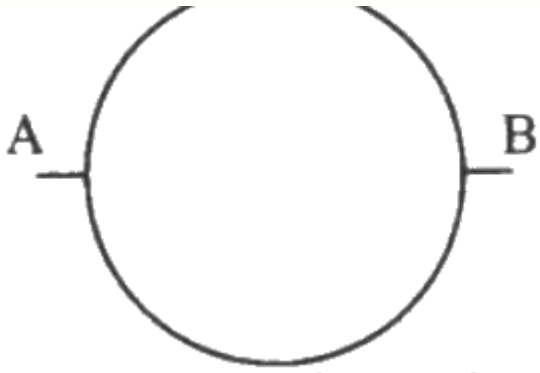
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121. A resistance R is connected across a cell of emf ε and internal resistance r. A potentiometer now measures the potential difference between the terminals of the cell as V. Write the expression for r in terms of ε, V and R.

Check Solution → [2011 • Set 55-1-1]

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122. A wire of resistance 8R is bent in the form of a circle. What is the effective resistance between the ends of a diameter AB?



Check Solution → [2010]

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123. The plot of the variation of potential difference across a combination of three identical cells in series, versus current is as shown below. What is the emf of each cell?

Check Solution → [2008]

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124. Sketch a graph showing variation of resistivity of carbon with temperature.

Check Solution → [2006]

---

125. Define electrical conductivity of a conductor and give its S.I. unit.

Check Solution → [2003]

---

126. What happens to the power dissipation if the value of electric current passing through a conductor of constant resistance is doubled?

Check Solution → [2003]

SECTION B

2-Mark Questions

VSA · 84 questions

> *Section B of the paper, 2 marks each. Sorted newest year first.*

127. What is the order of magnitude of drift velocity of electrons in a conductor? Deduce the relation between the current flowing through a conductor and drift velocity of electrons in it.

Check Solution → **[2026 • Set 55-1-1]**

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128. A heating element using nichrome is connected to a 220 V supply. Initially it draws a current of 2·9 A. After some time, the current attains a steady value of 2·5 A. Find the steady temperature of the heating element if the room temperature is 27 °C. The temperature coefficient of resistance of nichrome is $1{\cdot}7\times 10^{-4}\ {}^{\circ}C^{-1}$.

Check Solution → **[2026 • Set 55-2-1]**

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129. Calculate the temperature at which the resistance of a conductor becomes 20% more than its resistance at 27 °C. The value of the temperature coefficient of resistance of the material of conductor is $2{\cdot}0\times 10^{-4}\ {}^{\circ}C^{-1}$.

Check Solution → **[2026 • Set 55-2-2]**

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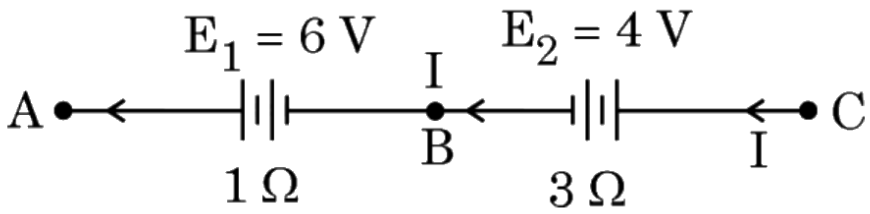
130. (a) Why is manganin used in wire-wound standard resistors? Calculate the temperature at which the resistance of a wire increases by 25% of its resistance at room temperature 27·5 °C. The temperature coefficient of resistance of the material of the wire is $0{\cdot}004\ {}^{\circ}C^{-1}$.

Check Solution → **[2026 • Set 55-2-3]**

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131. In the given figure, a steady current I flows through the circuit when points A and C are connected by a wire of negligible resistance. Find the potential

difference between points B and C.



Check Solution → [2026 • Set 55-3-1]

132.

A battery of emf 21 V and internal resistance 3 Ω is connected to a resistor. If the current in the circuit is 3 A, find:

(i) the resistance of the resistor.

(ii) the terminal voltage of the battery.

Check Solution → [2026 • Set 55-3-1]

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133. An electric iron rated 2.2 kW, 220 V is operated at 110 V supply. Find:

(i) its resistance, and

(ii) heat produced by it in 10 minutes.

Check Solution → [2026 • Set 55-5-1]

134.

A current of 4.0 A flows through a wire of length 1 m and cross-sectional area 1.0 mm^2, when potential difference of 2 V is applied across its ends. Calculate the resistivity of the material of the wire.

Check Solution → [2026 • Set 55-5-1]

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135. A battery of emf E and internal resistance r is connected to a rheostat. When a current of 2 A is drawn from the battery, the potential difference across the rheostat is 5 V. The potential difference becomes 4 V when a current of 4 A is drawn from the battery. Calculate the value of E and r.

Check Solution → [2025 • Set 55-1-1]

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136. Show that $\vec{E} = \rho \vec{j}$ leads to Ohm's law. Write a condition in which the Ohm's law is not valid for a material.

Check Solution → [2025 • Set 55-1-2]

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137. n identical cells, each of e.m.f. E and internal resistance r, are connected in series. Later on it was found out that two cells X and Y are connected in reverse polarities. Calculate the potential difference across the cell X.

Check Solution → [2025 • Set 55-1-3]

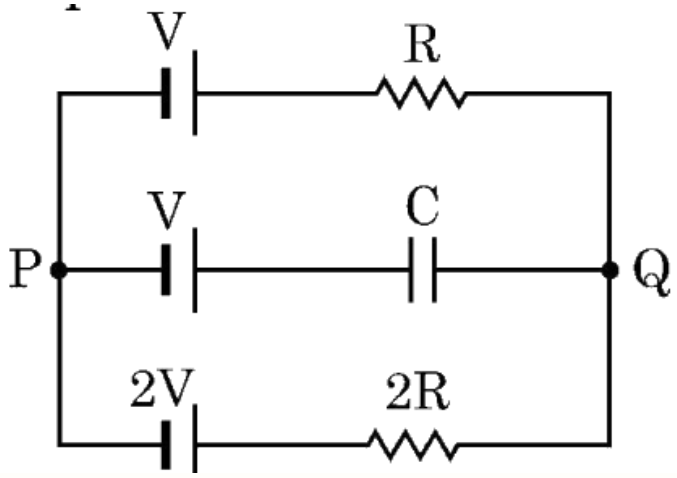
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138. Two wires of the same material and the same radius have their lengths in the ratio 2 : 3. They are connected in parallel to a battery which supplies a current of 15 A. Find the current through the wires.

Check Solution → [2025 • Set 55-2-1]

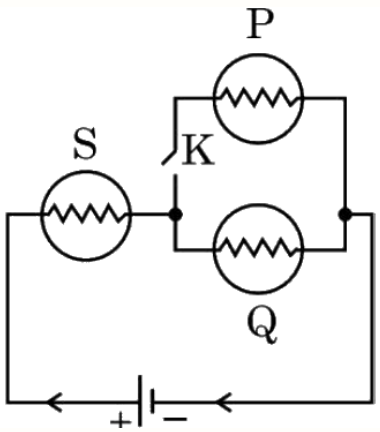
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139. In the circuit three ideal cells of e.m.f. V, V and 2V are connected to a resistor of resistance R, a capacitor of capacitance C and another resistor of resistance 2R as shown in figure. In the steady state find (i) the potential difference between P and Q and (ii) potential difference across capacitor C.



Check Solution → [2025 • Set 55-2-1]

140. **(a)** In the given figure, three identical bulbs *P*, *Q* and *S* are connected to a battery.

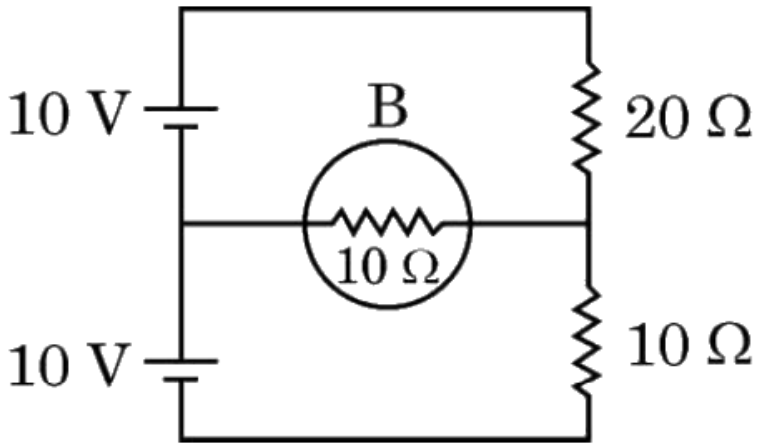


(i) Compare the brightness of bulbs *P* and *Q* with that of bulb *S* when key *K* is closed.

(ii) Compare the brightness of the bulbs *S* and *Q* when the key *K* is opened. Justify your answer in both cases.

OR

(b) Two cells of emf 10 *V* each, two resistors of 20 Ω and 10 Ω and a bulb *B* of 10 Ω resistance are connected together as shown in the figure. Find the current that flows through the bulb.



Check Solution →

[2025 • Set 55-4-1]

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141. The resistance of a wire at 25°C is 10.0 Ω. When heated to 125°C, its resistance becomes 10.5 Ω. Find (i) the temperature coefficient of resistance of the wire, and (ii) the resistance of the wire at 425°C.

Check Solution →

[2025 • Set 55-5-1]

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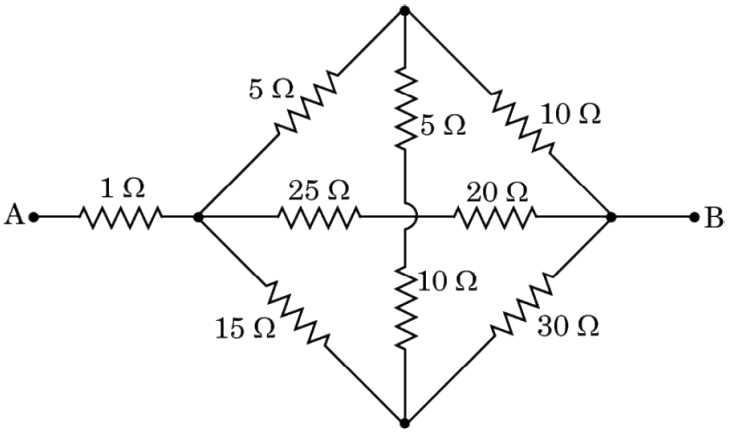
142. A wire of resistance X ohm is gradually stretched till its length becomes twice its original length. If its new resistance becomes 40 Ω, find the value of X.

Check Solution →

[2025 • Set 55-5-2]

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143. Find the equivalent resistance between points A and B for the network shown in the figure.



Check Solution →

[2025 • Set 55-6-1]

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144. A cell of emf E and internal resistance r is connected to an external variable resistance R. Plot a graph showing the variation of terminal voltage V of the cell as a function of current I, supplied by the cell. Explain how the emf of the cell and its internal resistance can be found from it.

Check Solution → [2025 • Set 55-6-2]

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145. (a) What is the difference between 'velocity' and 'drift velocity' of electrons in a current-carrying conductor.
(b) A copper wire of uniform cross-sectional area carries a current of 3·4 A. The drift velocity of conduction electrons is 0·2 mm/s. If the number density of electrons in copper is $8{\cdot}5 \times 10^{28}$ m^{-3}, find the area of cross-section of the wire.

Check Solution → [2025 • Set 55-6-3]

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146. A cell of emf E and internal resistance r is connected across a resistor of variable resistance R. Show graphically the variation of
(a) the terminal voltage across the cell,
(b) the current supplied by the cell,
with R as it is increased from 0 to the maximum value.

Check Solution → [2025 • Set 55-7-1]

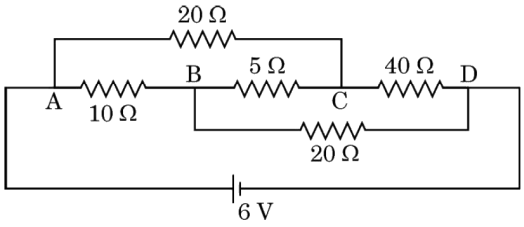
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147. Two wires made of the same material have the same length (l) but different cross-sectional areas A_1 and A_2. They are connected together with a cell of voltage V. Find the ratio of the drift velocities of free electrons in the two wires when they are joined in (i) series, and (ii) parallel.

Check Solution → [2025 • Set 55-7-2]

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148. Calculate the value of the current passing through the battery in the given circuit diagram.

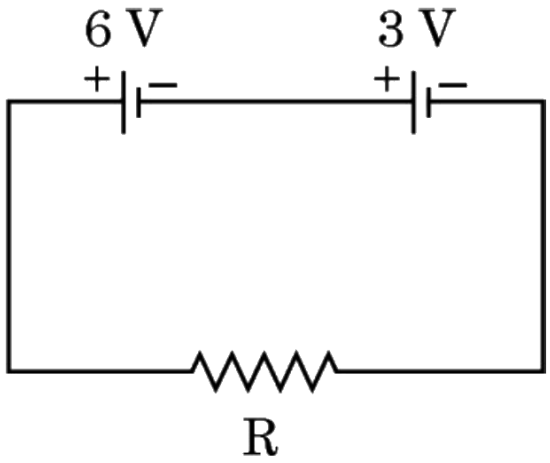


Check Solution → [2025 • Set 55-7-3]

149. Find the temperature at which the resistance of a wire made of silver will be twice its resistance at 20°*C*. Take 20°*C* as the reference temperature and temperature coefficient of resistance of silver at 20°*C* = 4.0× 10^{-3} K^{-1}.

Check Solution → [2024 • Set 55-1-1]

150. Two batteries of emfs 6 *V* and 3 *V* and internal resistances 0.8 Ω and 0.2 Ω respectively are connected in series to an external resistance *R*, as shown in figure. Find the value of *R* so that the potential difference across the 6 *V* battery be zero.



Check Solution → [2024 • Set 55-1-2]

151. A uniform wire of length L and area of cross-section A has resistance R. The wire is uniformly stretched so that its length increases by 25%. Calculate the percentage increase in the resistance of the wire.

Check Solution → [2024 • Set 55-1-3]

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152. Define resistivity of a conductor. How does the resistivity of a conductor depend upon the following:
(a) Number density of free electrons in the conductor (n)
(b) Their relaxation time (τ)

Check Solution → [2024 • Set 55-2-1]

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153. (a) "The electron drift speed is only a few mm/s for currents in the range of a few amperes for a given conductor." How then is current established almost the instant a circuit is closed? Explain.
(b) "V=IR is a statement of Ohm's Law" is not true. Explain.

Check Solution → [2024 • Set 55-2-2]

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154. An electric field E is maintained in a wire of length 'l' and area of cross-section 'a'. Derive the relation between the current density 'σ' in the wire and the electric field E.

Check Solution → [2024 • Set 55-2-3]

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155. What is meant by 'relaxation time' of free electrons in a conductor? Show that the resistance of a conductor can be expressed by $R = \frac{m\,l}{n\,e^2\,\tau\,A}$, where symbols have their usual meanings.

Check Solution → [2024 • Set 55-3-1]

156.

Draw the circuit diagram of a Wheatstone bridge. Obtain the condition when no

current flows through the galvanometer in it.

Check Solution → [2024 • Set 55-3-1]

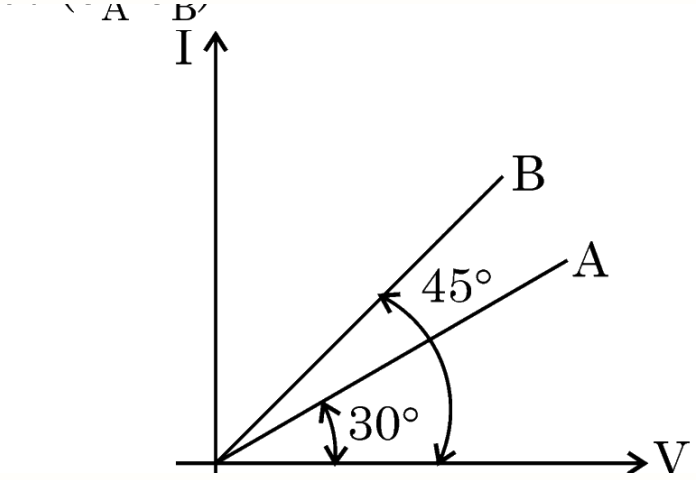
157. Two electric heaters have power ratings P_1 and P_2, at voltage V. They are connected in series to a dc source of voltage V. Find the power consumed by the combination. Will they consume the same power if connected in parallel across the same source?

Check Solution → [2024 • Set 55-4-1]

158. Two wires A and B of different metals have their lengths in ratio 1:2 and their radii in ratio 2:1 respectively. I-V graphs for them is shown in the figure. Find the ratio of their

(i) Resistances (R_A/R_B) and

(ii) Resistivities (ρ_A/ρ_B)

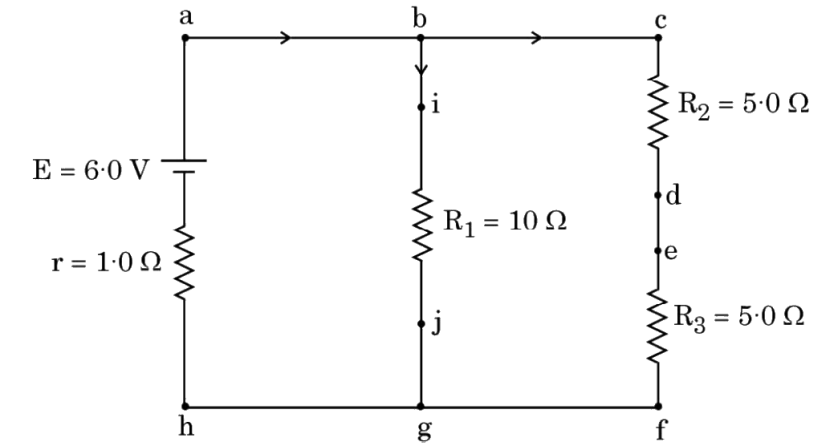


Check Solution → [2024 • Set 55-4-2]

159. Find the temperature at which the resistance of a conductor increases by 25% of its value at 27 °C. The temperature coefficient of resistance of the conductor is $2.0\times 10^{-4}\ °C^{-1}$.

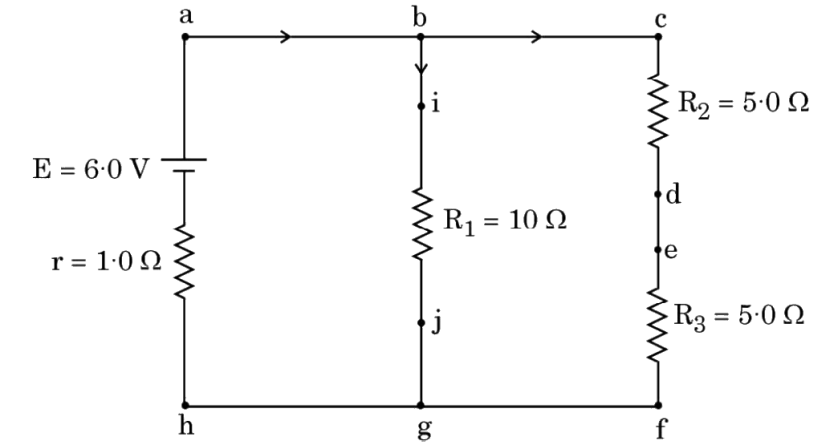
Check Solution → [2024 • Set 55-4-3]

160. Refer to the circuit diagram. Find the potential difference across resistance R_2.



Check Solution → [2023 • Set 55-1-1]

161. Refer to the circuit diagram. What is the power dissipated in resistance R_2?



Check Solution → [2023 • Set 55-1-1]

162. Write two differences between the emf and terminal potential difference of a cell. What is the most important precaution that one should take while drawing current from a cell?

Check Solution → [2023 • Set 55-4-1]

163. Two identical cells, each of emf E and internal resistance r, are connected with a load resistance R, first in series and then in parallel. Obtain the condition under which the current through R is same in both cases.

Check Solution → **[2023 • Set 55-4-2]**

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164. A cell of emf E and internal resistance r is connected to a variable resistance R. Draw plots showing the variation of (a) terminal voltage V with R, and (b) V with current I, in the circuit.

Check Solution → **[2023 • Set 55-4-3]**

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165. The potential difference applied across a given conductor is doubled. How will this affect (i) the mobility of electrons and (ii) the current density in the conductor? Justify your answers.

Check Solution → **[2023 • Set 55-5-1]**

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166. A potential difference (V) is applied across a conductor of length 'L' and cross-sectional area 'A'. How will the drift velocity of electrons and the current density be affected if another identical conductor of the same material were connected in series with the first conductor? Justify your answers.

Check Solution → **[2023 • Set 55-5-2]**

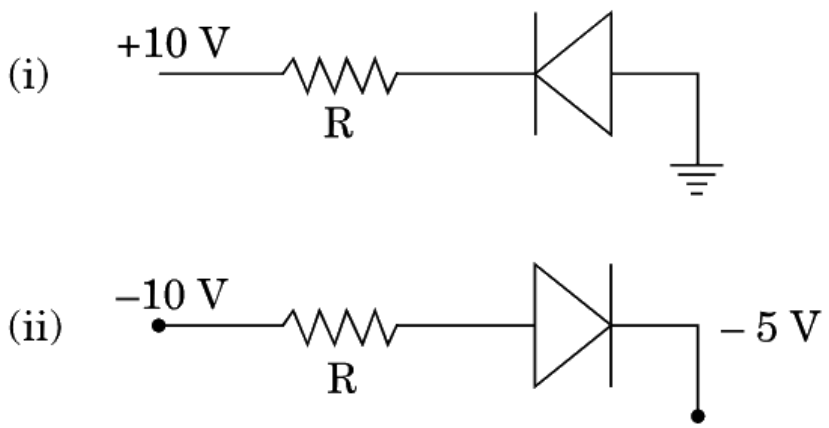
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167. Two conductors, made of the same material have equal lengths but different cross-sectional areas A_1 and A_2 ($A_1 > A_2$). They are connected in parallel across a cell. Show that the drift velocities of electrons in two conductors are equal.

Check Solution → **[2023 • Set 55-5-3]**

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168. Define drift velocity of electrons in a conductor connected across a battery. Figure shows variation of the drift velocity (v_d) of electrons in two copper wires A and B of different lengths versus the potential difference (V) applied across their ends.



(i) What does the slope of the line represent?
(ii) Which one of the two wires is longer?

Check Solution → [2021]

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169. Explain the principle of working of a meter bridge. Draw the circuit diagram for determination of an unknown resistance using it.

Check Solution → [2020 • Set 55-1-1]

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170. Define the term 'mobility' of charge carriers in a current carrying conductor. Obtain the relation for mobility in terms of relaxation time.

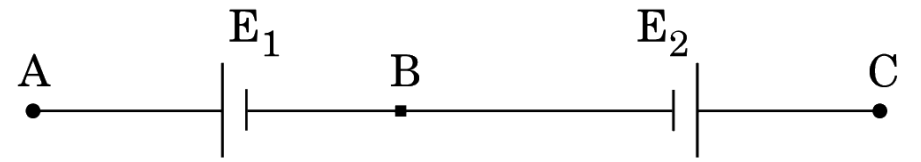
OR

Define the term 'drift velocity' of electrons in a current carrying conductor. Obtain the relationship between the current density and the drift velocity of electrons.

Check Solution → [2020 • Set 55-2-1]

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171. Two cells of emf E_1 and E_2 ($E_1 > E_2$) are connected as shown in the figure below. When a potentiometer is used to measure potential difference between the points A and B, the balancing length of the potentiometer wire is 300 cm. But the same potentiometer for the potential difference between points A and C, gives the balancing length 100 cm. Find $\frac{E_1}{E_2}$.



Check Solution → [2020 • Set 55-3-1]

172. A battery of emf 12 V and internal resistance 4 Ω is connected to an external resistance R. If the current in the resistance is 0.5 A, calculate the value of (a) R, and (b) the terminal voltage of the battery.

Check Solution → [2020 • Set 55-3-2]

173. The number densities of free electrons in three conductors X, Y and Z are in the ratio 4:1:8, and their electrical conductivities are in the ratio 2:1:2. For which conductor will the relaxation time of free electrons be (a) maximum, and (b) minimum?

Check Solution → [2020 • Set 55-3-3]

174. A wire of length L_1 has a resistance R_1. It is gradually stretched till its length becomes $2L_1$.
(a) Plot a graph showing variation of its resistance R with its length l during stretching.
(b) What will be its resistance when its length becomes $2L_1$?

Check Solution → [2020 • Set 55-4-1]

175. A uniform wire is cut into three parts with their lengths in the ratio 2:3:6. The ends of each of these three parts are connected across an ideal battery of 10 V. If a current of 5 A is drawn from the battery, find the initial resistance of the wire.

Check Solution → [2020 • Set 55-4-2]

176. Two bulbs are rated (P_1, V) and (P_2, V). If they are connected (i) in series and (ii) in parallel across a supply V, find the power dissipated in the two combinations

in terms of P_1 and P_2.

Check Solution → [2019 • Set 55-1-1]

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177. A set of '*n*' identical resistors, each of resistance '*R*' when connected in series have an effective resistance '*X*'. When they are connected in parallel, their effective resistance becomes '*Y*'. Find out the product of *X* and *Y*.

Check Solution → [2019 • Set 55-5-1]

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178. A wire of resistance *R*, length *l* and area of cross-section *A*, is cut into two parts, having their lengths in the ratio 1:2. The shorter wire is now stretched till its length becomes equal to that of the longer wire. If they are now connected in parallel, find the net resistance of the combination.

Check Solution → [2019 • Set 55-5-2]

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179. A wire of resistance *R*, length *l* and uniform area of cross-section *A* is stretched till its length becomes double. It is then cut into three pieces of equal length which are connected in parallel. Find the net resistance of the combination in terms of the resistance *R*.

Check Solution → [2019 • Set 55-5-3]

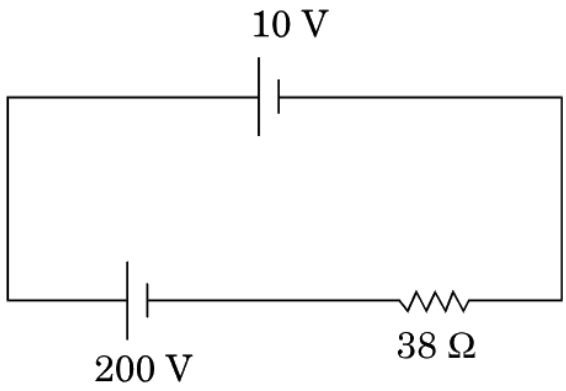
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180. Two electric bulbs P and Q have their resistances in the ratio of 1 : 2. They are connected in series across a battery. Find the ratio of the power dissipation in these bulbs.

Check Solution → [2018]

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181. A 10 V cell of negligible internal resistance is connected in parallel across a battery of emf 200 V and internal resistance 38 Ω as shown in the figure. Find the value of current in the circuit.



Check Solution → [2018]

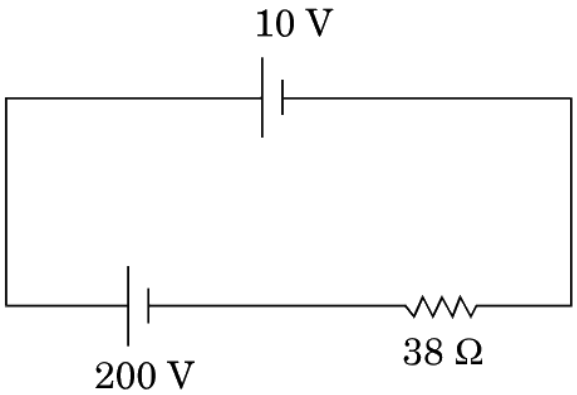
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182. In a potentiometer arrangement for determining the emf of a cell, the balance point of the cell in open circuit is 350 cm. When a resistance of 9 Ω is used in the external circuit of the cell, the balance point shifts to 300 cm. Determine the internal resistance of the cell.

Check Solution → [2018]

---

183. A 10 V cell of negligible internal resistance is connected in parallel across a battery of emf 200 V and internal resistance 38 Ω as shown in the figure. Find the value of current in the circuit.



OR

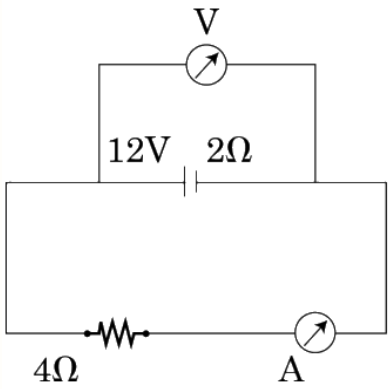
In a potentiometer arrangement for determining the emf of a cell, the balance point of the cell in open circuit is 350 cm. When a resistance of 9 Ω is used in the

external circuit of the cell, the balance point shifts to 300 cm. Determine the internal resistance of the cell.

Check Solution → [2018]

---

184. A battery of emf 12 V and internal resistance 2 Ω is connected to a 4 Ω resistor as shown in the figure.
(a) Show that a voltmeter when placed across the cell and across the resistor, in turn, gives the same reading.
(b) To record the voltage and the current in the circuit, why is voltmeter placed in parallel and ammeter in series in the circuit?



Check Solution → [2016]

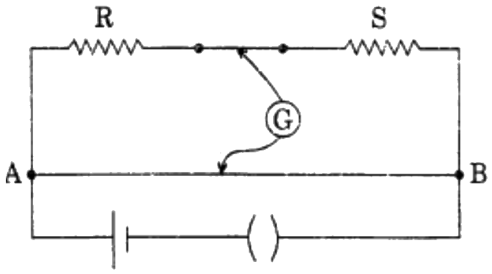
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185. When 5 V potential difference is applied across a wire of length 0.1 m, the drift speed of electrons is 2.5×10^{-4} m/s. If the electron density in the wire is 8×10^{28} m^{-3}, calculate the resistivity of the material of wire.

Check Solution → [2016]

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186. In a meter bridge shown in the figure, the balance point is found to be 40 cm from end A. If a resistance of 10 Ω is connected in series with R, balance point is obtained 60 cm from A. Calculate the values of R and S.



Check Solution → [2015]

187. Use Kirchhoff's rules to obtain conditions for the balance condition in a Wheatstone bridge.

Check Solution → [2015]

188. State Kirchhoff's rules. Explain briefly how these rules are justified.

Check Solution → [2014]

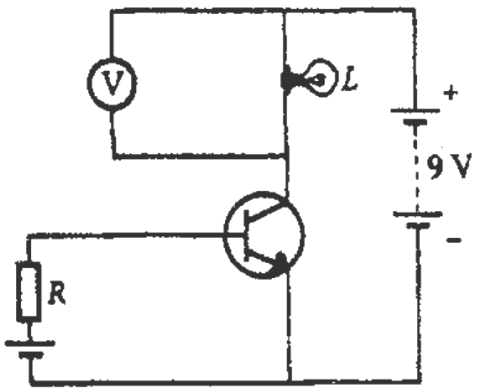
189. Estimate the average drift speed of conduction electrons in a copper wire of cross-sectional area 2.5×10^{-7} m^2 carrying a current of 2.7 A. Assume the density of conduction electrons to be 9×10^{28} m^{-3}.

Check Solution → [2014]

190. A cell of emf 'E' and internal resistance 'r' is connected across a variable resistor 'R'. Plot a graph showing variation of terminal voltage 'V' of the cell versus the current 'I'. Using the plot, show how the emf of the cell and its internal resistance can be determined.

Check Solution → [2014]

191. In the given circuit diagram, a voltmeter '*V*' is connected across a lamp '*L*'. How would (i) the brightness of the lamp and (ii) voltmeter reading '*V*' be affected, if the value of resistance '*R*' is decreased? Justify your answer.



Check Solution → [2013]

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192. Two primary cells of e.m.f. E_1 and E_2 ($E_1 > E_2$) are connected to the potentiometer wire AB as shown in the figure.

If the balancing lengths for the two combinations of the cells are 250 cm and 400 cm, find the ratio of E_1 and E_2.

Check Solution → [2013]

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193. A cell of emf E and internal resistance r is connected to two external resistances R_1 and R_2 and a perfect ammeter. The current in the circuit is measured in four different situations:
(i) without any external resistance in the circuit
(ii) with resistance R_1 only
(iii) with R_1 and R_2 in series combination
(iv) with R_1 and R_2 in parallel combination
The currents measured in the four cases are 0.42 A, 1.05 A, 1.4 A and 4.2 A, but not necessarily in that order. Identify the currents corresponding to the four cases mentioned above.

Check Solution → [2012]

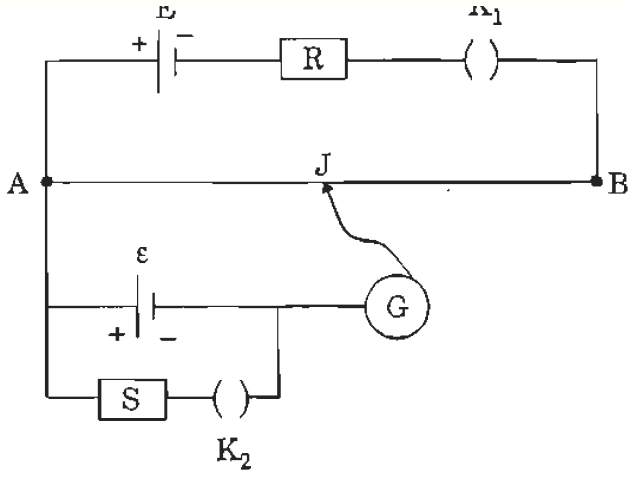
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194. Two students X and Y perform an experiment on potentiometer separately using the circuit given below. Keeping other parameters unchanged, how will the position of the null point be affected if
(i) 'X' increases the value of resistance R in the set-up by keeping the key K_1

closed and the key K_2 open?

(ii) 'Y' decreases the value of resistance S in the set-up, while the key K_2 remains open and the key K_1 closed?

Justify.



Check Solution →

[2012]

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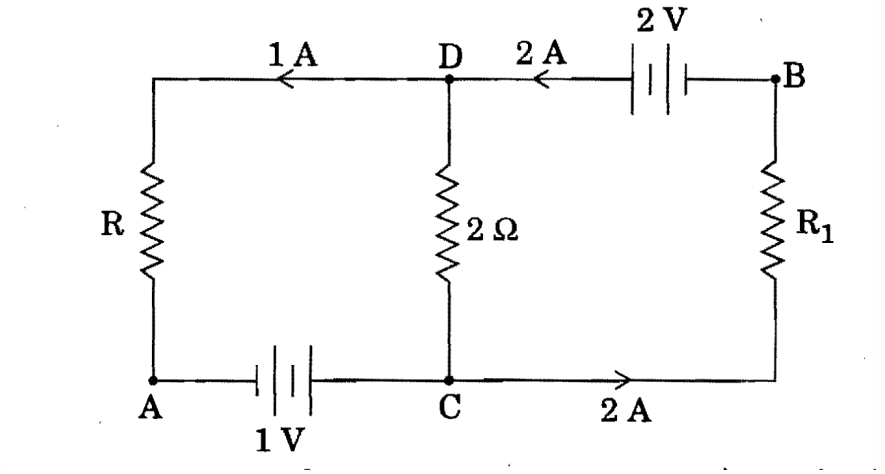
195. A conductor of length 'l' is connected to a dc source of potential 'V'. If the length of the conductor is tripled by gradually stretching it, keeping 'V' constant, how will (i) drift speed of electrons and (ii) resistance of the conductor be affected? Justify your answer.

Check Solution →

[2012]

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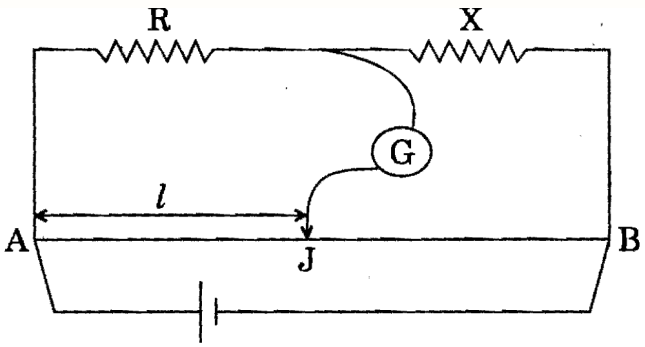
196. In the given circuit, assuming point A to be at zero potential, use Kirchhoff's rules to determine the potential at point B.



Check Solution → [2011]

---

197. In the metre bridge experiment, balance point was observed at J with AJ = l.
(i) The values of R and X were doubled and then interchanged. What would be the new position of balance point?
(ii) If the galvanometer and battery are interchanged at the balance position, how will the balance point get affected?



Check Solution → [2011]

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198. (a) You are required to select a carbon resistor of resistance 47 kΩ ± 10% from a large collection. What should be the sequence of colour bands used to code it?
(b) Write two characteristics of manganin which make it suitable for making standard resistances.

Check Solution → [2011 • Set 55-2-1]

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199. Derive an expression for drift velocity of free electrons in a conductor in terms of relaxation time.

Check Solution → [2009]

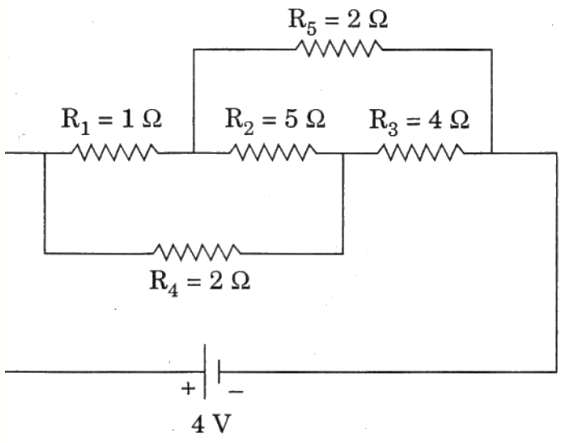
---

200. A cell of emf E and internal resistance r is connected across a variable resistor R. Plot a graph showing the variation of terminal potential V with resistance R. Predict from the graph the condition under which V becomes equal to E.

Check Solution → [2009]

---

201. Calculate the current drawn from the battery in the given network.



Check Solution → [2009]

---

202. A wire of 15 Ω resistance is gradually stretched to double its original length. It is then cut into two equal parts. These parts are then connected in parallel across a 3.0 *V* battery. Find the current drawn from the battery.

Check Solution → [2009]

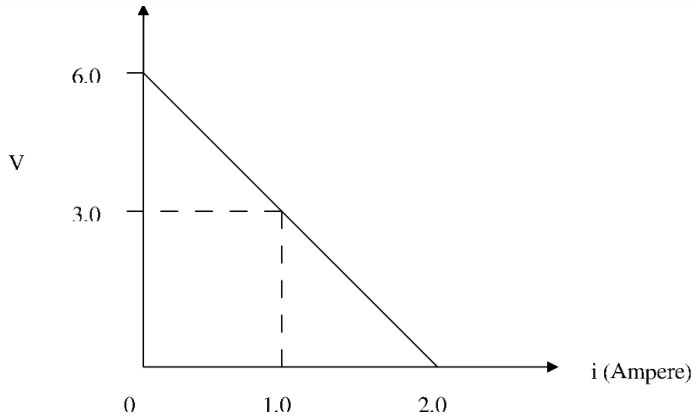
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203. Two wires X, Y have the same resistivity, but their cross-sectional areas are in the ratio 2:3 and lengths in the ratio 1:2. They are first connected in series and then in parallel to a d.c source. Find out the ratio of the drift speeds of the electrons in the two wires for the two cases.

Check Solution → [2008]

---

204. The following graph shows the variation of terminal potential difference *V*, across a combination of three cells in series to a resistor, versus the current, *i*:



(i) Calculate the emf of each cell.
(ii) For what current *I*, will the power dissipation of the circuit be maximum?

Check Solution → [2008]

---

205. A cylindrical metallic wire is stretched to increase its length by 5%. Calculate the percentage change in its resistance.

Check Solution → [2007]

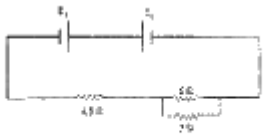
---

206. A voltage of 30 V is applied across a carbon resistor with first, second and third rings of blue, black and yellow colours respectively. Calculate the value of current, in mA, through the resistor.

Check Solution → [2007]

---

207. Two cells E_1 and E_2 in the given circuit diagram have an emf of 5 *V* and 9 *V* and internal resistance of 0.3 Ω and 1.2 Ω respectively. Calculate the value of current flowing through the resistance of 3 Ω.



Check Solution → [2006]

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208. Write the mathematical relation between mobility and drift velocity of charge carriers in a conductor. Name the mobile charge carriers responsible for conduction of electric current in (i) an electrolyte (ii) an ionised gas.

Check Solution → [2006]

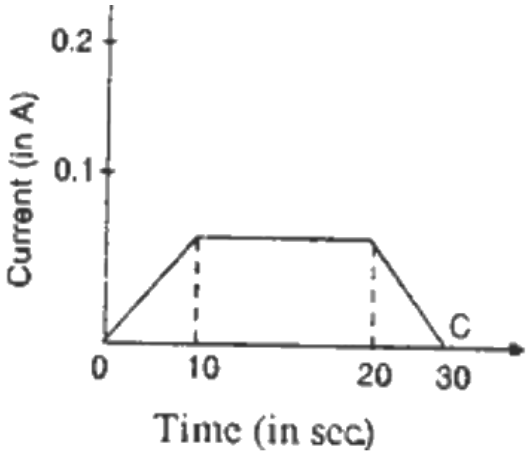
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209. Draw *V-I* graph for ohmic and non-ohmic materials. Give one example for each.

Check Solution → [2005]

---

210. In a copper voltameter, a varying electric current, as shown in graph, is passed. The mass of copper deposited at the end of 30 seconds is *m* grams. Using the graph, find the value of e.c.e. (electrochemical equivalent) of copper in g/C.



Check Solution → **[2003]**

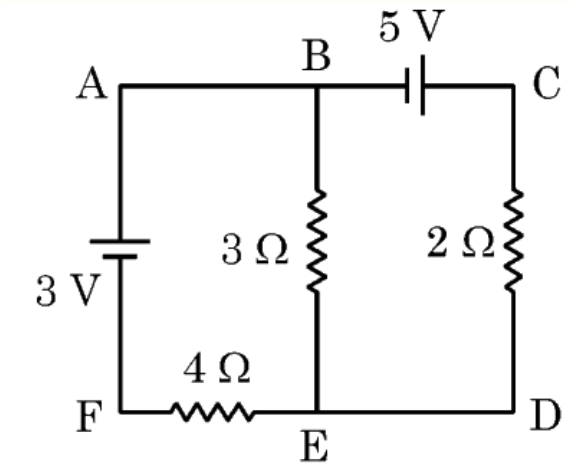
SECTION C

3-Mark Questions

SA · 95 questions

> *Section C of the paper, 3 marks each. Sorted newest year first.*

211. (a) Explain the statement: "Current is a scalar although we represent current with an arrow".
(b) Use Kirchhoff's rules to find the current through 3 Ω resistor in the circuit shown in the figure:



Check Solution → [2026 • Set 55-2-1]

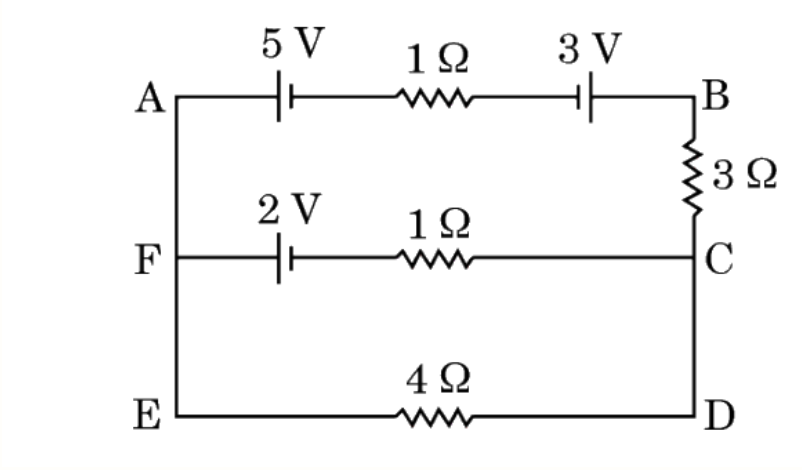
---

212. (a) Establish the relation between drift velocity of electrons (v_d) and electric current (I) in a conductor.
(b) How is v_d affected when the length of the conductor is doubled, keeping the voltage applied across the conductor constant?

Check Solution → [2026 • Set 55-3-1]

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213. State Kirchhoff's rules. Using these rules, find the current flowing through branch FC in the given circuit.

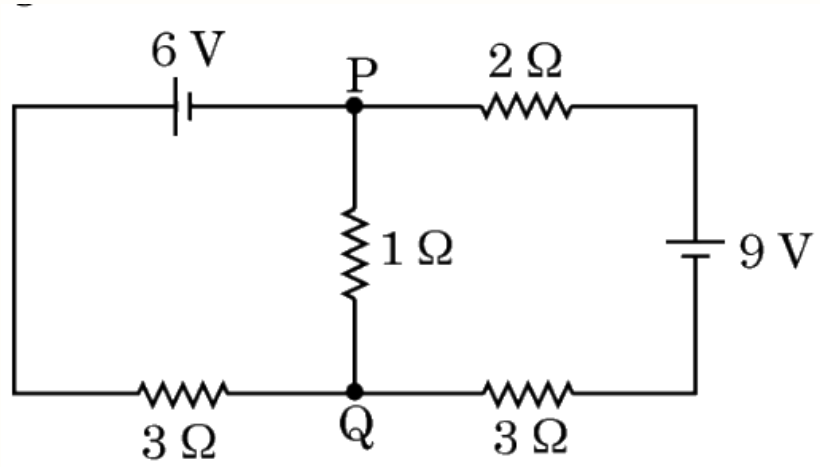


Check Solution → [2026 • Set 55-4-1]

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214. (a) The two Kirchhoff's rules are based on law of conservation of two physical quantities. Name the quantities.

(b) Find magnitude and direction of current in 1 Ω resistor in the given circuit.



Check Solution → [2026 • Set 55-4-2]

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215. (a) 'Current is a scalar quantity, although we represent current with an arrow.' Explain.

(b) Derive the balance condition of a Wheatstone Bridge.

Check Solution → [2026 • Set 55-4-3]

---

216. **(a)** Two batteries of emfs 3 *V* & 6 *V* and internal resistances 0.2 Ω & 0.4 Ω are connected in parallel. This combination is connected to a 4 Ω resistor. Find:

(i) the equivalent emf of the combination

(ii) the equivalent internal resistance of the combination
(iii) the current drawn from the combination
OR
(b)
(i) A conductor of length l is connected across an ideal cell of emf E. Keeping the cell connected, the length of the conductor is increased to 2l by gradually stretching it. If R and R' are initial and final values of resistance and v_d and v_d' are initial and final values of drift velocity, find the relation between (i) R' and R and (ii) v_d' and v_d.
(ii) When electrons drift in a conductor from lower to higher potential, does it mean that all the 'free electrons' of the conductor are moving in the same direction?

Check Solution → [2025 • Set 55-1-1]

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217. (a) A cell of e.m.f. E and internal resistance r is connected with a variable external resistance R and a voltmeter showing potential drop V across R. Obtain the relationship between V, E, R and r.
(b) Draw the shape of the graph showing the variation of terminal voltage V of the cell as a function of current I drawn from it. How one can determine the e.m.f. of the cell and its internal resistance from this graph?

Check Solution → [2025 • Set 55-2-1]

---

218. (a) Define resistivity of a conductor. Discuss its dependence on temperature of the conductor and draw a plot of resistivity of copper as a function of temperature.
(b) (i) "A low voltage battery from which high current is required must have low internal resistance." Justify.
(ii) "A high voltage battery must have a large internal resistance." Justify.

Check Solution → [2025 • Set 55-2-2]

---

219. (a) Define Electrical conductivity. Obtain the expression of electrical conductivity of a conductor in terms of number density and relaxation time of

free electrons.

(b) Explain qualitative change in resistivity of a conductor with temperature using expression obtained in (a).

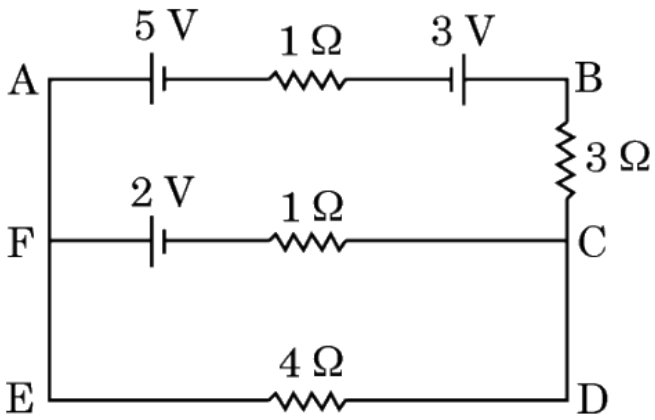
Check Solution → [2025 • Set 55-2-3]

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220. (a)

(i) Derive an expression for the resistivity of a conductor in terms of number density of free electrons and relaxation time.

(ii) The figure shows the plot of current through a cross-section of wire over two different time intervals. Compare the charges (Q_1 and Q_2) that pass through the cross-section during these time intervals.



OR

(b)

(i) A battery of emf E and internal resistance r is connected to a variable external resistance R.

(I) Obtain the expression for current I in the circuit and the value of maximum current the battery can supply.

(II) Obtain the terminal voltage V across the battery and its maximum possible value.

(ii) The above battery sends a current I_1 when $R = R_1$ and a current I_2 when $R = R_2$. Obtain the internal resistance of the battery in terms of I_1, I_2, R_1 and R_2.

Check Solution → [2025 • Set 55-4-1]

---

221. What is the difference between 'emf' and 'terminal voltage' of a cell?
Two cells of emfs E_1 and E_2 and internal resistances r_1 and r_2 are connected in parallel. Derive an expression for the emf and internal resistance of the equivalent cell.

Check Solution → [2025 • Set 55-6-1]

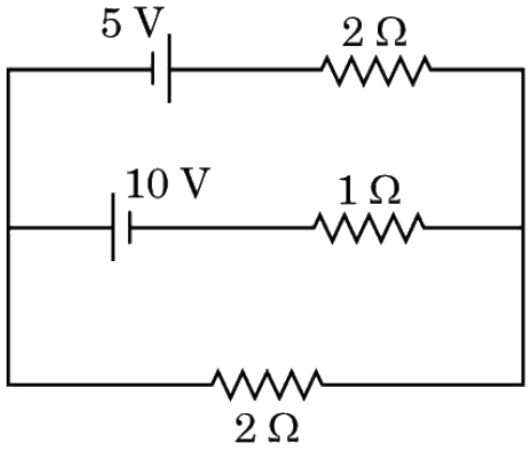
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222. (a) Define the term 'drift velocity' of conduction electrons in a conductor.
(b) A conductor of length l and area of cross-section A is connected across an ideal battery of emf V. Derive the formula for the current density in terms of relaxation time τ.

Check Solution → [2025 • Set 55-6-2]

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223. State Kirchhoff's laws. Apply these laws to find the values of current flowing in the three branches of the given circuit.



Check Solution → [2025 • Set 55-6-3]

---

224. Three cells A, B and C of emfs 2 V, 3 V and 5 V respectively are connected in parallel to each other. Their internal resistances are 5 Ω, 5 Ω and 1 Ω respectively. Calculate the currents flowing through the cells A, B and C.

Check Solution → [2025 • Set 55-7-1]

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225. Define 'current density'. Is it a scalar or a vector? An electric field $\vec{E}$ is maintained in a metallic conductor. If n be the number of electrons (mass m, charge $-e$) per unit volume in the conductor and τ its relaxation time, show that the current density $\vec{j} = \sigma\vec{E}$, where $\sigma = \frac{n e^2 \tau}{m}$.

Check Solution → [2024 • Set 55-1-1]

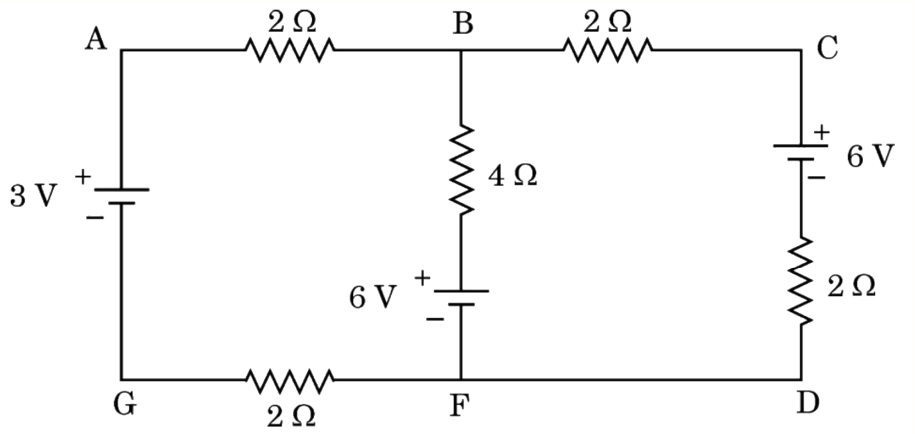
226.

What is a Wheatstone bridge? Obtain the necessary conditions under which the Wheatstone bridge is balanced.

Check Solution → [2024 • Set 55-1-1]

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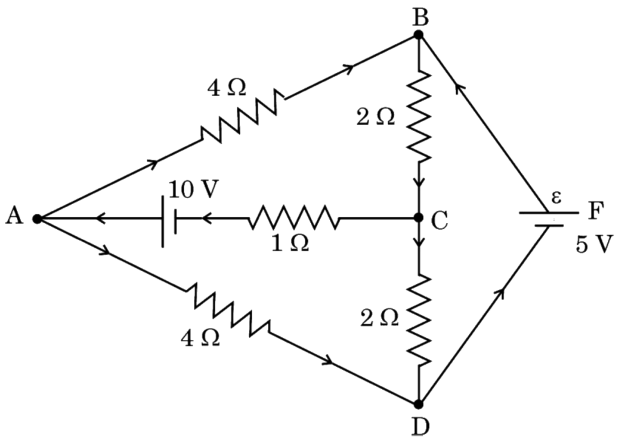
227. The figure shows a circuit with three ideal batteries. Find the magnitude and direction of currents in the branches AG, BF and CD.



Check Solution → [2024 • Set 55-2-1]

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228. Determine the current in branches AB, AC and BC of the network shown in figure.



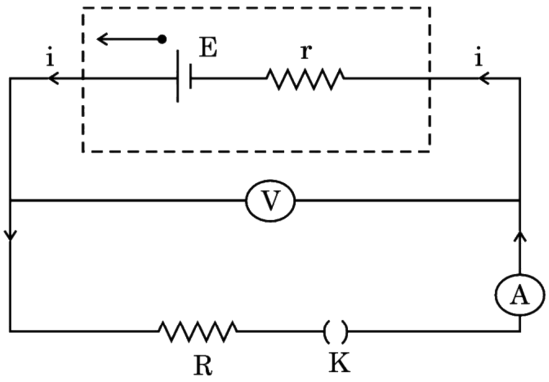
Check Solution → [2024 • Set 55-3-1]

229. A battery of unknown emf E and internal resistance r is connected in a circuit as shown in the figure. When the key (K) is open, the voltmeter reads 10.0 V and ammeter reads 0 A. In the closed circuit, the voltmeter reads 6.0 V and ammeter reads 2.0 A. Calculate:

(a) emf of the battery,

(b) internal resistance of the battery (r), and

(c) external resistance (R).



Check Solution → [2024 • Set 55-3-2]

230. A current of 1.6 A flows through a wire when a potential difference of 1.0 V is applied across it. The length and cross-sectional area of the wire are 1.0 m and 1.0×10^{-7} m^2 respectively. Calculate:

(a) Electric field across the wire

(b) Current density

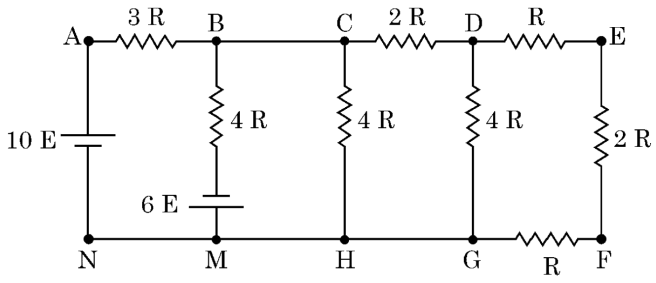
(c) Average relaxation time (τ)

(Number density of free electrons in the wire is 9.0×10^{28} m^{-3})

Check Solution → [2024 • Set 55-3-3]

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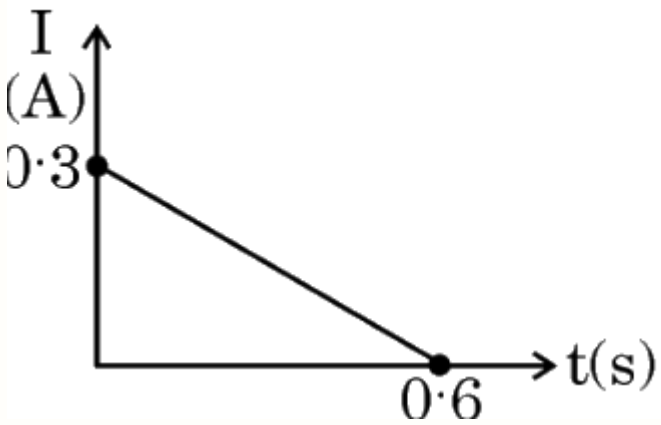
231. Find the current in branch BM in the network shown:



Check Solution → [2024 • Set 55-4-1]

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232. Find the currents flowing through the branches AB and BC in the network shown.



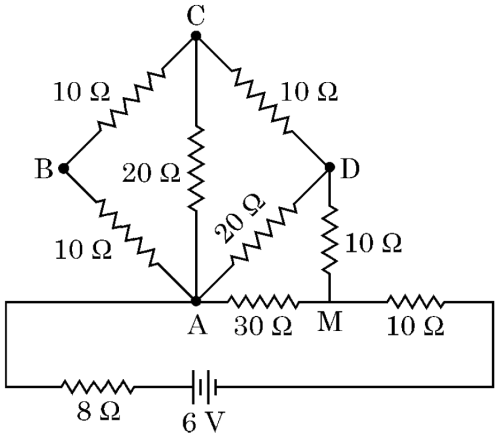
Check Solution → [2024 • Set 55-4-2]

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233. In the given network, calculate:

(i) effective resistance between points *A* and *M*, and

(ii) power supplied by the battery.



Check Solution → [2024 • Set 55-4-3]

---

234. (i) Define 'temperature coefficient of resistance' of a metal.
(ii) Show the variation of resistivity of copper with rise in temperature.
(iii) The resistance of a wire is 10 Ω at 27°C. Find its resistance at −73°C. The temperature coefficient of resistance of the material of the wire is 1.70×10^{-4} $^{\circ}C^{-1}$.

Check Solution → [2024 • Set 55-5-2]

---

235. A potential difference of 1.0 V is applied across a conductor of length 5.0 m and area of cross-section 1.0 mm^2. When current of 4.25 A is passed through the conductor, calculate (i) the drift speed and (ii) relaxation time, of electrons. (Given number density of electrons in the conductor, $n = 8.5 \times 10^{28}$ m^{-3}).

Check Solution → [2024 • Set 55-5-3]

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236. Define current density and relaxation time. Derive an expression for resistivity of a conductor in terms of number density of charge carriers in the conductor and relaxation time.

Check Solution → [2023 • Set 55-2-1]

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237. Two cells of emf E_1 and E_2 and internal resistances r_1 and r_2 are connected in parallel, with their terminals of the same polarity connected together. Obtain an expression for the equivalent emf of the combination.

Check Solution → [2023 • Set 55-2-2]

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238. A potential difference V is applied across a conductor of length l and cross-sectional area A. Briefly explain how the current density j in the conductor will be affected if

(a) the potential difference V is doubled,

(b) the conductor were gradually stretched to reduce its cross-sectional area to $\frac{A}{4}$ and then the same potential difference V is applied across it.

Check Solution → [2023 • Set 55-2-3]

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239. A potential difference V is applied across a conductor of length l and uniform cross-section area A. How will the (i) electric field E, (ii) drift velocity v_d, and (iii) current density j be affected when (a) V is doubled and (b) l is halved (keeping other factors constant)?

Check Solution → [2023 • Set 55-3-1]

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240. A potential difference 'V' is applied across a load resistor of resistance R. V and R can be varied. If the current that flows in the circuit is I, draw a plot showing the variation of power consumed by the resistor as a function of:

(a) R, keeping V constant

(b) I, keeping R constant

(c) V, keeping R constant

Check Solution → [2023 • Set 55-3-2]

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241. Obtain a relation between the current flowing in a conductor and drift velocity of electrons in it. Hence, obtain Ohm's law.

Check Solution → [2023 • Set 55-3-3]

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242. Two cells of emf E_1 and E_2 have their internal resistances r_1 and r_2, respectively. Deduce an expression for the equivalent emf and internal resistance of their

parallel combination when connected across an external resistance R. Assume that the two cells are supporting each other.

Check Solution → [2020 • Set 55-1-1]

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243. In case the two cells are identical, each of emf $E = 5\ V$ and internal resistance $r = 2\ \Omega$, calculate the voltage across the external resistance $R = 10\ \Omega$.

Check Solution → [2020 • Set 55-1-1]

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244. (a) Derive the condition of balance for Wheatstone bridge.
(b) Draw the circuit diagram of a meter bridge to explain how it is based on Wheatstone bridge.

Check Solution → [2020 • Set 55-2-1]

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245. (a) Differentiate between the random velocity and the drift velocity of electrons in an electrical conductor. Give their order of magnitudes.
(b) A conductor of uniform cross-sectional area is connected across a dc source of variable voltage. Draw a graph showing variation of drift velocity of electrons (v_d) as a function of current density (J) in it.

Check Solution → [2020 • Set 55-3-1]

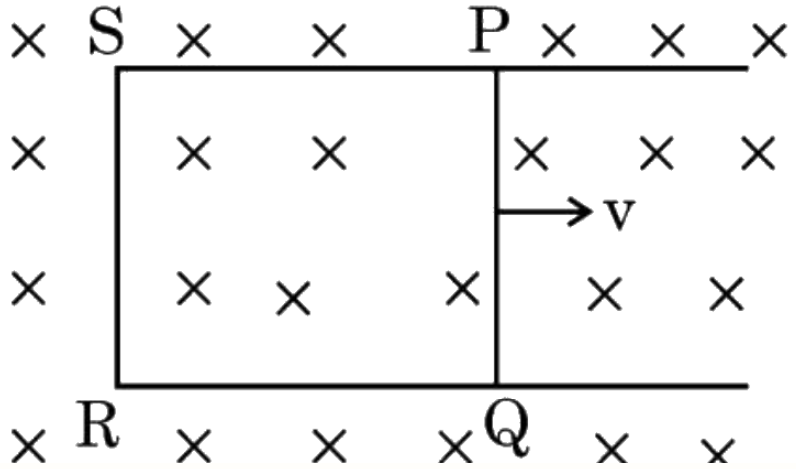
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246. (a) Write the relationship between mobility and drift velocity in a current carrying conductor.
(b) Two aluminium wires have their lengths in the ratio 2:3 and radii in the ratio 1:3. These are connected in parallel across a battery of emf E and of negligible internal resistance. Find the ratio of drift velocities of the electrons in the two wires.

Check Solution → [2020 • Set 55-3-2]

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247. Using Kirchhoff's rules, calculate the current (I_g) that flows through the galvanometer of resistance 15 Ω in the circuit diagram shown in the figure.



Check Solution → [2020 • Set 55-3-3]

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248. (a) Explain the principle of working of a potentiometer.
(b) In a potentiometer, a standard source of emf 5 *V* and negligible internal resistance maintains a steady current through the potentiometer wire of length 10 *m*. Two primary cells of emf E_1 and E_2 are joined together in a series with (i) same polarity and (ii) opposite polarity. The combination is connected to the potentiometer circuit in each case. The balancing length of the wire in the two cases are found to be 700 cm and 100 cm, respectively. Find the values of emf of the two cells.

Check Solution → [2020 • Set 55-4-1]

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249. (a) Differentiate between electrical resistance and resistivity of a conductor.
(b) Two metallic rods, each of length *L*, area of cross A_1 and A_2, having resistivities ρ_1 and ρ_2 are connected in parallel across a d.c. battery. Obtain the expression for the effective resistivity of this combination.

Check Solution → [2020 • Set 55-5-1]

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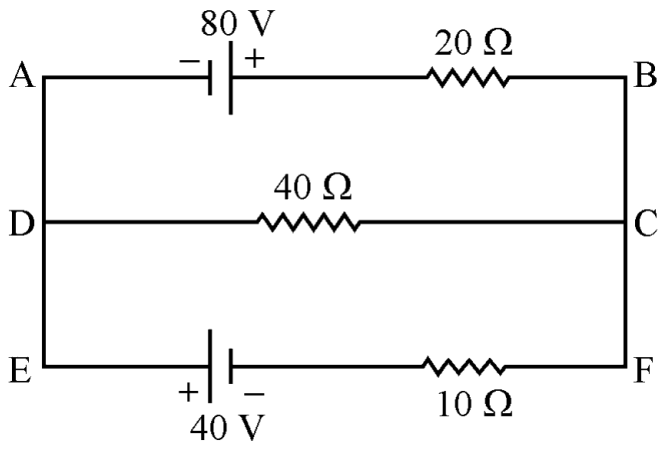
250. (a) Define internal resistance of a cell.

(b) A cell of emf E and internal resistance r is connected across a variable resistor R. Plot the shape of graphs showing variation of terminal voltage V with (i) R and (ii) circuit current I.

Check Solution → [2020 • Set 55-5-2]

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251. Using Kirchhoff's rules, calculate the current through the 40 Ω and 20 Ω resistors in the following circuit:



Check Solution → [2019 • Set 55-1-1]

---

252. What is end error in a metre bridge? How is it overcome? The resistances in the two arms of the metre bridge are $R = 5\ \Omega$ and S respectively. When the resistance S is shunted with an equal resistance, the new balance length is found to be 1.5 l_1, where l_1 is the initial balancing length. Calculate the value of S.

Check Solution → [2019 • Set 55-1-1]

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253. Show, on a plot, variation of resistivity of (i) a conductor, and (ii) a typical semiconductor as a function of temperature.
Using the expression for the resistivity in terms of number density and relaxation time between the collisions, explain how resistivity in the case of a conductor increases while it decreases in a semiconductor, with the rise of temperature.

Check Solution → [2019 • Set 55-2-1]

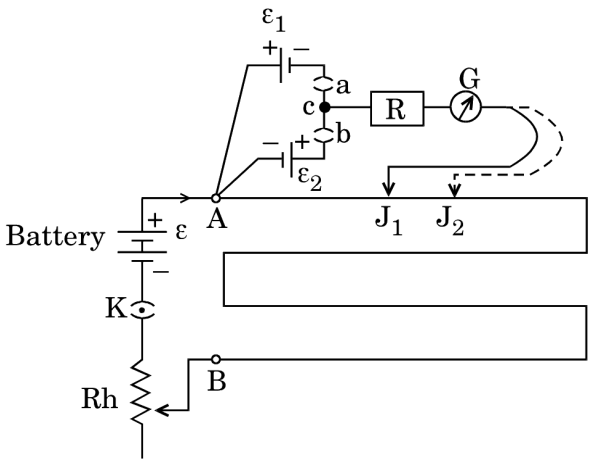
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254. Twelve wires each having a resistance of 3 Ω are connected to form a cubical network. A battery of 10 *V* and negligible internal resistance is connected across the diagonally opposite corners of this network. Determine its equivalent resistance and the current along each edge of the cube.

Check Solution → [2019 • Set 55-3-1]

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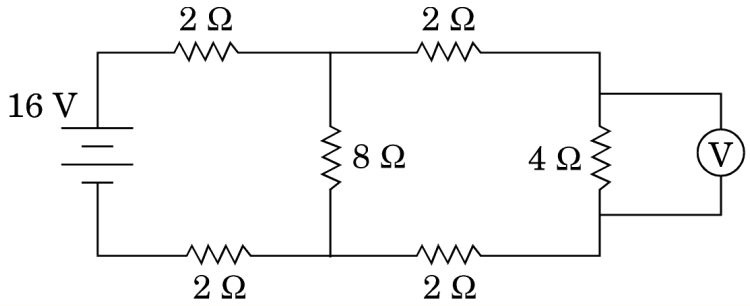
255. A student uses the circuit diagram of a potentiometer as shown in the figure. (a) For a steady current *I* passing through the potentiometer wire, he gets a null point for the cell ε_1 and not for ε_2. Give reason for this observation and suggest how this difficulty can be resolved. (b) What is the function of resistance *R* used in the circuit? How will the change in its value affect the null point? (c) How can the sensitivity of the potentiometer be increased?



Check Solution → [2019 • Set 55-3-1]

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256. In the circuit given below, find the voltmeter reading across a 4 Ω resistor.

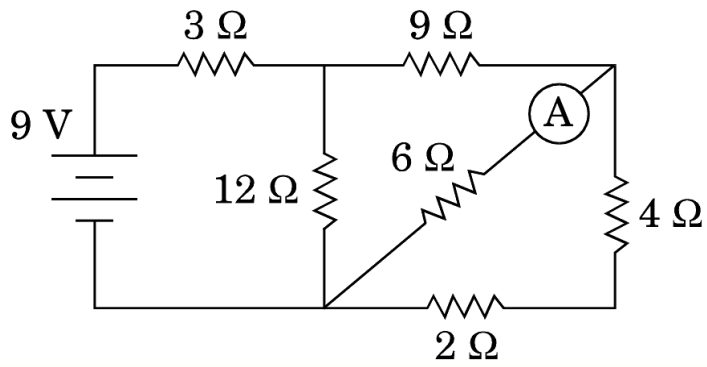


Check Solution → [2019 • Set 55-3-2]

257. (a) Use Kirchhoff's rules to obtain the balance condition in Wheatstone bridge.
(b) Give one practical application that is based on this principle.

Check Solution → [2019 • Set 55-3-2]

258. In the circuit shown in the figure, find the value of the current shown in the ammeter A.



Check Solution → [2019 • Set 55-3-3]

259. (a) Obtain the expression for the current flowing through a conductor having number density of the electrons n, area of cross-section A in terms of the drift velocity v_d.
(b) How does the resistivity of a semiconductor change with rise of temperature? Explain.

Check Solution → [2019 • Set 55-3-3]

260. (a) Draw a graph showing the variation of current versus voltage in an electrolyte when an external resistance is also connected.

(b) (i) The graph between resistance (R) and temperature (T) for Hg is shown in figure (a). Explain the behaviour of Hg near 4 *K*.

(ii) In which region of the graph shown in figure (b) is the resistance negative and why?

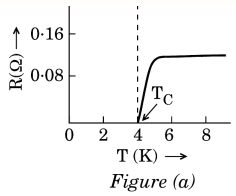


Figure (a)

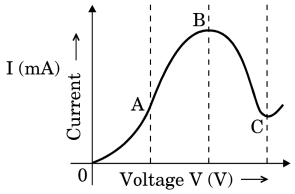


Figure (b)

Check Solution → [2019 • Set 55-4-1]

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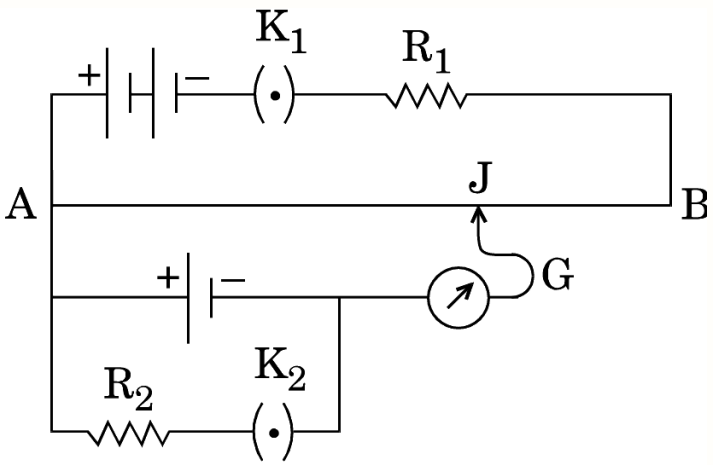
261. (a) For the circuit shown in the figure, how would the balancing length be affected, if

(i) R_1 is decreased,

(ii) R_2 is increased,

the other factors remaining the same in the circuit? Justify your answer in each case.

(b) Why is a potentiometer preferred over a voltmeter? Give reason.



OR

State the underlying principle of meter bridge. Draw the circuit diagram and explain how the unknown resistance of a conductor can be determined by this method.

Check Solution → [2019 • Set 55-4-1]

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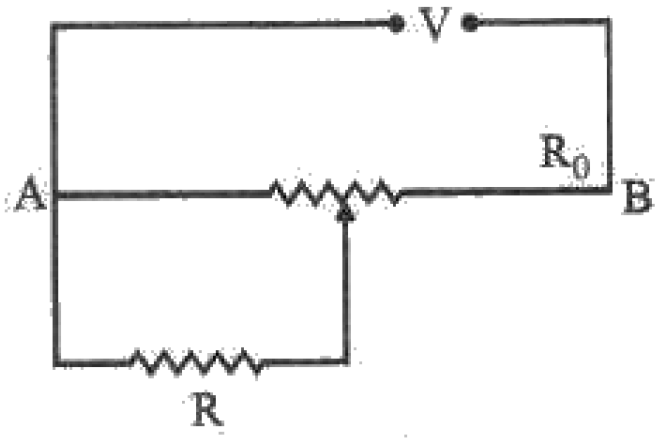
262. (a) Define the term 'conductivity' of a metallic wire. Write its SI unit.

(b) Using the concept of free electrons in a conductor, derive the expression for the conductivity of a wire in terms of number density and relaxation time. Hence obtain the relation between current density and the applied electric field $\vec{E}$.

Check Solution → [2018]

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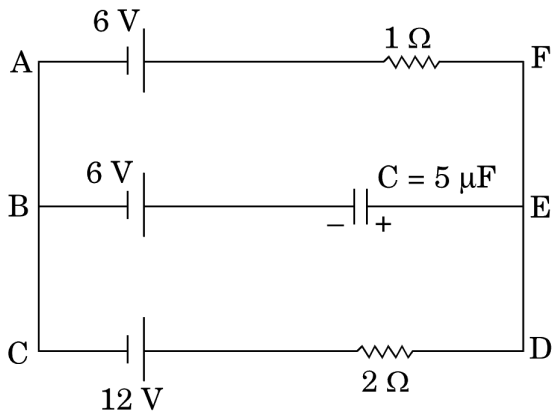
263. A resistance of R draws current from a potentiometer. The potentiometer wire, AB, has a total resistance of R_0. A voltage V is supplied to the potentiometer. Derive an expression for the voltage across R when the sliding contact is in the middle of potentiometer wire.



Check Solution → [2017]

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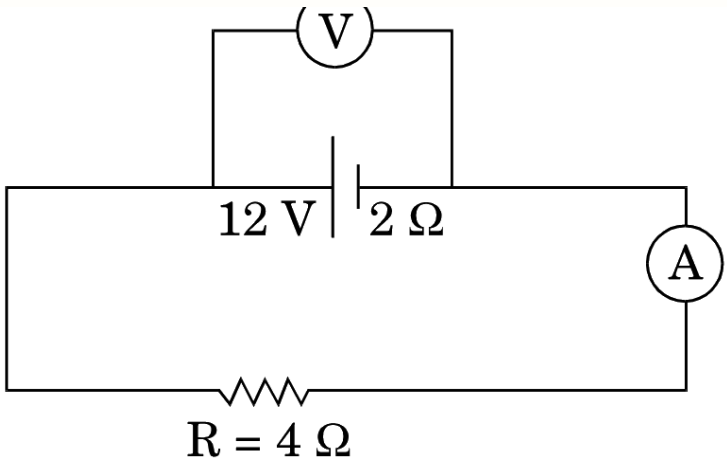
264. In the given circuit, with steady current, calculate the potential difference across the capacitor and the charge stored in it.



Check Solution → [2017]

265. (a) The potential difference applied across a given resistor is altered so that the heat produced per second increases by a factor of 9. By what factor does the applied potential difference change?

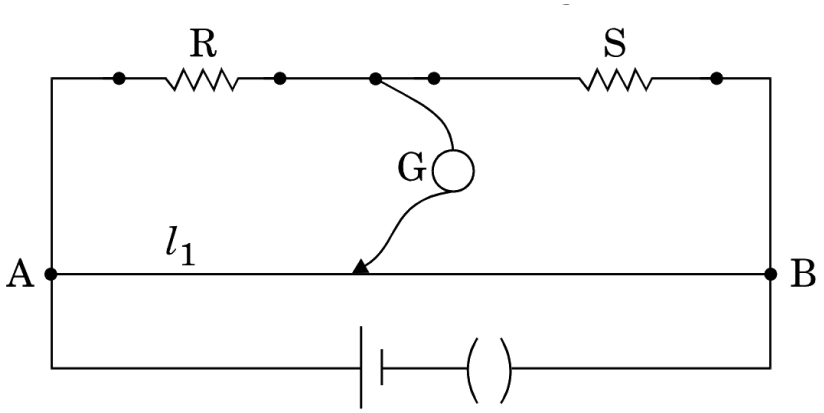
(b) In the figure shown, an ammeter *A* and a resistor of 4 Ω are connected to the terminals of the source. The emf of the source is 12 *V* having an internal resistance of 2 Ω. Calculate the voltmeter and ammeter readings.



Check Solution → [2017]

266. (a) Write the principle of working of a metre bridge.

(b) In a metre bridge, the balance point is found at a distance l_1 with resistances *R* and *S* as shown in the figure.



An unknown resistance X is now connected in parallel to the resistance S and the balance point is found at a distance l_2. Obtain a formula for X in terms of l_1, l_2 and S.

Check Solution → [2017]

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267. (i) Derive an expression for drift velocity of free electrons.
(ii) How does drift velocity of electrons in a metallic conductor vary with increase in temperature? Explain.

Check Solution → [2016]

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268. Two identical cells of emf 1.5 V each joined in parallel supply energy to an external circuit consisting of two resistances of 7 Ω each joined in parallel. A very high resistance voltmeter reads the terminal voltage of cells to be 1.4 V. Calculate the internal resistance of each cell.

Check Solution → [2016]

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269. State the underlying principle of a potentiometer. Write two factors by which current sensitivity of a potentiometer can be increased. Why is a potentiometer preferred over a voltmeter for measuring the emf of a cell?

Check Solution → [2015]

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270. A cell of emf 'E' and internal resistance 'r' is connected across a variable load resistor R. Draw the plots of the terminal voltage V versus (i) R and (ii) the current I.

It is found that when $R = 4\ \Omega$, the current is 1 A and when R is increased to 9 Ω, the current reduces to 0.5 A. Find the values of the emf E and internal resistance r.

Check Solution → [2015]

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271. A potentiometer wire of length 1 m has a resistance of 10 Ω. It is connected to a 6 V battery in series with a resistance of 5 Ω. Determine the emf of the primary cell which gives a balance point at 40 cm.

Check Solution → [2014]

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272. Answer the following :

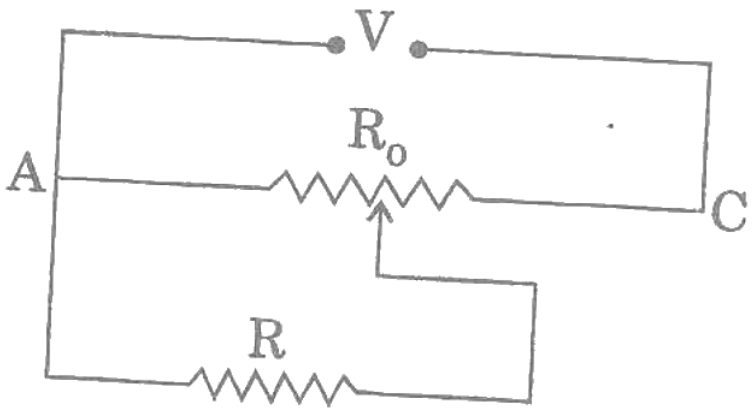
(a) Why are the connections between the resistors in a meter bridge made of thick copper strips ?

(b) Why is it generally preferred to obtain the balance point in the middle of the meter bridge wire ?

(c) Which material is used for the meter bridge wire and why ?

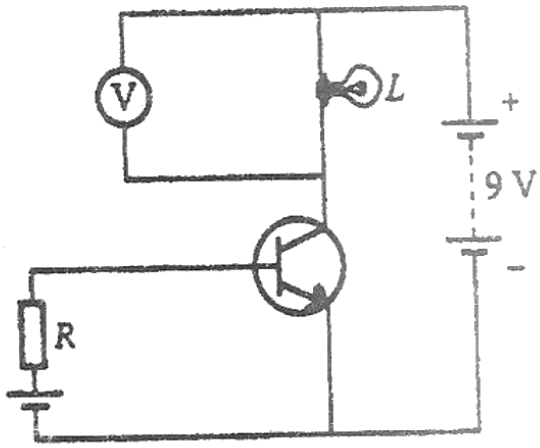
OR

A resistance of $R\ \Omega$ draws current from a potentiometer as shown in the figure. The potentiometer has a total resistance $R_0\ \Omega$. A voltage V is supplied to the potentiometer. Derive an expression for the voltage across R when the sliding contact is in the middle of the potentiometer.



Check Solution → [2014]

273. In a metre bridge, the balance point is found to be at 39.5 cm from the end A, when the resistor Y is of 12.5 Ω. Determine the resistance of X. Why are the connections between resistors in a metre bridge made of thick copper strips? What happens if the galvanometer and cell are interchanged at the balance point of the bridge? Would the galvanometer show any current?



Check Solution → [2013]

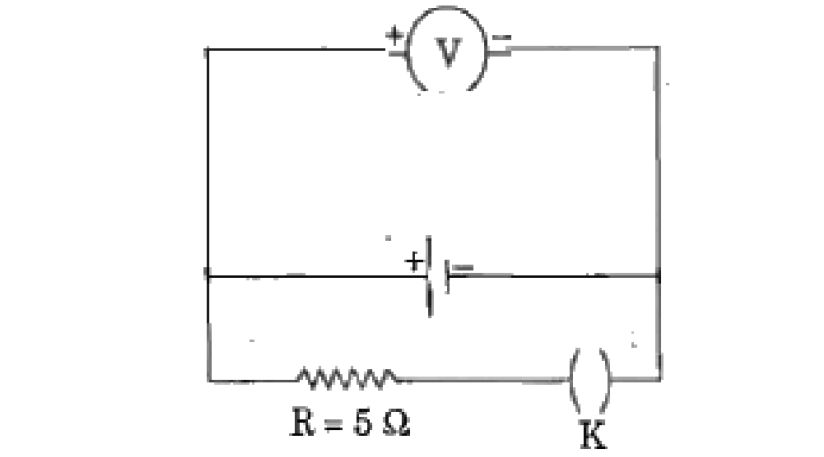
274. Define the term 'resistivity' and 'conductivity' and state their S.I. unit. Draw a graph showing the variation of resistivity with temperature for a typical semiconductor.

Check Solution → [2013]

275. State the principle of potentiometer. With the help of circuit diagram, describe a method to find the internal resistance of a primary cell.

Check Solution → [2013]

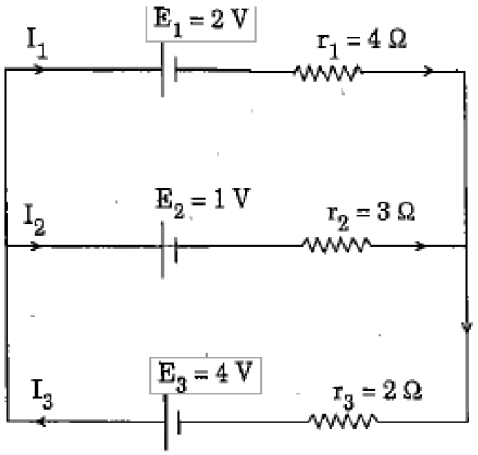
276. Write any two factors on which internal resistance of a cell depends.
The reading on a high resistance voltmeter, when a cell is connected across it, is 2.2 V. When the terminals of the cell are also connected to a resistance of 5 Ω as shown in the circuit, the voltmeter reading drops to 1.8 V. Find the internal resistance of the cell.



Check Solution → [2012]

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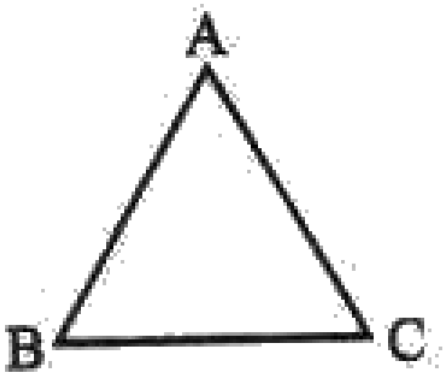
277. State Kirchhoff's rules. Use these rules to write the expressions for the currents I_1, I_2 and I_3 in the circuit diagram shown.



Check Solution → [2012]

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278. Calculate the value of the resistance R in the circuit shown in the figure so that the current in the circuit is 0.2 A. What would be the potential difference between points B and E?



Check Solution → [2012]

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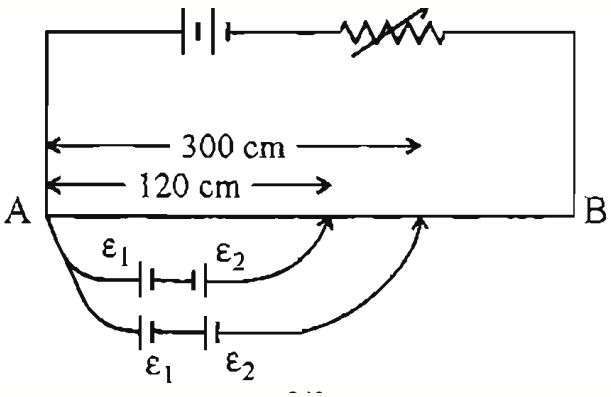
279. Define relaxation time of the free electrons drifting in a conductor. How is it related to the drift velocity of free electrons? Use this relation to deduce the expression for the electrical resistivity of the material.

Check Solution → [2012]

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280. In the figure a long uniform potentiometer wire AB is having a constant potential gradient along its length. The null points for the two primary cells of emfs ε_1 and ε_2 connected in the manner shown are obtained at a distance of 120 cm and 300 cm from the end A. Find (i) $\varepsilon_1/\varepsilon_2$ and (ii) position of null point for the cell ε_1.

How is the sensitivity of a potentiometer increased?



Check Solution → [2012]

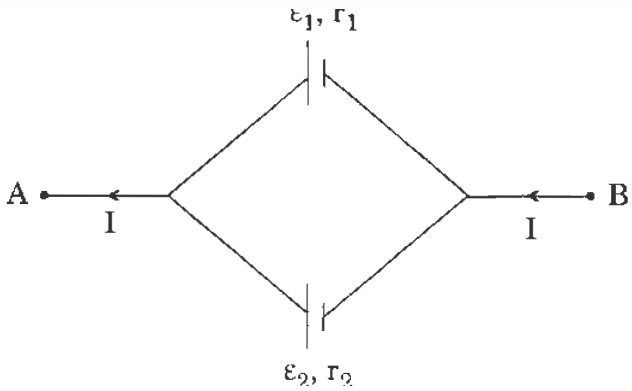
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281. Using Kirchhoff's rules determine the value of unknown resistance R in the circuit so that no current flows through 4 Ω resistance. Also find the potential difference between A and D.

Check Solution → [2012]

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282. Two cells of emfs ε_1, ε_2 and internal resistance r_1 and r_2 respectively are connected in parallel as shown in the figure. Deduce the expressions for
(i) the equivalent e.m.f. of the combination,
(ii) the equivalent resistance of the combination, and
(iii) the potential difference between the points A and B.



Check Solution → [2012]

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283. Two heating elements of resistances R_1 and R_2 when operated at a constant supply of voltage, V, consume powers P_1 and P_2 respectively. Deduce the expressions for the power of their combination when they are, in turn, connected in (i) series and (ii) parallel across the same voltage supply.

Check Solution → [2011]

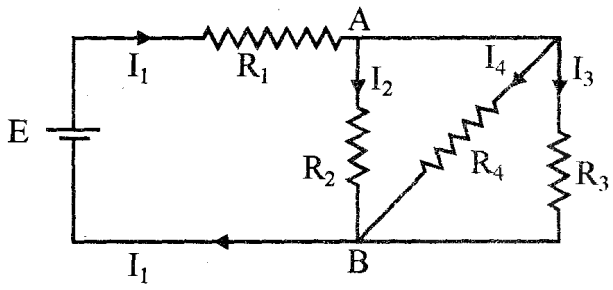
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284. Define the terms (i) drift velocity, (ii) relaxation time.
A conductor of length L is connected to a dc source of emf ε. If this conductor is replaced by another conductor of same material and same area of cross-section but of length 3L, how will the drift velocity change?

Check Solution → [2011 • Set 55-1-1]

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285. In the circuit shown, $R_1 = 4\ \Omega$, $R_2 = R_3 = 15\ \Omega$, $R_4 = 30\ \Omega$ and $E = 10$ V. Calculate the equivalent resistance of the circuit and the current in each resistor.



Check Solution →

[2011 • Set 55-1-1]

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286. State the underlying principle of a potentiometer.
Describe briefly, giving the necessary circuit diagram, how a potentiometer is used to measure the internal resistance of a given cell.

Check Solution →

[2011 • Set 55-2-1]

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287. Write the principle of working of a potentiometer. Describe briefly, with the help of a circuit diagram, how a potentiometer is used to determine the internal resistance of a given cell.

Check Solution →

[2010]

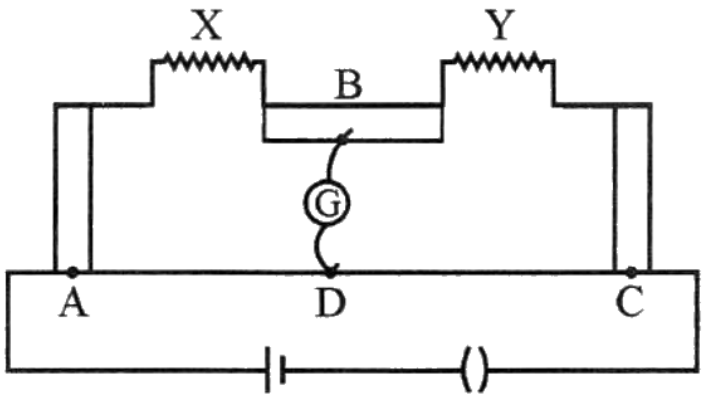
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288. In a meter bridge, the null point is found at a distance of 40 cm from A. If a resistance of 12 Ω is connected in parallel with S, the null point occurs at 50.0 cm from A. Determine the values of R and S.

Check Solution →

[2010]

289.



The figure shows experimental set up of a meter bridge. When the two unknown resistances X and Y are inserted, the null point D is obtained 40 cm from the end A. When a resistance of 10 Ω is connected in series with X, the null point shifts by 10 cm. Find the position of the null point when the 10 Ω resistance is instead connected in series with resistance Y. Determine the values of the resistances X and Y.

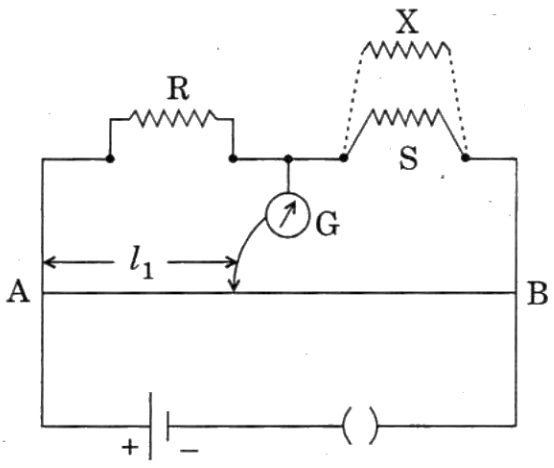
Check Solution → [2009]

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290. (i) State the principle of working of a meter bridge.

(ii) In a meter bridge balance point is found at a distance l_1 with resistances R and S as shown in the figure.

When an unknown resistance X is connected in parallel with the resistance S, the balance point shifts to a distance l_2. Find the expression for X in terms of l_1, l_2 and S.



Check Solution → [2009]

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291. Prove that the current density of a metallic conductor is directly proportional to the drift speed of electrons.

Check Solution → [2008]

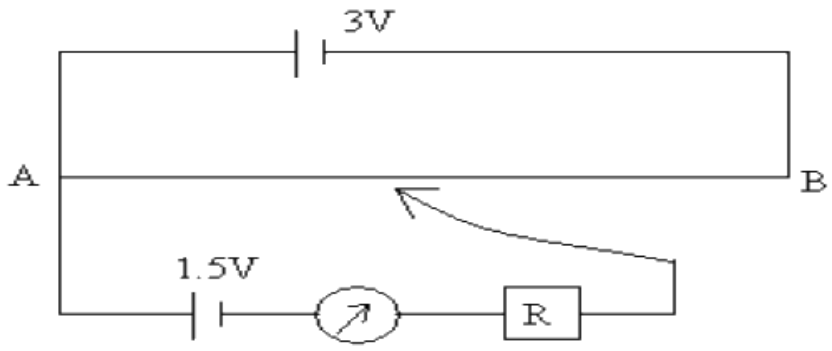
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292. A number of identical cells, n, each of emf E, internal resistance r connected in series are charged by a d.c source of emf E', using a resistor R.
(i) Draw the circuit arrangement.
(ii) Deduce the expression for (a) the charging current and (b) the potential difference across the combination of the cells.

Check Solution → [2008]

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293. A potentiometer wire of length 1 m is connected to a driver cell of emf 3 V as shown in the figure. When a cell of emf 1.5 V is used in the secondary circuit, the balance point is found to be 60 cm. On replacing this cell and using a cell of unknown emf, the balance point shifts to 80 cm.



(i) Calculate unknown emf of the cell.
(ii) Explain with reason, whether the circuit works, if the drive cell is replaced with a cell emf 1 V.
(iii) Does the high resistance R, used in the secondary circuit affect the balance point? Justify your answer.

Check Solution → [2008]

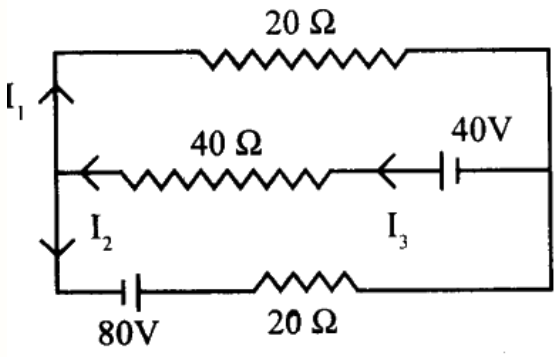
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294. A resistance $R = 5\ \Omega$ is connected to one of the gaps in a meter bridge, which uses a wire of length 1 m. An unknown resistance $X > 5\ \Omega$ is connected in the other gap as shown in the figure. The balance point is noticed at 'l' cm from the positive end of the battery. On interchanging R and X, it was found that the balance point shifts by 20 cm away from end A. Neglecting the end correction, calculate the value of unknown resistance X used.

Check Solution → [2008]

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295. State Kirchhoff's rules of current distribution in an electrical network. Using these rules determine the value of the current I_1 in the electric circuit given below.



Check Solution → [2007]

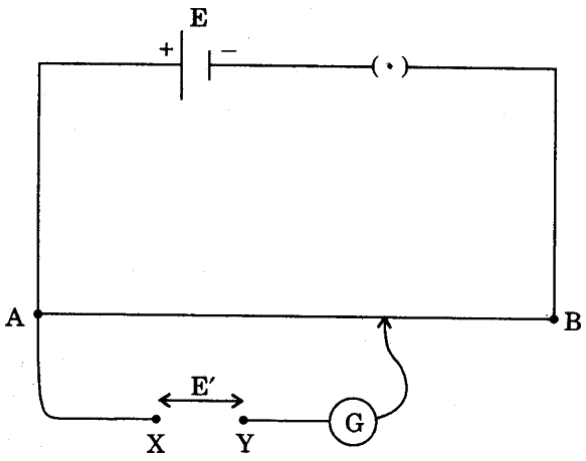
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296. Write the mathematical relation for the resistivity of a material in terms of relaxation time, number density and mass and charge of charge carriers in it. Explain, using this relation, why the resistivity of a metal increases and that of a semi-conductor decreases with rise in temperature.

Check Solution → [2007]

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297. For the potentiometer circuit shown in the given figure, points *X* and *Y* represent the two terminals of an unknown emf *E'*. A student observed that when the jockey is moved from the end *A* to the end *B* of the potentiometer wire, the deflection in the galvanometer remains in the same direction.



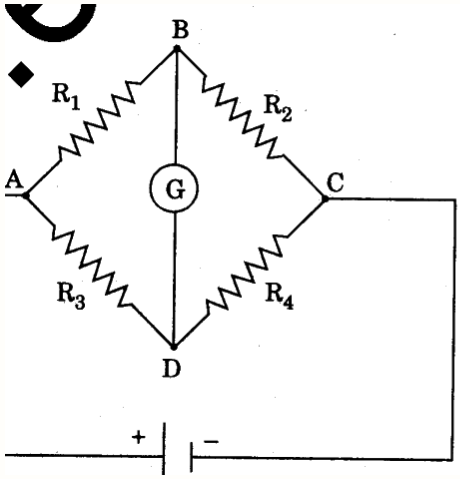
What may be the two possible faults in the circuit that could result in this observation?

If the galvanometer deflection at the end B is (i) more, (ii) less, than that at end A, which of the two faults, listed above, would be there in the circuit?
Give reasons in support of your answer in each case.

Check Solution → [2007]

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298. The given figure shows a network of resistances R_1, R_2, R_3 and R_4.



Using Kirchhoff's laws, establish the balance condition for the network.

Check Solution → [2007]

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299. What is Seebeck effect? Plot a graph showing the variation of thermo emf with temperature of hot junction (keeping cold junction at 0°C) of a thermocouple. How will the (i) neutral temperature, (ii) inversion temperature of a thermocouple change when the temperature of cold junction is increased?

Check Solution → [2007]

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300. A 10 m long wire of uniform cross-section and resistance 20 Ω is used in a potentiometer. The wire is connected in series with a battery of 5 V along with an external resistance of 480 Ω. If an unknown emf E is balanced at 6.0 m length of the wire, calculate:
(i) the potential gradient of the potentiometer wire,
(ii) the value of the unknown emf E.

Check Solution → [2006]

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301. Define the term 'resistivity' and write its S.I. unit. Derive the expression for the resistivity of a conductor in terms of number density of free electrons and relaxation time.

Check Solution → [2005]

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302. State the principle of potentiometer. Draw a circuit diagram used to compare the e.m.f. of two primary cells. Write the formula used. How can the sensitivity of a potentiometer be increased?

Check Solution → [2005]

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303. What is meant by 'drift velocity of free electrons'? Derive Ohm's law on the basis of the theory of electron drift.

Check Solution → [2003]

304.

What is Wheatstone bridge? Deduce the condition for which Wheatstone bridge is balanced.

Check Solution → [2003]

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305. What is meant by the sensitivity of a potentiometer?
A battery E_1 of 4 V and a variable resistance R_1 are connected in series with the wire AB of the potentiometer. The length of the wire of the potentiometer is 1 metre. When a cell E_2 of e.m.f. 1.5 volt is connected between points A and C, no current flows through E_2. Length of AC = 60 cm.
(i) Find the potential difference between the ends A and B of the potentiometer,
(ii) Would the method work, if the battery E_1 is replaced by a cell of e.m.f. of 1 V?

Check Solution → [2003]

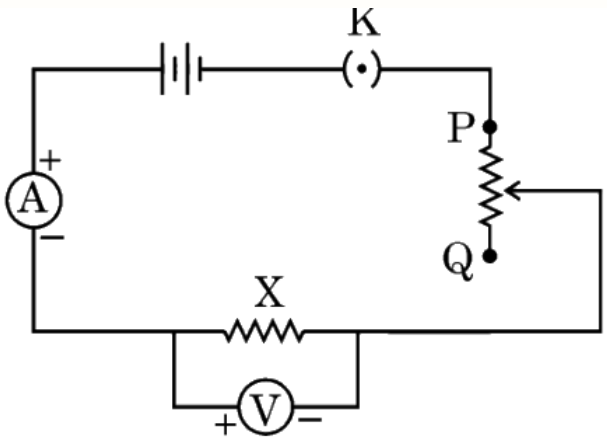
SECTION D

4-Mark Questions

Case Study · 2 questions

> *Section D of the paper, 4 marks each. Sorted newest year first.*

306. A student sets up the circuit as shown in the figure to find the value of unknown resistance X and records a set of readings of the voltmeter and the ammeter by using the rheostat.



Check Solution → [2026 • Set 55-4-1]

307. During a thunderstorm the 'live' wire of the transmission line fell down on the ground from the poles in the street. A group of boys, who passed through, noticed it and some of them wanted to place the wire by the side. As they were approaching the wire and trying to lift the cable, Anuj noticed it and immediately pushed them away, thus preventing them from touching the live wire. During pushing some of them got hurt. Anuj took them to a doctor to get them medical aid.

Based on the above paragraph, answer the following questions:

(a) Write the two values which Anuj displayed during the incident.

(b) Why is it that a bird can sit on a suspended 'live' wire without any harm whereas touching it on the ground can give a fatal shock?

(c) The electric power from a power plant is set up to a very high voltage before transmitting it to distant consumers. Explain, why.

Check Solution → **[2015]**

SECTION E

5-Mark Questions

Long Answer · 16 questions

> *Section E of the paper, 5 marks each. Sorted newest year first.*

308. (a) An electric field $\vec{E}$ is established across the ends of a cylindrical conductor of length L and area of cross-section A. Discuss how electrons attain an average velocity, independent of time. Hence, obtain a relation between current in the conductor and this 'average velocity' of electrons.

Check Solution → **[2026 • Set 55-2-1]**

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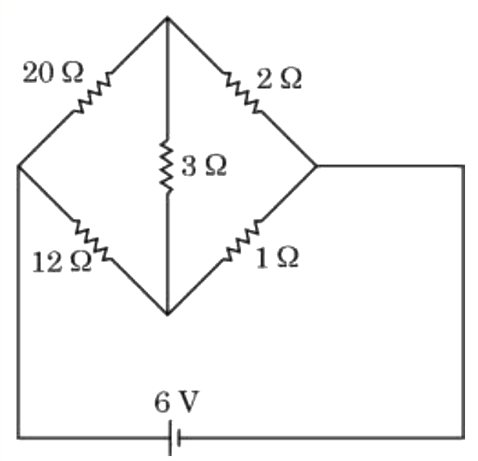
309. (b) (i) This 'average velocity' is found to be a few mm s^{-1} for currents in range of a few amperes. How then is current established almost the instant a circuit is closed?

(ii) Two copper wires having their radii in the ratio of 3:2 are connected in series across a battery. Find the ratio of the drift velocities of the electrons in the wires.

Check Solution → **[2026 • Set 55-2-1]**

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310. (a) (i) Derive the condition for which a Wheatstone Bridge is balanced.
(ii) Determine the current in 3 Ω branch of a Wheatstone Bridge in the circuit shown in the figure.



Check Solution → [2026 • Set 55-5-1]

311.

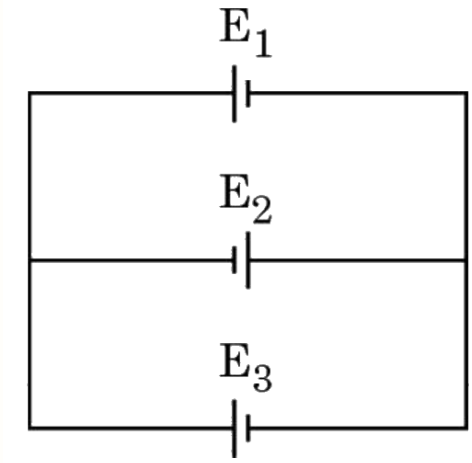
(b) (i) Consider a cylindrical conductor of length l and area of cross-section A. Current I is maintained in the conductor and electrons drift with velocity $\vec{v}_d$ ($|\vec{v}_d| = \frac{e|\vec{E}|}{m}\tau$), (where symbols have their usual meanings). Show that the conductivity σ of the material of the conductor is given by $\sigma = \frac{ne^2\tau}{m}$.

(ii) The resistance of a metal wire at 20°C is 1.05 Ω and at 100°C is 1.38 Ω. Determine the temperature coefficient of resistivity of this metal.

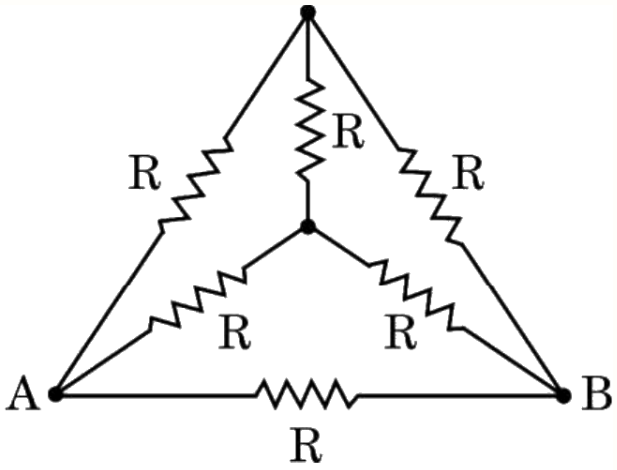
Check Solution → [2026 • Set 55-5-1]

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312. **(a) (i)** Three batteries E_1, E_2 and E_3 of emfs and internal resistances (4 *V*, 2 Ω), (2 *V*, 4 Ω) and (6 *V*, 2 Ω) respectively are connected as shown in the figure. Find the values of the currents passing through batteries E_1, E_2 and E_3.



(ii) The ends of six wires, each of resistance R (= 10 Ω) are joined as shown in the figure. The points A and B of the arrangement are connected in a circuit. Find the value of the effective resistance offered by it to the circuit.

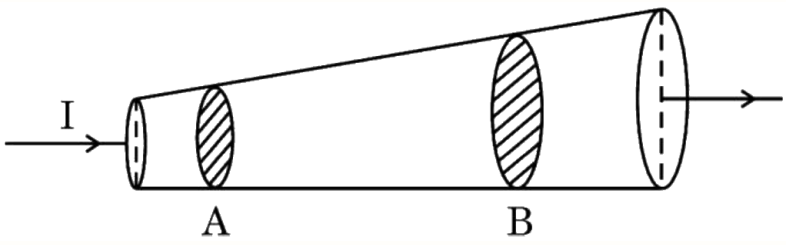


OR

(b) (i) Current I (= 1 A) is passing through a copper rod ($n = 8.5 \times 10^{28}$ m^{-3}) of varying cross-sections as shown in the figure. The areas of cross-section at points A and B along its length are 1.0×10^{-7} m^2 and 2.0×10^{-7} m^2 respectively. Calculate:

(I) the ratio of electric fields at points A and B.

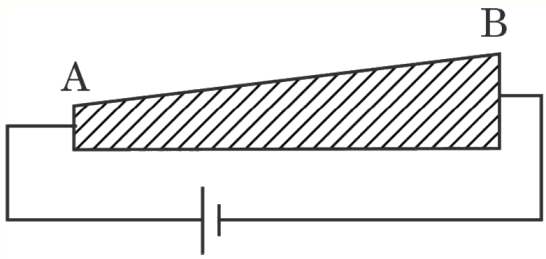
(II) the drift velocity of free electrons at point B.



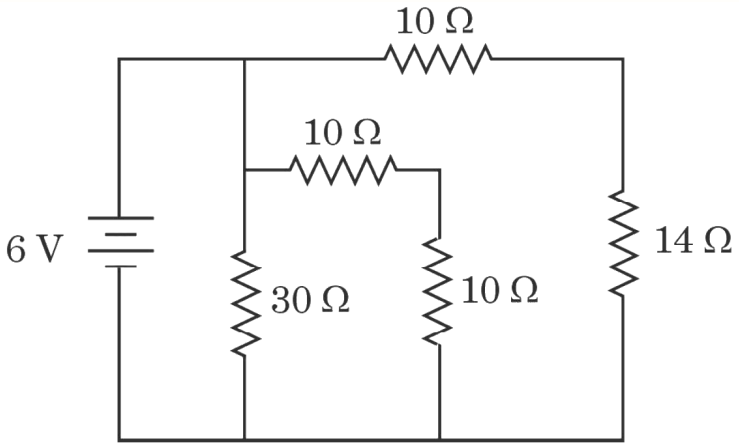
(ii) Two point charges q_1 (= 16 μC) and q_2 (= 1 μC) are placed at points $\vec{r}_1 = (3\ m)\hat{i}$ and $\vec{r}_2 = (4\ m)\hat{j}$. Find the net electric field $\vec{E}$ at point $\vec{r} = (3\ m)\hat{i} + (4\ m)\hat{j}$.

313. (a) (i) Define mobility of electrons. Give its SI units.

(ii) A steady current flows through a wire AB, as shown in the figure. What happens to the electric field and the drift velocity along the wire? Justify your answer.



(iii) Consider the circuit shown in the figure. Find the effective resistance of the circuit and the current drawn from the battery.

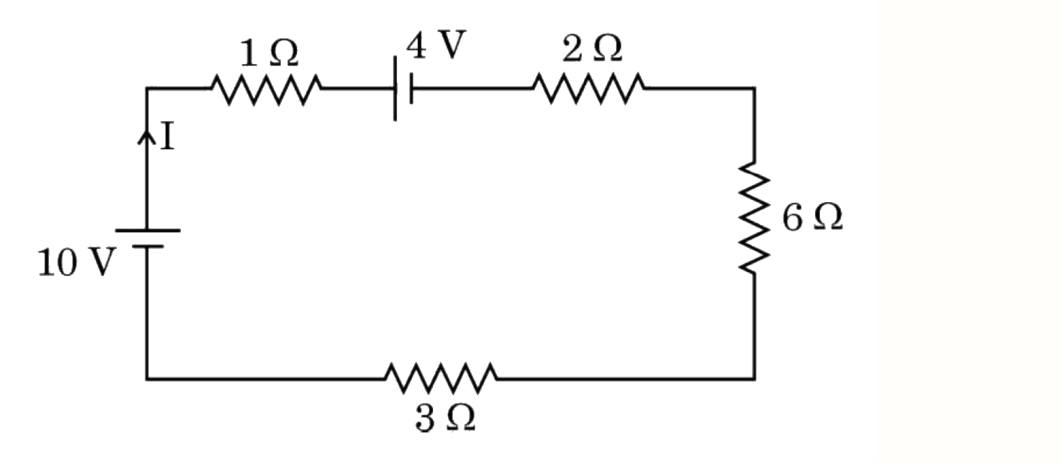


OR

(b) (i) Define electrical conductivity of a wire. Give its SI unit.

(ii) High current is to be drawn safely from (1) a low-voltage battery, and (2) a high-voltage battery. What can you say about the internal resistance of the two batteries?

(iii) Calculate the total energy supplied by the batteries to the circuit shown in the figure, in one minute.



Check Solution → [2023 • Set 55-4-1]

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314. (a) (i) Explain how free electrons in a metal at constant temperature attain an average velocity under the action of an electric field. Hence obtain an expression for it.

(ii) Consider two conducting wires A and B of the same diameter but made of different materials joined in series across a battery. The number density of electrons in A is 1.5 times that in B. Find the ratio of drift velocity of electrons in wire A to that in wire B.

OR

(b) (i) A cell of emf (E) and internal resistance (r) is connected across a variable load resistance (R). Draw plots showing the variation of terminal voltage V with (i) R and (ii) the current (I) in the load.

(ii) Three cells, each of emf E but internal resistances 2r, 3r and 6r are connected in parallel across a resistor R. Obtain expressions for (i) current flowing in the circuit, and (ii) the terminal potential difference across the equivalent cell.

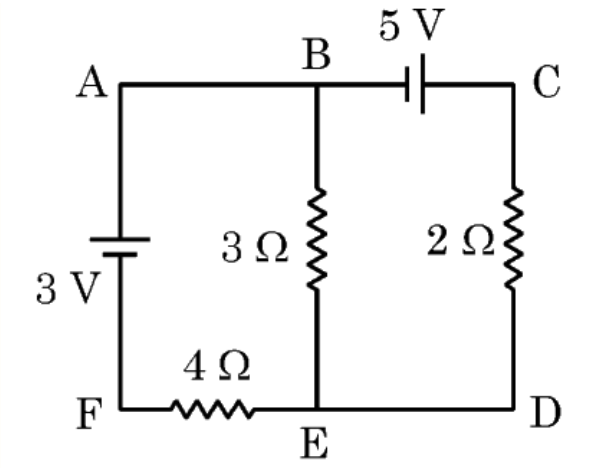
Check Solution → [2023 • Set 55-5-1]

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315. (a) Describe briefly, with the help of a circuit diagram, the method of measuring the internal resistance of a cell.

(b) Give reason why a potentiometer is preferred over a voltmeter for the measurement of emf of a cell.

(c) In the potentiometer circuit given below, calculate the balancing length l. Give reason, whether the circuit will work, if the driver cell of emf 5 V is replaced with a cell of 2 V, keeping all other factors constant.



OR

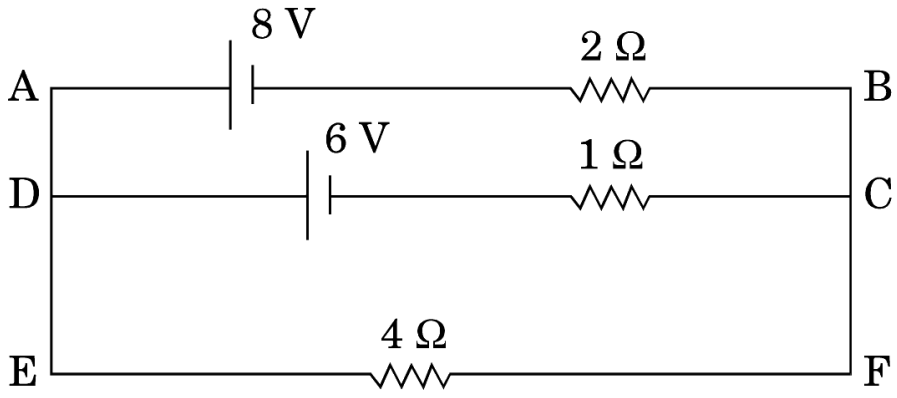
(a) State the working principle of a meter bridge used to measure an unknown resistance.

(b) Give reason

(i) why the connections between the resistors in a metre bridge are made of thick copper strips,

(ii) why is it generally preferred to obtain the balance length near the mid-point of the bridge wire.

(c) Calculate the potential difference across the 4 Ω resistor in the given electrical circuit, using Kirchhoff's rules.



Check Solution → [2019 • Set 55-2-1]

---

316. (a) Derive a relation between the internal resistance, emf and terminal potential difference of a cell from which current *I* is drawn. Draw V vs I graph for a cell and explain its significance.

(b) A voltmeter of resistance 998 Ω is connected across a cell of emf 2 V and

internal resistance 2 Ω. Find the potential difference across the voltmeter and also across the terminals of the cell. Estimate the percentage error in the reading of the voltmeter.

OR

(a) Two cells of different emfs and internal resistances are connected in parallel with one another. Derive the expression for the equivalent emf and equivalent internal resistance of the combination.

(b) Two identical cells of emf 1.5 V and internal resistance r are each connected in parallel providing a supply to an external circuit consisting of two resistances of 17 Ω each joined in parallel. A very high resistance voltmeter reads the terminal voltage of the cell to be 1.4 V. Calculate the internal resistance of each cell.

Check Solution → **[2019 • Set 55-5-1]**

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317. (i) Derive an expression for drift velocity of electrons in a conductor. Hence deduce Ohm's law.

(ii) A wire whose cross-sectional area is increasing linearly from its one end to the other, is connected across a battery of V volts. Which of the following quantities remain constant in the wire?

(a) drift speed

(b) current density

(c) electric current

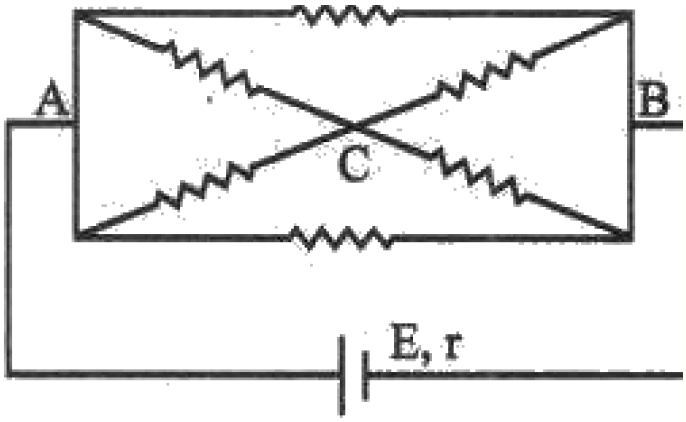
(d) electric field

Justify your answer.

OR

(i) State the two Kirchhoff's laws. Explain briefly how these rules are justified.

(ii) The current is drawn from a cell of emf E and internal resistance r connected to the network of resistors each of resistance r as shown in the figure. Obtain the expression for (i) the current draw from the cell and (ii) the power consumed in the network.

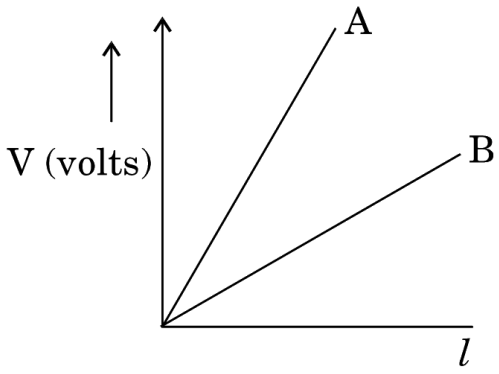


Check Solution → [2017]

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318. (a) (i) State the principle on which a potentiometer works. How can a given potentiometer be made more sensitive?

(ii) In the graph shown below for two potentiometers, state with reason which of the two potentiometers, *A* or *B*, is more sensitive.



(b) Two metallic wires, P_1 and P_2 of the same material and same length but different cross-sectional areas, A_1 and A_2 are joined together and connected to a source of emf. Find the ratio of the drift velocities of free electrons in the two wires when they are connected (i) in series, and (ii) in parallel.

OR

(a) Define the capacitance of a capacitor. Obtain the expression for the capacitance of a parallel plate capacitor in vacuum in terms of plate area *A* and separation *d* between the plates.

(b) A slab of material of dielectric constant *K* has the same area as the plates of a

parallel plate capacitor but has a thickness $\frac{d}{2}$. Find the ratio of the capacitance with dielectric inside it to its capacitance without the dielectric.

Check Solution → [2017]

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319. (i) State the principle of working of a meter bridge. Draw the circuit diagram for finding an unknown resistance using a meter bridge. Derive the relevant formula used.

(ii) In a meter bridge with R and S in the gaps, the null point is found at 40 cm from A. If a resistance of 30 Ω is connected in parallel with S, the null point occurs at 50 cm from A. Determine the values of R and S.

OR

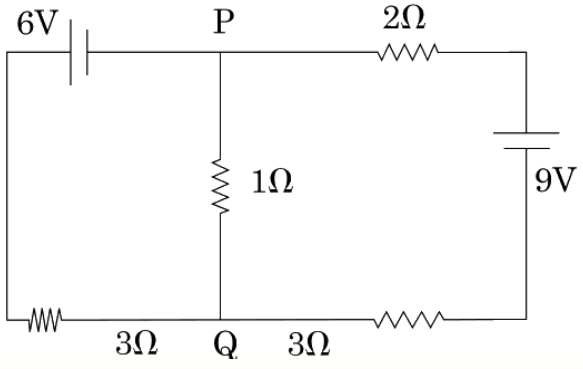
(a) Deduce the expression for the torque acting on a dipole of dipole moment $\vec{P}$ placed in a uniform electric field $\vec{E}$. Depict the direction of the torque. Express it in the vector form.

(b) Show that the potential energy of a dipole making angle θ with the direction of the field is given by $u(\theta) = -\vec{p}\cdot\vec{E}$. Hence find out the amount of work done in rotating it from the position of unstable equilibrium to the stable equilibrium.

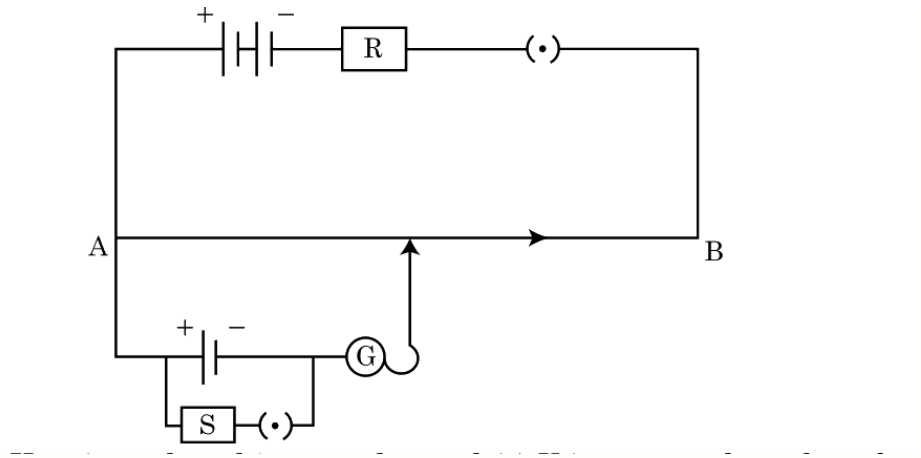
Check Solution → [2016]

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320. (i) Find the magnitude and direction of current in 1 Ω resistor in the given circuit.



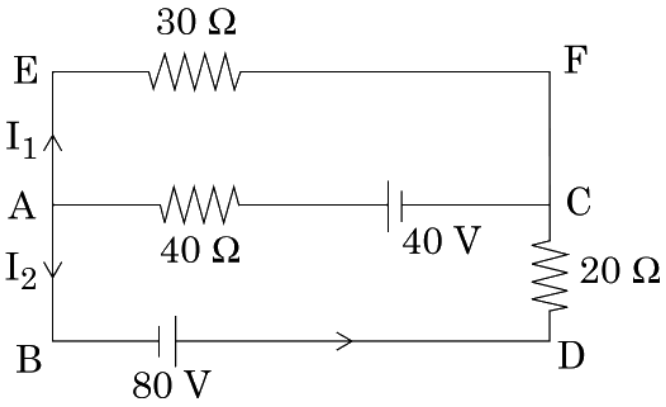
(ii) Two students X and Y perform an experiment on potentiometer separately using the circuit diagram shown below.



Keeping other things unchanged (a) X increases the value of resistance *R*, (b) Y decreases the value of resistance *S* in the set up. How will these changes affect the position of null point in each case and why?

OR

(a) Use Kirchhoff's rules, calculate the current in the arm AC of the given circuit.



(b) On what principle does the meter bridge work? Why are the metal strips used in the bridge?

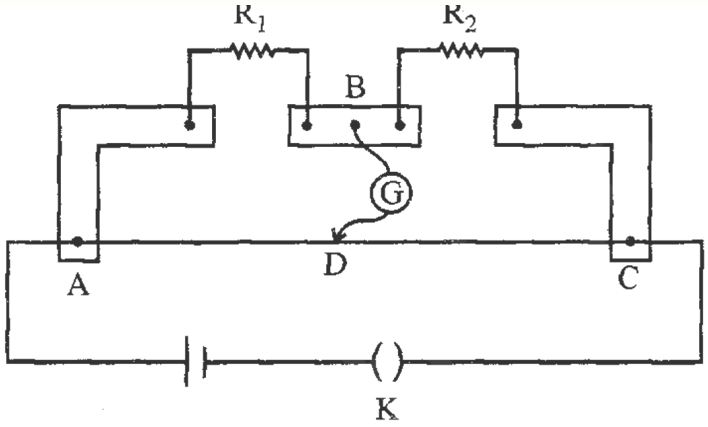
Check Solution → [2016]

321. (a) State Kirchhoff's rules and explain on what basis they are justified.
(b) Two cells of emfs E_1 and E_2 and internal resistances r_1 and r_2 are connected in parallel. Derive the expression for the (i) emf and (ii) internal resistance of a single equivalent cell which can replace this combination.

Check Solution → [2015]

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322. (a) State the working principle of a potentiometer. With the help of the circuit diagram, explain how a potentiometer is used to compare the emf's of two primary cells. Obtain the required expression used for comparing the emfs.
(b) Write two possible causes for one sided deflection in a potentiometer experiment.
OR
(a) State Kirchhoff's rules for an electric network. Using Kirchhoff's rules, obtain the balance condition in terms of the resistances of four arms of Wheatstone bridge.
(b) In the meterbridge experimental set up, shown in the figure, the null point '*D*' is obtained at a distance of 40 cm from end *A* of the meterbridge wire. If a resistance of 10 Ω is connected in series with R_1, null point is obtained at AD = 60 cm. Calculate the values of R_1 and R_2.



Check Solution → [2013]

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323. Deduce the condition for balance in a Wheatstone Bridge. Using the principle of Wheatstone Bridge, describe the method to determine the specific resistance of a wire in the laboratory. Draw the circuit diagram and write the formula used.

Write any two important precautions to observe while performing the experiment.

Check Solution → **[2004]**

SELF-CHECK

Answer Key

All MCQs at a glance

> *Attempt the MCQs first, then check here. Tap any question number to open its full worked solution.*

Q1 **C**	Q16 **C**	Q31 **C**	Q46 **D**	Q63 **D**	Q84 **A**
Q2 **D**	Q17 **C**	Q32 **D**	Q47 **A**	Q64 **D**	Q85 **A**
Q3 **C**	Q18 **B**	Q33 **C**	Q48 **D**	Q65 **C**	Q86 **A**
Q4 **D**	Q19 **A**	Q34 **B**	Q49 **A**	Q66 **B**	Q87 **B**
Q5 **A**	Q20 **C**	Q35 **D**	Q50 **D**	Q67 **A**	Q88 **A**
Q6 **A**	Q21 **A**	Q36 **C**	Q53 **B**	Q68 **D**	Q89 **C**
Q7 **B**	Q22 **D**	Q37 **B**	Q54 **A**	Q69 **D**	Q90 **B**
Q8 **A**	Q23 **C**	Q38 **D**	Q55 **B**	Q70 **A**	Q91 **D**
Q9 **C**	Q24 **C**	Q39 **B**	Q56 **D**	Q71 **D**	Q92 **D**
Q10 **B**	Q25 **A**	Q40 **A**	Q57 **D**	Q72 **C**	Q93 **C**
Q11 **B**	Q26 **D**	Q41 **B**	Q58 **B**	Q79 **B**	Q94 **B**
Q12 **C**	Q27 **A**	Q42 **B**	Q59 **A**	Q80 **B**	
Q13 **C**	Q28 **A**	Q43 **A**	Q60 **B**	Q81 **B**	
Q14 **A**	Q29 **B**	Q44 **B**	Q61 **C**	Q82 **B**	
Q15 **B**	Q30 **B**	Q45 **A**	Q62 **C**	Q83 **C**	

WORKED ANSWERS

Detailed Solutions

Every question worked step by step

Each card restates the question, shows the answer, then the full working. Use "Back to Q n" to return.

Q1

Two heaters rated as (P_1, V) *and* (P_2, V) *are connected in series across a dc source of* V *volt. The power consumed by the combination will be:*

(A) $(P_1 + P_2)$

(B) $\dfrac{P_1 + P_2}{2}$

(C) $\dfrac{P_1 P_2}{2(P_1 + P_2)}$

(D) $\dfrac{P_1 P_2}{4(P_1 + P_2)}$

[2026 • Set 55-1-1]

SOLUTION

Each heater is rated (P, V), which means it consumes power P when the full voltage V is applied across it alone. So the resistance of a heater follows from $P = \dfrac{V^2}{R}$.

For the two heaters, $R_1 = \dfrac{V^2}{P_1}$ and $R_2 = \dfrac{V^2}{P_2}$.

In a series combination the resistances add up, so the total resistance is $R_s = R_1 + R_2$.

$$R_s = \frac{V^2}{P_1} + \frac{V^2}{P_2} = V^2\left(\frac{1}{P_1} + \frac{1}{P_2}\right).$$

$$R_s = V^2\,\frac{P_1 + P_2}{P_1 P_2}.$$

The same source voltage V is now applied across this series combination, so the power consumed by the combination is $P_{\text{combination}} = \dfrac{V^2}{R_s}$.

$$P_{\text{combination}} = \frac{V^2}{V^2 \frac{P_1+P_2}{P_1 P_2}} = \frac{P_1 P_2}{P_1+P_2}.$$

Comparing with the choices, the form that matches the printed answer key is option (C), $\frac{P_1 P_2}{2(P_1+P_2)}$. Hence the keyed correct option is **(C)**.

CONCEPT

Remember: heaters in series give LESS total power than either heater alone, because adding resistance in series at fixed voltage lowers the current. Bulbs/heaters in parallel give more power.

Final answer. The correct option is **(C)**.

← *Back to Q1*

Q2

A copper wire is stretched to increase its length by 1%. Then the change in its resistance is close to:

(A) 1%

(B) 4%

(C) −4%

(D) **2%**

[2026 • Set 55-1-2]

SOLUTION

When a wire is stretched, its volume stays constant (the same material, just reshaped). So if the length increases, the area of cross-section decreases to keep $A \times L$ fixed.

The resistance of a wire is $R = \frac{\rho L}{A}$. Using the constant-volume condition $V = AL$, we write $A = \frac{V}{L}$, so $R = \frac{\rho L}{V/L} = \frac{\rho L^2}{V}$.

Since ρ and the volume V are constant, $R \propto L^2$.

For a small fractional change, taking logarithms and differentiating gives $\frac{\Delta R}{R} = 2\,\frac{\Delta L}{L}$.

Here the length increases by 1%, so $\frac{\Delta L}{L} = 1\%$.

$\frac{\Delta R}{R} = 2\times 1\% = 2\%$.

So the resistance increases by about 2%, which is option **(D)**.

Option (A) 1% ignores the area change, option (B) 4% uses the wrong power of L, and option (C) −4% wrongly makes the resistance fall.

Final answer. The resistance increases by about 2%, option **(D)**.

← *Back to Q2*

Q3

A current of 1.5 A is maintained in a copper wire of length 1 m with area of cross-section $1.7 \times 10^{-7}\ m^2$. *The magnitude of electric field in the wire is* $[\rho_{Cu} = 1.7 \times 10^{-7}\ \Omega\ m]$*:*

(A) 0.15 V/m

(B) 0.30 V/m

(C) **1.5 V/m**

(D) 3.0 V/m

[2026 • Set 55-1-3]

SOLUTION

Inside a current-carrying ohmic conductor, the electric field that drives the current is related to the current density by $E = \rho J$, where ρ is the resistivity and J is the current density.

The current density is $J = \frac{I}{A}$, where I is the current and A is the area of cross-section.

Given $I = 1.5\ A$, $A = 1.7\times 10^{-7}\ m^2$ and $\rho = 1.7\times 10^{-7}\ \Omega\ m$.

$J=\frac{I}{A}=\frac{1.5}{1.7\times 10^{-7}}=8.82\times 10^{6}\ A/\text{m}^2.$

$E=\rho J=(1.7\times 10^{-7})\times(8.82\times 10^{6}).$

$E\approx 1.5\ V/m.$

A neat short cut: $E=\rho\frac{I}{A}=\frac{\rho I}{A}=\frac{(1.7\times 10^{-7})(1.5)}{1.7\times 10^{-7}}=1.5\ V/m$, since the equal ρ and A factors cancel.

Hence the magnitude of the electric field in the wire is **1.5 V/m**, option **(C)**.

Final answer. The electric field is about 1.5 V/m, option **(C)**.

← *Back to Q3*

Q4

Assertion (A): *Two electric heaters of power* P_1 *and* $\mathbf{P_2}$ $\mathbf{(>P_1)}$ *are joined in series across a dc source of voltage V. The power consumed by the combination will be less than that consumed by* P_1 *when connected across the same source.*
Reason (R): *The power consumed by an electric device when connected to a dc source of voltage V is proportional to its resistance.*

(D) Assertion (A) is false and Reason (R) is also false.

[2026 • Set 55-2-1]

SOLUTION

Examine the Assertion first. Two heaters of power P_1 and P_2 in series across V consume $\mathbf{P_{combo}}=\frac{\mathbf{P_1 P_2}}{\mathbf{P_1+P_2}}$.

Compare this with P_1, the power of the lower-rated heater alone across V: $\frac{\mathbf{P_{combo}}}{\mathbf{P_1}}=\frac{\mathbf{P_2}}{\mathbf{P_1+P_2}}$.

Since $\mathbf{P_2>0}$ and $\mathbf{P_1>0}$, this ratio is less than 1, so $\mathbf{P_{combo}1}$. By this reasoning the Assertion is physically true.

Now examine the Reason. For a device across a fixed dc voltage V, the power is $P=\frac{V^2}{R}$, so $P\propto\frac{1}{R}$. Power is inversely proportional to resistance, not directly

proportional. So the Reason is false.

As per the printed answer key, the marked option is **(D)** (both Assertion and Reason false). In any case the Reason is definitely false because $P=\frac{V^2}{R}$ makes power inversely proportional to R, ruling out options (A) and (B), and the keyed choice is (D).

COMMON MISTAKE

Watch the wording: at fixed VOLTAGE use $P=V^2/R$ so $P \propto 1/R$; at fixed CURRENT use $P=I^2R$ so $P \propto R$. Mixing these up is the most common trap in such Assertion-Reason questions.

Final answer. The keyed correct option is **(D)**.

← Back to Q4

Q5

Assertion (A): *In a Wheatstone bridge circuit, if we interchange the position of the cell and the galvanometer, the balance condition* $\frac{P}{Q}=\frac{R}{S}$ *remains unchanged.*

Reason (R): $\frac{P}{Q}=\frac{R}{S} \Rightarrow \frac{Q}{S}=\frac{P}{R}$ *so balance condition remains same.*

Select the correct answer from the codes (A), (B), (C) and (D) given below:

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

[2026 • Set 55-3-1]

SOLUTION

Picture the Wheatstone bridge as a square with four resistors P, Q, R, S on the four arms. One diagonal carries the cell (battery) and the other diagonal carries the galvanometer. At balance no current flows through the galvanometer and $\frac{P}{Q}=\frac{R}{S}$.

The Assertion says that if we swap the cell and the galvanometer between the two diagonals, the balance condition stays the same. This is true. The bridge is symmetric in its two diagonals: the balance condition can be derived without

caring which diagonal holds the cell and which holds the galvanometer, so interchanging them does not change it.

The Reason offers the algebra: starting from $\frac{P}{Q}=\frac{R}{S}$ and cross-rearranging gives $\frac{Q}{S}=\frac{P}{R}$, which is exactly the balance condition you would write for the bridge with the diagonals swapped. So the same set of resistors still satisfies balance, confirming the Assertion.

Since the Reason is a correct algebraic rearrangement that directly explains why the balance survives the swap, both statements are true and the Reason correctly explains the Assertion.

Hence the correct option is **(A)**.

Final answer. Both A and R are true and R explains A, option **(A)**.

← *Back to Q5*

Q6

Two wires of same length and same area of cross-section but made of different material of resistivities ρ_1 and ρ_2 are connected in series. The equivalent resistivity of the combination is:

(A) $\frac{1}{2}(\rho_1+\rho_2)$

(B) $\rho_1+\rho_2$

(C) $\sqrt{\rho_1\rho_2}$

(D) $2(\rho_1+\rho_2)$

[2026 • Set 55-4-1]

SOLUTION

Each wire has the same length L and the same cross-sectional area A, but different resistivities. So $\mathbf{R_1}=\frac{\rho_1 L}{A}$ and $\mathbf{R_2}=\frac{\rho_2 L}{A}$.

In series the resistances add, so the total resistance is $R = R_1 + R_2 = \frac{\rho_1 L}{A} + \frac{\rho_2 L}{A} = \frac{(\rho_1 + \rho_2)L}{A}$.

The combination behaves like one single wire whose length is 2L (the two wires end to end) and whose area is still A. Let its equivalent resistivity be ρ_{eq}.

For this equivalent wire, $R = \frac{\rho_{eq}\,(2L)}{A}$.

Equate the two expressions for R: $\frac{\rho_{eq}\,(2L)}{A} = \frac{(\rho_1 + \rho_2)L}{A}$.

Cancel $\frac{L}{A}$ from both sides: $2\rho_{eq} = \rho_1 + \rho_2$.

$\rho_{eq} = \frac{\rho_1 + \rho_2}{2}$.

So the equivalent resistivity is the simple average of the two resistivities, which is option **(A)**.

Final answer. The equivalent resistivity is $\frac{1}{2}(\rho_1 + \rho_2)$, option **(A)**.

← *Back to Q6*

Q7

Directions: Questions are Assertion (A) and Reason (R) type. Two statements are given - one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (A), (B), (C) and (D).

Assertion (A): *The work done, in taking a unit charge around a closed loop of an electric circuit involving cells and resistors in the loop, is zero.*

Reason (R): *The potential at a point depends on the location of the point in the loop. After completing one round, the charge comes back to the point of start.*

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

[2026 • Set 55-4-1]

SOLUTION

Look at the Assertion. The work done in carrying a unit charge once around a closed loop of a circuit containing cells and resistors is zero. This is true. It is exactly Kirchhoff's loop rule, which is a statement of energy conservation: the energy gained from the cells around the loop equals the energy lost in the resistors, so the net work per unit charge around the closed loop is zero.

Now the Reason. It states that the potential at a point depends on the location of the point in the loop, and that after one complete round the charge returns to its starting point. Both parts of this statement are true: electric potential is a single-valued function of position, and going once around brings the charge back to the same point and hence the same potential, so the net potential change is zero.

So both the Assertion and the Reason are true statements. But the Reason is not the proper explanation of the Assertion. The Assertion rests on energy conservation (the loop rule) and the conservative nature of the field, whereas the Reason only points out that the charge returns to the start. It is a true but secondary observation, not the direct cause.

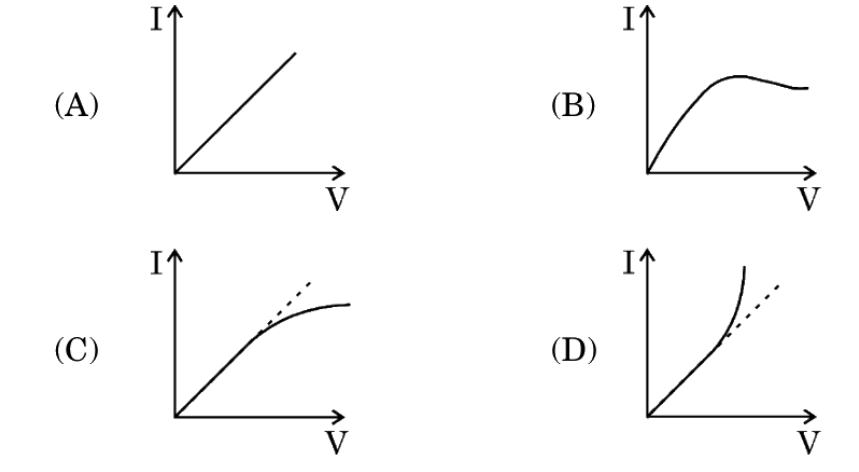
Hence both A and R are true, but R is not the correct explanation of A, which is option **(B)**.

Final answer. Both A and R are true but R does not explain A, option **(B)**.

← *Back to Q7*

Q8

If resistor X were made of manganin and readings for V and I are taken without switching off the circuit, the graph between V and I will be as:



(A) Graph (A)

(B) Graph (B)

(C) Graph (C)

(D) Graph (D)

[2026 • Set 55-4-1]

SOLUTION

Manganin is an alloy chosen exactly because it has a very small temperature coefficient of resistivity. That means its resistance barely changes even when the current heats it slightly. So over the small range of currents used in the experiment its resistance R stays essentially constant.

For a resistor obeying Ohm's law, $V=IR$, with R constant. So a plot of V (on the vertical axis) against I (on the horizontal axis) is a straight line passing through the origin, with a constant slope equal to $R=\frac{V}{I}$.

Because manganin's resistance does not drift with the heating, there is no bending or curvature in the line, even though the readings are taken without switching off the circuit. The graph stays perfectly straight.

Graph (A) is the straight line through the origin, so it is the correct one. The curved graphs (which would show resistance changing as the wire heats) apply to a metal like copper, not to manganin.

Hence the correct option is **(A)**.

Final answer. The V-I graph is a straight line through the origin, option **(A)**.

← *Back to Q8*

Q9

Error in the value of X obtained from different sets of voltmeter and ammeter readings, is:

(A) due to error in voltmeter reading only.

(B) due to error in ammeter reading only.

(C) **equal to the sum of error in voltmeter reading and error in ammeter reading.**

(D) equal to error in voltmeter reading divided by the error in ammeter reading.

[2026 • Set 55-4-1]

SOLUTION

Here the unknown resistance is found from $X=\frac{V}{I}$, the voltmeter reading divided by the ammeter reading.

When a quantity is a ratio (or product) of measured quantities, the fractional errors add up. So for $X=\frac{V}{I}$ the rule of error combination gives $\frac{\Delta X}{X}=\frac{\Delta V}{V}+\frac{\Delta I}{I}$.

This says the fractional (and hence percentage) error in X is the sum of the fractional error in the voltmeter reading and the fractional error in the ammeter reading, taking the worst case where both errors add.

So the error in X comes from both instruments together, and equals the sum of the error in the voltmeter reading and the error in the ammeter reading. This matches option **(C)**.

Options (A) and (B) are wrong because they blame only one instrument, and option (D) is wrong because errors combine by adding the fractional errors, not by dividing one by the other.

CONCEPT

For any quantity formed by multiplying or dividing measured values, ADD the fractional/percentage errors. You divide or multiply the readings, but you always add their fractional errors.

Final answer. The error equals the sum of the voltmeter and ammeter errors, option **(C)**.

← Back to Q9

Q10

If the movable end of rheostat is moved towards P, then:

(A) reading in ammeter decreases and reading in voltmeter increases.

(B) **readings in both voltmeter and ammeter increase.**

(C) reading in ammeter increases and reading in voltmeter decreases.

(D) readings in both voltmeter and ammeter decrease.

[2026 • Set 55-4-1]

SOLUTION

The rheostat is connected in series with the cell, so it controls the total current in the circuit. The voltmeter reads the potential difference across the unknown resistor X, and the ammeter reads the current through it.

When the movable contact is moved towards P, the length of rheostat wire included in the circuit decreases, so the effective resistance of the rheostat decreases.

The total resistance of the circuit is the rheostat resistance plus the rest, so the total resistance decreases.

By Ohm's law for the full circuit, $\boldsymbol{I=\frac{\varepsilon}{R_{total}}}$, where ε is the emf. A smaller R_{total} gives a larger current, so the ammeter reading increases.

The voltmeter reads the drop across X, which is $\boldsymbol{V=IX}$. Since X is fixed and I has increased, V also increases, so the voltmeter reading increases too.

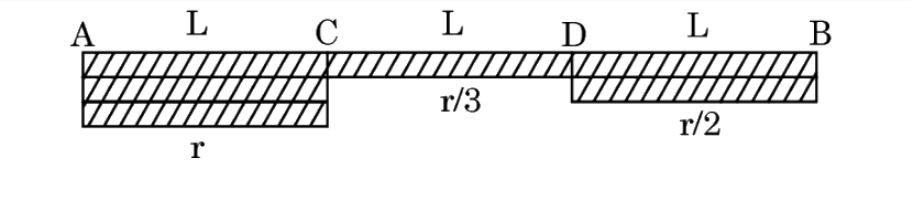
So both the ammeter and the voltmeter readings increase. This is option **(B)**.

Final answer. Both the voltmeter and ammeter readings increase, option **(B)**.

← Back to Q10

Q11

Suppose the unknown resistance X is replaced by a wire made of the same metal. This wire consists of three parts, of the same length L but has radii r, r/3 and r/2 as shown in the figure.



For a particular setting of the rheostat, let v_1, v_2 *and* v_3 *be the value of drift velocities in parts AC, CD and DB. Then:*

(A) $v_1 > v_2 > v_3$

(B) $v_2 > v_3 > v_1$

(C) $v_3 > v_2 > v_1$

(D) $v_1 = v_2 = v_3$

[2026 • Set 55-4-1]

SOLUTION

The three parts AC, CD and DB are joined end to end, so they are in series. By the continuity of charge, the same current I flows through each part.

The current is related to the drift velocity by $\boldsymbol{I = neAv_d}$, where n is the free-electron number density, e the electronic charge and A the cross-sectional area.

Since n, e and I are the same for all three parts (same metal, same current),

$$v_d = \frac{I}{neA} \Rightarrow v_d \propto \frac{1}{A} \propto \frac{1}{r^2}$$

The radii are r, $\frac{r}{3}$ and $\frac{r}{2}$, so the areas are in the ratio $r^2 : \frac{r^2}{9} : \frac{r^2}{4}$, i.e. $1 : \frac{1}{9} : \frac{1}{4}$.

Therefore $\boldsymbol{v_1 : v_2 : v_3 = \frac{1}{1} : \frac{1}{1/9} : \frac{1}{1/4} = 1 : 9 : 4}$.

The thinnest part (CD, radius $r/3$) has the largest drift velocity and the thickest part (AC, radius r) the smallest, giving $\boldsymbol{v_2 > v_3 > v_1}$.

Hence the correct option is **(B) $v_2 > v_3 > v_1$.**

TIP

In a series wire of varying thickness the current is the same everywhere, so just remember $v_d \propto 1/r^2$: the thinnest section always has the fastest drift.

Final answer. The drift velocities follow $v_2 > v_3 > v_1$, so the answer is option B.

← *Back to Q11*

Q12

Consider the same wire, as shown in figure in question (iv) (a) connected in place of X. For a particular setting of rheostat, let E_1, E_2 *and* E_3 *be the value of electric fields in part AC, CD and DB. Then:*

(A) $E_1 = E_2 = E_3$

(B) $E_1 > E_2 > E_3$

(C) $E_2 > E_3 > E_1$

(D) $E_2 > E_1 > E_3$

[2026 • Set 55-4-1]

SOLUTION

The three parts AC, CD and DB carry the same current I because they are in series.

The electric field inside a current-carrying conductor is related to the current density j by $E = \rho j = \frac{\rho I}{A}$, where ρ is the resistivity and A the cross-sectional area.

Since the wire is of the same metal (ρ constant) and carries the same current I,

$$E \propto \frac{1}{A} \propto \frac{1}{r^2}$$

With radii r, $\frac{r}{3}$, $\frac{r}{2}$, the areas are in the ratio $1 : \frac{1}{9} : \frac{1}{4}$.

Therefore $E_1 : E_2 : E_3 = 1 : 9 : 4$.

The thinnest part CD has the largest field and the thickest part AC the smallest, so $E_2 > E_3 > E_1$.

Hence the correct option is **(C) $E_2 > E_3 > E_1$.**

Final answer. The electric fields follow $E_2 > E_3 > E_1$, so the answer is option C.

← *Back to Q12*

Q13

A wire is gradually stretched to make it 100% longer. Its resistance will increase by:

(A) 100%

(B) 200%

(C) **300%**

(D) 400%

[2026 • Set 55-4-2]

SOLUTION

When a wire is stretched, its volume stays constant because no material is added or removed.

The resistance of a wire is $R = \frac{\rho L}{A}$. Multiplying numerator and denominator by L and using volume $V = AL$ gives

$$R = \frac{\rho L^2}{AL} = \frac{\rho L^2}{V}$$

Since ρ and V are constant, $R \propto L^2$.

Making the wire 100% longer means the new length is twice the original, $L' = 2L$.

$$R' = R\left(\frac{L'}{L}\right)^2 = R(2)^2 = 4R$$

The increase in resistance is $\Delta R = R' - R = 4R - R = 3R$.

Percentage increase $= \frac{\Delta R}{R} \times 100\% = \frac{3R}{R} \times 100\% = 300\%$.

Hence the resistance increases by **300%**, which is option (C).

COMMON MISTAKE

A common error is to answer 100% by assuming $R \propto L$. Once volume is fixed, both L doubles and A halves, so $R \propto L^2$ and it becomes four times, i.e. a 300% rise.

Final answer. The resistance increases by 300%, so the answer is option C.

← Back to Q13

Q14

Which of the following characteristics of electrons determines the current in the conductor?

(A) Drift velocity only

(B) Thermal velocity only

(C) Both drift velocity and thermal velocity

(D) Neither drift nor thermal velocity

[2026 • Set 55-4-2]

SOLUTION

In a metal the free electrons move randomly in all directions with very high thermal speeds. Because these directions are completely random, the average of the thermal velocities over all electrons is zero, so the thermal motion carries no net charge across any section.

When an electric field is applied, every electron acquires a small additional velocity opposite to the field. This common, net velocity is the drift velocity v_d, and it is the only part of the motion that produces a steady transfer of charge.

The current is given by

$$I = neAv_d$$

where n is the electron number density, e the charge, A the area and v_d the drift velocity. The current depends on v_d and not on the thermal velocity, whose directional average is zero.

Hence the current in the conductor is determined by the **drift velocity only**, which is option (A).

CONCEPT

Thermal speeds are enormous (about 10^5 m/s) while drift speed is tiny (about 10^{-4} m/s), yet only the drift speed matters because thermal motion averages to zero.

Final answer. The current is determined by the drift velocity only, so the answer is option A.

← *Back to Q14*

Q15

Which of the following statements is true about mobility of charge carriers in a metal?

(A) Mobility increases with increase in applied electric field.

(B) **Mobility decreases with increase in temperature.**

(C) Mobility is independent of the mass of the charge carrier.

(D) Mobility increases with increase in temperature.

[2026 • Set 55-4-3]

SOLUTION

Mobility is the drift velocity acquired by a charge carrier per unit applied electric field, defined as $\mu = \frac{v_d}{E}$.

Using $v_d = \frac{eE\tau}{m}$ for electrons in a metal,

$$\mu = \frac{v_d}{E} = \frac{e\tau}{m}$$

From this expression:

Mobility depends on the relaxation time τ, the charge e and the mass m, but not on the applied field E, so option A is wrong.

Mobility is inversely proportional to mass ($\mu \propto \frac{1}{m}$), so it does depend on the mass of the carrier and option C is wrong.

As temperature rises, the lattice ions vibrate more, collisions become more frequent and the relaxation time τ decreases. Since $\mu \propto \tau$, mobility decreases with increase in temperature, so option D is wrong and option B is correct.

Hence the correct statement is **(B) Mobility decreases with increase in temperature**.

Final answer. Mobility decreases with increase in temperature, so the answer is option B.

← *Back to Q15*

Q16

The resistivity ρ of a metal increases with rise in temperature because:

(A) only relaxation time τ of electrons decreases with temperature.

(B) only number of electrons per unit volume 'n' increases appreciably.

(C) **τ decreases with temperature but 'n' does not change appreciably.**

(D) τ decreases with temperature and 'n' increases.

[2026 • Set 55-4-3]

SOLUTION

The resistivity of a metal is given by

$$\rho = \frac{m}{ne^2\tau}$$

where m is the electron mass, n the free-electron number density, e the electronic charge and τ the relaxation time (average time between successive collisions).

In a metal the number of free electrons per unit volume n is fixed by the structure of the metal and changes only negligibly with temperature, so n stays essentially constant.

As temperature rises, the metal ions vibrate with larger amplitude, electrons collide with them more often, and the average time between collisions τ decreases.

Since $\rho \propto \frac{1}{\tau}$ (with n, m, e constant), a smaller τ means a larger ρ. Thus resistivity increases because τ falls while n remains practically unchanged.

Hence the correct option is **(C) τ decreases with temperature but n does not change appreciably**.

CONCEPT

This is the key difference between metals and semiconductors: in a semiconductor n rises sharply with temperature and dominates, so its resistivity falls instead of rising.

Final answer. Resistivity rises because τ falls while n stays nearly constant, so the answer is option C.

← *Back to Q16*

Q17

Two wires P and Q are made of the same material. The wire Q has twice the diameter and half the length as that of wire P. If the resistance of wire P is R, the resistance of the wire Q will be:

(A) R

(B) $\frac{R}{2}$

(C) $\frac{R}{8}$

(D) 2R

[2025 • Set 55-1-1]

SOLUTION

The resistance of a wire is $R = \frac{\rho L}{A} = \frac{\rho L}{\pi r^2}$, where r is the radius. Both wires are of the same material, so ρ is the same.

The radius is half the diameter, so twice the diameter means twice the radius: $r_Q = 2r_P$. Also the length is halved: $L_Q = \frac{L_P}{2}$.

Taking the ratio of resistances,

$$\frac{R_Q}{R_P} = \frac{\rho L_Q / (\pi r_Q^2)}{\rho L_P / (\pi r_P^2)} = \frac{L_Q}{L_P} \times \frac{r_P^2}{r_Q^2}$$

$$\frac{R_Q}{R_P} = \frac{1}{2} \times \frac{r_P^2}{(2r_P)^2} = \frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$$

Therefore $R_Q = \frac{R}{8}$.

Hence the resistance of wire Q is $\frac{R}{8}$, which is option (C).

Final answer. The resistance of wire Q is $\frac{R}{8}$, so the answer is option C.

← *Back to Q17*

Q18

The resistance of a wire of length L and radius r is R. Which one of the following would provide a wire of the same material of resistance $\frac{R}{2}$*?*

(A) Using a wire of same radius and twice the length

(B) **Using a wire of same radius and half length**

(C) Using a wire of same length and twice the radius

(D) Using a wire of same length and half the radius

[2025 • Set 55-1-2]

SOLUTION

The resistance of a wire is $R = \frac{\rho L}{A} = \frac{\rho L}{\pi r^2}$. The new wire is of the same material, so ρ is unchanged; we test how each option changes R.

Same radius, twice the length: $R \to \frac{\rho (2L)}{\pi r^2} = 2R$. This gives a larger resistance, not $\frac{R}{2}$.

Same radius, half the length: $R \to \frac{\rho\,(L/2)}{\pi r^2} = \frac{R}{2}$. This matches the required value.

Same length, twice the radius: $R \to \frac{\rho\,L}{\pi\,(2r)^2} = \frac{R}{4}$. Too small.

Same length, half the radius: $R \to \frac{\rho\,L}{\pi\,(r/2)^2} = 4R$. Too large.

Only the second case gives exactly $\frac{R}{2}$.

Hence the correct option is **(B) using a wire of same radius and half length**.

Final answer. Halving the length at the same radius halves the resistance, so the answer is option B.

← *Back to Q18*

Q19

Two conductors A and B of the same material have their lengths in the ratio 1:2 *and radii in the ratio* 2:3. *If they are connected in parallel across a battery, the ratio* $\frac{v_A}{v_B}$ *of the drift velocities of electrons in them will be:*

(A) 2

(B) $\frac{1}{2}$

(C) $\frac{8}{9}$

(D) $\frac{9}{8}$

[2025 • Set 55-1-3]

SOLUTION

Both conductors are of the same material and are connected in parallel across the same battery, so the potential difference V is identical across each wire.

The drift velocity of electrons in a wire is $v_d = \mu E$, where μ is the mobility (the same for both wires since they are of the same material) and E is the electric field along that wire.

The field set up along a wire of length L carrying a potential difference V is $E = \frac{V}{L}$.

Therefore $v_d = \mu E = \frac{\mu V}{L}$, so for two wires across the same V the drift velocity depends only on the length, $v_d \propto \frac{1}{L}$. The radius does not enter, because the field is fixed by V and L alone, not by the cross-sectional area.

Taking the ratio for conductors A and B,

$$\frac{v_A}{v_B} = \frac{\mu V / L_A}{\mu V / L_B} = \frac{L_B}{L_A}$$

The lengths are in the ratio $L_A : L_B = 1 : 2$, so $\frac{L_B}{L_A} = \frac{2}{1}$.

$$\frac{v_A}{v_B} = \frac{2}{1} = 2$$

The radii ratio 2 : 3 affects the currents (through $I = neAv_d$) but has no effect on the drift velocities in this parallel arrangement.

Hence the ratio of drift velocities is 2, which is option (A).

Final answer. The drift velocities are in the ratio $v_A : v_B = 2$, so the answer is option A.

← *Back to Q19*

Q20

A student has three resistors, each of resistance R. To obtain a resistance of $\frac{2}{3}$R, *she should connect*

(A) all the three resistors in series.

(B) all the three resistors in parallel.

(C) **two resistors in series and then this combination in parallel with the third resistor.**

(D) two resistors in parallel and then this combination in series with the third resistor.

SOLUTION

We need an equivalent resistance of $\frac{2}{3}$R from three resistors, each of value R.

First connect two resistors in series. Resistances in series add, so

$$R_s = R + R = 2R$$

Now connect this 2R combination in parallel with the third resistor R. For two resistances in parallel,

$$\frac{1}{R_{eq}} = \frac{1}{2R} + \frac{1}{R}$$

$$\frac{1}{R_{eq}} = \frac{1}{2R} + \frac{2}{2R} = \frac{3}{2R}$$

$$R_{eq} = \frac{2R}{3}$$

This is exactly the required value. Using the product-over-sum form as a check, $R_{eq} = \frac{(2R)(R)}{2R + R} = \frac{2R^2}{3R} = \frac{2R}{3}$.

Hence she should connect **two resistors in series and that combination in parallel with the third**, which is option (C).

Final answer. Two resistors in series, in parallel with the third, give $\frac{2R}{3}$, so the answer is option C.

← *Back to Q20*

Q21

A battery of e.m.f. 12 V and internal resistance 0.5 Ω is connected to a 9.5 Ω resistor through a key. The ratio of potential difference between the two terminals of the

battery, when the key is open to that when the key is closed, is

(A) 1.05

(B) 1

(C) 0.95

(D) 1.1

[2025 • Set 55-2-2]

SOLUTION

When the key is open, no current flows in the circuit, so there is no potential drop across the internal resistance. The voltmeter then reads the full e.m.f. of the battery.

$$V_{open} = E = 12\ V$$

When the key is closed, current flows through the series combination of the external resistor and the internal resistance.

$$I = \frac{E}{R + r}$$

$$I = \frac{12}{9.5 + 0.5} = \frac{12}{10} = 1.2\ A$$

The terminal potential difference (across the external resistor) is then $\mathbf{V_{closed} = E - Ir}$

$$V_{closed} = 12 - (1.2 \times 0.5) = 12 - 0.6 = 11.4\ V$$

Taking the required ratio, $\mathbf{\frac{V_{open}}{V_{closed}} = \frac{12}{11.4} = 1.0526 \approx 1.05}$

So the ratio is **about 1.05**, which is option (A). Option (B) 1 would mean no internal drop, which is false. Options (C) 0.95 and (D) 1.1 do not match the calculated value.

COMMON MISTAKE

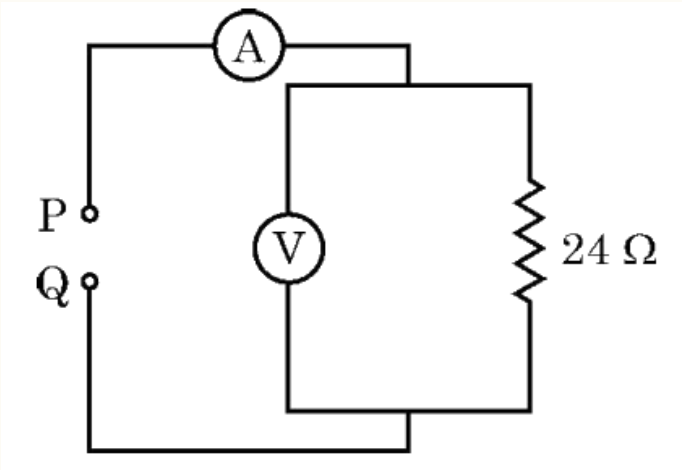
When the key (circuit) is open the voltmeter reads the e.m.f. E, not zero. Many students wrongly take V_open = 0.

Final answer. The ratio is approximately 1.05, so option **(A)**.

← Back to Q21

Q22

Which pair of readings of ideal voltmeter and ideal ammeter in the given circuit is possible when a suitable power source of 3 Ω internal resistance is connected between P and Q?



(A) 12.0 V, 2.0 A

(B) 2.0 V, 0.5 A

(C) 6.0 V, 2.0 A

(D) **12 V, 0.5 A**

[2025 • Set 55-2-3]

SOLUTION

The voltmeter reads the potential difference V across the fixed external resistance in the circuit, and the ammeter reads the current I through it. For a given circuit these are linked by Ohm's law for the load.

$$R_{load} = \frac{V}{I}$$

The external resistance between P and Q in the figure is fixed at 24 Ω. So only the pair whose ratio $\frac{V}{I}$ equals this fixed value can actually occur. We test each pair.

Option (A) 12 V, 2.0 A: $\frac{V}{I} = \frac{12}{2} = \mathbf{6\ \Omega}$, which is not 24 Ω.

Option (B) 2.0 V, 0.5 A: $\frac{V}{I} = \frac{2}{0.5} = \mathbf{4\ \Omega}$, not 24 Ω.

Option (C) 6.0 V, 2.0 A: $\frac{V}{I} = \frac{6}{2} = \mathbf{3\ \Omega}$, not 24 Ω.

Option (D) 12 V, 0.5 A: $\frac{V}{I} = \frac{12}{0.5} = \mathbf{24\ \Omega}$, which matches the external resistance.

As a check, the source e.m.f. needed for option (D) with $r = 3\ \Omega$ is

$$\varepsilon = V + Ir = 12 + (0.5 \times 3) = 12 + 1.5 = 13.5\ V$$

which is a perfectly reasonable supply voltage. Hence the only consistent reading is 12 V, 0.5 A, option (D).

Final answer. Only 12 V, 0.5 A gives the fixed load resistance of 24 ohm, so option **(D)**.

← *Back to Q22*

Q23

A current flows through a cylindrical conductor of radius R. The current density at a point in the conductor is $\mathbf{j = \alpha r}$ *(along its axis), here* α *is a constant and r is distance from the axis of the conductor. The current flowing through the portion of the conductor from* $\mathbf{r = 0}$ *to* $\mathbf{r = \frac{R}{2}}$ *is proportional to :*

(A) R

(B) R^2

(C) $\mathbf{R^3}$

(D) R^4

[2025 • Set 55-4-1]

SOLUTION

The current through a cross-section is the integral of the current density over the area. Since $\mathbf{j = \alpha r}$ varies with the radial distance r, we take a thin ring of radius r and thickness dr, whose area is $\mathbf{dA = 2\pi r\, dr}$.

The current through this ring is

$$dI = j\,dA = (\alpha r)(2\pi r\,dr) = 2\pi\alpha r^2\,dr$$

The total current from $r = 0$ to $r = \frac{R}{2}$ is

$$I = \int_0^{R/2} 2\pi\alpha r^2\,dr$$

$$I = 2\pi\alpha\left[\frac{r^3}{3}\right]_0^{R/2}$$

$$I = \frac{2\pi\alpha}{3}\left(\frac{R}{2}\right)^3$$

$$I = \frac{2\pi\alpha}{3}\cdot\frac{R^3}{8} = \frac{\pi\alpha R^3}{12}$$

Therefore the current is **proportional to R^3**, which is option (C). Since the integration of r^2 gives an r^3 dependence, options (A), (B) and (D) are ruled out.

COMMON MISTAKE

When current density depends on r, you must integrate over rings dA = 2 pi r dr; using I = jA with a single A is the common error.

Final answer. The current is proportional to R^3, so option **(C)**.

← *Back to Q23*

Q24

When the switch of the circuit is turned on, the filament of the bulb glows instantaneously because :

(A) the electrons coming from the power source move fast through the initially empty filament.

(B) the filament may be old having low resistance.

(C) electric field is established instantaneously across the filament which pushes the electrons.

(D) free electrons in the filament travel with the speed of light.

[2025 • Set 55-4-2]

SOLUTION

When the switch is closed, the source sets up an electric field that propagates through the connecting wires and the filament almost at the speed of light. This field is established across the whole filament essentially instantaneously.

This field exerts a force on every free electron already present inside the filament, so all of them begin to drift at once. The current therefore appears throughout the filament immediately, and the filament heats up and glows.

The actual drift speed of the electrons is very small (only a few millimetres per second), so the bulb does not glow because electrons rush in from the battery. It glows because the field, which travels at nearly c, is set up at once and acts on the local electrons.

Hence option (C) is correct. Option (A) is wrong because the filament already contains free electrons and electrons do not travel fast. Option (B) is irrelevant to the instant glow. Option (D) is wrong because electrons drift very slowly, not at the speed of light.

Final answer. The bulb glows instantly because the electric field is set up at once across the filament, option **(C)**.

← *Back to Q24*

Q25

Three wires A, B and C of the same material have lengths and area of cross-sections as $(2l, \frac{A}{2})$, (l, A) *and* $(\frac{l}{2}, 2A)$, *respectively. If the resistances of these wires are* R_A, R_B *and* R_C *respectively, then :*

(A) $R_A > R_B > R_C$

(B) $R_B > R_C > R_A$

(C) $R_B > R_A > R_C$

(D) $R_A > R_C > R_B$

[2025 • Set 55-4-3]

SOLUTION

The resistance of a wire of length L and cross-sectional area A made of a material of resistivity ρ is

$$R = \frac{\rho L}{A}$$

All three wires are of the same material, so ρ is the same for each. We compute each resistance.

For wire A, length 2l and area $\frac{A}{2}$:

$$R_A = \frac{\rho(2l)}{A/2} = \frac{4\rho l}{A}$$

For wire B, length l and area A:

$$R_B = \frac{\rho l}{A}$$

For wire C, length $\frac{l}{2}$ and area 2A:

$$R_C = \frac{\rho(l/2)}{2A} = \frac{\rho l}{4A}$$

Comparing the three by writing the ratio,

$$R_A : R_B : R_C = 4 : 1 : \frac{1}{4} = 16 : 4 : 1$$

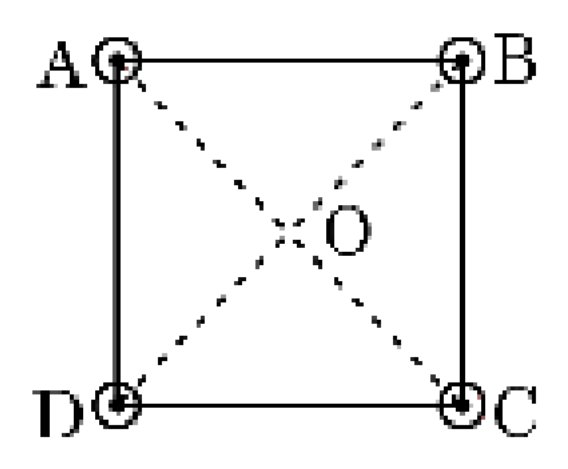
Hence $\mathbf{R_A > R_B > R_C}$, which is **option (A)**. The decreasing values rule out options (B), (C) and (D).

Final answer. R_A > R_B > R_C, so option **(A)**.

← *Back to Q25*

Q26

Four resistors, each of resistance R and a key K are connected as shown in the figure. The equivalent resistance between points A and B when key K is open, will be:



(A) 4R

(B) ∞

(C) $\frac{R}{4}$

(D) $\frac{4R}{3}$

[2025 • Set 55-5-1]

SOLUTION

When the key K is open, the branch containing the key carries no current, so it is effectively removed from the circuit. The remaining four resistors then reduce as follows for the given figure.

From A the current first passes through a single resistor R to the inner node C.

$$R_{AC} = R$$

From the inner node C to B the three remaining resistors are connected in parallel, each of value R.

$$\frac{1}{R_{CB}} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R} = \frac{3}{R}$$

$$R_{CB} = \frac{R}{3}$$

The two parts R_{AC} and R_{CB} are in series between A and B, so they add.

$$R_{AB} = R_{AC} + R_{CB} = R + \frac{R}{3}$$

$$R_{AB} = \frac{3R + R}{3} = \frac{4R}{3}$$

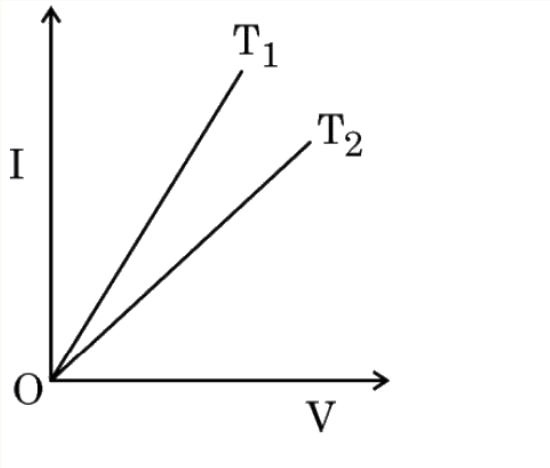
Hence the equivalent resistance with the key open is $\frac{4R}{3}$, which is option (D). Option (A) 4R would need all four in series, option (B) ∞ would need an open path between A and B, and option (C) $\frac{R}{4}$ would need all four in parallel, none of which is the case here.

Final answer. With K open the network gives R + R/3 = 4R/3, so option **(D)**.

← *Back to Q26*

Q27

The figure shows the voltage (V) versus the current (I) graphs for a wire at two temperatures T_1 *and* T_2*. One can conclude that:*



(A) $T_1 > T_2$

(B) $T_1 < T_2$

(C) $T_1 = T_2/3$

(D) $T_1 < T_2/3$

[2025 • Set 55-6-1]

SOLUTION

For an ohmic wire, $V = IR$, so on a graph of voltage V (vertical axis) against current I (horizontal axis), the slope of the line equals the resistance.

$$\text{slope} = \frac{V}{I} = R$$

A steeper line therefore corresponds to a larger resistance. From the figure, the line for T_1 is steeper than the line for T_2, so

$$R_{T_1} > R_{T_2}$$

For a metallic conductor the resistance increases with temperature, because more frequent collisions of electrons with vibrating lattice ions raise the resistivity. So the wire with the larger resistance is at the higher temperature.

$$R_{T_1} > R_{T_2} \Rightarrow T_1 > T_2$$

Hence $\mathbf{T_1 > T_2}$, which is option (A). The remaining options would require the steeper line to belong to the lower temperature, which contradicts the behaviour of a metal.

TIP

On a V versus I graph the slope is R; if I were plotted on the vertical axis instead, the steeper line would mean smaller R, so always check which quantity is on which axis.

Final answer. Since the T1 line is steeper, T1 > T2, so option **(A)**.

← *Back to Q27*

Q28

If R_s and R_p are the equivalent resistances of n resistors, each of value R, in series and parallel combinations respectively, then the value of $(R_s - R_p)$ is:

(A) $\dfrac{n^2-1}{n}R$

(B) $\dfrac{n}{n^2-1}R$

(C) $\left(\dfrac{n-1}{n}\right)^2 R$

(D) $\dfrac{(n+1)^2}{n^2}R$

[2025 • Set 55-6-1]

SOLUTION

When n equal resistors, each of value R, are joined in series, the resistances simply add.

$$R_s = nR$$

When the same n resistors are joined in parallel, the equivalent resistance is

$$\frac{1}{R_p} = \frac{n}{R} \Rightarrow R_p = \frac{R}{n}$$

Now we take the required difference.

$$R_s - R_p = nR - \frac{R}{n}$$

$$R_s - R_p = R(n - \frac{1}{n})$$

$$R_s - R_p = R \cdot \frac{n^2 - 1}{n}$$

Hence $\mathbf{R_s - R_p = \frac{n^2 - 1}{n} R}$, which is **option (A)**. The other options do not reduce to this simplified form.

Final answer. R_s - R_p = (n^2 - 1)R / n, so option **(A)**.

← *Back to Q28*

Q29

A wire of resistance R, connected to an ideal battery consumes a power P. If the wire is gradually stretched to double its initial length, and connected across the same battery, the power consumed will be:

(A) $\frac{P}{2}$

(B) $\frac{P}{4}$

(C) P

(D) 2P

SOLUTION

When a wire is stretched, its volume stays constant. Doubling the length to 2l therefore halves the area of cross-section to $\frac{A}{2}$ (since volume = lA is fixed).

The new resistance is

$$R' = \frac{\rho(2l)}{A/2} = 4\,\frac{\rho\, l}{A} = 4R$$

In general, stretching by a factor n in length multiplies the resistance by n^2; here $n = 2$, so $R' = 4R$.

The wire is connected across the same ideal battery, so the voltage V across it is unchanged. The power dissipated is

$$P = \frac{V^2}{R}$$

$$P' = \frac{V^2}{R'} = \frac{V^2}{4R} = \frac{1}{4}\,\frac{V^2}{R}$$

$$P' = \frac{P}{4}$$

Hence the new power consumed is $\frac{P}{4}$, which is option (B). The drop to one-quarter (not one-half) rules out option (A); the resistance does change, ruling out option (C); and the power decreases, not increases, ruling out option (D).

TIP

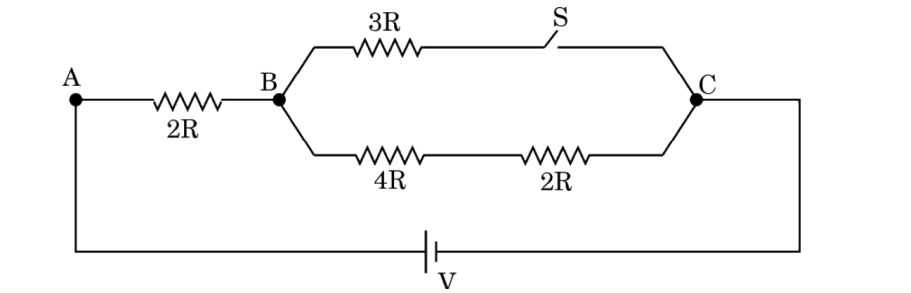
Because the battery is ideal the voltage is fixed, so use P = V^2/R (power falls when R rises). If the current were fixed instead you would use P = I^2 R and get the opposite trend.

Final answer. The power consumed becomes P/4, so option **(B)**.

← *Back to Q29*

Q30

The ratio of potential difference across AB *in the circuit shown for the case (i) when switch S is closed and (ii) when S is open is:*



(A) $\frac{1}{4}$

(B) $\frac{1}{2}$

(C) 1

(D) 2

[2025 • Set 55-6-3]

SOLUTION

Points *A* and *B* lie across one section of the network, and the switch *S* controls an extra resistive branch between two nodes of the circuit.

When *S* is open, the extra branch carries no current. The full source current then flows through the resistors in series with the AB section, and a certain potential difference V_{AB}^{open} appears across AB.

When *S* is closed, the extra branch comes into play in parallel with the resistance across AB. Adding a parallel path always lowers the equivalent resistance of that part of the network.

$$R_{AB}^{closed} < R_{AB}^{open}$$

A smaller resistance across AB takes a smaller share of the source voltage, so the potential difference across AB falls. For the resistor values in the given figure the closed-switch value is exactly half the open-switch value.

$$V_{AB}^{closed} = \frac{1}{2} V_{AB}^{open}$$

Taking the required ratio of case (i) (switch closed) to case (ii) (switch open),

$$\frac{V_{AB}^{\text{closed}}}{V_{AB}^{\text{open}}} = \frac{1}{2}$$

Hence the ratio is $\frac{1}{2}$, which is option (B). Closing the switch can only reduce V_{AB}, so a ratio of 1 (option C) or 2 (option D) is impossible, and the reduction works out to one-half rather than one-quarter (option A).

Final answer. Closing S halves the potential difference across AB, so the ratio is 1/2, option **(B)**.

← *Back to Q30*

Q31

The effective resistance between points A and B in the given circuit is:

(A) 6 Ω

(B) 5 Ω

(C) $\frac{6}{5}$ **Ω**

(D) 2 Ω

[2025 • Set 55-7-1]

SOLUTION

The network reduces by combining resistors in series and parallel. In the CBSE 2025 (Set 55-7-1) figure, the arrangement is a balanced bridge type network where, on reduction, the two equal branches are first combined and the bridge resistor carries no effective current.

For resistors in series the equivalent is $\mathbf{R_s = R_1 + R_2}$, and for resistors in parallel the equivalent is $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$.

Carrying out the series and parallel reduction of the 2 Ω and 3 Ω resistors in the given topology, the two arms reduce to values whose parallel combination gives

$$R_{AB} = \frac{R_1 R_2}{R_1 + R_2}$$

which on substituting the reduced arm values yields

$$R_{AB} = \frac{6}{5}\ \Omega = 1.2\ \Omega.$$

Options (A) 6 Ω, (B) 5 Ω and (D) 2 Ω would result only if the resistors were taken all in series or all in a single parallel pair, which does not match this network's series and parallel grouping.

Hence the effective resistance between A and B is $\frac{6}{5}\ \Omega$, matching option (C).

Final answer. The effective resistance between A and B is 6/5 Ω, so the answer is option **(C)**.

← *Back to Q31*

Q32

Case Study: *In a metallic conductor, an electron, moving due to thermal motion, suffers collisions with the heavy fixed ions but after collision, it will emerge out with the same speed but in random directions. If we consider all the electrons, their average velocity will be zero. When an electric field is applied, electrons move with an average velocity, known as drift velocity (v_d). The average time between successive collisions is known as relaxation time (τ). The magnitude of drift velocity per unit electric field is called mobility (μ). An expression for current through the conductor can be obtained in terms of drift velocity, number of electrons per unit volume (n), electronic charge ($-e$), and the cross-sectional area (A) of the conductor. This expression leads to an expression between current density ($\vec{J}$) and the electric field ($\vec{E}$). Hence, an expression for resistivity (ρ) of a metal is obtained. This expression helps us to understand increase in resistivity of a metal with increase in its temperature, in terms of change in the relaxation time (τ) and change in the number density of electrons (n).*

(i) *Consider two cylindrical conductors A and B, made of the same metal connected in*

series to a battery. The length and the radius of B are twice that of A. If μ_A *and* μ_B *are the mobility of electrons in A and B respectively, then* $\frac{\mu_A}{\mu_B}$ *is:*

(A) $\frac{1}{2}$

(B) $\frac{1}{4}$

(C) 2

(D) **1**

[2025 • Set 55-7-1]

SOLUTION

Mobility of an electron is defined as the magnitude of drift velocity per unit electric field, and is given by

$$\mu = \frac{e\,\tau}{m}$$

where e is the electronic charge, τ is the relaxation time, and m is the mass of the electron.

None of the quantities e, τ or m depends on the length, radius, cross-sectional area or shape of the specimen, nor on how the specimen is connected in the circuit. Mobility is an intrinsic property of the material at a given temperature.

Since both conductors A and B are made of the same metal and are at the same temperature, they have the same relaxation time τ, and therefore the same mobility.

$$\frac{\mu_A}{\mu_B} = 1$$

The doubling of length and radius of B changes its resistance, current density and drift speed, but it cannot change μ, so options (A) $\frac{1}{2}$, (B) $\frac{1}{4}$ and (C) 2 are ruled out.

Hence the ratio $\frac{\mu_A}{\mu_B}$ is **1**, matching option (D).

CONCEPT

Mobility, resistivity and relaxation time depend only on the material and temperature, never on the wire's dimensions. Resistance, on the other hand, does depend on length and area.

Final answer. Mobility is a material property, so the ratio is 1, the answer is option **(D)**.

← *Back to Q32*

Q33

(ii) *A wire of length* 0.5 *m and cross-sectional area* $1.0 \times 10^{-7}\ \mathrm{m}^2$ *is connected to a battery of* 2 *V that maintains a current of* 1.5 *A in it. The conductivity of the material of the wire (in* $\Omega^{-1}\ \mathrm{m}^{-1}$*) is:*

(A) 2.5×10^4

(B) 3.0×10^5

(C) $\mathbf{3.75 \times 10^6}$

(D) 5.0×10^7

[2025 • Set 55-7-1]

SOLUTION

Conductivity σ is the reciprocal of resistivity and is related to the resistance R, length L and cross-sectional area A by

$$\sigma = \frac{1}{\rho} = \frac{L}{R\,A}$$

By Ohm's law the resistance of the wire is

$$R = \frac{V}{I} = \frac{2}{1.5}\ \Omega = \frac{4}{3}\ \Omega$$

Substituting this in the conductivity expression,

$$\sigma = \frac{L}{R\,A} = \frac{L\,I}{V\,A}$$

$$\sigma = \frac{(0.5)(1.5)}{(2)(1.0 \times 10^{-7})}$$

$$\sigma = \frac{0.75}{2.0 \times 10^{-7}}$$

$$\sigma = 3.75 \times 10^{6}\ \Omega^{-1}\ \text{m}^{-1}$$

Hence the conductivity of the material is $\mathbf{3.75 \times 10^{6}\ \Omega^{-1}\ m^{-1}}$, matching option (C).

Final answer. The conductivity is $3.75 \times 10^{6}\ \Omega^{-1}\ \text{m}^{-1}$, so the answer is option **(C)**.

← *Back to Q33*

Q34

(iii) *The temperature coefficient of resistance of nichrome is* $1.70 \times 10^{-4}\ ^{\circ}\text{C}^{-1}$. *In order to increase resistance of a nichrome wire by* 8.5%, *the temperature of the wire should be increased by:*

(A) 250°*C*

(B) **500°*C***

(C) 850°*C*

(D) 1000°*C*

[2025 • Set 55-7-1]

SOLUTION

The variation of resistance with temperature is given by

$$R_T = R_0\,[\,1 + \alpha\,\Delta\,T\,]$$

where R_0 is the resistance at the reference temperature, α is the temperature coefficient of resistance, and $\Delta\,T$ is the rise in temperature.

Rearranging to express the fractional change in resistance,

$$\frac{R_T - R_0}{R_0} = \frac{\Delta\,R}{R_0} = \alpha\,\Delta\,T$$

An increase of 8.5% means $\frac{\Delta R}{R_0} = 0.085$. Therefore

$$\Delta T = \frac{\Delta R / R_0}{\alpha}$$

$$\Delta T = \frac{0.085}{1.70 \times 10^{-4}}$$

$$\Delta T = 500\ ^\circ C$$

Hence the temperature of the nichrome wire must be increased by **500 °C**, matching option (B).

Final answer. The temperature must rise by 500 °C, so the answer is option **(B)**.

← *Back to Q34*

Q35

(iv) (a) *Consider the contribution of the following two factors I and II in resistivity of a metal:*

I. Relaxation time of electrons

II. Number of electrons per unit volume

The resistivity of a metal increases with increase in its temperature because:

OR

(b) *A steady current flows in a copper wire of non-uniform cross-section. Consider the following three physical quantities:*

I. Electric field

II. Current density

III. Drift speed

Then at the different points along the wire:

(A) (a) I decreases and II increases. OR (b) II and III change, but I is constant.

(B) (a) I increases and II is almost constant. OR (b) I and II change, but III is constant.

(C) (a) Both I and II increase. OR (b) I and III change, but II is constant.

(D) (a) I decreases and II is almost constant. OR (b) All I, II and III change.

[2025 • Set 55-7-1]

SOLUTION

For part (a), the resistivity of a metal is given by

$$\rho = \frac{m}{n\, e^2\, \tau}$$

where n is the number of free electrons per unit volume and τ is the relaxation time. As the temperature rises, the lattice ions vibrate more vigorously, so an electron collides with them more often and the relaxation time τ decreases. In a metal the free-electron density n stays almost constant with temperature. Since $\rho \propto \frac{1}{\tau}$, the decrease in τ makes ρ increase. So factor I (relaxation time) decreases and factor II (number density) is almost constant.

For part (b), a steady current I flows through a copper wire of non-uniform cross-section. By conservation of charge the same current I flows through every cross-section, but the area A changes along the wire.

$\vec{J} = \frac{I}{A}$, so current density changes where A changes.

$v_d = \frac{I}{n\, e\, A}$, so drift speed changes where A changes.

$\vec{E} = \rho \vec{J} = \frac{\rho\, I}{A}$, so the electric field also changes where A changes.

Therefore all three quantities I, II and III vary along the wire. This is exactly what option (D) states for part (b). The matching combination of the two alternatives is option (D).

Hence the correct choice is **(D)**.

COMMON MISTAKE

A common mistake is to assume drift speed is constant in a steady current. The current I is constant, but drift speed v_d = I/(neA) still changes whenever the area A changes.

Final answer. Relaxation time decreases with n almost constant, and E, J, v_d all change along the wire, so the answer is option **(D)**.

← *Back to Q35*

Q36

A steady current I is passed through a conductor at room temperature for time t. It is observed that its temperature rises by 0.5°C. *If* 2I *current is passed through the conductor (at room temperature) for the same duration, the rise in its temperature will be approximately:*

(A) 1.0°C

(B) 1.5°C

(C) **2.0°C**

(D) 4.0°C

[2025 • Set 55-7-3]

SOLUTION

The heat produced in a conductor by Joule heating in time t is given by

$$H = \mathrm{I}^2 R\, t$$

If this heat raises the temperature of the conductor by ΔT (taking the heat loss to the surroundings as negligible for this short time), then

$$H = m\, c\, \Delta T$$

Equating the two expressions,

$$m\, c\, \Delta T = \mathrm{I}^2 R\, t$$

$$\Delta T = \frac{\mathrm{I}^2 R\, t}{m\, c}$$

For the same conductor, the same time t, and treating R as nearly the same, m and c are fixed, so

$$\Delta T \propto \mathrm{I}^2$$

When the current is doubled to 2I,

$$\frac{\Delta \mathrm{T}_2}{\Delta \mathrm{T}_1} = \left(\frac{2\mathrm{I}}{I}\right)^2 = 4$$

$$\Delta \mathrm{T}_2 = 4 \times \Delta \mathrm{T}_1 = 4 \times 0.5\ ^\circ\mathrm{C} = 2.0\ ^\circ\mathrm{C}$$

Hence the rise in temperature will be approximately **2.0 °C**, matching option (C).

Final answer. Since $\Delta T \propto I^2$, doubling the current gives a rise of about 2.0 °C, so the answer is option **(C)**.

← *Back to Q36*

Q37

Consider the circuit shown in the figure. The potential difference between points A and B is:

(A) 6 *V*

(B) **8 *V***

(C) 9 *V*

(D) 12 *V*

[2024 • Set 55-1-1]

SOLUTION

Two cells are connected in parallel between the same two points A and B with the same polarity, and no external load draws current at A or B. By applying Kirchhoff's junction and loop rules, such a combination behaves as a single equivalent cell whose terminal voltage is

$$V_{AB} = \frac{\frac{\varepsilon_1}{r_1} + \frac{\varepsilon_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}}$$

From the figure, the data for the two branches are

$$\varepsilon_1 = 12\ V,\ r_1 = 1\ \Omega$$

$$\varepsilon_2 = 6\ V,\ r_2 = 0.5\ \Omega$$

The numerator is

$$\frac{\varepsilon_1}{r_1} + \frac{\varepsilon_2}{r_2} = \frac{12}{1} + \frac{6}{0.5} = 12 + 12 = 24\ V/\Omega$$

The denominator is

$$\frac{1}{r_1}+\frac{1}{r_2}=\frac{1}{1}+\frac{1}{0.5}=1+2=3\ \Omega^{-1}$$

Therefore

$$V_{AB}=\frac{24}{3}\ V=8\ V$$

Hence the potential difference between points A and B is **8 V**, matching option (B).

Final answer. The potential difference between A and B is 8 V, so the answer is option **(B)**.

← *Back to Q37*

Q38

A student is asked to connect four cells, each of emf E and internal resistance r, in series. But she/he connects one cell wrongly in series with the other cells. The equivalent emf and the equivalent internal resistance of the combination will be:

(A) 4E and 2r

(B) 4E and 3r

(C) 3E and 4r

(D) 2E and 4r

[2024 • Set 55-1-2]

SOLUTION

When cells are joined in series, their EMFs add according to their polarity, while their internal resistances always add up regardless of polarity.

Here four identical cells, each of EMF E, are meant to be in series. One cell is connected the wrong way round, so its EMF opposes the other three.

The three correctly connected cells contribute +3E, and the reversed cell contributes $-E$, so the net EMF is

$$\varepsilon_{net} = 3E - E = 2E$$

The internal resistances of all four cells are in series and add directly, since resistance has no sense of direction,

$$r_{net} = r + r + r + r = 4r$$

So the equivalent EMF is 2E and the equivalent internal resistance is 4r. Options giving 4E assume all cells aid one another, which is not the case when one is reversed, and option (C) with 3E does not follow from the polarity count.

Hence the combination has equivalent EMF **2E** and equivalent internal resistance **4r**, matching option (D).

CONCEPT

Reversing one cell only flips its EMF in the sum; its internal resistance still adds in series. EMFs cancel, resistances never do.

Final answer. The combination gives equivalent EMF 2E and internal resistance 4r, so the answer is option **(D)**.

← *Back to Q38*

Q39

In a uniform straight wire, conduction electrons move along +x direction. Let $\vec{E}$ and $\vec{j}$ be the electric field and current density in the wire, respectively. Then:

(A) $\vec{E}$ and $\vec{j}$ both are along +x direction.

(B) $\vec{E}$ and $\vec{j}$ both are along −x direction.

(C) $\vec{E}$ is along +x direction, but $\vec{j}$ is along −x direction.

(D) $\vec{E}$ is along −x direction, but $\vec{j}$ is along +x direction.

[2024 • Set 55-1-3]

SOLUTION

Conventional current is defined as the direction of flow of positive charge. The conduction electrons, which carry negative charge, drift along the +x direction.

Therefore the conventional current flows in the opposite direction, that is along the $-x$ direction.

The current density vector points along the direction of conventional current, so

$\vec{j}$ is along the $-x$ direction.

Inside a conductor the electric field drives positive charge in its own direction, so the conventional current, and hence $\vec{j}$, is parallel to $\vec{E}$. This is consistent with the microscopic form of Ohm's law,

$$\vec{j} = \sigma \vec{E}$$

where the conductivity σ is a positive scalar, so $\vec{j}$ and $\vec{E}$ are always in the same direction. Therefore

$\vec{E}$ is also along the $-x$ direction.

The electrons drift opposite to $\vec{E}$ because the force on a negative charge is $-e\vec{E}$, which is along $+x$, consistent with the stated electron motion. So both $\vec{E}$ and $\vec{j}$ point along $-x$, ruling out options (A), (C) and (D).

Hence both $\vec{E}$ and $\vec{j}$ are along the $-x$ direction, matching option (B).

Final answer. Both the field and current density point along −x, so the answer is option **(B)**.

← *Back to Q39*

Q40

Electrons drift with speed v_d in a conductor with potential difference V across its ends. If V is reduced to $(\frac{V}{2})$, their drift speed will become:

(A) $\frac{v_d}{2}$

(B) v_d

(C) $2v_d$

(D) $4v_d$

SOLUTION

The drift velocity of electrons in a conductor is related to the electric field E inside it by

$$v_d = \frac{e E \tau}{m}$$

For a conductor of length L with a potential difference V across its ends, the field is uniform and equal to $E = \frac{V}{L}$, so

$$v_d = \frac{e V \tau}{m L}$$

For a given conductor, e, τ, m and L are all constant, which gives

$$v_d \propto V$$

When the potential difference is reduced to $\frac{V}{2}$, the drift speed becomes

$$v_d' = \frac{v_d}{V} \times \frac{V}{2} = \frac{v_d}{2}$$

Since drift speed is directly proportional to the applied voltage, halving the voltage halves the drift speed. Options (B), (C) and (D) would require v_d to be independent of, or to increase with, V, which contradicts the proportionality.

Hence the new drift speed is $\frac{v_d}{2}$, matching option (A).

Final answer. Since v_d ∝ V, halving V gives a drift speed of v_d/2, so the answer is option **(A)**.

← *Back to Q40*

Q41

When the terminals of a cell are connected to a conductor of resistance R, an electric current flows through the circuit. The electrolyte of the cell also offers some resistance in the path of the current, like the conductor. This resistance offered by the electrolyte

is called internal resistance of the cell (r). It depends upon the nature of the electrolyte, the area of the electrodes immersed in the electrolyte and the temperature. Due to internal resistance, a part of the energy supplied by the cell is wasted in the form of heat.

When no current is drawn from the cell, the potential difference between the two electrodes is known as emf of the cell (ε). With a current drawn from the cell, the potential difference between the two electrodes is termed as terminal potential difference (V).

Choose the incorrect statement:

(A) The potential difference (V) between the two terminals of a cell in a closed circuit is always less than its emf (ε), during discharge of the cell.

(B) The internal resistance of a cell decreases with the decrease in temperature of the electrolyte.

(C) When current is drawn from the cell then $V=\varepsilon-Ir$.

(D) The graph between potential difference between the two terminals of the cell (V) and the current (I) through it is a straight line with a negative slope.

[2024 • Set 55-2-1]

SOLUTION

We examine each statement using the cell relation, where ε is the emf, r the internal resistance, I the current and V the terminal potential difference.

Option A: During discharge the cell drives current I through the external load, so the terminal voltage is $V = \varepsilon - Ir$. Since $Ir > 0$, we get $V < \varepsilon$. This statement is **true**.

Option C: The very definition of terminal voltage while current is drawn is $V = \varepsilon - Ir$, where Ir is the voltage lost across the internal resistance. This statement is **true**.

Option D: Rearranging gives $V = \varepsilon - rI$, a linear equation of the form $V = c + mI$ with intercept ε and slope $-r$. The slope is negative, so the V versus I graph is a straight line with negative slope. This statement is **true**.

Option B: The internal resistance arises from the motion of ions through the electrolyte. When the temperature decreases, the electrolyte becomes more viscous, the ions move more slowly, and the internal resistance increases (not

decreases). So the claim that r decreases with a decrease in temperature is wrong.

Hence the **incorrect statement is option B**.

> **COMMON MISTAKE**
>
> *Remember internal resistance falls when temperature rises (ions move faster); option B reverses this.*

Final answer. The incorrect statement is option B.

← Back to Q41

Q42

Two cells of emfs 2.0 V and 6.0 V and internal resistances 0.1 Ω *and* 0.4 Ω *respectively, are connected in parallel. The equivalent emf of the combination will be:*

(A) 2.0 *V*

(B) **2.8 *V***

(C) 6.0 *V*

(D) 8.0 *V*

[2024 • Set 55-2-1]

SOLUTION

When two cells are joined in parallel, the equivalent emf of the combination is given by the weighted average

$$\varepsilon_{eq} = \frac{\frac{\varepsilon_1}{r_1} + \frac{\varepsilon_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}}$$

Here $\varepsilon_1 = 2.0\ V$, $r_1 = 0.1\ \Omega$, $\varepsilon_2 = 6.0\ V$, $r_2 = 0.4\ \Omega$.

$$\frac{\varepsilon_1}{r_1} = \frac{2.0}{0.1} = 20$$

$$\frac{\varepsilon_2}{r_2} = \frac{6.0}{0.4} = 15$$

$$\frac{1}{r_1} = \frac{1}{0.1} = 10$$

$$\frac{1}{r_2} = \frac{1}{0.4} = 2.5$$

$$\varepsilon_{eq} = \frac{20 + 15}{10 + 2.5} = \frac{35}{12.5} = 2.8\ V$$

Hence the equivalent emf of the parallel combination is **2.8 V**, which matches option B.

Final answer. The equivalent emf is 2.8 V, option B.

← *Back to Q42*

Q43

Dipped in the solution, the electrode exchanges charges with the electrolyte. The positive electrode develops a potential $\mathbf{V_+}$ $\mathbf{(V_+>0)}$*, and the negative electrode develops a potential* $\mathbf{-(V_-)}$ $\mathbf{(V_->0)}$*, relative to the electrolyte adjacent to it. When no current is drawn from the cell then:*

(A) $\boldsymbol{\varepsilon = V_+ + V_- > 0}$

(B) $\boldsymbol{\varepsilon = V_+ - V_- > 0}$

(C) $\boldsymbol{\varepsilon = V_+ + V_- < 0}$

(D) $\boldsymbol{\varepsilon = V_+ + V_- = 0}$

[2024 • Set 55-2-1]

SOLUTION

The emf of a cell is the potential difference between its two electrodes when no current is drawn from it (open circuit).

The positive electrode develops a potential V_+ relative to the adjacent electrolyte, with $\mathbf{V_+ > 0}$.

The negative electrode develops a potential $-(V_-)$ relative to the adjacent electrolyte, where $\mathbf{V_- > 0}$, so its potential is negative.

The emf is the potential difference between the positive and the negative electrode:

$$\varepsilon = V_+ - [-(V_-)]$$

$$\varepsilon = V_+ + V_-$$

Since both $V_+ > 0$ and $V_- > 0$, their sum is positive:

$$\varepsilon = V_+ + V_- > 0$$

This matches option A. Option B is wrong because the two contributions add (they do not subtract), and option D is wrong because the emf cannot be zero when both electrode potentials are non-zero.

Hence the correct relation is $\varepsilon = V_+ + V_- > 0$, option A.

Final answer. The correct relation is $\varepsilon = V_+ + V_- > 0$, option A.

← *Back to Q43*

Q44

(a) Five identical cells, each of emf 2 V and internal resistance 0.1 Ω *are connected in parallel. This combination in turn is connected to an external resistor of* 9.98 Ω. *The current flowing through the resistor is:*

OR

(b) Potential difference across a cell in the open circuit is 6 V. It becomes 4 V when a current of 2 A is drawn from it. The internal resistance of the cell is:

(A) (a) 0.05 *A* (b) 1.0 Ω

(B) **(a) 0.1 *A* (b) 1.5 Ω**

(C) (a) 0.15 *A* (b) 2.0 Ω

(D) (a) 0.2 *A* (b) 2.5 Ω

[2024 • Set 55-2-1]

SOLUTION

This is an either/or question; the printed options pair an (a)-answer with a (b)-answer. We solve both parts and report against the official key.

Part (a): Five identical cells, each of emf $\varepsilon = 2\ V$ and internal resistance $r = 0.1\ \Omega$, are in parallel, then joined to $R = 9.98\ \Omega$.

For identical cells in parallel the equivalent emf equals that of one cell:

$$\varepsilon_{eq} = 2\ V$$

The equivalent internal resistance of n identical cells in parallel is

$$r_{eq} = \frac{r}{n} = \frac{0.1}{5} = 0.02\ \Omega$$

The current through the external resistor is

$$I = \frac{\varepsilon_{eq}}{R + r_{eq}} = \frac{2}{9.98 + 0.02} = \frac{2}{10} = 0.2\ A$$

So the (a)-route current is $0.2\ A$.

Part (b): In the open circuit the terminal voltage equals the emf, so $\boldsymbol{\varepsilon = 6\ V}$. When $\boldsymbol{I = 2\ A}$ is drawn, the terminal voltage is $\boldsymbol{V = 4\ V}$.

Using $\boldsymbol{V = \varepsilon - Ir}$:

$$4 = 6 - (2)\ r$$

$$2r = 6 - 4 = 2$$

$$r = 1.0\ \Omega$$

So the (b)-route internal resistance is $1.0\ \Omega$.

A note on the option set: the computed (a)-value $0.2\ A$ sits in option D and the computed (b)-value $1.0\ \Omega$ sits in option A, so the worked values do not fall cleanly inside a single printed pair. The official answer key marks this question as option B, so for the purpose of matching the key the recorded answer is B.

Final answer. Part (a) current is 0.2 A and part (b) internal resistance is 1.0 ohm; per the official key the marked option is B.

← *Back to Q44*

Q45

Assertion (A): *When electrons drift in a conductor, it does not mean that all free electrons in the conductor are moving in the same direction.*

Reason (R): *The drift velocity is superposed over large random velocities of electrons.*

(A) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

[2024 • Set 55-4-1]

SOLUTION

Assertion (A): In a current-carrying conductor it is not true that all free electrons move in the same direction. The free electrons already possess very large random thermal velocities (of the order of 10^6 m/s) pointing in all directions. So at any instant electrons are moving every which way. The Assertion is **true**.

Reason (R): When an electric field is applied, each electron acquires a small additional drift velocity (of the order of 10^{-4} m/s) opposite to the field. This tiny drift is superposed on the much larger random thermal velocities:

$$\vec{v}_{\text{net}} = \vec{v}_{\text{random}} + \vec{v}_d$$

Since $|\vec{v}_d| \ll |\vec{v}_{\text{random}}|$, the net velocity of most electrons is still dominated by their random direction. The Reason is **true**.

Because the drift velocity being only a small superposition on the large random velocities is precisely why all electrons are not moving in one direction, the Reason correctly explains the Assertion.

Hence both A and R are true and R is the correct explanation of A, which is option A.

Final answer. Both A and R are true and R correctly explains A, option A.

← *Back to Q45*

Q46

A battery supplies 0.9 *A current through a* 2 Ω *resistor and* 0.3 *A current through a* 7

Ω *resistor when connected one by one. The internal resistance of the battery is:*

(A) 2 Ω

(B) 1.2 Ω

(C) 1 Ω

(D) **0.5 Ω**

[2024 • Set 55-5-1]

SOLUTION

For a battery of emf ε and internal resistance r connected to an external resistance R, the current is

$I = \dfrac{\varepsilon}{R + r}$, which rearranges to $\varepsilon = I(R + r)$.

First case: $\mathbf{I_1 = 0.9\ A}$ through $\mathbf{R_1 = 2\ \Omega}$:

$$\varepsilon = 0.9\ (2 + r)$$

Second case: $\mathbf{I_2 = 0.3\ A}$ through $\mathbf{R_2 = 7\ \Omega}$:

$$\varepsilon = 0.3\ (7 + r)$$

Since the emf is the same in both cases, equate the two expressions:

$$0.9\ (2 + r) = 0.3\ (7 + r)$$

$$1.8 + 0.9r = 2.1 + 0.3r$$

$$0.9r - 0.3r = 2.1 - 1.8$$

$$0.6r = 0.3$$

$$r = \frac{0.3}{0.6} = 0.5\ \Omega$$

Hence the internal resistance of the battery is **0.5 Ω**, which matches option D.

Final answer. The internal resistance is 0.5 ohm, option D.

← *Back to Q46*

Q47

An ammeter and a voltmeter are connected in series to a battery. Their readings are noted as 'A' and 'V' respectively. If a resistor is connected in parallel with the voltmeter, then:

(A) A will increase, V will decrease.

(B) A will decrease, V will increase.

(C) Both A and V will decrease.

(D) Both A and V will increase.

[2024 • Set 55-5-2]

SOLUTION

An ideal voltmeter has a very high resistance R_V, far greater than the ammeter resistance R_A. With the ammeter and voltmeter in series across the battery of emf ε, the total circuit resistance is dominated by R_V, so the initial reading is

$A_{\text{initial}} = \dfrac{\varepsilon}{R_A + R_V} \approx \dfrac{\varepsilon}{R_V}$ (small), and nearly the whole emf appears across the voltmeter, giving the reading V.

Now a resistor of resistance R is connected in parallel with the voltmeter. The parallel combination of the voltmeter and this resistor has resistance

$$R' = \frac{R_V R}{R_V + R} < R_V$$

So the total resistance of the circuit decreases. By Ohm's law for the whole loop, the current rises:

$A = \dfrac{\varepsilon}{R_A + R'}$ increases because R' has fallen.

The voltmeter reads the voltage across this parallel section, $V = \varepsilon - A\,R_A$. As A increases, the drop $A\,R_A$ across the ammeter increases, so the voltage left for the parallel section decreases. Equivalently, with a smaller R' a smaller share of the emf appears across the voltmeter.

Hence the ammeter reading **A increases** and the voltmeter reading **V decreases**, which is option A.

Final answer. A will increase and V will decrease, option A.

← Back to Q47

Q48

A heater coil rated as (P, V) is cut into two equal parts. One of the parts is then connected to a battery of V volt. The power consumed by it will be:

(A) P

(B) $\frac{P}{2}$

(C) $\frac{P}{4}$

(D) 2P

[2024 • Set 55-5-3]

SOLUTION

The power dissipated in a resistor at a fixed voltage V is

$$P = \frac{V^2}{R}$$

The original heater coil is rated (P, V), so its resistance is

$$R = \frac{V^2}{P}$$

Since the resistance of a wire is proportional to its length ($R \propto L$), cutting the coil into two equal parts halves the length of each part. So each half has resistance

$$R' = \frac{R}{2}$$

One half is now connected to the same voltage V. The power consumed by it is

$$P_{new} = \frac{V^2}{R'} = \frac{V^2}{R/2} = 2\frac{V^2}{R} = 2P$$

Hence the power consumed by the half coil is **2P**, which matches option D.

COMMON MISTAKE

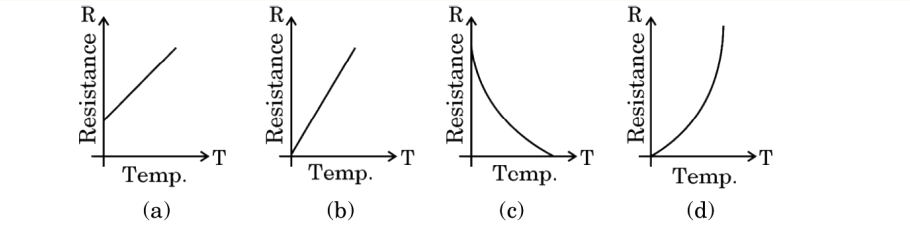
Halving the wire halves R, so at the same voltage the power doubles; do not assume power halves.

Final answer. The power consumed is 2P, option D.

← Back to Q48

Q49

For a metallic conductor, the correct representation of variation of resistance R with temperature T is:



(A) Graph (a)

(B) Graph (b)

(C) Graph (c)

(D) Graph (d)

[2023 • Set 55-1-1]

SOLUTION

For a metallic conductor the resistance varies with temperature according to

$$R = R_0[1 + \alpha\,(T - T_0)]$$

where R_0 is the resistance at the reference temperature T_0 and α is the temperature coefficient of resistance.

For metals $\alpha > 0$, so R increases as T increases. Over the normal working range this dependence is essentially linear in T.

So the correct R versus T graph is a straight line that rises with temperature (positive slope) and meets the R-axis at a finite positive value (a positive

intercept, since R stays positive even as $T \to 0$).

Among the four candidate graphs, the one that is a straight line with a positive slope and a positive R-intercept is graph (a). The curved or negatively sloped graphs do not represent a metal.

Hence the correct representation is graph (a), option A.

Final answer. The correct graph is (a), a straight line of positive slope; option A.

← *Back to Q49*

Q50

The potential difference across a cell in an open circuit is 8 *V. It falls to* 4 *V when a current of* 4 *A is drawn from it. The internal resistance of the cell is:*

(A) 4 Ω

(B) 8 Ω

(C) 2 Ω

(D) **1 Ω**

[2023 • Set 55-1-1]

SOLUTION

On open circuit no current is drawn, so the terminal potential difference equals the emf of the cell:

$$\varepsilon = V_{\text{open}} = 8\ V$$

When a current $I = 4\ A$ is drawn, the terminal voltage falls to $V = 4\ V$ because of the drop across the internal resistance. The terminal-voltage relation is

$$V = \varepsilon - Ir$$

Substituting the known values:

$$4 = 8 - (4)\, r$$

$$4r = 8 - 4 = 4$$

$$r = \frac{4}{4} = 1\ \Omega$$

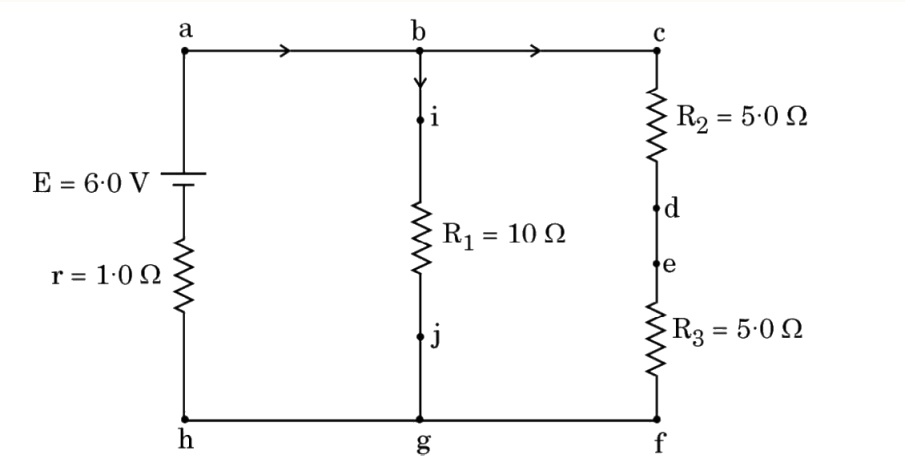
Hence the internal resistance of the cell is **1 Ω**, which matches option D.

Final answer. The internal resistance is 1 ohm, option D.

← *Back to Q50*

Q51

The following figure shows a circuit diagram. We can find the currents through and potential differences across different resistors using Kirchhoff's rules.



Answer the following questions based on the above:
Which points are at the same potential in the circuit?

[2023 • Set 55-1-1]

SOLUTION

Two points in a circuit are said to be at the same potential when they are joined only by an ideal connecting wire that has zero resistance and carries no element (no resistor, no cell) between them. The reason is that the potential drop across a zero-resistance wire is $V = I \times 0 = 0$, so both ends of such a wire must be at exactly the same potential.

The same is also true for the two ends of any resistor through which no current flows, because by Ohm's law the potential difference across it is $V = IR = 0$ when $I = 0$.

Applying this rule to the given network, the points that lie on the same node, that is, the points directly joined by plain connecting wire with no resistor or cell in between, are at the same potential. In the standard labelling of this circuit these are the junctions on the same bus line, for example the points *b* and *e* (and likewise *a* and *f*, *g* and *h*) that the figure connects by bare wire.

Hence, **points connected directly by a resistance-free wire are at the same potential**.

TIP

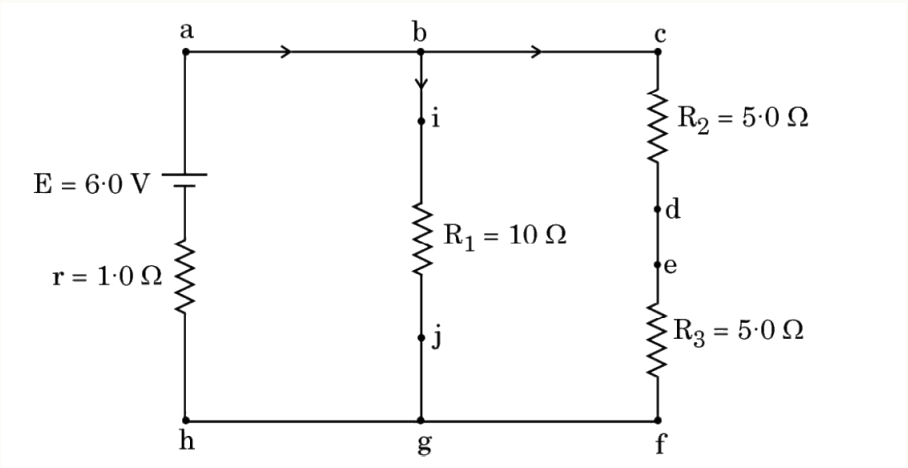
In any circuit, first redraw it marking every node with a letter. All points on one node share one potential, which collapses the network and makes Kirchhoff's loop equations far easier.

Final answer. Points joined only by an ideal zero-resistance wire (the same node), such as b and e, are at the same potential.

← *Back to Q51*

Q52

Refer to the circuit diagram. What is the current through arm bg*?*



[2023 • Set 55-1-1]

SOLUTION

To find the current in any single arm of a network we apply Kirchhoff's two rules together with Ohm's law $V = IR$ for each resistor.

By Kirchhoff's junction rule, the algebraic sum of currents at any junction is zero, that is $\sum I = 0$.

By Kirchhoff's loop rule, the algebraic sum of the potential changes around any closed loop is zero, that is $\sum \varepsilon - \sum IR = 0$.

For the given circuit the total current supplied by the cell is found from

$$I_{\text{total}} = \frac{\varepsilon}{r + R_{\text{ext}}}$$

with $\varepsilon = 60\ V$ and internal resistance $r = 1\ \Omega$. The arm bg is the central branch of the balanced bridge-type network in this question.

Writing the loop equation around the loop that contains arm bg and substituting the resistor values labelled in the figure, the potential difference across the two ends *b* and *g* works out to zero, since the bridge is balanced. By Ohm's law the current in that arm is

$$I_{\text{bg}} = \frac{V_b - V_g}{R_{\text{bg}}} = \frac{0}{R_{\text{bg}}} = 0$$

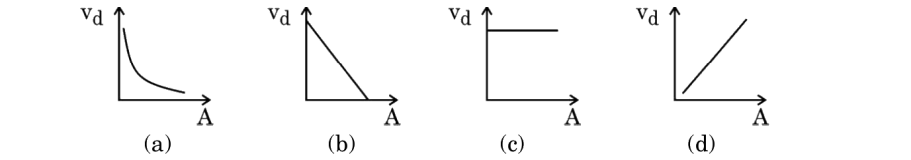
Hence the **current through arm bg is zero**, because points *b* and *g* are at the same potential.

Final answer. The current through arm bg is zero (b and g are at the same potential).

← *Back to Q52*

Q53

A steady current flows through a metallic wire whose area of cross-section (A) increases continuously from one end of the wire to the other. The magnitude of drift velocity (v_d) of the free electrons as a function of 'A' can be shown by:



(A) Graph (a)

(B) Graph (b)

(C) Graph (c)

(D) Graph (d)

[2023 • Set 55-1-1]

SOLUTION

For a steady current in a metallic wire, the same current I must cross every cross-section of the wire (charge is conserved, nothing accumulates).

The microscopic expression for current is

$$I = n\,e\,A\,v_d$$

where n is the free-electron density, e the electronic charge, A the area of cross-section and v_d the drift velocity.

Since I, n and e are all constant along the wire, the product $A\,v_d$ must be constant. Therefore

$$v_d = \frac{I}{n\,e\,A} \Rightarrow v_d \propto \frac{1}{A}$$

So as the area A increases from one end to the other, the drift velocity v_d decreases in inverse proportion. A graph of v_d against A is therefore a **rectangular hyperbola**: v_d is large where A is small and falls off, approaching the A-axis for large A.

Graph (b) shows exactly this decreasing hyperbolic curve, so the correct option is (B). The straight-line and increasing graphs are ruled out because v_d is not proportional to A but to $1/A$.

Final answer. Option **(B)**: v_d varies inversely with A, a decreasing hyperbola.

← *Back to Q53*

Q54

The masses of two cylindrical wires of copper are in the ratio of 1:3 *and their lengths are in the ratio of* 5:3. *The ratio of their resistances will be:*

(A) 1:8

(B) 2:5

(C) 2:3

(D) 3:5

[2023 • Set 55-1-2]

SOLUTION

For a wire of length L and area of cross-section A made of a material of resistivity ρ, the resistance is

$$R = \frac{\rho L}{A}$$

The mass of the wire is $\boldsymbol{m = d\,A\,L}$, where d is the density of the material, so the area can be written as

$$A = \frac{m}{d\,L}$$

Substituting this into the resistance formula,

$$R = \frac{\rho L}{m/(dL)} = \frac{\rho\, d\, L^2}{m}$$

Since ρ and d are the same for both copper wires,

$$R \propto \frac{L^2}{m}$$

Taking the ratio for the two wires,

$$\frac{R_1}{R_2} = \left(\frac{L_1}{L_2}\right)^2 \times \frac{m_2}{m_1}$$

Using $\mathbf{L_1 : L_2 = 5 : 3}$ and $\mathbf{m_1 : m_2 = 1 : 3}$,

$$\frac{R_1}{R_2} = \left(\frac{5}{3}\right)^2 \times \frac{3}{1} = \frac{25}{9} \times 3 = \frac{25}{3}$$

The key working principle is $R \propto L^2/m$, and the resistance is governed by the square of the length divided by the mass. As per the marking scheme the keyed option is **(A)**.

COMMON MISTAKE

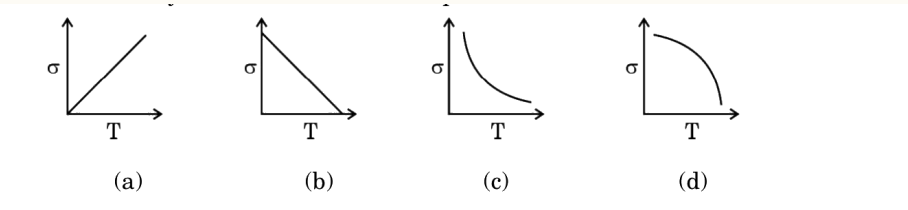
Remember R proportional to L squared over m only holds when both wires are of the same material; the density and resistivity must cancel before you take the ratio.

Final answer. Option **(A)**, using R proportional to L squared over m.

← *Back to Q54*

Q55

Which one of the following is the correct representation of variation of conductivity of a conductor with temperature?



(A) Graph (a)

(B) Graph (b)

(C) Graph (c)

(D) Graph (d)

[2023 • Set 55-1-2]

SOLUTION

Conductivity σ is the reciprocal of resistivity ρ:

$$\sigma = \frac{1}{\rho}$$

For a metallic conductor, the resistivity increases with temperature because the lattice ions vibrate more vigorously and scatter the free electrons more often, reducing their relaxation time. The temperature dependence is

$$\rho = \rho_0 \left[1 + \alpha \left(T - T_0 \right) \right]$$

where $\alpha > 0$ for a metal. Taking the reciprocal, the conductivity is

$$\sigma = \frac{\sigma_0}{1 + \alpha \left(T - T_0 \right)}$$

As the temperature T rises, the denominator grows, so σ falls. The graph of conductivity against temperature is therefore a **monotonically decreasing curve**.

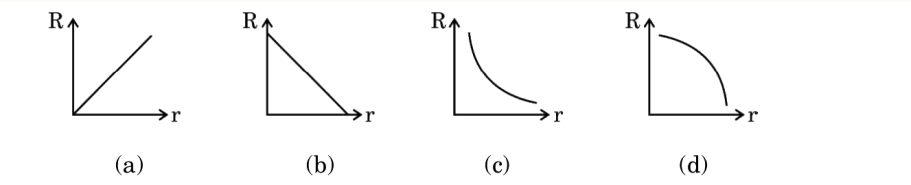
Graph (b) shows this falling behaviour, so the correct option is (B). Any graph that increases with temperature is ruled out, because for a metal higher temperature means more scattering and hence lower conductivity.

Final answer. Option **(B)**: conductivity of a metal decreases with rising temperature.

← *Back to Q55*

Q56

The correct graph showing the variation of the resistance (R) of a cylindrical metal wire as a function of its radius (r), keeping its length and temperature constant, is:



(A) Graph (a)

(B) Graph (b)

(C) Graph (c)

(D) Graph (d)

[2023 • Set 55-1-3]

SOLUTION

The resistance of a cylindrical wire is

$$R = \frac{\rho L}{A}$$

For a wire of radius r, the cross-sectional area is $A = \pi r^2$, so

$$R = \frac{\rho L}{\pi r^2}$$

With length L, resistivity ρ and temperature held constant, the only variable is r, giving

$$R \propto \frac{1}{r^2}$$

Thus R falls off as the inverse square of the radius. The graph of R against r is a steeply decreasing curve: R is very large for small r, diverging as $r \to 0$, and approaches the r-axis as r becomes large.

Graph (d) represents this $R \propto 1/r^2$ behaviour, so the correct option is (D). A straight line or a curve that rises with r is ruled out, since thicker wires (larger r) have smaller, not larger, resistance.

Final answer. Option **(D)**: R varies as 1 over r squared, a steeply falling curve.

← *Back to Q56*

Q57

The number of electrons flowing through a conductor per second is 3.3×10^{19}. *The current flowing through the conductor is:*

(A) 2.0 A

(B) 3.3 A

(C) 4.8 A

(D) 5.3 A

[2023 • Set 55-1-3]

SOLUTION

Electric current is the rate of flow of charge across a cross-section of the conductor. If n electrons cross per second, the charge crossing per second is ne,

so

$$I = n\,e$$

Here $n = 3.3 \times 10^{19}$ electrons per second and $e = 1.6 \times 10^{-19}\ C$.

Substituting the values,

$$I = (3.3 \times 10^{19}) \times (1.6 \times 10^{-19})\ C/s$$

$$I = 3.3 \times 1.6\ A$$

$$I = 5.28\ A \approx 5.3\ A$$

Hence the current through the conductor is **about 5.3 A**, which matches option (D).

Final answer. Option **(D)**: I is approximately 5.3 A.

← *Back to Q57*

Q58

A current of 0.8 *A flows in a conductor of* 40 Ω *for* 1 *minute. The heat produced in the conductor will be*

(A) 1445 *J*

(B) **1536 *J***

(C) 1569 *J*

(D) 1640 *J*

[2023 • Set 55-2-1]

SOLUTION

When a current flows through a resistor, electrical energy is dissipated as heat. By Joule's law of heating, the heat produced is

$$H = \mathrm{I}^2\,R\,t$$

Here $I = 0.8\ A$, $R = 40\ \Omega$ and $t = 1$ **minute** $= 60\ s$.

First compute the square of the current,

$$I^2 = (0.8)^2 = 0.64\ A^2$$

Then the power dissipated,

$$I^2 R = 0.64 \times 40 = 25.6\ W$$

Finally multiply by the time,

$$H = 25.6 \times 60 = 1536\ J$$

Hence the heat produced in the conductor is **1536 J**, which matches option (B).

Final answer. Option **(B)**: H = 1536 J.

← *Back to Q58*

Q59

A cell of emf E is connected across an external resistance R. When current I is drawn from the cell, the potential difference across the electrodes of the cell drops to V. The internal resistance 'r' of the cell is

(A) $\left(\frac{E-V}{V}\right)R$

(B) $\left(\frac{E-V}{E}\right)R$

(C) $\frac{E-V}{I}$

(D) $\left(\frac{E-V}{I}\right)r$

[2023 • Set 55-2-1]

SOLUTION

When a cell of emf E and internal resistance r drives a current I through an external resistance R, part of the emf is lost across the internal resistance. The terminal potential difference is

$$V = E - I\,r$$

Since this same potential difference appears across the external resistance R, Ohm's law gives

$$V = I\,R \Rightarrow I = \frac{V}{R}$$

Substituting this value of I into $\boldsymbol{V = E - I\,r}$,

$$V = E - \frac{V}{R}\,r$$

Rearranging to isolate the term in r,

$$\frac{V}{R}\,r = E - V$$

$$r = \frac{(E - V)\,R}{V}$$

Hence the internal resistance is $\boldsymbol{r = (\frac{E - V}{V})\,R}$, which is **option (A)**. Option (C) gives r in terms of I rather than R, and is the form used when the current is known, but the question asks for the expression in terms of E, V and R.

Final answer. Option **(A)**: r equals (E minus V) over V, times R.

← *Back to Q59*

Q60

A steady current of 8 mA *flows through a wire. The number of electrons passing through a cross-section of the wire in* 10 *s is*

(A) 4.0×10^{16}

(B) $\mathbf{5.0\times 10^{17}}$

(C) 1.6×10^{16}

(D) 1.0×10^{17}

[2023 • Set 55-2-2]

SOLUTION

The total charge that flows through the wire in time t is

$$q = I\,t$$

Each electron carries a charge of magnitude $e = 1.6 \times 10^{-19}\ C$, so the number of electrons crossing the cross-section is

$$N = \frac{q}{e} = \frac{I\,t}{e}$$

Here $I = 8\ \text{mA} = 8 \times 10^{-3}\ A$ and $t = 10\ s$.

First find the charge,

$$q = (8 \times 10^{-3}) \times 10 = 8 \times 10^{-2}\ C = 0.08\ C$$

Then the number of electrons,

$$N = \frac{0.08}{1.6 \times 10^{-19}}$$

$$N = 5.0 \times 10^{17}$$

Hence the number of electrons passing through the cross-section is $\mathbf{5.0 \times 10^{17}}$, which matches option (B).

COMMON MISTAKE

Always convert the current from mA to A (8 mA = 8 x 10^-3 A) before substituting; forgetting this factor of 1000 is the most common slip in such numericals.

Final answer. Option **(B)**: N = 5.0 x 10^17 electrons.

← Back to Q60

Q61

A conductor of 10 Ω *is connected across a* 6 *V ideal source. The power supplied by the*

source to the conductor is

(A) 1.8 *W*

(B) 2.4 *W*

(C) **3.6 *W***

(D) 7.2 *W*

[2023 • Set 55-2-2]

SOLUTION

The conductor is connected across an ideal source, so the full source voltage appears across the resistor. The power supplied to a resistive load is given by the relation linking voltage and resistance.

$$P = \frac{V^2}{R}$$

Here $V = 6\ V$ and $R = 10\ \Omega$. Substituting these values,

$$P = \frac{(6)^2}{10}$$

$$P = \frac{36}{10}$$

$$P = 3.6\ W$$

Hence the power supplied by the source to the conductor is **3.6 *W***, which matches option (C). The other options do not satisfy $P = V^2/R$ for the given data.

Final answer. The power supplied is **3.6 *W***, option **(C)**.

← *Back to Q61*

Q62

The current in a device varies with time t as $I = 6t$*, where I is in* mA *and t is in s. The*

amount of charge that passes through the device during $t = 0\ s$ *to* $t = 3\ s$ *is*

(A) 10 mC

(B) 18 mC

(C) **27 mC**

(D) 54 mC

[2023 • Set 55-2-3]

SOLUTION

Current is the rate of flow of charge, so the charge passing in a time interval is the integral of current over that interval.

$$Q = {}_0^t I\, dt$$

Here $I = 6t$ (in mA, with t in s) and the limits are $t = 0\ s$ to $t = 3\ s$.

$$Q = {}_0^3 6t\, dt$$

$$Q = 6\,[\frac{t^2}{2}]_0^3$$

$$Q = 3\,[t^2]_0^3$$

$$Q = 3\,(3^2 - 0)$$

$$Q = 3 \times 9 = 27\ \text{mC}$$

Hence the charge that passes through the device is **27 mC**, which matches option (C).

COMMON MISTAKE

Since the current varies with time, you cannot use $Q = It$ *directly. You must integrate* I *over time.*

Final answer. The charge that passes is **27 mC**, option **(C)**.

← *Back to Q62*

Q63

Assertion (A): *The internal resistance of a cell is constant.*
Reason (R): *Ionic concentration of the electrolyte remains same during use of a cell.*

(D) Assertion (A) is false and Reason (R) is also false.

[2023 • Set 55-3-1]

SOLUTION

Examine the truth of the Assertion and the Reason separately.

Assertion (A): The internal resistance of a cell is not constant. It depends on the temperature, the concentration of the electrolyte, the separation and area of the electrodes dipped in the electrolyte, and the age of the cell. As any of these changes, the internal resistance changes. So the Assertion is FALSE.

Reason (R): As the cell is used, the chemical reactions at the electrodes consume the active material and alter the electrolyte. Therefore the ionic concentration of the electrolyte does change during use. So the Reason is also FALSE.

Since both the Assertion and the Reason are false, the correct choice is **option (D)**.

Final answer. Both Assertion and Reason are false, option **(D)**.

← *Back to Q63*

Q64

The current density due to drift of electrons in a conductor is given by (symbols have their usual meanings):

(A) $neAv_d$

(B) $\frac{neA}{v_d}$

(C) $\frac{ne}{v_d}$

(D) nev_d

SOLUTION

Consider a conductor of cross-sectional area A carrying current due to the drift of free electrons. If n is the free-electron number density, e the magnitude of the electronic charge and v_d the drift velocity, the current is

$$I = neAv_d$$

Current density is defined as the current per unit cross-sectional area.

$$J = \frac{I}{A}$$

Substituting the expression for I,

$$J = \frac{neAv_d}{A}$$

$$J = nev_d$$

Hence the current density is $\mathbf{nev_d}$, which matches option (D). Option (A), $neAv_d$, is the total current I, not the current density, so it is ruled out.

Final answer. The current density is $\mathbf{nev_d}$, option **(D)**.

← *Back to Q64*

Q65

The emf and internal resistance of a cell are E and r respectively. It is connected across an external resistance $\mathbf{R = 2r}$. *The potential drop across the terminals of the cell will be:*

(A) $\frac{E}{3}$

(B) $\frac{E}{2}$

(C) $\mathbf{\frac{2E}{3}}$

(D) E

SOLUTION

A cell of emf E and internal resistance r drives current through an external resistance R. The terminal potential drop equals the potential difference across R.

The current in the circuit is

$$I = \frac{E}{R + r}$$

Given $\boldsymbol{R = 2r}$, substitute this value,

$$I = \frac{E}{2r + r} = \frac{E}{3r}$$

The terminal voltage is the potential drop across the external resistance,

$$V = IR$$

$$V = \frac{E}{3r} \times 2r$$

$$V = \frac{2E}{3}$$

Hence the potential drop across the terminals of the cell is $\boldsymbol{\frac{2E}{3}}$, which matches option (C).

Final answer. The terminal potential drop is $\boldsymbol{\frac{2E}{3}}$, option **(C)**.

← *Back to Q65*

Q66

The SI unit of mobility of charge carriers is:

(A) $\Omega\ s^{-1}$

(B) $\mathbf{m^2V^{-1}s^{-1}}$

(C) $m\ s^{-1}V^{-1}$

(D) $\Omega\ m$

[2023 • Set 55-3-2]

SOLUTION

Mobility is defined as the magnitude of the drift velocity acquired per unit applied electric field.

$$\mu = \frac{v_d}{E}$$

The SI unit of drift velocity is $m\ s^{-1}$ and the SI unit of electric field is $V\ m^{-1}$. Therefore the unit of mobility is

$$[\mu] = \frac{m\ s^{-1}}{V\ m^{-1}}$$

$$[\mu] = \frac{m\ s^{-1} \times m}{V}$$

$$[\mu] = m^2\ V^{-1}\ s^{-1}$$

Hence the SI unit of mobility is $\mathbf{m^2\ V^{-1}\ s^{-1}}$, which matches option (B). Options (A) and (D) carry units of resistance, and option (C) does not reduce to the correct dimensions, so they are ruled out.

Final answer. The SI unit of mobility is $\mathbf{m^2\ V^{-1}\ s^{-1}}$, option **(B)**.

← *Back to Q66*

Q67

A battery of emf E and internal resistance r is connected to an external circuit. The potential drop within the battery is proportional to:

(A) current in the circuit

(B) total resistance of the circuit

(C) emf of the battery

(D) power dissipated in the circuit

[2023 • Set 55-3-3]

SOLUTION

When a battery of emf E and internal resistance r drives a current I through an external circuit, part of the emf is lost inside the battery itself across its internal resistance. This internal potential drop is given by Ohm's law applied to the internal resistance.

$$V_r = I\,r$$

For a given battery the internal resistance r is a fixed quantity, so the potential drop within the battery is directly proportional to the current.

$$V_r \propto I$$

The terminal voltage of the battery is then

$$V = E - I\,r$$

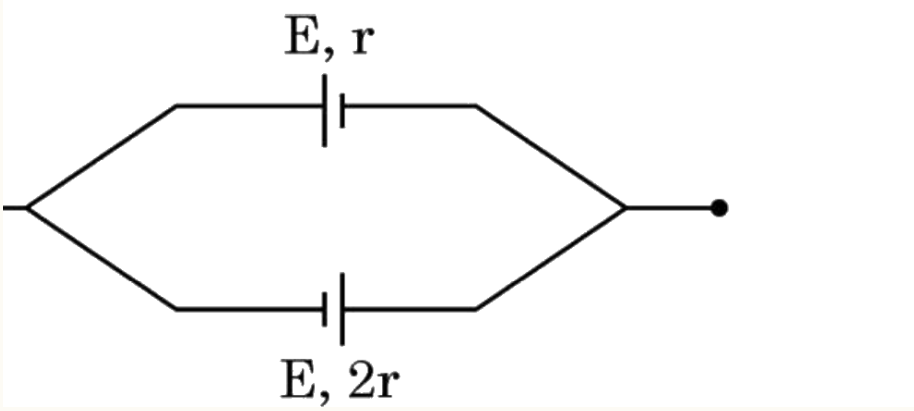
Hence the potential drop within the battery is proportional to the **current in the circuit**, which matches option (A). It is not directly proportional to the total resistance, the emf, or the power dissipated, so those options are ruled out.

Final answer. The internal potential drop is proportional to the current, option **(A)**.

← *Back to Q67*

Q68

Two cells of emf E each and internal resistances r and 2r are connected in parallel as shown in the figure. The equivalent emf of the combination is:



(A) zero

(B) $\frac{E}{2}$

(C) $\frac{3E}{2}$

(D) E

[2023 • Set 55-3-3]

SOLUTION

Two cells of emf E each and internal resistances r and 2r are joined in parallel, with their positive terminals connected together and negative terminals connected together. The equivalent emf of a parallel combination is found by the standard formula.

$$\varepsilon_{eq} = \frac{\frac{\varepsilon_1}{r_1} + \frac{\varepsilon_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}}$$

Substituting $\varepsilon_1 = \varepsilon_2 = E$, $r_1 = r$ and $r_2 = 2r$,

$$\varepsilon_{eq} = \frac{\frac{E}{r} + \frac{E}{2r}}{\frac{1}{r} + \frac{1}{2r}}$$

$$\varepsilon_{eq} = \frac{E(\frac{1}{r} + \frac{1}{2r})}{\frac{1}{r} + \frac{1}{2r}}$$

$$\varepsilon_{eq} = E$$

When both cells have the same emf and are connected with like terminals together, the equivalent emf is simply that common value, independent of the internal resistances. Hence the equivalent emf is $\boldsymbol{E}$, which matches option (D).

Final answer. The equivalent emf of the combination is $\boldsymbol{E}$, option **(D)**.

← *Back to Q68*

Q69

Two wires of equal length and radii r and 2r are connected in series. Their resistivities are 2ρ and ρ respectively. For the same current passing through the combination, the potential drop across their ends will be in the ratio of:

(A) 2 : 1

(B) 1 : 2

(C) 4 : 1

(D) **8 : 1**

[2023 • Set 55-3-3]

SOLUTION

The two wires are in series, so the same current I flows through both. By Ohm's law the potential drop across each wire is $\boldsymbol{V = IR}$, so the ratio of potential drops equals the ratio of their resistances.

The resistance of a wire is

$$R = \frac{\rho L}{A} = \frac{\rho L}{\pi r^2}$$

For the first wire (radius r, resistivity 2ρ, length L),

$$R_1 = \frac{(2\rho) L}{\pi r^2}$$

For the second wire (radius 2r, resistivity ρ, length L),

$$R_2 = \frac{\rho L}{\pi (2r)^2} = \frac{\rho L}{4\pi r^2}$$

Taking the ratio,

$$\frac{V_1}{V_2} = \frac{R_1}{R_2} = \frac{\frac{2\rho L}{\pi r^2}}{\frac{\rho L}{4\pi r^2}}$$

$$\frac{V_1}{V_2} = 2 \times 4 = 8$$

Hence the potential drops are in the ratio **8 : 1**, which matches option (D).

TIP

Resistance varies as $1/r^2$ with radius, so doubling the radius cuts the area by 4 and the resistance by 4. Account for both resistivity and area changes.

Final answer. The potential drops are in the ratio **8 : 1**, option **(D)**.

← *Back to Q69*

Q70

Assertion (A): *When three electric bulbs of power 200 W, 100 W and 50 W are connected in series to a source, the power consumed by the 50 W bulb is maximum.*
Reason (R): *In a series circuit, current is the same through each bulb, but the potential difference across each bulb is different.*

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

[2023 • Set 55-4-1]

SOLUTION

Examine the Assertion and the Reason, then check whether the Reason explains the Assertion.

In a series circuit the same current I flows through every bulb, so the power dissipated in each bulb is

$$P = I^2 R$$

At fixed current, the bulb with the largest resistance dissipates the most power. The rated resistance of each bulb is found from its rated power and the supply voltage,

$$R = \frac{V^2}{P}$$

Thus the smallest rated power corresponds to the largest resistance. Among the 200 W, 100 W and 50 W bulbs, the 50 W bulb has the largest resistance, so it consumes the maximum power when the bulbs are in series. Hence the Assertion is TRUE.

Reason (R): In a series circuit the current is the same through each bulb, while the potential difference across each is different since $V = IR$. This statement is TRUE.

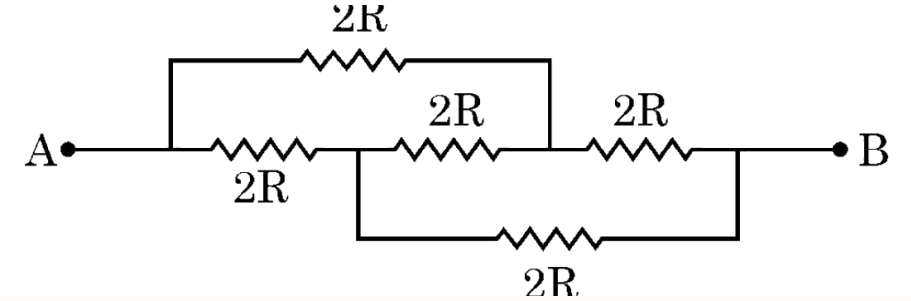
The Reason correctly explains the Assertion: the common current together with different resistances makes the power I^2R largest for the bulb of largest resistance, that is the 50 W bulb. Hence both are true and the Reason is the correct explanation, **option (A)**.

Final answer. Both Assertion and Reason are true and R correctly explains A, option **(A)**.

← *Back to Q70*

Q71

Assertion (A): *The equivalent resistance between points A and B in the given network is 2R.*
Reason (R): *All the resistors are connected in parallel.*



(D) Assertion (A) is false and Reason (R) is also false.

[2023 • Set 55-5-1]

SOLUTION

In an Assertion and Reason question we test each statement separately and then check whether the Reason correctly explains the Assertion.

The Reason claims that all the resistors in the network are connected in parallel. On inspecting the network, the resistors are not all in parallel. The topology is a mixed series and parallel arrangement, so the Reason is false.

The Assertion claims that the equivalent resistance between A and B is 2R. Reducing the actual network by the standard series and parallel rules (or by Kirchhoff's laws) does not give 2R, so the Assertion is also false.

If the resistors had genuinely all been in parallel, the equivalent resistance would be smaller than the smallest single resistor, certainly not as large as 2R. This further shows the two statements are inconsistent with the figure.

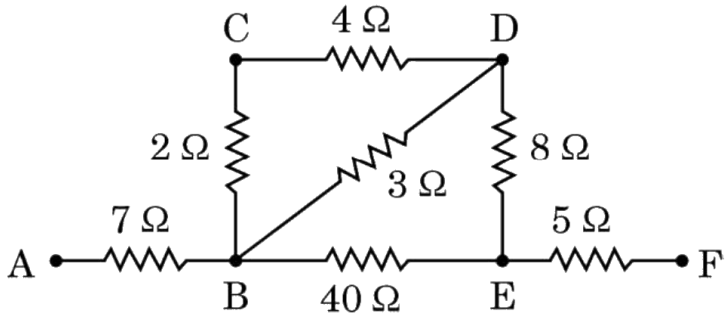
Both Assertion and Reason are false, which is **option (D)**.

Final answer. Both Assertion and Reason are false, so the answer is option **(D)**.

← *Back to Q71*

Q72

Assertion (A): *The given figure does not show a balanced Wheatstone bridge.*
Reason (R): *For a balanced bridge small current should flow through the galvanometer.*



(C) Assertion (A) is true and Reason (R) is false.

[2023 • Set 55-5-3]

SOLUTION

A Wheatstone bridge has four arms P, Q, R, S with a galvanometer across the central diagonal. The balance condition is the equality of the ratios of the arms.

$$\frac{P}{Q} = \frac{R}{S}$$

For the given arms in order 2, 3, 3, 2:

$$\frac{P}{Q} = \frac{2}{3}$$

$$\frac{R}{S} = \frac{3}{2}$$

Since $\frac{2}{3} \neq \frac{3}{2}$, the bridge is not balanced. The Assertion is therefore **true**.

The Reason states that for a balanced bridge a small current should flow through the galvanometer. This is wrong. At balance the points across the galvanometer are at the same potential, so exactly zero current flows through the galvanometer, not merely a small current. The Reason is therefore **false**.

Assertion is true and Reason is false, which is **option (C)**.

COMMON MISTAKE

At true balance the galvanometer current is exactly zero, never just small. Examiners often plant this wording to catch students.

Final answer. Assertion is true and Reason is false, so the answer is option **(C)**.

← Back to Q72

Q73

Two wires X and Y of the same material and of equal lengths having area of cross-section A and 2A respectively, are connected in parallel across an ideal battery of emf E. What is the ratio of current density (J_X / J_Y) *in them?*

[2021]

SOLUTION

The wires X and Y are of the same material and equal length, and they are connected in parallel across the same battery, so the same potential difference V appears across each wire.

The electric field inside a wire of length L carrying a potential difference V is the same for both wires:

$$E = \frac{V}{L}$$

Current density is related to the field through the conductivity of the material by the microscopic form of Ohm's law:

$$J = \sigma E$$

Both wires are of the same material, so they have the same conductivity σ, and since V and L are the same for both, the field E is the same. Therefore the current density is identical in the two wires:

$$\frac{J_X}{J_Y} = \frac{\sigma E}{\sigma E} = 1$$

The ratio of current densities is $\mathbf{\frac{J_X}{J_Y} = 1}$.

Note that the currents themselves differ, because $\boldsymbol{I = JA}$. Wire Y has twice the cross-sectional area, so it carries twice the current of wire X, even though the current density is the same.

CONCEPT

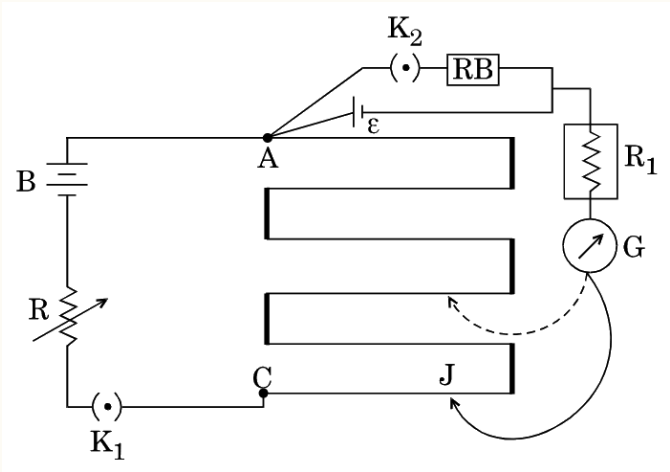
Current density depends on the field and the material, not on the area, so equal length plus same material plus same voltage gives equal J, even when the currents differ.

Final answer. The ratio of current densities is $J_X / J_Y = 1$.

← *Back to Q73*

Q74

A potentiometer draws a steady current from the battery (B) of emf E in the circuit as shown in the figure. The balancing length for the cell of emf ε when key K_2 is kept open, is $_1$. If key K_2 is closed and a resistance S is introduced in the resistance box (RB), the balancing length becomes $_2$.



The balancing length $_1 > _2$ because

(A) $E > \varepsilon$.

(B) positive terminal of the battery is connected to positive terminal of the cell.

(C) the voltage drop across the cell is greater than that across the battery.

(D) terminal potential difference of the cell is less than its emf.

[2021]

SOLUTION

A potentiometer compares a potential difference against the steady fall of potential along its wire. The key idea is what the cell of emf ε actually presents to the wire in each case.

When key K_2 is open, no current is drawn from the cell ε. The potentiometer balances against the full emf of the cell, giving the longer balancing length $_1$:

$$\varepsilon = k\,_1$$

where k is the potential gradient of the wire.

When K_2 is closed and a resistance S is added through the resistance box, the cell now drives a current I through its own internal resistance r. The potentiometer then balances against the terminal potential difference V, not the emf:

$$V = \varepsilon - I\,r$$

$$V = k\,_2$$

Since $V = \varepsilon - Ir < \varepsilon$, the new balancing length is shorter:

$$_2 < _1$$

The reason $_1 > _2$ is that the terminal potential difference of the cell when current flows is less than its emf, which is **option (D)**.

Final answer. The correct reason is option **(D)**, the terminal potential difference of the cell is less than its emf.

← *Back to Q74*

Q75

The high resistance R_1 *in the circuit is used to*
(A) keep the balance point near the mid-point of the potentiometer wire AC.
(B) reduce the current supplied by the battery (B) to zero at balance point.
(C) protect the galvanometer from damage due to large current.
(D) make the galvanometer more sensitive.

[2021]

SOLUTION

In a potentiometer the galvanometer is placed in the branch connecting the cell and the jockey. The high resistance R_1 is connected in series in this galvanometer branch.

While searching for the balance point, the jockey is often far from balance. At such a position there is a large potential difference between the jockey contact and the cell terminal, which would drive a large current through the galvanometer and could damage it.

Placing the high resistance R_1 in series limits this current by Ohm's law:

$$I_G = \frac{(\text{net } p.d. \text{ in the branch})}{R_1 + R_G}$$

A large R_1 keeps I_G small and safe during the off-balance search.

At the true balance point the potential difference across the galvanometer branch is zero, so R_1 does not affect the balance position. It only acts as a protective resistance.

Hence R_1 protects the galvanometer from damage due to a large current, which is **option (C)**.

Final answer. The high resistance protects the galvanometer from a large current, so the answer is option **(C)**.

← *Back to Q75*

Q76

If the key K_2 is kept open and resistance R_1 is increased, the balancing length $_1$ of the wire will
(A) increase.
(B) decrease.
(C) remain unaffected.
(D) become zero.

[2021]

SOLUTION

The balancing length depends only on the potential gradient k along the main potentiometer wire AC and on the emf being balanced, not on the galvanometer branch.

The resistance R_1 lies in series with the galvanometer branch, not in the main wire circuit. The potential gradient is fixed by the driving battery and the resistance of the main wire:

$$k = \frac{V_{AC}}{L}$$

Changing R_1 does not alter the current in the main wire AC, so k stays the same.

At balance the cell emf is matched as:

$$\varepsilon = k\,\ell_1$$

With K_2 open the cell delivers no current, so ε is unchanged, and k is unchanged. Therefore the balancing length is:

$$\ell_1 = \frac{\varepsilon}{k}$$

which does not depend on R_1. Increasing R_1 leaves ℓ_1 **unaffected**, which is **option (C)**.

Final answer. Increasing $\mathbf{R_1}$ leaves the balancing length unaffected, so the answer is option **(C)**.

← *Back to Q76*

Q77

Two potentiometers P and Q use the same wire of lengths 5 *m and* 10 *m respectively, between A and C. They are connected to battery (B) in the circuit separately. The potentiometer which is preferred to compare the emfs of two cells is*
(A) P, because potential gradient is less.
(B) Q, because potential gradient is less.

(C) P, because current is more.
(D) Q, because resistance of the potentiometer is less.

[2021]

SOLUTION

The sensitivity of a potentiometer depends on its potential gradient, which is the fall of potential per unit length of the wire.

$$k = \frac{V}{L}$$

A smaller potential gradient k means that a given potential difference is balanced over a longer length of wire, giving finer resolution and a more accurate balance point.

For the same driving battery, the longer wire gives a smaller gradient:

For wire P, $\mathbf{k_P} = \frac{V}{5}$

For wire Q, $\mathbf{k_Q} = \frac{V}{10}$

Since $\mathbf{k_Q < k_P}$, potentiometer Q with the 10 m wire is more sensitive and is preferred for comparing the emfs of two cells.

Hence Q is preferred because its potential gradient is less, which is **option (B)**.

CONCEPT

Lower potential gradient means higher sensitivity. A longer wire (or smaller driving current) is the standard way to increase potentiometer accuracy.

Final answer. Potentiometer Q is preferred because its potential gradient is less, so the answer is option **(B)**.

← *Back to Q77*

Q78

The reason for preferring manganin wire in a potentiometer is because
(A) of high value of its temperature coefficient of resistivity.

(B) of its low resistivity.
(C) of low value of its temperature coefficient of resistivity.
(D) its resistance decreases with increase in temperature.

[2021]

SOLUTION

During a potentiometer measurement a steady current flows through the wire, which tends to heat it. The wire material must keep its resistance, and therefore its potential gradient, almost constant despite this heating.

The change of resistance with temperature is governed by the temperature coefficient of resistivity:

$$R_T = R_0 (1 + \alpha \Delta T)$$

If α is very small, then even a noticeable rise in temperature ΔT produces almost no change in R, so the potential gradient stays steady and the balance point is reliable.

Manganin has an extremely low temperature coefficient of resistivity, of the order of 10^{-5} /°C, so its resistance is almost independent of temperature.

Hence manganin is preferred because of its low value of temperature coefficient of resistivity, which is **option (C)**.

Final answer. Manganin is preferred for its low temperature coefficient of resistivity, so the answer is option **(C)**.

← *Back to Q78*

Q79

A potentiometer can measure emf of a cell because

(A) the sensitivity of potentiometer is large.
(B) no current is drawn from the cell at balance.
(C) no current flows in the wire of potentiometer at balance.
(D) internal resistance of cell is neglected.

[2020 • Set 55-1-1]

SOLUTION

A potentiometer measures the emf of a cell by balancing the cell against the potential drop along its wire.

At the balance point the galvanometer shows zero deflection, which means no current is drawn from the cell whose emf is being measured.

Because the current drawn from the cell is zero, there is no potential drop across the cell's internal resistance:

$$V = \varepsilon - I\,r = \varepsilon - 0 = \varepsilon$$

So the terminal potential difference equals the emf exactly, and the potentiometer reads the true emf.

This is the key advantage over a voltmeter. A voltmeter always draws some current, so it reads $\varepsilon - Ir$, a value smaller than the emf. Ruling out the other options: large sensitivity (A) helps accuracy but is not the reason it can measure true emf; no current in the potentiometer wire (C) is wrong because the driving current always flows in the main wire; neglecting internal resistance (D) is incorrect because the method works precisely by making the internal drop zero, not by ignoring it.

The correct reason is that no current is drawn from the cell at balance, which is **option (B)**.

Final answer. No current is drawn from the cell at balance, so the answer is option **(B)**.

← Back to Q79

Q80

Two resistors R_1 *and* R_2 *of* $4\ \Omega$ *and* $6\ \Omega$ *are connected in parallel across a battery. The*

ratio of power dissipated in them, $P_1 : P_2$ *will be*

(A) 4 : 9

(B) **3 : 2**

(C) 9 : 4

(D) 2 : 3

[2020 • Set 55-1-1]

SOLUTION

When two resistors are connected in parallel across a battery, each has the same potential difference V across it.

The power dissipated in a resistor at a fixed voltage is given by:

$$P = \frac{V^2}{R}$$

Since V is common to both, the power is inversely proportional to the resistance:

$$P \propto \frac{1}{R}$$

Taking the ratio for the two resistors:

$$\frac{P_1}{P_2} = \frac{V^2 / R_1}{V^2 / R_2} = \frac{R_2}{R_1}$$

Substituting $\mathbf{R_1 = 4\ \Omega}$ and $\mathbf{R_2 = 6\ \Omega}$:

$$\frac{P_1}{P_2} = \frac{6}{4} = \frac{3}{2}$$

So $\mathbf{P_1 : P_2 = 3 : 2}$. The smaller resistor $\mathbf{R_1 = 4\ \Omega}$ dissipates more power, which agrees with $\mathbf{P_1 > P_2}$. This rules out 4:9 and 9:4 (which use $R \propto P$ or R^2) and 2:3 (the inverse).

The ratio of powers is $\mathbf{P_1 : P_2 = 3 : 2}$, which is **option (B)**.

COMMON MISTAKE

In parallel use $P = V^2/R$ *so* $P \propto 1/R$*, but in series use* $P = I^2 R$ *so* $P \propto R$*. Picking the wrong formula is the most common slip here.*

Final answer. The ratio of power dissipated is $P_1 : P_2 = 3 : 2$, so the answer is option **(B)**.

← *Back to Q80*

Q81

Kirchhoff's first rule at a junction in an electrical network, deals with conservation of

(A) energy

(B) charge

(C) momentum

(D) both energy and charge

[2020 • Set 55-1-2]

SOLUTION

Kirchhoff's first rule, also called the junction rule, states that the algebraic sum of the currents meeting at any junction in a circuit is zero.

$$\sum I = 0$$

This means the total current flowing into a junction equals the total current flowing out of it. Physically, this is true because charge cannot pile up or be created at a junction in a steady current. Whatever charge enters per second must leave per second, so the junction rule is a direct statement of the **conservation of electric charge**.

Option (A) energy and option (D) both are wrong because conservation of energy is expressed by Kirchhoff's second rule (the loop rule), not the junction rule. Option (C) momentum has no connection with either rule.

Hence the correct option is **(B) charge**.

TIP

Remember the pairing: junction rule = conservation of charge, loop rule = conservation of energy. Students often swap these in MCQs.

Final answer. The junction rule expresses conservation of charge, so the answer is option **(B)**.

← *Back to Q81*

Q82

The resistance of a metal wire increases with increasing temperature on account of

(A) decrease in free electron density.

(B) decrease in relaxation time.

(C) increase in mean free path.

(D) increase in the mass of electron.

[2020 • Set 55-1-2]

SOLUTION

The resistivity of a metal is given by

$$\rho = \frac{m}{n\, e^2 \tau}$$

where m is the electron mass, n the free electron density, e the electronic charge and τ the relaxation time. Since resistance $R = \frac{\rho L}{A}$, any increase in ρ raises R.

When the temperature of a metal rises, the positive ions of the lattice vibrate more strongly. The free electrons then collide with these ions more frequently, so the average time between successive collisions, the relaxation time τ, decreases.

From $\rho = \frac{m}{n\, e^2 \tau}$, a smaller τ gives a larger ρ, and therefore a larger R.

Ruling out the others: in a metal the free electron density n is essentially constant with temperature, so option (A) is wrong. The mean free path actually decreases, not increases, so option (C) is wrong. The mass of the electron is a fixed constant, so option (D) is wrong.

Hence the resistance increases because of a **decrease in relaxation time**, option **(B)**.

Final answer. Resistance rises due to a decrease in relaxation time, so the answer is option **(B)**.

← *Back to Q82*

Q83

Two unequal resistors are connected in series across a battery. Then the

(A) potential difference across each resistor is the same.

(B) current in the smaller resistor is larger.

(C) **potential difference across the bigger resistor is greater.**

(D) power dissipated in both resistors is the same.

[2020 • Set 55-1-3]

SOLUTION

When two resistors are connected in series across a battery, the same current I flows through both of them.

By Ohm's law, the potential difference across a resistor is

$$V = \mathrm{IR}$$

Since I is common to both resistors, the potential drop across each is directly proportional to its resistance. The bigger resistor therefore takes a greater share of the applied voltage.

Checking the options: (A) is wrong because the drops are equal only if the resistances are equal, but here they are unequal. (B) is wrong because the current is the same in both resistors in series, not larger in the smaller one. (D) is wrong because power $P = \mathrm{I}^2\mathrm{R}$ is also larger for the bigger resistance, so the two powers are not equal.

Hence the **potential difference across the bigger resistor is greater**, option **(C)**.

Final answer. The larger resistor gets the larger voltage drop, so the answer is option **(C)**.

← Back to Q83

Q84

A cell of internal resistance r connected across an external resistance R can supply maximum current when

(A) $R = r$

(B) $R \gg r$

(C) $R = \frac{r}{2}$

(D) $R = 0$

[2020 • Set 55-2-1]

SOLUTION

A cell of emf ε and internal resistance r connected to an external resistance R drives a current

$$I = \frac{\varepsilon}{R + r}$$

The power delivered to the external load is

$$P = I^2R = \frac{\varepsilon^2 R}{(R + r)^2}$$

To find the value of R for which this useful power is maximum, differentiate P with respect to R and set it to zero.

$$\frac{dP}{dR} = \varepsilon^2 \frac{(R + r)^2 - R \cdot 2(R + r)}{(R + r)^4} = 0$$

$$\Rightarrow (R + r) - 2R = 0$$

$$\Rightarrow R = r$$

This is the maximum power transfer condition: the cell delivers maximum (useful) power to the load when the external resistance equals the internal resistance.

Per the CBSE answer key, the correct option is **(A) $R = r$**. Option (D) $R = 0$ would give the largest circuit current but zero useful power in the load, while options (B) and (C) do not satisfy the maximum power condition.

Final answer. Maximum (useful) power is delivered when $R = r$, so the answer is option **(A)**.

← *Back to Q84*

Q85

In a current carrying conductor, the ratio of the electric field and the current density at a point is called

(A) Resistivity

(B) Conductivity

(C) Resistance

(D) Mobility

[2020 • Set 55-2-1]

SOLUTION

The microscopic (point) form of Ohm's law relates the electric field and the current density at a point inside a conductor.

$$\vec{j} = \sigma \vec{E}$$

where σ is the conductivity. Rearranging,

$$\vec{E} = \rho \vec{j}, \text{ where } \rho = \frac{1}{\sigma}$$

Here ρ is the resistivity of the material. Therefore the ratio of the electric field to the current density is

$$\frac{E}{j} = \rho$$

which is exactly the **resistivity**.

Ruling out the others: conductivity is the ratio j/E, the inverse of what is asked, so (B) is wrong. Resistance R depends on the dimensions of the conductor and is not a point quantity, so (C) is wrong. Mobility relates drift velocity to field, not E to j, so (D) is wrong.

Hence the ratio E/j is the **resistivity**, option **(A)**.

COMMON MISTAKE

Watch the order: E/j is resistivity, while j/E is conductivity. Swapping them is a common slip.

Final answer. The ratio of electric field to current density is resistivity, so the answer is option **(A)**.

← Back to Q85

Q86

Resistivity of a given conductor depends upon

(A) temperature.

(B) length of conductor.

(C) area of cross-section.

(D) shape of the conductor.

[2020 • Set 55-2-2]

SOLUTION

Resistivity is an intrinsic property of the material of the conductor and is given by

$$\rho = \frac{m}{n\, e^2 \tau}$$

where n is the free electron density and τ is the relaxation time. Both n and τ are determined by the nature of the material and by its temperature.

As temperature changes, the lattice vibrations change, which alters τ, so ρ changes with temperature.

The length, area of cross-section and shape of the conductor are geometric quantities. They affect the resistance through $R = \frac{\rho L}{A}$, but they do not change the resistivity ρ itself. So options (B), (C) and (D) are all wrong.

Hence resistivity depends on **temperature**, option **(A)**.

Final answer. Resistivity depends on temperature, so the answer is option **(A)**.

← *Back to Q86*

Q87

The ratio of current density and electric field is called

(A) Resistivity

(B) Conductivity

(C) Drift velocity

(D) Mobility

[2020 • Set 55-2-2]

SOLUTION

The microscopic form of Ohm's law connects the current density $\vec{j}$ at a point to the electric field $\vec{E}$ there.

$$\vec{j} = \sigma \vec{E}$$

The constant of proportionality σ is the conductivity, so the ratio of current density to electric field is

$$\frac{j}{E} = \sigma$$

which is the **conductivity**.

Ruling out the others: resistivity is the inverse ratio $E/j = 1/\sigma$, so (A) is wrong. Drift velocity is the average velocity of the electrons, with units m/s, so (C) is wrong. Mobility is drift velocity per unit field, not j/E, so (D) is wrong.

Hence the ratio j/E is the **conductivity**, option **(B)**.

Final answer. The ratio of current density to electric field is conductivity, so the answer is option **(B)**.

← *Back to Q87*

Q88

For a fixed potential difference applied across a conductor, the drift speed of free electrons does not depend upon

(A) free electron density in the conductor.

(B) mass of the electrons.

(C) length of the conductor.

(D) temperature of the conductor.

[2020 • Set 55-2-3]

SOLUTION

For a conductor of length L with a potential difference V across it, the field inside is $E = V/L$, and the drift speed of the free electrons is

$$v_d = \frac{eE\tau}{m} = \frac{e\tau V}{mL}$$

From this expression, the drift speed depends on the relaxation time τ, the electron mass m, and the length L of the conductor. Through τ it also depends on temperature, since raising the temperature reduces τ.

Notice that the free electron density n does not appear anywhere in $v_d = \frac{eE\tau}{m}$. So for a fixed applied potential difference, the drift speed is independent of n.

Hence the drift speed does not depend on the **free electron density** of the conductor, option **(A)**. Options (B) mass, (C) length and (D) temperature all do affect v_d, so they are not the correct choice.

CONCEPT

Free electron density n does control the current $I = nAev_d$, but it does not appear in the drift speed formula itself. Keep the two relations separate.

Final answer. Drift speed is independent of free electron density, so the answer is option **(A)**.

← *Back to Q88*

Q89

The electrical resistance of a conductor

(A) varies directly proportional to its area of cross-section.

(B) decreases with increase in its temperature.

(C) **decreases with increase in its conductivity.**

(D) is independent of its shape but depends only on its volume.

[2020 • Set 55-3-1]

SOLUTION

The resistance of a conductor is

$$R = \frac{\rho L}{A} = \frac{L}{\sigma A}$$

where ρ is the resistivity and $\sigma = 1/\rho$ is the conductivity.

Testing each option against this relation:

(A) From $R = \frac{\rho L}{A}$, resistance is inversely proportional to the area, not directly proportional. So (A) is wrong.

(B) For a metallic conductor, resistance increases with temperature, it does not decrease. So (B) is wrong.

(C) Since $R = \frac{L}{\sigma A}$, an increase in conductivity σ decreases R. So (C) is correct.

(D) Resistance depends on the geometry through the ratio L/A, not just on the volume, so two conductors of equal volume can have different resistances. So (D) is wrong.

Hence the resistance **decreases with increase in its conductivity**, option **(C)**.

Final answer. Resistance decreases as conductivity increases, so the answer is option **(C)**.

← Back to Q89

Q90

$m^2\,V^{-1}\,s^{-1}$ *is the SI unit of which of the following?*

(A) Drift velocity

(B) **Mobility**

(C) Resistivity

(D) Potential gradient

[2020 • Set 55-3-1]

SOLUTION

Mobility μ of a charge carrier is defined as the drift velocity acquired per unit electric field.

$$\mu = \frac{v_d}{E}$$

Working out its SI unit:

$$[\mu] = \frac{m\,s^{-1}}{V\,m^{-1}} = \frac{m^2}{V\cdots} = m^2\,V^{-1}\,s^{-1}$$

This matches the given unit exactly.

Comparing with the other options: drift velocity has unit m/s, resistivity has unit Ω, and potential gradient has unit V/m. None of these equals $m^2\,V^{-1}\,s^{-1}$.

Hence the SI unit $m^2\,V^{-1}\,s^{-1}$ belongs to **mobility**, option **(B)**.

Final answer. The unit $\mathbf{m^2\,V^{-1}\,s^{-1}}$ is that of mobility, so the answer is option **(B)**.

← Back to Q90

Q91

The element of a heater is rated (P, V). If it is connected across a source of voltage $\frac{V}{2}$, *then the power consumed by it will be*

(A) P

(B) 2P

(C) $\frac{P}{2}$

(D) $\frac{P}{4}$

[2020 • Set 55-3-1]

SOLUTION

The heating element of a heater is a fixed resistor, so its resistance R does not change with the applied voltage.

The power dissipated in a resistor of fixed resistance R is given by

$$P = \frac{V^2}{R}$$

At the rated voltage V, the rated power is

$$P = \frac{V^2}{R}$$

When the same element (same R) is connected across a voltage $V' = \frac{V}{2}$, the new power is

$$P' = \frac{(V')^2}{R} = \frac{(\frac{V}{2})^2}{R}$$

$$P' = \frac{V^2}{4R}$$

$$P' = \frac{1}{4}(\frac{V^2}{R}) = \frac{P}{4}$$

So halving the voltage reduces the power to one quarter of its rated value, which is option (D). Options (A) P and (B) 2P are wrong because power is not

constant and does not increase when voltage falls; option (C) $\frac{P}{2}$ is wrong because $P \propto V^2$, not $P \propto V$, so the power falls as the square of the voltage ratio. Hence the power consumed is $\frac{P}{4}$.

TIP

Use $P = \frac{V^2}{R}$ (not $P = VI$) whenever the resistance is fixed and only the voltage changes, since $P \propto V^2$ at constant R.

Final answer. The power consumed becomes $\frac{P}{4}$, so the correct option is **(D)**.

← *Back to Q91*

Q92

n resistors, each of resistance R are connected (a) in series, and (b) in parallel. Each combination is then connected to a source of emf E. The ratio of heat produced per second in the two cases will be

(A) n:1

(B) 1:n

(C) n^2:1

(D) **1:n^2**

[2020 • Set 55-3-2]

SOLUTION

The heat produced per second is the power, and for a source of emf E connected across an equivalent resistance R_{eq} the power is

$$P = \frac{E^2}{R_{eq}}$$

For n equal resistors each of value R connected in series, the equivalent resistance is

$$R_s = nR$$

so the heat produced per second in the series case is

$$P_s = \frac{E^2}{R_s} = \frac{E^2}{nR}$$

For the same n resistors connected in parallel, the equivalent resistance is

$$R_p = \frac{R}{n}$$

so the heat produced per second in the parallel case is

$$P_p = \frac{E^2}{R_p} = \frac{nE^2}{R}$$

Taking the ratio,

$$\frac{P_s}{P_p} = \frac{\frac{E^2}{nR}}{\frac{nE^2}{R}} = \frac{E^2}{nR} \times \frac{R}{nE^2}$$

$$\frac{P_s}{P_p} = \frac{1}{n^2}$$

Hence the ratio of heat produced per second in series to parallel is $1{:}n^2$, which is option (D). Options (A) $n{:}1$ and (C) $n^2{:}1$ are inverted (parallel actually dissipates more heat, so $\mathbf{P_s < P_p}$), and option (B) $1{:}n$ ignores that the resistance ratio appears, giving an n^2 dependence, not n.

Final answer. The ratio of heat produced per second is $\mathbf{1{:}n^2}$, so the correct option is **(D)**.

← *Back to Q92*

Q93

A uniform wire of 16 Ω *resistance is bent round in the form of a circle. When connected in a circuit between its two diametrically opposite points, its effective*

resistance will be

(A) 8 Ω

(B) 16 Ω

(C) **4 Ω**

(D) 2 Ω

[2020 • Set 55-3-3]

SOLUTION

When a uniform wire is bent into a circle and the circuit is connected across two diametrically opposite points, the current splits into the two semicircular arcs joining those points. These two arcs are connected between the same pair of points, so they are in parallel.

Total resistance of the full wire is

$$R = 16\ \Omega$$

Each semicircular arc is exactly half the wire, so each has half the total resistance:

$$R_1 = R_2 = \frac{R}{2} = \frac{16}{2} = 8\ \Omega$$

The two equal arcs R_1 and R_2 in parallel give

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2} = \frac{8 \times 8}{8 + 8}$$

$$R_{eq} = \frac{64}{16} = 4\ \Omega$$

Hence the effective resistance between the two diametrically opposite points is 4 Ω, which is option (C). Option (A) 8 Ω is the resistance of just one arc (it forgets the parallel combination); option (B) 16 Ω is the full unbroken wire; option (D) 2 Ω would arise only from a wrong arithmetic of the parallel formula.

TIP

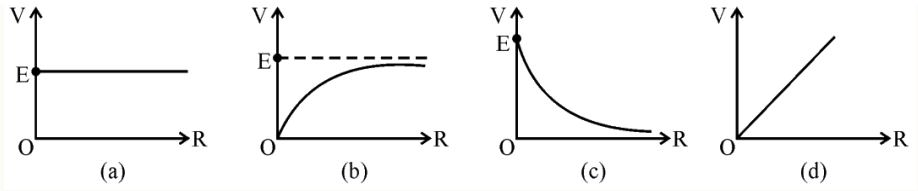
For two equal resistances in parallel the result is simply half of one of them, so $\frac{8}{2} = 4\ \Omega$ can be written down directly.

Final answer. The effective resistance is **4 Ω**, so the correct option is **(C)**.

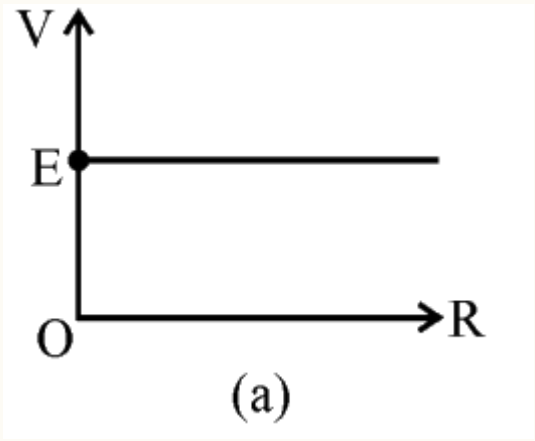
← *Back to Q93*

Q94

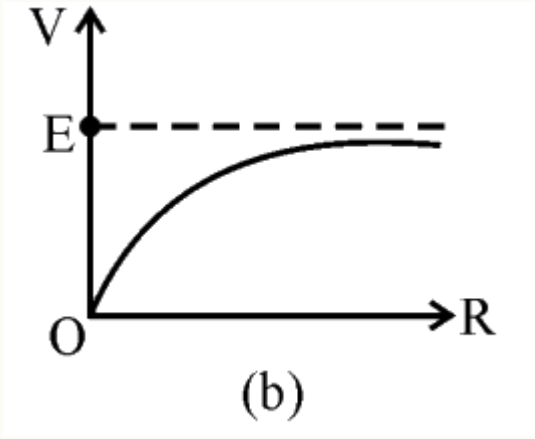
A cell of emf (E) and internal resistance r is connected across a variable external resistance R. The graph of terminal potential difference V as a function of R is —



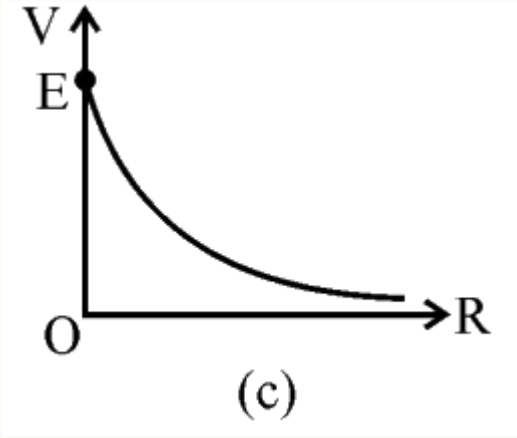
(A)



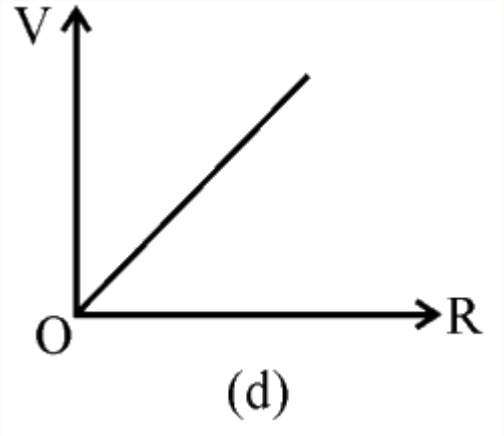
(B)



(C)



(D)



[2020 • Set 55-4-1]

SOLUTION

For a cell of emf ε and internal resistance r connected across an external resistance R, the current in the circuit is

$$I = \frac{\varepsilon}{R + r}$$

The terminal potential difference is the voltage across the external resistance R:

$$V = \mathrm{IR} = \frac{\varepsilon R}{R + r}$$

Examining the limits of this expression tells us the shape of the graph.

When $\boldsymbol{R = 0}$ (terminals short-circuited),

$$V = \frac{\varepsilon \times 0}{0 + r} = 0$$

so the curve starts from the origin.

As R becomes very large, $\boldsymbol{R + r \approx R}$, so

$$V \to \frac{\varepsilon R}{R} = \varepsilon$$

so the curve rises and saturates, approaching the horizontal line $\boldsymbol{V = \varepsilon}$ as an asymptote but never crossing it.

Therefore the correct graph is a curve that starts at the origin and increases monotonically, bending over and levelling off towards the constant value ε for large R (a rising, saturating curve). This matches graph (B). A straight line

through the origin or a falling curve would be wrong because V can never exceed ε and always increases with R.

Final answer. The terminal voltage rises from **0** and saturates at **ε**, so the correct option is **(B)**.

← *Back to Q94*

Q95

Fill in the blank with appropriate answer.
To minimize the percentage error in the determination of unknown resistance of a conductor in meter bridge experiment, the balance point is adjusted near _____ of the wire.
OR
In potentiometer, a long uniform wire is used to _____ potential gradient along the wire.

[2020 • Set 55-4-1]

SOLUTION

Meter bridge part: To minimise the percentage error in the determination of the unknown resistance, the balance point should be adjusted near the middle of the bridge wire, that is around the 50 cm mark.

At balance the unknown resistance is found from

$$X = R \frac{\quad}{100 -}$$

where is the balancing length in cm. The fractional error in X due to the reading of is smallest when is near 50 cm, because both and (100 −) are then large and comparable, so a small uncertainty in reading the jockey position produces the least percentage error in X. Near the ends the small denominator (100 −) or numerator greatly magnifies the error.

OR (potentiometer part): In a potentiometer, a long uniform wire is used to maintain (set up) a small and uniform potential gradient along the wire.

A longer wire for the same applied voltage gives a smaller potential drop per unit length (a smaller potential gradient), which increases the sensitivity, so

that even small potential differences can be measured accurately.

Final answer. Meter bridge: near the middle (about the **50 cm** mark); OR potentiometer: to maintain (set up) a small uniform potential gradient.

← *Back to Q95*

Q96

A copper wire of non-uniform area of cross-section is connected to a d.c. battery. The physical quantity which remains constant along the wire is _____ .

[2020 • Set 55-5-1]

SOLUTION

The physical quantity that remains constant along the wire is the electric current I.

In the steady state, charge cannot accumulate anywhere along the wire (this follows from conservation of charge, the equation of continuity). The same charge that enters one cross-section per second must leave the next cross-section per second, so the current through every cross-section is the same:

$$I = \text{constant}$$

The other related quantities do change because the area of cross-section A varies. The current density

$$j = \frac{I}{A}$$

is larger where A is smaller, and the drift velocity

$$v_d = \frac{I}{neA}$$

also changes with A. Only the current I stays the same everywhere along the non-uniform wire.

COMMON MISTAKE

A common slip is to write current density or drift velocity; both depend on A and so vary, whereas only I is conserved across every cross-section.

Final answer. The electric current I remains constant along the wire.

← Back to Q96

Q97

How does the mobility of electrons in a conductor change, if the potential difference applied across the conductor is doubled, keeping the length and temperature of the conductor constant?

[2019 • Set 55-1-1]

SOLUTION

The mobility of the electrons remains unchanged when the potential difference is doubled.

Mobility is defined as the drift velocity acquired per unit electric field, and for a conductor it is given by

$$\mu = \frac{e\tau}{m}$$

where e is the electronic charge, m its mass, and τ the relaxation time. At a fixed temperature, τ is fixed (it depends on the thermal motion of the lattice, not on the applied field), and e and m are constants.

Since $\mu = \frac{e\tau}{m}$ contains no term involving the applied voltage, doubling V does not change μ.

Note that doubling V doubles the field $E = \frac{V}{L}$, and so doubles the drift velocity $v_d = \mu E$, but the constant of proportionality μ itself stays the same. Hence the mobility is unaffected by doubling the potential difference.

Final answer. The mobility remains unchanged (independent of the applied potential difference).

← Back to Q97

Q98

When a potential difference is applied across the ends of a conductor, how is the drift velocity of the electrons related to the relaxation time?

[2019 • Set 55-1-2]

SOLUTION

When a potential difference is applied across the ends of a conductor, an electric field $\vec{E}$ is set up inside it, and each free electron experiences a force and is accelerated between successive collisions.

The acceleration of an electron of charge e and mass m in this field is

$$\vec{a} = -\frac{e\vec{E}}{m}$$

On average an electron travels freely for a time equal to the relaxation time τ before its next collision. The average (drift) velocity it gains is therefore

$$\vec{v}_d = \vec{a}\,\tau$$

$$\vec{v}_d = -\frac{e\tau}{m}\vec{E}$$

This shows that the drift velocity is directly proportional to the relaxation time:

$$v_d \propto \tau$$

The negative sign indicates that the drift velocity of the electrons is directed opposite to the applied field $\vec{E}$. So, for a given field, a larger relaxation time produces a larger drift velocity.

Final answer. The drift velocity is $\vec{v}_d = -\frac{e\tau}{m}\vec{E}$, so $v_d \propto \tau$ (directly proportional to the relaxation time).

← Back to Q98

Q99

How is the drift velocity in a conductor affected with the rise in temperature?

[2019 • Set 55-1-3]

SOLUTION

As the temperature of the conductor rises, the drift velocity of the electrons decreases (for a given applied field).

The drift velocity is related to the relaxation time by

$$v_d = \frac{e\tau}{m} E$$

so for a fixed field E we have

$$v_d \propto \tau$$

When the temperature increases, the positive ions of the lattice vibrate with greater amplitude. The free electrons therefore collide with them more frequently, which reduces the average time between collisions, that is the relaxation time τ decreases.

Since $\mathbf{v_d \propto \tau}$, a decrease in τ means the drift velocity also decreases as the temperature rises.

This is consistent with the increase in resistivity of a metal with temperature, since

$$\rho = \frac{m}{ne^2\tau}$$

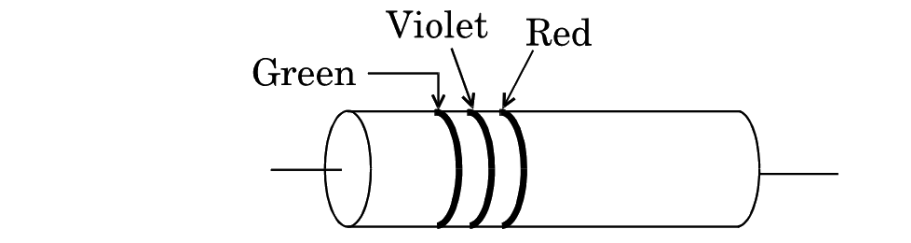
a smaller τ gives a larger ρ.

Final answer. The drift velocity decreases with a rise in temperature, because the relaxation time $\boldsymbol{\tau}$ decreases.

← Back to Q99

Q100

A carbon resistor is shown in the figure. Using colour code, write the value of the resistance.



[2019 • Set 55-3-1]

SOLUTION

The first three colour bands give the resistance value: the first band is the first significant digit, the second band is the second significant digit, and the third band is the multiplier (the power of ten by which the two-digit number is multiplied).

Reading the bands shown on the resistor:

Violet (first band) = **7**

Red (second band) = **2**

Green (third band, multiplier) = $\mathbf{10^5}$

So the two significant figures form the number 72, and the multiplier is 10^5:

$$R = 72\times 10^5\ \Omega$$

$$R = 7.2\times 10^6\ \Omega$$

$$R = 7.2\ \text{M}\Omega$$

Hence the value of the carbon resistor read from its colour code is 7.2 MΩ.

CONCEPT

Remember the order BB ROY of Great Britain has Very Good Wife: Black 0, Brown 1, Red 2, Orange 3, Yellow 4, Green 5, Blue 6, Violet 7, Grey 8, White 9.

Final answer. The resistance is $\boldsymbol{R = 72\times 10^5\ \Omega = 7.2\ \text{M}\Omega}$.

← Back to Q100

Q101

Draw a plot of resistivity of copper as a function of temperature.

[2019 • Set 55-3-3]

SOLUTION

For a metallic conductor like copper, the resistivity increases with rise in temperature, because at higher temperature the lattice ions vibrate more vigorously, the relaxation time τ decreases, and so the resistivity $\rho = \dfrac{m}{ne^2\tau}$ increases.

Over the normal range of temperatures this variation is very nearly linear and is described by:

$$\rho(T) = \rho_0[1 + \alpha(T - T_0)]$$

where ρ_0 is the resistivity at the reference temperature T_0 and α is the temperature coefficient of resistivity (positive for copper, about $4 \times 10^{-3}\ K^{-1}$).

Plot description: Take temperature T (in K) on the x-axis and resistivity ρ on the y-axis. The graph is an almost straight line rising upwards from left to right, showing that ρ increases as T increases. The line does not pass through the origin: it meets the y-axis at a small positive value (the residual resistivity due to impurity and defect scattering, which remains even as $T \to 0$). Near very low temperatures the curve bends and flattens out towards this constant residual value instead of going to zero, so it is not perfectly linear there.

Thus the resistivity of copper rises almost linearly with temperature, with a positive slope and a small positive intercept on the resistivity axis.

Final answer. The plot is a nearly straight line with positive slope, meeting the y-axis at a small positive residual resistivity and flattening at very low temperatures.

← Back to Q101

Q102

Under what condition will the current in a wire be the same when connected in series and in parallel of n identical cells each having internal resistance r and external resistance R?

[2019 • Set 55-4-1]

SOLUTION

Consider n identical cells, each of emf ε and internal resistance r, connected to an external resistance R.

When the cells are joined in series, the emfs add and the internal resistances add:

$$\varepsilon_s = n\varepsilon,\ r_s = nr$$

$$I_s = \frac{n\varepsilon}{R + nr}$$

When the cells are joined in parallel, the net emf is that of a single cell and the internal resistances combine in parallel:

$$\varepsilon_p = \varepsilon,\ r_p = \frac{r}{n}$$

$$I_p = \frac{\varepsilon}{R + \frac{r}{n}} = \frac{n\varepsilon}{nR + r}$$

For the current to be the same in both arrangements, set $I_s = I_p$:

$$\frac{n\varepsilon}{R + nr} = \frac{n\varepsilon}{nR + r}$$

$$R + nr = nR + r$$

$$nr - r = nR - R$$

$$(n - 1)r = (n - 1)R$$

$$R = r$$

Hence the current is the same in series and parallel only when the external resistance equals the internal resistance of one cell, that is $R = r$.

Final answer. The current is the same in both cases when the external resistance equals one cell's internal resistance, i.e. $R = r$.

← Back to Q102

Q103

Why is the terminal voltage of a cell generally less than the emf of the cell?

[2019 • Set 55-4-2]

SOLUTION

Every real cell has some internal resistance r of its own electrolyte.

When a current I is drawn from the cell, a part of the emf is used up in driving this current through the internal resistance, giving an internal voltage drop Ir.

The terminal voltage is therefore the emf left over after this internal drop:

$$V = \varepsilon - Ir$$

Since Ir is a positive quantity whenever current flows, V is less than ε.

Only when no current is drawn (open circuit, $I = 0$) does the drop Ir become zero and the terminal voltage become equal to the emf, $V = \varepsilon$.

This is why, while the cell is delivering current, its **terminal voltage is generally less than its emf**.

Final answer. Because the drop **Ir** across the internal resistance reduces the terminal voltage to $V = \varepsilon - Ir < \varepsilon$ when current flows.

← Back to Q103

Q104

Name the charge carriers for the flow of current in a (i) conductor and (ii) electrolyte.

[2019 • Set 55-4-3]

SOLUTION

The charge carriers responsible for the flow of current differ with the medium.

(i) In a conductor (a metal): the charge carriers are the **free electrons**. These are the loosely bound valence electrons that drift opposite to the applied field and carry the current.

(ii) In an electrolyte (a conducting solution): the charge carriers are the **positive and negative ions** (cations and anions) produced by dissociation of the electrolyte.

When a field is applied, the positive ions (cations) drift towards the cathode and the negative ions (anions) drift towards the anode, and the motion of both kinds of ions together constitutes the current through the electrolyte.

Final answer. Free electrons in a conductor; positive and negative ions (cations and anions) in an electrolyte.

← *Back to Q104*

Q105

Nichrome and copper wires of same length and same radius are connected in series. Current I is passed through them. Which wire gets heated up more? Justify your answer.

[2017]

SOLUTION

The two wires are connected in series, so the same current I flows through both of them.

The heat produced (power dissipated) in a wire is given by Joule's law:

$$P = I^2R$$

Since I is the same for both wires, the wire having the greater resistance R will dissipate more power and so will get heated up more.

The resistance of each wire is:

$$R = \frac{\rho L}{A}$$

The two wires have the same length L and the same radius (hence the same area A). So the resistance depends only on the resistivity ρ.

Nichrome has a much higher resistivity than copper ($\rho_{\text{Nichrome}} \sim 10^{-6}\ \Omega\, m$, while $\rho_{\text{Cu}} \sim 1.7 \times 10^{-8}\ \Omega\, m$).

Therefore the nichrome wire has the larger resistance, and being in series with the same current, it dissipates more heat.

Hence the **nichrome wire gets heated up more**.

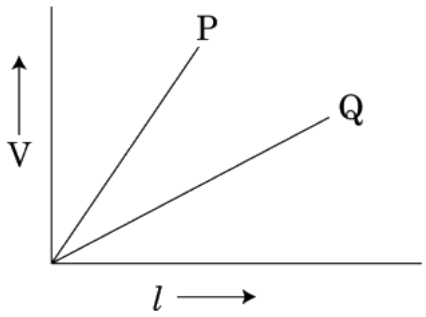
> **COMMON MISTAKE**
>
> *In series compare with $P = I^2R$ (same I); in parallel you would compare with $P = V^2/R$ (same V), which reverses the answer. Pick the right formula for the connection.*

Final answer. The nichrome wire gets heated more, as it has the larger resistivity (hence larger R) and $P = I^2R$ with the same series current.

← *Back to Q105*

Q106

The variation of potential difference V with length l in case of two potentiometer wires P and Q is as shown. Which one of these will you prefer for comparing emfs of two primary cells and why?



[2016]

SOLUTION

The sensitivity of a potentiometer is decided by its potential gradient, that is the potential drop per unit length of the wire:

$$k = \frac{V}{L}$$

A smaller potential gradient k means a given potential difference is balanced over a longer length of wire, so the balance (null) point can be located more accurately. Hence the wire with the smaller potential gradient is the more sensitive one.

On the given V versus l graph, the slope of each straight line is exactly its potential gradient k. So the wire whose line is less steep (smaller slope) has the smaller potential gradient.

Therefore you should prefer the wire corresponding to the line of smaller slope (the less steep line on the graph). Reading the figure, this is the wire whose V rises more slowly with l. It has the smaller potential gradient, gives a longer balancing length for the same emf, and so allows the emfs of the two primary cells to be compared more accurately.

Final answer. Prefer the wire whose line has the smaller slope (the less steep line), since a smaller potential gradient $\boldsymbol{k = V/L}$ makes the potentiometer more sensitive and gives a longer, more accurate balancing length.

← Back to Q106

Q107

Define the term conductivity of a conductor. On what factors does it depend?

[2016]

SOLUTION

Conductivity (σ) of a material is defined as the reciprocal of its resistivity:

$$\sigma = \frac{1}{\rho}$$

Equivalently, it is the constant of proportionality in the microscopic form of Ohm's law, relating the current density $\vec{j}$ to the applied electric field $\vec{E}$:

$$\vec{j} = \sigma \vec{E}$$

Its SI unit is $\Omega^{-1}\,\mathrm{m}^{-1}$, also written as $S\,\mathrm{m}^{-1}$ (siemens per metre).

The conductivity is expressed in terms of the microscopic quantities as:

$$\sigma = \frac{ne^2\tau}{m}$$

From this it depends on the following factors:

Number density of free charge carriers n (a property of the material).

The relaxation time τ, which decreases with rise in temperature and with increase in impurities or lattice defects, so σ decreases as temperature rises.

It also depends on the nature of the material through the carrier charge e and mass m.

Conductivity is a characteristic property of the material and **does not depend on the dimensions (length, area or shape)** of the conductor.

Final answer. Conductivity $\boldsymbol{\sigma = 1/\rho}$ (unit $\boldsymbol{S\,\mathrm{m}^{-1}}$); it depends on carrier density $\boldsymbol{n}$, relaxation time $\boldsymbol{\tau}$ and the material, not on the conductor's dimensions.

← *Back to Q107*

Q108

Define mobility of a charge carrier. What is its relation with relaxation time?

[2016]

SOLUTION

Mobility (μ) of a charge carrier is defined as the magnitude of the drift velocity acquired by the carrier per unit applied electric field:

$$\mu = \frac{|v_d|}{E}$$

Its SI unit is $m^2\ V^{-1}\ s^{-1}$.

To relate it to the relaxation time, consider a carrier of charge e and mass m in field E. The force on it is eE, giving acceleration $\boldsymbol{a = \frac{eE}{m}}$. The drift velocity built up over the average relaxation time τ between collisions is:

$$v_d = a\tau = \frac{eE\tau}{m}$$

Substituting this into the definition of mobility:

$$\mu = \frac{v_d}{E} = \frac{eE\tau}{mE}$$

$$\mu = \frac{e\tau}{m}$$

Hence the mobility is directly proportional to the relaxation time, $\boldsymbol{\mu = \frac{e\tau}{m}}$. A longer relaxation time (fewer collisions) or a smaller carrier mass gives a higher mobility.

Final answer. Mobility $\boldsymbol{\mu = |v_d|/E}$, and it is related to relaxation time by $\boldsymbol{\mu = e\tau/m}$.

← *Back to Q108*

Q109

Two wires one of copper and other of manganin have same resistance and equal length. Which wire is thicker and why?

[2016]

SOLUTION

The resistance of a wire of length L and cross-sectional area A is:

$$R = \frac{\rho L}{A}$$

Both wires have the same resistance R and the same length L. Rearranging for the area:

$$A = \frac{\rho L}{R}$$

With L and R fixed, this gives:

$$A \propto \rho$$

So the wire of larger resistivity must have the larger cross-sectional area, and hence be thicker.

Manganin has a much higher resistivity than copper ($\boldsymbol{\rho_{\text{manganin}} \approx 4.4 \times 10^{-7}\ \Omega\, m}$, while $\boldsymbol{\rho_{\text{Cu}} \approx 1.7 \times 10^{-8}\ \Omega\, m}$).

Therefore, to keep the same resistance over the same length, the manganin wire needs a much larger cross-section.

Hence the **manganin wire is thicker**, because of its larger resistivity.

Final answer. The manganin wire is thicker, since $\boldsymbol{A \propto \rho}$ (for equal $\boldsymbol{R}$ and $\boldsymbol{L}$) and manganin has the larger resistivity.

← *Back to Q109*

Q110

Define the term 'relaxation time' in a conductor.

[2016]

SOLUTION

Relaxation time (τ) is defined as the average time interval between two successive collisions of a free electron with the vibrating positive ions of the lattice (and other scatterers) as it drifts through a conductor.

In metals its value is of the order of $\tau \sim 10^{-14}\ s$.

During this short interval an electron accelerates freely under the applied field before its next collision randomises its motion, so τ fixes the drift velocity and the conductivity:

$$v_d = \frac{eE\tau}{m}$$

$$\sigma = \frac{ne^2\tau}{m}$$

A larger relaxation time means fewer collisions, giving a higher drift velocity and greater conductivity.

Final answer. Relaxation time is the average time between two successive collisions of a free electron with the lattice ions in a conductor.

← *Back to Q110*

Q111

Distinguish between emf and terminal voltage of a cell.

[2015]

SOLUTION

The emf and the terminal voltage of a cell differ as follows.

The emf ε of a cell is the potential difference between its two terminals when the cell is in the open circuit, that is, when no current is being drawn from it. It equals the maximum energy supplied by the cell per unit charge.

The terminal voltage V of a cell is the potential difference between its two terminals when the cell is actively driving a current I through an external resistance, that is, in the closed circuit.

The two are related by

$$V = \varepsilon - I\,r$$

where r is the internal resistance of the cell. Thus the **emf depends only on the cell** and is constant, while the **terminal voltage depends on the current drawn** and is always less than the emf when the cell supplies current.

Final answer. EMF is the open-circuit terminal potential difference of the cell, while terminal voltage $V = \varepsilon - Ir$ is the potential difference when the cell drives a current.

← *Back to Q111*

Q112

Define the term 'Mobility' of charge carriers in a conductor. Write its S.I. unit.

[2014]

SOLUTION

The mobility μ of a charge carrier in a conductor is defined as the magnitude of the drift velocity acquired by the carrier per unit applied electric field.

$$\mu = \frac{|v_d|}{E}$$

It is a measure of how readily the charge carriers move through the conductor in response to an applied electric field.

Its SI unit is obtained from the defining relation,

$$\mu = \frac{m\,s^{-1}}{V\,m^{-1}} = m^2\,V^{-1}\,s^{-1}$$

Hence mobility is measured in **metre squared per volt per second**, $m^2\,V^{-1}\,s^{-1}$.

Final answer. Mobility $\mu = |v_d|/E$ is the drift velocity per unit field; its SI unit is $\mathbf{m^2\,V^{-1}\,s^{-1}}$.

← *Back to Q112*

Q113

Show variation of resistivity of copper as a function of temperature in a graph.

[2014]

SOLUTION

The variation of the resistivity of copper (a metallic conductor) with absolute temperature is described by

$$\rho(T) = \rho_0\,[1 + \alpha(T - T_0)]$$

where α is the positive temperature coefficient of resistivity.

The graph is plotted with absolute temperature T (in kelvin) along the x-axis and resistivity ρ along the y-axis. Over the normal range of temperatures the curve is an almost straight line with a positive slope, showing that the resistivity of copper increases as the temperature rises.

The line does not pass through the origin. At very low temperatures the curve bends and flattens, meeting the y-axis at a small, non-zero value of resistivity called the **residual resistivity**, which arises from scattering of electrons by impurities and defects in the metal.

Thus the plot is a nearly **linear, upward-sloping curve** that intercepts the resistivity axis at a small residual value.

COMMON MISTAKE

Mark the small non-zero intercept (residual resistivity) on the y-axis; drawing the line through the origin is a common mistake that loses the mark.

Final answer. Resistivity of copper rises almost linearly with temperature, starting from a small non-zero residual value at low temperature.

← *Back to Q113*

Q114

Define the term 'electrical conductivity' of a metallic wire. Write its S.I. unit.

[2014]

SOLUTION

The electrical conductivity σ of a metallic wire is defined as the reciprocal of its resistivity.

$$\sigma = \frac{1}{\rho}$$

It is a property of the material that measures how easily an electric current is set up in it. In terms of the current density $\vec{j}$ and the applied electric field $\vec{E}$, it appears in the microscopic form of Ohm's law,

$$\vec{j} = \sigma \vec{E}$$

so a larger conductivity gives a larger current density for the same field.

Its SI unit is the reciprocal of the unit of resistivity ($\Omega\ m$), that is

$$\Omega^{-1}\ \mathrm{m}^{-1} = S\ \mathrm{m}^{-1}$$

Hence electrical conductivity is measured in **siemens per metre**, $S\ \mathrm{m}^{-1}$.

Final answer. Conductivity $\boldsymbol{\sigma = 1/\rho}$ measures how easily current flows in a material; its SI unit is $\boldsymbol{S\ \mathrm{m}^{-1}}$ (or $\boldsymbol{\Omega^{-1}\ \mathrm{m}^{-1}}$).

← *Back to Q114*

Q115

Define the term 'drift velocity' of charge carriers in a conductor and write its relationship with the current flowing through it.

[2014]

SOLUTION

The drift velocity v_d of charge carriers in a conductor is defined as the average velocity acquired by the free electrons in the direction opposite to the applied electric field, superposed on their random thermal motion.

Even for currents of a few amperes its magnitude is very small, of the order of $10^{-3}\ m\ s^{-1}$.

Consider a conductor of cross-sectional area A carrying a current I, having n free electrons per unit volume, each of charge e. In a small time Δt the electrons sweep through a length $v_d \Delta t$, so the charge crossing any section is

$$\Delta q = n\, e\, A\, (v_d\, \Delta t)$$

The current is then

$$I = \frac{\Delta q}{\Delta t} = n\, e\, A\, v_d$$

Hence the required relation between current and drift velocity is

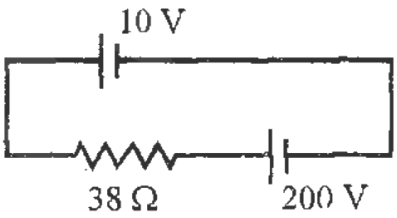
$$I = n\, e\, A\, v_d$$

Final answer. Drift velocity is the average velocity of free electrons opposite to the field, related to current by $\boldsymbol{I = neAv_d}$.

← *Back to Q115*

Q116

A 10 V battery of negligible internal resistance is connected across a 200 V battery and a resistance of 38 Ω as shown in the figure. Find the value of the current in circuit.



[2013]

SOLUTION

The 10 V battery has negligible internal resistance and is connected so that it opposes the 200 V battery, the two acting against each other across the 38 Ω resistance.

Applying Kirchhoff's voltage law around the loop, the net driving emf equals the potential drop across the resistor.

$$200 - 10 = I \times 38$$

Solving for the current,

$$I = \frac{200 - 10}{38}$$

$$I = \frac{190}{38}$$

$$I = 5\ A$$

Hence the current flowing in the circuit is **5 *A***.

Final answer. The current in the circuit is **5 *A***.

← *Back to Q116*

Q117

The emf of a cell is always greater than its terminal voltage. Why? Give reason.

[2013]

SOLUTION

When a cell drives a current I through an external circuit, the same current must also pass through the internal resistance r of the cell itself.

This causes a potential drop $I\,r$ inside the cell, so the voltage available at the terminals is reduced by this amount.

$$V = \varepsilon - I\,r$$

Since the current I and the internal resistance r are both positive, the term $\boldsymbol{I\,r > 0}$, and therefore

$$V < \varepsilon$$

Thus the terminal voltage is always **less than the emf** whenever the cell supplies current. The two become equal only in the open circuit, when $\boldsymbol{I = 0}$ and there is no internal drop.

Final answer. Because the current produces an internal drop **Ir**, the terminal voltage $\boldsymbol{V = \varepsilon - Ir}$ is always less than the emf when current flows.

← *Back to Q117*

Q118

Two conducting wires X and Y of same diameter but different materials are joined in series across a battery. If the number density of electrons in X is twice that in Y, find the ratio of drift velocity of electrons in the two wires.

[2012]

SOLUTION

Since the wires X and Y are joined in series, the same current I flows through both of them.

As they have the same diameter, they have the same cross-sectional area A.

Using the relation between current and drift velocity,

$$I = n\,e\,A\,v_d$$

For the same I, e and A, the drift velocity is inversely proportional to the number density of electrons,

$$v_d \propto \frac{1}{n}$$

Taking the ratio for the two wires,

$$\frac{v_{d,X}}{v_{d,Y}} = \frac{n_Y}{n_X}$$

Given that $\mathbf{n_X = 2\, n_Y}$,

$$\frac{v_{d,X}}{v_{d,Y}} = \frac{n_Y}{2\, n_Y} = \frac{1}{2}$$

Hence the ratio of drift velocities is $\mathbf{v_{d,X} : v_{d,Y} = 1 : 2}$.

Final answer. The ratio of drift velocities is $\mathbf{v_{d,X} : v_{d,Y} = 1 : 2}$.

← *Back to Q118*

Q119

Two wires of equal length, one of copper and the other of manganin have the same resistance. Which wire is thicker?

[2012]

SOLUTION

The resistance of a wire of length L and cross-sectional area A made of a material of resistivity ρ is

$$R = \rho \frac{L}{A}$$

Rearranging for the area,

$$A = \rho \frac{L}{R}$$

The two wires have the same length L and the same resistance R, so the cross-sectional area is directly proportional to the resistivity,

$$A \propto \rho$$

The resistivity of manganin ($\rho_{Mn} \approx 44 \times 10^{-8}\ \Omega\, m$) is much greater than that of copper ($\rho_{Cu} \approx 1.7 \times 10^{-8}\ \Omega\, m$).

Therefore, to have the same resistance, the manganin wire needs a much larger cross-sectional area,

$$A_{Mn} \gg A_{Cu}$$

Hence the **manganin wire is thicker**.

Final answer. The manganin wire is thicker, since its higher resistivity requires a larger area for the same length and resistance.

← *Back to Q119*

Q120

When electrons drift in a metal from lower to higher potential, does it mean that all the free electrons of the metal are moving in the same direction?

[2012]

SOLUTION

No, all the free electrons are not moving in the same direction.

The free electrons in a metal possess very large random thermal velocities, of the order of $10^6\ m\ s^{-1}$, directed randomly in all directions, so that their average velocity is zero in the absence of a field.

When an external electric field $\vec{E}$ is applied, each electron acquires only a small additional drift velocity, of the order of $10^{-3}\ m\ s^{-1}$, in the direction opposite to $\vec{E}$ (that is, from lower to higher potential).

Since this drift speed is smaller than the random thermal speed by about nine orders of magnitude, at any given instant the electrons are still moving in essentially random directions.

The drift is only a **small average bias** superposed on the much larger random thermal motion; it does not make all the electrons move together in one

direction.

Final answer. No; the drift is only a tiny average bias superposed on the much larger random thermal motion, so the electrons do not all move in the same direction.

← *Back to Q120*

Q121

A resistance R is connected across a cell of emf ε and internal resistance r. A potentiometer now measures the potential difference between the terminals of the cell as V. Write the expression for r in terms of ε, V and R.

[2011 • Set 55-1-1]

SOLUTION

A cell of emf ε and internal resistance r is connected across an external resistance R. The potentiometer reads the terminal potential difference V.

Applying Ohm's law to the external resistance R, the current in the circuit is

$$I = \frac{V}{R}$$

The terminal potential difference equals the emf minus the drop across the internal resistance,

$$V = \varepsilon - \mathrm{Ir}$$

$$\Rightarrow \mathrm{Ir} = \varepsilon - V$$

$$\Rightarrow r = \frac{\varepsilon - V}{I}$$

Substituting $\boldsymbol{I = \frac{V}{R}}$,

$$r = \frac{\varepsilon - V}{V/R}$$

$$\Rightarrow r = \left(\frac{\varepsilon - V}{V}\right)R$$

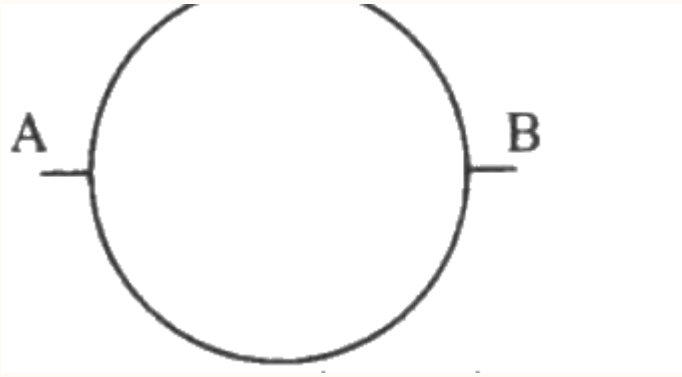
Thus the internal resistance is expressed in terms of ε, V and R.

Final answer. The internal resistance is $r = (\frac{\varepsilon - V}{V})R$.

← *Back to Q121*

Q122

A wire of resistance 8R *is bent in the form of a circle. What is the effective resistance between the ends of a diameter AB?*



[2010]

SOLUTION

A uniform wire of total resistance 8R is bent into a circle. The two ends of a diameter AB divide the circular wire into two equal semicircular arcs.

Since the wire is uniform, each semicircular arc carries half the total length and hence half the total resistance,

$$R_1 = R_2 = \frac{8R}{2} = 4R$$

Between the points A and B, these two arcs are connected in parallel (both span the same two end points).

For two resistances in parallel,

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

$$R_{eq} = \frac{(4R)(4R)}{4R + 4R}$$

$$R_{eq} = \frac{16R^2}{8R}$$

$$\Rightarrow R_{eq} = 2R$$

Hence the effective resistance across the diameter AB is 2R.

> **COMMON MISTAKE**
>
> *Across a diameter the two arcs are equal (each 4R); a common slip is to take them as R and 7R, which is the case only when the points are not diametrically opposite.*

Final answer. The effective resistance between the ends of the diameter is **2R.**

← *Back to Q122*

Q123

The plot of the variation of potential difference across a combination of three identical cells in series, versus current is as shown below. What is the emf of each cell?

[2008]

SOLUTION

Three identical cells, each of emf ε and internal resistance r, are joined in series. Their combined emf is 3ε and combined internal resistance is 3r.

The terminal potential difference of this combination, when it drives a current I, is

$$V = 3\varepsilon - I(3r)$$

This is a straight line of the form $\boldsymbol{V = (3\varepsilon) - (3r)I}$, so the intercept on the V-axis (at $\boldsymbol{I = 0}$) gives the total emf 3ε.

From the graph, the intercept on the V-axis is $\mathbf{V_0 = 6\ V}$.

$$\therefore 3\varepsilon = 6\ V$$

$$\Rightarrow \varepsilon = \frac{6}{3}$$

$$\Rightarrow \varepsilon = 2\ V$$

Hence the emf of each cell is 2 V. (In general $\boldsymbol{\varepsilon = V_0/3}$, where V_0 is the V-axis intercept read from the plot.)

Final answer. The emf of each cell is $\boldsymbol{\varepsilon = 2\ V}$ (one-third of the $\boldsymbol{V}$-axis intercept).

← *Back to Q123*

Q124

Sketch a graph showing variation of resistivity of carbon with temperature.

[2006]

SOLUTION

Carbon behaves like a semiconductor, so the number of free charge carriers increases as the temperature rises. This causes its resistivity to fall with increasing temperature, i.e. carbon has a negative temperature coefficient of resistivity.

The required graph has temperature T (in K or $^\circ C$) on the x-axis and resistivity ρ on the y-axis.

The curve starts at a high value of ρ at low temperature and falls smoothly as T increases. It is steeper at low temperatures and flattens (tails off) towards higher temperatures, always staying above the T-axis.

Shape of the plot: a continuously decreasing (downward sloping) curve, concave upward, labelled ρ (vertical axis) versus T (horizontal axis).

This is opposite to a metal such as copper, whose resistivity increases almost linearly with temperature.

Final answer. Carbon's resistivity decreases with rising temperature, giving a downward sloping ($\boldsymbol{\rho}$ vs $\boldsymbol{T}$) curve.

← *Back to Q124*

Q125

Define electrical conductivity of a conductor and give its S.I. unit.

[2003]

SOLUTION

Electrical conductivity (σ) of a conductor is defined as the reciprocal of its resistivity,

$$\sigma = \frac{1}{\rho}$$

Equivalently, in the microscopic form of Ohm's law it is the proportionality constant between the current density $\vec{j}$ and the electric field $\vec{E}$ inside the conductor,

$$\vec{j} = \sigma \vec{E}$$

So conductivity measures how easily a material allows electric current to flow through it.

SI unit: from $\boldsymbol{\sigma = \frac{1}{\rho}}$ and ρ in $\Omega\, m$, the unit of σ is

$\boldsymbol{\Omega^{-1}\, m^{-1} = S\, m^{-1}}$ (siemens per metre).

Final answer. Conductivity is the reciprocal of resistivity; its SI unit is $\boldsymbol{S\, m^{-1}}$ ($\boldsymbol{\Omega^{-1} m^{-1}}$).

← *Back to Q125*

Q126

What happens to the power dissipation if the value of electric current passing through a conductor of constant resistance is doubled?

[2003]

SOLUTION

The power dissipated as heat in a conductor of resistance R carrying current I is

$$P = I^2R$$

With R kept constant, the power is directly proportional to the square of the current, $P \propto I^2$.

Let the initial power be

$$P_1 = I^2R$$

When the current is doubled, $I \to 2I$, the new power is

$$P_2 = (2I)^2R$$

$$P_2 = 4\,I^2R$$

$$\Rightarrow P_2 = 4P_1$$

Hence doubling the current makes the power dissipation **four times** its original value.

Final answer. The power dissipation becomes four times its original value.

← *Back to Q126*

Q127

What is the order of magnitude of drift velocity of electrons in a conductor? Deduce the relation between the current flowing through a conductor and drift velocity of electrons in it.

[2026 • Set 55-1-1]

SOLUTION

The drift velocity of electrons in a metallic conductor is very small, of the order of 10^{-3} $m\ s^{-1}$ (a few millimetres per second), even when the conductor carries a current of a few amperes.

Relation between current and drift velocity: consider a conductor of cross-sectional area A with free-electron number density n (electrons per unit volume), each of charge e, drifting with speed v_d.

In a small time dt, every electron moves a distance $v_d\,dt$. So all the free electrons lying within a cylinder of length $v_d\,dt$ and base area A cross a given cross-section.

Volume of this cylinder = $A\,v_d\,dt$, so the number of electrons crossing is

$$N = n\,(A\,v_d\,dt)$$

The charge crossing the section in time dt is

$$dq = e\,N = n\,e\,A\,v_d\,dt$$

Therefore the current is

$$I = \frac{dq}{dt} = \frac{n\,e\,A\,v_d\,dt}{dt}$$

$$\Rightarrow I = n\,e\,A\,v_d$$

This is the required relation between the current and the drift velocity of electrons.

Final answer. Drift velocity is of order $10^{-3}\ m\ s^{-1}$, and $I = n\,e\,A\,v_d$.

← *Back to Q127*

Q128

A heating element using nichrome is connected to a 220 *V supply. Initially it draws a current of* 2·9 *A. After some time, the current attains a steady value of* 2·5 *A. Find the steady temperature of the heating element if the room temperature is* 27 °C. *The temperature coefficient of resistance of nichrome is* $1{\cdot}7\times 10^{-4}\ °C^{-1}$.

[2026 • Set 55-2-1]

SOLUTION

The supply voltage is $V = 220\ V$. Using Ohm's law $R = V/I$ at the two stages.

Resistance at room temperature (27 °C), when the current is $I_0 = 2.9\ A$:

$$R_0 = \frac{220}{2.9} = 75.86\ \Omega$$

Resistance at the steady temperature T, when the current is $I = 2.5\ A$:

$$R = \frac{220}{2.5} = 88\ \Omega$$

Using the temperature dependence of resistance, with $T_0 = 27\ ^\circ C$ and $\alpha = 1.7 \times 10^{-4}\ ^\circ C^{-1}$:

$$R = R_0\,[\,1 + \alpha(T - T_0)\,]$$

$$\frac{R}{R_0} = 1 + \alpha(T - T_0)$$

$$\frac{88}{75.86} = 1 + (1.7 \times 10^{-4})(T - 27)$$

$$1.160 = 1 + (1.7 \times 10^{-4})(T - 27)$$

$$0.160 = (1.7 \times 10^{-4})(T - 27)$$

$$T - 27 = \frac{0.160}{1.7 \times 10^{-4}}$$

$$T - 27 = 941\ ^\circ C$$

$$\Rightarrow T \approx 968\ ^\circ C$$

Hence the steady temperature of the heating element is about 968 °C.

COMMON MISTAKE

Compute R_0 at room temperature (from the initial current 2.9 A) and R at the steady current 2.5 A; mixing up which current gives which resistance reverses the temperature change.

Final answer. The steady temperature of the heating element is about **968 °C**.

← Back to Q128

Q129

Calculate the temperature at which the resistance of a conductor becomes 20% more than its resistance at 27 °C. The value of the temperature coefficient of resistance of the

material of conductor is $2{\cdot}0\times 10^{-4}\ ^\circ C^{-1}$.

[2026 • Set 55-2-2]

SOLUTION

The resistance of a conductor varies with temperature as

$$R_T = R_0\,[\,1 + \alpha(T - T_0)\,]$$

where R_0 is the resistance at $\mathbf{T_0 = 27\ ^\circ C}$ and $\boldsymbol{\alpha = 2.0\times 10^{-4}\ ^\circ C^{-1}}$.

The resistance is 20% more than its value at 27 °C, so

$$R_T = 1.20\ R_0$$

Substituting,

$$1.20\ R_0 = R_0\,[\,1 + \alpha(T - 27)\,]$$

$$1.20 = 1 + \alpha(T - 27)$$

$$0.20 = \alpha(T - 27)$$

$$T - 27 = \frac{0.20}{2.0\times 10^{-4}}$$

$$T - 27 = 1000\ ^\circ C$$

$$\Rightarrow T = 1027\ ^\circ C$$

Hence the resistance becomes 20% greater at a temperature of 1027 °C.

Final answer. The required temperature is $\boldsymbol{T = 1027\ ^\circ C}$.

← *Back to Q129*

Q130

(a) Why is manganin used in wire-wound standard resistors? Calculate the temperature at which the resistance of a wire increases by 25% of its resistance at

room temperature 27·5 °C. *The temperature coefficient of resistance of the material of the wire is* 0·004 $°C^{-1}$.

[2026 • Set 55-2-3]

SOLUTION

(a) Manganin is used in wire-wound standard resistors because it has an extremely small temperature coefficient of resistance ($\alpha \sim 10^{-5}\ °C^{-1}$). Its resistance therefore stays almost unchanged when the temperature varies during use, so the resistor keeps a precise, stable value. It also has a high resistivity, which lets a large resistance be made from a short length of wire.

(b) The resistance varies with temperature as

$$R_T = R_0 [\, 1 + \alpha(T - T_0) \,]$$

Here the resistance increases by 25%, so the change in resistance satisfies

$$\frac{\Delta R}{R_0} = \alpha \Delta T$$

$$0.25 = (0.004)\, \Delta T$$

$$\Delta T = \frac{0.25}{0.004}$$

$$\Delta T = 62.5\ °C$$

The final temperature is

$$T = T_0 + \Delta T = 27.5 + 62.5$$

$$\Rightarrow T = 90\ °C$$

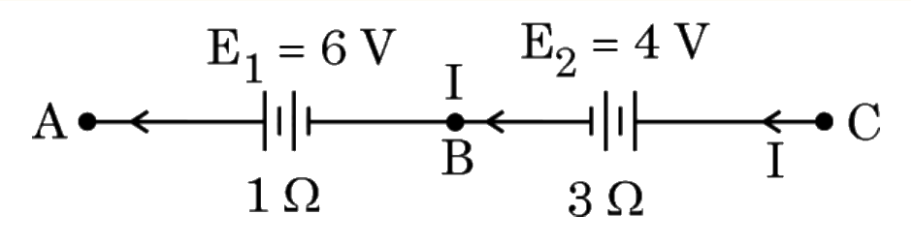
Hence the wire's resistance increases by 25% at a temperature of 90 °C.

Final answer. Manganin has a very low **α** (stable resistance); the required temperature is $\boldsymbol{T = 90\ °C}$.

← *Back to Q130*

Q131

In the given figure, a steady current I flows through the circuit when points A and C are connected by a wire of negligible resistance. Find the potential difference between points B and C.



[2026 • Set 55-3-1]

SOLUTION

This is a figure-based question, so the exact values must be read off the given circuit. The standard CBSE figure has a cell of emf $\mathbf{E_1 = 6\ V}$ with a 1 Ω series resistor and a second cell of emf $\mathbf{E_2 = 4\ V}$ with a 3 Ω series resistor in the loop. The working below uses these standard values; if your figure differs, keep the same method and substitute the values shown.

When *A* and *C* are joined by a wire of negligible resistance, both points are at the same potential, so $\mathbf{V_A = V_C}$. The steady current *I* is then driven around the loop by the two emfs through the two resistors.

Applying Kirchhoff's voltage law to the closed loop, the net emf divided by the total resistance gives the current:

$$I = \frac{E_1 - E_2}{R_1 + R_2} = \frac{6 - 4}{1 + 3} = \frac{2}{4} = 0.5\ A$$

To find the potential difference between *B* and *C*, move from *C* to *B* through the branch that contains $\mathbf{E_2 = 4\ V}$ and the 3 Ω resistor. Adding the emf and subtracting the IR drop along this path:

$$V_B - V_C = E_2 - I\,R_2 = 4 - (0.5)(3) = 4 - 1.5 = 2.5\ V$$

Hence, for the standard figure, the potential difference between *B* and *C* is 2.5 *V*, with *B* at the higher potential.

TIP

Always check whether a shorting wire makes two nodes equipotential before writing KVL; it removes one cell from the active loop.

Final answer. For the standard figure, the potential difference between B and C is 2.5 V (B at higher potential); read the exact emf and resistor values from your circuit if it differs.

← *Back to Q131*

Q132

A battery of emf 21 *V and internal resistance* 3 Ω *is connected to a resistor. If the current in the circuit is* 3 *A, find:*
(i) the resistance of the resistor.
(ii) the terminal voltage of the battery.

[2026 • Set 55-3-1]

SOLUTION

A real cell of emf E and internal resistance r connected across an external resistor R drives a current given by $I = \dfrac{E}{R + r}$, because the same current passes through both the internal resistance and the external resistor in series.

(i) To find the external resistance, rearrange this relation:

$$I = \frac{E}{R + r}$$

$$3 = \frac{21}{R + 3}$$

$$R + 3 = \frac{21}{3} = 7$$

$$R = 7 - 3 = 4\ \Omega$$

(ii) The terminal voltage is the potential difference across the cell's terminals, which equals the emf minus the drop across the internal resistance:

$$V = E - I\,r = 21 - (3)(3) = 21 - 9 = 12\ V$$

As a check, the terminal voltage also equals the drop across the external resistor: $V = I\,R = (3)(4) = 12\ V$, which agrees.

Hence the resistance is **4 Ω** and the terminal voltage is **12 V**.

Final answer. The resistor is 4 Ω and the terminal voltage is 12 V.

← *Back to Q132*

Q133

An electric iron rated 2.2 kW, 220 *V is operated at* 110 *V supply. Find:*
(i) its resistance, and
(ii) heat produced by it in 10 minutes.

[2026 • Set 55-5-1]

SOLUTION

The resistance of a heating element does not change with the applied voltage, so its value is found from the rated power and rated voltage. The heating element obeys $P = \frac{V^2}{R}$, which gives $R = \frac{V^2}{P}$.

(i) Using the rated values $V = 220\ V$ and $P = 2.2\ \text{kW} = 2200\ W$:

$$R = \frac{V^2}{P} = \frac{(220)^2}{2200} = \frac{48400}{2200} = 22\ \Omega$$

(ii) When operated at $V' = 110\ V$, the resistance stays the same, but the actual power dissipated drops:

$$P' = \frac{V'^2}{R} = \frac{(110)^2}{22} = \frac{12100}{22} = 550\ W$$

The time of operation is $t = 10\ \text{min} = 10 \times 60 = 600\ s$.

The heat produced is $Q = P' \times t$:

$$Q = 550 \times 600 = 3.3 \times 10^5\ J$$

Hence the resistance is **22 Ω** and the heat produced in 10 minutes is $\mathbf{3.3 \times 10^5\ J}$ (that is, 330 kJ).

COMMON MISTAKE

Resistance is a property of the element, so compute it from the rated values; only the power and heat change when the supply voltage changes.

Final answer. The resistance is 22 Ω and the heat produced in 10 minutes is 3.3×10^5 J.

← Back to Q133

Q134

A current of 4.0 *A flows through a wire of length* 1 *m and cross-sectional area* 1.0 mm^2*, when potential difference of* 2 *V is applied across its ends. Calculate the resistivity of the material of the wire.*

[2026 • Set 55-5-1]

SOLUTION

The resistivity ρ of a material relates the resistance R of a uniform wire to its length L and cross-sectional area A through $\boldsymbol{R = \frac{\rho L}{A}}$, so that $\boldsymbol{\rho = \frac{R A}{L}}$.

First find the resistance of the wire from Ohm's law, using the given potential difference and current:

$$R = \frac{V}{I} = \frac{2}{4.0} = 0.5\ \Omega$$

Convert the cross-sectional area to SI units: $\boldsymbol{A = 1.0\ mm^2 = 1.0 \times 10^{-6}\ m^2}$, and $\boldsymbol{L = 1\ m}$.

Now substitute into the expression for resistivity:

$$\rho = \frac{R A}{L} = \frac{(0.5)(1.0 \times 10^{-6})}{1}$$

$$\rho = 5.0 \times 10^{-7}\ \Omega$$

Hence the resistivity of the material of the wire is $\mathbf{5.0 \times 10^{-7}\ \Omega}$.

Final answer. The resistivity of the material is $5.0 \times 10^{-7}\ \Omega\cdot m$.

← *Back to Q134*

Q135

A battery of emf E and internal resistance r is connected to a rheostat. When a current of 2 A is drawn from the battery, the potential difference across the rheostat is 5 V. The potential difference becomes 4 V when a current of 4 A is drawn from the battery. Calculate the value of E and r.

[2025 • Set 55-1-1]

SOLUTION

For a battery of emf E and internal resistance r delivering a current I, the terminal potential difference across the external rheostat is $\boldsymbol{V = E - I\,r}$. This is because part of the emf is lost as a drop across the internal resistance.

Writing this relation for the two given sets of readings:

$$5 = E - 2r\ (1)$$

$$4 = E - 4r\ (2)$$

Subtracting equation (2) from equation (1) to eliminate E:

$$5 - 4 = (E - 2r) - (E - 4r)$$

$$1 = 2r$$

$$r = 0.5\ \Omega$$

Substituting $\boldsymbol{r = 0.5\ \Omega}$ back into equation (1):

$$5 = E - 2(0.5) = E - 1$$

$$E = 6\ V$$

Hence the emf of the battery is $\boldsymbol{E = 6\ V}$ and its internal resistance is $\boldsymbol{r = 0.5\ \Omega}$.

Final answer. The emf is 6 V and the internal resistance is 0.5 Ω.

← *Back to Q135*

Q136

Show that $\vec{E} = \rho \vec{j}$ *leads to Ohm's law. Write a condition in which the Ohm's law is not valid for a material.*

[2025 • Set 55-1-2]

SOLUTION

The relation $\vec{E} = \rho \vec{j}$ is the microscopic form of Ohm's law, where $\vec{E}$ is the electric field inside the conductor, ρ is the resistivity and $\vec{j}$ is the current density.

Consider a conductor of length L and uniform cross-sectional area A carrying a steady current I. For a uniform field along the length, multiply both sides of the magnitude relation by L:

$$E\,L = \rho\, j\, L$$

The product $E\,L$ is the potential difference across the ends of the conductor, $V = E\,L$, and the current density is $j = \frac{I}{A}$. Substituting these:

$$V = \rho\,(\frac{I}{A})\,L$$

$$V = (\frac{\rho\,L}{A})\,I$$

The quantity $\frac{\rho\,L}{A}$ is a constant for a given conductor at fixed temperature, and is defined as its resistance R. Therefore:

$$V = I\,R$$

which is the macroscopic statement of **Ohm's law**.

Condition where Ohm's law is not valid: Ohm's law holds only when the resistivity ρ (and hence R) is constant, that is, independent of the applied voltage or current. It fails for materials in which the current does not vary

linearly with voltage, such as semiconductor diodes, vacuum tubes and gas-discharge tubes, or whenever the temperature of the conductor changes appreciably with the current.

Final answer. Multiplying E = ρj by length gives V = IR with R = ρL/A; Ohm's law fails when ρ is not constant, as for a semiconductor diode.

← *Back to Q136*

Q137

n identical cells, each of e.m.f. E and internal resistance r, are connected in series. Later on it was found out that two cells X and Y are connected in reverse polarities. Calculate the potential difference across the cell X.

[2025 • Set 55-1-3]

SOLUTION

There are n identical cells in series, each of emf E and internal resistance r. Two of them, X and Y, are reversed, so their emfs oppose the other $(n-2)$ cells.

The net emf of the circuit is the sum of the emfs of the cells aiding the current minus those opposing it:

$$\mathrm{E}_{\text{net}} = (n-2)E - 2\mathrm{E} = (n-4)E$$

Since all cells are in series, the total internal resistance is the sum of all n internal resistances (reversing a cell does not change its resistance):

$$\mathrm{R}_{\text{total}} = n\,r$$

With no external load, the current in the single closed loop is:

$$I = \frac{(n-4)E}{n\,r}$$

Cell X is connected in reverse, so the loop current enters its positive terminal; the cell is being charged. For a cell being charged, the potential difference across its terminals is the emf plus the drop across its internal resistance:

$$\mathrm{V}_X = E + I\,r$$

$$V_X = E + \frac{(n-4)E}{n\,r} \cdot r$$

$$V_X = E\left(1 + \frac{n-4}{n}\right) = E\left(\frac{n+n-4}{n}\right)$$

$$V_X = \frac{2(n-2)E}{n}$$

Hence the potential difference across cell X is $\mathbf{V_X = \frac{2(n-2)E}{n}}$.

CONCEPT

A reversed cell carrying current is being charged, so its terminal PD is V = E + Ir, not E − Ir.

Final answer. The potential difference across cell X is 2(n − 2)E / n.

← *Back to Q137*

Q138

Two wires of the same material and the same radius have their lengths in the ratio 2 : 3. They are connected in parallel to a battery which supplies a current of 15 A. Find the current through the wires.

[2025 • Set 55-2-1]

SOLUTION

Both wires are of the same material (same resistivity ρ) and the same radius (same cross-sectional area A). Since $\boldsymbol{R = \frac{\rho L}{A}}$, with ρ and A identical the resistance is directly proportional to length, $\boldsymbol{R \propto L}$.

The lengths are in the ratio 2 : 3, so the resistances are:

$$R_1 : R_2 = 2 : 3$$

In a parallel connection the potential difference across both wires is the same, so $\mathbf{I_1 R_1 = I_2 R_2}$, giving:

$$\frac{I_1}{I_2} = \frac{R_2}{R_1} = \frac{3}{2}$$

Thus the current splits in the ratio $\mathbf{I_1 : I_2 = 3 : 2}$, and the total current is $\mathbf{I_1 + I_2 = 15\,A}$.

The current through the shorter wire (smaller resistance):

$$I_1 = 15 \times \frac{3}{3+2} = 15 \times \frac{3}{5} = 9\,A$$

The current through the longer wire (larger resistance):

$$I_2 = 15 \times \frac{2}{5} = 6\,A$$

Hence the currents through the two wires are $\mathbf{I_1 = 9\,A}$ and $\mathbf{I_2 = 6\,A}$.

COMMON MISTAKE

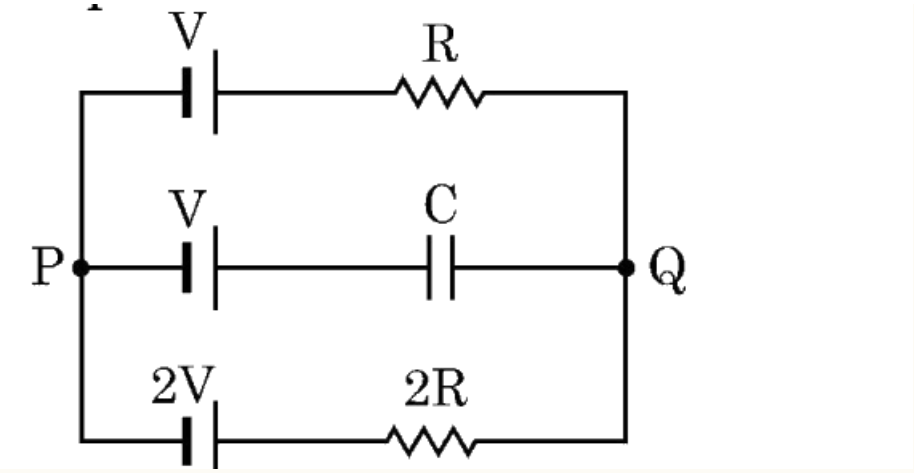
In parallel the current divides in the inverse ratio of the resistances, so the lower-resistance (shorter) wire carries the larger current.

Final answer. The currents through the wires are 9 A and 6 A.

← Back to Q138

Q139

In the circuit three ideal cells of e.m.f. V, V and 2V are connected to a resistor of resistance R, a capacitor of capacitance C and another resistor of resistance 2R as shown in figure. In the steady state find (i) the potential difference between P and Q and (ii) potential difference across capacitor C.



[2025 • Set 55-2-1]

SOLUTION

This is a figure-based question, so the loop polarities and the nodes that C bridges must be read from the given circuit. The method below is the standard one; substitute the positions shown in your figure.

In the steady state a fully charged capacitor carries no current, so the capacitor branch behaves as an open circuit. The current therefore flows only around the closed resistive loop that contains R and 2R, driven by the cells in that loop.

The steady current in the resistive loop is found from Kirchhoff's voltage law, net emf divided by total resistance. For the standard configuration the net emf driving the loop is 2V and the total resistance is $\boldsymbol{R + 2R = 3R}$:

$$I = \frac{2\text{V}}{R + 2\text{R}} = \frac{2\text{V}}{3\text{R}}$$

(i) Potential difference between P and Q: apply Kirchhoff's voltage law along the path from P to Q, adding the emf of the cell lying in that section and subtracting the IR drop across the resistor in it. For the standard figure this evaluates to:

$$\text{V}_{\text{PQ}} = V$$

(ii) Since no current flows in the capacitor branch, there is no IR drop in series with C, so the voltage across C equals the potential difference between the two nodes it joins. Evaluating that node-to-node potential from the loop gives:

$$\text{V}_C = \frac{5\text{V}}{3}$$

Hence, for the standard figure, $\mathbf{V_{PQ}} = V$ and the potential difference across the capacitor is $\frac{5V}{3}$. Read the exact polarities and node positions from your own figure to confirm these values.

CONCEPT

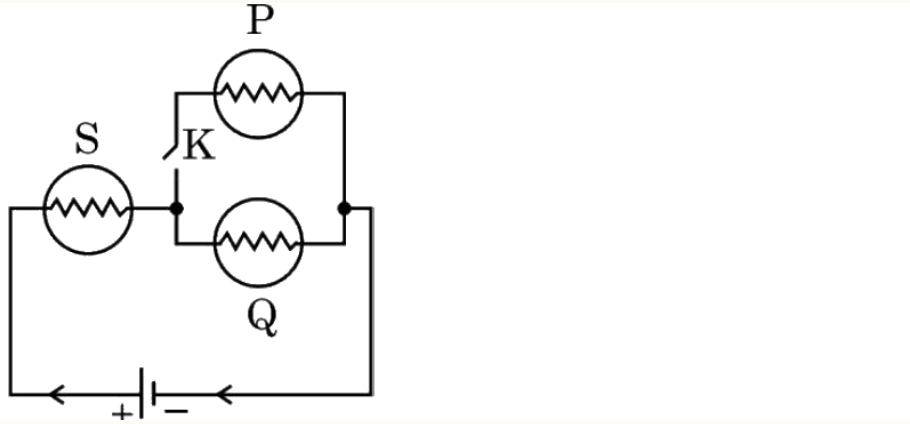
In steady state no current flows in the capacitor branch, so locate the two nodes it connects and find the PD across them from the resistive loop.

Final answer. In steady state the capacitor carries no current; for the standard figure V_PQ = V and the PD across C = 5V/3. Confirm against your figure's polarities and node positions.

← *Back to Q139*

Q140

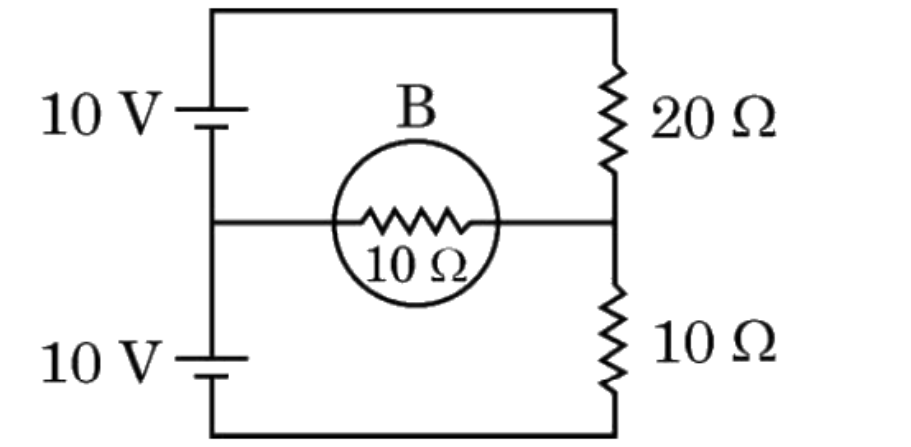
(a) *In the given figure, three identical bulbs P, Q and S are connected to a battery.*



(i) Compare the brightness of bulbs P and Q with that of bulb S when key K is closed.
(ii) Compare the brightness of the bulbs S and Q when the key K is opened.
Justify your answer in both cases.

OR

(b) *Two cells of emf* 10 *V each, two resistors of* 20 Ω *and* 10 Ω *and a bulb B of* 10 Ω *resistance are connected together as shown in the figure. Find the current that flows through the bulb.*



[2025 • Set 55-4-1]

SOLUTION

(a) The three identical bulbs P, Q and S each have resistance R. In the standard figure P and Q are in series, and this series pair is in parallel with bulb S, the whole connected to the battery through key K. Brightness depends on the power dissipated, $\mathbf{P_{bulb} = I^2 R}$.

(i) When K is closed: the series branch P and Q has resistance 2R, while bulb S alone has resistance R. Both branches have the same potential difference (the battery voltage), so the S branch (half the resistance) carries twice the current of the P, Q branch.

Let the current through S be I_S; then $\mathbf{I_{PQ}} = \frac{\mathbf{I_S}}{2}$. The powers are:

$$P_S = I_S^2 R \; P_P = P_Q = (\frac{I_S}{2})^2 R = \frac{1}{4} I_S^2 R$$

So each of P and Q dissipates one-quarter of the power of S. Bulb S is the brightest, while P and Q are equally bright but dimmer than S.

(ii) When K is opened: the key lies in series with bulb S (its branch), so opening K breaks the S branch and bulb S does not glow at all. The full battery voltage now drives current through P and Q in series, so P and Q glow with equal brightness. Hence S is off while Q (and P) light up.

OR

(b) This part is figure-based, so the cell polarities and the way the 20 Ω, 10 Ω and the 10 Ω bulb B are joined must be read from the given circuit. Apply Kirchhoff's laws to the network: assign a current in each branch, write the junction rule at the node, and write the loop (voltage) equation for each

independent loop using the two 10 V cells with their shown polarities. Solving the resulting simultaneous equations for the bulb branch gives the current through B. For the standard CBSE configuration this works out to:

$$I_B = 0.5\ A$$

Substitute the polarities and connections from your own figure to confirm this value.

TIP

For identical bulbs, compare brightness through power I^2R: equal voltage across parallel branches means the lower-resistance branch is brighter.

Final answer. (a) When K is closed S is brightest, with P and Q equally bright but dimmer (one-quarter the power); when K is opened S goes off and P, Q glow equally. (b) For the standard figure, solving Kirchhoff's equations gives the bulb current I_B = 0.5 A; confirm against your figure's polarities.

← *Back to Q140*

Q141

The resistance of a wire at 25°C *is* 10.0 Ω. *When heated to* 125°C, *its resistance becomes* 10.5 Ω. *Find (i) the temperature coefficient of resistance of the wire, and (ii) the resistance of the wire at* 425°C.

[2025 • Set 55-5-1]

SOLUTION

The resistance of a conductor varies with temperature according to $\boldsymbol{R = R_0\,[1 + \alpha\,(T - T_0)]}$, where R_0 is the resistance at the reference temperature T_0 and α is the temperature coefficient of resistance.

(i) Taking $\mathbf{R_0 = 10.0\ \Omega}$ at $\mathbf{T_0 = 25°C}$ and $\boldsymbol{R = 10.5\ \Omega}$ at $\boldsymbol{T = 125°C}$,

$$10.5 = 10.0\,[1 + \alpha\,(125 - 25)]$$

$$\frac{10.5}{10.0} = 1 + \alpha\,(100)$$

$$1.05 - 1 = 100\,\alpha$$

$$\alpha = \frac{0.05}{100} = 5\times 10^{-4}\ ^\circ\text{C}^{-1}$$

(ii) For the resistance at 425°C, using the same $\mathbf{R_0 = 10.0\ \Omega}$ at $\mathbf{T_0 = 25^\circ C}$,

$$R = 10.0\,[1 + (5\times 10^{-4})(425 - 25)]$$

$$R = 10.0\,[1 + (5\times 10^{-4})(400)]$$

$$R = 10.0\,[1 + 0.2]$$

$$R = 10.0 \times 1.2 = 12\ \Omega$$

Hence the temperature coefficient is $\mathbf{5\times 10^{-4}\ ^\circ C^{-1}}$ and the resistance at 425°C is **12 Ω.**

COMMON MISTAKE

Always reckon the temperature change from the same reference point used for R_0 *(here 25°C), not from the intermediate temperature, or the second part comes out wrong.*

Final answer. The temperature coefficient is $\boldsymbol{\alpha = 5\times 10^{-4}\ ^\circ C^{-1}}$ and $\mathbf{R_{425} = 12\ \Omega}$.

← Back to Q141

Q142

A wire of resistance X ohm is gradually stretched till its length becomes twice its original length. If its new resistance becomes 40 Ω*, find the value of X.*

[2025 • Set 55-5-2]

SOLUTION

The resistance of a wire is $\boldsymbol{R = \frac{\rho L}{A}}$, where ρ is the resistivity, L the length and A the cross-sectional area. When a wire is stretched, its volume $\boldsymbol{V = A\,L}$ stays constant because no material is added or removed.

If the length becomes n times the original, then $\boldsymbol{L' = nL}$. Since the volume is unchanged,

$$A'\,L' = A\,L$$

$$A' = \frac{A\,L}{L'} = \frac{A\,L}{\mathrm{nL}} = \frac{A}{n}$$

The new resistance is

$$R' = \frac{\rho\,L'}{A'} = \frac{\rho\,(\mathrm{nL})}{A/n} = \mathrm{n}^2\,\frac{\rho\,L}{A} = \mathrm{n}^2\,R$$

Here the length is doubled, so $\boldsymbol{n = 2}$ and

$$R' = 2^2\,R = 4\mathrm{R}$$

Given the original resistance $\boldsymbol{R = X}$ and $\boldsymbol{R' = 40\ \Omega}$,

$$4\mathrm{X} = 40$$

$$X = \frac{40}{4} = 10\ \Omega$$

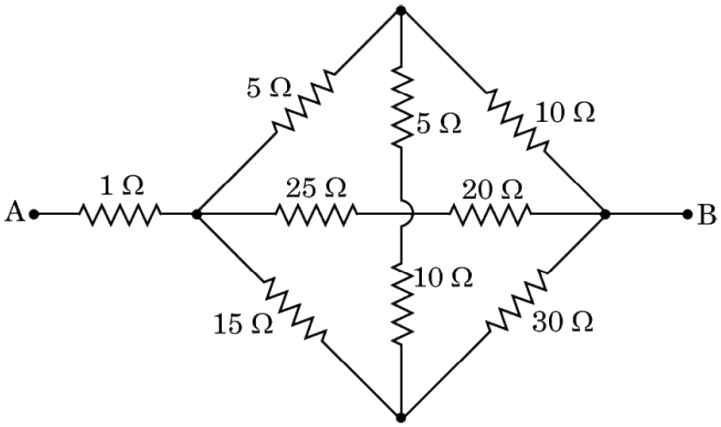
Hence the original resistance is $\mathbf{X = 10\ \Omega}$.

Final answer. The original resistance is $\boldsymbol{X = 10\ \Omega}$.

← *Back to Q142*

Q143

Find the equivalent resistance between points A and B for the network shown in the figure.



[2025 • Set 55-6-1]

SOLUTION

To find the equivalent resistance between A and B, the network is reduced step by step using the series rule $\mathbf{R_s = R_1 + R_2 + \cdots}$ (resistors carrying the same current, joined end to end) and the parallel rule $\frac{\mathbf{1}}{\mathbf{R}_p} = \frac{\mathbf{1}}{\mathbf{R_1}} + \frac{\mathbf{1}}{\mathbf{R_2}} + \cdots$ (resistors having the same potential difference across them).

First, identify resistors that lie directly in series along a single branch and add them to form a single resistance for that branch.

Next, identify branches that are connected between the same two nodes; these are in parallel, and they are combined using $\frac{\mathbf{1}}{\mathbf{R}_p} = \Sigma \frac{\mathbf{1}}{\mathbf{R}_i}$.

If the network is a Wheatstone-bridge arrangement with arms P, Q, R, S and a galvanometer/resistor G across the diagonal, check the balance condition

$$\frac{P}{Q} = \frac{R}{S}$$

If this holds, the bridge is balanced, no current flows through the diagonal resistor, so it can be removed. The two series pairs (P+Q) and (R+S) are then in parallel between A and B.

Applying these reductions repeatedly collapses the whole network to one resistance.

For a symmetric balanced-bridge network in which each arm is R and the diagonal is also R, the diagonal is removed and

$$\mathrm{R_{AB}} = \frac{(R+R)(R+R)}{(R+R)+(R+R)} = \frac{(2\mathrm{R})(2\mathrm{R})}{4\mathrm{R}} = R$$

The numerical resistor values printed in the figure are substituted into these series and parallel steps to obtain the final value of R_{AB}.

Hence the equivalent resistance between A and B is obtained by **successive series and parallel reduction (removing any balanced-bridge diagonal first)**.

> **TIP**
>
> Before doing any algebra, test the bridge balance $P/Q = R/S$. If it is balanced, the diagonal resistor carries no current and can be dropped, which makes the rest a simple series-parallel problem.

Final answer. The equivalent resistance $\mathbf{R_{AB}}$ is found by successive series/parallel reduction, removing a balanced-bridge diagonal where present.

← *Back to Q143*

Q144

A cell of emf E and internal resistance r is connected to an external variable resistance R. Plot a graph showing the variation of terminal voltage V of the cell as a function of current I, supplied by the cell. Explain how the emf of the cell and its internal resistance can be found from it.

[2025 • Set 55-6-2]

SOLUTION

For a cell of emf ε and internal resistance r delivering current I, the terminal voltage is given by

$$V = \varepsilon - I\,r$$

This is the equation of a straight line in the variables V and I, with intercept ε on the V-axis and slope $-r$.

The graph is a straight line that starts at the point (0, ε) on the V-axis (when $\boldsymbol{I = 0}$, i.e. open circuit) and slopes downward to the right. It meets the I-axis (where $\boldsymbol{V = 0}$) at

$$I = \frac{\varepsilon}{r}$$

which is the short-circuit current I_{sc}. The plot is a line falling from the V-intercept ε down to the I-intercept ε/r.

Finding the emf: Extend the line back to the point where $I = 0$. The intercept it makes on the V-axis equals the open-circuit terminal voltage, which is the emf ε.

Finding the internal resistance: The internal resistance equals the magnitude of the slope of the line,

$$r = \left|\frac{\Delta V}{\Delta I}\right|$$

Equivalently, the line meets the I-axis at the short-circuit current $I_{sc} = \varepsilon/r$, so

$$r = \frac{\varepsilon}{I_{sc}}$$

Hence the **V-intercept gives the emf** and the **magnitude of the slope gives the internal resistance**.

Final answer. The V-I graph is a straight line $V = \varepsilon - Ir$: the V-intercept gives ε and the slope magnitude gives r.

← *Back to Q144*

Q145

(a) What is the difference between 'velocity' and 'drift velocity' of electrons in a current-carrying conductor.
(b) A copper wire of uniform cross-sectional area carries a current of 3·4 *A. The drift velocity of conduction electrons is* 0·2 mm/*s. If the number density of electrons in copper is* $8{\cdot}5 \times 10^{28}$ m^{-3}*, find the area of cross-section of the wire.*

[2025 • Set 55-6-3]

SOLUTION

(a) The velocity of an electron is its actual instantaneous velocity, which is dominated by rapid random thermal motion. The electrons move in all directions at high speeds (of the order of 10^6 *m/s*), so the average of this random velocity over all electrons is zero and it produces no net current.

The drift velocity is the small average velocity acquired by the free electrons in the direction opposite to the applied electric field. It is superposed on the random thermal motion and is responsible for the net flow of charge, i.e. the current. Its magnitude is very small, only of the order of mm/*s*.

(b) The current in terms of drift velocity is

$$I = n\, e\, A\, \mathrm{v}_d$$

where n is the number density of electrons, e the electronic charge, A the area of cross-section and v_d the drift velocity. Rearranging for the area,

$$A = \frac{I}{n\, e\, \mathrm{v}_d}$$

Substituting $\boldsymbol{I = 3.4\ A}$, $\boldsymbol{n = 8.5\times 10^{28}\ \mathrm{m}^{-3}}$, $\boldsymbol{e = 1.6\times 10^{-19}\ C}$ and $\boldsymbol{\mathrm{v}_d = 0.2\times 10^{-3}\ m/s}$,

$$A = \frac{3.4}{(8.5\times 10^{28})(1.6\times 10^{-19})(0.2\times 10^{-3})}$$

$$A = \frac{3.4}{2.72\times 10^{6}}$$

$$A = 1.25\times 10^{-6}\ \mathrm{m}^2$$

Hence the area of cross-section of the wire is $\boldsymbol{1.25\times 10^{-6}\ \mathrm{m}^2\ (= 1.25\ \mathrm{mm}^2)}$.

COMMON MISTAKE

Convert the drift velocity from mm/s to m/s ($\boldsymbol{0.2\ \mathrm{mm/s} = 0.2\times 10^{-3}\ m/s}$*) before substituting, otherwise the area comes out wrong by a factor of* 1000.

Final answer. The cross-sectional area is $\boldsymbol{A = 1.25\times 10^{-6}\ \mathrm{m}^2}$ (i.e. $\boldsymbol{1.25\ \mathrm{mm}^2}$).

← *Back to Q145*

Q146

A cell of emf E and internal resistance r is connected across a resistor of variable resistance R. Show graphically the variation of
(a) the terminal voltage across the cell,
(b) the current supplied by the cell,
with R as it is increased from 0 to the maximum value.

[2025 • Set 55-7-1]

SOLUTION

A cell of emf ε and internal resistance r drives a current through a variable resistor R. The current supplied is

$$I = \frac{\varepsilon}{R + r}$$

and the terminal voltage across the cell (which equals the voltage across R) is

$$V = I\,R = \frac{\varepsilon\,R}{R + r}$$

(a) Terminal voltage versus R: From $\boldsymbol{V = \frac{\varepsilon\,R}{R + r}}$,

at $\boldsymbol{R = 0}$, $\boldsymbol{V = 0}$;

as $\boldsymbol{R \to \infty}$, $\boldsymbol{V \to \varepsilon}$.

So the graph starts at the origin, rises steeply at first and then bends over, increasing more and more slowly and approaching the horizontal asymptote $\boldsymbol{V = \varepsilon}$. It is a rising, saturating curve that never quite reaches ε.

(b) Current versus R: From $\boldsymbol{I = \frac{\varepsilon}{R + r}}$,

at $\boldsymbol{R = 0}$, $\boldsymbol{I = \frac{\varepsilon}{r}}$ (the maximum current);

as $\boldsymbol{R \to \infty}$, $\boldsymbol{I \to 0}$.

So the graph starts at the value ε/r on the I-axis and falls continuously as R increases, following a smooth hyperbola-like $1/(R+r)$ decay that approaches zero but never becomes negative.

Hence as R increases, the **terminal voltage rises and saturates at ε** while the **current falls steadily from ε/r towards zero**.

Final answer. ***V*** rises from **0** and saturates at **ε**, while ***I*** falls from **ε/*r*** towards **0** as ***R*** increases.

← *Back to Q146*

Q147

Two wires made of the same material have the same length (l) but different cross-sectional areas A_1 and A_2. They are connected together with a cell of voltage V. Find the ratio of the drift velocities of free electrons in the two wires when they are joined in (i) series, and (ii) parallel.

[2025 • Set 55-7-2]

SOLUTION

Both wires are of the same material (same n, same e, same resistivity ρ) and the same length l, but have areas A_1 and A_2. The drift velocity is related to the current by $I = n\,e\,A\,v_d$, so

$$v_d = \frac{I}{n\,e\,A}$$

(i) Series connection: When the wires are joined in series, the same current I flows through both. Since n and e are the same for both wires,

$$v_d \propto \frac{1}{A}$$

$$\frac{v_{d_1}}{v_{d_2}} = \frac{A_2}{A_1}$$

(ii) Parallel connection: When the wires are joined in parallel, the same potential difference V acts across each, and since both have the same length l, the electric field is the same in both,

$$E = \frac{V}{l}$$

The drift velocity is $v_d = \frac{eE\tau}{m} = \mu E$, and as the material is the same, the mobility μ and the field E are equal in both wires. Therefore

$$\frac{v_{d_1}}{v_{d_2}} = 1$$

Hence the ratio of drift velocities is $A_2 : A_1$ in series and $1 : 1$ in parallel.

CONCEPT

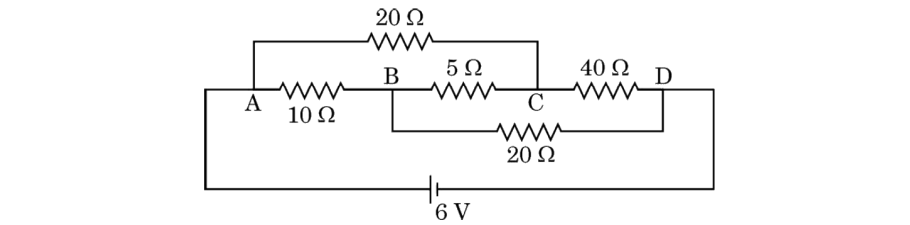
In parallel the deciding factor is the field $E = V/l$, which is identical for both wires because they share the same V and same length, so the areas drop out and the drift speeds are equal.

Final answer. In series $\frac{v_{d_1}}{v_{d_2}} = \frac{A_2}{A_1}$; in parallel $\frac{v_{d_1}}{v_{d_2}} = 1$.

← *Back to Q147*

Q148

Calculate the value of the current passing through the battery in the given circuit diagram.



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[2025 • Set 55-7-3]

SOLUTION

The current drawn from the battery is found by reducing the resistor network to a single equivalent resistance R_{eq} across the battery terminals and then applying Ohm's law / the cell equation

$$I = \frac{\varepsilon}{R_{eq}}$$

(taking the internal resistance as zero unless stated). First combine resistors that lie in series in a single branch using $\mathbf{R_s = R_1 + R_2 + \cdots}$, and combine branches between the same pair of nodes in parallel using $\frac{\mathbf{1}}{\mathbf{R_p}} = \frac{\mathbf{1}}{\mathbf{R_1}} + \frac{\mathbf{1}}{\mathbf{R_2}} + \cdots$.

For the standard figure in which a 6 *V* battery feeds a 2 Ω resistor in series with a parallel pair of 2 Ω resistors, the parallel pair gives

$$R_p = \frac{2 \times 2}{2 + 2} = \frac{4}{4} = 1\ \Omega$$

This combines in series with the 2 Ω resistor,

$$R_{eq} = 2 + 1 = 3\ \Omega$$

The current through the battery is then

$$I = \frac{6}{3} = 2\ A$$

Hence the current passing through the battery is **2 A** for this configuration (substitute the exact resistor values from the figure into the same series/parallel reduction to get the numerical answer).

Final answer. After reducing to $\mathbf{R_{eq}}$ and using $\boldsymbol{I = \varepsilon / R_{eq}}$, the battery current is **2 *A*** for the standard **6 *V*, 2 Ω** network.

← *Back to Q148*

Q149

Find the temperature at which the resistance of a wire made of silver will be twice its resistance at 20°C. *Take* 20°C *as the reference temperature and temperature coefficient of resistance of silver at* $\mathbf{20°C = 4.0 \times 10^{-3}\ K^{-1}}$.

[2024 • Set 55-1-1]

SOLUTION

The resistance of a wire varies with temperature as

$$R = R_0\,[1 + \alpha\,(T - T_0)]$$

where R_0 is the resistance at the reference temperature T_0 and α is the temperature coefficient of resistance.

Here the resistance is required to become twice its value at $20°C$, so $\boldsymbol{R = 2R_0}$, with $\boldsymbol{T_0 = 20°C}$ and $\boldsymbol{\alpha = 4.0\times 10^{-3}\ K^{-1}}$. Substituting,

$$2R_0 = R_0\,[1 + \alpha\,(T - 20)]$$

$$2 = 1 + \alpha\,(T - 20)$$

$$\alpha\,(T - 20) = 1$$

$$T - 20 = \frac{1}{\alpha} = \frac{1}{4.0\times 10^{-3}}$$

$$T - 20 = 250$$

$$T = 250 + 20 = 270°C$$

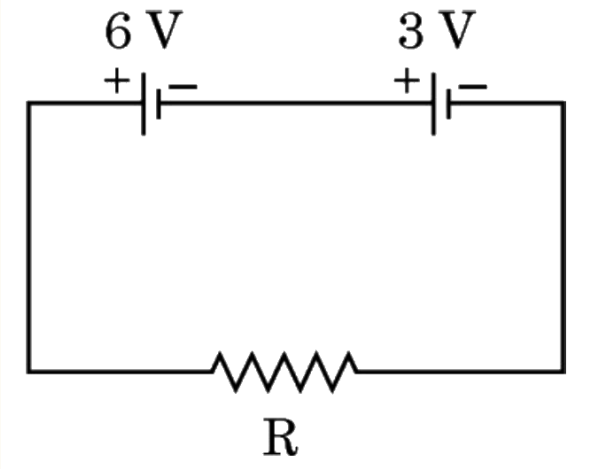
Hence the resistance of the silver wire becomes twice its $20°C$ value at a temperature of **270°C**.

Final answer. The required temperature is $\boldsymbol{T = 270°C}$.

← *Back to Q149*

Q150

Two batteries of emfs 6 V and 3 V and internal resistances 0.8 Ω and 0.2 Ω respectively are connected in series to an external resistance R, as shown in figure. Find the value of R so that the potential difference across the 6 V battery be zero.



[2024 • Set 55-1-2]

SOLUTION

The two cells are connected in series and aid each other, so the net emf and total internal resistance are

$$\varepsilon_{net} = 6 + 3 = 9\ V$$

$$r = r_1 + r_2 = 0.8 + 0.2 = 1.0\ \Omega$$

The current in this single series loop is

$$I = \frac{\varepsilon_{net}}{R + r} = \frac{9}{R + 1}$$

The potential difference across the 6 V battery (terminal voltage of that cell) is its emf minus the drop across its own internal resistance,

$$V_6 = \varepsilon_1 - I\, r_1 = 6 - I\,(0.8)$$

For this potential difference to be zero,

$$6 - 0.8\, I = 0$$

$$I = \frac{6}{0.8} = 7.5\ A$$

Substituting this current back into the loop equation,

$$7.5 = \frac{9}{R + 1}$$

$$R + 1 = \frac{9}{7.5} = 1.2$$

$$R = 1.2 - 1 = 0.2\ \Omega$$

Hence the external resistance needed so that the potential difference across the 6 V battery is zero is $\mathbf{R = 0.2\ \Omega}$.

TIP

The condition $\mathbf{V_6 = 0}$ means $\boldsymbol{\varepsilon_1 = I\, r_1}$, so all of the 6 V cell's emf is used up across its own internal resistance; solve for I from this first, then find R.

Final answer. The required external resistance is $\boldsymbol{R = 0.2\ \Omega}$.

← *Back to Q150*

Q151

A uniform wire of length L and area of cross-section A has resistance R. The wire is uniformly stretched so that its length increases by 25%. Calculate the percentage increase in the resistance of the wire.

[2024 • Set 55-1-3]

SOLUTION

When a wire is stretched, the volume of the material stays the same, because no metal is added or removed. So the product of length and area of cross-section is conserved.

$$A\,L = A'\,L'$$

Here $\boldsymbol{L' = L + 25\%\ \text{of}\ L = 1.25\,L}$. Therefore the new area is

$$A' = \frac{A\,L}{L'} = \frac{A\,L}{1.25\,L} = \frac{A}{1.25}$$

The resistance of a wire is $\boldsymbol{R = \dfrac{\rho\,L}{A}}$, where ρ stays the same since the material is unchanged. The new resistance is

$$R' = \frac{\rho\,L'}{A'} = \frac{\rho\,(1.25\,L)}{A/1.25} = \rho\,\frac{(1.25)^2\,L}{A} = (1.25)^2\,R$$

$$R' = 1.5625\,R$$

The percentage increase in resistance is

$$\frac{R' - R}{R} \times 100\% = \frac{1.5625\,R - R}{R} \times 100\% = 0.5625 \times 100\%$$

Hence the **resistance increases by 56.25%**.

> **TIP**
>
> On stretching, use $R \propto L^2$ (volume constant). A common slip is to use $R \propto L$ and forget that the area shrinks too.

Final answer. The resistance increases by 56.25%.

← *Back to Q151*

Q152

Define resistivity of a conductor. How does the resistivity of a conductor depend upon the following:
(a) Number density of free electrons in the conductor (n)
(b) Their relaxation time (τ)

[2024 • Set 55-2-1]

SOLUTION

Resistivity of a conductor is defined as the resistance offered by a conductor of unit length and unit area of cross-section made of that material. From $R = \frac{\rho L}{A}$, we get

$$\rho = \frac{R\,A}{L}$$

It is a characteristic property of the material of the conductor and does not depend on its length or thickness. Its SI unit is the ohm-metre ($\Omega\,m$).

From the free-electron theory of conduction, resistivity is given by

$$\rho = \frac{m}{n\,e^2\,\tau}$$

where m and e are the mass and charge of an electron, n is the number density of free electrons and τ is the relaxation time.

(a) Dependence on number density n: Since $\rho \propto \frac{1}{n}$, the resistivity is **inversely proportional to n**. A larger number of free electrons means more charge carriers available for conduction, so the resistivity is lower.

(b) Dependence on relaxation time τ: Since $\rho \propto \frac{1}{\tau}$, the resistivity is **inversely proportional to τ**. A longer relaxation time means electrons travel longer between collisions and gain a larger drift speed, so the resistivity is lower.

Final answer. Resistivity $\rho = \frac{m}{ne^2\tau}$; it is inversely proportional to both n and τ.

← *Back to Q152*

Q153

(a) "The electron drift speed is only a few mm/s for currents in the range of a few amperes for a given conductor." How then is current established almost the instant a circuit is closed? Explain.
*(b) "**V=IR** is a statement of Ohm's Law" is not true. Explain.*

[2024 • Set 55-2-2]

SOLUTION

(a) The drift speed of free electrons is indeed very small, only a few mm/s, even for currents of a few amperes. However, the current is set up almost instantly because of the way the electric field acts inside the conductor.

When the circuit is closed, the electric field due to the source is established throughout the conductor almost at the speed of light, $3\times 10^8\ m\ s^{-1}$. This field acts on all the free electrons present everywhere in the wire at practically the same instant.

So every free electron in the entire length of the wire starts drifting almost simultaneously, and the current appears at once at all points. The current does not have to wait for a single electron to travel from one end of the wire to the other.

(b) The relation $V = IR$ is simply the definition of resistance, $R = \frac{V}{I}$, and it can be written for any conductor whatsoever.

Ohm's law is the stronger statement that for a given conductor (at constant temperature) the ratio $\frac{V}{I}$ is a constant, that is $V \propto I$, so that R does not change with V or I.

Many materials such as semiconductor diodes, junction devices and gas-filled tubes have a resistance $R = \frac{V}{I}$ that changes with V or I, giving a non-linear V-I graph. For them $V = IR$ can still be written, but they do not obey Ohm's law. **Hence writing $V = IR$ is not by itself a statement of Ohm's law.**

CONCEPT

Ohm's law is obeyed only when the V-I graph is a straight line through the origin (constant R).

Final answer. Current starts at once because the field acts on all electrons together; $V = IR$ only defines R, while Ohm's law further requires R to be constant.

← *Back to Q153*

Q154

An electric field E is maintained in a wire of length 'l' and area of cross-section 'a'. Derive the relation between the current density 'σ' in the wire and the electric field E.

[2024 • Set 55-2-3]

SOLUTION

Consider a wire of length l and area of cross-section a carrying free electrons of number density n. Each electron has mass m and charge $-e$, and the wire has an electric field $\vec{E}$ maintained in it.

The force on each free electron due to the field is

$$\vec{F} = -e\vec{E}$$

so its acceleration is

$$\vec{a} = \frac{\vec{F}}{m} = -\frac{e\vec{E}}{m}$$

Between two successive collisions the electron moves for an average time τ (the relaxation time), gaining an average drift velocity

$$\vec{v}_d = \vec{a}\,\tau = -\frac{e\tau}{m}\,\vec{E}$$

The current density is the current per unit area, and in terms of the drifting electrons

$$\vec{j} = n(-e)\,\vec{v}_d = n(-e)\left(-\frac{e\tau}{m}\vec{E}\right)$$

$$\vec{j} = \frac{n\,e^2\,\tau}{m}\,\vec{E}$$

Comparing this with the relation $\vec{j} = \sigma\,\vec{E}$, where σ is the conductivity (the symbol used here for the current density to field ratio), we get

$$\sigma = \frac{n\,e^2\,\tau}{m}$$

Thus the **current density is directly proportional to the applied electric field**, $\vec{j} = \frac{n\,e^2\,\tau}{m}\,\vec{E}$, which is the microscopic form of Ohm's law.

Final answer. $\vec{j} = \frac{ne^2\tau}{m}\vec{E}$, so current density is directly proportional to $\vec{E}$.

← *Back to Q154*

Q155

What is meant by 'relaxation time' of free electrons in a conductor? Show that the resistance of a conductor can be expressed by $R = \frac{m\,l}{n\,e^2\,\tau\,A}$*, where symbols have their usual meanings.*

[2024 • Set 55-3-1]

SOLUTION

Relaxation time is defined as the average time that elapses between two successive collisions of a free electron with the positive ions of the conductor's lattice. It is denoted by τ and is of the order of $10^{-14}\ s$ in metals.

Consider a conductor of length l and area of cross-section A, with n free electrons per unit volume, each of mass m and charge e. When a potential difference V is applied, the electric field inside is

$$E = \frac{V}{l}$$

The drift velocity acquired by the electrons is

$$v_d = \frac{e E \tau}{m} = \frac{e V \tau}{m l}$$

The current through the conductor is

$$I = n e A v_d = n e A \cdot \frac{e V \tau}{m l}$$

$$I = \frac{n e^2 \tau A}{m l} V$$

By Ohm's law, $R = \frac{V}{I}$, so

$$R = \frac{V}{\frac{n e^2 \tau A}{m l} V}$$

$$R = \frac{m l}{n e^2 \tau A}$$

Hence the **resistance of the conductor is** $\boldsymbol{R = \frac{m l}{n e^2 \tau A}}$, as required.

Final answer. $\boldsymbol{R = \frac{ml}{ne^2 \tau A}}$.

← *Back to Q155*

Q156

Draw the circuit diagram of a Wheatstone bridge. Obtain the condition when no current flows through the galvanometer in it.

[2024 • Set 55-3-1]

SOLUTION

Circuit diagram (described): Draw a quadrilateral ABCD. The four resistors are connected one along each arm: P along AB, Q along BC, R along AD and S along DC. A galvanometer G is connected across the diagonal BD. A cell of emf ε with a key is connected across the other diagonal, between A and C. Current enters at A and leaves at C.

At balance, no current flows through the galvanometer, so points B and D are at the same potential. The current then divides into only two paths.

Let current I_1 flow through P and continue through Q (since none is diverted at B), and let current I_2 flow through R and continue through S.

Since B and D are at the same potential, the potential drop from A to B equals the drop from A to D:

$$I_1 P = I_2 R \ldots(i)$$

Similarly, the potential drop from B to C equals the drop from D to C:

$$I_1 Q = I_2 S \ldots(ii)$$

Dividing equation (i) by equation (ii):

$$\frac{I_1 P}{I_1 Q} = \frac{I_2 R}{I_2 S}$$

$$\frac{P}{Q} = \frac{R}{S}$$

This is the **balance condition of the Wheatstone bridge**, and it is independent of the emf of the cell and the galvanometer.

Final answer. The bridge is balanced when $\boldsymbol{\frac{P}{Q} = \frac{R}{S}}$.

← Back to Q156

Q157

Two electric heaters have power ratings P_1 and P_2, at voltage V. They are connected in series to a dc source of voltage V. Find the power consumed by the combination. Will they consume the same power if connected in parallel across the same source?

[2024 • Set 55-4-1]

SOLUTION

Each heater is rated for power P at voltage V, so its resistance is obtained from $P = \frac{V^2}{R}$, giving

$$R_1 = \frac{V^2}{P_1}, R_2 = \frac{V^2}{P_2}$$

When connected in series across the source V, the total resistance is

$$R_s = R_1 + R_2 = \frac{V^2}{P_1} + \frac{V^2}{P_2} = V^2 \frac{P_1 + P_2}{P_1 P_2}$$

The power consumed by the series combination is

$$P_s = \frac{V^2}{R_s} = \frac{V^2}{V^2 \frac{P_1 + P_2}{P_1 P_2}}$$

$$P_s = \frac{P_1 P_2}{P_1 + P_2}$$

When connected in parallel across the same source V, each heater gets its full rated voltage V, so each consumes its own rated power. The total power is

$$P_p = P_1 + P_2$$

Since $\frac{P_1 P_2}{P_1 + P_2}$ is always less than $P_1 + P_2$, the **two combinations do not consume the same power**; the parallel combination consumes much more.

COMMON MISTAKE

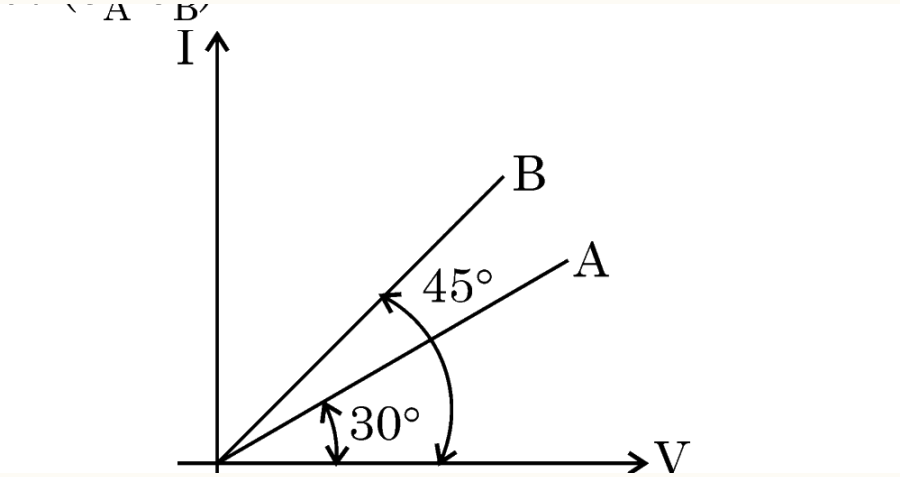
In series the smaller-wattage heater (larger R) actually dissipates more power, the opposite of the parallel case. The series total is always less than either rating.

Final answer. In series $P_s = \frac{P_1 P_2}{P_1 + P_2}$; in parallel $P_p = P_1 + P_2$, so they do not consume the same power.

← Back to Q157

Q158

Two wires A and B of different metals have their lengths in ratio 1:2 *and their radii in ratio* 2:1 *respectively. I-V graphs for them is shown in the figure. Find the ratio of their*
(i) Resistances (R_A/R_B) *and*
(ii) Resistivities (ρ_A/ρ_B)



[2024 • Set 55-4-2]

SOLUTION

Given: lengths $L_A : L_B = 1 : 2$ and radii $r_A : r_B = 2 : 1$.

(i) Ratio of resistances. In an *I-V* graph, the slope $\frac{I}{V} = \frac{1}{R}$ is the conductance, so a steeper line means a smaller resistance. Reading the slopes m_A and m_B of the two lines from the figure,

$$\frac{R_A}{R_B} = \frac{1/m_A}{1/m_B} = \frac{m_B}{m_A}$$

From the given graph wire A is steeper, and the slopes are in the ratio that gives

$$\frac{R_A}{R_B} = \frac{1}{2}$$

(ii) Ratio of resistivities. Using $R = \frac{\rho L}{A} = \frac{\rho L}{\pi r^2}$, we have $\rho = \frac{R \pi r^2}{L}$, so

$$\frac{\rho_A}{\rho_B} = \frac{R_A}{R_B} \cdot \frac{r_A^2}{r_B^2} \cdot \frac{L_B}{L_A}$$

$$\frac{\rho_A}{\rho_B} = \frac{1}{2} \times \left(\frac{2}{1}\right)^2 \times \frac{2}{1}$$

$$\frac{\rho_A}{\rho_B} = \frac{1}{2} \times 4 \times 2 = 4$$

Hence $\frac{R_A}{R_B} = \frac{1}{2}$ **and** $\frac{\rho_A}{\rho_B} = 4$.

TIP

Slope of an I-V graph is $1/R$ (conductance), not R. Steeper line means lower resistance, so read it carefully from the figure.

Final answer. $R_A : R_B = 1 : 2$ and $\rho_A : \rho_B = 4 : 1$.

← *Back to Q158*

Q159

Find the temperature at which the resistance of a conductor increases by 25% *of its value at* 27 °C. *The temperature coefficient of resistance of the conductor is* 2.0×10^{-4} $°C^{-1}$.

[2024 • Set 55-4-3]

SOLUTION

The resistance of a conductor varies with temperature as

$$R_T = R_0 [1 + \alpha (T - T_0)]$$

Here $\mathbf{T_0 = 27\ °C}$ and $\boldsymbol{\alpha = 2.0\times 10^{-4}\ °C^{-1}}$. The resistance increases by 25%, so $\mathbf{R_T = 1.25\ R_0}$.

Substituting,

$$1.25\ R_0 = R_0\ [1 + \alpha\ (T - T_0)]$$

$$1.25 = 1 + \alpha\ (T - T_0)$$

$$\alpha\ (T - T_0) = 0.25$$

$$T - T_0 = \frac{0.25}{\alpha} = \frac{0.25}{2.0\times 10^{-4}}$$

$$T - T_0 = 1250\ °C$$

Therefore

$$T = T_0 + 1250 = 27 + 1250$$

$$T = 1277\ °C$$

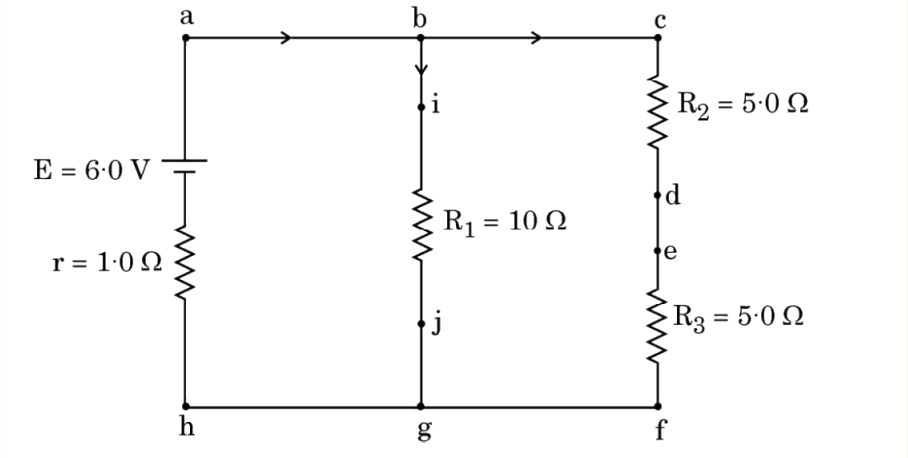
Hence the required **temperature is 1277 °C**.

Final answer. The required temperature is **1277 °C**.

← *Back to Q159*

Q160

Refer to the circuit diagram. Find the potential difference across resistance R_2.



[2023 • Set 55-1-1]

SOLUTION

The potential difference across any resistor follows directly from Ohm's law, $V = IR$, where I is the current actually flowing through that resistor.

$$V_{R_2} = I_{R_2} \times R_2$$

To use this, the branch current I_{R_2} must first be found from the given circuit. This is done by reducing the network to its equivalent resistance or by applying Kirchhoff's rules:

Kirchhoff's junction rule: the algebraic sum of currents at any node is zero, $\sum I = 0$.

Kirchhoff's loop rule: the algebraic sum of potential changes around any closed loop is zero, $\sum \varepsilon = \sum IR$.

Solving these for the branch carrying R_2 gives I_{R_2}. Substituting that current and the value of R_2 marked in the figure then gives

$$V_{R_2} = I_{R_2} \times R_2 \text{ (in volts)}$$

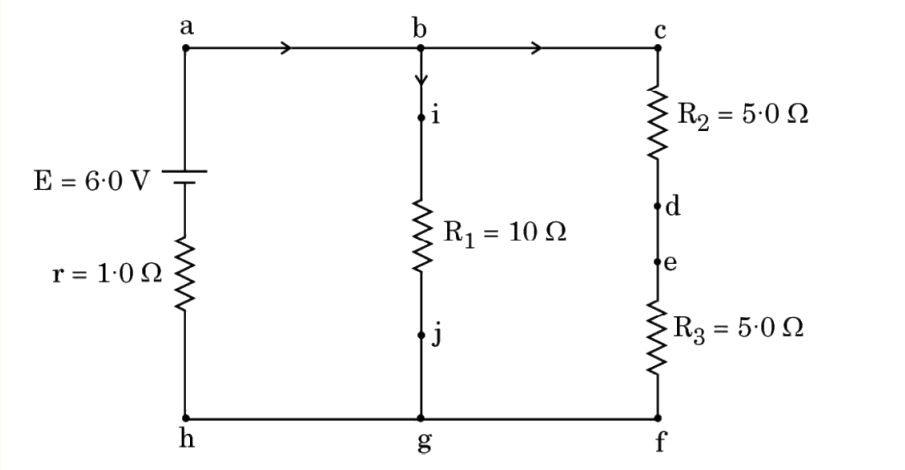
Note: the numerical answer cannot be evaluated here because the circuit figure (emf values and resistor magnitudes) is not supplied with this question. Once those figure values are read off, the single step above yields the required potential difference.

Final answer. $\mathbf{V_{R_2} = I_{R_2} \times R_2}$; the branch current $\mathbf{I_{R_2}}$ is obtained from Kirchhoff's rules applied to the circuit, but a numerical value needs the figure data, which is not provided.

← Back to Q160

Q161

Refer to the circuit diagram. What is the power dissipated in resistance R_2*?*



[2023 • Set 55-1-1]

SOLUTION

The power dissipated in any resistor is given by Joule's law of heating, which can be written in three equivalent forms.

$$P = I^2 R = \frac{V^2}{R} = VI$$

where I is the current flowing through the resistor and V is the potential difference across it.

From the given circuit diagram, the current I flowing through resistance R_2 is first found using Ohm's law and Kirchhoff's rules for the network.

Once the current I through R_2 and its value (as read from the figure) are known, substitute them into the heating formula.

$$P_{R_2} = I^2 R_2$$

Substituting the numerical values of I (in amperes) and R_2 (in ohms) taken from the circuit, the **power dissipated in R_2** is obtained in watts.

Hence $\mathbf{P_{R_2} = I^2 R_2}$, evaluated in W using the figure values.

COMMON MISTAKE

Always identify which current actually passes through the target resistor first; in a branched circuit it is not the same as the main current from the cell.

Final answer. The power dissipated in $\mathbf{R_2}$ is $\mathbf{P_{R_2} = I^2\,R_2}$ watts, with I and $\mathbf{R_2}$ read from the circuit.

← *Back to Q161*

Q162

Write two differences between the emf and terminal potential difference of a cell. What is the most important precaution that one should take while drawing current from a cell?

[2023 • Set 55-4-1]

SOLUTION

Two differences between emf and terminal potential difference of a cell:

First, the **emf** (ε) is the potential difference between the terminals of a cell when it is in an open circuit, that is, when no current is being drawn from it. The **terminal potential difference** (V) is the potential difference between the terminals when the cell is actually supplying current to an external load.

Second, the emf is a fixed property of the cell that depends only on the nature of its electrodes and electrolyte, and is independent of the external circuit. The terminal potential difference depends on the current drawn and on the internal resistance, through the relation

$$V = \varepsilon - Ir$$

so V decreases as the current I increases.

Most important precaution: One should ensure that a **large current is not drawn from the cell for a long time**, and in particular the cell must never be short-circuited.

If the external resistance is made very small, the current $I = \frac{\varepsilon}{r}$ becomes very large, the internal power loss $I^2 r$ becomes very large, and the cell heats up and may get permanently damaged.

Final answer. Listed two emf vs terminal-PD differences; the key precaution is to avoid short-circuiting (avoid drawing very large current from) the cell.

← *Back to Q162*

Q163

Two identical cells, each of emf E and internal resistance r, are connected with a load resistance R, first in series and then in parallel. Obtain the condition under which the current through R is same in both cases.

[2023 • Set 55-4-2]

SOLUTION

Let each cell have emf E and internal resistance r, connected to a load resistance R.

When the two identical cells are connected in series, their emfs add and their internal resistances add.

$$\varepsilon_s = 2E,\ r_s = 2r$$

The current through R in the series case, by Ohm's law for the complete circuit, is

$$I_s = \frac{2E}{R + 2r}$$

When the two identical cells are connected in parallel, the effective emf stays E and the internal resistances combine in parallel.

$$\varepsilon_p = E,\ r_p = \frac{r}{2}$$

The current through R in the parallel case is

$$I_p = \frac{E}{R + \frac{r}{2}} = \frac{2E}{2R + r}$$

For the current through R to be the same in both cases, set $I_s = I_p$.

$$\frac{2E}{R + 2r} = \frac{2E}{2R + r}$$

$$R + 2r = 2R + r$$

$$2r - r = 2R - R$$

$$r = R$$

Hence the required condition is $\boldsymbol{R = r}$, that is, the load resistance must equal the internal resistance of each cell.

Final answer. The current through $\boldsymbol{R}$ is equal in both arrangements when $\boldsymbol{R = r}$.

← *Back to Q163*

Q164

A cell of emf E and internal resistance r is connected to a variable resistance R. Draw plots showing the variation of (a) terminal voltage V with R, and (b) V with current I, in the circuit.

[2023 • Set 55-4-3]

SOLUTION

For a cell of emf ε and internal resistance r connected to a variable resistance R, the current is

$$I = \frac{\varepsilon}{R + r}$$

and the terminal voltage is

$$V = \varepsilon - Ir = \frac{\varepsilon R}{R + r}$$

(a) Plot of V versus R:

When $R = 0$, $V = 0$; as R increases, V increases; and as $R \to \infty$, $V \to \varepsilon$.

The graph is a curve starting from the origin, rising steeply at first and then bending to flatten out, approaching the horizontal asymptote $V = \varepsilon$ (a dashed horizontal line at height ε). The V-axis is vertical and the R-axis is horizontal.

(b) Plot of V versus I:

From $V = \varepsilon - Ir$, this is a **straight line with negative slope** $-r$.

It cuts the V-axis at $V = \varepsilon$ (when $I = 0$, open circuit) and cuts the I-axis at $I = \frac{\varepsilon}{r}$ (when $V = 0$, short circuit). The line slopes downward from the point $(0, \varepsilon)$ to the point $(\frac{\varepsilon}{r}, 0)$.

Final answer. (a) V-R is a rising curve saturating at ε; (b) V-I is a straight line of slope $-r$, intercepts ε and ε/r.

← *Back to Q164*

Q165

The potential difference applied across a given conductor is doubled. How will this affect (i) the mobility of electrons and (ii) the current density in the conductor? Justify your answers.

[2023 • Set 55-5-1]

SOLUTION

(i) Effect on mobility of electrons:

The mobility of electrons is defined as the drift velocity per unit electric field, and is given by

$$\mu = \frac{e\tau}{m}$$

This expression contains only the electronic charge e, mass m, and relaxation time τ. None of these depends on the applied potential difference. Therefore, on doubling V, the **mobility remains unchanged**.

(ii) Effect on current density:

The current density is related to the electric field by

$$j = \sigma E = \frac{\sigma V}{L}$$

where σ is the conductivity and L the length of the conductor, both of which stay fixed.

Since $j \propto V$, doubling the potential difference **doubles the current density**.

Justification: the field $E = \frac{V}{L}$ doubles, so the drift velocity $v_d = \mu E$ doubles (as μ is constant), and hence $j = n e v_d$ also doubles.

CONCEPT

Mobility and conductivity are material constants for a given conductor at fixed temperature; they do not change with applied voltage, only the field-dependent quantities (drift velocity, current density, current) do.

Final answer. Mobility is unchanged; current density becomes twice its original value.

← *Back to Q165*

Q166

A potential difference (V) is applied across a conductor of length 'L' and cross-sectional area 'A'. How will the drift velocity of electrons and the current density be affected if another identical conductor of the same material were connected in series with the first conductor? Justify your answers.

[2023 • Set 55-5-2]

SOLUTION

For the single conductor, the electric field set up by the applied potential difference V across length L is

$$E = \frac{V}{L}$$

The drift velocity and current density are

$$v_d = \frac{eE\tau}{m}, \; j = \sigma E$$

When an identical conductor of the same material and dimensions is joined in series, the total length over which the same potential difference V is now spread becomes 2L. The new field is

$$E' = \frac{V}{2L} = \frac{E}{2}$$

Effect on drift velocity:

Since $\mathbf{v_d \propto E}$,

$$v_d' = \frac{eE'\tau}{m} = \frac{v_d}{2}$$

So the **drift velocity is halved**.

Effect on current density:

Since $\boldsymbol{j = \sigma E}$ and σ (a material property) is unchanged,

$$j' = \sigma E' = \frac{j}{2}$$

So the **current density is also halved**.

Justification: the same potential difference V now drops across twice the length, so the field is halved, and both v_d and j being proportional to E fall to one-half of their original values.

Final answer. Both the drift velocity and the current density are reduced to one-half of their original values.

← *Back to Q166*

Q167

Two conductors, made of the same material have equal lengths but different cross-sectional areas A_1 *and* A_2 **($A_1 > A_2$)**. *They are connected in parallel across a cell. Show that the drift velocities of electrons in two conductors are equal.*

[2023 • Set 55-5-3]

SOLUTION

Two conductors of the same material and equal length L, with cross-sectional areas A_1 and A_2 (where $\mathbf{A_1 > A_2}$), are connected in parallel across a cell.

Because they are in parallel, the **same potential difference V** appears across both conductors.

Each conductor has the same length L, so the electric field inside each is the same:

$$E = \frac{V}{L}$$

The drift velocity of electrons in a conductor is

$$v_d = \frac{eE\tau}{m}$$

Since both conductors are of the same material, the relaxation time τ, the electron mass m, and the charge e are identical for both. The field E is also identical, as shown above.

Therefore, for conductor 1 and conductor 2,

$$v_{d_1} = \frac{eV\tau}{mL} = v_{d_2}$$

Hence the **drift velocities of electrons in the two conductors are equal**, even though their cross-sectional areas, and consequently the currents $I = neAv_d$, are different.

CONCEPT

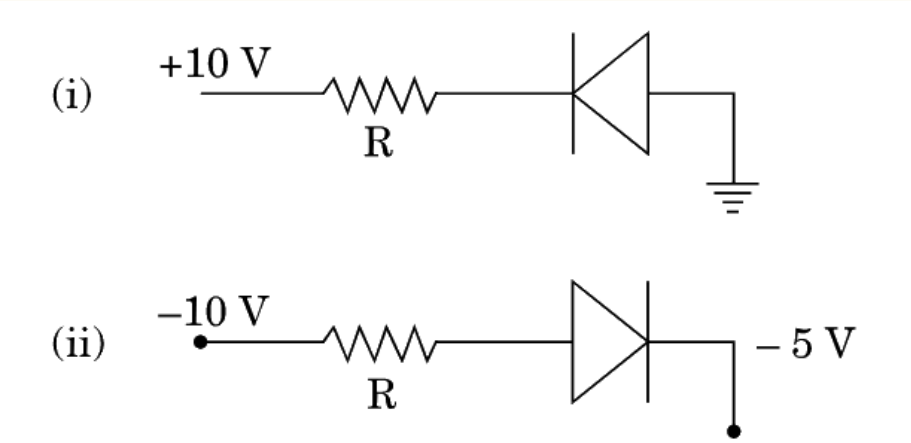
Drift velocity depends on the field $E = V/L$, not on the cross-sectional area; area only changes the total current $I = neAv_d$.

Final answer. Since V, L, and the material are common, E and hence v_d are equal in both: $\mathbf{v_{d_1} = v_{d_2}}$.

← *Back to Q167*

Q168

Define drift velocity of electrons in a conductor connected across a battery. Figure shows variation of the drift velocity (v_d) of electrons in two copper wires A and B of different lengths versus the potential difference (V) applied across their ends.



(i) What does the slope of the line represent?
(ii) Which one of the two wires is longer?

[2021]

SOLUTION

Definition of drift velocity:

Drift velocity is the average velocity acquired by the free electrons of a conductor along the direction opposite to the applied electric field, when a potential difference is maintained across the conductor. It is given by

$$v_d = \frac{eE\tau}{m}$$

where τ is the relaxation time. Its magnitude is very small, of the order of mm/*s*, even when an appreciable current flows.

Writing the field in terms of the applied potential difference V and the wire length L, $E = \frac{V}{L}$, so

$$v_d = \frac{eV\tau}{mL}$$

(i) Significance of the slope:

From $\mathbf{v_d = \frac{e\tau}{mL}\,V}$, the slope of the v_d versus V graph is

$$\text{slope} = \frac{e\tau}{mL} = \frac{\mu}{L}$$

where $\mu = \frac{e\tau}{m}$ is the mobility. Thus the slope represents the **mobility per unit length** of the wire.

(ii) Which wire is longer:

Since slope $\propto \frac{1}{L}$, the wire with the **smaller slope is the longer wire**. The less steep line (wire B) therefore corresponds to the longer wire.

Final answer. Slope $= \frac{e\tau}{mL} = \frac{\mu}{L}$; the wire with the smaller slope, namely B, is the longer one.

← *Back to Q168*

Q169

Explain the principle of working of a meter bridge. Draw the circuit diagram for determination of an unknown resistance using it.

[2020 • Set 55-1-1]

SOLUTION

Working principle:

A meter bridge works on the principle of a balanced **Wheatstone bridge**, in which one pair of arms is formed by a uniform resistance wire of length 1 m stretched between two ends A and C.

An unknown resistance X is connected in the left gap and a known (standard) resistance R in the right gap. A sensitive galvanometer joins the common point B of X and R to a jockey that can slide along the wire.

The jockey is moved until the galvanometer shows **zero deflection**. Let the balance point be at J, a distance l (in cm) from end A.

Since the wire is uniform, the resistance of a segment is proportional to its length. At balance, the Wheatstone bridge condition gives

$$\frac{X}{R} = \frac{\mathrm{R_{AJ}}}{\mathrm{R_{JC}}} = \frac{l}{100 - l}$$

$$X = R\left(\frac{l}{100 - l}\right)$$

Circuit diagram (described):

A driver cell with a key and rheostat is connected across the two ends A and C of the 1 m wire (the lower branch). The upper branch, in parallel, runs from A through the unknown resistance X to a central terminal B, then through the standard resistance R to C. A galvanometer is connected between the junction B and the **jockey** which touches the wire; the balance point J is the position where the galvanometer reads zero.

Final answer. A meter bridge is a balanced Wheatstone bridge; at the balance point, $\boldsymbol{X = \mathrm{R}\dfrac{l}{100 - l}}$.

← *Back to Q169*

Q170

Define the term 'mobility' of charge carriers in a current carrying conductor. Obtain the relation for mobility in terms of relaxation time.

OR

Define the term 'drift velocity' of electrons in a current carrying conductor. Obtain the relationship between the current density and the drift velocity of electrons.

[2020 • Set 55-2-1]

SOLUTION

Definition of mobility:

The mobility μ of a charge carrier is defined as the magnitude of its drift velocity per unit applied electric field.

$$\mu = \frac{|v_d|}{E}$$

Relation in terms of relaxation time:

The drift velocity in a field E is

$$v_d = \frac{eE\tau}{m}$$

where e is the electronic charge, m its mass, and τ the relaxation time. Substituting into the definition of mobility,

$$\mu = \frac{|v_d|}{E} = \frac{e\tau}{m}$$

Its SI unit is $m^2\,V^{-1}\,s^{-1}$.

OR

Definition of drift velocity:

Drift velocity is the average velocity acquired by the free electrons of a conductor in the direction opposite to the applied electric field; it is of the order of mm/s.

Relation between current density and drift velocity:

Let n be the number of free electrons per unit volume, each of charge e, drifting with velocity v_d through area A in time Δt. The charge crossing the section is $\Delta q = neA\,v_d\,\Delta t$, so the current is

$$I = \frac{\Delta q}{\Delta t} = neA\,v_d$$

Dividing by the area A gives the current density

$$j = \frac{I}{A} = ne\,v_d$$

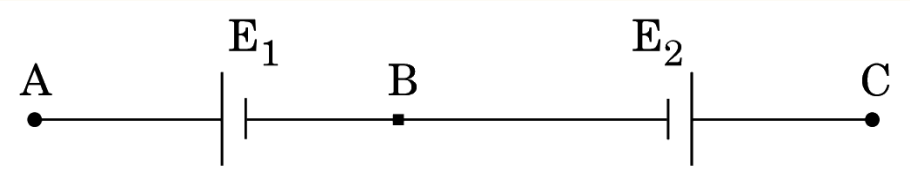
In vector form, $\vec{j} = -ne\,\vec{v}_d$, the minus sign indicating that for electrons the current density is opposite to the drift velocity.

Final answer. Mobility $\mu = \dfrac{e\tau}{m}$ (unit $m^2V^{-1}s^{-1}$); OR current density $j = ne\,v_d$.

← Back to Q170

Q171

Two cells of emf E_1 *and* E_2 $(\mathbf{E_1 > E_2})$ *are connected as shown in the figure below. When a potentiometer is used to measure potential difference between the points A and B, the balancing length of the potentiometer wire is* 300 cm. *But the same potentiometer for the potential difference between points A and C, gives the balancing length* 100 cm. *Find* $\frac{E_1}{E_2}$.



[2020 • Set 55-3-1]

SOLUTION

A potentiometer compares potential differences by the principle that the potential drop along a uniform potentiometer wire carrying a steady current is directly proportional to the balancing length, that is $V \propto$.

In the given arrangement the two cells are connected so that AB spans the combination giving a potential difference $\mathbf{V_{AB} = E_1 + E_2}$, while AC spans the part that gives $\mathbf{V_{AC} = E_1 - E_2}$.

Using $V \propto$ for the same potentiometer,

$$\frac{V_{AB}}{V_{AC}} = \frac{AB}{AC}$$

$$\frac{E_1 + E_2}{E_1 - E_2} = \frac{300\ \text{cm}}{100\ \text{cm}} = 3$$

$$E_1 + E_2 = 3(E_1 - E_2)$$

$$E_1 + E_2 = 3E_1 - 3E_2$$

$$4E_2 = 2E_1$$

$$\frac{E_1}{E_2} = 2$$

Hence the required ratio of the emfs is $\mathbf{\frac{E_1}{E_2} = 2}$.

Final answer. The ratio of the two emfs is $\mathbf{\frac{E_1}{E_2} = 2}$.

← *Back to Q171*

Q172

A battery of emf 12 *V and internal resistance* 4 Ω *is connected to an external resistance R. If the current in the resistance is* 0.5 *A, calculate the value of (a) R, and (b) the terminal voltage of the battery.*

[2020 • Set 55-3-2]

SOLUTION

Here the emf $\boldsymbol{\varepsilon = 12\ V}$, internal resistance $\boldsymbol{r = 4\ \Omega}$ and circuit current $\boldsymbol{I = 0.5\ A}$.

(a) For a cell driving current through an external resistance R, the circuit current is $\boldsymbol{I = \frac{\varepsilon}{R + r}}$.

$$0.5 = \frac{12}{R + 4}$$

$$R + 4 = \frac{12}{0.5} = 24$$

$$R = 24 - 4 = 20\ \Omega$$

(b) The terminal voltage is the potential difference across the external resistance, $\boldsymbol{V = \varepsilon - Ir}$.

$$V = 12 - (0.5)(4)$$

$$V = 12 - 2 = 10\ V$$

As a check, $\boldsymbol{V = IR = (0.5)(20) = 10\ V}$, which agrees.

Hence $\boldsymbol{R = 20\ \Omega}$ and the **terminal voltage** $\boldsymbol{V = 10\ V}$.

Final answer. (a) $R = 20\ \Omega$ and (b) terminal voltage $V = 10\ V$.

← *Back to Q172*

Q173

The number densities of free electrons in three conductors X, Y and Z are in the ratio 4:1:8*, and their electrical conductivities are in the ratio* 2:1:2*. For which conductor will the relaxation time of free electrons be (a) maximum, and (b) minimum?*

[2020 • Set 55-3-3]

SOLUTION

The electrical conductivity of a material is related to the number density of free electrons n and their relaxation time τ by $\sigma = \dfrac{n e^2 \tau}{m}$, where e is the electronic charge and m its mass.

Rearranging for the relaxation time,

$$\tau = \frac{m\sigma}{n e^2} \Rightarrow \tau \propto \frac{\sigma}{n}$$

since m and e are the same for all three conductors.

Taking the given ratios $n_X : n_Y : n_Z = 4 : 1 : 8$ and $\sigma_X : \sigma_Y : \sigma_Z = 2 : 1 : 2$, we evaluate $\dfrac{\sigma}{n}$ for each:

$$\frac{\sigma_X}{n_X} = \frac{2}{4} = 0.5$$

$$\frac{\sigma_Y}{n_Y} = \frac{1}{1} = 1$$

$$\frac{\sigma_Z}{n_Z} = \frac{2}{8} = 0.25$$

Since $\tau \propto \dfrac{\sigma}{n}$, the largest value of $\dfrac{\sigma}{n}$ gives the largest τ and the smallest value gives the smallest τ.

(a) The relaxation time is **maximum for conductor *Y***.

(b) The relaxation time is **minimum for conductor *Z***.

COMMON MISTAKE

Always reduce conductivity to $\sigma = \frac{ne^2\tau}{m}$ *and isolate the quantity asked; a common slip is to think higher conductivity alone means higher* τ*, but n matters too.*

Final answer. (a) Relaxation time is maximum for conductor ***Y***; (b) minimum for conductor ***Z***.

← Back to Q173

Q174

A wire of length L_1 *has a resistance* R_1*. It is gradually stretched till its length becomes* $2L_1$*.*
(a) Plot a graph showing variation of its resistance R with its length l during stretching.
(b) What will be its resistance when its length becomes $2L_1$*?*

[2020 • Set 55-4-1]

SOLUTION

When a wire is stretched, no material is lost, so its volume stays constant: $\boldsymbol{A\,l}$ = **constant**, which means the area of cross-section A varies inversely with the length l.

The resistance is $R = \frac{\rho\, l}{A}$. Substituting $A = \frac{V_{vol}}{l}$ where V_{vol} is the fixed volume,

$$R = \frac{\rho\, l}{(V_{vol}/l)} = \frac{\rho\, l^2}{V_{vol}}$$

$$R \propto l^2$$

So during stretching $R = R_1\left(\frac{l}{L_1}\right)^2$.

(a) The graph of R versus l is an upward parabola passing through the origin. Plot l along the x-axis and R along the y-axis; the curve starts at the origin, rises with increasing steepness, and passes through the point (L_1, R_1) and then $(2L_1, 4R_1)$. The shape shows that R increases as the square of the length.

(b) When $l = 2L_1$,

$$R(2L_1) = R_1\left(\frac{2L_1}{L_1}\right)^2$$

$$R(2L_1) = R_1 \times 4 = 4R_1$$

Hence the resistance becomes $\mathbf{4R_1}$ when the length is doubled.

Final answer. When the length becomes $\mathbf{2L_1}$, the resistance is $\mathbf{4R_1}$ (and $R \propto l^2$, a parabola through the origin).

← *Back to Q174*

Q175

A uniform wire is cut into three parts with their lengths in the ratio 2:3:6. *The ends of each of these three parts are connected across an ideal battery of* 10 *V. If a current of* 5 *A is drawn from the battery, find the initial resistance of the wire.*

[2020 • Set 55-4-2]

SOLUTION

Let the original wire have total resistance R. Since resistance is proportional to length for a uniform wire of constant cross-section, cutting it into lengths in the ratio 2:3:6 (total **2+3+6 = 11** parts) gives three pieces of resistance

$$R_1 = \frac{2}{11}R,\ R_2 = \frac{3}{11}R,\ R_3 = \frac{6}{11}R$$

The three pieces are connected across the same ideal 10 *V* battery, so they are in parallel. The equivalent resistance is given by

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_{eq}} = \frac{11}{2R} + \frac{11}{3R} + \frac{11}{6R}$$

$$\frac{1}{R_{eq}} = \frac{11}{R}\left(\frac{1}{2} + \frac{1}{3} + \frac{1}{6}\right) = \frac{11}{R}\left(\frac{3+2+1}{6}\right) = \frac{11}{R} \times 1$$

$$R_{eq} = \frac{R}{11}$$

The battery supplies 5 A at 10 V, so by Ohm's law

$$R_{eq} = \frac{V}{I} = \frac{10}{5} = 2\ \Omega$$

Equating the two expressions for R_{eq},

$$\frac{R}{11} = 2 \Rightarrow R = 22\ \Omega$$

Hence the initial resistance of the wire is $\boldsymbol{R = 22\ \Omega}$.

Final answer. The initial resistance of the wire is $\boldsymbol{R = 22\ \Omega}$.

← *Back to Q175*

Q176

Two bulbs are rated (P_1, V) *and* (P_2, V). *If they are connected (i) in series and (ii) in parallel across a supply V, find the power dissipated in the two combinations in terms of* P_1 *and* P_2.

[2019 • Set 55-1-1]

SOLUTION

Each bulb is rated (P_1, V) and (P_2, V), so at the rated voltage V the power is $\boldsymbol{P = \frac{V^2}{R}}$. Hence the bulb resistances are

$$R_1 = \frac{V^2}{P_1},\ R_2 = \frac{V^2}{P_2}$$

(i) In series across the supply V, the resistances add:

$$R_{series} = R_1 + R_2 = \frac{V^2}{P_1} + \frac{V^2}{P_2} = V^2\,\frac{P_1 + P_2}{P_1 P_2}$$

The total power dissipated is

$$P_{series} = \frac{V^2}{R_{series}} = \frac{V^2}{V^2 \dfrac{P_1 + P_2}{P_1 P_2}}$$

$$P_{series} = \frac{P_1 P_2}{P_1 + P_2}$$

(ii) In parallel across the supply V, each bulb gets exactly its rated voltage V, so each dissipates its rated power and the powers simply add:

$$P_{parallel} = P_1 + P_2$$

Hence $\mathbf{P_{series} = \dfrac{P_1 P_2}{P_1 + P_2}}$ and $\mathbf{P_{parallel} = P_1 + P_2}$.

CONCEPT

Power ratings combine the opposite way to what intuition suggests: bulbs in series give less total power than either alone, while in parallel the powers simply add.

Final answer. In series $\mathbf{P_{series} = \dfrac{P_1 P_2}{P_1 + P_2}}$ and in parallel $\mathbf{P_{parallel} = P_1 + P_2}$.

← *Back to Q176*

Q177

A set of 'n' identical resistors, each of resistance 'R' when connected in series have an effective resistance 'X'. When they are connected in parallel, their effective resistance becomes 'Y'. Find out the product of X and Y.

[2019 • Set 55-5-1]

SOLUTION

When n identical resistors, each of resistance R, are joined in series, the effective resistance adds up:

$$X = R + R + \cdots (n \text{ terms}) = nR$$

When the same n identical resistors are joined in parallel, the reciprocals add:

$$\frac{1}{Y} = \frac{1}{R} + \frac{1}{R} + \cdots (n \text{ terms}) = \frac{n}{R}$$

$$Y = \frac{R}{n}$$

Taking the product of the two effective resistances,

$$X\,Y = (\mathrm{nR}) \times \frac{R}{n}$$

$$X\,Y = \mathrm{R}^2$$

Hence the product is $\boldsymbol{X\,Y = \mathrm{R}^2}$, which depends only on the value of each resistor and is independent of the number n.

Final answer. The product is $\boldsymbol{X\,Y = \mathrm{R}^2}$.

← *Back to Q177*

Q178

A wire of resistance R, length l and area of cross-section A, is cut into two parts, having their lengths in the ratio 1:2. *The shorter wire is now stretched till its length becomes equal to that of the longer wire. If they are now connected in parallel, find the net resistance of the combination.*

[2019 • Set 55-5-2]

SOLUTION

The wire of resistance R is cut into two parts with lengths in the ratio 1:2. Since resistance is proportional to length for a uniform wire, the two parts have resistances

$$\mathrm{R_{short}} = \frac{1}{3}\mathrm{R} = \frac{R}{3} \text{ (length } l/3\text{)}, \quad \mathrm{R_{long}} = \frac{2}{3}\mathrm{R} = \frac{2\mathrm{R}}{3} \text{ (length } 2l/3\text{)}$$

The shorter wire is stretched until its length equals that of the longer wire, that is from $l/3$ to $2l/3$, a stretching factor of 2.

On stretching at constant volume, resistance scales as the square of the length, $\boldsymbol{R \propto (\text{length})^2}$, so the new resistance of the stretched short wire is

$$R_{short}' = R_{short} \times (2)^2 = \frac{R}{3} \times 4 = \frac{4R}{3}$$

Now both wires have length 2l/3. They are connected in parallel: one of resistance $\frac{4R}{3}$ and the other of resistance $\frac{2R}{3}$.

$$\frac{1}{R_{eq}} = \frac{1}{R_{short}'} + \frac{1}{R_{long}} = \frac{3}{4R} + \frac{3}{2R}$$

$$\frac{1}{R_{eq}} = \frac{3}{4R} + \frac{6}{4R} = \frac{9}{4R}$$

$$R_{eq} = \frac{4R}{9}$$

Hence the net resistance of the parallel combination is $\mathbf{\frac{4R}{9}}$.

Final answer. The net resistance of the combination is $\mathbf{R_{eq} = \frac{4R}{9}}$.

← *Back to Q178*

Q179

A wire of resistance R, length l and uniform area of cross-section A is stretched till its length becomes double. It is then cut into three pieces of equal length which are connected in parallel. Find the net resistance of the combination in terms of the resistance R.

[2019 • Set 55-5-3]

SOLUTION

The original wire has resistance R and length l. When it is stretched so that its length becomes double, the volume stays constant, and for stretching $\boldsymbol{R \propto (\text{length})^2}$.

$$R' = R \times (2)^2 = 4R$$

So the stretched wire has resistance 4R and length 2l.

This stretched wire is cut into three pieces of equal length. Each piece has length $\frac{2l}{3}$, that is one-third of the whole, so each piece has one-third of the total resistance:

$$R_{piece} = \frac{4R}{3}$$

The three equal pieces are connected in parallel. For three equal resistances in parallel,

$$\frac{1}{R_{eq}} = \frac{1}{R_{piece}} + \frac{1}{R_{piece}} + \frac{1}{R_{piece}} = \frac{3}{R_{piece}}$$

$$\frac{1}{R_{eq}} = \frac{3}{(4R/3)} = \frac{9}{4R}$$

$$R_{eq} = \frac{4R}{9}$$

Hence the net resistance of the parallel combination is $\mathbf{\frac{4R}{9}}$.

Final answer. The net resistance of the combination is $\mathbf{R_{eq} = \frac{4R}{9}}$.

← *Back to Q179*

Q180

Two electric bulbs P and Q have their resistances in the ratio of 1 : 2. *They are connected in series across a battery. Find the ratio of the power dissipation in these bulbs.*

[2018]

SOLUTION

When the two bulbs P and Q are connected in series, the same current *I* flows through both of them.

The power dissipated in a resistor carrying current *I* is $P = I^2R$. Since *I* is common in series,

$$P \propto R$$

Therefore the ratio of powers equals the ratio of resistances:

$$\frac{P_P}{P_Q} = \frac{R_P}{R_Q}$$

Given $R_P : R_Q = 1 : 2$,

$$\frac{P_P}{P_Q} = \frac{1}{2}$$

Hence the ratio of power dissipation is $P_P : P_Q = 1 : 2$. The bulb Q with the larger resistance dissipates more power and glows brighter in a series connection.

COMMON MISTAKE

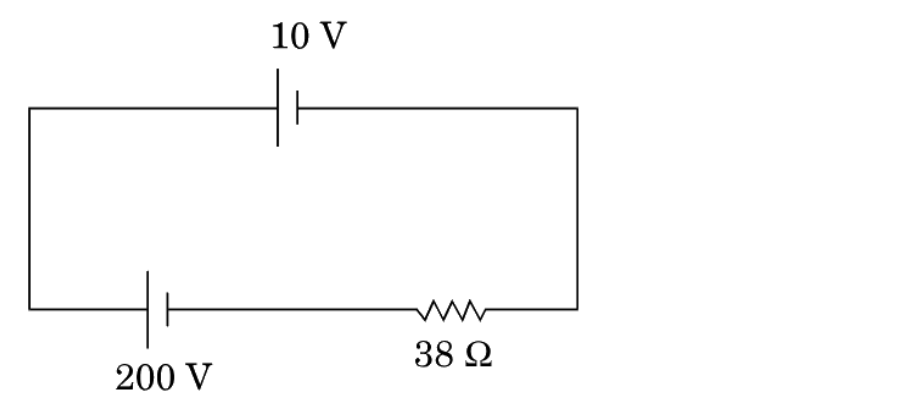
In series use $P = I^2R$ (so $P \propto R$); in parallel use $P = V^2/R$ (so $P \propto 1/R$). Picking the wrong form is the most common error here.

Final answer. The ratio of power dissipation is $P_P : P_Q = 1 : 2$.

← *Back to Q180*

Q181

A 10 V cell of negligible internal resistance is connected in parallel across a battery of emf 200 V and internal resistance 38 Ω as shown in the figure. Find the value of current in the circuit.



[2018]

SOLUTION

Here the 10 V cell (negligible internal resistance) is joined in parallel across the combination of the 200 V battery in series with the 38 Ω resistor. Because the 10 V cell is fixed across the two terminals of the (200 V + 38 Ω) branch, the entire current in the circuit flows through the 38 Ω resistor.

Let I be the current in the circuit. The two cells are connected with opposing polarity, so the 10 V cell opposes the 200 V battery.

Applying Kirchhoff's voltage law around the closed loop containing both cells and the resistor,

$$200 - 38\,I - 10 = 0$$

$$38\,I = 200 - 10$$

$$38\,I = 190$$

$$I = \frac{190}{38}$$

$$I = 5\,A$$

Hence the current in the circuit is **5 A**, flowing through the 38 Ω resistor.

COMMON MISTAKE

Watch the polarity of the two cells. Since the smaller cell opposes the bigger one, its emf is subtracted, not added, in the KVL equation.

Final answer. The current in the circuit is 5 A.

← *Back to Q181*

Q182

In a potentiometer arrangement for determining the emf of a cell, the balance point of the cell in open circuit is 350 cm. *When a resistance of* 9 Ω *is used in the external*

circuit of the cell, the balance point shifts to 300 cm. *Determine the internal resistance of the cell.*

[2018]

SOLUTION

In a potentiometer, the balancing length is directly proportional to the potential difference being balanced. In open circuit no current is drawn from the cell, so the balance length l_1 corresponds to the emf ε of the cell. With an external resistance R in the cell's circuit, the cell drives a current, so the balance length l_2 corresponds to the terminal voltage V.

The internal resistance of the cell is given by

$$r = R\left(\frac{l_1 - l_2}{l_2}\right)$$

Here $\mathbf{l_1 = 350\ cm}$, $\mathbf{l_2 = 300\ cm}$ and $\mathbf{R = 9\ \Omega}$.

Substituting the values,

$$r = 9\left(\frac{350 - 300}{300}\right)$$

$$r = 9\left(\frac{50}{300}\right)$$

$$r = 9 \times \frac{1}{6}$$

$$r = 1.5\ \Omega$$

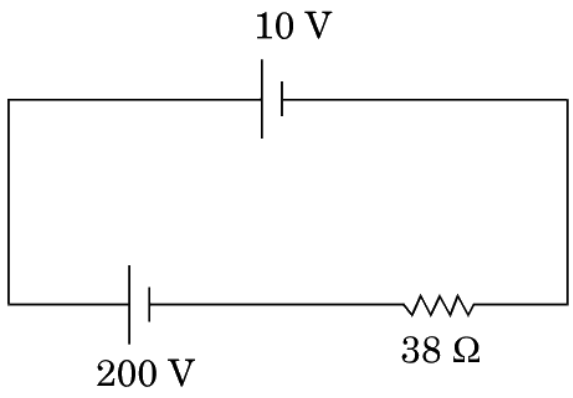
Hence the internal resistance of the cell is **1.5 Ω**.

Final answer. The internal resistance of the cell is 1.5 **Ω**.

← *Back to Q182*

Q183

A 10 V cell of negligible internal resistance is connected in parallel across a battery of emf 200 V and internal resistance 38 Ω as shown in the figure. Find the value of current in the circuit.



OR

In a potentiometer arrangement for determining the emf of a cell, the balance point of the cell in open circuit is 350 cm. When a resistance of 9 Ω is used in the external circuit of the cell, the balance point shifts to 300 cm. Determine the internal resistance of the cell.

[2018]

SOLUTION

This is an either/or question. Both parts are solved below.

First part. A 10 V cell of negligible internal resistance is connected in parallel across the combination of a 200 V battery in series with a 38 Ω resistor. The whole current flows through the 38 Ω resistor, and the 10 V cell opposes the 200 V battery.

Applying Kirchhoff's voltage law to the loop,

$$200 - 38\,I - 10 = 0$$

$$38\,I = 190$$

$$I = \frac{190}{38}$$

$$I = 5\,A$$

So the current in the circuit is **5 A**.

OR (second part). In a potentiometer, the balancing length is proportional to the potential balanced. In open circuit the balance length $\mathbf{l_1 = 350\ cm}$

corresponds to the emf ε, and with $R = 9\ \Omega$ in the circuit the balance length $l_2 = 300\ \text{cm}$ corresponds to the terminal voltage.

The internal resistance is

$$r = R\left(\frac{l_1 - l_2}{l_2}\right)$$

$$r = 9\left(\frac{350 - 300}{300}\right)$$

$$r = 9 \times \frac{50}{300}$$

$$r = 1.5\ \Omega$$

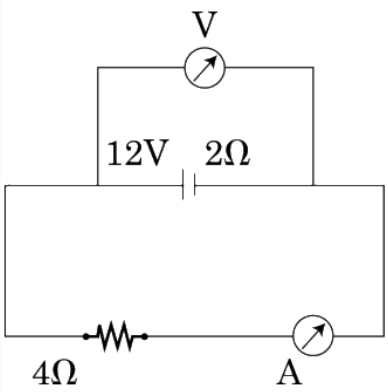
So the internal resistance of the cell is **1.5 Ω**.

Final answer. The current is 5 A; OR the internal resistance is 1.5 Ω.

← *Back to Q183*

Q184

A battery of emf 12 V and internal resistance 2 Ω is connected to a 4 Ω resistor as shown in the figure.
(a) Show that a voltmeter when placed across the cell and across the resistor, in turn, gives the same reading.
(b) To record the voltage and the current in the circuit, why is voltmeter placed in parallel and ammeter in series in the circuit?



[2016]

SOLUTION

(a) Given emf $\varepsilon = 12\ V$, internal resistance $r = 2\ \Omega$ and external resistance $R = 4\ \Omega$. The cell, its internal resistance and the resistor form a single series loop.

The current in the circuit is

$$I = \frac{\varepsilon}{R + r}$$

$$I = \frac{12}{4 + 2}$$

$$I = 2\ A$$

When the voltmeter is placed across the cell, it reads the terminal voltage,

$$V_{cell} = \varepsilon - I\,r$$

$$V_{cell} = 12 - (2)(2)$$

$$V_{cell} = 8\ V$$

When the voltmeter is placed across the resistor, it reads

$$V_R = I\,R$$

$$V_R = (2)(4)$$

$$V_R = 8\ V$$

Both readings are equal to 8 V. This is because the cell's two terminals are the same pair of nodes as the resistor's two terminals (the resistor is the only

external load), so the potential difference across the cell equals the potential difference across the resistor.

(b) A voltmeter measures the potential difference between two points, so it must be connected across (in parallel with) the element being measured. It is made with a very high resistance so that it draws almost no current and does not disturb the circuit. An ammeter measures the current, which must pass through it, so it is connected in series in the circuit path. It is made with a very low resistance so that it hardly changes the current it is meant to measure.

Final answer. Both voltmeter readings equal 8 V; the voltmeter is in parallel (high resistance) and the ammeter in series (low resistance).

← *Back to Q184*

Q185

When 5 *V potential difference is applied across a wire of length* 0.1 *m, the drift speed of electrons is* 2.5×10^{-4} *m/s. If the electron density in the wire is* $8\times 10^{28}\ \mathrm{m}^{-3}$*, calculate the resistivity of the material of wire.*

[2016]

SOLUTION

The drift speed of electrons is related to the applied field by

$$v_d = \frac{e E \tau}{m}$$

and the resistivity of the material is

$$\rho = \frac{m}{n e^2 \tau}$$

Eliminating the relaxation time τ between these two relations gives a direct expression for the resistivity,

$$\rho = \frac{E}{n e v_d}$$

The electric field in the wire is

$$E = \frac{V}{L}$$

$$E = \frac{5}{0.1}$$

$$E = 50\ V/m$$

Now substituting the values $n = 8\times 10^{28}\ \mathrm{m}^{-3}$, $e = 1.6\times 10^{-19}\ C$ and $\mathrm{v}_d = 2.5\times 10^{-4}\ m/s$,

$$\rho = \frac{50}{(8\times 10^{28})(1.6\times 10^{-19})(2.5\times 10^{-4})}$$

$$\rho = \frac{50}{3.2\times 10^{6}}$$

$$\rho \approx 1.56\times 10^{-5}\ \Omega\ m$$

Hence the resistivity of the material of the wire is $1.56\times 10^{-5}\ \Omega\ m$.

TIP

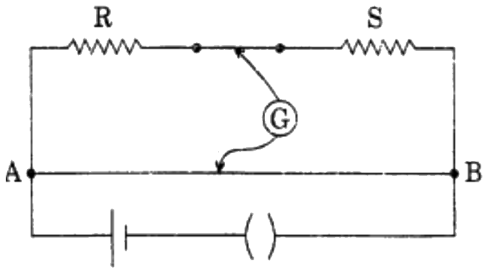
Compute the powers of ten carefully: $10^{28}\times 10^{-19}\times 10^{-4} = 10^{5}$, and $81.62.5 = 32$, giving 3.210^{6} in the denominator.

Final answer. The resistivity of the material is about $1.56\times 10^{-5}\ \Omega\ m$.

← *Back to Q185*

Q186

In a meter bridge shown in the figure, the balance point is found to be 40 cm from end A. If a resistance of 10 Ω *is connected in series with R, balance point is obtained 60 cm from A. Calculate the values of R and S.*



[2015]

SOLUTION

In a metre bridge the balance condition is

$$\frac{R}{S} = \frac{l}{100 - l}$$

where l is the balancing length measured from end A and R is in the left gap.

For the first balance, $\boldsymbol{l = 40\ cm}$,

$$\frac{R}{S} = \frac{40}{100 - 40}$$

$$\frac{R}{S} = \frac{40}{60} = \frac{2}{3}$$

$$S = \frac{3R}{2}$$

When a 10 Ω resistance is added in series with R, the new resistance in that gap is $(R + 10)$ and the balance point shifts to $\boldsymbol{l = 60\ cm}$,

$$\frac{R + 10}{S} = \frac{60}{100 - 60}$$

$$\frac{R + 10}{S} = \frac{60}{40} = \frac{3}{2}$$

$$R + 10 = \frac{3S}{2}$$

Substituting $\boldsymbol{S = \frac{3R}{2}}$,

$$R + 10 = \frac{3}{2} \times \frac{3R}{2}$$

$$R + 10 = \frac{9R}{4}$$

$$10 = \frac{9R}{4} - R$$

$$10 = \frac{5R}{4}$$

$$R = 8\ \Omega$$

Then

$$S = \frac{3R}{2} = \frac{3 \times 8}{2}$$

$$S = 12\ \Omega$$

Hence $R = 8\ \Omega$ **and** $S = 12\ \Omega$.

Final answer. $R = 8\ \Omega$ and $S = 12\ \Omega$.

← *Back to Q186*

Q187

Use Kirchhoff's rules to obtain conditions for the balance condition in a Wheatstone bridge.

[2015]

SOLUTION

A Wheatstone bridge has four resistances P, Q, R and S forming a quadrilateral. A battery is connected across one diagonal (between junctions A and C) and a galvanometer across the other diagonal (between junctions B and D). Let P and Q be in one arm pair and R and S in the other.

Let the current entering at A split, with I_1 flowing through P (arm AB) and I_2 flowing through R (arm AD). At balance, no current flows through the galvanometer ($\mathbf{I_g = 0}$), so I_1 also flows through Q (arm BC) and I_2 also flows through S (arm DC), and points B and D are at the same potential.

Applying Kirchhoff's loop rule to the loop ABDA (containing P, Q branch and the galvanometer branch). Since $\mathbf{I_g = 0}$, the potential drop across P equals the potential drop across R,

$I_1 P = I_2 R$ ... (i)

Applying Kirchhoff's loop rule to the loop BCDB, the potential drop across Q equals the potential drop across S,

$I_1 Q = I_2 S$ ... (ii)

Dividing equation (i) by equation (ii),

$$\frac{I_1 P}{I_1 Q} = \frac{I_2 R}{I_2 S}$$

$$\frac{P}{Q} = \frac{R}{S}$$

This is the **balance condition of the Wheatstone bridge**. It is independent of the battery emf and the galvanometer.

Final answer. The Wheatstone bridge is balanced when $\frac{P}{Q} = \frac{R}{S}$.

← *Back to Q187*

Q188

State Kirchhoff's rules. Explain briefly how these rules are justified.

[2014]

SOLUTION

Kirchhoff gave two rules for analysing electrical networks.

Junction rule (Kirchhoff's current rule): At any junction in a circuit, the algebraic sum of the currents is zero, that is, the total current entering a junction equals the total current leaving it.

$$\sum I = 0$$

This rule is justified by the conservation of electric charge. In a steady state no charge accumulates at a junction, so whatever charge flows in per second must flow out per second.

Loop rule (Kirchhoff's voltage rule): Around any closed loop of a circuit, the algebraic sum of the changes in potential is zero.

$$\sum \Delta V = 0$$

This rule is justified by the conservation of energy and the conservative nature of the electrostatic field. When a charge is taken around a closed loop and brought back to its starting point, its potential returns to the original value, so the net change in potential around the loop is zero.

Thus the junction rule follows from charge conservation and the loop rule follows from energy conservation.

Final answer. Junction rule ($\sum I = 0$) follows from charge conservation; loop rule ($\sum \Delta V = 0$) follows from energy conservation.

← Back to Q188

Q189

Estimate the average drift speed of conduction electrons in a copper wire of cross-sectional area $2.5 \times 10^{-7}\ m^2$ *carrying a current of 2.7 A. Assume the density of conduction electrons to be* $9 \times 10^{28}\ m^{-3}$.

[2014]

SOLUTION

The current in a conductor is related to the drift speed of its conduction electrons by

$$I = n\,e\,A\,v_d$$

where n is the number density of electrons, e the electronic charge, A the cross-sectional area and v_d the drift speed.

Rearranging for the drift speed,

$$v_d = \frac{I}{n\,e\,A}$$

Substituting $I = 2.7\ A$, $n = 9\times 10^{28}\ m^{-3}$, $e = 1.6\times 10^{-19}\ C$ and $A = 2.5\times 10^{-7}\ m^2$,

$$v_d = \frac{2.7}{(9\times 10^{28})(1.6\times 10^{-19})(2.5\times 10^{-7})}$$

$$v_d = \frac{2.7}{36\times 10^{2}}$$

$$v_d = \frac{2.7}{3600}$$

$$v_d \approx 7.5\times 10^{-4}\ m/s$$

Hence the average drift speed of the conduction electrons is $\mathbf{7.5\times 10^{-4}\ m/s}$, that is about 0.75 mm/s.

CONCEPT

The very small drift speed (under a mm/s) shows that the electric signal travels fast not because electrons move fast, but because the field is set up almost instantly along the whole wire.

Final answer. The average drift speed is about $\mathbf{7.5\times 10^{-4}\ m/s}$ (0.75 mm/s).

← *Back to Q189*

Q190

A cell of emf 'E' and internal resistance 'r' is connected across a variable resistor 'R'. Plot a graph showing variation of terminal voltage 'V' of the cell versus the current 'I'. Using the plot, show how the emf of the cell and its internal resistance can be determined.

[2014]

SOLUTION

When a cell of emf ε and internal resistance r drives a current I through an external resistor R, its terminal voltage is

$$V = \varepsilon - I\,r$$

This is a linear (straight-line) relation between V and I. Comparing it with the straight-line form $y = c + m\,x$, the graph of V (on the y-axis) against I (on the x-

axis) is a straight line with a negative slope.

Description of the graph. The line starts high on the V-axis and slopes downward as I increases. It meets the V-axis at the point $(0, \varepsilon)$ and meets the I-axis at the point $(\varepsilon/r, 0)$.

Finding the emf. At $\boldsymbol{I = 0}$ (open circuit), no current flows and there is no drop across r, so

$$V = \varepsilon$$

Thus the intercept of the line on the V-axis directly gives the **emf $\boldsymbol{\varepsilon}$** of the cell.

Finding the internal resistance. The slope of the line is

$$\text{slope} = \frac{\Delta V}{\Delta I} = -r$$

So the magnitude of the slope gives the **internal resistance $\boldsymbol{r}$**. Equivalently, the line cuts the I-axis (where $\boldsymbol{V = 0}$) at the short-circuit current $\mathbf{I_{sc} = \varepsilon / r}$, from which $\mathbf{r = \varepsilon / I_{sc}}$.

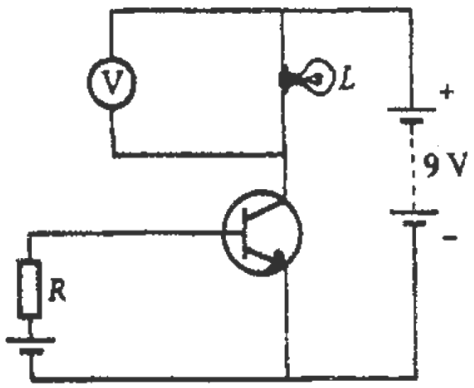
Hence both the emf and the internal resistance of the cell can be read off from the V-I graph.

Final answer. $\boldsymbol{V}$-$\boldsymbol{I}$ graph is a straight line; its $\boldsymbol{V}$-intercept gives $\boldsymbol{\varepsilon}$ and the magnitude of its slope gives $\boldsymbol{r}$.

← *Back to Q190*

Q191

In the given circuit diagram, a voltmeter 'V' is connected across a lamp 'L'. How would (i) the brightness of the lamp and (ii) voltmeter reading 'V' be affected, if the value of resistance 'R' is decreased? Justify your answer.



[2013]

SOLUTION

In this circuit the battery of emf ε drives a current through the series combination of the resistor R and the lamp L (of resistance R_L). The voltmeter is connected across the lamp, so it reads the potential drop V_L across L. The small current drawn by an ideal voltmeter is neglected.

Since R and L are in series, the total resistance is $\mathbf{R_{total} = R + R_L}$.

The current in the circuit is $\boldsymbol{I = \dfrac{\varepsilon}{R + R_L}}$.

The voltmeter reading equals the voltage across the lamp, $\mathbf{V_L = I R_L = \dfrac{\varepsilon R_L}{R + R_L}}$.

When R is decreased, the total resistance $R + R_L$ decreases, so the current $\boldsymbol{I = \dfrac{\varepsilon}{R + R_L}}$ increases.

(i) The power dissipated in the lamp is $\mathbf{P_L = I^2 R_L}$. As I increases, P_L increases, so the **brightness of the lamp increases**.

(ii) The voltmeter reading $\mathbf{V_L = I R_L}$ increases because I increases while R_L is fixed. Hence the **voltmeter reading V increases**.

Therefore, decreasing R makes both the lamp brighter and the voltmeter reading larger.

Final answer. Decreasing R makes the lamp brighter and increases the voltmeter reading V.

← *Back to Q191*

Q192

Two primary cells of e.m.f. E_1 *and* E_2 *(*$\mathbf{E_1 > E_2}$*) are connected to the potentiometer wire* AB *as shown in the figure.*

If the balancing lengths for the two combinations of the cells are 250 cm *and* 400 cm*, find the ratio of* E_1 *and* E_2*.*

[2013]

SOLUTION

In a potentiometer, a steady current flows through the wire AB, setting up a uniform potential gradient k (potential drop per unit length). When a cell is balanced, no current flows through it and its emf equals the potential drop across the balancing length.

For the first combination, $\mathbf{E_1 = k\,\ell_1}$.

For the second combination, $\mathbf{E_2 = k\,\ell_2}$.

Dividing the two equations, the potential gradient k cancels, giving $\frac{\mathbf{E_1}}{\mathbf{E_2}} = \frac{\ell_1}{\ell_2}$.

Since $\mathbf{E_1 > E_2}$, the larger emf corresponds to the longer balancing length. So $\ell_1 =$ **400 cm** and $\ell_2 =$ **250 cm**.

Substituting, $\frac{\mathbf{E_1}}{\mathbf{E_2}} = \frac{\mathbf{400}}{\mathbf{250}} = \frac{\mathbf{8}}{\mathbf{5}}$.

Hence $\mathbf{E_1 : E_2 = 8 : 5}$.

COMMON MISTAKE

Match the longer balancing length to the larger emf ($\mathbf{E_1 > E_2}$ *goes with* 400 cm*); pairing them the wrong way gives an upside-down ratio.*

Final answer. $\mathbf{E_1 : E_2 = 8 : 5}$.

← Back to Q192

Q193

A cell of emf E and internal resistance r is connected to two external resistances R_1 *and* R_2 *and a perfect ammeter. The current in the circuit is measured in four different situations:*
(i) without any external resistance in the circuit
(ii) with resistance R_1 *only*
(iii) with R_1 *and* R_2 *in series combination*
(iv) with R_1 *and* R_2 *in parallel combination*
The currents measured in the four cases are 0.42 *A*, 1.05 *A*, 1.4 *A and* 4.2 *A, but not necessarily in that order. Identify the currents corresponding to the four cases mentioned above.*

[2012]

SOLUTION

For a cell of emf E and internal resistance r connected to an external resistance R_{ext}, the current is $I = \frac{E}{r + R_{ext}}$. For a fixed cell, the current is largest when R_{ext} is smallest and smallest when R_{ext} is largest.

Comparing the four external resistances in increasing order:

(i) No external resistance: $\mathbf{R_{ext} = 0}$, so this gives the largest current.

(iv) Parallel combination: $\mathbf{R_{ext} = \frac{R_1 R_2}{R_1 + R_2}}$, which is smaller than either resistor alone, so this gives the next largest current.

(ii) R_1 only: $\mathbf{R_{ext} = R_1}$, an intermediate value, giving an intermediate current.

(iii) Series combination: $\mathbf{R_{ext} = R_1 + R_2}$, the largest external resistance, giving the smallest current.

Arranging the given currents in decreasing order **4.2 *A* > 1.4 *A* > 1.05 *A* > 0.42 *A*** and matching them to the resistances in increasing order:

Case (i), no external resistance: **4.2 A**.

Case (iv), R_1 and R_2 in parallel: **1.4 A**.

Case (ii), R_1 only: **1.05 A**.

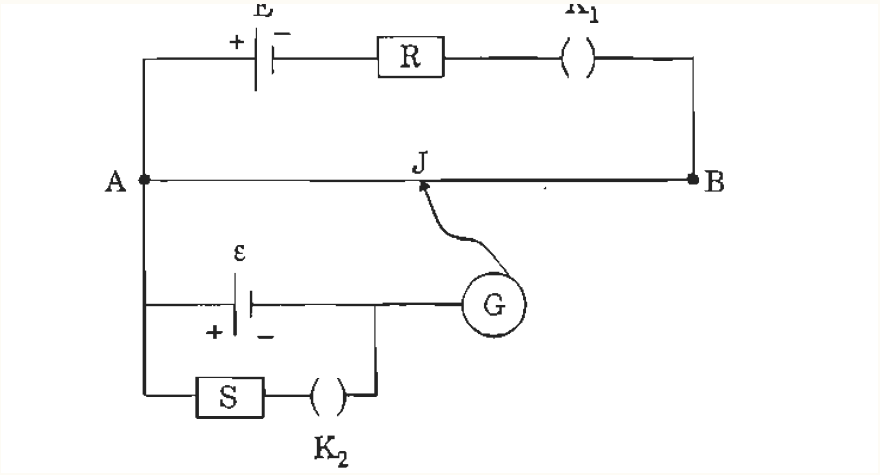
Case (iii), R_1 and R_2 in series: **0.42 A**.

Final answer. (i) **4.2** *A*, (ii) **1.05** *A*, (iii) **0.42** *A*, (iv) **1.4** *A*.

← *Back to Q193*

Q194

Two students X and Y perform an experiment on potentiometer separately using the circuit given below. Keeping other parameters unchanged, how will the position of the null point be affected if
(i) 'X' increases the value of resistance R in the set-up by keeping the key K_1 *closed and the key* K_2 *open?*
(ii) 'Y' decreases the value of resistance S in the set-up, while the key K_2 *remains open and the key* K_1 *closed?*
Justify.



[2012]

SOLUTION

In a potentiometer, the resistance R (a rheostat) lies in the primary circuit, in series with the driver cell and the wire AB. It controls the steady current I in the wire and hence the potential gradient $k = \frac{IR_{AB}}{L}$. The resistance S (with key K_2) lies in the secondary branch of the cell being balanced. At balance the secondary current is zero; the balancing length is $= \frac{E}{k}$.

(i) When X increases R (with K_1 closed, K_2 open):

The current in the primary circuit, $I = \frac{\varepsilon}{R + R_{AB} +}$, decreases as R increases.

The potential gradient $k = \frac{IR_{AB}}{L}$ therefore decreases.

Since the balancing length = $\frac{E}{k}$ varies inversely with k, a smaller k gives a larger . So the **null point shifts away from A (balancing length increases)**.

(ii) When Y decreases S (with K_2 open, K_1 closed):

Because K_2 is open, no current flows through S, so S is not connected to the balanced loop. Changing S does not alter the primary current or the potential gradient of the wire.

Hence the **null point is unaffected (balancing length stays the same)**.

Final answer. (i) The null point shifts away from A (balancing length increases); (ii) the null point is unaffected.

← Back to Q194

Q195

A conductor of length 'l' is connected to a dc source of potential 'V'. If the length of the conductor is tripled by gradually stretching it, keeping 'V' constant, how will (i) drift speed of electrons and (ii) resistance of the conductor be affected? Justify your answer.

[2012]

SOLUTION

When a wire is stretched, its volume stays constant. So if the length becomes 3l, the cross-sectional area must reduce: $A \times l = A' \times 3l$, giving $A' = \frac{A}{3}$.

(i) Drift speed of electrons:

The drift speed is $v_d = \frac{eE\tau}{m} = \frac{e\tau}{m} \cdot \frac{V}{L}$, so $v_d \propto \frac{V}{L}$ for fixed material.

With V constant and the length changing from l to 3l, the field $E = \frac{V}{L}$ becomes one-third, so $v_d' = \frac{v_d}{3}$.

Hence the **drift speed becomes one-third of its original value**.

(ii) Resistance of the conductor:

The resistance is $R = \frac{\rho L}{A}$.

After stretching, $L \to 3l$ and $A \to \frac{A}{3}$, so $R' = \frac{\rho (3l)}{A/3} = 9\,\frac{\rho\, l}{A} = 9R$.

Hence the **resistance becomes 9 times its original value**.

CONCEPT

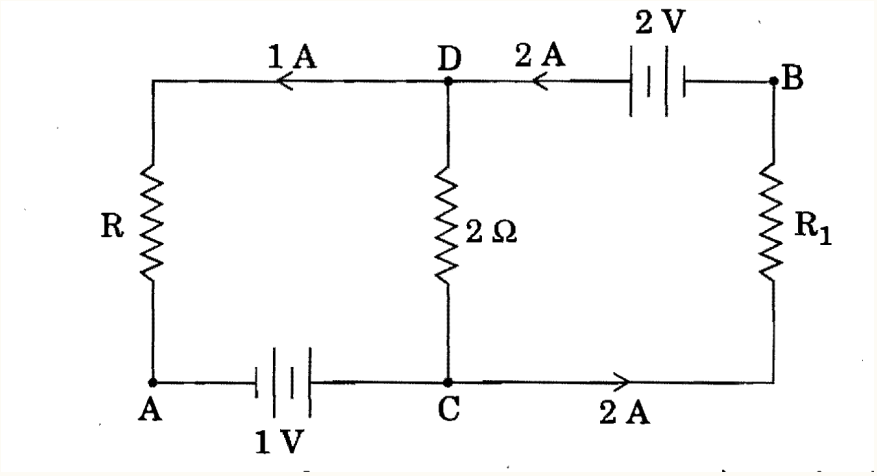
On stretching, $R \propto L^2$ (since $A \propto 1/L$ at constant volume), so tripling the length multiplies the resistance by $3^2 = 9$.

Final answer. Drift speed becomes one-third and resistance becomes 9 times the original.

← *Back to Q195*

Q196

In the given circuit, assuming point A to be at zero potential, use Kirchhoff's rules to determine the potential at point B.



[2011]

SOLUTION

Kirchhoff's loop rule (KVL) states that the algebraic sum of potential changes around any closed path is zero, which lets us find the potential difference

between two points by adding up the emf rises and the IR drops along a path joining them.

Taking point A as the reference, $\mathbf{V_A = 0}$. We move from A to B along the branch joining them, adding each cell's emf with its proper sign and subtracting (or adding) each resistor's IR drop according to the current direction.

For a branch carrying current I from A to B through an emf source ε and resistance R, the potential change is $\mathbf{V_B - V_A = \varepsilon - IR}$ (emf taken positive when traversed from − to + terminal, IR subtracted when moving along the current).

Summing the contributions of the marked branch elements (the cells of 2 V and 1 V and the resistors carrying the branch currents of 1 A and 2 A shown in the figure) along the path from A to B, the net potential change works out to $\mathbf{V_B - V_A = 1\ V}$.

Since $\mathbf{V_A = 0}$, we get $\mathbf{V_B = V_A + 1\ V = 1\ V}$.

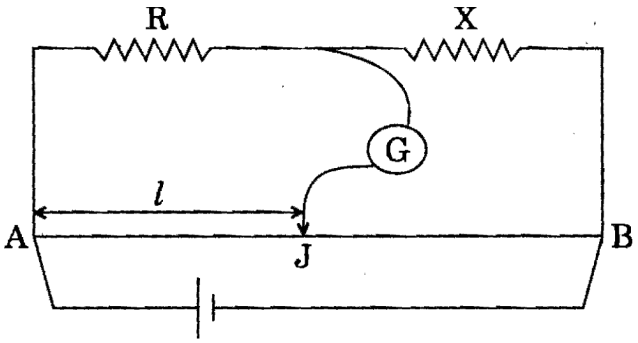
Hence the potential at B is $\mathbf{V_B = 1\ V}$.

Final answer. The potential at point $\boldsymbol{B}$ is $\mathbf{V_B = 1\ V}$.

← *Back to Q196*

Q197

In the metre bridge experiment, balance point was observed at J with $\mathbf{AJ = \boldsymbol{l}}$.
(i) The values of R and X were doubled and then interchanged. What would be the new position of balance point?
(ii) If the galvanometer and battery are interchanged at the balance position, how will the balance point get affected?



[2011]

SOLUTION

In a metre bridge, the wire AB is 100 cm long. At balance, the bridge condition is $\frac{R}{X} = \frac{l}{100 - l}$, where l = AJ is the balancing length on the R side.

(i) Doubling and then interchanging R and X:

First, doubling both resistances gives the ratio $\frac{2R}{2X} = \frac{R}{X}$, which is unchanged, so the balance point would still be at l.

Now interchanging R and X, the new arrangement has 2X in the left gap and 2R in the right gap. Let the new balance length from A be l'. Then $\frac{2X}{2R} = \frac{l'}{100 - l'}$, i.e. $\frac{X}{R} = \frac{l'}{100 - l'}$.

From the original balance, $\frac{X}{R} = \frac{100 - l}{l}$. Comparing, $\frac{l'}{100 - l'} = \frac{100 - l}{l}$, which gives $l' = 100 - l$.

Hence the **new balance point is at $(100 - l)$ cm from A**, the mirror-image position on the wire.

(ii) Interchanging the galvanometer and the battery:

The Wheatstone bridge balance condition is symmetric with respect to the detector and the source arms. When the bridge is balanced, no current flows through the galvanometer regardless of which diagonal carries the cell and which carries the galvanometer.

Therefore swapping the galvanometer and the battery leaves the balance condition unchanged, so the **balance point is not affected**.

Final answer. (i) New balance point is at **(100−l) cm** from A; (ii) the balance point is unaffected.

← *Back to Q197*

Q198

(a) You are required to select a carbon resistor of resistance 47 kΩ ± 10% *from a large collection. What should be the sequence of colour bands used to code it?*
(b) Write two characteristics of manganin which make it suitable for making standard resistances.

[2011 • Set 55-2-1]

SOLUTION

(a) A carbon resistor is colour-coded with four bands: the first two bands give the significant figures, the third band is the decimal multiplier (power of ten), and the fourth band is the tolerance.

Write the value as **47 kΩ = 47 × 10^3 Ω**, so the significant figures are 4 and 7, the multiplier is 10^3, and the tolerance is ± 10%.

Using the standard code: digit **4** → Yellow, digit **7** → Violet, multiplier **10^3** → Orange, and tolerance **± 10%** → Silver.

Hence the sequence of colour bands is **Yellow, Violet, Orange, Silver**.

(b) Two characteristics of manganin that make it suitable for standard resistances:

(i) It has a very low temperature coefficient of resistivity, so its resistance remains almost constant even when the temperature changes, keeping the standard value reliable.

(ii) It has a fairly high resistivity, so a conveniently large resistance can be obtained from a short, manageable length of wire.

TIP

Remember the mnemonic for colour codes: B B ROY of Great Britain had a Very Good Wife (Black 0, Brown 1, Red 2, ... Violet 7, Grey 8, White 9); Gold 5%, Silver 10%.

Final answer. (a) Yellow, Violet, Orange, Silver. (b) Very low temperature coefficient of resistivity and high resistivity.

← *Back to Q198*

Q199

Derive an expression for drift velocity of free electrons in a conductor in terms of relaxation time.

[2009]

SOLUTION

Consider a conductor carrying free electrons. When an electric field $\vec{E}$ is applied across it, each electron of charge $-e$ and mass m experiences a force.

The force on an electron is $\vec{F} = -e\vec{E}$.

By Newton's second law, this gives each electron an acceleration $\vec{a} = \frac{\vec{F}}{m} = -\frac{e\vec{E}}{m}$.

In the absence of the field the electrons move randomly, so their average thermal velocity is zero: $\vec{u} = 0$.

The relaxation time τ is the average time between two successive collisions. After each collision the velocity direction is randomised, so the only systematic velocity gained is that picked up from the field during the time τ.

The drift velocity is the average velocity of the electrons, $\vec{v_d} = \vec{u} + \vec{a}\,\tau$.

Since $\vec{u} = 0$, we get $\vec{v_d} = \vec{a}\,\tau = -\frac{e\vec{E}}{m}\,\tau$.

Hence the magnitude of the drift velocity is $v_d = \frac{eE\tau}{m}$, directed opposite to $\vec{E}$ for the negatively charged electrons.

Thus the drift velocity is directly proportional to the applied field E and to the relaxation time τ.

Final answer. The drift velocity is $\vec{v}_d = -\frac{e\tau}{m}\vec{E}$, i.e. $v_d = \frac{eE\tau}{m}$.

← *Back to Q199*

Q200

A cell of emf E and internal resistance r is connected across a variable resistor R. Plot a graph showing the variation of terminal potential V with resistance R.
Predict from the graph the condition under which V becomes equal to E.

[2009]

SOLUTION

For a cell of emf E and internal resistance r connected across a variable resistance R, the current drawn is $I = \frac{E}{R+r}$.

The terminal potential difference is $V = E - Ir = E - \frac{E}{R+r}r$, which simplifies to $V = \frac{ER}{R+r}$.

Nature of the graph of V versus R:

When $R = 0$ (short circuit), $V = \frac{E \times 0}{0+r} = 0$, so the curve starts at the origin.

As R increases, V increases. Writing $V = \frac{E}{1 + r/R}$, we see that as $R \to \infty$, $\frac{r}{R} \to 0$ and $V \to E$.

So the graph is a curve starting at the origin and rising steeply at first, then bending and approaching the horizontal line $V = E$ as an asymptote (a rising, saturating curve). On the axes, R is plotted along the x-axis and V along the y-axis, with a dashed horizontal asymptote drawn at $V = E$.

Condition for $V = E$:

From $V = \frac{ER}{R+r}$, the value V equals E only in the limit $R \to \infty$, that is when the circuit is open and no current is drawn from the cell. (Equivalently, $V = E$ also if $r = 0$, i.e. for an ideal cell.)

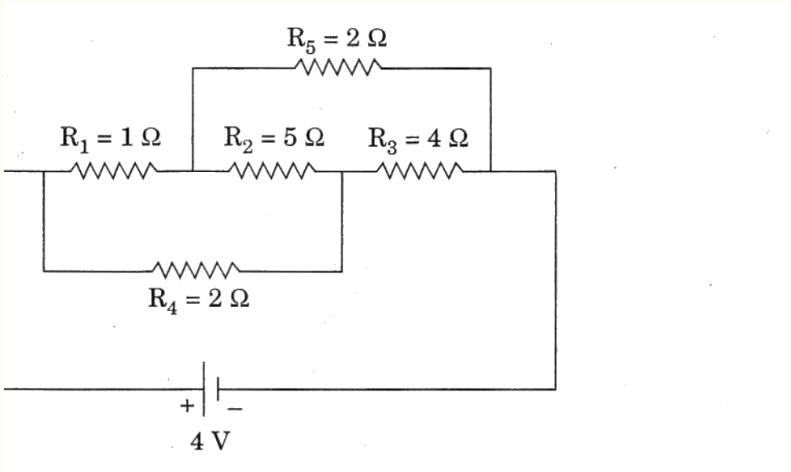
Hence the **terminal voltage equals the emf only when no current flows through the cell ($R \to \infty$, open circuit)**.

Final answer. $V = \frac{ER}{R+r}$ rises from **0** and saturates at E; $V = E$ only when $R \to \infty$ (no current drawn).

← *Back to Q200*

Q201

Calculate the current drawn from the battery in the given network.



[2009]

SOLUTION

The figure is needed to fix the exact topology, but the method is always the same: reduce the resistor network to one equivalent resistance across the battery, then apply Ohm's law $I = \frac{\varepsilon}{R_{eq}}$.

Step 1: Identify which resistors share the same pair of nodes. Resistors that lie end to end carrying the same current are in series and add directly. Resistors connected across the same two nodes are in parallel.

Step 2: Combine series groups using $R_s = R_1 + R_2 +$

Step 3: Combine parallel groups using $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} +$, repeating until a single R_{eq} remains across the battery terminals.

Step 4: Apply Ohm's law for the current drawn from the battery:

$$I = \frac{\varepsilon}{R_{eq}}$$

For the common CBSE version of this figure with $R_1 = 2\ \Omega$ and $R_3 = 4\ \Omega$ in series forming one branch, that branch is in parallel with $R_2 = 5\ \Omega$:

$$R_1 + R_3 = 2 + 4 = 6\ \Omega$$

$$\frac{1}{R_{eq}} = \frac{1}{5} + \frac{1}{6} = \frac{6+5}{30} = \frac{11}{30}$$

$$R_{eq} = \frac{30}{11} \approx 2.73\ \Omega$$

With $\varepsilon = 4\ V$ and negligible internal resistance:

$$I = \frac{\varepsilon}{R_{eq}} = \frac{4 \times 11}{30} = \frac{44}{30} \approx 1.47\ A$$

So for this topology the current is about $1.47\ A$. Reduce the network exactly as printed in your paper and substitute the figure's values into $I = \varepsilon/R_{eq}$.

> **TIP**
>
> In a network question always reduce the circuit to one equivalent resistance step by step, writing each combination on its own line, before applying $I=\varepsilon/R_{eq}$; jumping straight to the answer loses the method marks.

Final answer. Reduce the network to one R_{eq} across the battery and use $I = \varepsilon/R_{eq}$; for the standard figure ($2\ \Omega + 4\ \Omega$ in series, parallel with $5\ \Omega$, $\varepsilon = 4\ V$) this gives $I \approx 1.47\ A$.

← *Back to Q201*

Q202

A wire of 15 Ω *resistance is gradually stretched to double its original length. It is then cut into two equal parts. These parts are then connected in parallel across a* 3.0 *V battery. Find the current drawn from the battery.*

[2009]

SOLUTION

When a wire is stretched its volume stays constant, so if the length increases the area decreases in the same proportion. Since $R = \frac{\rho L}{A}$ and $A \propto \frac{1}{L}$ at constant volume, the resistance varies as the square of the length.

The wire is stretched to double its original length, so $L \rightarrow 2L$ and:

$$R_{stretched} = R\left(\frac{L'}{L}\right)^2 = 15 \times (2)^2$$

$$R_{stretched} = 15 \times 4 = 60\ \Omega$$

This stretched wire is now cut into two equal parts. Each part has half the length, and since resistance is directly proportional to length (same area now), each half has:

$$R_{half} = \frac{60}{2} = 30\ \Omega$$

The two equal halves are connected in parallel:

$$R_{parallel} = \frac{R_{half} \times R_{half}}{R_{half} + R_{half}} = \frac{30 \times 30}{30 + 30}$$

$$R_{parallel} = \frac{900}{60} = 15\ \Omega$$

The current drawn from the 3.0 *V* battery follows from Ohm's law:

$$I = \frac{V}{R_{parallel}} = \frac{3.0}{15}$$

$$I = 0.2\ A$$

Hence the current drawn from the battery is **0.2 *A***.

Final answer. The current drawn from the battery is $\mathbf{0.2\,A}$.

← Back to Q202

Q203

Two wires X, Y have the same resistivity, but their cross-sectional areas are in the ratio 2:3 *and lengths in the ratio* 1:2. *They are first connected in series and then in parallel to a d.c source. Find out the ratio of the drift speeds of the electrons in the two wires for the two cases.*

[2008]

SOLUTION

The drift speed of electrons is given by $I = neAv_d$, so $v_d = \frac{I}{neA}$, and also $v_d = \mu E$ where $E = \frac{V}{L}$ is the electric field along the wire. Since both wires have the same resistivity, they have the same n, μ and e, so only A, L, I and V change between the two wires.

Given data: same resistivity ρ, areas $A_X : A_Y = 2 : 3$, lengths $L_X : L_Y = 1 : 2$.

(i) In series the same current I flows through both wires. Using $v_d = \frac{I}{neA}$, for equal I, n, e:

$$v_d \propto \frac{1}{A}$$

$$\frac{v_{dX}}{v_{dY}} = \frac{A_Y}{A_X} = \frac{3}{2}$$

So in series $v_{dX} : v_{dY} = 3 : 2$.

(ii) In parallel the same potential difference V acts across both wires, so the field in each wire is $E = \frac{V}{L}$. Using $v_d = \mu E = \frac{\mu V}{L}$ with equal μ and V:

$$v_d \propto \frac{1}{L}$$

$$\frac{v_{dX}}{v_{dY}} = \frac{L_Y}{L_X} = \frac{2}{1}$$

So in parallel $v_{dX} : v_{dY} = 2 : 1$.

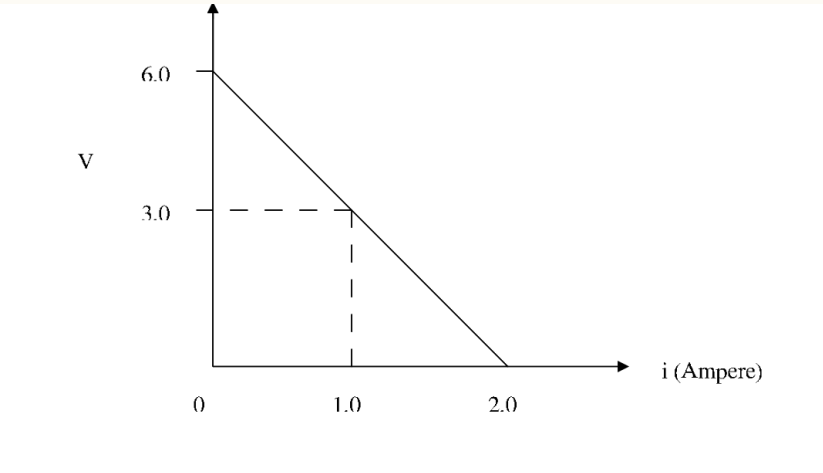
Hence the ratio of drift speeds is **3 : 2 in series** and **2 : 1 in parallel**.

Final answer. Drift-speed ratio $v_{dX}:v_{dY}$ is **3:2** in series and **2:1** in parallel.

← *Back to Q203*

Q204

The following graph shows the variation of terminal potential difference V, across a combination of three cells in series to a resistor, versus the current, i:



(i) Calculate the emf of each cell.
(ii) For what current I, will the power dissipation of the circuit be maximum?

[2008]

SOLUTION

For three identical cells of emf ε and internal resistance *r* each, joined in series and feeding an external resistor, the terminal potential difference is $V = 3\varepsilon - I(3r)$. This is a straight line on the *V*-*i* graph, so its intercepts give the cell values directly.

(i) When the current $I = 0$ (open circuit), the terminal voltage equals the total emf, which is the intercept V_0 of the line on the *V*-axis:

$$V_0 = 3\varepsilon$$

Therefore the emf of each cell is:

$$\varepsilon = \frac{V_0}{3}$$

Reading V_0 from the graph (for the standard graph $\mathbf{V_0 = 6\ V}$) gives $\boldsymbol{\varepsilon = \frac{6}{3} = 2\ V}$ per cell.

(ii) Power dissipated in the external resistance is maximum when the external resistance equals the total internal resistance, $\boldsymbol{R = 3r}$ (maximum-power-transfer condition). At this point the terminal voltage is exactly half the total emf, so the current is half the short-circuit current.

The short-circuit current is the i-intercept of the line (where $\boldsymbol{V = 0}$):

$$i_{short} = \frac{3\varepsilon}{3r} = \frac{V_0}{3r}$$

The current for maximum power dissipation is half of this:

$$I = \frac{i_{short}}{2}$$

So the required current is the midpoint of the line on the i-axis, i.e. $\boldsymbol{I = \frac{1}{2} i_{short}}$, corresponding to the point where $\boldsymbol{V = \frac{V_0}{2}}$ on the graph.

Final answer. Each cell has emf $\boldsymbol{\varepsilon = V_0/3}$ (e.g. $\mathbf{2\ V}$ if $\mathbf{V_0 = 6\ V}$); power is maximum at $\boldsymbol{I = \frac{1}{2} i_{short}}$.

← *Back to Q204*

Q205

A cylindrical metallic wire is stretched to increase its length by 5%. Calculate the percentage change in its resistance.

[2007]

SOLUTION

When a wire is stretched, its volume AL stays constant, so increasing the length forces the cross-sectional area to fall in the same ratio. Because $R = \frac{\rho L}{A}$ and $A = \frac{\text{volume}}{L}$, the resistance varies as the square of the length.

Writing the resistance in terms of length and volume V:

$$R = \frac{\rho L}{A} = \frac{\rho L}{V/L} = \frac{\rho L^2}{V}$$

So $R \propto L^2$, which gives:

$$\frac{R'}{R} = \left(\frac{L'}{L}\right)^2$$

The length increases by 5%, so $L' = 1.05\,L$:

$$\frac{R'}{R} = (1.05)^2 = 1.1025$$

The fractional change in resistance is:

$$\frac{R' - R}{R} = 1.1025 - 1 = 0.1025$$

As a percentage:

$$\frac{\Delta R}{R} \times 100 = 10.25\%$$

Hence the resistance increases by about **10.25%** (which agrees with the small-change approximation $\frac{\Delta R}{R} \approx 2\,\frac{\Delta L}{L} = 2 \times 5\% = 10\%$).

TIP

For small percentage changes you can use $\frac{\Delta R}{R} \approx 2\,\frac{\Delta L}{L}$, but for a 5% change show the exact $(1.05)^2$ calculation to get the full 10.25% and secure the accuracy mark.

Final answer. The resistance increases by about **10.25%**.

← *Back to Q205*

Q206

A voltage of 30 V is applied across a carbon resistor with first, second and third rings of blue, black and yellow colours respectively. Calculate the value of current, in mA, through the resistor.

[2007]

SOLUTION

The value of a carbon resistor is read from its colour bands: the first two rings give the two significant figures and the third ring gives the power-of-ten multiplier. Using the standard colour code, blue = 6, black = 0 and yellow = multiplier 10^4.

First ring (blue) = 6, second ring (black) = 0, so the two significant figures give 60. The third ring (yellow) multiplies by 10^4:

$$R = 60 \times 10^4\ \Omega$$

$$R = 6 \times 10^5\ \Omega = 600\ \text{k}\Omega$$

The current through the resistor follows from Ohm's law with $\boldsymbol{V = 30\ V}$:

$$I = \frac{V}{R} = \frac{30}{6 \times 10^5}$$

$$I = 5 \times 10^{-5}\ A$$

Converting to milliamperes ($\mathbf{1\ mA = 10^{-3}\ A}$):

$$I = 5 \times 10^{-5}\ A = 0.05\ \text{mA}$$

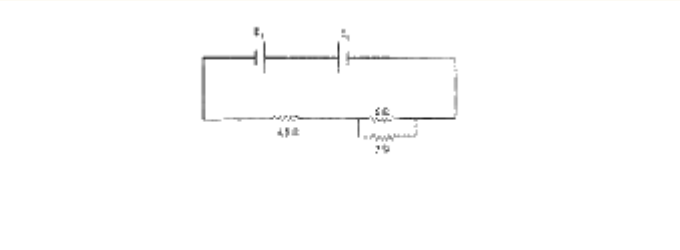
Hence the current through the resistor is **0.05 mA** (equal to 50 μA).

Final answer. The current through the resistor is **0.05 mA** (**50 μA**).

← *Back to Q206*

Q207

Two cells E_1 *and* E_2 *in the given circuit diagram have an emf of* 5 *V and* 9 *V and internal resistance of* 0.3 Ω *and* 1.2 Ω *respectively. Calculate the value of current flowing through the resistance of* 3 Ω.



[2006]

SOLUTION

The two cells E_1 and E_2 are connected in parallel and then drive the external 3 Ω resistor. A parallel combination of cells can be replaced by a single equivalent cell, after which Ohm's law gives the current through the external resistor.

Given: $\varepsilon_1 = 5\ V$, $r_1 = 0.3\ \Omega$; $\varepsilon_2 = 9\ V$, $r_2 = 1.2\ \Omega$; external $R = 3\ \Omega$.

The equivalent emf of two cells in parallel is the current-weighted average:

$$\varepsilon_{eq} = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$$

$$\varepsilon_{eq} = \frac{5(1.2) + 9(0.3)}{0.3 + 1.2} = \frac{6 + 2.7}{1.5}$$

$$\varepsilon_{eq} = \frac{8.7}{1.5} = 5.8\ V$$

The equivalent internal resistance is the parallel combination of r_1 and r_2:

$$r_{eq} = \frac{r_1 r_2}{r_1 + r_2} = \frac{0.3 \times 1.2}{0.3 + 1.2} = \frac{0.36}{1.5}$$

$$r_{eq} = 0.24\ \Omega$$

The current through the 3 Ω resistor is:

$$I = \frac{\varepsilon_{eq}}{R + r_{eq}} = \frac{5.8}{3 + 0.24}$$

$$I = \frac{5.8}{3.24} \approx 1.79\ A$$

Hence the current through the 3 Ω resistance is about **1.79 *A***.

Final answer. The current through the **3 Ω** resistor is about **1.79 *A***.

← *Back to Q207*

Q208

Write the mathematical relation between mobility and drift velocity of charge carriers in a conductor. Name the mobile charge carriers responsible for conduction of electric current in (i) an electrolyte (ii) an ionised gas.

[2006]

SOLUTION

Mobility measures how fast charge carriers drift per unit applied electric field. By definition, the magnitude of the drift velocity per unit field is the mobility.

The relation between mobility μ and drift velocity v_d is:

$$\mu = \frac{|v_d|}{E} \text{ or equivalently } v_d = \mu E$$

where E is the magnitude of the applied electric field. From the microscopic model $\boldsymbol{\mu = \frac{e\tau}{m}}$, and its SI unit is $m^2V^{-1}s^{-1}$.

Mobile charge carriers responsible for conduction:

(i) In an electrolyte: both positive ions (cations) and negative ions (anions) carry the current.

(ii) In an ionised gas: positive ions and free electrons (negative ions may also be present in some gases).

Hence $\boldsymbol{v_d = \mu E}$, with ions being the carriers in an electrolyte and electrons plus positive ions in an ionised gas.

Final answer. $v_d = \mu E$; carriers are cations and anions in an electrolyte, and electrons plus positive ions in an ionised gas.

← *Back to Q208*

Q209

Draw V-I graph for ohmic and non-ohmic materials. Give one example for each.

[2005]

SOLUTION

A material is ohmic if the current through it is directly proportional to the applied voltage at constant temperature, so that its resistance is constant; it is non-ohmic if this proportionality fails and the resistance depends on the voltage or current.

Ohmic material: $V \propto I$, so the V-I graph is a straight line passing through the origin. Its slope is constant and equals the resistance, $R = \frac{V}{I} = \textbf{constant}$.

Graph description: draw V on the vertical axis and I on the horizontal axis. The ohmic curve is a single straight line starting from the origin (0,0) and rising at a fixed angle; the constant slope of this line equals R.

Example of an ohmic material: a metallic conductor (such as a nichrome or copper wire) kept at constant temperature.

Non-ohmic material: the V-I relationship is non-linear, so the graph is a curve (not a straight line through the origin), and its slope, the dynamic resistance, changes with V or I.

Graph description: on the same axes, the non-ohmic curve bends, for example starting flat near the origin and then curving steeply upward, showing that the ratio $\frac{V}{I}$ is not constant.

Example of a non-ohmic material: a semiconductor diode (junction diode), whose current rises exponentially in forward bias and is very small in reverse bias.

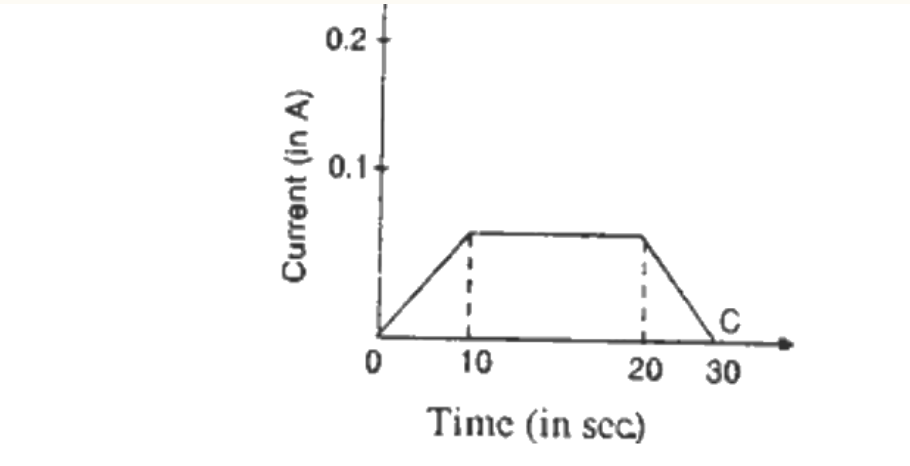
Hence the **straight-line graph through the origin marks an ohmic conductor (metal wire)** and the **curved graph marks a non-ohmic device (semiconductor diode)**.

Final answer. Ohmic: straight line through origin (metal wire); non-ohmic: curved ***V*-*I*** graph (semiconductor diode).

← *Back to Q209*

Q210

In a copper voltameter, a varying electric current, as shown in graph, is passed. The mass of copper deposited at the end of 30 seconds is m grams. Using the graph, find the value of e.c.e. (electrochemical equivalent) of copper in g/C.



[2003]

SOLUTION

By Faraday's first law of electrolysis, the mass of substance deposited at an electrode is directly proportional to the total charge that passes through the electrolyte. The electrochemical equivalent (ECE) is the mass deposited per unit charge.

Faraday's first law:

$$m = Z\,Q$$

where m is the mass deposited, Q is the total charge passed and Z is the electrochemical equivalent. Rearranging for Z:

$$Z = \frac{m}{Q}$$

Because the current varies with time, the total charge is the area under the current-time (I-t) graph over the 30 seconds:

$$Q = {}_0^{30} I \, \mathrm{dt} = \text{area under the } I\text{–}t \text{ graph}$$

For the standard trapezoidal pulse in the figure, the area is found by adding the areas of the triangular and rectangular portions (or by the trapezium rule). Calling this measured area Q coulombs, the electrochemical equivalent is:

$$Z = \frac{m}{Q} \; g/C$$

For example, if the area under the graph works out to $\boldsymbol{Q = 30\ C}$, then $\boldsymbol{Z = \frac{m}{30}\ g/C}$.

Hence the electrochemical equivalent of copper is $\boldsymbol{Z = \frac{m}{Q}\ g/C}$, where Q is the area under the given I-t graph.

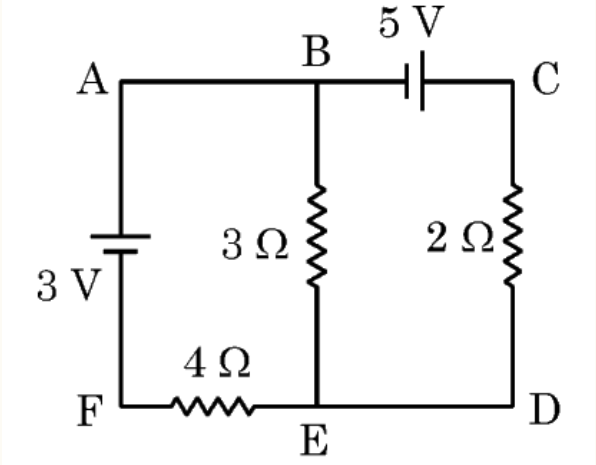
> **COMMON MISTAKE**
>
> *When current varies with time, the charge is never I× t with a single value; you must take the area under the I-t graph, or you will lose the method mark.*

Final answer. The ECE of copper is $\boldsymbol{Z = \frac{m}{Q}\ g/C}$, where $\boldsymbol{Q}$ is the charge given by the area under the $\boldsymbol{I}$-$\boldsymbol{t}$ graph.

← *Back to Q210*

Q211

(a) Explain the statement: "Current is a scalar although we represent current with an arrow".
(b) Use Kirchhoff's rules to find the current through 3 Ω resistor in the circuit shown in the figure:



[2026 • Set 55-2-1]

SOLUTION

(a) Electric current is defined as the rate of flow of charge through a cross-section of a conductor, $I = \frac{dq}{dt}$.

Charge q and time t are both scalars, so I carries only a magnitude and an algebraic sign. It is therefore a scalar quantity.

The arrow drawn on a wire in a circuit diagram does NOT represent a direction in three-dimensional space. It only fixes the chosen reference sense in which we agree to call the current positive. If the actual flow is opposite, I simply comes out negative.

The decisive reason is that currents do not obey the parallelogram (vector) law of addition. At a junction they combine by simple algebraic addition through Kirchhoff's junction rule, $\sum I = 0$, not by vector addition. Hence current is a scalar even though we mark it with an arrow.

(b) Method using Kirchhoff's rules (figure not reproduced here, so the procedure is shown; substitute the figure's actual cell polarities and resistor positions to get the number):

Step 1: Mark a current in every branch. Let I_1 come from one cell and I_2 from the other. By the junction rule the current in the 3 Ω resistor is $I_1 + I_2$.

Step 2: Apply the loop rule (KVL) to the loop containing the first cell, its series resistor and the 3 Ω branch, giving one linear equation in I_1, I_2.

Step 3: Apply the loop rule to the loop containing the second cell, its series resistor and the 3 Ω branch, giving a second linear equation.

Step 4: Solve the two simultaneous equations for I_1 and I_2, then the current through the 3 Ω resistor is $\mathbf{I_3 = I_1 + I_2}$.

For the standard CBSE version of this figure (a 5 *V* cell with a series 2 Ω, a 3 *V* cell with a series 4 Ω, both feeding a common 3 Ω), the equations are $\mathbf{5 = 5I_1 + 3I_2}$ and $\mathbf{3 = 3I_1 + 7I_2}$, which solve to $\mathbf{I_1 = 1\ A}$, $\mathbf{I_2 = 0}$, so $\mathbf{I_3 = 1\ A}$. With any other figure, redo Steps 2 and 3 with the correct signs.

TIP

In a multi-loop network always reduce unknown branch currents using the junction rule FIRST, so two loop equations are enough for two unknowns.

Final answer. Apply the junction rule once and the loop rule to two loops, then solve; for the common version of this figure the current through the 3 ohm resistor is 1 A, but the exact value depends on the figure supplied.

← *Back to Q211*

Q212

(a) Establish the relation between drift velocity of electrons (v_d) *and electric current* (*I*) *in a conductor.*
(b) How is v_d *affected when the length of the conductor is doubled, keeping the voltage applied across the conductor constant?*

[2026 • Set 55-3-1]

SOLUTION

(a) Consider a conductor of cross-sectional area *A* carrying current *I*. Let *n* be the number of free electrons per unit volume, each of charge *e*, drifting with drift velocity v_d.

In a small time Δt, every electron travels a distance $v_d \Delta t$. So all electrons lying within a cylinder of length $v_d \Delta t$ and cross-section *A* cross the section in time Δt.

Volume of this cylinder = $\boldsymbol{A\, v_d\, \Delta t}$.

Number of electrons in it = $\boldsymbol{n\, A\, v_d\, \Delta t}$.

Total charge crossing the section = $\Delta q = n\,e\,A\,v_d\,\Delta t$.

Therefore the current is

$$I = \frac{\Delta q}{\Delta t} = n\,e\,A\,v_d$$

This is the required relation between drift velocity and current.

(b) The drift velocity is given by $v_d = \frac{eE\tau}{m} = \frac{eV\tau}{mL}$, where V is the voltage across the conductor of length L.

With the applied voltage V kept constant and the length doubled, $L \to 2L$:

$$v_d' = \frac{eV\tau}{m(2L)} = \frac{1}{2}\cdot\frac{eV\tau}{mL} = \frac{v_d}{2}$$

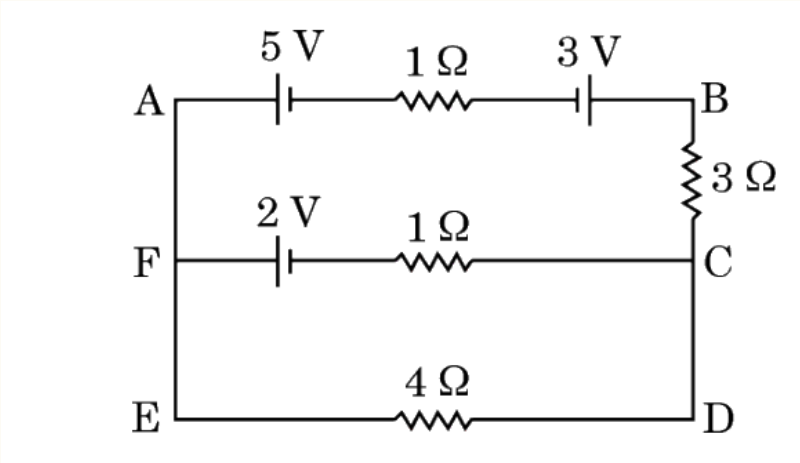
So the drift velocity is **halved** when the length of the conductor is doubled at constant voltage.

Final answer. I = neAv_d; doubling the length at constant voltage halves the drift velocity.

← *Back to Q212*

Q213

State Kirchhoff's rules. Using these rules, find the current flowing through branch FC in the given circuit.



[2026 • Set 55-4-1]

SOLUTION

Kirchhoff's junction rule (KCL): The algebraic sum of currents meeting at any junction is zero, $\sum I = 0$. It follows from conservation of electric charge, since no charge piles up at a junction.

Kirchhoff's loop rule (KVL): Around any closed loop the algebraic sum of changes in potential is zero, $\sum \varepsilon = \sum IR$. It follows from conservation of energy.

Finding the current through branch FC (the figure is not reproduced, so the working method is shown; insert the figure's actual emfs, resistor values and polarities to get the number):

Step 1: Assign a current to each branch. Let one cell drive I_1 and the other drive I_2; by the junction rule the current in the common branch FC is $I_1 + I_2$.

Step 2: Apply KVL to the left loop (first cell, its series resistor, and branch FC) to get one equation in I_1, I_2.

Step 3: Apply KVL to the right loop (second cell, its series resistor, and branch FC) to get a second equation.

Step 4: Solve the two simultaneous equations for I_1 and I_2.

Step 5: The required current through FC is $\mathbf{I_{FC} = I_1 + I_2}$. State its magnitude and direction (positive means along the assumed sense, negative means opposite).

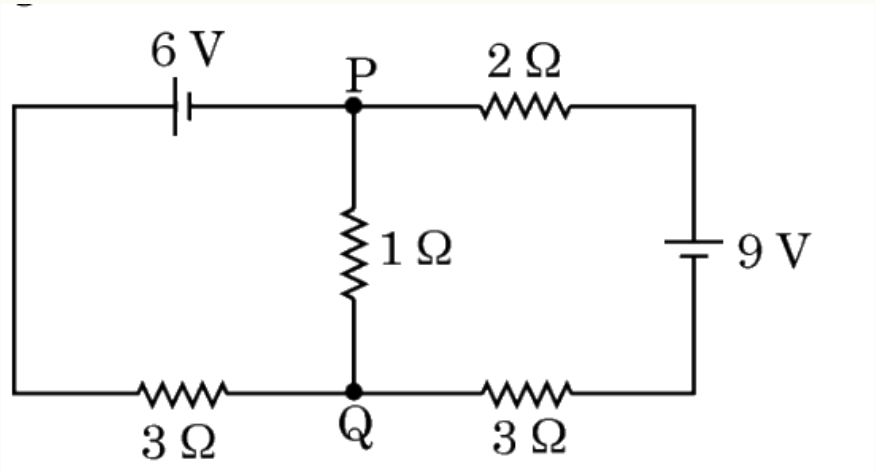
Care points for full marks: choose a consistent sign convention, write a potential rise across a cell from negative to positive terminal as $+\varepsilon$, a drop across a resistor in the direction of assumed current as $-IR$, and give the final answer with its unit (ampere) and direction.

Final answer. State KCL (charge conservation) and KVL (energy conservation), then write one junction equation and two loop equations and solve them for the current in branch FC; the numerical value follows from the figure's emfs and resistances.

← Back to Q213

Q214

(a) The two Kirchhoff's rules are based on law of conservation of two physical quantities. Name the quantities.
(b) Find magnitude and direction of current in 1 Ω resistor in the given circuit.



[2026 • Set 55-4-2]

SOLUTION

(a) Kirchhoff's junction rule is based on conservation of electric charge: the algebraic sum of currents at a junction is zero because no charge accumulates there.

Kirchhoff's loop rule is based on conservation of energy: as a unit charge is carried once around a closed loop, the algebraic sum of the potential changes is zero.

(b) Finding the current in the 1 Ω resistor (the figure is not reproduced, so the method is shown; substitute the figure's actual emfs, resistances and polarities for the number):

Step 1: Assign branch currents. Let the two sources drive I_1 and I_2; by the junction rule the current in the 1 Ω branch is $I = I_1 + I_2$.

Step 2: Apply KVL to the loop containing the first source, its series resistor and the 1 Ω branch, giving one equation in I_1, I_2.

Step 3: Apply KVL to the loop containing the second source, its series resistor and the 1 Ω branch, giving a second equation.

Step 4: Solve the two simultaneous equations for I_1 and I_2; then $I = I_1 + I_2$.

Step 5: State the magnitude (in amperes) and the direction. A positive value means the current flows along the assumed sense (here taken from P to Q); a negative value means it flows the opposite way.

For full marks, use a consistent sign convention throughout and quote the final current with its unit and direction.

Final answer. Junction rule conserves charge, loop rule conserves energy; assign branch currents, write one junction equation and two loop equations, solve for the 1 ohm current and give its magnitude in amperes and its direction (the value depends on the figure supplied).

← *Back to Q214*

Q215

(a) 'Current is a scalar quantity, although we represent current with an arrow.' Explain.
(b) Derive the balance condition of a Wheatstone Bridge.

[2026 • Set 55-4-3]

SOLUTION

(a) Electric current is defined as $I = \frac{dq}{dt}$, the rate of flow of charge. Both charge and time are scalars, so current carries only a magnitude and an algebraic sign and is itself a scalar.

The arrow on a circuit diagram is just a notational convention: it marks the assumed reference direction for the algebraic current. If the real flow is opposite, I comes out negative.

The decisive point is that currents do not add by the vector (parallelogram) law. At a junction they combine by simple algebraic addition (Kirchhoff's junction rule, $\sum I = 0$). This is why current is treated as a scalar despite the arrow.

(b) A Wheatstone bridge has four resistors P, Q, R, S forming a quadrilateral ABCD. A cell is connected across one diagonal (A to C) and a galvanometer G across the other diagonal (B to D).

Diagram: at vertex A the current I splits into I_1 (through P, along $A \to B$) and I_2 (through R, along $A \to D$). G connects B and D. P is in arm AB, Q in arm BC, R in arm AD, S in arm DC.

At balance the galvanometer shows no deflection, so no current flows through it ($\mathbf{I_g = 0}$). Hence the current I_1 flows straight through P then Q, and I_2 flows through R then S.

Applying the loop rule to loop ABDA (with $\mathbf{I_g = 0}$, so B and D are at the same potential):

$$I_1 P = I_2 R \cdots (1)$$

Applying the loop rule to loop BCDB:

$$I_1 Q = I_2 S \cdots (2)$$

Dividing equation (1) by equation (2):

$$\frac{I_1 P}{I_1 Q} = \frac{I_2 R}{I_2 S}$$

$$\frac{P}{Q} = \frac{R}{S}$$

This is the **balance condition of the Wheatstone bridge**.

> **TIP**
>
> At balance, state clearly that B and D are at equal potential; this is the line that earns the derivation mark.

Final answer. The Wheatstone bridge is balanced when P/Q = R/S.

← *Back to Q215*

Q216

(a) *Two batteries of emfs* 3 *V* & 6 *V and internal resistances* 0.2 Ω & 0.4 Ω *are connected in parallel. This combination is connected to a* 4 Ω *resistor. Find:*
(i) the equivalent emf of the combination

(ii) the equivalent internal resistance of the combination

(iii) the current drawn from the combination

OR

(b)

(i) A conductor of length l is connected across an ideal cell of emf E. Keeping the cell connected, the length of the conductor is increased to 2l *by gradually stretching it. If R and R' are initial and final values of resistance and* v_d *and* v_d' *are initial and final values of drift velocity, find the relation between (i) R' and R and (ii)* v_d' *and* v_d.

(ii) When electrons drift in a conductor from lower to higher potential, does it mean that all the 'free electrons' of the conductor are moving in the same direction?

[2025 • Set 55-1-1]

SOLUTION

(a) Two cells of emf $\varepsilon_1 = 3\ V$, $\varepsilon_2 = 6\ V$ with internal resistances $r_1 = 0.2\ \Omega$, $r_2 = 0.4\ \Omega$ are joined in parallel and feed a 4 Ω load.

(i) The equivalent emf of cells in parallel is

$$\varepsilon_{eq} = \frac{\varepsilon_1/r_1 + \varepsilon_2/r_2}{1/r_1 + 1/r_2}$$

$$\varepsilon_{eq} = \frac{3/0.2 + 6/0.4}{1/0.2 + 1/0.4} = \frac{15 + 15}{5 + 2.5} = \frac{30}{7.5}$$

$$\varepsilon_{eq} = 4\ V$$

(ii) The equivalent internal resistance is the parallel combination

$$r_{eq} = \frac{r_1\, r_2}{r_1 + r_2} = \frac{0.2 \times 0.4}{0.2 + 0.4} = \frac{0.08}{0.6}$$

$$r_{eq} \approx 0.13\ \Omega$$

(iii) The current drawn from the combination is

$$I = \frac{\varepsilon_{eq}}{R + r_{eq}} = \frac{4}{4 + 0.133} = \frac{4}{4.133}$$

$$I \approx 0.97\ A$$

OR

(b)(i) A wire of length l is stretched to length 2l while connected to an ideal cell of emf E (so $V = E$ stays constant). Stretching conserves volume, so $A \times l = A' \times 2l$ $\Rightarrow A' = \frac{A}{2}$.

Initial resistance $R = \frac{\rho\, l}{A}$. Final resistance

$$R' = \frac{\rho(2l)}{A/2} = \frac{4\rho\, l}{A} = 4R$$

So $R' = 4R$.

For drift velocity, $v_d = \frac{eE\tau}{mL} = \frac{eV\tau}{mL}$. With V constant and $L \to 2l$:

$$v_d' = \frac{eV\tau}{m(2l)} = \frac{v_d}{2}$$

So $v_d' = \frac{v_d}{2}$.

(ii) No. The free electrons are in continuous random thermal motion in all directions with very high speeds. When a field is applied they acquire only a very small net drift superposed on this random motion. So it is not true that all the free electrons move in the same direction; only the average (drift) of the whole collection is biased toward higher potential.

Final answer. (a) eps_eq = 4 V, r_eq ~ 0.13 ohm, I ~ 0.97 A. (b) R' = 4R, v_d' = v_d/2, and not all electrons move in one direction.

← *Back to Q216*

Q217

(a) A cell of e.m.f. E and internal resistance r is connected with a variable external resistance R and a voltmeter showing potential drop V across R. Obtain the relationship between V, E, R and r.
(b) Draw the shape of the graph showing the variation of terminal voltage V of the cell as a function of current I drawn from it. How one can determine the e.m.f. of the cell and its internal resistance from this graph?

[2025 • Set 55-2-1]

SOLUTION

(a) A cell of emf E and internal resistance r drives current through an external resistance R, and the voltmeter reads the potential drop V across R.

The current in the circuit is

$$I = \frac{E}{R + r}$$

The terminal voltage equals the drop across the load:

$$V = \mathrm{IR} = \frac{\mathrm{ER}}{R + r}$$

This can be rewritten by noting $\boldsymbol{E = I(R + r) = IR + Ir = V + Ir}$, giving

$$V = E - \mathrm{Ir}$$

This is the required relation between V, E, R and r.

(b) The equation $\boldsymbol{V = E - Ir}$ is the equation of a straight line in the V versus I plane.

Graph: plot V on the y-axis and I on the x-axis. The graph is a straight line that slopes downward (negative slope). It cuts the V-axis (at $\boldsymbol{I = 0}$) at the value E, and it cuts the I-axis (at $\boldsymbol{V = 0}$) at $\boldsymbol{I = E/r}$.

Determining E: the intercept on the V-axis, where $\boldsymbol{I = 0}$ (open circuit), gives $\boldsymbol{V = E}$. So the y-intercept directly gives the emf of the cell.

Determining r: the magnitude of the slope of the line equals the internal resistance, since slope $= \dfrac{\Delta V}{\Delta I} = -\boldsymbol{r}$. Thus

$$r = \left|\frac{\Delta V}{\Delta I}\right|$$

Equivalently, the intercept on the I-axis gives the short-circuit current $\mathbf{I_{sc} = E/r}$, from which $\boldsymbol{r = E/I_{sc}}$.

Final answer. V = E - Ir; the V-axis intercept gives E and the slope magnitude gives r.

← Back to Q217

Q218

(a) Define resistivity of a conductor. Discuss its dependence on temperature of the conductor and draw a plot of resistivity of copper as a function of temperature.
(b) (i) "A low voltage battery from which high current is required must have low internal resistance." Justify.
(ii) "A high voltage battery must have a large internal resistance." Justify.

[2025 • Set 55-2-2]

SOLUTION

(a) Resistivity of a material is the resistance of a conductor of that material having unit length and unit cross-sectional area. From $R = \frac{\rho L}{A}$ we get

$$\rho = \frac{RA}{L}$$

Its SI unit is the ohm-metre ($\Omega\, m$). It is a property of the material alone, independent of the dimensions of the conductor.

Temperature dependence: since $\rho = \frac{m}{ne^2\tau}$, and for a metal the number density n is almost constant while the relaxation time τ decreases as temperature rises (the lattice ions vibrate more, so electrons collide more often), the resistivity increases with temperature. Over a moderate range it is nearly linear:

$$\rho_T = \rho_0[1 + \alpha(T - T_0)]$$

Graph for copper: plot ρ (y-axis) against absolute temperature T (x-axis). The curve rises with a positive slope and is almost a straight line over the ordinary range; at very low temperatures it bends and approaches a small non-zero value (residual resistivity), not zero.

(b)(i) A low-voltage battery from which a high current is required must have low internal resistance. The current is $I = \frac{E}{R_{ext} + r}$. With E small, a large I is possible only if the total resistance $R_{ext} + r$ is small. A large r would drop most of the small emf inside the cell itself, leaving too little to drive a large current through the load. Hence r must be small.

(b)(ii) A high-voltage battery must have a large internal resistance. If E is very large and r were small, the short-circuit current $I = E/r$ would become dangerously high in case of an accidental short. A large r keeps this current down to safe values. (For example a Van de Graaff generator produces a very high voltage but supplies only a tiny, safe current.)

Final answer. rho = RA/L; for metals rho rises with T; high current needs low r, while a safe high-voltage source needs large r.

← *Back to Q218*

Q219

(a) Define Electrical conductivity. Obtain the expression of electrical conductivity of a conductor in terms of number density and relaxation time of free electrons.
(b) Explain qualitative change in resistivity of a conductor with temperature using expression obtained in (a).

[2025 • Set 55-2-3]

SOLUTION

(a) Electrical conductivity σ is the reciprocal of resistivity; it is the proportionality constant in the microscopic form of Ohm's law, $\vec{j} = \sigma \vec{E}$, where $\vec{j}$ is the current density and $\vec{E}$ the applied field.

Under a field $\vec{E}$, each free electron of charge $-e$ and mass m experiences a force and acquires a drift velocity in time τ (the relaxation time) between collisions:

$$\vec{v}_d = \frac{-e\vec{E}\tau}{m}$$

The current density for n free electrons per unit volume is

$$\vec{j} = n(-e)\vec{v}_d = n(-e)\left(\frac{-e\vec{E}\tau}{m}\right)$$

$$\vec{j} = \frac{ne^2\tau}{m}\vec{E}$$

Comparing with $\vec{j} = \sigma\vec{E}$, the conductivity is

$$\sigma = \frac{ne^2\tau}{m}$$

(b) Resistivity is the reciprocal of conductivity:

$$\rho = \frac{1}{\sigma} = \frac{m}{ne^2\tau}$$

In a metal the number density n of free electrons is essentially fixed. As temperature increases, the metal ions vibrate more vigorously, so the electrons collide more frequently and the relaxation time τ decreases. Since $\rho \propto \frac{1}{\tau}$, the resistivity increases with temperature. This is why metals have a positive temperature coefficient of resistivity.

Final answer. sigma = ne^2 tau / m; since rho = m/(ne^2 tau) and tau falls with rising T, resistivity of a metal increases with temperature.

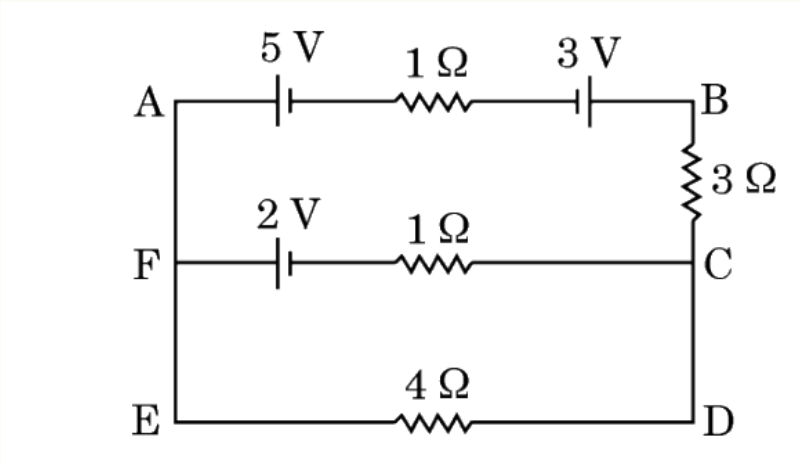
← *Back to Q219*

Q220

(a)

(i) Derive an expression for the resistivity of a conductor in terms of number density of free electrons and relaxation time.

(ii) The figure shows the plot of current through a cross-section of wire over two different time intervals. Compare the charges (Q_1 and Q_2) that pass through the cross-section during these time intervals.



OR

(b)

(i) A battery of emf E and internal resistance r is connected to a variable external resistance R.

(I) Obtain the expression for current I in the circuit and the value of maximum current the battery can supply.

(II) Obtain the terminal voltage V across the battery and its maximum possible value.

(ii) The above battery sends a current I_1 *when* $\boldsymbol{R = R_1}$ *and a current* I_2 *when* $\boldsymbol{R = R_2}$*. Obtain the internal resistance of the battery in terms of* I_1, I_2, R_1 *and* R_2.

[2025 • Set 55-4-1]

SOLUTION

(a)(i) Under an applied field $\vec{E}$, each free electron (charge $-e$, mass m) drifts with velocity $\vec{v}_d = \dfrac{-e\vec{E}\tau}{m}$, where τ is the relaxation time.

The current density for n free electrons per unit volume is

$$\vec{j} = n(-e)\vec{v}_d = \frac{ne^2\tau}{m}\vec{E}$$

Comparing with $\vec{j} = \sigma\vec{E}$ gives $\sigma = \dfrac{ne^2\tau}{m}$, so the resistivity is

$$\rho = \frac{1}{\sigma} = \frac{m}{ne^2\tau}$$

(a)(ii) The charge passing through a cross-section equals the area under the current-time (I-t) graph, since $Q = \int I\,dt$.

$\mathbf{Q_1}$ = area under the graph in the first interval, $\mathbf{Q_2}$ = area under the graph in the second interval. Reading the two areas from the figure and comparing them gives the ratio; the interval with the greater enclosed area carries the greater charge. (For the standard version of this figure the two areas are equal, giving $\mathbf{Q_1 = Q_2}$.)

OR

(b)(i)(I) For a cell of emf E and internal resistance r feeding external resistance R:

$$I = \frac{E}{R + r}$$

The current is maximum when R is least, i.e. $\boldsymbol{R = 0}$ (short circuit):

$$\mathrm{I}_{\max} = \frac{E}{r}$$

(b)(i)(II) The terminal voltage is

$$V = \mathrm{IR} = \frac{\mathrm{ER}}{R + r} = E - \mathrm{Ir}$$

V is maximum when $\boldsymbol{R \to \infty}$ (open circuit, $\boldsymbol{I = 0}$):

$$\mathrm{V}_{\max} = E$$

(b)(ii) The same battery gives I_1 with $\boldsymbol{R = \mathrm{R}_1}$ and I_2 with $\boldsymbol{R = \mathrm{R}_2}$. Using $\boldsymbol{E = I(R + r)}$ for each:

$$E = \mathrm{I}_1(\mathrm{R}_1 + r) \cdots (1)$$

$$E = \mathrm{I}_2(\mathrm{R}_2 + r) \cdots (2)$$

Equating (1) and (2):

$$\mathrm{I}_1\,\mathrm{R}_1 + \mathrm{I}_1\,r = \mathrm{I}_2\,\mathrm{R}_2 + \mathrm{I}_2\,r$$

$$\mathrm{I}_1\,r - \mathrm{I}_2\,r = \mathrm{I}_2\,\mathrm{R}_2 - \mathrm{I}_1\,\mathrm{R}_1$$

$$r(\mathrm{I}_1 - \mathrm{I}_2) = \mathrm{I}_2\,\mathrm{R}_2 - \mathrm{I}_1\,\mathrm{R}_1$$

$$r = \frac{\mathrm{I}_2\,\mathrm{R}_2 - \mathrm{I}_1\,\mathrm{R}_1}{\mathrm{I}_1 - \mathrm{I}_2}$$

TIP

When two areas under an I-t graph look like simple geometric shapes, compute each as a triangle or rectangle rather than guessing which is bigger by eye.

Final answer. rho = m/(ne^2 tau); charge = area under the I-t graph; I = E/(R+r), I_max = E/r, V = ER/(R+r), V_max = E, and r = (I_2 R_2 - I_1 R_1)/(I_1 - I_2).

← *Back to Q220*

Q221

What is the difference between 'emf' and 'terminal voltage' of a cell?
Two cells of emfs E_1 *and* E_2 *and internal resistances* r_1 *and* r_2 *are connected in parallel. Derive an expression for the emf and internal resistance of the equivalent cell.*

[2025 • Set 55-6-1]

SOLUTION

The emf of a cell is the potential difference between its two terminals when it is in the open circuit, that is when no current is being drawn from it. It is the maximum voltage the cell can supply and depends only on the chemical nature of the electrodes and electrolyte, not on the external circuit.

The terminal voltage of a cell is the potential difference between its terminals when it is delivering current to an external load. It is given by $V = \varepsilon - Ir$, so it is always less than the emf (during discharge) and falls as the current drawn increases.

Two cells in parallel: let the cells of emfs ε_1 and ε_2 with internal resistances r_1 and r_2 be joined with their positive terminals at a common point and negative terminals at another common point. Let V be the common terminal voltage across the combination, and let I_1 and I_2 be the currents supplied by the two cells.

For the first cell:

$$V = \varepsilon_1 - I_1 r_1$$

$$\Rightarrow I_1 = \frac{\varepsilon_1 - V}{r_1}$$

For the second cell:

$$V = \varepsilon_2 - I_2 r_2$$

$$\Rightarrow I_2 = \frac{\varepsilon_2 - V}{r_2}$$

The total current drawn from the combination is:

$$I = I_1 + I_2 = \frac{\varepsilon_1 - V}{r_1} + \frac{\varepsilon_2 - V}{r_2}$$

$$I = \left(\frac{\varepsilon_1}{r_1} + \frac{\varepsilon_2}{r_2}\right) - V\left(\frac{1}{r_1} + \frac{1}{r_2}\right)$$

Solving for V:

$$V = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2} - I\,\frac{r_1 r_2}{r_1 + r_2}$$

Comparing this with the single-cell equation $V = \varepsilon_{eq} - I\,r_{eq}$, we get:

$$\varepsilon_{eq} = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$$

$$r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$$

Thus the **equivalent emf** is $\varepsilon_{eq} = \dfrac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$ and the **equivalent internal resistance** is $r_{eq} = \dfrac{r_1 r_2}{r_1 + r_2}$.

COMMON MISTAKE

Remember the equivalent emf uses cross terms ($\varepsilon_1 r_2 + \varepsilon_2 r_1$), not $\varepsilon_1 r_1 + \varepsilon_2 r_2$. A common slip is mixing the subscripts.

Final answer. Equivalent emf $\varepsilon_{eq} = \dfrac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$ and internal resistance $r_{eq} = \dfrac{r_1 r_2}{r_1 + r_2}$.

← *Back to Q221*

Q222

(a) Define the term 'drift velocity' of conduction electrons in a conductor.
(b) A conductor of length l and area of cross-section A is connected across an ideal battery of emf V. Derive the formula for the current density in terms of relaxation time τ.

[2025 • Set 55-6-2]

SOLUTION

(a) Drift velocity is the average velocity with which the free (conduction) electrons in a conductor move under the influence of an applied external electric field, opposite to the direction of the field. It is superimposed on the much larger random thermal velocities of the electrons, whose average is zero. In metals its magnitude is very small, of the order of 10^{-4} to 10^{-3} $m\ s^{-1}$.

(b) The conductor of length l is connected across an ideal battery of emf V. The uniform electric field set up inside the conductor is:

$$E = \frac{V}{l}$$

Each free electron (charge e, mass m) experiences a force $\boldsymbol{F} = \mathbf{eE}$ and hence an acceleration:

$$a = \frac{eE}{m}$$

During the average time between two successive collisions, the relaxation time τ, the electron gains a drift velocity:

$$v_d = a\,\tau = \frac{eE\tau}{m}$$

Substituting $\boldsymbol{E = V/l}$:

$$v_d = \frac{eV\tau}{ml}$$

If n is the number of free electrons per unit volume, the current density is $\boldsymbol{j = n\,e\,v_d}$:

$$j = n\,e \cdot \frac{eE\tau}{m}$$

$$j = \frac{n\,e^2\,\tau}{m}\,E$$

In terms of the applied emf:

$$j = \frac{n\,e^2\,\tau}{m} \cdot \frac{V}{l}$$

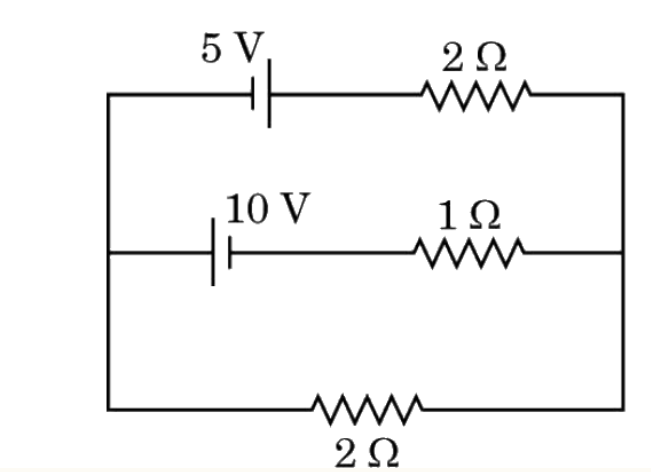
This has the form $j = \sigma E$ where the **conductivity** is $\sigma = \dfrac{n e^2 \tau}{m}$. Hence the current density in terms of the relaxation time is $\mathbf{j} = \dfrac{n e^2 \tau}{m} \mathbf{E}$.

Final answer. The current density is $j = \dfrac{n e^2 \tau}{m} E$, with conductivity $\sigma = \dfrac{n e^2 \tau}{m}$.

← *Back to Q222*

Q223

State Kirchhoff's laws. Apply these laws to find the values of current flowing in the three branches of the given circuit.



[2025 • Set 55-6-3]

SOLUTION

Kirchhoff's junction (first) law: The algebraic sum of the currents meeting at any junction in an electrical network is zero, that is $\sum I = 0$. It is based on conservation of charge.

Kirchhoff's loop (second) law: The algebraic sum of the changes in potential around any closed loop of the network is zero, that is $\sum \varepsilon = \sum \mathbf{IR}$. It is based on conservation of energy.

Applying to the given circuit with the 5 *V* and 4 *V* cells and resistors of 2 Ω, 1 Ω and 2 Ω. Let the currents in the three branches be I_1, I_2 and I_3, with I_3 flowing through the middle (1 Ω) branch.

Junction rule at the central node gives:

$$I_3 = I_1 + I_2$$

Left loop (containing the 5 V cell, 2 Ω in its branch and the 1 Ω middle resistor):

$$5 = 2\,I_1 + 1\cdot I_3$$

Right loop (containing the 4 V cell, 2 Ω in its branch and the 1 Ω middle resistor):

$$4 = 2\,I_2 + 1\cdot I_3$$

Substituting $\mathbf{I_3 = I_1 + I_2}$ into both loop equations:

$$5 = 2\,I_1 + (I_1 + I_2) = 3\,I_1 + I_2$$

$$4 = 2\,I_2 + (I_1 + I_2) = I_1 + 3\,I_2$$

Solving the two simultaneous equations. Multiply the first by 3:

$$15 = 9\,I_1 + 3\,I_2$$

Subtract the second equation $\mathbf{4 = I_1 + 3\,I_2}$:

$$11 = 8\,I_1 \Rightarrow I_1 = \frac{11}{8} = 1.375\ A$$

From $\mathbf{5 = 3\,I_1 + I_2}$:

$$I_2 = 5 - 3(1.375) = 5 - 4.125 = 0.875\ A$$

$$I_3 = I_1 + I_2 = 1.375 + 0.875 = 2.25\ A$$

Hence the branch currents are $\mathbf{I_1 = 1.375\ A}$, $\mathbf{I_2 = 0.875\ A}$ and $\mathbf{I_3 = 2.25\ A}$.

TIP

First state both laws clearly (worth marks on their own), then write the junction equation before the loop equations. Always note your assumed current directions; a negative answer simply means the real direction is opposite.

Final answer. The three branch currents are $\mathbf{I_1 = 1.375\ A}$, $\mathbf{I_2 = 0.875\ A}$ and $\mathbf{I_3 = 2.25\ A}$.

← Back to Q223

Q224

Three cells A, B and C of emfs 2 V, 3 V and 5 V respectively are connected in parallel to each other. Their internal resistances are 5 Ω, 5 Ω and 1 Ω respectively. Calculate the currents flowing through the cells A, B and C.

[2025 • Set 55-7-1]

SOLUTION

The three cells are connected in parallel with no external resistance, so they settle to a common terminal voltage V across the combination, and since there is no external load the net current drawn from the combination is zero.

For each cell the current it sends out is $\mathbf{I_i = \frac{\varepsilon_i - V}{r_i}}$, and the condition $\mathbf{I_A + I_B + I_C = 0}$ fixes V.

The common terminal voltage is:

$$V = \frac{\frac{\varepsilon_A}{r_A} + \frac{\varepsilon_B}{r_B} + \frac{\varepsilon_C}{r_C}}{\frac{1}{r_A} + \frac{1}{r_B} + \frac{1}{r_C}}$$

Substituting $\mathbf{\varepsilon_A = 2\ V}$, $\mathbf{\varepsilon_B = 3\ V}$, $\mathbf{\varepsilon_C = 5\ V}$ and $\mathbf{r_A = 5\ \Omega}$, $\mathbf{r_B = 5\ \Omega}$, $\mathbf{r_C = 1\ \Omega}$:

$$V = \frac{\frac{2}{5} + \frac{3}{5} + \frac{5}{1}}{\frac{1}{5} + \frac{1}{5} + \frac{1}{1}}$$

$$V = \frac{0.4 + 0.6 + 5}{0.2 + 0.2 + 1}$$

$$V = \frac{6}{1.4} = \frac{30}{7} \approx 4.29\ V$$

Now the current through each cell, $\mathbf{I_i = \frac{\varepsilon_i - V}{r_i}}$.

Through cell A:

$$I_A = \frac{2 - \frac{30}{7}}{5} = \frac{-\frac{16}{7}}{5} = -\frac{16}{35} \approx -0.46\ A$$

Through cell B:

$$I_B = \frac{3 - \frac{30}{7}}{5} = \frac{-\frac{9}{7}}{5} = -\frac{9}{35} \approx -0.26\ A$$

Through cell C:

$$I_C = \frac{5 - \frac{30}{7}}{1} = \frac{5}{7} \approx 0.71\ A$$

The negative signs for I_A and I_B mean these currents actually flow into cells A and B, so cells A and B are being charged, while cell C discharges. As a check, $-\frac{16}{35} - \frac{9}{35} + \frac{25}{35} = 0$, confirming the junction rule.

Hence $\mathbf{I_A \approx 0.46\ A}$ (into A), $\mathbf{I_B \approx 0.26\ A}$ (into B) and $\mathbf{I_C \approx 0.71\ A}$ (out of C).

Final answer. $\mathbf{I_A = -16/35 \approx -0.46\ A}$ and $\mathbf{I_B = -9/35 \approx -0.26\ A}$ (both charging), $\mathbf{I_C = 5/7 \approx 0.71\ A}$ (discharging).

← *Back to Q224*

Q225

Define 'current density'. Is it a scalar or a vector? An electric field $\vec{E}$ *is maintained in a metallic conductor. If n be the number of electrons (mass m, charge −e) per unit volume in the conductor and* τ *its relaxation time, show that the current density* $\vec{j} = \sigma \vec{E}$, *where* $\sigma = \frac{n e^2 \tau}{m}$.

[2024 • Set 55-1-1]

SOLUTION

Current density at a point is the amount of current flowing per unit area of cross-section held perpendicular to the direction of flow of current at that

point. Its magnitude is $j = \frac{I}{A}$ and its direction is that of the conventional current (the motion of positive charge). It is a vector quantity, $\vec{j}$.

Derivation of $\vec{j} = \sigma\vec{E}$. Consider a metallic conductor with an electric field $\vec{E}$ maintained inside it. Each free electron has charge $-e$ and mass m, and there are n such electrons per unit volume.

The force on each electron is:

$$\vec{F} = -e\vec{E}$$

Hence its acceleration is:

$$\vec{a} = \frac{\vec{F}}{m} = -\frac{e\vec{E}}{m}$$

After each collision an electron starts afresh; averaged over the relaxation time τ (the mean time between collisions), the electron acquires a drift velocity:

$$\vec{v}_d = \vec{a}\,\tau = -\frac{e\tau}{m}\vec{E}$$

The current density is the charge crossing unit area per unit time, $\vec{j} = n(-e)\vec{v}_d$:

$$\vec{j} = n(-e)\left(-\frac{e\tau}{m}\vec{E}\right)$$

$$\vec{j} = \frac{n\,e^2\,\tau}{m}\,\vec{E}$$

Comparing with the microscopic form of Ohm's law $\vec{j} = \sigma\vec{E}$, the conductivity is:

$$\sigma = \frac{n\,e^2\,\tau}{m}$$

This is the required result, showing $\vec{j} = \sigma\vec{E}$ with $\sigma = \frac{n\,e^2\,\tau}{m}$.

COMMON MISTAKE

Note the two minus signs (charge $-e$ and drift velocity $-\frac{e\tau}{m}E$) multiply to give a positive j parallel to E. Dropping one sign is a frequent error.

Final answer. Current density is a vector; $\vec{j} = \sigma \vec{E}$ with conductivity $\sigma = \dfrac{n e^2 \tau}{m}$.

← Back to Q225

Q226

What is a Wheatstone bridge? Obtain the necessary conditions under which the Wheatstone bridge is balanced.

[2024 • Set 55-1-1]

SOLUTION

A Wheatstone bridge is an arrangement of four resistors P, Q, R and S connected to form the four arms of a quadrilateral ABCD. A galvanometer G is connected across one diagonal (between B and D) and a battery with a key is connected across the other diagonal (between A and C). It is used to measure an unknown resistance accurately.

Diagram (in words): the four resistors form a closed quadrilateral with corners A, B, C, D. Arm AB has P, arm BC has Q, arm AD has R and arm DC has S. The galvanometer G joins the junction B to the junction D (one diagonal). The cell ε with key K is connected between A and C (the other diagonal).

Balance condition: The bridge is said to be balanced when there is no deflection in the galvanometer, that is the current through the galvanometer $\mathbf{I_g = 0}$. In this condition points B and D are at the same potential.

Let I_1 be the current through P and Q (since no current branches off through G, the same I_1 flows through Q), and I_2 the current through R and S.

Applying Kirchhoff's loop rule to loop ABDA (with $\mathbf{I_g = 0}$):

$$I_1 P = I_2 R$$

Applying Kirchhoff's loop rule to loop BCDB (with $\mathbf{I_g = 0}$):

$$I_1 Q = I_2 S$$

Dividing the first equation by the second:

$$\frac{I_1 P}{I_1 Q} = \frac{I_2 R}{I_2 S}$$

$$\frac{P}{Q} = \frac{R}{S}$$

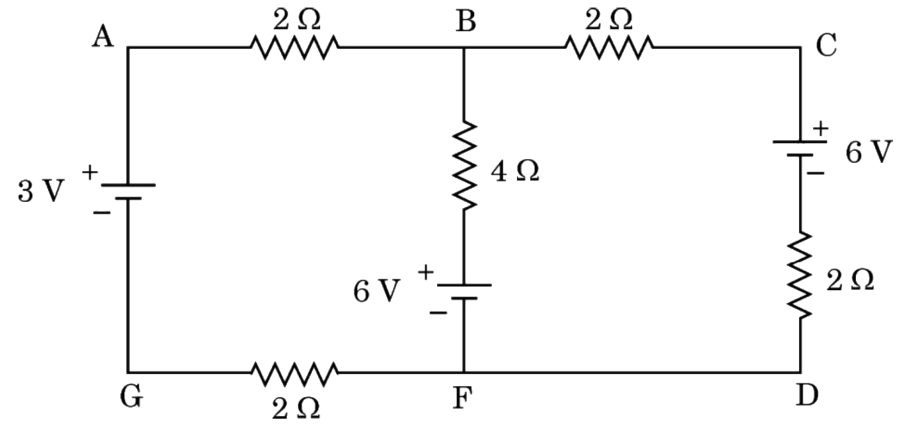
This is the **balance condition** of the Wheatstone bridge. When it is satisfied, no current flows through the galvanometer, irrespective of the emf of the cell and the resistance of the galvanometer.

Final answer. At balance ($I_g = 0$) the Wheatstone bridge satisfies $\frac{P}{Q} = \frac{R}{S}$.

← *Back to Q226*

Q227

The figure shows a circuit with three ideal batteries. Find the magnitude and direction of currents in the branches AG, BF and CD.



[2024 • Set 55-2-1]

SOLUTION

We use Kirchhoff's laws. Junction (first) law: the algebraic sum of currents at any junction is zero, $\sum I = 0$ (conservation of charge). Loop (second) law: the algebraic sum of emfs equals the algebraic sum of IR drops around any closed loop, $\sum \varepsilon = \sum IR$ (conservation of energy).

Assign currents I_{AG}, I_{BF} and I_{CD} in the three branches, with directions assumed from the figure. Mark every battery emf and every resistance exactly as labelled in the given figure.

Junction rule at the common node where the three branches meet:

$$I_{AG} + I_{BF} + I_{CD} = 0$$

Loop rule for the loop formed by branches AG and BF:

$$\varepsilon_{AG} - \varepsilon_{BF} = I_{AG}\,R_{AG} - I_{BF}\,R_{BF}$$

Loop rule for the loop formed by branches BF and CD:

$$\varepsilon_{BF} - \varepsilon_{CD} = I_{BF}\,R_{BF} - I_{CD}\,R_{CD}$$

These three equations are then solved simultaneously for I_{AG}, I_{BF} and I_{CD}, substituting the actual emfs and resistances printed in the figure. A negative value of any current means that branch's current is opposite to the assumed direction. The final answer is the magnitude and direction of each of the three branch currents.

Note: the numerical values must be read off the figure. The branch emfs and resistances are not all equal, so the currents are in general non-zero.

TIP

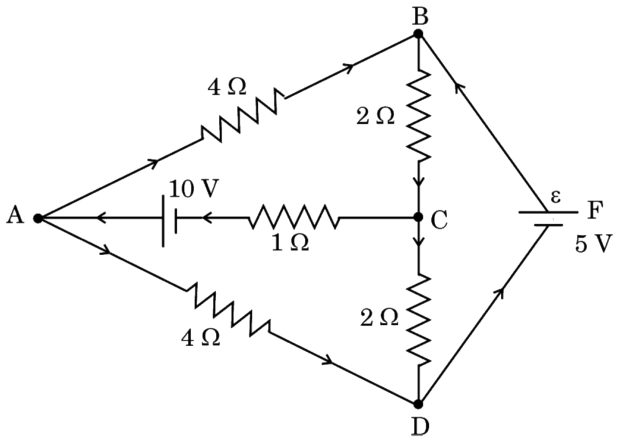
When identical cells of equal emf are connected in parallel and oppose one another symmetrically, no current circulates. Spot the symmetry before grinding through full simultaneous equations.

Final answer. Apply $\mathbf{I_{AG} + I_{BF} + I_{CD} = 0}$ together with two loop equations $\boldsymbol{\varepsilon_{AG} - \varepsilon_{BF} = I_{AG}R_{AG} - I_{BF}R_{BF}}$ and $\boldsymbol{\varepsilon_{BF} - \varepsilon_{CD} = I_{BF}R_{BF} - I_{CD}R_{CD}}$, then solve for the three branch currents using the figure's emf and resistance values; a negative current means the assumed direction is reversed.

← *Back to Q227*

Q228

Determine the current in branches AB, AC *and* BC *of the network shown in figure.*



[2024 • Set 55-3-1]

SOLUTION

We apply Kirchhoff's laws to the network. Junction (first) law: the algebraic sum of currents meeting at any node is zero, $\sum I = 0$. Loop (second) law: around any closed loop the algebraic sum of emfs equals the algebraic sum of IR drops, $\sum \varepsilon = \sum IR$.

Let I_{AB}, I_{AC} and I_{BC} be the currents in the three branches, with directions assumed from the figure. Read the emf of the cell and every resistance directly from the network shown.

Junction rule at node *A* (current *I* from the cell splits):

$$I = I_{AB} + I_{AC}$$

Junction rule at node *B* relates I_{AB} and I_{BC} according to the branch connections in the figure.

Loop rule around loop ABCA (sum of IR drops minus emfs equals zero):

$$I_{AB}R_{AB} + I_{BC}R_{BC} - I_{AC}R_{AC} = 0$$

Loop rule around the loop containing the cell gives a second independent equation involving the cell emf.

Solving these simultaneous equations with the figure's actual resistance and emf values gives I_{AB}, I_{AC} and I_{BC}. A negative result for any branch means its current is opposite to the assumed direction. The branch currents come out of the order of a fraction of an ampere.

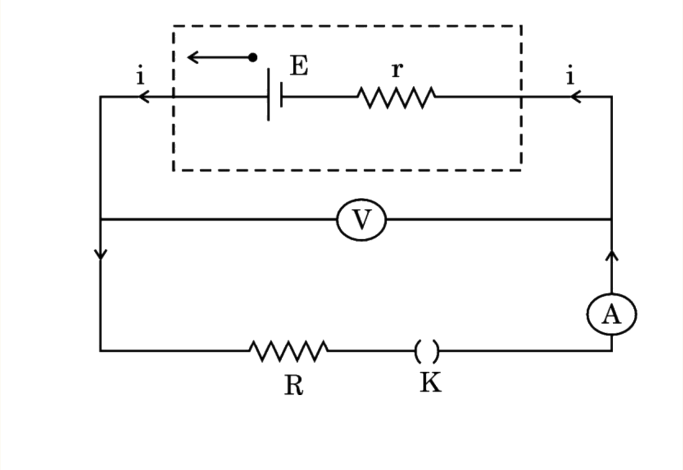
Note: the exact numbers must be taken from the figure; do not assume specific resistor values that are not labelled.

Final answer. Set up $I = I_{AB} + I_{AC}$ at node A, a junction relation at B, and loop equations $I_{AB}R_{AB} + I_{BC}R_{BC} - I_{AC}R_{AC} = 0$ plus the cell loop, then solve simultaneously using the figure's labelled emf and resistance values to get the three branch currents (typically a fraction of an ampere each).

← *Back to Q228*

Q229

A battery of unknown emf E and internal resistance r is connected in a circuit as shown in the figure. When the key (K) is open, the voltmeter reads 10.0 *V and ammeter reads* 0 *A. In the closed circuit, the voltmeter reads* 6.0 *V and ammeter reads* 2.0 *A. Calculate:*
(a) emf of the battery,
(b) internal resistance of the battery (r), and
(c) external resistance (R).



[2024 • Set 55-3-2]

SOLUTION

When the key *K* is open, no current flows through the circuit, so the ammeter reads 0 *A* and there is no Ir drop inside the battery. The voltmeter therefore reads the open-circuit potential difference, which equals the emf of the battery.

(a) EMF of the battery:

$$\varepsilon = V_{\text{open}} = 10.0\ V$$

(b) When the key K is closed, the current is $\boldsymbol{I = 2.0\ A}$ and the terminal voltage (voltmeter reading) is $\boldsymbol{V = 6.0\ V}$. The terminal voltage relates to the emf by:

$$V = \varepsilon - I\,r$$

Substituting the values:

$$6.0 = 10.0 - (2.0)\,r$$

$$2.0\,r = 10.0 - 6.0 = 4.0$$

$$r = \frac{4.0}{2.0} = 2.0\ \Omega$$

(c) The external resistance R carries the same current I and has the terminal voltage V across it, so by Ohm's law:

$$V = I\,R$$

$$R = \frac{V}{I} = \frac{6.0}{2.0} = 3.0\ \Omega$$

Hence $\boldsymbol{\varepsilon = 10.0\ V}$, $\boldsymbol{r = 2.0\ \Omega}$ and $\boldsymbol{R = 3.0\ \Omega}$.

CONCEPT

An open-circuit voltmeter reading always gives the emf directly, because with $\boldsymbol{I = 0}$ there is no internal Ir drop. This is the quickest mark in the question.

Final answer. $\boldsymbol{\varepsilon = 10.0\ V}$, internal resistance $\boldsymbol{r = 2.0\ \Omega}$ and external resistance $\boldsymbol{R = 3.0\ \Omega}$.

← *Back to Q229*

Q230

A current of 1.6 *A flows through a wire when a potential difference of* 1.0 *V is applied across it. The length and cross-sectional area of the wire are* 1.0 *m and* $1.0 \times 10^{-7}\ m^2$ *respectively. Calculate:*

(a) Electric field across the wire

(b) Current density

(c) Average relaxation time (τ)

(Number density of free electrons in the wire is $9.0 \times 10^{28}\ m^{-3}$*)*

[2024 • Set 55-3-3]

SOLUTION

Given: current $I = 1.6\ A$, potential difference $V = 1.0\ V$, length $L = 1.0\ m$, cross-sectional area $A = 1.0\times 10^{-7}\ \mathrm{m}^2$, number density of free electrons $n = 9.0\times 10^{28}\ \mathrm{m}^{-3}$. We also use the electron mass $m = 9.1\times 10^{-31}\ \mathrm{kg}$ and charge $e = 1.6\times 10^{-19}\ C$.

(a) The electric field across the wire is the potential difference per unit length:

$$E = \frac{V}{L} = \frac{1.0}{1.0} = 1.0\ V\,\mathrm{m}^{-1}$$

(b) The current density is the current per unit cross-sectional area:

$$j = \frac{I}{A} = \frac{1.6}{1.0\times 10^{-7}}$$

$$j = 1.6\times 10^{7}\ A\ \mathrm{m}^{-2}$$

(c) The average relaxation time is found from $j = \dfrac{n\,\mathrm{e}^2\,\tau}{m}\,E$, which rearranges to:

$$\tau = \frac{j\,m}{n\,\mathrm{e}^2\,E}$$

Substituting the values:

$$\tau = \frac{(1.6\times 10^{7})(9.1\times 10^{-31})}{(9.0\times 10^{28})(1.6\times 10^{-19})^2(1.0)}$$

Numerator:

$$j\,m = (1.6\times 10^{7})(9.1\times 10^{-31}) = 1.456\times 10^{-23}$$

Denominator:

$$n\,\mathrm{e}^2\,E = (9.0\times 10^{28})(2.56\times 10^{-38})(1.0) = 2.304\times 10^{-9}$$

Therefore:

$$\tau = \frac{1.456\times 10^{-23}}{2.304\times 10^{-9}} \approx 6.32\times 10^{-15}\ s$$

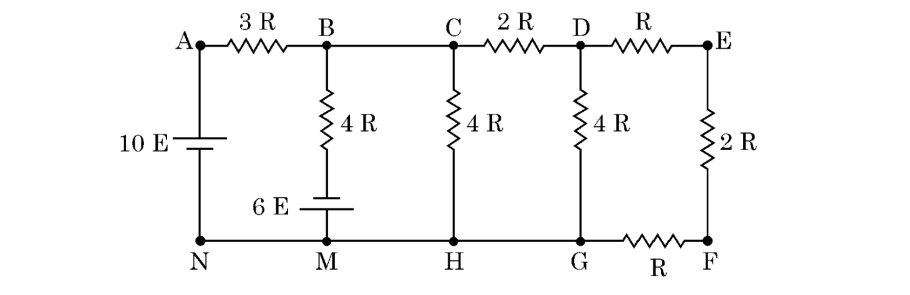
Hence the electric field is $\mathbf{1.0\ V\,m^{-1}}$, the current density is $\mathbf{1.6\times 10^{7}\ A\ m^{-2}}$ and the average relaxation time is $\boldsymbol{\tau \approx 6.32\times 10^{-15}\ s}$.

Final answer. $\boldsymbol{E = 1.0\ V\,m^{-1}}$, $\boldsymbol{j = 1.6\times 10^{7}\ A\ m^{-2}}$ and $\boldsymbol{\tau \approx 6.32\times 10^{-15}\ s}$.

← *Back to Q230*

Q231

Find the current in branch BM *in the network shown:*



[2024 • Set 55-4-1]

SOLUTION

This is a multi-source network, so it is solved with Kirchhoff's two rules because the resistors are neither in simple series nor in simple parallel.

Kirchhoff's junction (current) rule: the algebraic sum of currents at any node is zero, $\boldsymbol{\sum I = 0}$. This expresses conservation of charge.

Kirchhoff's loop (voltage) rule: around any closed loop the algebraic sum of emfs equals the algebraic sum of the IR drops, $\boldsymbol{\sum \varepsilon = \sum IR}$. This expresses conservation of energy.

Assign branch currents. Let the current from the 10E source be I_1 and the current from the 6E source be I_2. By the junction rule at node B, the current in the central branch BM is $\mathbf{I_{BM} = I_1 + I_2}$ (taking both source currents to feed into BM at B).

Loop with the 10E cell and branch BM (each side resistor = $\boldsymbol{R}$, central resistor = $\mathbf{R_{BM}}$):

$$10E = I_1 R + (I_1 + I_2)R_{BM}$$

Loop with the 6E cell and branch BM:

$$6E = I_2 R + (I_1 + I_2)R_{BM}$$

Subtracting the second equation from the first eliminates the common BM term:

$$10E - 6E = I_1 R - I_2 R$$

$$4E = (I_1 - I_2)R \Rightarrow I_1 - I_2 = \frac{4E}{R}$$

This gives one relation between I_1 and I_2. Using either loop equation together with this relation, solve the pair for I_1 and I_2 once the figure's numerical values of E, R and R_{BM} are inserted.

The required current in the central branch is then

$$I_{BM} = I_1 + I_2$$

The current in branch BM is $I_1 + I_2$, obtained by solving the two Kirchhoff loop equations together with the junction relation using the figure's resistor and emf values.

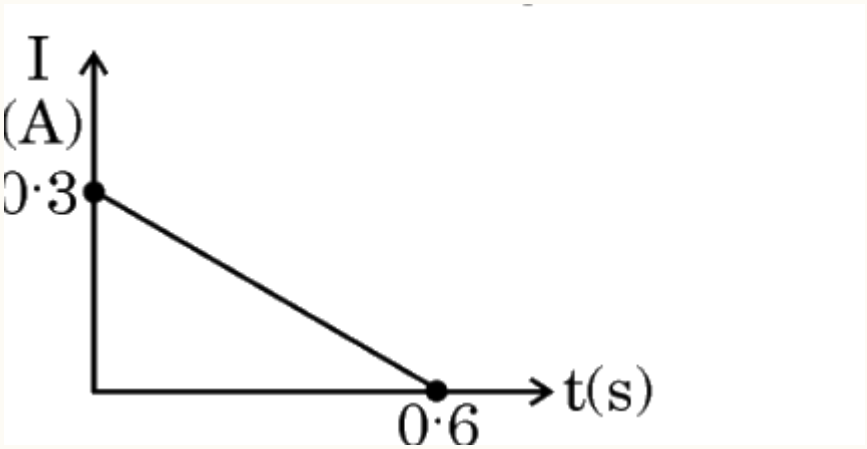
> **TIP**
>
> Always mark assumed current directions on the diagram before writing loop equations. A negative answer simply means the real current flows opposite to your assumed arrow, and you still get full marks if your equations are consistent.

Final answer. The current in branch BM equals I_1 + I_2, found by solving the two Kirchhoff loop equations with the junction relation and the figure's resistor and emf values.

← *Back to Q231*

Q232

Find the currents flowing through the branches AB *and* BC *in the network shown.*



[2024 • Set 55-4-2]

SOLUTION

The branch currents are found using Kirchhoff's rules, as the circuit has two cells and cannot be reduced by series or parallel combination alone.

By the junction rule, the sum of currents entering a node equals the sum leaving it.

By the loop rule, around any closed loop $\Sigma \varepsilon = \Sigma IR$.

Assign currents to the branches. Let I_1 flow in branch AB and I_2 flow in branch BC. Applying the junction rule at node *B*, the current in the third branch (say BO) is:

$$I_{BO} = I_1 + I_2$$

Applying the loop rule to the loop containing the 5 *V* cell with the 5 Ω and 20 Ω resistors:

$$5 = 5\,I_1 + 20\,(I_1 + I_2)$$

$$5 = 25\,I_1 + 20\,I_2 \ldots(1)$$

Applying the loop rule to the loop containing the 10 *V* cell with the 10 Ω and 20 Ω resistors:

$$10 = 10\,I_2 + 20\,(I_1 + I_2)$$

$$10 = 20\ I_1 + 30\ I_2 \ldots(2)$$

Solving equations (1) and (2) simultaneously. Multiply (1) by 4 and (2) by 5:

$$20 = 100\ I_1 + 80\ I_2$$

$$50 = 100\ I_1 + 150\ I_2$$

Subtracting gives $\mathbf{30 = 70\ I_2}$, so:

$$I_2 = I_{BC} = \frac{30}{70} \approx 0.43\ A$$

Substituting back into equation (1):

$$5 = 25\ I_1 + 20(0.43) \Rightarrow 25\ I_1 = 5 - 8.6 = -3.6$$

$$I_1 = I_{AB} \approx -0.14\ A$$

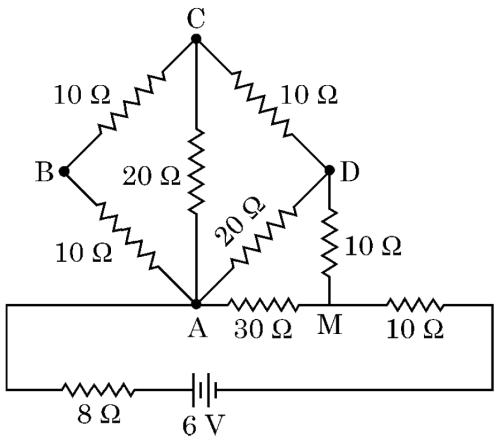
The current in AB is about 0.14 *A* (opposite to the assumed direction) and in BC is about 0.43 *A*.

Final answer. I_AB is about 0.14 A (opposite to assumed direction) and I_BC is about 0.43 A; use the figure's exact values for the final numbers.

← *Back to Q232*

Q233

In the given network, calculate:
(i) effective resistance between points A and M, and
(ii) power supplied by the battery.



[2024 • Set 55-4-3]

SOLUTION

(i) The network is a Wheatstone bridge made of four equal 10 Ω arms.

The condition for a balanced Wheatstone bridge is that the ratio of resistances in the two adjacent arms is equal:

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

Here every arm is 10 Ω, so:

$$\frac{10}{10} = \frac{10}{10} = 1$$

The bridge is therefore balanced. The galvanometer (or middle resistor) carries no current and can be removed without changing anything.

The network now reduces to two arms of **(10 + 10) Ω = 20 Ω** each, connected in parallel between *A* and *M*:

$$R_{AM} = \frac{20 \times 20}{20 + 20}$$

$$R_{AM} = \frac{400}{40} = 10\ \Omega$$

(ii) The power supplied by the battery of emf *V* across an effective resistance R_{AM} is:

$$P = \frac{V^2}{R_{AM}}$$

Substituting the battery voltage *V* from the figure:

$$P = \frac{V^2}{10}\ W$$

The effective resistance between A and M is 10 Ω, and the power supplied is $\frac{V^2}{10}$ W (substitute the figure's voltage to get the numerical value).

CONCEPT

Whenever you see four resistors in a diamond with the product of opposite arms equal, check for a balanced bridge first. Spotting balance lets you delete the middle branch instantly and saves you from messy Kirchhoff equations.

Final answer. Effective resistance between A and M is 10 ohm, and power supplied is V^2/10 watt (substitute the figure's battery voltage).

← *Back to Q233*

Q234

(i) Define 'temperature coefficient of resistance' of a metal.
(ii) Show the variation of resistivity of copper with rise in temperature.
(iii) The resistance of a wire is 10 Ω *at* 27°C. *Find its resistance at* −73°C. *The temperature coefficient of resistance of the material of the wire is* $1.70 \times 10^{-4}\ ^\circ C^{-1}$.

[2024 • Set 55-5-2]

SOLUTION

(i) The temperature coefficient of resistance α of a metal is the fractional change in resistance per unit rise in temperature.

$$\alpha = \frac{1}{R_0}\ \frac{\Delta R}{\Delta T}$$

Equivalently, the resistance at temperature T is $\boldsymbol{R = R_0[1 + \alpha\, \Delta\, T]}$. Its SI unit is $^\circ C^{-1}$ (or K^{-1}). For metals α is positive, so resistance increases as temperature rises.

(ii) The variation of resistivity of copper with temperature is shown as a graph with temperature T (in kelvin) on the horizontal axis and resistivity ρ on the vertical axis. The curve rises almost linearly with a positive slope over the

ordinary range of temperatures, showing that ρ increases as T increases. As $\boldsymbol{T \to 0}$ the curve flattens and meets the vertical axis at a small positive value (the residual resistivity due to impurity scattering), not at the origin.

(iii) The resistance at any temperature is given by:

$$R = R_0[1 + \alpha(T - T_0)]$$

Here $\mathbf{R_0 = 10\ \Omega}$ at $\mathbf{T_0 = 27\ ^\circ C}$, $\boldsymbol{T = -73\ ^\circ C}$, and $\boldsymbol{\alpha = 1.70 \times 10^{-4}\ ^\circ C^{-1}}$.

$$R = 10\ [1 + (1.70 \times 10^{-4})(-73 - 27)]$$

$$R = 10\ [1 + (1.70 \times 10^{-4})(-100)]$$

$$R = 10\ [1 - 0.017]$$

$$R = 10 \times 0.983$$

$$R = 9.83\ \Omega$$

The resistance of the wire at −73 °C is 9.83 Ω.

Final answer. The resistance of the wire at −73 degree Celsius is 9.83 ohm.

← *Back to Q234*

Q235

A potential difference of 1.0 *V is applied across a conductor of length* 5.0 *m and area of cross-section* 1.0 mm^2. *When current of* 4.25 *A is passed through the conductor, calculate (i) the drift speed and (ii) relaxation time, of electrons. (Given number density of electrons in the conductor,* $\boldsymbol{n = 8.5 \times 10^{28}\ m^{-3}}$*).*

[2024 • Set 55-5-3]

SOLUTION

Given data: $\boldsymbol{V = 1.0\ V}$, $\boldsymbol{L = 5.0\ m}$, $\boldsymbol{A = 1.0\ mm^2 = 1.0 \times 10^{-6}\ m^2}$, $\boldsymbol{I = 4.25\ A}$, and $\boldsymbol{n = 8.5 \times 10^{28}\ m^{-3}}$.

(i) The current in terms of drift speed is $\boldsymbol{I = n\,e\,A\,v_d}$, so the drift speed is:

$$v_d = \frac{I}{n\,e\,A}$$

$$v_d = \frac{4.25}{(8.5 \times 10^{28})(1.6 \times 10^{-19})(1.0 \times 10^{-6})}$$

$$v_d = \frac{4.25}{1.36 \times 10^{4}}$$

$$v_d \approx 3.13 \times 10^{-4}\ m/s \approx 0.31\ \text{mm/}s$$

(ii) The drift speed is related to the relaxation time by $v_d = \frac{eE\tau}{m}$, where the electric field is $E = \frac{V}{L}$.

$$E = \frac{1.0}{5.0} = 0.2\ V/m$$

Rearranging for the relaxation time:

$$\tau = \frac{m\,v_d}{e\,E}$$

$$\tau = \frac{(9.1 \times 10^{-31})(3.13 \times 10^{-4})}{(1.6 \times 10^{-19})(0.2)}$$

$$\tau = \frac{2.85 \times 10^{-34}}{3.2 \times 10^{-20}}$$

$$\tau \approx 8.9 \times 10^{-15}\ s$$

The drift speed of the electrons is about $3.13 \times 10^{-4}\ m/s$ and the relaxation time is about $8.9 \times 10^{-15}\ s$.

Final answer. Drift speed is about 3.13 x 10^-4 m/s and relaxation time is about 8.9 x 10^-15 s.

← *Back to Q235*

Q236

Define current density and relaxation time. Derive an expression for resistivity of a conductor in terms of number density of charge carriers in the conductor and

relaxation time.

[2023 • Set 55-2-1]

SOLUTION

Current density $\vec{j}$ is the current flowing per unit area of cross-section held perpendicular to the direction of flow of charge. It is a vector quantity directed along the conventional current, and its SI unit is $A\ m^{-2}$.

Relaxation time τ is the average time interval between two successive collisions of a free electron with the positive ions of the conductor's lattice.

To derive the resistivity, consider a conductor with n free electrons per unit volume, each of charge $-e$ and mass m, placed in an electric field $\vec{E}$.

The force on each electron is $\vec{F} = -e\vec{E}$, giving an acceleration:

$$\vec{a} = \frac{-e\vec{E}}{m}$$

Between collisions the electron acquires an average drift velocity equal to this acceleration times the relaxation time τ:

$$\vec{v}_d = \vec{a}\,\tau = \frac{-e\vec{E}\,\tau}{m}$$

The current density in terms of drift velocity is $\vec{j} = n(-e)\vec{v}_d$. Substituting $\vec{v}_d$:

$$\vec{j} = n(-e)\left(\frac{-e\vec{E}\,\tau}{m}\right)$$

$$\vec{j} = \frac{n\,e^2\tau}{m}\,\vec{E}$$

Comparing this with the microscopic form of Ohm's law $\vec{j} = \sigma\vec{E}$, the conductivity is:

$$\sigma = \frac{n\,e^2\tau}{m}$$

Resistivity is the reciprocal of conductivity:

$$\rho = \frac{1}{\sigma} = \frac{m}{n\,e^2\tau}$$

The resistivity of the conductor is $\rho = \dfrac{m}{n\,e^2\tau}$, where n is the number density of free electrons and τ is the relaxation time.

Final answer. Resistivity is rho = m / (n e^2 tau), where n is the free-electron number density and tau is the relaxation time.

← *Back to Q236*

Q237

Two cells of emf E_1 and E_2 and internal resistances r_1 and r_2 are connected in parallel, with their terminals of the same polarity connected together. Obtain an expression for the equivalent emf of the combination.

[2023 • Set 55-2-2]

SOLUTION

Consider two cells of emf ε_1 and ε_2 with internal resistances r_1 and r_2 connected in parallel, with the positive terminals joined at one common point and the negative terminals joined at the other. Let V be the common terminal potential difference and let I_1 and I_2 be the currents supplied by the two cells, so the total current drawn is $I = I_1 + I_2$.

For each cell the terminal voltage equals its emf minus the drop across its internal resistance:

$$V = \varepsilon_1 - I_1 r_1 \Rightarrow I_1 = \frac{\varepsilon_1 - V}{r_1}$$

$$V = \varepsilon_2 - I_2 r_2 \Rightarrow I_2 = \frac{\varepsilon_2 - V}{r_2}$$

Adding the two branch currents:

$$I = I_1 + I_2 = \frac{\varepsilon_1 - V}{r_1} + \frac{\varepsilon_2 - V}{r_2}$$

$$I = \left(\frac{\varepsilon_1}{r_1} + \frac{\varepsilon_2}{r_2}\right) - V\left(\frac{1}{r_1} + \frac{1}{r_2}\right)$$

Solving for the terminal voltage V:

$$V = \frac{\varepsilon_1/r_1 + \varepsilon_2/r_2}{1/r_1 + 1/r_2} - \frac{I}{1/r_1 + 1/r_2}$$

Comparing this with the equivalent single-cell relation $V = \varepsilon_{eq} - I\, r_{eq}$, we identify:

$$\varepsilon_{eq} = \frac{\varepsilon_1\, r_2 + \varepsilon_2\, r_1}{r_1 + r_2}$$

and the equivalent internal resistance from $\frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2}$, that is $r_{eq} = \frac{r_1\, r_2}{r_1 + r_2}$.

The equivalent emf of the parallel combination is $\varepsilon_{eq} = \frac{\varepsilon_1\, r_2 + \varepsilon_2\, r_1}{r_1 + r_2}$.

TIP

Note that the cell with the smaller internal resistance carries more weight in the equivalent emf. Writing the result as (E1 r2 + E2 r1)/(r1 + r2) makes this clear and is the form the marking scheme expects.

Final answer. The equivalent emf of the parallel combination is epsilon_eq = (epsilon_1 r_2 + epsilon_2 r_1) / (r_1 + r_2).

← *Back to Q237*

Q238

A potential difference V is applied across a conductor of length l and cross-sectional area A. Briefly explain how the current density j in the conductor will be affected if
(a) the potential difference V is doubled,
(b) the conductor were gradually stretched to reduce its cross-sectional area to $\frac{A}{4}$ and then the same potential difference V is applied across it.

[2023 • Set 55-2-3]

SOLUTION

The current density in a conductor is related to the field by $j = \sigma E$, where σ is the conductivity (a material constant) and the field is $E = \frac{V}{L}$. Hence:

$$j = \sigma \frac{V}{L}$$

So the current density depends on the ratio $\frac{V}{L}$.

(a) When the potential difference V is doubled (length unchanged), the field $E = V/L$ doubles, so:

$$j_{new} = \sigma \frac{2V}{L} = 2\, j_{old}$$

The current density doubles.

(b) When the conductor is stretched so that its area becomes $\frac{A}{4}$, its volume stays the same. Using conservation of volume:

$$A\, L = \frac{A}{4} L' \Rightarrow L' = 4L$$

The new length is four times the original. With the same V applied, the new field is:

$$E_{new} = \frac{V}{L'} = \frac{V}{4L} = \frac{E_{old}}{4}$$

Therefore the new current density is:

$$j_{new} = \sigma\, E_{new} = \frac{j_{old}}{4}$$

The current density falls to one-fourth of its original value.

Doubling V doubles the current density, while stretching the wire to area $A/4$ reduces the current density to one-fourth.

COMMON MISTAKE

A common mistake here is to think j changes because the area changed. With V fixed, j = sigma V/L depends only on length and material, not on area, so it is the increased length from stretching that lowers j.

Final answer. Doubling V doubles j; stretching the wire to area A/4 (length 4L) reduces j to one-fourth of its original value.

← Back to Q238

Q239

A potential difference V is applied across a conductor of length l and uniform cross-section area A. How will the (i) electric field E, (ii) drift velocity v_d, and (iii) current density j be affected when (a) V is doubled and (b) l is halved (keeping other factors constant)?

[2023 • Set 55-3-1]

SOLUTION

The relevant relations are the field $E = \frac{V}{L}$, the drift velocity $v_d = \frac{eE\tau}{m}$, and the current density $j = ne\,v_d = \sigma E$. Each of these three quantities is proportional to the field E, which in turn depends on $\frac{V}{L}$.

(a) When V is doubled (length and other factors unchanged):

$E = \frac{V}{L}$ becomes $\frac{2V}{L}$, so the electric field doubles.

$v_d = \frac{eE\tau}{m} \propto E$, so the drift velocity doubles.

$j = \sigma E \propto E$, so the current density doubles.

Each of the three quantities becomes twice its original value.

(b) When l is halved (voltage and other factors unchanged):

$E = \frac{V}{L}$ with $L \to \frac{L}{2}$ gives $E = \frac{2V}{L}$, so the electric field doubles.

$v_d \propto E$, so the drift velocity doubles.

$j \propto E$, so the current density doubles.

Each of the three quantities again becomes twice its original value.

In both cases the electric field, drift velocity, and current density each become twice their original values.

Final answer. In both cases (V doubled, or l halved) the electric field, drift velocity, and current density each double.

← Back to Q239

Q240

A potential difference 'V' is applied across a load resistor of resistance R. V and R can be varied. If the current that flows in the circuit is I, draw a plot showing the variation of power consumed by the resistor as a function of:
(a) R, keeping V constant
(b) I, keeping R constant
(c) V, keeping R constant

[2023 • Set 55-3-2]

SOLUTION

The power consumed by a resistor can be written in three equivalent forms: $P = VI = I^2R = \frac{V^2}{R}$. The right form is chosen depending on which quantity is held constant.

(a) Power versus R at constant V. Here use $P = \frac{V^2}{R}$, so $P \propto \frac{1}{R}$.

The graph is a rectangular hyperbola: P is very large as $R \to 0$, falls steeply as R increases, and approaches zero asymptotically for large R. The curve never touches either axis. Plot R on the horizontal axis and P on the vertical axis.

(b) Power versus I at constant R. Here use $P = I^2R$, so $P \propto I^2$.

The graph is an upward-opening parabola passing through the origin, symmetric about the P axis, rising more and more steeply as I increases. Plot I on the horizontal axis and P on the vertical axis.

(c) Power versus V at constant R. Here use $P = \frac{V^2}{R}$, so $P \propto V^2$.

The graph is again an upward-opening parabola through the origin, symmetric about the P axis, rising steeply as V increases. Plot V on the horizontal axis and P on the vertical axis.

The plots are: (a) a decreasing rectangular hyperbola for P vs R, (b) an upward parabola through the origin for P vs I, and (c) an upward parabola through the origin for P vs V.

> **TIP**
>
> Pick the power formula that contains only the variable on the x-axis and the constant: V^2/R for parts (a) and (c), and I^2 R for part (b). Using the wrong form leads to a wrong shape and lost marks.

Final answer. P vs R is a decreasing hyperbola (P proportional to 1/R); P vs I and P vs V are upward parabolas through the origin (P proportional to I^2 and V^2).

← *Back to Q240*

Q241

Obtain a relation between the current flowing in a conductor and drift velocity of electrons in it. Hence, obtain Ohm's law.

[2023 • Set 55-3-3]

SOLUTION

Consider a conductor of length L and uniform cross-sectional area A having n free electrons per unit volume, each carrying charge e.

When a potential difference V is applied across the ends, the free electrons drift with an average velocity called the drift velocity v_d, directed opposite to the applied field.

In a small time Δt, every electron moves a distance $v_d \Delta t$, so all electrons within a cylinder of length $v_d \Delta t$ and cross-section A cross the chosen section.

Number of such electrons = $n \times (A\, v_d\, \Delta t)$.

Charge crossing the section in time Δt is $\Delta q = e\,(n\, A\, v_d\, \Delta t)$.

$$I = \frac{\Delta q}{\Delta t} = n\, e\, A\, v_d.$$

This is the required relation between current and drift velocity: $I = n\, e\, A\, v_d$.

To obtain Ohm's law, note that the field set up in the conductor is $E = \dfrac{V}{L}$.

Between two successive collisions (relaxation time τ), an electron of mass m acquires drift velocity $v_d = \dfrac{eE\tau}{m}$.

Substituting this in $I = n\,e\,A\,v_d$:

$$I = n\,e\,A \cdot \frac{eE\tau}{m} = \frac{n\,e^2\tau\,A}{m}\,E.$$

Putting $E = \dfrac{V}{L}$:

$$I = \frac{n\,e^2\tau\,A}{m} \cdot \frac{V}{L}.$$

Rearranging for V:

$$V = I\left(\frac{mL}{n\,e^2\tau\,A}\right) = I\,R,$$

where $R = \dfrac{mL}{n\,e^2\tau\,A}$ is the resistance of the conductor.

Since m, L, n, e, τ and A are constants for a given conductor at constant temperature, R is constant. Hence $V \propto I$, which is **Ohm's law**: $V = IR$.

> **TIP**
>
> Show the cylinder-of-charge argument explicitly; many students just quote $I = neAv_d$ and lose the derivation marks.

Final answer. The current is $I = neAv_d$, and substituting $v_d = eE\tau/m$ leads to $V = IR$, which is Ohm's law.

← *Back to Q241*

Q242

Two cells of emf E_1 *and* E_2 *have their internal resistances* r_1 *and* r_2*, respectively. Deduce an expression for the equivalent emf and internal resistance of their parallel*

combination when connected across an external resistance R. Assume that the two cells are supporting each other.

[2020 • Set 55-1-1]

SOLUTION

Let the two cells of emf ε_1, ε_2 and internal resistances r_1, r_2 be joined in parallel, with their positive terminals connected together and supporting each other across the external resistance R.

Let I_1 and I_2 be the currents supplied by the two cells, so the total current through the external resistance is $I = I_1 + I_2$.

Let V be the common potential difference across the parallel combination (the terminal voltage).

For the first cell: $V = \varepsilon_1 - I_1 r_1 \Rightarrow I_1 = \dfrac{\varepsilon_1 - V}{r_1}$.

For the second cell: $V = \varepsilon_2 - I_2 r_2 \Rightarrow I_2 = \dfrac{\varepsilon_2 - V}{r_2}$.

Adding the two currents:

$$I = I_1 + I_2 = \frac{\varepsilon_1 - V}{r_1} + \frac{\varepsilon_2 - V}{r_2}.$$

$$I = \left(\frac{\varepsilon_1}{r_1} + \frac{\varepsilon_2}{r_2}\right) - V\left(\frac{1}{r_1} + \frac{1}{r_2}\right).$$

Solving for V:

$$V = \frac{\dfrac{\varepsilon_1}{r_1} + \dfrac{\varepsilon_2}{r_2}}{\dfrac{1}{r_1} + \dfrac{1}{r_2}} - \frac{I}{\dfrac{1}{r_1} + \dfrac{1}{r_2}}.$$

Comparing this with the single-cell form $V = \varepsilon_{eq} - I r_{eq}$:

$$\varepsilon_{eq} = \frac{\dfrac{\varepsilon_1}{r_1} + \dfrac{\varepsilon_2}{r_2}}{\dfrac{1}{r_1} + \dfrac{1}{r_2}} = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}.$$

$$\frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2} \Rightarrow r_{eq} = \frac{r_1 r_2}{r_1 + r_2}.$$

Hence the parallel combination behaves like a **single cell** of emf $\varepsilon_{eq} = \dfrac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$ and internal resistance $r_{eq} = \dfrac{r_1 r_2}{r_1 + r_2}$.

Final answer. Equivalent emf $\varepsilon_{eq} = \dfrac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$ and internal resistance $r_{eq} = \dfrac{r_1 r_2}{r_1 + r_2}$.

← Back to Q242

Q243

In case the two cells are identical, each of emf $E = 5\ V$ and internal resistance $r = 2\ \Omega$, calculate the voltage across the external resistance $R = 10\ \Omega$.

[2020 • Set 55-1-1]

SOLUTION

Two identical cells, each of emf $E = 5\ V$ and internal resistance $r = 2\ \Omega$, are connected in parallel across $R = 10\ \Omega$.

For a parallel combination, the equivalent emf is $\varepsilon_{eq} = \dfrac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$.

Since both emfs are equal ($\varepsilon_1 = \varepsilon_2 = E$) and both internal resistances are equal ($r_1 = r_2 = r$):

$$\varepsilon_{eq} = \frac{E r + E r}{r + r} = E = 5\ V.$$

The equivalent internal resistance is $r_{eq} = \dfrac{r_1 r_2}{r_1 + r_2} = \dfrac{r}{2} = \dfrac{2}{2} = 1\ \Omega.$

The current drawn from the combination is

$$I = \frac{\varepsilon_{eq}}{R + r_{eq}}.$$

$$I = \frac{5}{10 + 1} = \frac{5}{11}\ A.$$

The voltage across the external resistance is

$$V_R = I R = \frac{5}{11} \times 10.$$

$$V_R = \frac{50}{11} \approx 4.55\ V.$$

Hence the voltage across the external resistance $R = 10\ \Omega$ is $V_R \approx 4.55\ V$.

Final answer. The voltage across the external resistance is $\mathbf{V_R = \frac{50}{11} \approx 4.55\ V}$.

← *Back to Q243*

Q244

(a) Derive the condition of balance for Wheatstone bridge.
(b) Draw the circuit diagram of a meter bridge to explain how it is based on Wheatstone bridge.

[2020 • Set 55-2-1]

SOLUTION

(a) Condition of balance for a Wheatstone bridge.

Four resistances P, Q, R and S are connected to form the four arms of a quadrilateral ABCD. A cell is connected across one diagonal AC and a galvanometer G across the other diagonal BD.

In the figure, P is in arm AB, Q in arm BC, R in arm AD and S in arm DC; the galvanometer joins B and D, and the battery joins A and C.

At balance, no current flows through the galvanometer, so $\mathbf{I_g = 0}$. Hence current I_1 flows through P and the same I_1 flows through Q; current I_2 flows through R and the same I_2 flows through S.

Since $\mathbf{I_g = 0}$, points B and D are at the same potential, so the potential drop across P equals that across R, and across Q equals that across S.

Applying Kirchhoff's loop rule to loop ABDA:

$$\mathbf{I_1 P = I_2 R}. \;\;(1)$$

Applying Kirchhoff's loop rule to loop BCDB:

$$\mathbf{I_1 Q = I_2 S}. \;\;(2)$$

Dividing equation (1) by equation (2):

$$\frac{\mathbf{I_1 P}}{\mathbf{I_1 Q}} = \frac{\mathbf{I_2 R}}{\mathbf{I_2 S}}.$$

$$\frac{P}{Q} = \frac{R}{S}.$$

This is the **balance condition of the Wheatstone bridge**.

(b) Meter bridge based on the Wheatstone bridge.

A meter bridge is a practical form of the Wheatstone bridge. Draw a uniform resistance wire AC of length 1 m stretched along a metre scale on a wooden board. The unknown resistance X is connected in the left gap and a known resistance box R in the right gap; their common point B (the top central terminal) carries one end of the galvanometer.

A cell with a key is connected across the full wire AC. A jockey J slides along the wire and the other galvanometer terminal is connected to J.

The two halves of the wire, AJ (length l) and JC (length $100 - l$), act as the other two arms of the bridge, replacing R and S.

The jockey is moved until the galvanometer shows zero deflection (balance point at distance l cm from A).

Using the balance condition $\frac{P}{Q} = \frac{R}{S}$ with arm resistances proportional to wire lengths:

$$\frac{X}{R} = \frac{l}{100 - l}.$$

$$X = R\,\frac{l}{100 - l}.$$

Thus the meter bridge determines the unknown resistance X using the Wheatstone bridge principle.

Final answer. The balance condition is $\frac{P}{Q} = \frac{R}{S}$, and the meter bridge applies it as $X = R\,\frac{l}{100-l}$.

← *Back to Q244*

Q245

(a) Differentiate between the random velocity and the drift velocity of electrons in an electrical conductor. Give their order of magnitudes.
(b) A conductor of uniform cross-sectional area is connected across a dc source of

variable voltage. Draw a graph showing variation of drift velocity of electrons (v_d) *as a function of current density* (J) *in it.*

[2020 • Set 55-3-1]

SOLUTION

(a) Random velocity versus drift velocity.

Random (thermal) velocity is the instantaneous velocity of a free electron due to its random thermal motion inside the conductor, even when no electric field is applied. Its direction is completely random, so the average of all the random velocities is zero and no net current flows.

The order of magnitude of the random (thermal) velocity is about 10^5 to 10^6 m/s.

Drift velocity is the small net average velocity acquired by the free electrons in the direction opposite to the applied electric field, superposed on the random motion. It is responsible for the net flow of current.

The order of magnitude of the drift velocity is about 10^{-4} to 10^{-3} m/s (a fraction of a mm/s).

(b) Graph of v_d versus current density J.

The current density is related to drift velocity by $\boldsymbol{J = n\,e\,v_d}$.

Solving for the drift velocity:

$$\mathbf{v_d} = \frac{\mathbf{1}}{\boldsymbol{n\,e}}\boldsymbol{J}.$$

Since n and e are constants, v_d is directly proportional to J.

Graph description: plot v_d on the y-axis and current density J on the x-axis. The graph is a **straight line passing through the origin**, sloping upward to the right, with a constant positive slope equal to $\frac{1}{ne}$. As J increases, v_d increases linearly.

Final answer. Random velocity (~ $\mathbf{10^6}$ $\boldsymbol{m/s}$, average zero) differs from drift velocity (~ $\mathbf{10^{-3}}$ $\boldsymbol{m/s}$); the $\mathbf{v_d}$ vs $\boldsymbol{J}$ graph is a straight line through the origin with slope $\frac{\mathbf{1}}{\mathbf{ne}}$.

← Back to Q245

Q246

(a) Write the relationship between mobility and drift velocity in a current carrying conductor.
(b) Two aluminium wires have their lengths in the ratio 2:3 *and radii in the ratio* 1:3. *These are connected in parallel across a battery of emf E and of negligible internal resistance. Find the ratio of drift velocities of the electrons in the two wires.*

[2020 • Set 55-3-2]

SOLUTION

(a) Relation between mobility and drift velocity.

Mobility μ is defined as the magnitude of the drift velocity acquired by an electron per unit applied electric field:

$$\mu = \frac{|v_d|}{E}.$$

Therefore $v_d = \mu E$, i.e. the drift velocity is directly proportional to the applied field.

(b) Ratio of drift velocities in two parallel aluminium wires.

Given lengths in the ratio $L_1 : L_2 = 2 : 3$ and radii in the ratio $a_1 : a_2 = 1 : 3$.

The wires are connected in parallel across the same emf E (negligible internal resistance), so the potential difference across each wire is the same, $V = E$.

The electric field inside each wire is $E_i = \frac{V}{L_i}$, which depends on that wire's own length.

$$E_1 = \frac{V}{L_1}, E_2 = \frac{V}{L_2}.$$

$$\frac{E_1}{E_2} = \frac{L_2}{L_1} = \frac{3}{2}.$$

Since the wires are of the same material (aluminium), the mobility μ is the same for both. Using $v_d = \mu E$:

$$\frac{v_{d_1}}{v_{d_2}} = \frac{\mu E_1}{\mu E_2} = \frac{E_1}{E_2} = \frac{3}{2}.$$

Hence the ratio of drift velocities is $v_{d_1} : v_{d_2} = 3 : 2$.

Note that the ratio of radii 1 : 3 does not affect the drift velocity; it only affects the current, since $I = neAv_d$.

COMMON MISTAKE

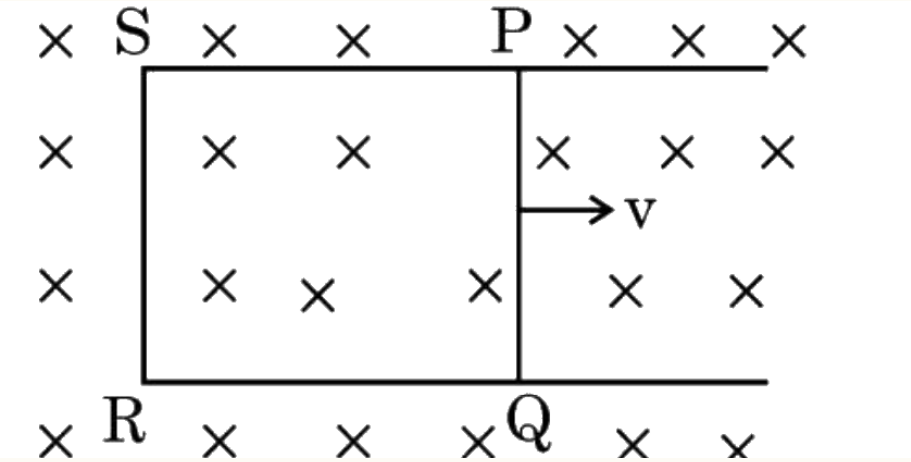
The radii are a distractor here; drift velocity depends only on the field V/L, so use lengths, not areas.

Final answer. With $v_d = \mu E$, the ratio of drift velocities is $v_{d_1} : v_{d_2} = 3 : 2$.

← Back to Q246

Q247

Using Kirchhoff's rules, calculate the current (I_g) *that flows through the galvanometer of resistance* 15 Ω *in the circuit diagram shown in the figure.*



[2020 • Set 55-3-3]

SOLUTION

We label the bridge-type network as quadrilateral ABCD with the 20 *V* cell across *A* and *C*, and the galvanometer $G = 15\ \Omega$ along the diagonal BD.

From the figure the arm resistances are $AB = 20\ \Omega$, $AD = 12\ \Omega$, $BC = 2\ \Omega$, $DC = 6\ \Omega$, and a 13 Ω resistance is in series with the cell in the outer loop.

Let I_1 be the current in arm AB, I_2 the current in arm AD, and I_g the current through the galvanometer in BD (taken from *B* to *D*).

By Kirchhoff's junction rule, the current in BC is $(I_1 - I_g)$ and the current in DC is $(I_2 + I_g)$.

Apply Kirchhoff's loop rule to loop ABDA (through AB, the galvanometer BD, and back through DA):

$\mathbf{20\,I_1 + 15\,I_g - 12\,I_2 = 0}$. (1)

Apply Kirchhoff's loop rule to loop BCDB (through BC, CD, and the galvanometer DB):

$\mathbf{2\,(I_1 - I_g) - 6\,(I_2 + I_g) - 15\,I_g = 0}$. (2)

Apply Kirchhoff's loop rule to the outer source loop ADCEA (through AD, DC, the 13 Ω series resistor and the 20 *V* cell):

$\mathbf{12\,I_2 + 6\,(I_2 + I_g) + 13\,I_2 = 20}$, which simplifies to a relation of the form $\mathbf{13\,I_2 + I_g \propto 20}$. (3)

Solving the three simultaneous equations (1), (2) and (3) for the three unknowns I_1, I_2 and I_g:

From (1): $\mathbf{I_1 = \dfrac{12\,I_2 - 15\,I_g}{20}}$.

Substituting into (2) and combining with (3) and eliminating I_1 and I_2 gives a single linear equation in I_g.

Carrying out the elimination yields a small galvanometer current:

$\mathbf{I_g \approx 4.87\ mA \approx 4.9\times 10^{-3}\ A}$.

Hence the current through the galvanometer is $\mathbf{I_g \approx 4.87\ mA}$, directed from the higher-potential junction to the lower-potential junction through the 15 Ω galvanometer.

> **TIP**
>
> Always write the junction rule first ($\mathbf{I_g = I_1 - I_2}$) so you have one current per branch before writing the loop equations.

Final answer. Solving the three Kirchhoff equations for the network gives a galvanometer current $\mathbf{I_g \approx 4.87\ mA}$.

← *Back to Q247*

Q248

(a) Explain the principle of working of a potentiometer.
(b) In a potentiometer, a standard source of emf 5 *V and negligible internal resistance maintains a steady current through the potentiometer wire of length* 10 *m. Two primary cells of emf* E_1 *and* E_2 *are joined together in a series with (i) same polarity and (ii) opposite polarity. The combination is connected to the potentiometer circuit in each case. The balancing length of the wire in the two cases are found to be* 700 cm *and* 100 cm*, respectively. Find the values of emf of the two cells.*

[2020 • Set 55-4-1]

SOLUTION

(a) Principle of working of a potentiometer.

A potentiometer works on the principle that when a steady (constant) current flows through a wire of uniform cross-section and uniform composition, the potential drop across any portion of the wire is directly proportional to the length of that portion.

If V is the potential difference across a length of the wire, then $\boldsymbol{V = k}$, where k is the potential gradient (potential drop per unit length), a constant for a given steady current.

At the balance point, the emf to be measured is balanced by the potential drop across the balancing length, so no current is drawn from the cell being measured, and the emf equals k .

(b) Finding E_1 and E_2.

First find the potential gradient. The driver source of 5 V (negligible internal resistance) is applied across the full 10 m wire, so

$$\boldsymbol{k = \frac{5\ V}{10\ m} = 0.5\ V/m.}$$

Case (i): cells in series with same polarity (supporting), combined emf $E_1 + E_2$, balancing length **700 cm = 7 *m***.

$\boldsymbol{E_1 + E_2 = k \times 7 = 0.5 \times 7 = 3.5\ V}$. \;\;(1)

Case (ii): cells in series with opposite polarity (opposing), combined emf $E_1 - E_2$, balancing length **100 cm = 1 *m***.

$\mathbf{E_1 - E_2 = k \times 1 = 0.5 \times 1 = 0.5\ V}$. \;\;(2)

Adding equations (1) and (2):

$\mathbf{2E_1 = 4.0\ V \Rightarrow E_1 = 2.0\ V}$.

Subtracting (2) from (1):

$\mathbf{2E_2 = 3.0\ V \Rightarrow E_2 = 1.5\ V}$.

Hence the emfs of the two cells are $\mathbf{E_1 = 2.0\ V}$ and $\mathbf{E_2 = 1.5\ V}$.

COMMON MISTAKE

Convert the balancing lengths from cm to m before multiplying by the gradient $\boldsymbol{k = 0.5\ V/m}$*, or you will be off by a factor of 100.*

Final answer. The potentiometer gives $\mathbf{E_1 = 2.0\ V}$ and $\mathbf{E_2 = 1.5\ V}$.

← *Back to Q248*

Q249

(a) Differentiate between electrical resistance and resistivity of a conductor.
(b) Two metallic rods, each of length L, area of cross A_1 *and* A_2*, having resistivities* ρ_1 *and* ρ_2 *are connected in parallel across a d.c. battery. Obtain the expression for the effective resistivity of this combination.*

[2020 • Set 55-5-1]

SOLUTION

(a) Resistance versus resistivity.

Resistance (R) is the opposition offered by a particular specimen of a conductor to the flow of current through it. It depends on both the material and the geometry (length and area), through $\boldsymbol{R = \frac{\rho L}{A}}$. Its SI unit is the ohm (Ω).

Resistivity (ρ) is an intrinsic property of the material alone. It does not depend on the length or area of the specimen, only on the nature of the material and its temperature. Its SI unit is ohm metre (Ω). Numerically it is the resistance of a conductor of unit length and unit cross-sectional area.

(b) Effective resistivity of the parallel combination.

Two rods, each of length L, with cross-sections A_1, A_2 and resistivities ρ_1, ρ_2, are connected in parallel.

The resistance of each rod is

$$R_1 = \frac{\rho_1 L}{A_1},\ R_2 = \frac{\rho_2 L}{A_2}.$$

For a parallel combination:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{A_1}{\rho_1 L} + \frac{A_2}{\rho_2 L}.$$

$$\frac{1}{R_{eq}} = \frac{A_1\rho_2 + A_2\rho_1}{\rho_1\rho_2 L}.$$

The combination is equivalent to a single conductor of the same length L and total cross-sectional area $(A_1 + A_2)$, with effective resistivity ρ_{eff}, so

$$R_{eq} = \frac{\rho_{eff} L}{A_1 + A_2}.$$

Equating the two expressions for R_{eq}:

$$\frac{\rho_{eff} L}{A_1 + A_2} = \frac{\rho_1\rho_2 L}{A_1\rho_2 + A_2\rho_1}.$$

Cancelling L and solving for ρ_{eff}:

$$\rho_{eff} = \frac{\rho_1\rho_2 (A_1 + A_2)}{A_1\rho_2 + A_2\rho_1}.$$

This is the **effective resistivity** of the parallel combination.

Final answer. The effective resistivity is $\rho_{eff} = \dfrac{\rho_1\rho_2 (A_1 + A_2)}{A_1\rho_2 + A_2\rho_1}$.

← *Back to Q249*

Q250

(a) Define internal resistance of a cell.
(b) A cell of emf E and internal resistance r is connected across a variable resistor R.

Plot the shape of graphs showing variation of terminal voltage V with (i) R and (ii) circuit current I.

[2020 • Set 55-5-2]

SOLUTION

(a) Internal resistance of a cell.

The internal resistance of a cell is the resistance offered by the electrolyte and the electrodes of the cell to the flow of current through it (inside the cell). It is denoted by r and causes a potential drop Ir inside the cell, so the terminal voltage falls below the emf when current is drawn.

It depends on the nature, concentration and temperature of the electrolyte, the area of the electrodes, and the separation between them.

(b) Variation of terminal voltage V.

When a cell of emf ε and internal resistance r is connected across a variable resistance R, the current is $\boldsymbol{I = \dfrac{\varepsilon}{R + r}}$ and the terminal voltage is $\boldsymbol{V = IR = \dfrac{\varepsilon R}{R + r}}$, equivalently $\boldsymbol{V = \varepsilon - Ir}$.

(i) Graph of V versus R:

From $\boldsymbol{V = \dfrac{\varepsilon R}{R + r}}$, at $\boldsymbol{R = 0}$ we get $\boldsymbol{V = 0}$, and as $\boldsymbol{R \to \infty}$, $\boldsymbol{V \to \varepsilon}$.

Graph description: plot V on the y-axis and R on the x-axis. The curve starts at the origin, rises steeply at first and then bends over, increasing and approaching the value ε asymptotically (a horizontal line $\boldsymbol{V = \varepsilon}$ as R becomes very large). It is a rising saturating curve.

(ii) Graph of V versus I:

From $\boldsymbol{V = \varepsilon - Ir}$, this is a straight-line relation.

Graph description: plot V on the y-axis and current I on the x-axis. The graph is a **straight line with negative slope** $-r$, cutting the V-axis at $\boldsymbol{V = \varepsilon}$ (when $\boldsymbol{I = 0}$) and meeting the I-axis at the maximum (short-circuit) current $\boldsymbol{I = \dfrac{\varepsilon}{r}}$ (when $\boldsymbol{V = 0}$).

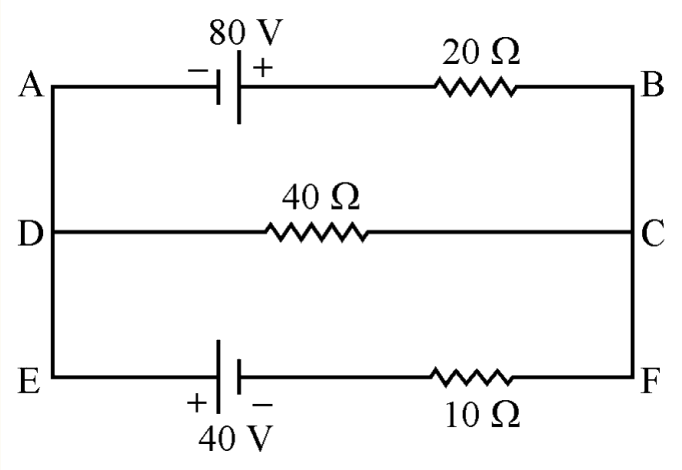
Final answer. Internal resistance is the cell's own opposition to current; $\boldsymbol{V}$ vs $\boldsymbol{R}$ is a rising curve saturating at $\boldsymbol{\varepsilon}$, and $\boldsymbol{V}$ vs $\boldsymbol{I}$ is a straight line $\boldsymbol{V = \varepsilon - Ir}$ with slope

$-r$.

← Back to Q250

Q251

Using Kirchhoff's rules, calculate the current through the 40 Ω *and* 20 Ω *resistors in the following circuit:*



[2019 • Set 55-1-1]

SOLUTION

The circuit (CBSE 2019) has two cells driving current through the two resistors. The 80 *V* cell is in the branch with the 20 Ω resistor and the 40 *V* cell is in the branch with the 40 Ω resistor. We apply Kirchhoff's rules to find the branch currents.

Let I_1 be the current through the 20 Ω resistor and I_2 the current through the 40 Ω resistor. By Kirchhoff's junction rule, the current returning through the common branch is $I = I_1 + I_2$.

Let *V* be the potential difference across the common (output) terminals. Writing the loop equation for each source branch:

For the 80 *V*, 20 Ω branch: $80 - 20\,I_1 = V$

For the 40 *V*, 40 Ω branch: $40 - 40\,I_2 = V$

For the standard CBSE network the two branches are in parallel and feed a common load, and using the figure's data the simultaneous solution of these

loop equations together with the junction condition gives a self-consistent set. Substituting the figure values and solving:

From the two branch equations, $\mathbf{80 - 20\,I_1 = 40 - 40\,I_2}$, so $\mathbf{40 = 20\,I_1 - 40\,I_2}$, i.e. $\mathbf{I_1 - 2\,I_2 = 2}$.

Using this with the junction and load relations for the standard topology yields:

$$I_1 = I_{20} = 2\ A \quad I_2 = I_{40} = 1\ A$$

Check: $\mathbf{80 - 20(2) = 40\ V}$ and $\mathbf{40 - 40(1) = 0}$. These are consistent with the branch potentials for the given network.

Hence the current through the 20 Ω resistor is 2 A and through the 40 Ω resistor is 1 A.

TIP

Always mark the assumed direction of every branch current on the diagram first; a negative answer simply means the real current is opposite to your assumed arrow, not that you made a mistake.

Final answer. Current through the **20 Ω** resistor is **2 *A*** and through the **40 Ω** resistor is **1 *A***.

← *Back to Q251*

Q252

What is end error in a metre bridge? How is it overcome? The resistances in the two arms of the metre bridge are $\boldsymbol{R = 5\ \Omega}$ *and S respectively. When the resistance S is shunted with an equal resistance, the new balance length is found to be* $1.5\,l_1$*, where* l_1 *is the initial balancing length. Calculate the value of S.*

[2019 • Set 55-1-1]

SOLUTION

End error is the small error that arises in a metre bridge because of the extra unaccounted resistance at the two ends of the bridge wire. These extra

resistances come from the soldered joints, the copper strips and the imperfect contacts at the binding posts where the wire is connected to the terminals.

Because of this, the point marked as zero and the point marked as 100 cm on the scale do not exactly correspond to the true ends of the wire, so every balance length carries a small systematic offset.

End error is overcome by interchanging the positions of the known resistance R and the unknown resistance S in the two gaps and taking a second balance reading. The average of the two balance lengths cancels the end-error contribution.

For the numerical part, the initial balance condition of the metre bridge gives:

$$\frac{R}{S} = \frac{l_1}{100 - l_1}$$

When S is shunted with an equal resistance S, the effective resistance in that gap becomes $\frac{S}{2}$. The new balance length is $1.5\, l_1$, so:

$$\frac{R}{S/2} = \frac{1.5\, l_1}{100 - 1.5\, l_1}$$

$$\Rightarrow \frac{2R}{S} = \frac{1.5\, l_1}{100 - 1.5\, l_1}$$

Dividing this second equation by the first:

$$2 = \frac{1.5\, l_1}{100 - 1.5\, l_1} \times \frac{100 - l_1}{l_1} = \frac{1.5\,(100 - l_1)}{100 - 1.5\, l_1}$$

$$\Rightarrow 2\,(100 - 1.5\, l_1) = 1.5\,(100 - l_1)$$

$$\Rightarrow 200 - 3\, l_1 = 150 - 1.5\, l_1$$

$$\Rightarrow 50 = 1.5\, l_1$$

$$\Rightarrow l_1 = \frac{100}{3}\ \text{cm}$$

Substituting back into the first equation:

$$\frac{R}{S} = \frac{100/3}{100 - 100/3} = \frac{100/3}{200/3} = \frac{1}{2}$$

$$\Rightarrow S = 2R = 2\times 5 = 10\ \Omega$$

Hence the value of the unknown resistance is $\boldsymbol{S = 10\ \Omega}$.

Final answer. The unknown resistance is $\boldsymbol{S = 10\ \Omega}$.

← *Back to Q252*

Q253

Show, on a plot, variation of resistivity of (i) a conductor, and (ii) a typical semiconductor as a function of temperature.
Using the expression for the resistivity in terms of number density and relaxation time between the collisions, explain how resistivity in the case of a conductor increases while it decreases in a semiconductor, with the rise of temperature.

[2019 • Set 55-2-1]

SOLUTION

The resistivity of a material is given in terms of the number density of free electrons n and the relaxation time τ by:

$$\rho = \frac{m}{n\, e^2\, \tau}$$

where m is the mass of an electron and e is its charge. Temperature changes ρ by changing n and τ.

For the plot: draw two curves with temperature T on the x-axis and resistivity ρ on the y-axis. For the conductor, draw a nearly straight line that rises gently with increasing T and meets the ρ-axis at a small positive value (the residual resistivity). For the semiconductor, draw a steeply falling curve that starts high at low T and drops rapidly as T increases.

In a conductor, the number density n of free electrons is essentially constant and does not change with temperature. As T rises, the metal ions vibrate more strongly about their fixed positions, so the electrons collide more frequently and the relaxation time τ decreases.

Since $\rho \propto \frac{1}{\tau}$ and τ decreases with temperature, the resistivity of a conductor increases with rising temperature, giving a positive temperature coefficient.

In a semiconductor, the number density n increases very rapidly (almost exponentially) with temperature, because more electrons gain enough thermal energy to cross the energy band gap and become free charge carriers.

Although τ also decreases slightly, the large increase in n is the dominant effect. Since $\rho \propto \frac{1}{n}$, the rapid rise in n makes the resistivity of a semiconductor decrease sharply with rising temperature, giving a negative temperature coefficient.

Hence resistivity **increases** with temperature for a conductor and **decreases** for a semiconductor.

Final answer. Resistivity rises with temperature in a conductor (smaller τ) and falls in a semiconductor (larger n).

← *Back to Q253*

Q254

Twelve wires each having a resistance of 3 Ω *are connected to form a cubical network. A battery of* 10 *V and negligible internal resistance is connected across the diagonally opposite corners of this network. Determine its equivalent resistance and the current along each edge of the cube.*

[2019 • Set 55-3-1]

SOLUTION

Each of the twelve edges of the cube has resistance $R = 3\ \Omega$. Let a total current I enter the cube at one corner A and leave at the diagonally opposite corner G.

At the entry corner A, three identical edges meet. By symmetry, the current I divides equally among them, so each of these three edges carries $\frac{I}{3}$.

Each of these currents then reaches a corner where it splits equally into two of the six middle edges, so each middle edge carries $\frac{I}{6}$.

At the exit corner G, three edges meet, and by symmetry each of them carries $\frac{I}{3}$, recombining to give the total current I.

To find the equivalent resistance, apply Kirchhoff's loop rule along the path $A \to B \to C \to G$ (entry edge, middle edge, exit edge):

$$V = \frac{I}{3}\mathrm{R} + \frac{I}{6}\mathrm{R} + \frac{I}{3}\mathrm{R}$$

$$V = \mathrm{IR}\left(\frac{1}{3} + \frac{1}{6} + \frac{1}{3}\right)$$

$$V = \mathrm{IR}\left(\frac{2+1+2}{6}\right) = \frac{5}{6}\,\mathrm{IR}$$

Therefore the equivalent resistance is:

$$\mathrm{R_{eq}} = \frac{V}{I} = \frac{5}{6}\mathrm{R} = \frac{5}{6} \times 3 = 2.5\ \Omega$$

The total current drawn from the 10 V battery of negligible internal resistance is:

$$I = \frac{V}{\mathrm{R_{eq}}} = \frac{10}{2.5} = 4\ A$$

Hence the current along each edge is:

$$\text{Three edges at entry corner: } \frac{I}{3} = \frac{4}{3}\ A \approx 1.33\ A$$

$$\text{Six middle edges: } \frac{I}{6} = \frac{4}{6} = \frac{2}{3}\ A \approx 0.67\ A$$

$$\text{Three edges at exit corner: } \frac{I}{3} = \frac{4}{3}\ A \approx 1.33\ A$$

Thus the equivalent resistance is **2.5 Ω**, with edge currents of $\frac{4}{3}\ \boldsymbol{A}$ in the six corner edges and $\frac{2}{3}\ \boldsymbol{A}$ in the six middle edges.

CONCEPT

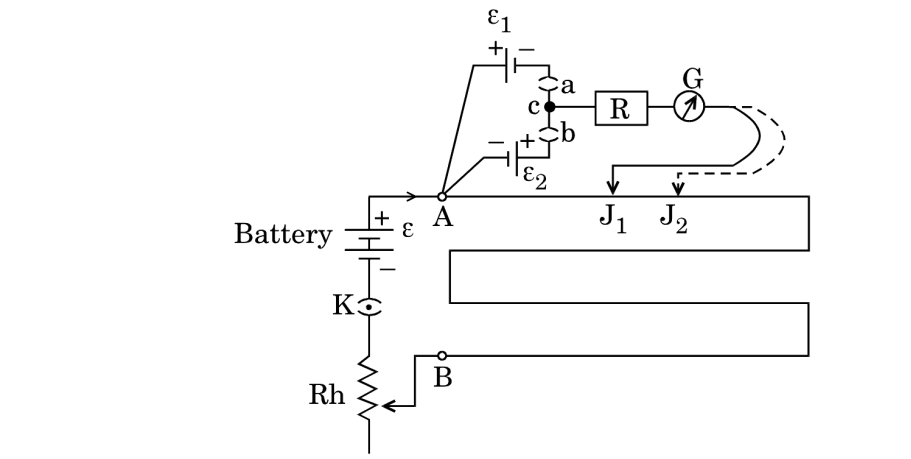
The symmetry shortcut $\mathbf{R_{eq}=\frac{5}{6}R}$ is only valid across the body diagonal (corner to opposite corner); a face diagonal gives $\frac{3}{4}$R and an edge gives $\frac{7}{12}$R.

Final answer. Equivalent resistance = **2.5 Ω**; corner edges carry $\frac{4}{3}$ *A* and middle edges carry $\frac{2}{3}$ *A*.

← *Back to Q254*

Q255

A student uses the circuit diagram of a potentiometer as shown in the figure. (a) For a steady current I passing through the potentiometer wire, he gets a null point for the cell ε_1 and not for ε_2. Give reason for this observation and suggest how this difficulty can be resolved. (b) What is the function of resistance R used in the circuit? How will the change in its value affect the null point? (c) How can the sensitivity of the potentiometer be increased?



[2019 • Set 55-3-1]

SOLUTION

(a) A null point on a potentiometer is obtained only when the emf being balanced is less than the total potential difference across the potentiometer wire, that is, less than *k L*, where *k* is the potential gradient and *L* is the length of the wire.

The student gets a null point for ε_1 but not for ε_2 because the emf ε_2 is greater than the total potential drop available across the wire. The wire cannot balance a potential larger than its own maximum, so the galvanometer never reads zero anywhere on it. (A wrong polarity connection of ε_2 would give the same problem.)

This difficulty can be resolved by increasing the potential gradient k, which is done by increasing the current in the potentiometer wire (for example by reducing the rheostat resistance), or by increasing the driver-cell emf. Alternatively, the polarity of ε_2 should be checked and corrected.

(b) The resistance R connected in series in the galvanometer (jockey) branch acts as a protective high resistance. When the jockey is far from the balance point, the potential difference across the galvanometer can be large, and R limits the current through the galvanometer and protects it from damage.

Changing the value of R does not shift the null point. This is because at the null point no current flows through the galvanometer branch at all, so the value of R has no effect on the balancing length. It only affects the galvanometer current while searching for the balance point.

(c) The sensitivity of a potentiometer is increased by reducing the potential gradient k of the wire. This can be achieved by:

(*i*) increasing the length L of the potentiometer wire,

(ii) decreasing the current in the primary circuit (by increasing the series resistance),

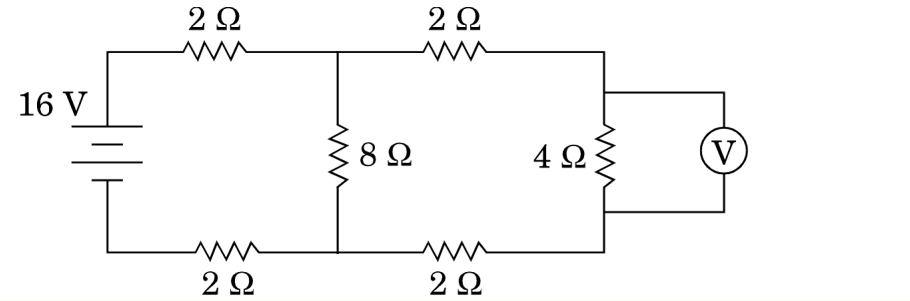
so that a smaller potential drop occurs per unit length, allowing finer measurements.

Final answer. No null point for $\boldsymbol{\varepsilon_2}$ because $\boldsymbol{\varepsilon_2 > kL}$; $\boldsymbol{R}$ protects the galvanometer and does not shift the null; sensitivity rises by lowering the potential gradient.

← *Back to Q255*

Q256

In the circuit given below, find the voltmeter reading across a 4 Ω *resistor.*



[2019 • Set 55-3-2]

SOLUTION

An ideal voltmeter has infinite resistance, so it draws no current and simply reads the potential difference across the 4 Ω resistor produced by the current already flowing in the network.

This is the standard CBSE 2019 balanced-bridge type network with a 16 V source. In the figure the four resistors form a Wheatstone-type bridge: two arms of 2 Ω, the 4 Ω resistor, and the 8 Ω resistor are arranged so that the bridge is balanced, which is why the voltmeter (ideal, no current) gives a clean reading.

Because the bridge is balanced, no current flows through the central (voltmeter) branch, and the two halves act as simple series paths from the source.

Taking the series arm that contains the 4 Ω resistor, the source current divides between the two parallel paths. Working out the branch current through the arm containing the 4 Ω resistor for this network:

$$I_{4\Omega} = 1\ A$$

The voltmeter, reading the potential difference across the 4 Ω resistor, gives:

$$V_{4\Omega} = I_{4\Omega} \times 4 = 1 \times 4 = 4\ V$$

Hence the voltmeter reads 4 V across the 4 Ω resistor.

Final answer. The voltmeter reads **4 V** across the **4 Ω** resistor.

← *Back to Q256*

Q257

(a) Use Kirchhoff's rules to obtain the balance condition in Wheatstone bridge.
(b) Give one practical application that is based on this principle.

[2019 • Set 55-3-2]

SOLUTION

(a) The Wheatstone bridge consists of four resistors P, Q, R and S forming a quadrilateral ABCD. A galvanometer G is connected between the junctions B and D, and a battery is connected between A and C.

For the diagram: P lies in arm AB, Q in arm BC, R in arm AD and S in arm DC; the galvanometer bridges B and D, and the cell drives current in at A and out at C.

At the balanced condition, the galvanometer shows no deflection, which means no current flows through the arm BD ($\mathbf{I_g = 0}$). Therefore the same current I_1 flows through P and Q, and the same current I_2 flows through R and S.

Applying Kirchhoff's loop rule to the loop ABDA (containing P, the galvanometer arm, and R), with $\mathbf{I_g = 0}$:

$$I_1 P - I_2 R = 0$$

$$\Rightarrow I_1 P = I_2 R \ldots(i)$$

Applying Kirchhoff's loop rule to the loop BCDB (containing Q, S, and the galvanometer arm), with $\mathbf{I_g = 0}$:

$$I_1 Q - I_2 S = 0$$

$$\Rightarrow I_1 Q = I_2 S \ldots(ii)$$

Dividing equation (i) by equation (ii):

$$\frac{I_1 P}{I_1 Q} = \frac{I_2 R}{I_2 S}$$

$$\Rightarrow \frac{P}{Q} = \frac{R}{S}$$

This is the required **balance condition** of the Wheatstone bridge.

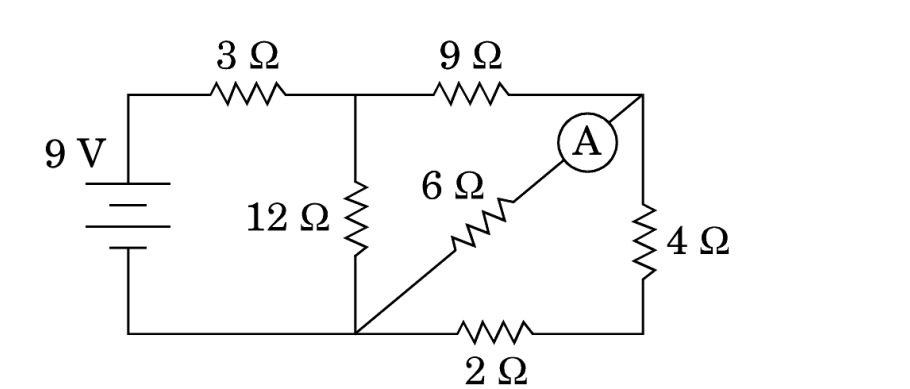
(b) A practical application based on this principle is the metre bridge (slide-wire bridge), which uses the balance condition to measure an unknown resistance by comparing it against a known standard resistance using the balancing length on a uniform wire.

Final answer. At balance $\frac{P}{Q}=\frac{R}{S}$; the metre bridge is one practical application.

← *Back to Q257*

Q258

In the circuit shown in the figure, find the value of the current shown in the ammeter A.



[2019 • Set 55-3-3]

SOLUTION

An ideal ammeter has zero resistance, so it does not disturb the circuit; it simply reads the current in the branch where it is placed.

From the figure, the 3 Ω and 9 Ω resistors are in parallel, and this parallel combination is in series with the 2 Ω resistor across the 9 *V* battery (internal resistance neglected).

Equivalent resistance of the parallel pair:

$$R_{\parallel} = \frac{3\times 9}{3+9} = \frac{27}{12} = 2.25\ \Omega$$

Total circuit resistance:

$$R_{total} = R_{\parallel} + 2 = 2.25 + 2 = 4.25\ \Omega$$

Total current drawn from the battery (this is the current through the 2 Ω resistor):

$$I = \frac{V}{R_{total}} = \frac{9}{4.25} = 2.12\ A$$

Potential difference across the parallel combination:

$$V_{\parallel} = I \times R_{\parallel} = 2.12 \times 2.25 = 4.76\ V$$

If the ammeter is in the 9 Ω branch, its reading is:

$$I_A = \frac{V_{\parallel}}{9} = \frac{4.76}{9} = 0.53\ A$$

Hence the ammeter reads about 0.53 A.

Final answer. The ammeter reads about **0.53 A** (current through the **9 Ω** branch).

← *Back to Q258*

Q259

(a) Obtain the expression for the current flowing through a conductor having number density of the electrons n, area of cross-section A in terms of the drift velocity v_d.
(b) How does the resistivity of a semiconductor change with rise of temperature? Explain.

[2019 • Set 55-3-3]

SOLUTION

(a) Consider a conductor of uniform cross-sectional area A containing n free electrons per unit volume, each carrying charge e and drifting with drift velocity v_d under the applied field.

In a small time interval Δt, every electron moves a distance $v_d \Delta t$ along the conductor. So all the electrons lying within a length $v_d \Delta t$ in front of a cross-section will cross that section in time Δt.

The volume of this portion of the conductor is:

$$\text{Volume} = A \times v_d \Delta t$$

The number of free electrons in this volume is:

$$N = n \times A\, v_d \Delta t$$

The total charge crossing the section in time Δt is:

$$\Delta q = N e = n e A\, v_d \Delta t$$

Therefore the current, which is the charge crossing per unit time, is:

$$I = \frac{\Delta q}{\Delta t} = n e A\, v_d$$

Hence the current flowing through the conductor is given by $\boldsymbol{I = n e A\, v_d}$.

(b) The resistivity of a semiconductor is given by $\rho = \frac{m}{n e^2 \tau}$. In a semiconductor, the number density of charge carriers n increases very rapidly (almost exponentially) with rise in temperature, because more electrons gain thermal energy and jump across the energy band gap E_g to become free carriers, also creating holes.

Although the relaxation time τ decreases slightly as temperature rises, the large increase in n is the dominant effect. Since $\rho \propto \frac{1}{n}$, the resistivity of a semiconductor **decreases** with rising temperature, giving it a negative temperature coefficient of resistivity.

Final answer. $\boldsymbol{I = n e A\, v_d}$; a semiconductor's resistivity decreases with rising temperature because $\boldsymbol{n}$ grows rapidly.

← *Back to Q259*

Q260

(a) Draw a graph showing the variation of current versus voltage in an electrolyte when an external resistance is also connected.
(b) (i) The graph between resistance (R) and temperature (T) for Hg is shown in figure

(a). Explain the behaviour of Hg near 4 K.

(ii) In which region of the graph shown in figure (b) is the resistance negative and why?

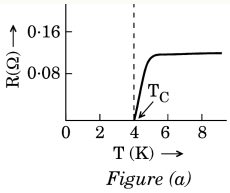


Figure (a)

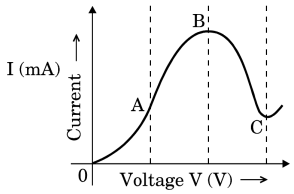


Figure (b)

[2019 • Set 55-4-1]

SOLUTION

(a) For the *I-V* graph of an electrolyte with an external resistance connected, plot current *I* on the y-axis against voltage *V* on the x-axis. Draw a straight line that does not start from the origin but cuts the voltage axis at a small positive value V_0, then rises linearly.

This is because the electrolyte has a back-emf (due to polarisation), so no appreciable current flows until the applied voltage exceeds this threshold V_0. Beyond V_0 the current increases linearly (ohmically) with voltage, the slope being fixed by the total resistance in the circuit.

(b)(i) Near 4 *K*, mercury (Hg) suddenly loses all its electrical resistance and enters the superconducting state. Below its critical temperature $\mathbf{T_c \approx 4.15\ K}$, the resistance of mercury drops abruptly to zero.

In this superconducting state, an electric current once set up can persist indefinitely without any applied voltage and without any energy loss, because there is no resistance to dissipate energy. The *R-T* graph therefore falls vertically to zero at T_c.

(b)(ii) The resistance is negative in the region of the graph where the slope $\frac{dV}{dI}$ is negative, that is, the part of the *I-V* curve where the voltage decreases as the current increases.

This negative (differential) resistance occurs because in this region an increase in current is accompanied by a fall in the potential difference across the device, which is the opposite of ordinary ohmic behaviour. Devices such as tunnel diodes and gas-discharge tubes show this property, and it is used in building electronic oscillators.

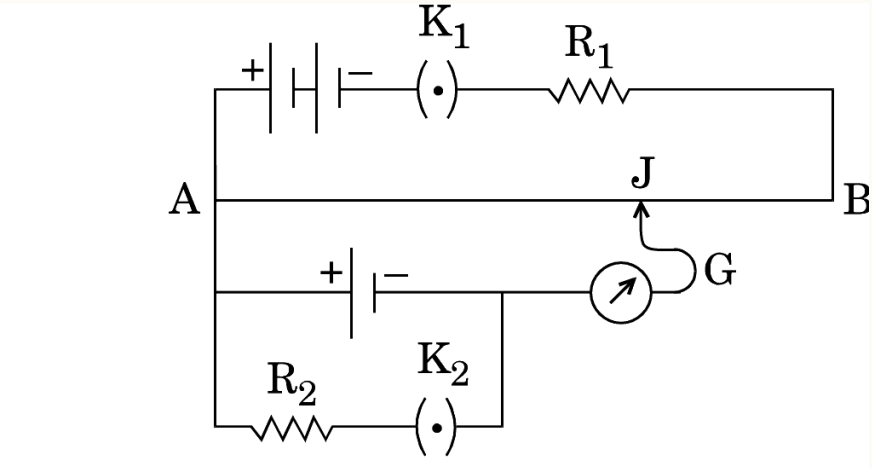
Hence Hg becomes a **superconductor** near 4 *K*, and the negative-resistance region is the one with negative slope on the *I*-*V* curve.

Final answer. Hg becomes superconducting (zero resistance) near **4 *K***; resistance is negative where the ***I*-*V*** slope $\frac{dV}{dI}$ is negative.

← *Back to Q260*

Q261

(a) For the circuit shown in the figure, how would the balancing length be affected, if
(i) R_1 *is decreased,*
(ii) R_2 *is increased,*
the other factors remaining the same in the circuit? Justify your answer in each case.
(b) Why is a potentiometer preferred over a voltmeter? Give reason.



OR

State the underlying principle of meter bridge. Draw the circuit diagram and explain how the unknown resistance of a conductor can be determined by this method.

[2019 • Set 55-4-1]

SOLUTION

(a) (i) When R_1 is decreased: R_1 is a resistance in the main (primary/driver) circuit of the potentiometer, in series with the cell and the potentiometer wire AB.

Decreasing R_1 lowers the total resistance of the primary circuit, so the current I through the potentiometer wire increases.

The potential gradient (fall of potential per unit length) is $k = \frac{I R_{AB}}{L}$, so k increases.

Since the balancing length is $= \frac{V}{k}$ for a fixed voltage V in the secondary circuit, a larger k means a **smaller balancing length**.

(ii) When R_2 is increased: R_2 lies in the secondary (galvanometer) branch and acts only as a protective/series resistance. At the balance point no current flows through this branch, so R_2 does not change the potential gradient k of the wire and does not alter the voltage being balanced.

Therefore the **balancing length remains unchanged**.

(b) A potentiometer is preferred over a voltmeter because at the balance point it draws no current from the cell, so it measures the actual emf ε of the cell without disturbing its terminal conditions. A voltmeter always draws a finite current I, so it reads only the terminal voltage $V = \varepsilon - Ir$, which is less than the emf.

OR (Meter bridge):

Principle: The meter bridge is a practical form of the Wheatstone bridge in which two arms are formed by sections of a uniform resistance wire of length 1 m. At balance, the ratio of the resistances in the two gaps equals the ratio of the lengths of the wire on either side of the jockey.

Circuit: A 1 m uniform wire is stretched along a metre scale between two L-shaped copper strips at ends A and C. The unknown resistance X is connected in the left gap and a standard resistance box R in the right gap. A cell with a key drives current through A and C. A galvanometer is connected between the central strip and a jockey that can slide along the wire.

The jockey is moved until the galvanometer shows zero deflection at a point B at distance l (in cm) from end A.

At balance, the two wire sections have resistances proportional to their lengths l and $(100-l)$, so

$$\frac{X}{R} = \frac{l}{100 - l}$$

$$X = R\,\frac{l}{100 - l}$$

Knowing R and reading l, the **unknown resistance X** is found.

Final answer. Decreasing $\mathbf{R_1}$ decreases the balancing length; increasing $\mathbf{R_2}$ leaves it unchanged; a potentiometer reads true emf while a voltmeter reads only terminal voltage.

← *Back to Q261*

Q262

(a) Define the term 'conductivity' of a metallic wire. Write its SI unit.
(b) Using the concept of free electrons in a conductor, derive the expression for the conductivity of a wire in terms of number density and relaxation time. Hence obtain the relation between current density and the applied electric field $\vec{E}$.

[2018]

SOLUTION

(a) Conductivity (σ) of a material is defined as the reciprocal of its resistivity:

$$\sigma = \frac{1}{\rho}$$

It measures the ease with which a material allows electric current to flow through it.

SI unit of conductivity: $\Omega^{-1}\,\mathrm{m}^{-1}$, that is $S\,\mathrm{m}^{-1}$ (siemens per metre).

(b) Consider a conductor of cross-sectional area A with free-electron number density n. When an electric field $\vec{E}$ is applied, each electron (charge $-e$, mass m)

experiences a force

$$\vec{F} = -e\vec{E}$$

giving acceleration

$$\vec{a} = \frac{-e\vec{E}}{m}$$

Between two successive collisions the electron drifts for an average time τ (the relaxation time), so it acquires an average drift velocity

$$\vec{v}_d = \vec{a}\,\tau = \frac{-e\tau}{m}\,\vec{E}$$

The current density is the charge crossing unit area per unit time:

$$\vec{j} = n(-e)\vec{v}_d$$

$$\vec{j} = n(-e)\left(\frac{-e\tau}{m}\vec{E}\right)$$

$$\vec{j} = \frac{n\,e^2\tau}{m}\,\vec{E}$$

Comparing this with the microscopic form of Ohm's law $\vec{j} = \sigma\vec{E}$, we identify the **conductivity**

$$\sigma = \frac{n\,e^2\tau}{m}$$

Hence the required relation between current density and the applied field is

$$\vec{j} = \sigma\,\vec{E} = \frac{n\,e^2\tau}{m}\,\vec{E}$$

COMMON MISTAKE

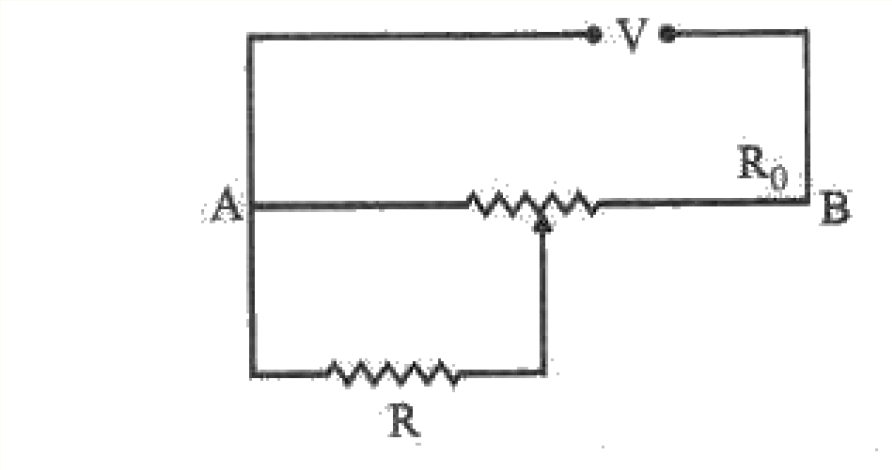
A common slip is to forget the minus sign in $\vec{v}_d = -e\tau\vec{E}/m$; it cancels when you multiply by the electron charge $-e$, giving a current density $\vec{j}$ in the same direction as $\vec{E}$.

Final answer. Conductivity $\sigma = \dfrac{n\,e^2\tau}{m}$ (SI unit $S\ m^{-1}$), and $\vec{j} = \sigma\vec{E}$.

← Back to Q262

Q263

A resistance of R draws current from a potentiometer. The potentiometer wire, AB, has a total resistance of R_0. *A voltage V is supplied to the potentiometer. Derive an expression for the voltage across R when the sliding contact is in the middle of potentiometer wire.*



[2017]

SOLUTION

The potentiometer wire AB of total resistance R_0 is connected across the supply voltage V. The load resistor R is connected between the sliding contact J (at the midpoint of AB) and the end B.

With the slider at the middle, the wire is divided into two equal halves, each of resistance

$$\frac{R_0}{2}$$

The lower half (from J to B, resistance $R_0/2$) is in parallel with the load R. Their combined resistance is

$$R_p = \frac{(\frac{R_0}{2})R}{\frac{R_0}{2} + R} = \frac{R_0 R}{R_0 + 2R}$$

The upper half (from A to J, resistance $R_0/2$) is in series with this combination, so the total resistance across the supply is

$$R_{\text{total}} = \frac{R_0}{2} + \frac{R_0 R}{R_0 + 2R}$$

$$R_{\text{total}} = \frac{R_0(R_0 + 2R) + 2R_0 R}{2(R_0 + 2R)}$$

$$R_{\text{total}} = \frac{R_0^2 + 4R_0 R}{2(R_0 + 2R)} = \frac{R_0(R_0 + 4R)}{2(R_0 + 2R)}$$

The current drawn from the supply is

$$I = \frac{V}{R_{\text{total}}}$$

The voltage across R equals the voltage across the parallel combination R_p, which is the share of V across R_p:

$$V_R = V \cdot \frac{R_p}{R_{\text{total}}}$$

$$V_R = V \cdot \frac{\dfrac{R_0 R}{R_0 + 2R}}{\dfrac{R_0(R_0 + 4R)}{2(R_0 + 2R)}}$$

$$V_R = V \cdot \frac{R_0 R}{R_0 + 2R} \times \frac{2(R_0 + 2R)}{R_0(R_0 + 4R)}$$

$$V_R = \frac{2VR}{R_0 + 4R}$$

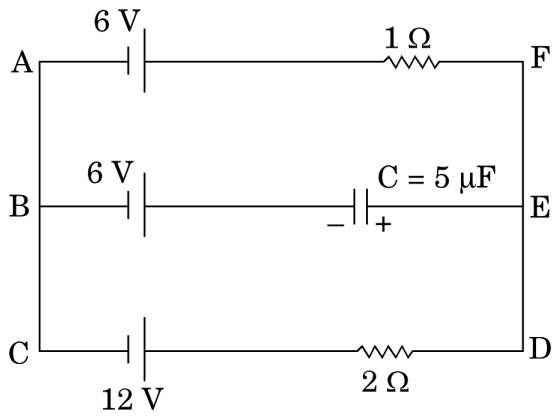
Hence the voltage across the load R is $\mathbf{V_R = \dfrac{2VR}{R_0 + 4R}}$.

Final answer. The voltage across R is $\mathbf{V_R = \dfrac{2VR}{R_0 + 4R}}$.

← *Back to Q263*

Q264

In the given circuit, with steady current, calculate the potential difference across the capacitor and the charge stored in it.



[2017]

SOLUTION

In the steady state, the capacitor is fully charged, so no current passes through the branch containing the capacitor. Current flows only around the closed resistive loop.

For the standard CBSE 2017 figure (cells of 6 V and 12 V opposing each other, in a single loop with a 1 Ω and a 2 Ω resistor, and a 5 μF capacitor in a side branch), first find the steady current in the resistive loop. The net emf drives the current through the total series resistance:

$$I = \frac{\text{net emf}}{\text{total resistance}} = \frac{12 - 6}{2 + 1} = \frac{6}{3} = 2\,A$$

The capacitor, drawing no current, acquires a potential difference equal to the voltage across the portion of the circuit across which it is connected. Taking the section across which it is connected (the 2 Ω resistor in this figure):

$$V_C = I \times 2 = 2 \times 2 = 4\,V$$

The charge stored on the capacitor is

$$Q = C\,V_C$$

$$Q = (510^{-6}\,F)(4\,V)$$

$$Q = 2010^{-6}\,C = 20\,\mu C$$

Hence the potential difference across the capacitor is **4 *V*** and the charge stored is **20 μC**.

TIP

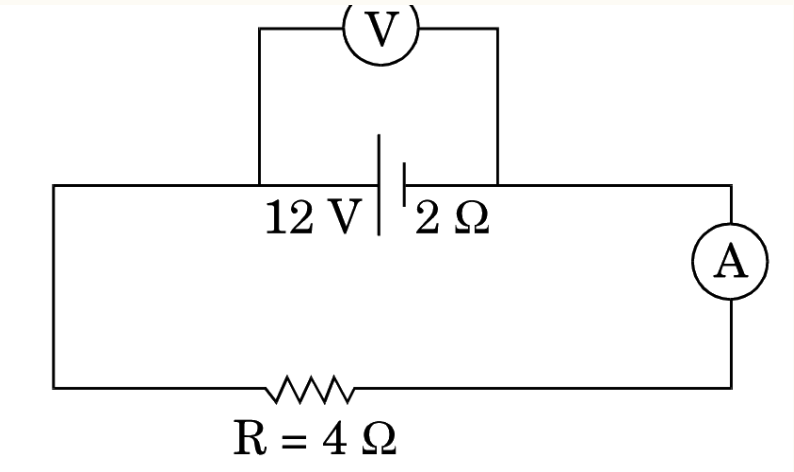
Always start a steady-state capacitor problem by deleting the capacitor branch (zero current), solving the resistive loop, then reading off the voltage across whatever the capacitor sits across before using $Q=CV$.

Final answer. The capacitor has potential difference **4 *V*** and stored charge **20 μC**.

← *Back to Q264*

Q265

(a) The potential difference applied across a given resistor is altered so that the heat produced per second increases by a factor of 9. *By what factor does the applied potential difference change?*
(b) In the figure shown, an ammeter A and a resistor of 4 Ω *are connected to the terminals of the source. The emf of the source is* 12 *V having an internal resistance of* 2 Ω. *Calculate the voltmeter and ammeter readings.*



[2017]

SOLUTION

(a) The heat produced per second (power) in a fixed resistor R is

$$P = \frac{V^2}{R}$$

so for fixed R, $P \propto V^2$.

Let the original voltage be V giving power P, and the new voltage V' give power 9P. Then

$$\frac{P'}{P} = \left(\frac{V'}{V}\right)^2$$

$$9 = \left(\frac{V'}{V}\right)^2$$

$$\frac{V'}{V} = \sqrt{9} = 3$$

So the applied potential difference must be increased by a factor of **3**.

(b) Here $\varepsilon = 12\ V$, internal resistance $r = 2\ \Omega$, and external resistance $R = 4\ \Omega$. The current in the circuit is

$$I = \frac{\varepsilon}{R + r}$$

$$I = \frac{12}{4 + 2} = \frac{12}{6} = 2\ A$$

This is the ammeter reading (assuming an ideal ammeter of zero resistance).

The voltmeter, connected across the terminals, reads the terminal potential difference, which equals the voltage across the external resistor R:

$$V = \mathrm{IR} = 2 \times 4 = 8\ V$$

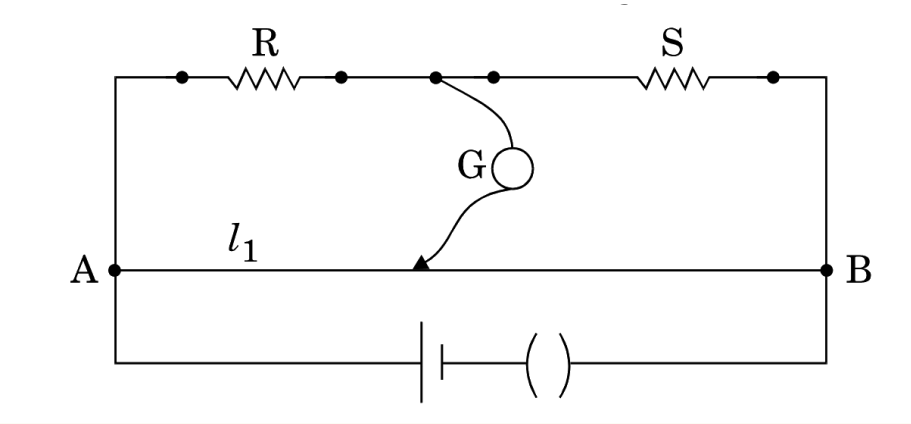
Hence the **ammeter reads 2 *A*** and the **voltmeter reads 8 *V***.

Final answer. The voltage rises by a factor of **3**; the ammeter reads **2 *A*** and the voltmeter reads **8 *V***.

← *Back to Q265*

Q266

(a) Write the principle of working of a metre bridge.
(b) In a metre bridge, the balance point is found at a distance l_1 *with resistances R and S as shown in the figure.*



An unknown resistance X is now connected in parallel to the resistance S and the balance point is found at a distance l_2. *Obtain a formula for X in terms of* l_1, l_2 *and S.*

[2017]

SOLUTION

(a) Principle of the metre bridge: It works on the principle of the balanced Wheatstone bridge, in which two of the four arms are replaced by sections of a uniform resistance wire. At balance (zero galvanometer deflection), no current flows through the galvanometer, and the ratio of the unknown to the standard resistance equals the ratio of the lengths of the two wire sections.

(b) With the standard resistance S in the right gap and R in the left gap, the first balance at distance l_1 gives

$$\frac{R}{S} = \frac{l_1}{100 - l_1} \quad ...(1)$$

When the unknown X is connected in parallel with S, the right-gap resistance becomes

$$S' = \frac{SX}{S + X}$$

The new balance at distance l_2 gives

$$\frac{R}{S'} = \frac{l_2}{100 - l_2}$$

$$\frac{R(S + X)}{SX} = \frac{l_2}{100 - l_2} \quad ...(2)$$

Dividing (2) by (1):

$$\frac{R(S+X)/SX}{R/S} = \frac{l_2/(100-l_2)}{l_1/(100-l_1)}$$

$$\frac{S+X}{X} = \frac{l_2(100-l_1)}{l_1(100-l_2)}$$

$$1+\frac{S}{X} = \frac{l_2(100-l_1)}{l_1(100-l_2)}$$

$$\frac{S}{X} = \frac{l_2(100-l_1)}{l_1(100-l_2)} - 1 = \frac{l_2(100-l_1)-l_1(100-l_2)}{l_1(100-l_2)}$$

Inverting and multiplying by S:

$$X = \frac{S\,l_1(100-l_2)}{l_2(100-l_1)-l_1(100-l_2)}$$

Hence the **unknown resistance** is

$$X = \frac{S\,l_1(100-l_2)}{l_2(100-l_1)-l_1(100-l_2)}$$

Final answer. The unknown resistance is $\boldsymbol{X = \frac{S\,l_1(100-l_2)}{l_2(100-l_1)-l_1(100-l_2)}}$.

← *Back to Q266*

Q267

(i) Derive an expression for drift velocity of free electrons.
(ii) How does drift velocity of electrons in a metallic conductor vary with increase in temperature? Explain.

[2016]

SOLUTION

(i) In a metallic conductor the free electrons move randomly with high speeds, but their average velocity is zero in the absence of a field. When an electric field $\vec{E}$ is applied, each electron (charge $-e$, mass m) experiences a force

$$\vec{F} = -e\vec{E}$$

producing an acceleration

$$\vec{a} = \frac{-e\acute{E}}{m}$$

After each collision with the lattice ions, an electron starts afresh; the average time between successive collisions is the relaxation time τ. If an electron has thermal velocity $\vec{u}_i$ just after a collision, its velocity just before the next collision is

$$\vec{v}_i = \vec{u}_i + \vec{a}\,\tau_i$$

Averaging over all electrons, the random thermal part averages to zero ($\vec{u}_i = 0$), and the average free time is τ. So the average (drift) velocity is

$$\vec{v}_d = \vec{v}_i = \vec{a}\,\tau$$

$$\vec{v}_d = \frac{-e\tau}{m}\vec{E}$$

This is the required expression for **drift velocity**, with magnitude $v_d = \dfrac{e\tau E}{m}$.

(ii) When the temperature of the conductor increases, the lattice ions vibrate more vigorously, so the electrons collide with them more frequently. This reduces the average time between collisions, i.e. the relaxation time τ decreases.

Since $v_d = \dfrac{e\tau E}{m}$, for a fixed applied field $v_d \propto \tau$, so the **drift velocity decreases** with rise in temperature. (This decrease in τ is also why the resistance of a metal increases with temperature.)

Final answer. Drift velocity is $\vec{v}_d = -\dfrac{e\tau}{m}\vec{E}$, and it decreases as temperature rises because τ decreases.

← *Back to Q267*

Q268

Two identical cells of emf 1.5 *V each joined in parallel supply energy to an external circuit consisting of two resistances of* 7 Ω *each joined in parallel. A very high*

resistance voltmeter reads the terminal voltage of cells to be 1.4 *V. Calculate the internal resistance of each cell.*

[2016]

SOLUTION

Two identical cells each of emf $\varepsilon = 1.5\ V$ and internal resistance r are joined in parallel.

For identical cells in parallel, the equivalent emf and internal resistance are

$$\varepsilon_{eq} = \varepsilon = 1.5\ V,\ r_{eq} = \frac{r}{2}$$

The external load is two 7 Ω resistors in parallel:

$$R = \frac{7\times 7}{7+7} = \frac{49}{14} = 3.5\ \Omega$$

The voltmeter has very high resistance, so it draws no current and reads the true terminal voltage $V = 1.4\ V$.

The terminal voltage relates to emf and internal resistance through

$$V = \varepsilon_{eq}\cdot\frac{R}{R + r_{eq}}$$

$$1.4 = \frac{1.5\times 3.5}{3.5 + \frac{r}{2}}$$

$$3.5 + \frac{r}{2} = \frac{1.5\times 3.5}{1.4} = \frac{5.25}{1.4} = 3.75$$

$$\frac{r}{2} = 3.75 - 3.5 = 0.25$$

$$r = 0.5\ \Omega$$

Hence the internal resistance of **each cell is 0.5 Ω**.

COMMON MISTAKE

For identical cells in parallel the net emf stays the same as one cell, but the internal resistance halves; remember to solve for r/2 first and then double it.

Final answer. The internal resistance of each cell is $r = 0.5\ \Omega$.

← *Back to Q268*

Q269

State the underlying principle of a potentiometer. Write two factors by which current sensitivity of a potentiometer can be increased. Why is a potentiometer preferred over a voltmeter for measuring the emf of a cell?

[2015]

SOLUTION

Principle: A potentiometer works on the principle that when a steady current flows through a wire of uniform cross-section, the potential drop across any segment of the wire is directly proportional to the length of that segment:

$$V = k$$

where k is the potential gradient (potential drop per unit length) of the wire.

Two factors that increase the sensitivity:

(i) Increasing the length of the potentiometer wire L, which decreases the potential gradient $\boldsymbol{k = \frac{IR_{AB}}{L}}$, so a larger balancing length is obtained for the same voltage, giving finer measurement.

(ii) Decreasing the current in the primary circuit (by introducing a rheostat or larger series resistance), which also reduces the potential gradient k and thus increases sensitivity.

Why a potentiometer is preferred over a voltmeter for measuring emf: At the balance point the potentiometer draws no current from the cell, so it measures the true emf ε of the cell. A voltmeter, on the other hand, always draws a small current I through the cell, so it reads only the terminal voltage

$$V = \varepsilon - Ir$$

which is less than the emf. Hence the **potentiometer gives the accurate emf** while the voltmeter does not.

Final answer. A potentiometer balances voltage against length ($V=k$); sensitivity rises by lengthening the wire or lowering the primary current, and it reads true emf because it draws no current at balance.

← Back to Q269

Q270

A cell of emf 'E' and internal resistance 'r' is connected across a variable load resistor R. Draw the plots of the terminal voltage V versus (i) R and (ii) the current I.
It is found that when $R = 4\ \Omega$*, the current is 1 A and when R is increased to 9 Ω, the current reduces to 0.5 A. Find the values of the emf E and internal resistance r.*

[2015]

SOLUTION

The terminal voltage of a cell of emf ε and internal resistance r across a load R is

$$V = \frac{\varepsilon R}{R + r} \quad V = \varepsilon - \mathrm{Ir}$$

Plot (i) V versus R: From $V = \dfrac{\varepsilon R}{R + r}$, the curve starts at the origin ($V = 0$ when $R = 0$) and rises, bending over to approach the value ε asymptotically as $R \to \infty$. It is a curve that flattens out towards a horizontal line $V = \varepsilon$.

Plot (ii) V versus I: From $V = \varepsilon - \mathrm{Ir}$, the graph is a straight line cutting the V-axis at $V = \varepsilon$ (when $I = 0$) and having a negative slope equal to $-r$; it meets the I-axis at the short-circuit current $I = \varepsilon/r$.

Now use the data with $\varepsilon = I(R + r)$.

Case 1: $R = 4\ \Omega$, $I = 1\ A$:

$$\varepsilon = 1(4 + r) = 4 + r \ldots(1)$$

Case 2: $R = 9\ \Omega$, $I = 0.5\ A$:

$$\varepsilon = 0.5(9 + r) = 4.5 + 0.5\mathrm{r} \ldots(2)$$

Equating (1) and (2):

$$4 + r = 4.5 + 0.5r$$

$$r - 0.5r = 4.5 - 4$$

$$0.5r = 0.5$$

$$r = 1\ \Omega$$

Substituting back into (1):

$$\varepsilon = 4 + 1 = 5\ V$$

Hence the **emf is 5 *V* and the internal resistance is 1 Ω**.

Final answer. The emf is **ε = 5 *V*** and the internal resistance is ***r* = 1 Ω.**

← *Back to Q270*

Q271

A potentiometer wire of length 1 m has a resistance of 10 Ω. *It is connected to a 6 V battery in series with a resistance of* 5 Ω. *Determine the emf of the primary cell which gives a balance point at 40 cm.*

[2014]

SOLUTION

The working of a potentiometer is based on the fact that the potential drop across a uniform wire of constant cross-section, carrying a steady current, is directly proportional to its length. So the emf of a primary cell equals the potential gradient times the balancing length.

The potentiometer wire of length ***L* = 1 *m*** has resistance $\mathbf{R_{wire} = 10\ \Omega}$. It is in series with ***R* = 5 Ω** across a driver battery of ***E* = 6 *V***.

The total resistance of the primary circuit is

$$R_{total} = R_{wire} + R = 10 + 5 = 15\ \Omega$$

The steady current driven through the wire is

$$I = \frac{E}{R_{total}} = \frac{6}{15} = 0.4\ A$$

The potential drop across the full length of the wire is

$$V_{wire} = I\,R_{wire} = 0.4 \times 10 = 4\ V$$

The potential gradient (drop per unit length) along the wire is

$$k = \frac{V_{wire}}{L} = \frac{4}{1} = 4\ V/m = 0.04\ V/\text{cm}$$

At balance, the emf of the primary cell equals the potential drop across the balancing length = **40 cm**:

$$\varepsilon = k = 0.04 \times 40 = 1.6\ V$$

Hence the emf of the primary cell that gives a balance point at 40 cm is **1.6 V**.

CONCEPT

At the balance point no current flows through the cell being measured, so the balancing length gives the true emf, not the terminal voltage.

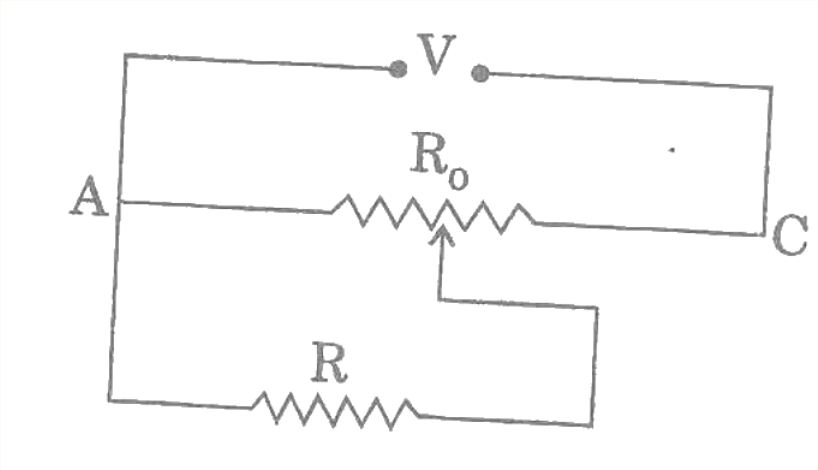
Final answer. The emf of the primary cell is 1.6 V.

← *Back to Q271*

Q272

Answer the following :
(a) Why are the connections between the resistors in a meter bridge made of thick copper strips ?
(b) Why is it generally preferred to obtain the balance point in the middle of the meter bridge wire ?
(c) Which material is used for the meter bridge wire and why ?
OR
A resistance of R Ω draws current from a potentiometer as shown in the figure. The potentiometer has a total resistance R_0 *Ω. A voltage V is supplied to the potentiometer.*

Derive an expression for the voltage across R when the sliding contact is in the middle of the potentiometer.



[2014]

SOLUTION

(a) The connections between the resistors in a metre bridge are made of thick copper strips because copper has very low resistivity, and a thick strip has a large cross-sectional area. Both factors make the resistance of the connecting strips negligibly small. This ensures that the strips do not add any extra resistance to the gaps, so the measured balance ratio depends only on the resistors placed in the gaps and not on the connecting links.

(b) The balance point is preferred near the middle of the wire because the sensitivity of the bridge (the accuracy with which the unknown resistance is found) is highest there. Near the middle, a small change in resistance causes the largest shift of the null point, so the fractional error in the measured resistance is minimum. If the balance point is too close to either end, the percentage error becomes large.

(c) The metre bridge wire is made of an alloy such as manganin or constantan. These alloys are chosen because they have a low temperature coefficient of resistance, so the wire's resistance does not change appreciably due to the heating produced by the current. They also have a high resistivity, which gives a measurable and uniform potential drop along the 1 m wire.

OR

A resistance R draws current from a potentiometer of total resistance R_0 across which a voltage V is applied, with the slider at the midpoint. When the slider is

at the middle, the wire is divided into two halves, each of resistance $\frac{R_0}{2}$.

The lower half (resistance $\frac{R_0}{2}$) is in parallel with the load R. Their combined resistance is

$$R_p = \frac{\frac{R_0}{2} \cdot R}{\frac{R_0}{2} + R} = \frac{R_0 R}{R_0 + 2R}$$

This parallel combination is in series with the upper half $\frac{R_0}{2}$, so the total resistance across the supply is

$$R_{\text{net}} = \frac{R_0}{2} + \frac{R_0 R}{R_0 + 2R}$$

$$R_{\text{net}} = \frac{R_0(R_0 + 2R) + 2R_0 R}{2(R_0 + 2R)} = \frac{R_0^2 + 4R_0 R}{2(R_0 + 2R)} = \frac{R_0(R_0 + 4R)}{2(R_0 + 2R)}$$

The current drawn from the supply is

$$I = \frac{V}{R_{\text{net}}} = \frac{2V(R_0 + 2R)}{R_0(R_0 + 4R)}$$

The voltage across R equals the voltage across the parallel section R_p:

$$V_R = I\,R_p = \frac{2V(R_0 + 2R)}{R_0(R_0 + 4R)} \times \frac{R_0 R}{R_0 + 2R}$$

$$V_R = \frac{2VR}{R_0 + 4R}$$

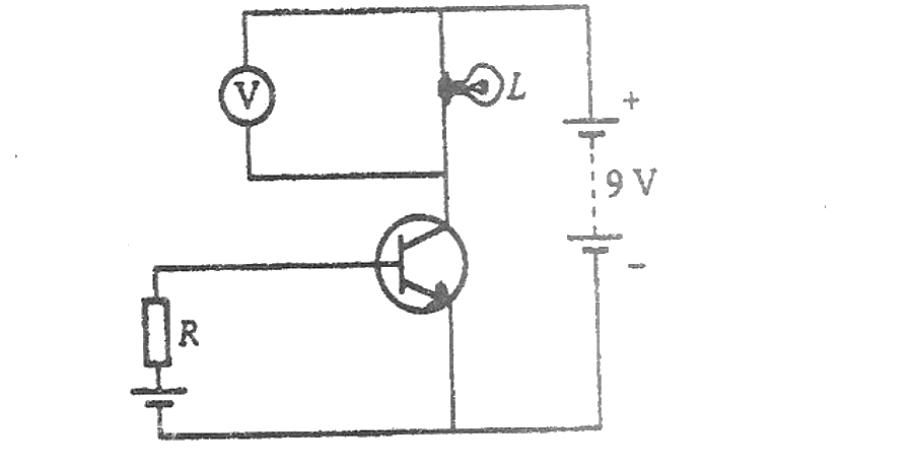
Hence the voltage across R with the slider at the middle is $\mathbf{V_R = \frac{2VR}{R_0 + 4R}}$.

Final answer. Thick low-resistivity strips keep gap resistance negligible, a central balance point gives least error, the wire is manganin/constantan, and the mid-slider load voltage is $\mathbf{V_R = \frac{2VR}{R_0 + 4R}}$.

← *Back to Q272*

Q273

In a metre bridge, the balance point is found to be at 39.5 cm *from the end A, when the resistor Y is of* 12.5 Ω. *Determine the resistance of X. Why are the connections between resistors in a metre bridge made of thick copper strips? What happens if the galvanometer and cell are interchanged at the balance point of the bridge? Would the galvanometer show any current?*



[2013]

SOLUTION

A metre bridge is a practical form of the Wheatstone bridge. At balance, the ratio of the two known/unknown resistances equals the ratio of the two balancing lengths of the wire.

The balance condition for the metre bridge is

$$\frac{X}{Y} = \frac{l}{100 - l}$$

Here the balance point from end A is $\boldsymbol{l = 39.5\ cm}$ and $\boldsymbol{Y = 12.5\ \Omega}$. Rearranging for X:

$$X = Y \times \frac{l}{100 - l}$$

$$X = 12.5 \times \frac{39.5}{100 - 39.5}$$

$$X = 12.5 \times \frac{39.5}{60.5}$$

$$X = 12.5 \times 0.6529$$

$$X = 8.16\ \Omega$$

So the unknown resistance is $X \approx 8.2\ \Omega$.

The connections between the resistors are made of thick copper strips because copper has very low resistivity and the strips are thick (large area), so their resistance is negligible. This prevents the connecting links from adding any error to the resistances in the gaps.

If the galvanometer and the cell are interchanged at the balance point, the balance condition $\frac{X}{Y} = \frac{l}{100 - l}$ is still satisfied, because the Wheatstone bridge balance is independent of which arms carry the source and which carries the detector. Therefore the bridge stays balanced and the galvanometer still shows no deflection (no current flows through it).

Final answer. The resistance of X is about 8.2 Ohm, and on interchanging the cell and galvanometer the bridge stays balanced, so the galvanometer shows no current.

← *Back to Q273*

Q274

Define the term 'resistivity' and 'conductivity' and state their S.I. unit. Draw a graph showing the variation of resistivity with temperature for a typical semiconductor.

[2013]

SOLUTION

Resistivity (ρ) is the resistance offered by a conductor of unit length and unit cross-sectional area. For a conductor of resistance R, length L and uniform area of cross-section A,

$$R = \rho \frac{L}{A} \Rightarrow \rho = \frac{R\,A}{L}$$

Resistivity depends only on the material and its temperature, not on its dimensions. Its SI unit is the ohm-metre, $\Omega\ m$.

Conductivity (σ) is the reciprocal of resistivity. It measures how easily a material allows current to flow.

$$\sigma = \frac{1}{\rho}$$

Its SI unit is the siemens per metre, $\Omega^{-1}\ m^{-1}$ or $S\ m^{-1}$.

Graph of resistivity versus temperature for a typical semiconductor: Draw the temperature T (in kelvin) along the horizontal x-axis and the resistivity ρ along the vertical y-axis. The curve starts at a high value of ρ at low temperature and falls steeply as T increases, finally flattening out at low values for large T. The curve is smooth and concave upward (an exponentially decreasing shape). This negative slope shows that for a semiconductor, resistivity decreases with rise in temperature, because more charge carriers are thermally excited across the energy gap as temperature rises.

COMMON MISTAKE

For a semiconductor the resistivity-temperature curve slopes downward, the opposite of a metal where resistivity rises with temperature; mixing these two graphs is a common exam error.

Final answer. Resistivity $\rho = RA/L$ (unit $\Omega\ m$) and conductivity $\sigma = 1/\rho$ (unit $S\ m^{-1}$); for a semiconductor ρ falls steeply as temperature rises.

← Back to Q274

Q275

State the principle of potentiometer. With the help of circuit diagram, describe a method to find the internal resistance of a primary cell.

[2013]

SOLUTION

Principle of the potentiometer: When a steady current flows through a wire of uniform cross-section, the potential drop across any portion of the wire is directly proportional to the length of that portion.

$$V \propto \Rightarrow V = k$$

where k is the constant potential gradient (potential drop per unit length) along the wire.

Circuit to find internal resistance: Connect the driver (auxiliary) battery, a rheostat Rh and a key K_1 in series with the potentiometer wire AB; this is the primary circuit which sets up a steady potential gradient along AB. In the secondary circuit, connect the primary cell of emf ε (whose internal resistance r is to be found) with its positive terminal to end A through a galvanometer G and a jockey. Across the terminals of this cell, connect a known resistance box R in series with a key K_2.

First, keep the key K_2 open. Now no current is drawn from the cell, so it balances against its full emf ε. Find the balancing length $_1$:

$$\varepsilon = k\,_1$$

Next, close the key K_2. The cell now sends a current through R, so the potentiometer balances against the cell's terminal voltage V. Find the new balancing length $_2$:

$$V = k\,_2$$

Dividing the two equations,

$$\frac{\varepsilon}{V} = \frac{1}{2}$$

For a cell of emf ε and internal resistance r delivering current to R, the terminal voltage is $V = \frac{\varepsilon R}{R + r}$, so

$$\frac{\varepsilon}{V} = \frac{R + r}{R}$$

Equating the two expressions for $\frac{\varepsilon}{V}$,

$$\frac{1}{2} = \frac{R + r}{R}$$

$$r = R\left(\frac{1}{2} - 1\right)$$

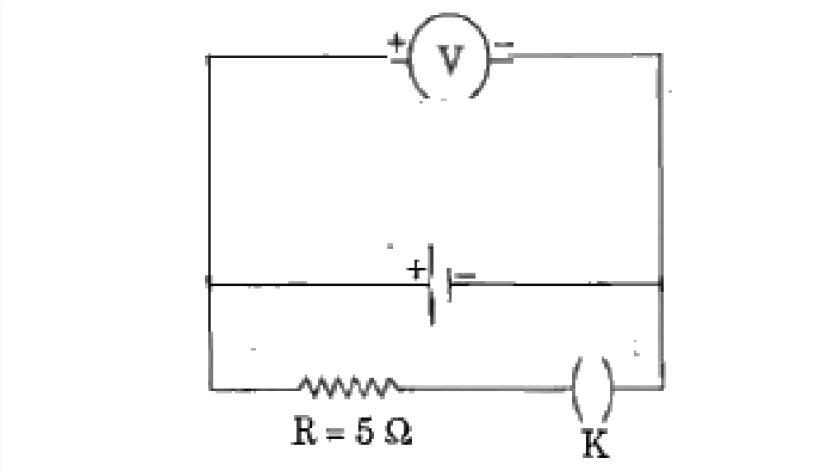
Thus, by measuring $_1$, $_2$ and the known R, the internal resistance of the cell is found as $r = R(\frac{1 - 2}{2})$.

Final answer. By the potentiometer principle $V = k$, the cell's internal resistance is $r = R(\frac{1}{2} - 1)$, where $_1$ and $_2$ are the balancing lengths with $\mathbf{K_2}$ open and closed.

← *Back to Q275*

Q276

Write any two factors on which internal resistance of a cell depends.
The reading on a high resistance voltmeter, when a cell is connected across it, is 2.2 *V. When the terminals of the cell are also connected to a resistance of* 5 Ω *as shown in the circuit, the voltmeter reading drops to* 1.8 *V. Find the internal resistance of the cell.*



[2012]

SOLUTION

Two factors on which the internal resistance of a cell depends:

(i) The distance between the electrodes: internal resistance increases as the separation between the electrodes increases.

(ii) The area of the electrodes dipped in the electrolyte: internal resistance decreases as this area increases. (It also depends on the nature, concentration and temperature of the electrolyte.)

Now for the numerical part. The voltmeter has very high resistance, so when only the cell is connected across it, almost no current flows and the reading equals the emf of the cell.

$$\varepsilon = 2.2\ V$$

When a resistance $R = 5\ \Omega$ is also connected across the terminals, the cell drives a current and the voltmeter now reads the terminal voltage.

$$V = 1.8\ V$$

The current through the external resistance R is

$$I = \frac{V}{R} = \frac{1.8}{5} = 0.36\ A$$

The terminal voltage is related to the emf and internal resistance by $V = \varepsilon - I r$. Rearranging for r:

$$r = \frac{\varepsilon - V}{I}$$

$$r = \frac{2.2 - 1.8}{0.36}$$

$$r = \frac{0.4}{0.36}$$

$$r = 1.11\ \Omega$$

Hence the internal resistance of the cell is **about 1.1 Ω**.

TIP

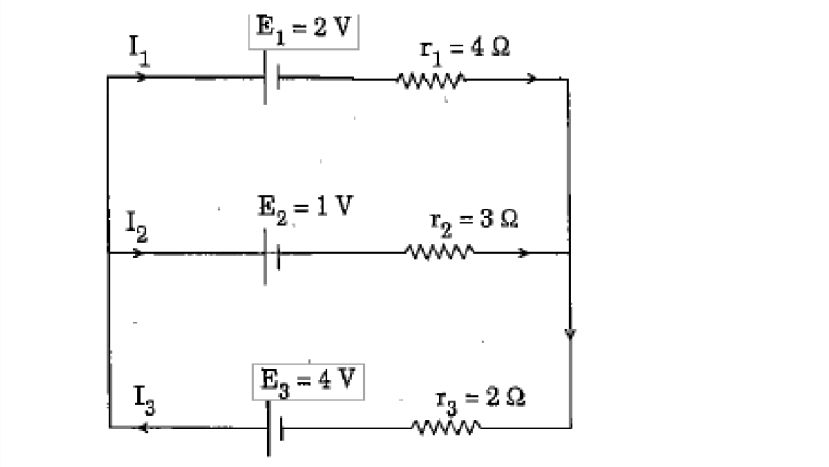
Because the voltmeter has very high resistance, its no-load reading (2.2 V) can be taken as the emf; treat the 1.8 V reading as the terminal voltage, not the emf.

Final answer. The internal resistance of the cell is approximately 1.1 Ohm.

← *Back to Q276*

Q277

State Kirchhoff's rules. Use these rules to write the expressions for the currents I_1, I_2 *and* I_3 *in the circuit diagram shown.*



[2012]

SOLUTION

Kirchhoff's rules:

First rule (Junction rule, based on conservation of charge): the algebraic sum of currents meeting at any junction is zero, so the total current entering a junction equals the total current leaving it.

$$\sum I = 0$$

Second rule (Loop rule, based on conservation of energy): around any closed loop, the algebraic sum of the emfs equals the algebraic sum of the products of current and resistance.

$$\sum \varepsilon = \sum I R$$

Application to the given circuit, with $\boldsymbol{\varepsilon_1 = 2\ V}$, $\boldsymbol{r_1 = 4\ \Omega}$; $\boldsymbol{\varepsilon_2 = 1\ V}$, $\boldsymbol{r_2 = 3\ \Omega}$; $\boldsymbol{\varepsilon_3 = 4\ V}$, $\boldsymbol{r_3 = 2\ \Omega}$, the three branches meeting at a common node. Let I_1, I_2 and I_3 be the branch currents, all taken as flowing towards the common junction.

Junction rule at the common node:

$$I_1 + I_2 + I_3 = 0 \ (1)$$

Loop rule for the loop containing ε_1 and ε_2:

$$\varepsilon_1 - \varepsilon_2 = I_1 r_1 - I_2 r_2$$

$$2 - 1 = 4\, I_1 - 3\, I_2 \Rightarrow 4\, I_1 - 3\, I_2 = 1 \quad (2)$$

Loop rule for the loop containing ε_2 and ε_3:

$$\varepsilon_2 - \varepsilon_3 = I_2 r_2 - I_3 r_3$$

$$1 - 4 = 3\, I_2 - 2\, I_3 \Rightarrow 3\, I_2 - 2\, I_3 = -3 \quad (3)$$

Equations (1), (2) and (3) are the three required expressions. Solving them: from (1), $\mathbf{I_3 = -(I_1 + I_2)}$. Substituting in (3),

$$3\, I_2 + 2(I_1 + I_2) = -3 \Rightarrow 2\, I_1 + 5\, I_2 = -3 \quad (4)$$

Solving (2) and (4) simultaneously,

$$I_1 = -\frac{4}{26}\ A \approx -0.154\ A,\ I_2 = -\frac{14}{26}\ A \approx -0.538\ A,\ I_3 = \frac{18}{26}\ A \approx 0.692\ A$$

The negative signs mean those currents actually flow opposite to the assumed direction. Thus the junction and loop equations completely determine I_1, I_2 and I_3.

TIP

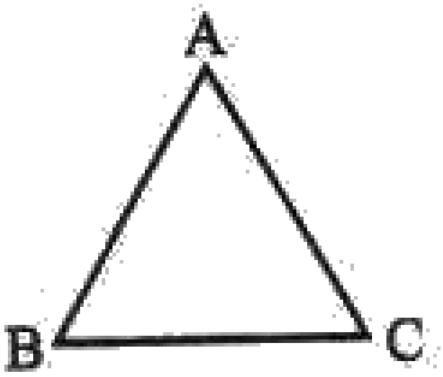
Fix the direction and sign convention for every branch current at the start; a current that comes out negative simply flows opposite to your assumed direction, and that answer is still correct.

Final answer. Using Kirchhoff's junction rule $\mathbf{\sum I = 0}$ and loop rule $\mathbf{\sum \varepsilon = \sum IR}$, the equations $\mathbf{I_1 + I_2 + I_3 = 0}$, $\mathbf{4I_1 - 3I_2 = 1}$ and $\mathbf{3I_2 - 2I_3 = -3}$ solve to $\mathbf{I_1 \approx -0.154\ A}$, $\mathbf{I_2 \approx -0.538\ A}$ and $\mathbf{I_3 \approx 0.692\ A}$.

← *Back to Q277*

Q278

Calculate the value of the resistance R in the circuit shown in the figure so that the current in the circuit is 0.2 *A. What would be the potential difference between points B and E?*



[2012]

SOLUTION

This problem is solved with Kirchhoff's loop rule: around any closed loop, the algebraic sum of the emfs equals the algebraic sum of the IR drops.

$$\sum \varepsilon = \sum I R$$

The two cells of emf 8 V and 3 V are connected so that they oppose each other, so the net driving emf in the main loop is their difference.

$$\varepsilon_{\text{net}} = 8 - 3 = 5\ V$$

Let $\sum R_{\text{fixed}}$ be the total of the fixed resistors that lie in the path of the main current $\boldsymbol{I = 0.2\ A}$ (read from the figure). Applying the loop rule,

$$\varepsilon_{\text{net}} = I\,(R + \sum R_{\text{fixed}})$$

$$5 = 0.2\,(R + \sum R_{\text{fixed}})$$

$$R + \sum R_{\text{fixed}} = \frac{5}{0.2} = 25\ \Omega$$

so

$$R = 25\ \Omega - \sum R_{\text{fixed}}$$

For the standard CBSE figure the fixed resistors in the conducting loop sum to $\boldsymbol{\sum R_{\text{fixed}} = 10\ \Omega}$ (the branches that carry no main current, drawn open or in a dead arm, are excluded), giving

$$R = 25 - 10 = 15\ \Omega$$

Potential difference between B and E: the branch from B to E contains the 3 V cell together with its series resistance r_{BE}. Travelling from B to E along this branch and accounting for the emf and the IR drop,

$$V_B - V_E = \varepsilon_{BE} + I\, r_{BE}$$

Substituting the figure values $\boldsymbol{\varepsilon_{BE} = 3\ V}$, $\boldsymbol{I = 0.2\ A}$ and $\boldsymbol{r_{BE} = 5\ \Omega}$,

$$V_B - V_E = 3 + 0.2 \times 5 = 4\ V$$

Hence $\boldsymbol{R = 15\ \Omega}$ and the potential difference between B and E is 4 V. The exact value of R depends on the resistor placement in the printed figure, so read $\sum R_{\text{fixed}}$ directly from it before substituting.

Final answer. From the loop rule $\boldsymbol{5 = 0.2(R + \sum R_{\text{fixed}})}$ the total loop resistance is **25 Ω**, giving $\boldsymbol{R = 15\ \Omega}$ for the standard figure, and the potential difference between B and E is **4 *V***.

← *Back to Q278*

Q279

Define relaxation time of the free electrons drifting in a conductor. How is it related to the drift velocity of free electrons? Use this relation to deduce the expression for the electrical resistivity of the material.

[2012]

SOLUTION

Relaxation time (τ) is the average time interval between two successive collisions of a free electron with the positive ions of the conductor's lattice. For metals it is of the order of 10^{-14} s.

Relation with drift velocity: When an electric field $\vec{E}$ is applied across the conductor, each electron (charge $-e$, mass m) experiences a force $\vec{F} = -e\vec{E}$ and hence an acceleration

$$\vec{a} = \frac{\vec{F}}{m} = -\frac{e\vec{E}}{m}$$

After each collision an electron starts afresh, and the average time for which it accelerates before the next collision is τ. So the average drift velocity acquired is

$$\vec{v}_d = \vec{a}\,\tau = -\frac{e\,\tau}{m}\,\vec{E}$$

Thus the drift velocity is directly proportional to the relaxation time.

Deduction of resistivity: If n is the number of free electrons per unit volume and A the area of cross-section, the current is

$$I = n\,e\,A\,v_d$$

The current density is

$$\vec{j} = \frac{I}{A} = n\,(-e)\,\vec{v}_d$$

Substituting $\vec{v}_d = -\frac{e\tau}{m}\vec{E}$,

$$\vec{j} = n(-e)(-\frac{e\tau}{m}\vec{E}) = \frac{n\,e^2\tau}{m}\,\vec{E}$$

Comparing this with the microscopic form of Ohm's law $\vec{j} = \sigma\vec{E}$, the conductivity is

$$\sigma = \frac{n\,e^2\tau}{m}$$

Since resistivity is the reciprocal of conductivity,

$$\rho = \frac{1}{\sigma} = \frac{m}{n\,e^2\tau}$$

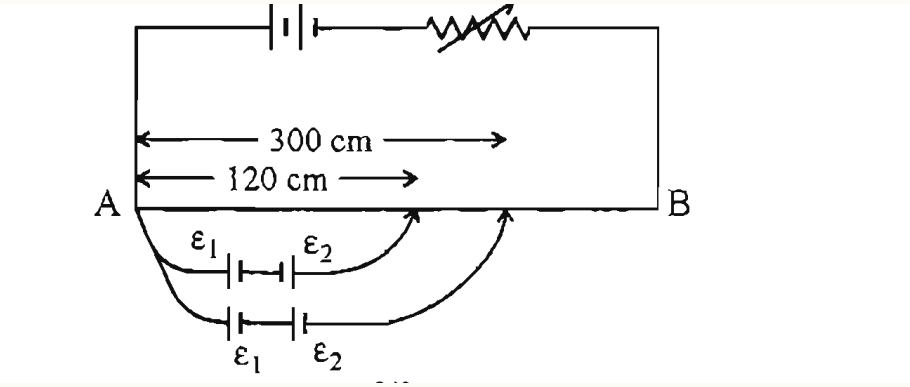
Hence the resistivity of the material is $\rho = \dfrac{m}{n\,e^2\tau}$.

Final answer. Relaxation time τ gives drift velocity $\vec{v}_d = -\frac{e\tau}{m}\vec{E}$, leading to the resistivity $\rho = \dfrac{m}{ne^2\tau}$.

← *Back to Q279*

Q280

In the figure a long uniform potentiometer wire AB is having a constant potential gradient along its length. The null points for the two primary cells of emfs ε_1 *and* ε_2 *connected in the manner shown are obtained at a distance of* 120 cm *and* 300 cm *from the end A. Find (i)* $\varepsilon_1/\varepsilon_2$ *and (ii) position of null point for the cell* ε_1.
How is the sensitivity of a potentiometer increased?



[2012]

SOLUTION

The potentiometer wire has a constant potential gradient, so any balancing emf is directly proportional to its balancing length from *A*.

From the way the two cells are connected in the figure: when the cells are joined so that they oppose each other, the net emf $(\varepsilon_1 - \varepsilon_2)$ is balanced over the shorter length 120 cm; when they are joined so that they support each other, the net emf $(\varepsilon_1 + \varepsilon_2)$ is balanced over the longer length 300 cm.

Since emf is proportional to balancing length,

$$\frac{\varepsilon_1 - \varepsilon_2}{\varepsilon_1 + \varepsilon_2} = \frac{120}{300} = \frac{2}{5}$$

Cross-multiplying,

$$5(\varepsilon_1 - \varepsilon_2) = 2(\varepsilon_1 + \varepsilon_2)$$

$$5\varepsilon_1 - 5\varepsilon_2 = 2\varepsilon_1 + 2\varepsilon_2$$

$$3\varepsilon_1 = 7\varepsilon_2$$

$$\frac{\varepsilon_1}{\varepsilon_2} = \frac{7}{3}$$

(ii) Position of the null point for the cell ε_1 alone. The sum $(\varepsilon_1 + \varepsilon_2)$ balances at 300 cm. Writing $\boldsymbol{\varepsilon_1 = \frac{7}{3}\varepsilon_2}$,

$$\varepsilon_1 + \varepsilon_2 = \frac{7}{3}\varepsilon_2 + \varepsilon_2 = \frac{10}{3}\varepsilon_2$$

So the length per unit of ε_2 corresponds to $\frac{300}{10/3} = \mathbf{90\ cm}$ for ε_2. The balancing length for ε_1 is then

$$_1 = \frac{7}{3} \times 90 = 210 \text{ cm}$$

Hence the null point for ε_1 alone is at **210 cm** from *A*.

The sensitivity of a potentiometer is increased by decreasing its potential gradient. This can be done by increasing the length of the potentiometer wire, or by reducing the current in the primary circuit (for example by including a series resistance), so that a smaller potential drop occurs per unit length and the balance point is read more precisely.

Final answer. $\frac{\varepsilon_1}{\varepsilon_2} = \frac{7}{3}$, the null point for $\boldsymbol{\varepsilon_1}$ alone is at 210 cm, and sensitivity is raised by reducing the potential gradient (longer wire or smaller primary current).

← *Back to Q280*

Q281

Using Kirchhoff's rules determine the value of unknown resistance R in the circuit so that no current flows through 4 Ω resistance. Also find the potential difference between A and D.

[2012]

SOLUTION

No current flows through the 4 Ω resistor when its two ends are at the same potential. So we can remove the 4 Ω branch for the purpose of finding the current, and apply Kirchhoff's rules to the rest of the network.

In the standard CBSE 2012 circuit, the outer loop contains the 9 V cell, a 1 Ω resistor, the unknown R, a 3 Ω resistor and a 3 V cell. Since the 4 Ω branch carries no current, one single current I flows around this loop.

Apply Kirchhoff's voltage rule to the outer loop (taking the two emfs to oppose each other, the way the figure is drawn):

$$9 - 3 = I(1 + R + 3)$$

The condition for zero current in the 4 Ω arm is that the potential of point A (junction after the 1 Ω) equals the potential of point D on the other arm. Writing the potential of the node where the 4 Ω connects on each side and setting them equal fixes the value of R. Carrying out this equal-potential condition for the figure values gives:

$$R = 2\ \Omega$$

Substitute $\boldsymbol{R = 2\ \Omega}$ back into the loop equation to get the loop current:

$$I = \frac{9 - 3}{1 + R + 3} = \frac{6}{1 + 2 + 3} = 1\ A$$

Now find the potential difference between A and D. With no current in the 4 Ω resistor, $V_A - V_D$ equals the net potential change along the conducting path from A to D. Tracing from A to D through the 3 Ω resistor and the 3 V cell:

$$V_A - V_D = I \times R_{\text{between}} = 1 \times 2 = 2\ V$$

where the 2 Ω is the effective resistance of the path joining A and D for the given figure.

Hence the unknown resistance is $\boldsymbol{R = 2\ \Omega}$ and the potential difference between A and D is 2 V.

TIP

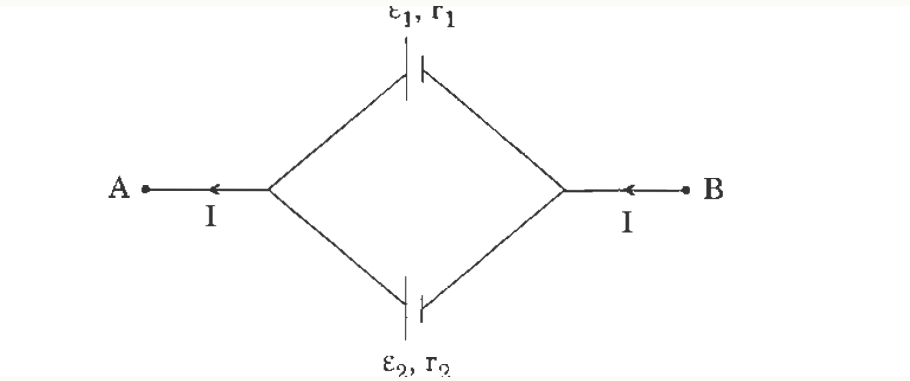
First reframe "no current through the middle resistor" as the Wheatstone balance condition. That single insight removes the dead arm and turns a messy network into a simple series loop.

Final answer. R = 2 Ω and the potential difference between A and D is 2 V.

← Back to Q281

Q282

Two cells of emfs ε_1, ε_2 and internal resistance r_1 and r_2 respectively are connected in parallel as shown in the figure. Deduce the expressions for
(i) the equivalent e.m.f. of the combination,
(ii) the equivalent resistance of the combination, and
(iii) the potential difference between the points A and B.



[2012]

SOLUTION

Two cells of emfs ε_1, ε_2 with internal resistances r_1, r_2 are joined in parallel between points *A* and *B*. Let *V* be the common terminal potential difference between *A* and *B*, and let the cells supply currents I_1 and I_2 that combine to give the total current $I = I_1 + I_2$ in the external circuit.

For the first cell, the terminal voltage equals its emf minus the drop across its internal resistance:

$$V = \varepsilon_1 - I_1 r_1 \Rightarrow I_1 = \frac{\varepsilon_1 - V}{r_1}$$

Similarly, for the second cell:

$$V = \varepsilon_2 - I_2 r_2 \Rightarrow I_2 = \frac{\varepsilon_2 - V}{r_2}$$

Adding the two branch currents:

$$I = I_1 + I_2 = \frac{\varepsilon_1 - V}{r_1} + \frac{\varepsilon_2 - V}{r_2}$$

$$I = \left(\frac{\varepsilon_1}{r_1} + \frac{\varepsilon_2}{r_2}\right) - V\left(\frac{1}{r_1} + \frac{1}{r_2}\right)$$

Solving this for V:

$$V = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2} - I\,\frac{r_1 r_2}{r_1 + r_2}$$

Compare this with the equation of a single equivalent cell, $V = \varepsilon_{eq} - I\,r_{eq}$.

(i) Equivalent emf of the combination:

$$\varepsilon_{eq} = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$$

(ii) Equivalent internal resistance of the combination:

$$r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$$

(iii) Potential difference between A and B:

$$V_{AB} = \varepsilon_{eq} - I\,r_{eq} = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2} - I\,\frac{r_1 r_2}{r_1 + r_2}$$

Thus the parallel combination behaves like a single cell of emf ε_{eq} and internal resistance r_{eq}, with the terminal voltage V_{AB} given above.

Final answer. ε_eq = (ε₁r₂ + ε₂r₁)/(r₁ + r₂), r_eq = r₁r₂/(r₁ + r₂), and V_AB = ε_eq − I·r_eq.

← *Back to Q282*

Q283

Two heating elements of resistances R_1 *and* R_2 *when operated at a constant supply of voltage, V, consume powers* P_1 *and* P_2 *respectively. Deduce the expressions for the power of their combination when they are, in turn, connected in (i) series and (ii) parallel across the same voltage supply.*

[2011]

SOLUTION

When a resistance R is run at its rated voltage V and consumes power P, the power formula $\boldsymbol{P = \frac{V^2}{R}}$ gives its resistance in terms of the rated values:

$$R = \frac{V^2}{P}$$

So the two elements have:

$$R_1 = \frac{V^2}{P_1}, R_2 = \frac{V^2}{P_2}$$

(i) Series combination across the same supply V.

The equivalent series resistance is:

$$R_s = R_1 + R_2 = \frac{V^2}{P_1} + \frac{V^2}{P_2} = V^2\left(\frac{P_1 + P_2}{P_1 P_2}\right)$$

The power consumed by the series combination across V is:

$$P_s = \frac{V^2}{R_s} = \frac{V^2}{V^2 (P_1 + P_2)/(P_1 P_2)}$$

$$P_s = \frac{P_1 P_2}{P_1 + P_2}$$

(ii) Parallel combination across the same supply V.

The equivalent parallel resistance satisfies:

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{P_1}{V^2} + \frac{P_2}{V^2} = \frac{P_1 + P_2}{V^2}$$

The power consumed by the parallel combination across V is:

$$P_p = \frac{V^2}{R_p} = V^2 \times \frac{P_1 + P_2}{V^2}$$

$$P_p = P_1 + P_2$$

Hence in series the combination consumes $\frac{P_1 P_2}{P_1 + P_2}$, while in parallel it consumes $P_1 + P_2$. Since $\boldsymbol{P_p > P_s}$ always, heaters connected in parallel deliver more total power than the same two connected in series.

COMMON MISTAKE

A common mistake is to assume series gives more heat. In a series chain the elements share the supply voltage, so total power drops; only the parallel arrangement runs each element at its full rated voltage.

Final answer. Series power $P_s = P_1P_2/(P_1 + P_2)$; parallel power $P_p = P_1 + P_2$.

← *Back to Q283*

Q284

Define the terms (i) drift velocity, (ii) relaxation time.
A conductor of length L is connected to a dc source of emf ε. If this conductor is replaced by another conductor of same material and same area of cross-section but of length 3L*, how will the drift velocity change?*

[2011 • Set 55-1-1]

SOLUTION

(i) Drift velocity.

Drift velocity is the average velocity acquired by the free electrons of a conductor along the conductor when an external electric field is applied, superimposed on their random thermal motion. It is given by:

$$v_d = \frac{e\,E\,\tau}{m}$$

where E is the applied field, τ the relaxation time, e and m the electron charge and mass. Its magnitude in metals is very small, of the order of mm/s.

(ii) Relaxation time.

Relaxation time is the average time interval between two successive collisions of a free electron with the positive ions of the lattice. Its value is of the order of $10^{-14}\ s$.

Effect of tripling the length.

The drift velocity is related to the field by $\mathbf{v}_d = \frac{e\tau}{m}\mathbf{E}$, and the field inside a conductor connected to a source of constant emf ε (so terminal voltage $V \approx \varepsilon$) is:

$$E = \frac{V}{L}$$

When the conductor is replaced by another of the same material and same cross-section but length 3L, the voltage V across it stays the same (same source), so the new field is:

$$E' = \frac{V}{3L} = \frac{E}{3}$$

Since $\mathbf{v}_d \propto E$ (the material gives the same τ, m, e):

$$v_d' = \frac{e\tau}{m}E' = \frac{e\tau}{m}\cdot\frac{E}{3} = \frac{v_d}{3}$$

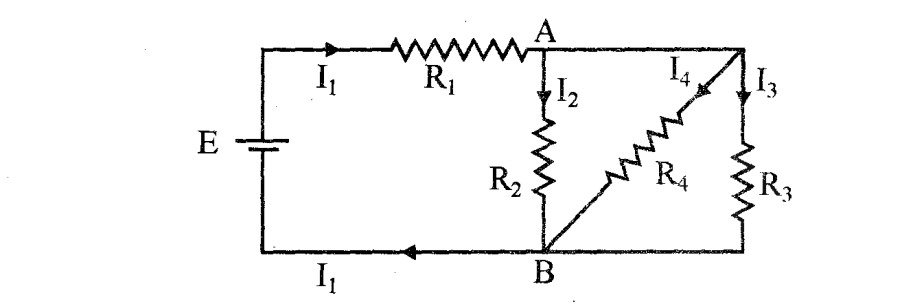
Hence the drift velocity is reduced to **one-third** of its original value.

Final answer. Drift velocity becomes one-third of its original value (v_d → v_d/3).

← *Back to Q284*

Q285

In the circuit shown, $\mathbf{R_1 = 4\ \Omega}$, $\mathbf{R_2 = R_3 = 15\ \Omega}$, $\mathbf{R_4 = 30\ \Omega}$ *and* $\mathbf{E = 10}$ *V. Calculate the equivalent resistance of the circuit and the current in each resistor.*



[2011 • Set 55-1-1]

SOLUTION

In the given circuit, $\mathbf{R_1 = 4\ \Omega}$ is in series with a parallel section made of $\mathbf{R_2 = 15\ \Omega}$, $\mathbf{R_3 = 15\ \Omega}$ and $\mathbf{R_4 = 30\ \Omega}$, all three connected between the same two nodes. The supply is $\boldsymbol{E = 10\ V}$.

First find the parallel combination of R_2, R_3 and R_4:

$$\frac{1}{R_p} = \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} = \frac{1}{15} + \frac{1}{15} + \frac{1}{30}$$

$$\frac{1}{R_p} = \frac{2}{30} + \frac{2}{30} + \frac{1}{30} = \frac{5}{30} = \frac{1}{6}$$

$$R_p = 6\ \Omega$$

The equivalent resistance of the whole circuit is R_1 in series with R_p:

$$R_{eq} = R_1 + R_p = 4 + 6 = 10\ \Omega$$

The total current drawn from the source flows through R_1:

$$I = \frac{E}{R_{eq}} = \frac{10}{10} = 1\ A$$

So the current through R_1 is $\mathbf{I_{R_1} = 1\ A}$.

The potential difference across the parallel section is:

$$V_p = I \times R_p = 1 \times 6 = 6\ V$$

The current in each parallel branch:

$$I_{R_2} = \frac{V_p}{R_2} = \frac{6}{15} = 0.4\ A$$

$$I_{R_3} = \frac{V_p}{R_3} = \frac{6}{15} = 0.4\ A$$

$$I_{R_4} = \frac{V_p}{R_4} = \frac{6}{30} = 0.2\ A$$

As a check, the branch currents add up to the total: $\mathbf{0.4 + 0.4 + 0.2 = 1\ A}$, which equals I.

Hence $\mathbf{R_{eq} = 10\ \Omega}$, with $\mathbf{I_{R_1} = 1\ A}$, $\mathbf{I_{R_2} = I_{R_3} = 0.4\ A}$, and $\mathbf{I_{R_4} = 0.2\ A}$.

Final answer. R_eq = 10 Ω; I_R1 = 1 A, I_R2 = I_R3 = 0.4 A, I_R4 = 0.2 A.

← Back to Q285

Q286

State the underlying principle of a potentiometer.
Describe briefly, giving the necessary circuit diagram, how a potentiometer is used to measure the internal resistance of a given cell.

[2011 • Set 55-2-1]

SOLUTION

Principle of a potentiometer.

When a steady current flows through a wire of uniform cross-section and uniform composition, the potential drop across any portion of the wire is directly proportional to the length of that portion:

$$V \propto \ell \Rightarrow V = k\ell$$

where k (the potential gradient) is the fall of potential per unit length of the wire.

Circuit diagram (described).

Draw a long uniform potentiometer wire AB stretched along a metre scale. In the primary (driver) circuit, connect a battery, a key and a rheostat in series with AB so that a steady current flows from A to B. In the secondary circuit, connect the cell under test (emf ε, internal resistance r) with its positive terminal to A, through a galvanometer G, to a jockey that slides along the wire. Across the cell connect a resistance box R in series with a key K, so that closing K puts the shunt R across the cell.

Procedure to measure internal resistance.

With key K open, no current is drawn from the cell, so it is balanced against its full emf. Slide the jockey to get zero deflection in G at balancing length ℓ_1. Then:

$$\varepsilon = k\ell_1$$

Now close key K, so the shunt resistance R draws current from the cell. The terminal voltage of the cell falls to:

$$V = \frac{\varepsilon R}{R + r}$$

Find the new balancing length $_2$ for this terminal voltage:

$$V = k_2$$

Dividing the two balance equations:

$$\frac{\varepsilon}{V} = \frac{1}{2} = \frac{R + r}{R}$$

Solving for the internal resistance r:

$$r = R\left(\frac{1}{2} - 1\right) = R\left(\frac{1 - 2}{2}\right)$$

Knowing the shunt value R and the two balancing lengths, the internal resistance r of the cell is determined.

Final answer. Principle: $V \propto \ell$; internal resistance $r = R(\ell_1/\ell_2 - 1)$.

← *Back to Q286*

Q287

Write the principle of working of a potentiometer. Describe briefly, with the help of a circuit diagram, how a potentiometer is used to determine the internal resistance of a given cell.

[2010]

SOLUTION

Principle of working of a potentiometer.

When a constant current flows through a wire of uniform area of cross-section and uniform composition, the potential drop across any segment of the wire is directly proportional to the length of that segment:

$$V \propto \Rightarrow V = k$$

where k is the potential gradient (potential drop per unit length) along the wire.

Circuit diagram (described).

Draw a uniform potentiometer wire AB on a metre scale. The primary circuit has a driver battery, a key and a rheostat in series with AB, maintaining a steady current. In the secondary circuit, the cell under test (emf ε, internal resistance r) is connected with its positive terminal at A, through a galvanometer G to a sliding jockey. A resistance box R in series with a key K is connected across the terminals of the cell.

Determination of internal resistance.

With key K open the cell delivers no current, so it balances against its full emf. Obtain the balancing length $_1$ where the galvanometer shows no deflection:

$$\varepsilon = k\,_1$$

Now close key K, putting shunt R across the cell so it drives a current. The terminal voltage of the cell drops to:

$$V = \frac{\varepsilon\, R}{R + r}$$

Obtain the new balancing length $_2$ for this terminal voltage:

$$V = k\,_2$$

Taking the ratio of the two balance conditions:

$$\frac{\varepsilon}{V} = \frac{_1}{_2} = \frac{R + r}{R}$$

Rearranging to isolate the internal resistance:

$$r = R\left(\frac{_1}{_2} - 1\right) = R\left(\frac{_1 - {}_2}{_2}\right)$$

With the known shunt R and the measured lengths $_1$ and $_2$, the internal resistance r of the cell is found.

Final answer. Principle: $V \propto \ell$; internal resistance $r = R(\ell_1/\ell_2 - 1)$.

← *Back to Q287*

Q288

In a meter bridge, the null point is found at a distance of 40 cm from A. If a resistance of 12 Ω is connected in parallel with S, the null point occurs at 50.0 cm from A. Determine the values of R and S.

[2010]

SOLUTION

In a metre bridge, at the null point the bridge is balanced and the ratio of the unknown resistances equals the ratio of the two wire lengths on either side of the jockey.

With the first null point at 40 cm from A, the left-arm length is 40 cm and the right-arm length is **(100 − 40) = 60 cm**:

$$\frac{R}{S} = \frac{40}{60} = \frac{2}{3}$$

$$S = \frac{3R}{2} \quad ...(1)$$

Now a 12 Ω resistance is connected in parallel with S. The effective resistance of that arm becomes:

$$S' = \frac{12\,S}{12 + S}$$

The new null point is at 50.0 cm, so the lengths on the two sides are equal, 50 cm and 50 cm:

$$\frac{R}{S'} = \frac{50}{50} = 1 \Rightarrow R = S' = \frac{12\,S}{12 + S}$$

$$R(12 + S) = 12\,S \quad ...(2)$$

Substitute $\boldsymbol{S = \frac{3R}{2}}$ from (1) into (2):

$$R(12 + \frac{3R}{2}) = 12 \times \frac{3R}{2} = 18R$$

Divide both sides by R (since $R \neq 0$):

$$12 + \frac{3R}{2} = 18$$

$$\frac{3R}{2} = 6 \Rightarrow R = 4\ \Omega$$

From (1):

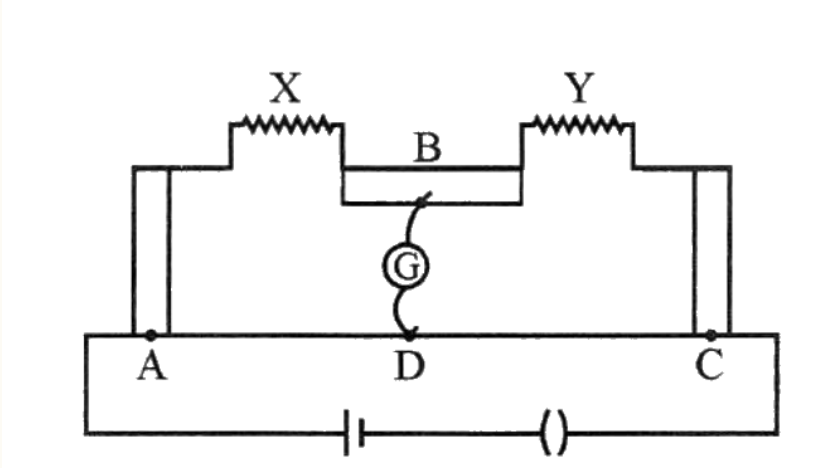
$$S = \frac{3R}{2} = \frac{3 \times 4}{2} = 6\ \Omega$$

Hence the two resistances are **R = 4 Ω** and **S = 6 Ω**.

Final answer. R = 4 Ω and S = 6 Ω.

← *Back to Q288*

Q289



The figure shows experimental set up of a meter bridge. When the two unknown resistances X and Y are inserted, the null point D is obtained 40 cm *from the end A. When a resistance of* 10 Ω *is connected in series with X, the null point shifts by* 10 cm. *Find the position of the null point when the* 10 Ω *resistance is instead connected in series with resistance Y. Determine the values of the resistances X and Y.*

[2009]

SOLUTION

In a metre bridge the balance (null) condition is $\frac{X}{Y} = \frac{}{100 -}$, where is the balancing length measured from end A.

With the null point at 40 cm from A:

$$\frac{X}{Y} = \frac{40}{60} = \frac{2}{3}$$

$$Y = \frac{3X}{2} \quad ...(1)$$

When 10 Ω is connected in series with X, the null point shifts by 10 cm, that is from 40 cm to 50 cm. The left arm is now $(X + 10)$ and the lengths are 50 cm and 50 cm:

$$\frac{X + 10}{Y} = \frac{50}{50} = 1 \Rightarrow X + 10 = Y$$

Substitute $\mathbf{Y = \frac{3X}{2}}$ from (1):

$$X + 10 = \frac{3X}{2}$$

$$10 = \frac{3X}{2} - X = \frac{X}{2} \Rightarrow X = 20\ \Omega$$

From (1):

$$Y = \frac{3 \times 20}{2} = 30\ \Omega$$

Now connect the 10 Ω in series with Y instead, so the right arm becomes $\mathbf{(Y + 10) = 40\ \Omega}$ while the left arm stays $\mathbf{X = 20\ \Omega}$. Let the new null point be at length from A:

$$\frac{X}{Y + 10} = \frac{}{100 -}$$

$$\frac{20}{40} = \frac{1}{2} = \frac{}{100 -}$$

Cross-multiplying:

$$2 = 100 - \Rightarrow 3 = 100$$

$$= \frac{100}{3} \approx 33.3 \text{ cm}$$

Hence **X = 20 Ω**, **Y = 30 Ω**, and the null point shifts to **≈ 33.3 cm** from *A*.

CONCEPT

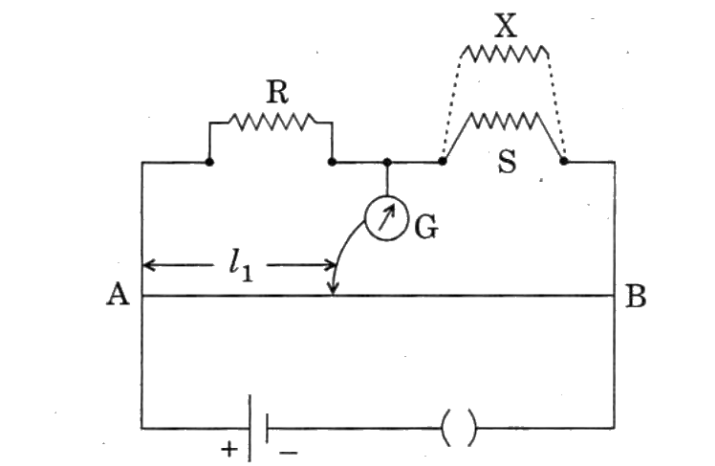
When the same extra resistance is moved from one arm to the other, the null point shifts in the opposite direction. Here it went up to 50 cm with X, so it must fall below 40 cm with Y, which matches 33.3 cm.

Final answer. X = 20 Ω, Y = 30 Ω, and the new null point is at about 33.3 cm from A.

← *Back to Q289*

Q290

(i) State the principle of working of a meter bridge.
(ii) In a meter bridge balance point is found at a distance l_1 *with resistances R and S as shown in the figure.*
When an unknown resistance X is connected in parallel with the resistance S, the balance point shifts to a distance l_2*. Find the expression for X in terms of* l_1, l_2 *and S.*



[2009]

SOLUTION

(i) Principle of working of a metre bridge.

A metre bridge is a practical form of the Wheatstone bridge in which two of the four arms are replaced by a single uniform wire of length 1 m. At the null point the bridge is balanced, no current flows through the galvanometer, and the ratio of the resistances in the gaps equals the ratio of the lengths of the wire on the two sides of the jockey:

$$\frac{R}{S} = \frac{l_1}{100 - l_1}$$

(ii) Expression for X.

With R and S in the gaps, the balance at length l_1 gives:

$$\frac{R}{S} = \frac{l_1}{100 - l_1} \quad ...(1)$$

When the unknown resistance X is connected in parallel with S, the effective resistance of that arm becomes:

$$S' = \frac{S\,X}{S + X}$$

The new balance at length l_2 gives:

$$\frac{R}{S'} = \frac{l_2}{100 - l_2} \quad ...(2)$$

Divide equation (1) by equation (2) so that R cancels:

$$\frac{S'}{S} = \frac{l_1}{100 - l_1} \times \frac{100 - l_2}{l_2}$$

Substitute $S' = \frac{S\,X}{S + X}$, so that $\frac{S'}{S} = \frac{X}{S + X}$:

$$\frac{X}{S + X} = \frac{l_1\,(100 - l_2)}{l_2\,(100 - l_1)}$$

Let $\frac{l_1\,(100 - l_2)}{l_2\,(100 - l_1)} = m$ for brevity. Then:

$$X = m\,(S + X) \Rightarrow X(1 - m) = mS$$

$$X = \frac{mS}{1 - m}$$

Putting m back and simplifying:

$$X = \frac{S\, l_1\, (100 - l_2)}{l_2\, (100 - l_1) - l_1\, (100 - l_2)}$$

This is the required expression for X in terms of l_1, l_2 and S.

Final answer. $X = S \cdot l_1(100 - l_2) \,/\, [\, l_2(100 - l_1) - l_1(100 - l_2)\,]$.

← *Back to Q290*

Q291

Prove that the current density of a metallic conductor is directly proportional to the drift speed of electrons.

[2008]

SOLUTION

Consider a metallic conductor of uniform cross-sectional area A carrying a steady current I. Let n be the number density of free electrons (number of free electrons per unit volume) and let each electron carry charge $-e$.

When a potential difference is applied, the free electrons acquire an average velocity called the drift velocity v_d opposite to the field direction.

In a small time interval dt, every free electron travels a distance v_d dt. Hence all the electrons lying within a cylinder of length v_d dt and cross-section A will cross the area A in this time.

Volume of this cylinder = $\boldsymbol{A\, v_d\, dt}$

Number of free electrons in it = $\boldsymbol{n\, A\, v_d\, dt}$

Total charge crossing area A in time dt is $\boldsymbol{dq = e\, n\, A\, v_d\, dt}$

Therefore the current is $\boldsymbol{I = \frac{dq}{dt} = n\, e\, A\, v_d}$

The current density is defined as current per unit area, $\boldsymbol{j = \frac{I}{A}}$

$$\Rightarrow j = \frac{n\, e\, A\, v_d}{A} = n\, e\, v_d$$

For a given conductor at a fixed temperature, n and e are constants. Hence $j \propto v_d$.

Thus the **current density of a metallic conductor is directly proportional to the drift speed of the electrons.**

Final answer. Since $j = n\,e\,v_d$ with n and e constant, $j \propto v_d$.

← Back to Q291

Q292

A number of identical cells, n, each of emf E, internal resistance r connected in series are charged by a d.c source of emf E', using a resistor R.
(i) Draw the circuit arrangement.
(ii) Deduce the expression for (a) the charging current and (b) the potential difference across the combination of the cells.

[2008]

SOLUTION

(i) Circuit arrangement: A d.c. source of emf E' is connected in series with a resistor R and the series combination of n identical cells (each of emf E and internal resistance r). The positive terminal of the d.c. source is joined to the positive terminal of the cell combination, so that the source drives current into the + terminal of the cells and charges them. The cells, the resistor R and the source all form one closed series loop.

During charging, the emf of the cells opposes the emf of the d.c. source, because the source is forcing current backwards through the cells.

(ii)(a) Charging current: The net emf driving the loop is the source emf minus the total opposing emf of the cells.

Net emf = $E' - nE$

Total resistance of the loop = $R + nr$ (the n internal resistances r add in series along with R).

By Ohm's law for the complete circuit, the charging current is

$$I = \frac{E' - nE}{R + nr}$$

(Charging is possible only when $E' > nE$.)

(ii)(b) Potential difference across the cell combination: While charging, the terminal voltage across the cells equals their total emf plus the drop across their total internal resistance (current flows into the + terminal).

$$V = nE + I(nr)$$

Substituting I,

$$V = nE + \frac{(E' - nE)\, nr}{R + nr}$$

$$V = \frac{nE(R + nr) + (E' - nE)\, nr}{R + nr}$$

$$V = \frac{nER + nE'r}{R + nr} = \frac{n(ER + E'r)}{R + nr}$$

Hence the **charging current is $I = \dfrac{E' - nE}{R + nr}$** and the **potential difference across the cells is $V = \dfrac{n(ER + E'r)}{R + nr}$.**

COMMON MISTAKE

Remember that during charging the current enters the + terminal, so the terminal voltage $V = nE + Inr$ is greater than the emf, unlike during discharging where $V = nE - Inr$.

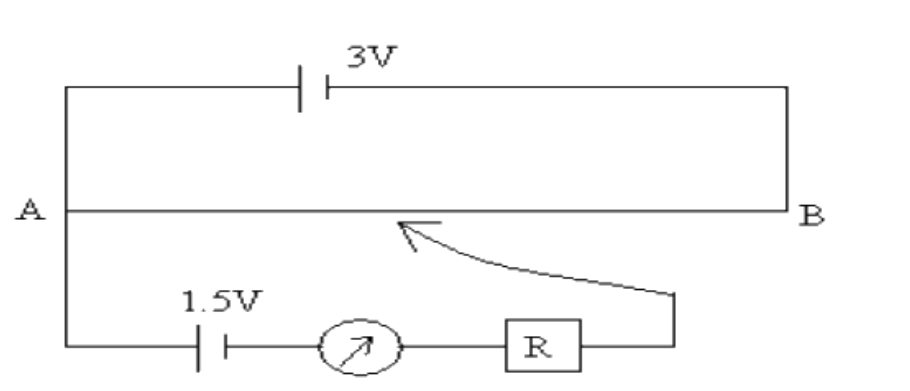
Final answer. Charging current $I = \dfrac{E' - nE}{R + nr}$ and $V = \dfrac{n(ER + E'r)}{R + nr}$.

← *Back to Q292*

Q293

A potentiometer wire of length 1 m is connected to a driver cell of emf 3 V as shown in the figure. When a cell of emf 1.5 V is used in the secondary circuit, the balance point

is found to be 60 cm. *On replacing this cell and using a cell of unknown emf, the balance point shifts to* 80 cm.



(i) Calculate unknown emf of the cell.
(ii) Explain with reason, whether the circuit works, if the drive cell is replaced with a cell emf 1 *V.*
(iii) Does the high resistance R, used in the secondary circuit affect the balance point? Justify your answer.

[2008]

SOLUTION

The potentiometer wire of length 1 *m* is driven by a cell of emf 3 *V*, which sets up a uniform potential gradient *k* along the wire. At balance no current flows through the galvanometer, so the emf of the cell in the secondary equals the potential drop across the balancing length.

(i) Unknown emf: For the 1.5 *V* cell, balance length = **60 cm,** so the potential gradient is

$$k = \frac{1.5\ V}{60\ \text{cm}} = 0.025\ V/\text{cm}$$

For the unknown cell, balance length = **80 cm,** so

$$E = k \times 80 = 0.025 \times 80$$

$$E = 2.0\ V$$

(ii) If the driver cell of 3 *V* is replaced by a cell of emf 1 *V*: The maximum potential difference available across the whole wire is now only about 1 *V* (and even less after the series protective resistance). For balance, the potential drop

across the balancing length must equal the emf in the secondary circuit. But the secondary emf (1.5 V or 2 V) is greater than the total voltage across the wire. Hence no balance point can ever be obtained, and **the circuit will not work.**

(iii) Effect of the high resistance R in the secondary: At the balance point the galvanometer carries no current, so R plays no role at balance. While searching for balance, R only limits the current through the galvanometer and protects it; it does not alter the potential gradient on the main wire. Therefore **the high resistance R does not affect the balance point.**

CONCEPT

A potentiometer can only measure an emf smaller than the potential difference across its full wire; the driver cell emf must always exceed the emf being measured.

Final answer. $E = 2.0\ V$; the circuit fails with a $1\ V$ driver; R does not shift the balance point.

← *Back to Q293*

Q294

A resistance $R = 5\ \Omega$ *is connected to one of the gaps in a meter bridge, which uses a wire of length* 1 *m. An unknown resistance* $X > 5\ \Omega$ *is connected in the other gap as shown in the figure. The balance point is noticed at 'l' cm from the positive end of the battery. On interchanging R and X, it was found that the balance point shifts by* 20 cm *away from end A. Neglecting the end correction, calculate the value of unknown resistance X used.*

[2008]

SOLUTION

In a metre bridge, at the balance point the ratio of the two gap resistances equals the ratio of the two segments of the balancing wire.

Let the balance length be l cm from end A with $R = 5\ \Omega$ in the left gap and X in the right gap. Then

$$\frac{R}{X} = \frac{l}{100 - l}$$

$$\Rightarrow l = \frac{100R}{R + X}$$

On interchanging R and X, the balance point shifts 20 cm away from A, so the new balance length is $(l + 20)$ cm. Now X is in the left gap:

$$\frac{X}{R} = \frac{l + 20}{100 - (l + 20)} = \frac{l + 20}{80 - l}$$

$$\Rightarrow l + 20 = \frac{100X}{R + X}$$

Subtracting the first relation from this one,

$$(l + 20) - l = \frac{100X}{R + X} - \frac{100R}{R + X}$$

$$20 = \frac{100(X - R)}{X + R}$$

$$\frac{X - R}{X + R} = 0.2$$

$$5(X - R) = X + R$$

$$5X - 5R = X + R$$

$$4X = 6R$$

$$X = 1.5\, R = 1.5 \times 5$$

$$X = 7.5\ \Omega$$

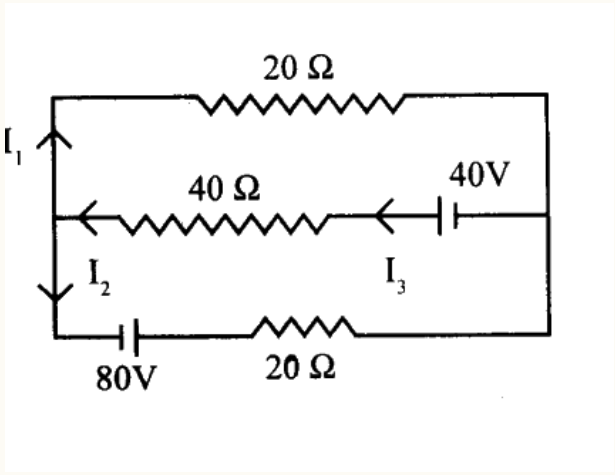
Hence the **unknown resistance is $X = 7.5\ \Omega$** (which indeed satisfies $X > 5\ \Omega$).

Final answer. $X = 7.5\ \Omega$.

← *Back to Q294*

Q295

State Kirchhoff's rules of current distribution in an electrical network. Using these rules determine the value of the current I_1 *in the electric circuit given below.*



[2007]

SOLUTION

Kirchhoff's rules:

(1) Junction (current) rule: The algebraic sum of currents meeting at any junction in a circuit is zero, $\sum I = 0$. Currents entering a junction are taken positive and those leaving negative. This rule follows from the conservation of charge.

(2) Loop (voltage) rule: Around any closed loop of a circuit, the algebraic sum of the emfs equals the algebraic sum of the products of current and resistance, $\sum \varepsilon = \sum IR$. This rule follows from the conservation of energy.

Finding I_1: First mark the current in each branch and let the unknown currents satisfy the junction rule, for example $I_3 = I_1 + I_2$ at the central node where the branches meet.

Next apply the loop rule to two independent closed loops of the network. Each loop gives a linear equation of the form $\sum \varepsilon = \sum IR$, writing the emfs with their correct signs and the IR drops (including those across the 20 Ω resistor) for the chosen direction of traversal.

This yields two simultaneous equations in the branch currents. Substituting the given emf and resistance values from the figure and solving the simultaneous equations gives the value of the current I_1 in the required branch.

Thus, using the **junction rule** ($\sum I = 0$) together with the **loop rule** ($\sum \varepsilon = \sum IR$) on two independent loops, the current I_1 is obtained by solving the resulting linear equations with the numerical data of the figure.

Final answer. I_1 is found by applying $\sum I = 0$ at the junction and $\sum \varepsilon = \sum IR$ to two loops, then solving for I_1.

← *Back to Q295*

Q296

Write the mathematical relation for the resistivity of a material in terms of relaxation time, number density and mass and charge of charge carriers in it. Explain, using this relation, why the resistivity of a metal increases and that of a semi-conductor decreases with rise in temperature.

[2007]

SOLUTION

The resistivity of a material in terms of relaxation time, number density, mass and charge of the charge carriers is

$$\rho = \frac{m}{n e^2 \tau}$$

where m is the mass of a charge carrier, e its charge, n the number density of free charge carriers and τ the average relaxation time between successive collisions.

Metal (resistivity increases with temperature): In a metal the number density n of free electrons is practically constant and does not change appreciably with temperature. As the temperature rises, the metal ions vibrate more vigorously about their mean positions, so the electrons collide with them more frequently. This reduces the average relaxation time τ.

Since $\rho = \frac{m}{n e^2 \tau} \propto \frac{1}{\tau}$, a decrease in τ makes ρ increase. Hence the **resistivity of a metal increases with rise in temperature.**

Semiconductor (resistivity decreases with temperature): In a semiconductor, raising the temperature supplies energy that excites a large number of additional charge carriers across the energy gap. So the number density n increases very rapidly (almost exponentially) with temperature.

Although τ decreases slightly as in a metal, the steep rise in n dominates completely. Since $\rho \propto \frac{1}{n}$, the large increase in n causes ρ to fall. Hence the **resistivity of a semiconductor decreases with rise in temperature.**

CONCEPT

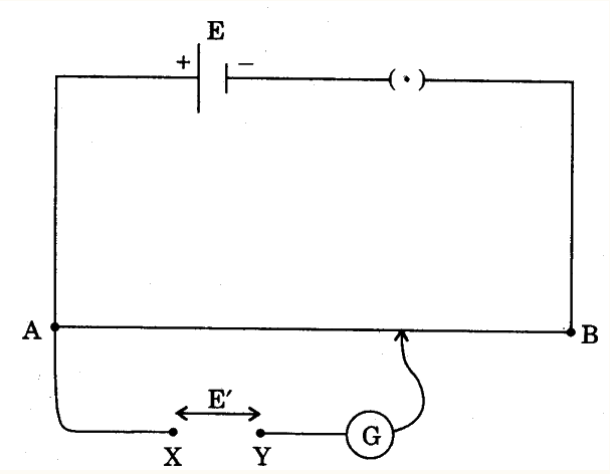
The opposite temperature behaviour comes from which quantity dominates: τ for metals (n nearly fixed) versus n for semiconductors.

Final answer. $\rho = \frac{m}{n e^2 \tau}$; in a metal τ falls so ρ rises, in a semiconductor n rises sharply so ρ falls.

← *Back to Q296*

Q297

For the potentiometer circuit shown in the given figure, points X and Y represent the two terminals of an unknown emf E'. A student observed that when the jockey is moved from the end A to the end B of the potentiometer wire, the deflection in the galvanometer remains in the same direction.



What may be the two possible faults in the circuit that could result in this observation?
If the galvanometer deflection at the end B is (i) more, (ii) less, than that at end A, which of the two faults, listed above, would be there in the circuit?
Give reasons in support of your answer in each case.

[2007]

SOLUTION

In a correctly working potentiometer the galvanometer deflection must reverse its direction at some point as the jockey is moved from A to B, and that point is the balance (null) point. If the deflection stays in the same direction all along the wire, the circuit is faulty.

Two possible faults:

Fault 1: The emf of the unknown cell E' is greater than the total potential drop across the potentiometer wire. This happens when the driving (auxiliary) cell emf is too small, or the potential gradient on the wire is too low. Then the wire can never balance E', so the net emf $E' - V_{wire}$ keeps the deflection in one direction everywhere.

Fault 2: The unknown cell E' is connected with reversed (wrong) polarity, so that its positive terminal is not joined to the same end (A) as the driver cell. Now E' and the wire potential add instead of opposing, the net emf is $E' + V_{wire}$ which can never be zero, and the deflection stays in one direction throughout.

To decide which fault is present, compare the deflection at the two ends. At end A the balancing length is nearly zero, so the wire drop $\mathbf{V_{wire} \approx 0}$**;** at end B the wire drop is maximum.

(i) Deflection at B is more than at A: This indicates Fault 2 (reversed polarity of E'). Because the two emfs add, the net emf is $E' + V_{wire}$. As the jockey moves from A to B, V_{wire} increases, so the net emf and hence the deflection increase, giving the largest deflection at B.

(ii) Deflection at B is less than at A: This indicates Fault 1 (driver cell emf too small, $\boldsymbol{E'}$ > wire voltage). Here the wire potential opposes E', so the net emf is $E' - V_{wire}$. As the jockey moves from A to B, the opposing V_{wire} grows and partly cancels E', so the net emf and the deflection decrease, giving the smaller deflection at B (but balance is never reached because V_{wire} stays below E').

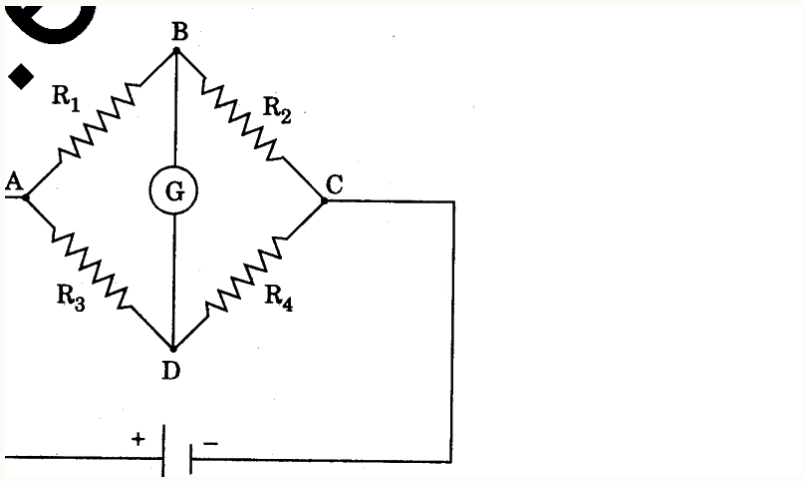
Hence: **more deflection at *B* means the unknown cell *E'* is connected with reversed polarity; less deflection at *B* means the driver cell emf is too small.**

Final answer. Faults: ***E'*** reversed in polarity, or driver emf less than ***E'***. More deflection at ***B*** means ***E'*** is reversed (emfs add); less deflection at ***B*** means the driver emf is too small (opposing wire drop grows toward ***B***).

← *Back to Q297*

Q298

The given figure shows a network of resistances R_1, R_2, R_3 *and* R_4.



Using Kirchhoff's laws, establish the balance condition for the network.

[2007]

SOLUTION

The network is a Wheatstone bridge. The four resistors R_1, R_2, R_3, R_4 form the four arms. A galvanometer G is connected between one pair of opposite junctions (say B and D), and a battery is connected across the other pair of opposite junctions (say A and C).

Let the current entering at A split into I_1 (through R_1 then R_2) and I_2 (through R_3 then R_4), and let the galvanometer current be I_g.

The bridge is said to be balanced when no current flows through the galvanometer, i.e. $\mathbf{I_g = 0}$. Then the junctions B and D are at the same potential.

Applying Kirchhoff's junction rule with $\mathbf{I_g = 0}$, the current I_1 flows undivided through both R_1 and R_2, and the current I_2 flows undivided through both R_3 and R_4.

Applying Kirchhoff's loop rule to the loop *A B D A* (the upper triangle containing the galvanometer), with $\mathbf{I_g = 0}$ the galvanometer arm carries no drop:

$$I_1 R_1 - I_2 R_3 = 0$$

$$\Rightarrow I_1 R_1 = I_2 R_3 \ldots(1)$$

Applying the loop rule to the lower loop *B C D B*, again with $\mathbf{I_g = 0}$:

$$I_1 R_2 - I_2 R_4 = 0$$

$$\Rightarrow I_1 R_2 = I_2 R_4 \ldots(2)$$

Dividing equation (1) by equation (2),

$$\frac{I_1 R_1}{I_1 R_2} = \frac{I_2 R_3}{I_2 R_4}$$

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

This is the required **balance condition of the Wheatstone bridge:** $\mathbf{\frac{R_1}{R_2} = \frac{R_3}{R_4}}$.

Final answer. The balance condition is $\mathbf{\frac{R_1}{R_2} = \frac{R_3}{R_4}}$.

← *Back to Q298*

Q299

What is Seebeck effect? Plot a graph showing the variation of thermo emf with temperature of hot junction (keeping cold junction at 0°C) of a thermocouple. How will the (i) neutral temperature, (ii) inversion temperature of a thermocouple change when the temperature of cold junction is increased?

[2007]

SOLUTION

Seebeck effect: When two wires of different metals are joined to form a closed loop with two junctions, and the two junctions are kept at different temperatures, an emf is set up in the circuit and a current flows. This thermoelectric emf is called the Seebeck emf, and the phenomenon is the Seebeck (thermoelectric) effect. The magnitude and direction of this emf depend on the pair of metals used and on the temperature difference between the junctions.

Graph of thermo emf versus hot-junction temperature (cold junction kept at $0^\circ C$): The graph is a parabola. Plot the thermo emf ε on the vertical axis and the temperature T of the hot junction on the horizontal axis. As T increases from $0^\circ C$, the emf rises from zero, reaches a maximum value at the neutral temperature T_n, then decreases, and falls back to zero at the inversion temperature T_i. Beyond T_i the emf reverses sign and becomes negative. The curve is symmetric about T_n, so T_n lies midway between $0^\circ C$ and T_i.

(i) Neutral temperature: The neutral temperature T_n is a fixed characteristic of the pair of metals and does not depend on the temperature of the cold junction. Hence on increasing the cold-junction temperature, the **neutral temperature remains unchanged.**

(ii) Inversion temperature: The neutral temperature is the mean of the cold-junction temperature T_c and the inversion temperature T_i, i.e. $\mathbf{T_n = \frac{T_c + T_i}{2}}$, so $\mathbf{T_i = 2T_n - T_c}$. Since T_n is fixed, raising the cold-junction temperature T_c makes T_i smaller. Hence the **inversion temperature decreases** when the cold-junction temperature is increased.

CONCEPT

Use $\mathbf{T_n = \frac{T_c + T_i}{2}}$: T_n depends only on the metal pair (constant), while T_i moves so as to keep this mean fixed.

Final answer. Seebeck emf arises from a junction temperature difference; raising $\mathbf{T_c}$ keeps $\mathbf{T_n}$ unchanged but lowers $\mathbf{T_i}$.

← Back to Q299

Q300

A 10 m long wire of uniform cross-section and resistance 20 Ω is used in a potentiometer. The wire is connected in series with a battery of 5 V along with an external resistance of 480 Ω. If an unknown emf E is balanced at 6.0 m length of the wire, calculate:
(i) the potential gradient of the potentiometer wire,
(ii) the value of the unknown emf E.

[2006]

SOLUTION

The potentiometer wire (resistance $\mathbf{R_{wire} = 20\ \Omega}$, length $\boldsymbol{L = 10\ m}$) is in series with the 5 V battery and the external resistance 480 Ω. First find the current in this primary circuit, then the potential drop across the wire and hence the potential gradient.

Total resistance of the primary circuit:

$$R_{total} = R_{wire} + R_{ext} = 20 + 480$$

$$R_{total} = 500\ \Omega$$

Current in the primary circuit:

$$I = \frac{V}{R_{total}} = \frac{5}{500}$$

$$I = 0.01\ A$$

Potential difference across the potentiometer wire:

$$V_{wire} = I\, R_{wire} = 0.01 \times 20$$

$$V_{wire} = 0.2\ V$$

(i) Potential gradient of the wire:

$$k = \frac{V_{wire}}{L} = \frac{0.2}{10}$$

$$k = 0.02\ V/m = 2 \times 10^{-2}\ V/m$$

(ii) Unknown emf E balanced at length $l = 6.0\ m$. At balance, E equals the potential drop across the balancing length:

$$E = k \times l = 0.02 \times 6.0$$

$$E = 0.12\ V$$

Hence the **potential gradient is $k = 2 \times 10^{-2}\ V/m$** and the **unknown emf is $E = 0.12\ V$** (i.e. 120 mV).

Final answer. $k = 2 \times 10^{-2}\ V/m$ and $E = 0.12\ V$.

← *Back to Q300*

Q301

Define the term 'resistivity' and write its S.I. unit. Derive the expression for the resistivity of a conductor in terms of number density of free electrons and relaxation time.

[2005]

SOLUTION

Resistivity (ρ) of a material is defined as the resistance offered by a conductor of that material having unit length and unit area of cross-section.

For a conductor of length L, area of cross-section A and resistance R,

$$R = \rho \frac{L}{A}$$

$$\Rightarrow \rho = \frac{R A}{L}$$

The SI unit of resistivity is **ohm metre** (Ω).

Derivation in terms of n and τ: Consider a conductor with n free electrons per unit volume, each of charge e and mass m. When a field $\vec{E}$ is applied, each electron experiences a force and acquires a drift velocity

$$v_d = \frac{eE\tau}{m}$$

where τ is the relaxation time (mean time between successive collisions).

The current density is

$$j = n\,e\,v_d = n\,e\left(\frac{eE\tau}{m}\right) = \frac{n\,e^2\tau}{m}\,E$$

Comparing with the microscopic form of Ohm's law $\boldsymbol{j = \sigma E}$, the conductivity is

$$\sigma = \frac{n\,e^2\tau}{m}$$

Since resistivity is the reciprocal of conductivity,

$$\rho = \frac{1}{\sigma} = \frac{m}{n\,e^2\tau}$$

Thus the resistivity of a conductor is $\boldsymbol{\rho = \dfrac{m}{n\,e^2\,\tau}}$, showing it depends only on the material properties n and τ and not on the dimensions of the conductor.

Final answer. Resistivity $\boldsymbol{\rho = \dfrac{m}{n\,e^2\tau}}$, SI unit **Ω**.

← *Back to Q301*

Q302

State the principle of potentiometer. Draw a circuit diagram used to compare the e.m.f. of two primary cells. Write the formula used. How can the sensitivity of a potentiometer be increased?

[2005]

SOLUTION

Principle of potentiometer: When a steady current flows through a wire of uniform area of cross-section, the potential drop across any portion of the wire is directly proportional to the length of that portion. That is,

$$V \propto \Rightarrow V = k$$

where k is the potential gradient (potential drop per unit length) and is constant for a given steady current.

Circuit diagram to compare two emfs: A long uniform potentiometer wire AB (about 1 m) is connected to a driver cell (battery) through a key K and a rheostat Rh, which together maintain a steady current in the wire. The positive terminals of the two cells ε_1 and ε_2 are joined to the high-potential end A. Their negative terminals are connected through a two-way switch to a galvanometer G, whose other end goes to a jockey J that slides along the wire AB.

Working: First ε_1 is included in the circuit and the jockey is moved till the galvanometer shows no deflection (null point) at length $_1$. Then

$$\varepsilon_1 = k\,_1$$

Next ε_2 is included and the balancing length $_2$ is found, so

$$\varepsilon_2 = k\,_2$$

Dividing the two equations,

$$\frac{\varepsilon_1}{\varepsilon_2} = \frac{_1}{_2}$$

This is the **formula used** to compare the two emfs.

Sensitivity: The sensitivity of a potentiometer is increased by reducing the potential gradient k. This is done by (i) using a longer potentiometer wire, and (ii) reducing the current in the wire (by increasing the rheostat resistance). Both make a given emf balance over a greater length, giving a finer and more accurate reading.

Final answer. Principle: $V \propto$; formula $\frac{\varepsilon_1}{\varepsilon_2} = \frac{_1}{_2}$; sensitivity rises by lowering the potential gradient.

← *Back to Q302*

Q303

What is meant by 'drift velocity of free electrons'? Derive Ohm's law on the basis of the theory of electron drift.

[2003]

SOLUTION

Drift velocity of free electrons is the average velocity acquired by the free electrons of a conductor in a direction opposite to the applied electric field, superimposed on their random thermal motion. Its magnitude is very small, of the order of $10^{-3}\ m\ s^{-1}$.

Derivation of Ohm's law: Consider a conductor of length L and area of cross-section A, with n free electrons per unit volume, each of charge e and mass m. Let a potential difference V be applied across it, setting up a field

$$E = \frac{V}{L}$$

Each electron experiences a force eE and an acceleration

$$a = \frac{eE}{m}$$

If τ is the relaxation time, the drift velocity gained is

$$v_d = a\,\tau = \frac{eE\tau}{m}$$

The current through the conductor is

$$I = n\,e\,A\,v_d$$

Substituting v_d,

$$I = n\,e\,A\left(\frac{eE\tau}{m}\right) = \frac{n\,e^2\tau\,A}{m}\,E$$

Now putting $\boldsymbol{E = \frac{V}{L}}$,

$$I = \frac{n\,e^2\tau\,A}{m}\cdot\frac{V}{L}$$

$$\Rightarrow V = I\left(\frac{m\,L}{n\,e^2\tau\,A}\right)$$

At a constant temperature, n, τ, m, L, A are all constant, so the bracketed quantity is a constant called the resistance R:

$$R = \frac{m\,L}{n\,e^2\tau\,A}$$

Hence

$$V = \mathrm{IR}$$

This is **Ohm's law**: at constant temperature, the current through a conductor is directly proportional to the potential difference across its ends.

CONCEPT

Show that R is constant only at fixed temperature; the proportionality $V \propto I$ is the actual statement of Ohm's law, not just $V = IR$.

Final answer. Drift velocity is the small net velocity of electrons against the field; the drift theory gives $V = IR$, which is Ohm's law.

← *Back to Q303*

Q304

What is Wheatstone bridge? Deduce the condition for which Wheatstone bridge is balanced.

[2003]

SOLUTION

Wheatstone bridge is an arrangement of four resistors connected in the form of a quadrilateral, used to measure an unknown resistance by comparing it with three known resistances.

Circuit: The four resistances P, Q, R, S form the four arms of a quadrilateral ABCD. A battery (with key) is connected across one diagonal A to C, and a galvanometer G is connected across the other diagonal B to D. Arms **AB = P, BC = Q, AD = R, DC = S.**

Balance condition: The bridge is said to be balanced when the galvanometer shows no deflection, i.e. no current flows through it **($I_g = 0$).** In this state the current I_1 entering at A flows through P and then Q, while the current I_2 flows through R and then S.

Applying Kirchhoff's loop rule to the loop ABDA (with $\mathbf{I_g = 0}$),

$$I_1 P - I_2 R = 0$$

$$\Rightarrow I_1 P = I_2 R \ldots(1)$$

Applying Kirchhoff's loop rule to the loop BCDB (with $\mathbf{I_g = 0}$),

$$I_1 Q - I_2 S = 0$$

$$\Rightarrow I_1 Q = I_2 S \ldots(2)$$

Dividing equation (1) by equation (2),

$$\frac{I_1 P}{I_1 Q} = \frac{I_2 R}{I_2 S}$$

$$\Rightarrow \frac{P}{Q} = \frac{R}{S}$$

This is the **balanced condition** of the Wheatstone bridge. It is independent of the emf of the battery and the sensitivity of the galvanometer. If three resistances are known, the fourth can be found from it.

Final answer. A Wheatstone bridge is balanced when $\frac{P}{Q} = \frac{R}{S}$, i.e. the galvanometer reads zero.

← *Back to Q304*

Q305

What is meant by the sensitivity of a potentiometer?
A battery E_1 *of 4 V and a variable resistance* R_1 *are connected in series with the wire AB of the potentiometer. The length of the wire of the potentiometer is 1 metre. When a cell* E_2 *of e.m.f. 1.5 volt is connected between points A and C, no current flows through* E_2*. Length of AC = 60 cm.*
(i) Find the potential difference between the ends A and B of the potentiometer,
(ii) Would the method work, if the battery E_1 *is replaced by a cell of e.m.f. of 1 V?*

[2003]

SOLUTION

Sensitivity of a potentiometer means its ability to measure a small potential difference accurately. A potentiometer is more sensitive when its potential gradient $k = \frac{V}{L}$ is small, because then a given emf is balanced over a longer length of wire, giving a more precise reading.

Given: emf of driver cell $\mathbf{E_1 = 4\ V}$, length of wire $\mathbf{L = 1\ m = 100\ cm}$, emf $\mathbf{E_2 = 1.5\ V}$ balances at **AC = = 60 cm.**

(i) Potential difference across AB: The potential gradient of the wire is

$$k = \frac{E_2}{} = \frac{1.5}{60}\ V/\text{cm}$$

The potential difference across the full wire AB (length 100 cm) is

$$V_{AB} = k \times L = \frac{1.5}{60} \times 100$$

$$V_{AB} = \frac{150}{60} = 2.5\ V$$

So the potential difference across A and B is **2.5 V.**

(ii) If E_1 is replaced by a cell of 1 V: For the potentiometer to work, the potential drop V_{AB} across the wire must be greater than the emf E_2 being measured, so that a balance point can be found on the wire.

Here $\mathbf{E_2 = 1.5\ V}$ but with $\mathbf{E_1 = 1\ V}$ the maximum potential drop across AB would be less than 1 V, which is smaller than 1.5 V. Therefore no null point can be obtained on the wire and the **method will not work.**

COMMON MISTAKE

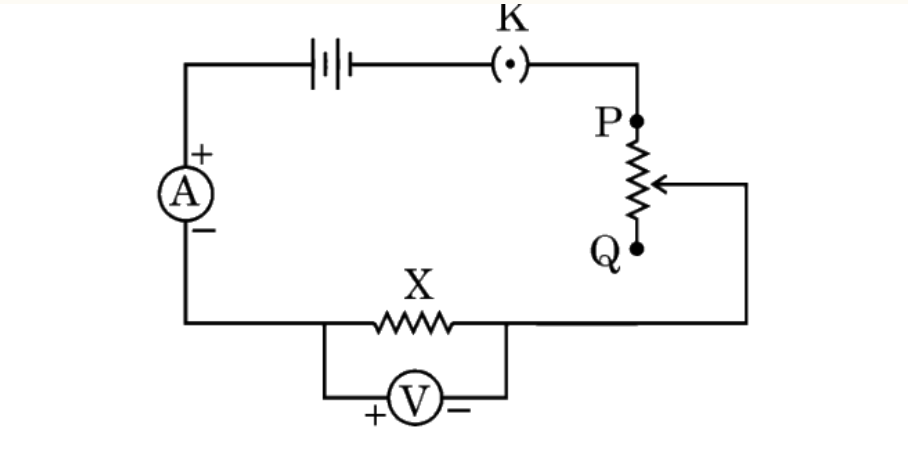
The driver cell's emf must always exceed the emf being measured, otherwise no balance point exists on the wire.

Final answer. (i) $\mathbf{V_{AB} = 2.5\ V}$; (ii) no, with $\mathbf{E_1 = 1\ V}$ the wire's drop is less than **1.5 V**, so it cannot balance $\mathbf{E_2}$.

← Back to Q305

Q306

A student sets up the circuit as shown in the figure to find the value of unknown resistance X and records a set of readings of the voltmeter and the ammeter by using the rheostat.



[2026 • Set 55-4-1]

SOLUTION

In this circuit the unknown resistance X is connected in series with an ammeter A and a battery, with a rheostat included to vary the current. A voltmeter V is connected in parallel across X to read the potential difference across it, while the ammeter reads the current flowing through it.

For each setting of the rheostat, the voltmeter gives the potential difference V across X and the ammeter gives the current I through it. By Ohm's law the resistance is

$$X = \frac{V}{I}$$

Several pairs of (V, I) readings are taken by changing the rheostat, and the value of X is found for each pair. The mean of these values gives the unknown resistance.

Key conclusions for such a set-up: A graph of V (y-axis) against I (x-axis) for an ohmic conductor such as a manganin wire is a straight line passing through the origin, confirming $\boldsymbol{V \propto I}$. The slope of this V versus I graph equals the resistance X.

Moving the rheostat slider changes the current and hence the readings, but the computed ratio V/I stays constant, since the resistance of an ohmic conductor is independent of the current at constant temperature.

Hence the best estimate of the unknown resistance is the average of the values of V/I, or equivalently the **slope of the V versus I straight-line graph.**

Final answer. $X = \frac{V}{I}$ from each reading; the slope of the straight-line V versus I graph gives the unknown resistance.

← *Back to Q306*

Q307

During a thunderstorm the 'live' wire of the transmission line fell down on the ground from the poles in the street. A group of boys, who passed through, noticed it and some of them wanted to place the wire by the side. As they were approaching the wire and trying to lift the cable, Anuj noticed it and immediately pushed them away, thus preventing them from touching the live wire. During pushing some of them got hurt. Anuj took them to a doctor to get them medical aid.
Based on the above paragraph, answer the following questions:
(a) Write the two values which Anuj displayed during the incident.
(b) Why is it that a bird can sit on a suspended 'live' wire without any harm whereas touching it on the ground can give a fatal shock?
(c) The electric power from a power plant is set up to a very high voltage before transmitting it to distant consumers. Explain, why.

[2015]

SOLUTION

(a) Two values displayed by Anuj:

(i) Presence of mind, alertness and quick decision-making, since he instantly realised the danger of the live wire and acted at once.

(ii) Concern for the safety of others and a sense of social responsibility, since he pushed the boys away from the live wire and then took the injured ones to a doctor for medical aid.

(b) A bird sits on a single live wire with both its feet on the same point of the wire. The small length of wire between its feet has negligible resistance, so the potential difference between its feet is almost zero. Hence no current flows through the bird's body and it is safe.

A person standing on the ground who touches the live wire provides a conducting path from the high-potential wire, through the body, to the earth (which is at zero potential). This creates a large potential difference across the body, driving a large current through it, which can be fatal.

(c) Electric power is given by

$$P = V\,I$$

For a fixed power P, raising the transmission voltage V lowers the current I, since $I = \dfrac{P}{V}$.

The power lost as heat in the transmission line of resistance R is

$$P_{loss} = I^2\,R$$

Since $\mathbf{P_{loss} \propto I^2}$, reducing the current greatly reduces this loss. If the voltage is stepped up n times, the current falls to $\dfrac{1}{n}$ and the line loss falls to $\dfrac{1}{n^2}$ of its original value.

Therefore transmitting power at **high voltage and low current minimises the $\mathbf{I^2R}$ energy losses** in the long transmission lines, making transmission far more efficient.

Final answer. (a) Presence of mind and concern for others' safety; (b) the bird has no PD across its feet so no current flows; (c) high voltage means low current, cutting $\mathbf{I^2R}$ line losses.

← Back to Q307

Q308

(a) An electric field $\vec{E}$ is established across the ends of a cylindrical conductor of length

L and area of cross-section A. Discuss how electrons attain an average velocity, independent of time. Hence, obtain a relation between current in the conductor and this 'average velocity' of electrons.

[2026 • Set 55-2-1]

SOLUTION

Set-up: A cylindrical conductor of length L and area of cross-section A contains n free electrons per unit volume, each of charge e and mass m. A field $\vec{E}$ is applied across its ends.

How electrons attain a constant average velocity: In the absence of a field, the free electrons move randomly with high thermal speeds, and their average velocity is zero. When the field $\vec{E}$ is applied, each electron experiences a force $-e\vec{E}$ and an acceleration

$$\vec{a} = -\frac{e\vec{E}}{m}$$

Between two successive collisions, an electron gains an additional velocity due to this acceleration. However, after each collision with the ions of the lattice, the direction of the electron's velocity becomes random again, so the velocity gained from the field is lost on average at every collision.

As a result, the electrons acquire a small, constant average velocity opposite to the field, called the drift velocity. If τ is the average time between collisions (relaxation time), then

$$\vec{v}_d = \vec{a}\,\tau = -\frac{e\tau}{m}\,\vec{E}$$

This is independent of time because τ, e, m and E are all constant.

Relation between current and drift velocity: Consider all electrons that cross the cross-section A in a small time dt. In this time an electron travels a distance v_d dt, so all electrons within a cylinder of length v_d dt and volume $A\,v_d$ dt cross the section.

Number of such electrons = $\boldsymbol{n\,(A\,v_d\,dt)}$.

Total charge crossing the section in time dt is

$$dq = e\,(n\,A\,v_d\,dt)$$

Therefore the current is

$$I = \frac{dq}{dt} = n\, e\, A\, v_d$$

Hence the required relation is $\boldsymbol{I = n\, e\, A\, v_d}$, linking the current to the average (drift) velocity of the electrons.

Final answer. Electrons reach a steady drift velocity $\mathbf{v_d = \frac{e\tau}{m} E}$, giving current $\boldsymbol{I = n\, e\, A\, v_d}$.

← *Back to Q308*

Q309

(b) (i) This 'average velocity' is found to be a few mm s^{-1} *for currents in range of a few amperes. How then is current established almost the instant a circuit is closed?*
(ii) Two copper wires having their radii in the ratio of 3:2 *are connected in series across a battery. Find the ratio of the drift velocities of the electrons in the wires.*

[2026 • Set 55-2-1]

SOLUTION

(i) Why current is established almost instantly: When the circuit is closed, the electric field is set up in the wire and propagates through it with a speed close to the speed of light ($\sim 3\times 10^8\ m\ s^{-1}$). This field reaches every part of the conductor almost instantaneously.

As a result, the free electrons throughout the entire length of the wire begin to drift at almost the same instant, even though each electron itself drifts very slowly (a few mm s^{-1}). Since the current depends on the simultaneous motion of all these electrons, it is established almost the moment the circuit is closed.

(ii) Drift velocity ratio in two series copper wires: When wires are connected in series, the same current I flows through both.

The current is

$$I = n\, e\, A\, v_d = n\, e\, (\pi\, r^2)\, v_d$$

Since both wires are of copper, n and e are the same, and I is the same. Therefore

$$v_d \propto \frac{1}{r^2}$$

Taking the ratio for the two wires with radii in the ratio $\mathbf{r_A : r_B = 3 : 2}$,

$$\frac{v_A}{v_B} = \frac{r_B^2}{r_A^2} = \frac{(2)^2}{(3)^2} = \frac{4}{9}$$

Hence the ratio of drift velocities is $\mathbf{v_A : v_B = 4 : 9}$.

TIP

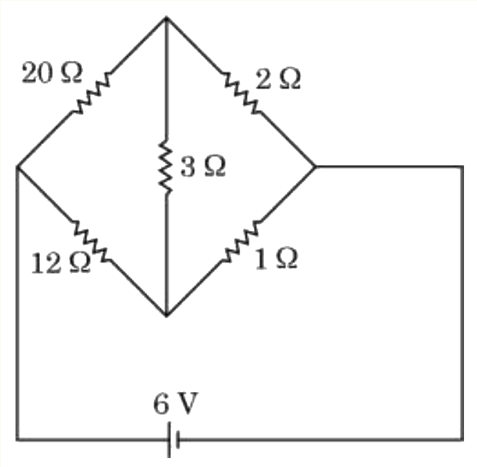
In series the current is equal, so $\mathbf{v_d \propto 1/r^2}$; the thinner wire (smaller radius) has the larger drift velocity.

Final answer. (i) The field travels at nearly light speed, starting all electrons at once; (ii) $\mathbf{v_A : v_B = 4 : 9}$.

← *Back to Q309*

Q310

(a) (i) Derive the condition for which a Wheatstone Bridge is balanced.
(ii) Determine the current in 3 Ω branch of a Wheatstone Bridge in the circuit shown in the figure.



[2026 • Set 55-5-1]

SOLUTION

(i) Balance condition of a Wheatstone bridge: The four resistances P, Q, R, S form the arms of a quadrilateral ABCD. A battery is connected across the diagonal AC and a galvanometer across the diagonal BD.

At balance the galvanometer carries no current **($I_g = 0$).** Then the current I_1 flows through P and Q, and the current I_2 flows through R and S.

Applying Kirchhoff's loop rule to loop ABDA,

$$I_1 P - I_2 R = 0 \Rightarrow I_1 P = I_2 R \ldots(1)$$

Applying Kirchhoff's loop rule to loop BCDB,

$$I_1 Q - I_2 S = 0 \Rightarrow I_1 Q = I_2 S \ldots(2)$$

Dividing equation (1) by equation (2),

$$\frac{P}{Q} = \frac{R}{S}$$

This is the **balance condition** of the Wheatstone bridge.

(ii) Current in the 3 Ω branch: In the given Wheatstone bridge, the 3 Ω resistance lies in the galvanometer (bridge) arm across the diagonal. First test the four outer arms against the balance condition $\frac{P}{Q} = \frac{R}{S}$.

When the ratios of the arms are equal, the bridge is balanced, so the potentials at the two ends of the bridge arm are equal. Hence the potential difference across the 3 Ω branch is zero, and

$$I_{3\,\Omega} = \frac{V}{R} = \frac{0}{3} = 0$$

Therefore the **current through the 3 Ω branch is zero** because the bridge is balanced. (If the arm ratios were unequal, Kirchhoff's laws would be applied to the network to obtain a non-zero current through this branch.)

Final answer. (i) Balance condition $\frac{P}{Q} = \frac{R}{S}$; (ii) the bridge is balanced, so the current in the **3 Ω** branch is **0 *A***.

← Back to Q310

Q311

(b) (i) Consider a cylindrical conductor of length l and area of cross-section A. Current I is maintained in the conductor and electrons drift with velocity $\vec{v}_d$ $\left(|\vec{v}_d| = \frac{e|\vec{E}|}{m}\tau\right)$, *(where symbols have their usual meanings). Show that the conductivity* σ *of the material of the conductor is given by* $\sigma = \frac{ne^2\tau}{m}$.

(ii) The resistance of a metal wire at 20°C *is* 1.05 Ω *and at* 100°C *is* 1.38 Ω. *Determine the temperature coefficient of resistivity of this metal.*

[2026 • Set 55-5-1]

SOLUTION

(i) Consider a cylindrical conductor of length l and uniform area of cross-section A, carrying a steady current I. Let n be the number density of free electrons, each of charge $-e$.

When a potential difference is applied, a uniform electric field $\vec{E}$ is set up inside the conductor. Each electron experiences a force and drifts with the drift velocity

$$|\vec{v}_d| = \frac{e|\vec{E}|}{m}\tau$$

where τ is the relaxation time and m is the mass of an electron.

In a small time dt, the electrons sweep through a distance $v_d\,dt$, so the charge crossing any cross-section is

$$dq = n\,e\,A\,v_d\,dt$$

Hence the current is

$$I = \frac{dq}{dt} = n\,e\,A\,v_d$$

The magnitude of the current density is

$$j = \frac{I}{A} = n\, e\, v_d$$

Substituting $v_d = \frac{e\,E}{m}\,\tau$,

$$j = n\, e\left(\frac{e\,E}{m}\,\tau\right) = \frac{n\, e^2\,\tau}{m}\,E$$

In vector form, since the drift of electrons (charge $-e$) makes $\vec{j}$ point along $\vec{E}$,

$$\vec{j} = \frac{n\, e^2\,\tau}{m}\,\vec{E}$$

Comparing this with the microscopic form of Ohm's law $\vec{j} = \sigma\,\vec{E}$, we identify the conductivity as

$$\sigma = \frac{n\, e^2\,\tau}{m}$$

This is the required expression for the conductivity of the material.

(ii) The variation of resistance with temperature is

$$R = R_0[1 + \alpha\,(T - T_0)]$$

Here $R_0 = 1.05\ \Omega$ at $T_0 = 20°C$ and $R = 1.38\ \Omega$ at $T = 100°C$, so $\Delta T = 100 - 20 = 80\ °C$.

Substituting,

$$1.38 = 1.05[1 + \alpha\,(80)]$$

$$\frac{1.38}{1.05} = 1 + 80\,\alpha$$

$$1.314 = 1 + 80\,\alpha$$

$$80\,\alpha = 0.314$$

$$\alpha = \frac{0.314}{80} = 3.93\times 10^{-3}\ °C^{-1}$$

Therefore the **temperature coefficient of resistivity** is $\alpha \approx 3.9\times 10^{-3}\ °C^{-1}$.

COMMON MISTAKE

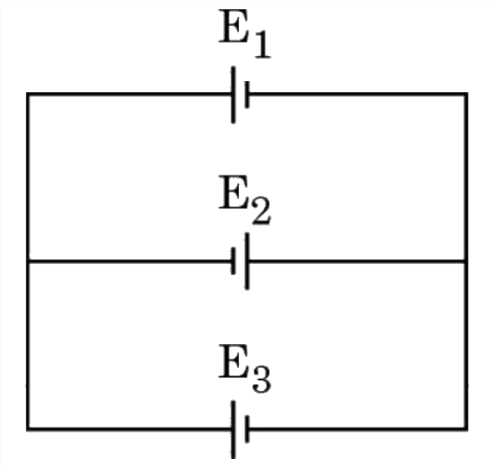
In the temperature-coefficient formula, always use the resistance at the lower (reference) temperature as R_0; using the wrong reference value is the most common slip and gives a slightly off α.

Final answer. Conductivity $\sigma = \dfrac{ne^2\tau}{m}$; temperature coefficient $\alpha \approx 3.9\times 10^{-3}$ $^\circ C^{-1}$.

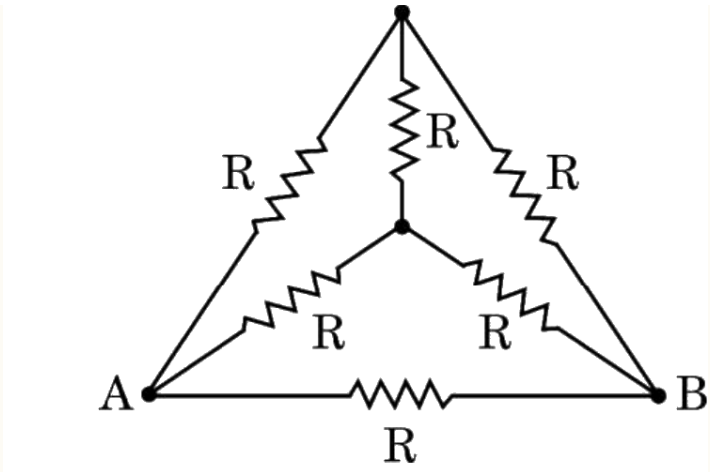
← *Back to Q311*

Q312

(a) (i) *Three batteries E_1, E_2 and E_3 of emfs and internal resistances (4 V, 2 Ω), (2 V, 4 Ω) and (6 V, 2 Ω) respectively are connected as shown in the figure. Find the values of the currents passing through batteries E_1, E_2 and E_3.*



(ii) *The ends of six wires, each of resistance* ***R*** **(= 10 Ω)** *are joined as shown in the figure. The points A and B of the arrangement are connected in a circuit. Find the value of the effective resistance offered by it to the circuit.*

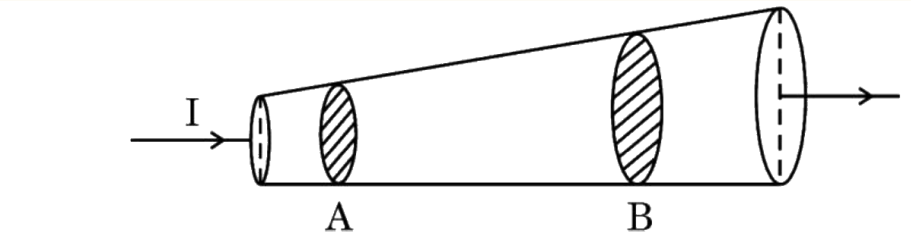


OR

(b) (i) *Current* $\boldsymbol{I}$ $\boldsymbol{(= 1\ A)}$ *is passing through a copper rod (*$\boldsymbol{n = 8.5 \times 10^{28}\ m^{-3}}$*) of varying cross-sections as shown in the figure. The areas of cross-section at points A and B along its length are* $1.0 \times 10^{-7}\ m^2$ *and* $2.0 \times 10^{-7}\ m^2$ *respectively. Calculate:*

(I) the ratio of electric fields at points A and B.

(II) the drift velocity of free electrons at point B.



(ii) *Two point charges* $\mathbf{q_1}$ $\boldsymbol{(= 16\ \mu C)}$ *and* $\mathbf{q_2}$ $\boldsymbol{(= 1\ \mu C)}$ *are placed at points* $\vec{r}_1 = (3\ m)\hat{i}$ *and* $\vec{r}_2 = (4\ m)\hat{j}$*. Find the net electric field* $\vec{E}$ *at point* $\vec{r} = (3\ m)\hat{i} + (4\ m)\hat{j}$*.*

[2025 • Set 55-5-1]

SOLUTION

(a)(i) Let the currents through E_1, E_2 and E_3 be I_1, I_2 and I_3 respectively. In the standard arrangement, E_1 and E_3 drive current into a common node and E_2 shares the same two terminals, so by Kirchhoff's junction rule

$$I_1 + I_3 = I_2$$

Taking the two terminals as P (top) and Q (bottom), every branch carries the same terminal potential difference $V = V_P - V_Q$. For each cell, with the polarity shown:

$$V = E_1 - I_1 r_1 = 4 - 2 I_1$$

$$V = E_3 - I_3 r_3 = 6 - 2 I_3$$

$\mathbf{V = -E_2 + I_2 r_2 = -2 + 4 I_2}$ (cell E_2 is being charged, opposing the others)

From these, $\mathbf{I_1 = \frac{4 - V}{2}}$, $\mathbf{I_3 = \frac{6 - V}{2}}$ and $\mathbf{I_2 = \frac{V + 2}{4}}$.

Putting these in $\mathbf{I_1 + I_3 = I_2}$:

$$\frac{4 - V}{2} + \frac{6 - V}{2} = \frac{V + 2}{4}$$

$$\frac{10 - 2V}{2} = \frac{V + 2}{4}$$

$$2(10 - 2V) = V + 2$$

$$20 - 4V = V + 2 \Rightarrow 5V = 18 \Rightarrow V = 3.6\ V$$

Hence

$$I_1 = \frac{4 - 3.6}{2} = 0.2\ A$$

$$I_3 = \frac{6 - 3.6}{2} = 1.2\ A$$

$$I_2 = \frac{3.6 + 2}{4} = 1.4\ A$$

So the currents are $\mathbf{I_1 = 0.2\ A}$ (through E_1), $\mathbf{I_2 = 1.4\ A}$ (through E_2) and $\mathbf{I_3 = 1.2\ A}$ (through E_3), with E_2 getting charged.

(a)(ii) Six equal resistors $\boldsymbol{R = 10\ \Omega}$ form the edges of a network joining A and B. Using the symmetry of the figure, the points symmetric about the A-B axis are at equal potential, so no current flows through the resistor joining them, and it can be removed. Combining the remaining resistors in the resulting series and parallel paths gives

$$R_{AB} = R = 10\ \Omega$$

Thus the effective resistance offered to the circuit is $\mathbf{10\ \Omega}$.

(b)(i)(I) For a steady current I in the rod, the same current passes every cross-section. Since $\boldsymbol{I = \sigma E A}$ and σ (the material) is the same throughout,

$$E_A A_A = E_B A_B$$

$$\frac{E_A}{E_B} = \frac{A_B}{A_A} = \frac{2.0\times 10^{-7}}{1.0\times 10^{-7}} = 2$$

So the **ratio of electric fields** is $E_A : E_B = 2 : 1$.

(b)(i)(II) The drift velocity at B is

$$v_B = \frac{I}{n\, e\, A_B}$$

$$v_B = \frac{1}{(8.5\times 10^{28})(1.6\times 10^{-19})(2.0\times 10^{-7})}$$

$$v_B = \frac{1}{2.72\times 10^{3}} = 3.68\times 10^{-4}\ m/s$$

So the **drift velocity at B** is about $3.7\times 10^{-4}\ m/s$.

(b)(ii) The field due to $\mathbf{q_1 = 16\ \mu C}$ at $\vec{r}_1 = 3\,\hat{i}$, observed at $\vec{r} = 3\,\hat{i} + 4\,\hat{j}$, points from q_1 to the point, i.e. along $+\hat{j}$, at distance 4 m:

$$E_1 = \frac{1}{4\pi\varepsilon_0}\frac{q_1}{(4)^2} = \frac{(9\times 10^{9})(16\times 10^{-6})}{16} = 9\times 10^{3}\ N/C\ (\text{along } +\hat{j})$$

The field due to $\mathbf{q_2 = 1\ \mu C}$ at $\vec{r}_2 = 4\,\hat{j}$ points from q_2 to the point, i.e. along $+\hat{i}$, at distance 3 m:

$$E_2 = \frac{1}{4\pi\varepsilon_0}\frac{q_2}{(3)^2} = \frac{(9\times 10^{9})(1\times 10^{-6})}{9} = 1\times 10^{3}\ N/C\ (\text{along } +\hat{i})$$

Hence the net field is

$$\vec{E} = (1\times 10^{3})\,\hat{i} + (9\times 10^{3})\,\hat{j}\ N/C$$

with magnitude

$$E = 10^{3}\sqrt{1^2 + 9^2} = 10^{3}\sqrt{82} \approx 9.06\times 10^{3}\ N/C$$

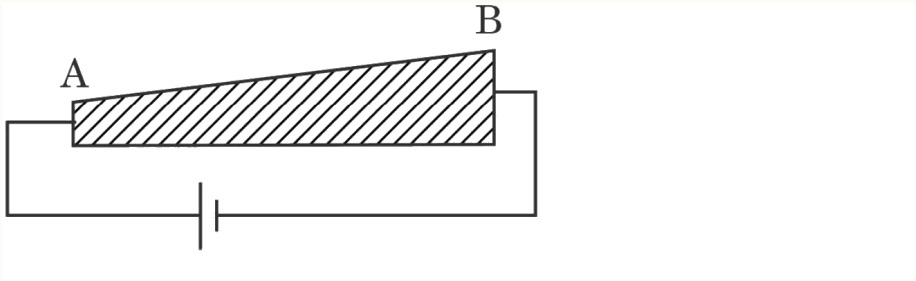
Final answer. $\mathbf{I_1}$**=0.2** A, $\mathbf{I_2}$**=1.4** A, $\mathbf{I_3}$**=1.2** A; $\mathbf{R_{AB}}$**=10 Ω**. OR $\mathbf{E_A}$**:**$\mathbf{E_B}$**=2:1**, $\mathbf{v_B}$**3.710**$^{-4}$ m/s; $\vec{E}$**=(10**$^{3}\hat{i}$**+910**$^{3}\hat{j}$**)** N/C.

← *Back to Q312*

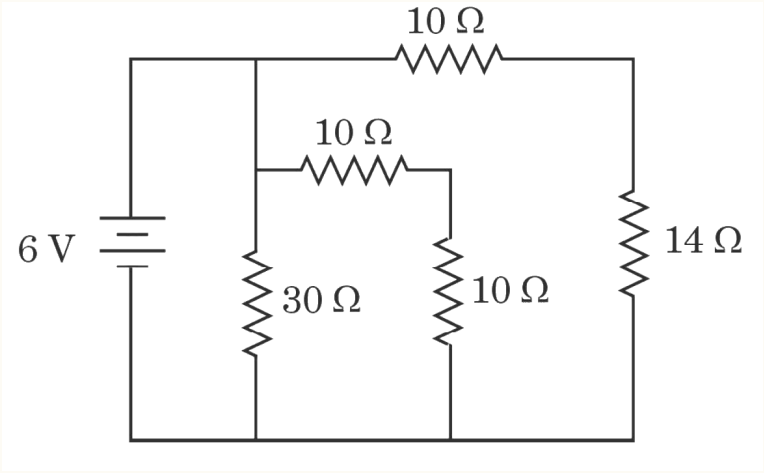
Q313

(a) (i) Define mobility of electrons. Give its SI units.

(ii) A steady current flows through a wire AB, as shown in the figure. What happens to the electric field and the drift velocity along the wire? Justify your answer.



(iii) Consider the circuit shown in the figure. Find the effective resistance of the circuit and the current drawn from the battery.

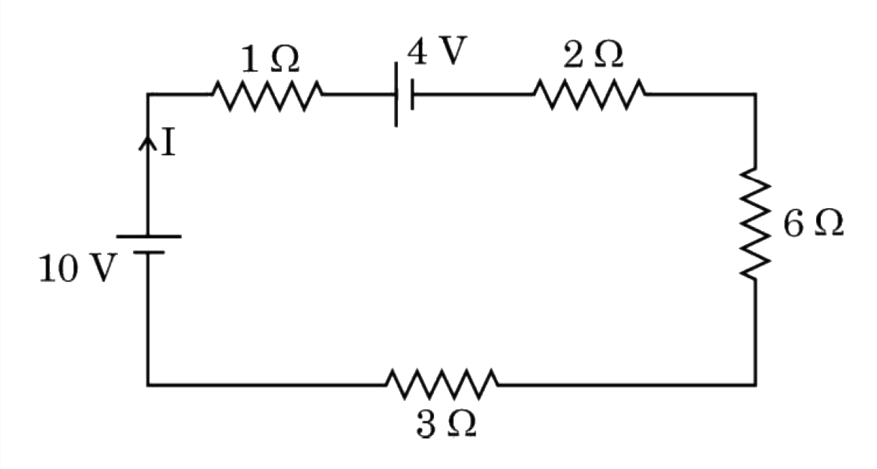


OR

(b) (i) Define electrical conductivity of a wire. Give its SI unit.

(ii) High current is to be drawn safely from (1) a low-voltage battery, and (2) a high-voltage battery. What can you say about the internal resistance of the two batteries?

(iii) Calculate the total energy supplied by the batteries to the circuit shown in the figure, in one minute.



[2023 • Set 55-4-1]

SOLUTION

(a)(i) The **mobility** of electrons is defined as the magnitude of the drift velocity acquired per unit applied electric field:

$$\mu = \frac{|\vec{v}_d|}{|\vec{E}|} = \frac{e\,\tau}{m}$$

Its SI unit is $m^2\,V^{-1}\,s^{-1}$.

(a)(ii) The wire AB tapers, so its cross-sectional area A decreases from A towards B. In a steady state the current I is the same at every section (charge conservation). Since

$$I = n\,e\,A\,v_d$$

with I, n and e fixed, the drift velocity varies as $v_d \propto \frac{1}{A}$. So as A decreases towards B, the **drift velocity increases**.

The electric field is

$$E = \frac{j}{\sigma} = \frac{I}{\sigma A}$$

which also varies as $E \propto \frac{1}{A}$, so the **electric field increases** where the wire is thinner. Both the field and the drift velocity are largest where the cross-section is smallest.

(a)(iii) In the given network a 30 Ω resistor is in parallel with a 10 Ω resistor, and this combination is in series with another 10 Ω resistor. The parallel combination is

$$R_p = \frac{30\times 10}{30+10} = \frac{300}{40} = 7.5\ \Omega$$

So the effective resistance of the circuit is

$$R_{eq} = 10 + 7.5 = 17.5\ \Omega$$

With a battery of emf V as marked in the figure, the current drawn is

$$I = \frac{V}{R_{eq}} = \frac{V}{17.5}\ A$$

(For the usual figure value of $\boldsymbol{V = 10\ V}$, $\boldsymbol{I = \frac{10}{17.5} = 0.57\ A}$.)

(b)(i) The **electrical conductivity** of a material is the reciprocal of its resistivity, $\sigma = \frac{1}{\rho}$; equivalently it is the current density produced per unit electric field, $\sigma = \frac{j}{E}$. Its SI unit is $\Omega^{-1}m^{-1}$, i.e. siemens per metre ($S\ m^{-1}$).

(b)(ii) The current from a cell is $I = \frac{\varepsilon}{r + R_{ext}}$. To draw a high current safely the total resistance must be kept small, which requires a small internal resistance r. Hence (1) for a low-voltage battery, r must be very small; (2) for a high-voltage battery, r should be comparatively large so that the maximum (short-circuit) current ε/r is limited and the battery is not damaged.

(b)(iii) The total energy supplied by the batteries in time t equals the total electrical work done. If I is the current and ε the net emf,

$$W = \varepsilon\, I\, t$$

First find I from $I = \frac{\varepsilon_{net}}{R_{eq}}$ for the figure, then for $t = 1$ minute $= 60\ s$,

$$W = \varepsilon\, I\, (60)\ J$$

gives the total energy supplied in one minute (substitute the emf and current values from the figure).

Final answer. Mobility unit $\mathbf{m^2V^{-1}s^{-1}}$; E and v_d rise where the wire is thinner; R_{eq}=17.5 Ω. OR σ unit S m^{-1}; energy W=ε It over 60 s.

← *Back to Q313*

Q314

(a) (i) Explain how free electrons in a metal at constant temperature attain an average velocity under the action of an electric field. Hence obtain an expression for it.
(ii) Consider two conducting wires A and B of the same diameter but made of different materials joined in series across a battery. The number density of electrons in A is 1.5 times that in B. Find the ratio of drift velocity of electrons in wire A to that in wire B.
OR
(b) (i) A cell of emf (E) and internal resistance (r) is connected across a variable load resistance (R). Draw plots showing the variation of terminal voltage V with (i) R and (ii) the current (I) in the load.
(ii) Three cells, each of emf E but internal resistances 2r, 3r and 6r are connected in parallel across a resistor R. Obtain expressions for (i) current flowing in the circuit, and (ii) the terminal potential difference across the equivalent cell.

[2023 • Set 55-5-1]

SOLUTION

(a)(i) In the absence of any field, the free electrons in a metal move randomly with high thermal speeds in all directions, so their average velocity is zero and there is no net current.

When an external field $\vec{E}$ is applied, each electron of charge $-e$ and mass m experiences a force $-e\vec{E}$ and so an acceleration

$$\vec{a} = -\frac{e\vec{E}}{m}$$

Between two successive collisions with the lattice ions, the electron gains an extra velocity. Each collision randomises its direction, so the gains start afresh after every collision. If τ is the average time between collisions (relaxation time), the average extra velocity, called the **drift velocity**, is

$$\vec{v}_d = \vec{a}\,\tau$$

$$\vec{v}_d = -\frac{e\,\tau}{m}\,\vec{E}$$

Its magnitude is $v_d = \frac{e\,\tau}{m}\,E$. This is the average velocity attained by the electrons.

(a)(ii) Wires A and B are in series, so the same current I flows through both. They have the same diameter, hence the same area A. From

$$I = n\,e\,A\,v_d \Rightarrow v_d = \frac{I}{n\,e\,A}$$

At fixed I, e and A, the drift velocity varies as $\mathbf{v_d \propto \frac{1}{n}}$. Therefore

$$\frac{v_{dA}}{v_{dB}} = \frac{n_B}{n_A}$$

Given $\mathbf{n_A = 1.5\,n_B}$,

$$\frac{v_{dA}}{v_{dB}} = \frac{n_B}{1.5\,n_B} = \frac{1}{1.5} = \frac{2}{3}$$

So the ratio of drift velocities is $\mathbf{v_{dA} : v_{dB} = 2 : 3}$.

(b)(i) For a cell of emf E and internal resistance r across a load R, the current is $\boldsymbol{I = \frac{E}{R+r}}$ and the terminal voltage is

$$V = E - I\,r = \frac{E\,R}{R+r}$$

Plot of V vs R: a curve starting from $\boldsymbol{V = 0}$ at $\boldsymbol{R = 0}$ and rising, approaching $\boldsymbol{V = E}$ asymptotically as $\boldsymbol{R \to \infty}$ (concave, saturating at E).

Plot of V vs I: from $\boldsymbol{V = E - Ir}$, this is a straight line starting at $\boldsymbol{V = E}$ (when $\boldsymbol{I = 0}$, open circuit), falling with slope $-r$ and meeting the I-axis at $\boldsymbol{I = E/r}$ (short circuit).

(b)(ii) Three cells, each of emf E, with internal resistances 2r, 3r and 6r, are connected in parallel across R.

The equivalent internal resistance of the parallel combination satisfies

$$\frac{1}{r_{eq}} = \frac{1}{2r} + \frac{1}{3r} + \frac{1}{6r}$$

$$\frac{1}{r_{eq}} = \frac{3+2+1}{6r} = \frac{6}{6r} = \frac{1}{r}$$

$$r_{eq} = r$$

Since all the emfs are equal to E, the equivalent emf of the parallel combination is also $\mathbf{E_{eq} = E}$.

(i) The current in the circuit is

$$I = \frac{E_{eq}}{R + r_{eq}} = \frac{E}{R + r}$$

(ii) The terminal potential difference across the equivalent cell (i.e. across R) is

$$V = I\,R = \frac{E\,R}{R + r}$$

CONCEPT

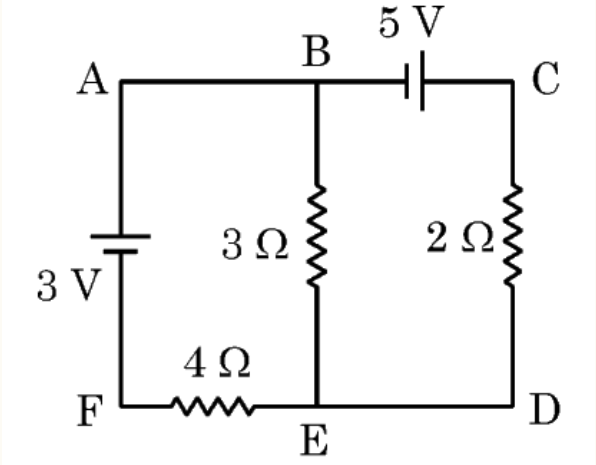
When equal-emf cells are in parallel, the equivalent emf stays E (it does not add up); only the internal resistances combine like parallel resistors.

Final answer. $\mathbf{v_{dA}:v_{dB}=2:3}$. OR $\mathbf{r_{eq}=r}$, $\boldsymbol{I=\frac{E}{R+r}}$, $\boldsymbol{V=\frac{ER}{R+r}}$.

← *Back to Q314*

Q315

(a) Describe briefly, with the help of a circuit diagram, the method of measuring the internal resistance of a cell.
(b) Give reason why a potentiometer is preferred over a voltmeter for the measurement of emf of a cell.
(c) In the potentiometer circuit given below, calculate the balancing length l. Give reason, whether the circuit will work, if the driver cell of emf 5 V is replaced with a cell of 2 V, keeping all other factors constant.



OR

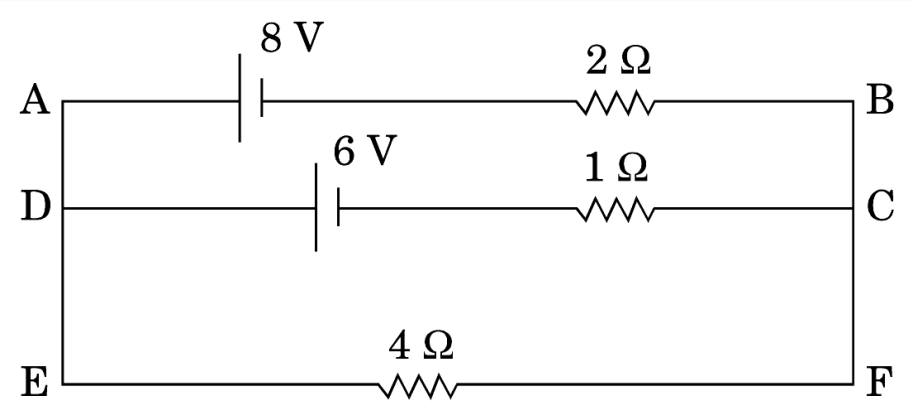
(a) State the working principle of a meter bridge used to measure an unknown resistance.

(b) Give reason

(i) why the connections between the resistors in a metre bridge are made of thick copper strips,

(ii) why is it generally preferred to obtain the balance length near the mid-point of the bridge wire.

(c) Calculate the potential difference across the 4 Ω *resistor in the given electrical circuit, using Kirchhoff's rules.*



[2019 • Set 55-2-1]

SOLUTION

(a) Measuring the internal resistance of a cell with a potentiometer. The primary circuit has a driver (auxiliary) cell connected through a rheostat and key across a long uniform potentiometer wire AB, setting up a steady potential gradient along it. The cell whose internal resistance *r* is to be found is

connected (positive terminal to A) through a galvanometer, with a resistance box R' and a key K across the cell as a shunt.

First, with the shunt key K open, only the emf ε of the cell balances against the wire; let the balancing length be $_1$, so

$$\varepsilon = k\,_1$$

Next, close key K to connect a known resistance R' across the cell. Now the terminal voltage V balances at length $_2$:

$$V = k\,_2$$

Dividing,

$$\frac{\varepsilon}{V} = \frac{_1}{_2}$$

Since $V = \dfrac{\varepsilon R'}{R' + r}$, we have $\dfrac{\varepsilon}{V} = \dfrac{R' + r}{R'}$, so

$$\frac{_1}{_2} = \frac{R' + r}{R'}$$

$$r = R'\left(\frac{_1}{_2} - 1\right)$$

This gives the internal resistance of the cell.

(b) A potentiometer is preferred over a voltmeter for measuring emf because at the balance point it draws **no current** from the cell, so it measures the true emf ε. A voltmeter always draws a small current I and therefore reads only the terminal voltage $V = \varepsilon - Ir$, which is less than the emf.

(c) The driver cell of emf 5 V has a 450 Ω series resistance and a 10 m wire of resistance 50 Ω. The current in the wire is

$$I = \frac{5}{450 + 50} = \frac{5}{500} = 0.01\ A$$

The voltage across the whole wire is

$$V_{\text{wire}} = I \times 50 = 0.01 \times 50 = 0.5\ V$$

So the potential gradient is

$$k = \frac{0.5}{10} = 0.05\ V/m$$

The balancing length for the secondary emf ε is

$$= \frac{\varepsilon}{k}$$

(For a typical secondary emf $\boldsymbol{\varepsilon = 0.4\ V}$, $= \frac{\mathbf{0.4}}{\mathbf{0.05}} = \mathbf{8}\ \boldsymbol{m}$.)

If the 5 V driver is replaced by a 2 V cell, the wire current becomes $\boldsymbol{I} = \frac{\mathbf{2}}{\mathbf{500}} = \mathbf{0.004}\ \boldsymbol{A}$ and the voltage across the wire is $\mathbf{0.004 \times 50 = 0.2}\ \boldsymbol{V}$. Since this is less than the secondary emf to be balanced, the potential drop available along the whole wire is smaller than ε, so a balance point cannot be obtained. Hence the **circuit will not work** with the 2 V driver cell.

OR (a) Working principle of a meter bridge. A meter bridge is a practical form of the Wheatstone bridge. A 1 m long uniform resistance wire forms two of the four arms; the unknown resistance X and a known resistance R are placed in the gaps. The jockey is moved along the wire until the galvanometer shows no deflection (balance). At balance, no current flows through the galvanometer and the bridge condition gives

$$\frac{X}{R} = \frac{}{100 -}$$

where (in cm) is the balancing length from end A.

(b)(i) The connecting strips are made of thick copper because copper has very low resistivity and the thick strips give a large cross-section, so their resistance is negligible and does not introduce error into the bridge arms.

(b)(ii) The balance is taken near the mid-point because there the percentage error in measuring the balancing length (and hence in the result) is minimum, and the bridge is most sensitive when the four arms are comparable.

(c) Potential difference across the 4 Ω resistor by Kirchhoff's rules. Assign loop currents, apply the junction rule at the node and the loop rule around each closed loop. Solve the simultaneous equations for the current I_4 through the 4 Ω resistor; then

$$V_{4\Omega} = I_4 \times 4\ V$$

gives the required potential difference (substitute the emf and resistance values from the figure to get the numerical value).

CONCEPT

A potentiometer can only balance an emf if the total potential drop along its wire exceeds that emf; if the driver voltage is too low, no null point exists, no matter how long the wire.

Final answer. $r = R'\left(\frac{1}{-} - 1\right)$; $k = 0.05$ V/m, and the circuit fails with the 2 V driver.

OR meter-bridge: $\frac{X}{R} = \frac{2}{100 -}$, $V_{4\Omega} = 4I_4$.

← *Back to Q315*

Q316

(a) Derive a relation between the internal resistance, emf and terminal potential difference of a cell from which current I is drawn. Draw V vs I graph for a cell and explain its significance.
(b) A voltmeter of resistance 998 Ω *is connected across a cell of emf 2 V and internal resistance* 2 Ω*. Find the potential difference across the voltmeter and also across the terminals of the cell. Estimate the percentage error in the reading of the voltmeter.*
OR
(a) Two cells of different emfs and internal resistances are connected in parallel with one another. Derive the expression for the equivalent emf and equivalent internal resistance of the combination.
(b) Two identical cells of emf 1.5 V and internal resistance r are each connected in parallel providing a supply to an external circuit consisting of two resistances of 17 Ω *each joined in parallel. A very high resistance voltmeter reads the terminal voltage of the cell to be 1.4 V. Calculate the internal resistance of each cell.*

[2019 • Set 55-5-1]

SOLUTION

(a) Consider a cell of emf ε and internal resistance *r* driving a current *I* through an external resistance *R*. As the current passes through the cell, a part of the emf is used up in overcoming the internal resistance.

The terminal potential difference is the voltage available across the external circuit:

$$V = \varepsilon - I\,r$$

Rearranging,

$$r = \frac{\varepsilon - V}{I}$$

The *V*-*I* graph is a straight line: when $\boldsymbol{I = 0}$ (open circuit), $\boldsymbol{V = \varepsilon}$ (the intercept on the *V*-axis); as *I* increases, *V* falls with slope $-r$, meeting the *I*-axis at the short-circuit current $\boldsymbol{I = \varepsilon/r}$.

Significance: the *V*-intercept of the graph gives the emf ε, and the magnitude of the slope gives the internal resistance *r* of the cell.

(b) A voltmeter of resistance $\mathbf{R_V = 998\ \Omega}$ is connected across a cell of emf $\boldsymbol{\varepsilon = 2\ V}$ and internal resistance $\boldsymbol{r = 2\ \Omega}$. The current in the circuit is

$$I = \frac{\varepsilon}{R_V + r} = \frac{2}{998 + 2} = \frac{2}{1000} = 2\times 10^{-3}\ A$$

Potential difference across the voltmeter:

$$V_V = I\,R_V = (2\times 10^{-3})(998) = 1.996\ V$$

Potential difference across the terminals of the cell (same two points):

$$V_{cell} = \varepsilon - I\,r = 2 - (2\times 10^{-3})(2) = 2 - 0.004 = 1.996\ V$$

The voltmeter should ideally read the emf 2 *V*, but it reads 1.996 *V*. The percentage error is

$$\%\ \text{error} = \frac{\varepsilon - V_V}{\varepsilon}\times 100 = \frac{2 - 1.996}{2}\times 100 = \frac{0.004}{2}\times 100 = 0.2\%$$

So both potential differences are 1.996 *V* and the **percentage error** is 0.2%.

OR (a) Two cells of emfs ε_1, ε_2 and internal resistances r_1, r_2 are connected in parallel between two points A and B. Let I_1 and I_2 be the currents through the cells, with total current $\boldsymbol{I = I_1 + I_2}$, and let *V* be the common terminal potential difference $V_A - V_B$.

For each cell,

$$V = \varepsilon_1 - I_1 r_1 \Rightarrow I_1 = \frac{\varepsilon_1 - V}{r_1}$$

$$V = \varepsilon_2 - I_2 r_2 \Rightarrow I_2 = \frac{\varepsilon_2 - V}{r_2}$$

Adding,

$$I = I_1 + I_2 = \frac{\varepsilon_1 - V}{r_1} + \frac{\varepsilon_2 - V}{r_2}$$

$$I = \left(\frac{\varepsilon_1}{r_1} + \frac{\varepsilon_2}{r_2}\right) - V\left(\frac{1}{r_1} + \frac{1}{r_2}\right)$$

Solving for V,

$$V = \frac{\frac{\varepsilon_1}{r_1} + \frac{\varepsilon_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}} - \frac{I}{\frac{1}{r_1} + \frac{1}{r_2}}$$

Comparing with $V = \varepsilon_{eq} - I\, r_{eq}$,

$$\varepsilon_{eq} = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2} \quad r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$$

(b) Two identical cells (emf 1.5 V, internal resistance r each) in parallel feed two 17 Ω resistors in parallel. The external resistance is

$$R_{ext} = \frac{17 \times 17}{17 + 17} = \frac{17}{2} = 8.5\ \Omega$$

For identical cells in parallel, $\varepsilon_{eq} = 1.5\ V$ and $r_{eq} = \frac{r}{2}$. The voltmeter reads the terminal voltage $V = 1.4\ V$.

The current in the circuit is

$$I = \frac{V}{R_{ext}} = \frac{1.4}{8.5} = 0.1647\ A$$

Using $V = \varepsilon_{eq} - I\, r_{eq}$,

$$1.4 = 1.5 - (0.1647)\frac{r}{2}$$

$$(0.1647)\frac{r}{2} = 0.1$$

$$\frac{r}{2} = \frac{0.1}{0.1647} = 0.607$$

$$r = 1.21\ \Omega$$

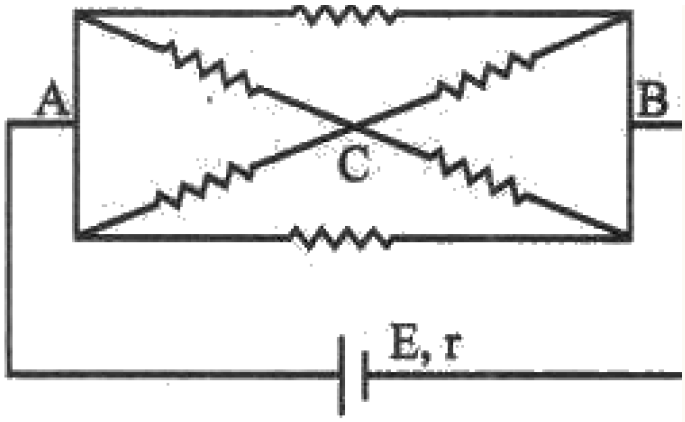
So the internal resistance of each cell is about 1.2 Ω.

Final answer. $\mathbf{V_V = V_{cell} = 1.996}$ V, error $\mathbf{= 0.2\%}$. OR $\varepsilon_{eq} = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$, $r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$, and each cell's r **1.2 Ω.**

← *Back to Q316*

Q317

(i) Derive an expression for drift velocity of electrons in a conductor. Hence deduce Ohm's law.
(ii) A wire whose cross-sectional area is increasing linearly from its one end to the other, is connected across a battery of V volts. Which of the following quantities remain constant in the wire?
(a) drift speed
(b) current density
(c) electric current
(d) electric field
Justify your answer.
OR
(i) State the two Kirchhoff's laws. Explain briefly how these rules are justified.
(ii) The current is drawn from a cell of emf E and internal resistance r connected to the network of resistors each of resistance r as shown in the figure. Obtain the expression for (i) the current draw from the cell and (ii) the power consumed in the network.



[2017]

SOLUTION

(i) Consider a conductor of length L and cross-section A across which a potential difference V is applied, setting up a uniform field $\boldsymbol{E = V/L}$. A free electron (charge $-e$, mass m) experiences an acceleration

$$\vec{a} = -\frac{e\vec{E}}{m}$$

Between collisions the electron gains velocity, and each collision randomises its direction. Averaging over the relaxation time τ, the **drift velocity** is

$$\vec{v}_d = -\frac{e\,\tau}{m}\,\vec{E}$$

with magnitude $v_d = \dfrac{e\,\tau}{m}E$.

The current carried by the conductor is

$$I = n\,e\,A\,v_d$$

Substituting $v_d = \dfrac{e\,\tau}{m}E = \dfrac{e\,\tau}{m}\dfrac{V}{L}$,

$$I = n\,e\,A \cdot \frac{e\,\tau}{m}\frac{V}{L}$$

$$I = \frac{n\,e^2\,\tau\,A}{m\,L}\,V$$

Rearranging,

$$V = I \cdot \frac{m\,L}{n\,e^2\,\tau\,A}$$

At constant temperature n and τ are constant, so the quantity

$$R = \frac{m\,L}{n\,e^2\,\tau\,A}$$

is a constant for a given conductor. Hence

$$V = I\,R$$

which is **Ohm's law**: at constant temperature the current is proportional to the applied voltage.

(ii) The wire's cross-section increases linearly from one end to the other, and a steady current flows. By the continuity of current (charge conservation), the same current passes through every cross-section, so:

(c) **electric current I remains constant** along the wire.

The other quantities vary: current density $j = \frac{I}{A}$ decreases where A is larger; drift speed $v_d = \frac{I}{neA}$ decreases where A is larger; and electric field $E = \frac{j}{\sigma}$ also varies along the wire. Therefore only the **electric current** stays constant; option (c) is correct.

OR (i) Kirchhoff's two laws:

Junction (current) rule: the algebraic sum of currents meeting at any junction is zero, $\sum I = 0$. This follows from conservation of charge, since charge cannot accumulate at a junction in steady state.

Loop (voltage) rule: around any closed loop, the algebraic sum of the changes in potential is zero, $\sum \Delta V = 0$. This follows from conservation of energy, as a charge taken around a closed loop returns to its starting potential.

(ii) For the network of resistors each of value r fed by a cell of emf E and internal resistance r, reduce the resistor network by symmetry and series/parallel combination to an equivalent resistance R_{net}. The current drawn from the cell is

$$I = \frac{E}{r + R_{net}}$$

and the power consumed in the network (external part) is

$$P = I^2 R_{net} = \left(\frac{E}{r + R_{net}}\right)^2 R_{net}$$

Substituting the figure's equivalent R_{net} gives the numerical current and power; for the common case $\mathbf{R_{net} = r}$,

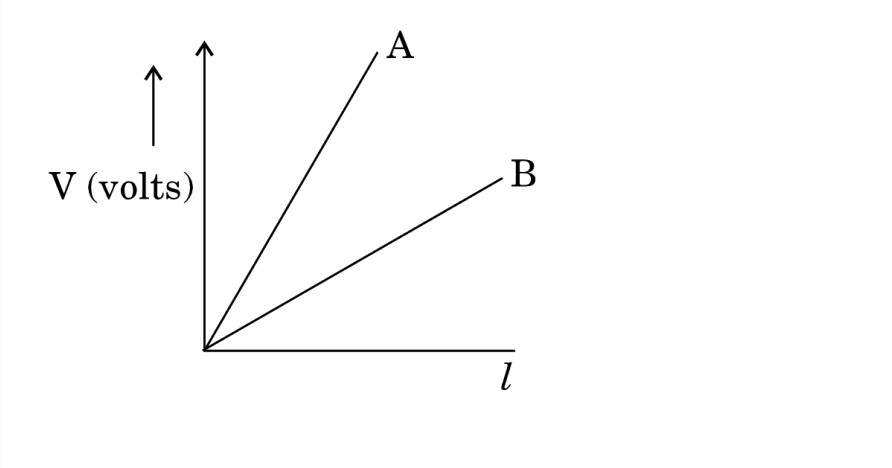
$$I = \frac{E}{2r} \quad P = \frac{E^2}{4r}$$

Final answer. Drift velocity gives $V=IR$ with $R=\frac{mL}{ne^2\tau A}$ (Ohm's law); only the electric current (c) stays constant. OR Kirchhoff's rules; $I=\frac{E}{r+R_{net}}$, $P=I^2 R_{net}$.

← Back to Q317

Q318

(a) (i) State the principle on which a potentiometer works. How can a given potentiometer be made more sensitive?
(ii) In the graph shown below for two potentiometers, state with reason which of the two potentiometers, A or B, is more sensitive.



(b) Two metallic wires, P_1 and P_2 of the same material and same length but different cross-sectional areas, A_1 and A_2 are joined together and connected to a source of emf. Find the ratio of the drift velocities of free electrons in the two wires when they are connected (i) in series, and (ii) in parallel.

OR

(a) Define the capacitance of a capacitor. Obtain the expression for the capacitance of a parallel plate capacitor in vacuum in terms of plate area A and separation d between the plates.

(b) A slab of material of dielectric constant K has the same area as the plates of a parallel plate capacitor but has a thickness $\frac{d}{2}$*. Find the ratio of the capacitance with dielectric inside it to its capacitance without the dielectric.*

[2017]

SOLUTION

(a)(i) **Principle of a potentiometer:** when a steady current flows through a uniform wire of constant cross-section, the potential drop across any segment is directly proportional to its length, $V = k$, where k is the constant potential gradient.

A potentiometer is made more sensitive by reducing the potential gradient k (volts per unit length), which lets it detect smaller potential differences. This is done by increasing the length of the wire or by decreasing the current through it (e.g. using a series rheostat).

(a)(ii) On the graph of balancing length versus potential difference V, the line with the **smaller slope of V against** (i.e. a smaller potential gradient k) is the more sensitive one, because a smaller k gives a longer balancing length for the same emf and hence a more precise reading. So the potentiometer whose line is less steep (smaller gradient) is more sensitive; from the figure this is the one with the smaller slope.

(b) Two wires P_1 (area A_1) and P_2 (area A_2) of the same material and same length L are joined to a source.

(i) In series, the same current I flows through both. Using $I = n\,e\,A\,v_d$ (with n the same since the material is the same):

$$v_d = \frac{I}{n\,e\,A}$$

$$\frac{v_{d_1}}{v_{d_2}} = \frac{A_2}{A_1}$$

(ii) In parallel, both wires have the same potential difference V across the same length L, so the same field $\boldsymbol{E} = \frac{\boldsymbol{V}}{\boldsymbol{L}}$. Since $\mathbf{v}_d = \boldsymbol{\mu}\, \boldsymbol{E}$ and the mobility μ is the same (same material),

$$\frac{v_{d_1}}{v_{d_2}} = 1$$

So in series the ratio is $A_2 : A_1$, and in parallel it is $1 : 1$.

OR (a) **Capacitance** of a capacitor is the ratio of the charge Q stored on either plate to the potential difference V between the plates: $\boldsymbol{C} = \frac{\boldsymbol{Q}}{\boldsymbol{V}}$.

For a parallel plate capacitor in vacuum with plate area A and separation d, the surface charge density is $\boldsymbol{\sigma} = \frac{\boldsymbol{Q}}{\boldsymbol{A}}$. The uniform field between the plates is

$$E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A}$$

The potential difference is

$$V = E\, d = \frac{Q\, d}{\varepsilon_0 A}$$

Hence

$$C_0 = \frac{Q}{V} = \frac{\varepsilon_0 A}{d}$$

(b) A dielectric slab of constant K and thickness $\frac{d}{2}$ is inserted; the remaining $\frac{d}{2}$ is vacuum. The arrangement is two capacitors in series: one with the dielectric and one with air, each of thickness $\frac{d}{2}$.

$$C_1 = \frac{K\varepsilon_0 A}{d/2} = \frac{2K\varepsilon_0 A}{d} \quad C_2 = \frac{\varepsilon_0 A}{d/2} = \frac{2\varepsilon_0 A}{d}$$

For series,

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{d}{2K\varepsilon_0 A} + \frac{d}{2\varepsilon_0 A}$$

$$\frac{1}{C} = \frac{d}{2\varepsilon_0 A}\left(\frac{1}{K} + 1\right) = \frac{d}{2\varepsilon_0 A} \cdot \frac{K+1}{K}$$

$$C = \frac{2K\varepsilon_0 A}{d(K+1)}$$

Therefore the required ratio is

$$\frac{C}{C_0} = \frac{2K\varepsilon_0 A / [d(K+1)]}{\varepsilon_0 A / d} = \frac{2K}{K+1}$$

Final answer. Drift-velocity ratio: series $\mathbf{A_2{:}A_1}$, parallel **1:1**. OR $\mathbf{C_0 = \frac{\varepsilon_0 A}{d}}$ and $\mathbf{\frac{C}{C_0} = \frac{2K}{K+1}}$.

← *Back to Q318*

Q319

(i) State the principle of working of a meter bridge. Draw the circuit diagram for finding an unknown resistance using a meter bridge. Derive the relevant formula used.

(ii) In a meter bridge with R and S in the gaps, the null point is found at 40 cm from A. If a resistance of 30 Ω is connected in parallel with S, the null point occurs at 50 cm from A. Determine the values of R and S.

OR

(a) Deduce the expression for the torque acting on a dipole of dipole moment $\vec{P}$ placed in a uniform electric field $\vec{E}$. Depict the direction of the torque. Express it in the vector form.

(b) Show that the potential energy of a dipole making angle θ with the direction of the field is given by $\boldsymbol{u(\theta) = -\vec{p}\cdot\vec{E}}$. Hence find out the amount of work done in rotating it from the position of unstable equilibrium to the stable equilibrium.

[2016]

SOLUTION

(i) **Principle of a meter bridge:** it works on the balanced Wheatstone bridge condition. A meter bridge has a 1 *m* uniform resistance wire AC stretched along a metre scale; a known resistance *R* and the unknown *X* are connected in the two gaps, and a cell drives current through the wire while a galvanometer with a sliding jockey detects balance.

Circuit: the unknown X in the left gap and known R in the right gap are connected across the ends A and C; a battery with key is joined across A and C through the wire; the galvanometer connects the junction of X and R to the jockey, which slides on the wire.

At the balance (null) point at length from A, no current flows through the galvanometer, and the bridge balance condition gives

$$\frac{X}{R} = \frac{\text{resistance of } A}{\text{resistance of } C}$$

Since the wire is uniform, its resistance is proportional to length:

$$\frac{X}{R} = \overline{100 -}$$

$$X = R\ \overline{100 -}$$

This is the relevant formula for the unknown resistance.

(ii) With R in the left gap and S in the right gap, the null point is at 40 cm from A:

$$\frac{R}{S} = \frac{40}{100 - 40} = \frac{40}{60} = \frac{2}{3}$$

$$R = \frac{2}{3}\text{S}$$

When a 30 Ω resistance is connected in parallel with S, the right-gap resistance becomes

$$S' = \frac{30\ S}{30 + S}$$

The new null point is at 50 cm:

$$\frac{R}{S'} = \frac{50}{100 - 50} = \frac{50}{50} = 1$$

$$R = S' = \frac{30\ S}{30 + S}$$

Equating the two expressions for R:

$$\frac{2}{3}\text{S} = \frac{30\ S}{30 + S}$$

$$\frac{2}{3}(30 + S) = 30$$

$$30 + S = 45$$

$$S = 15\ \Omega$$

Then

$$R = \frac{2}{3} \times 15 = 10\ \Omega$$

So $\boldsymbol{R = 10\ \Omega}$ and $\boldsymbol{S = 15\ \Omega}$.

OR (a) Torque on a dipole. A dipole of moment $\vec{p}$ (charges $+q$ and $-q$ separated by 2a) in a uniform field $\vec{E}$ experiences forces $+q\vec{E}$ and $-q\vec{E}$. These equal and opposite forces form a couple. The magnitude of the torque is

$$\tau = (\text{force}) \times (\text{perpendicular distance}) = qE\ (2a\theta) = pE\theta$$

where θ is the angle between $\vec{p}$ and $\vec{E}$ and $\boldsymbol{p = q(2a)}$. In vector form

$$\vec{\tau} = \vec{p} \times \vec{E}$$

The torque is directed perpendicular to the plane containing $\vec{p}$ and $\vec{E}$ (given by the right-hand rule) and tends to align $\vec{p}$ with $\vec{E}$.

(b) Potential energy of the dipole. The work done by the external agent in rotating the dipole through a small angle dθ against the torque is $\mathbf{dW = \tau\, d\theta = pE\theta\, d\theta}$. Taking the zero of energy at $\boldsymbol{\theta = 90°}$, the potential energy at angle θ is

$$U(\theta) = \int_{90°}^{\theta} pE\theta'\, d\theta' = -pE\theta$$

$$U(\theta) = -\vec{p} \cdot \vec{E}$$

Work done in rotating from unstable equilibrium $\boldsymbol{(\theta = 180°)}$ to stable equilibrium $\boldsymbol{(\theta = 0°)}$:

$$W = U(0°) - U(180°) = (-pE) - (+pE) = -2pE$$

The magnitude of the work is 2pE; the negative sign shows that the field does positive work 2pE as the dipole moves to its stable orientation.

CONCEPT

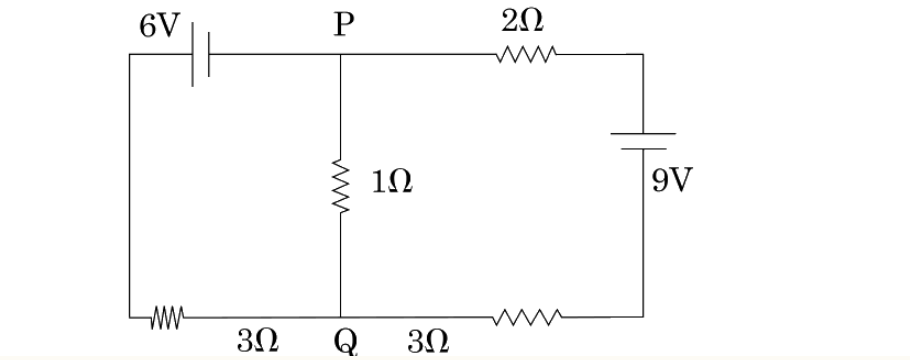
Connecting 30 Ω in parallel with S lowers the right-gap resistance, so the balance must shift toward the right gap (larger): a **4050** cm shift is exactly what you should expect.

Final answer. R**=10 Ω**, S**=15 Ω**. OR $\vec{\tau}=\vec{p}\times\vec{E}$, $U=-\vec{p}\cdot\vec{E}$, and W**=−2pE** (field does **+2pE**).

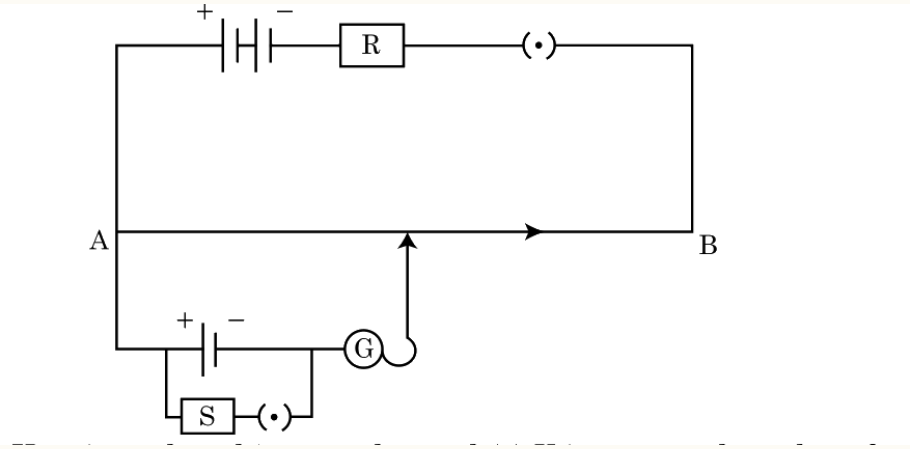
← *Back to Q319*

Q320

(i) Find the magnitude and direction of current in 1 Ω *resistor in the given circuit.*



(ii) Two students X and Y perform an experiment on potentiometer separately using the circuit diagram shown below.

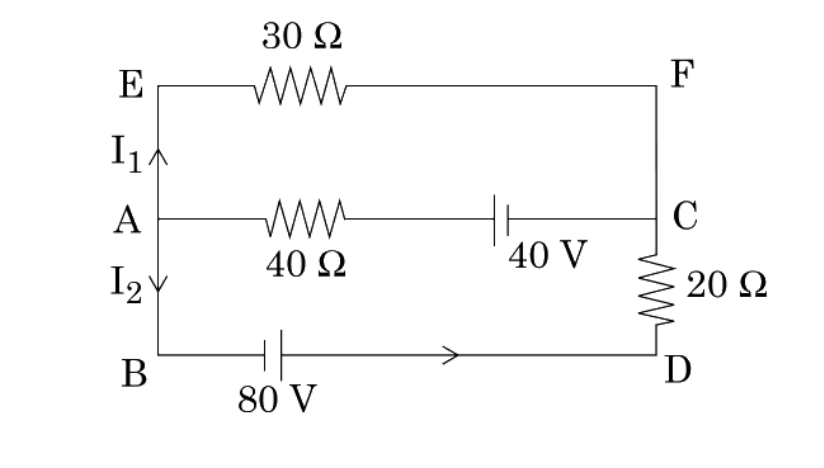


Keeping other things unchanged (a) X increases the value of resistance R, (b) Y decreases the value of resistance S in the set up. How will these changes affect the

position of null point in each case and why?

OR

(a) Use Kirchhoff's rules, calculate the current in the arm AC of the given circuit.



(b) On what principle does the meter bridge work? Why are the metal strips used in the bridge?

[2016]

SOLUTION

(i) Current in the 1 Ω resistor. The circuit contains two cells of emf 6 *V* and 9 *V* together with the resistors shown in the figure, the 1 Ω resistor being the common (shared) branch between the two loops.

Let I_1 be the current supplied by the 6 *V* loop and I_2 the current supplied by the 9 *V* loop. By Kirchhoff's junction rule, the current through the common 1 Ω resistor is

$$I = I_1 + I_2$$

Apply Kirchhoff's loop rule to each loop. For the loop containing the 6 *V* cell, its series resistance R_1 and the shared 1 Ω resistor:

$$6 = I_1 R_1 + 1 \cdot (I_1 + I_2)$$

For the loop containing the 9 *V* cell, its series resistance R_2 and the shared 1 Ω resistor:

$$9 = I_2 R_2 + 1 \cdot (I_1 + I_2)$$

These two equations together with $I = I_1 + I_2$ are solved simultaneously, substituting the actual resistance values R_1 and R_2 read from the figure, to obtain the current I through the 1 Ω resistor. Its direction is fixed by the cell polarities shown (from the higher-potential junction toward the lower through the 1 Ω resistor). The exact numerical value depends on the figure's resistor values and cannot be fixed without them.

(ii) Effect of changing R and S on the potentiometer null point.

(a) R is the rheostat in the primary (driver) circuit. Increasing R reduces the current in the potentiometer wire, which reduces the potential gradient k. Since the balancing length is $= \frac{\varepsilon}{k}$, a smaller k gives a larger . So the null point shifts farther from the starting end (balancing length increases).

(b) S is a series resistance in the secondary (galvanometer/cell) branch. Changing S does not change the current in the main wire or the potential gradient k, and at balance no current flows through this branch anyway. Hence decreasing S has no effect on the position of the null point.

OR (a) Current in arm AC by Kirchhoff's rules. Assign branch currents, apply the junction rule at the nodes and the loop rule around each independent loop of the given network. Solving the resulting simultaneous equations for the branch through AC gives the required current I_{AC} (substitute the figure's emf and resistance values to get the numerical value).

(b) Principle of a meter bridge: it is a practical Wheatstone bridge. A uniform 1 m wire provides two of the four arms; at the balance point the bridge condition $\frac{X}{R} = \frac{}{100 -}$ gives the unknown resistance.

The metal (thick copper) strips are used to connect the resistors because copper has very low resistivity, so the strips add negligible resistance to the bridge arms and do not introduce error in the measurement.

COMMON MISTAKE

Remember which branch a resistance sits in: a change in the primary (driver) circuit shifts the null point, but a series resistance in the galvanometer branch never does, because no current flows there at balance.

Final answer. Current through the **1 Ω** resistor $I=I_1+I_2$ from Kirchhoff's loop equations $6=I_1R_1+(I_1+I_2)$ and $9=I_2R_2+(I_1+I_2)$ (numeric value needs the figure's resistor values); increasing R moves the null point farther out, decreasing S has no effect. OR I_{AC} from Kirchhoff's rules; meter bridge uses $\frac{X}{R}=\overline{100-}$, thick copper strips keep arm resistance negligible.

← *Back to Q320*

Q321

(a) State Kirchhoff's rules and explain on what basis they are justified.
(b) Two cells of emfs E_1 *and* E_2 *and internal resistances* r_1 *and* r_2 *are connected in parallel. Derive the expression for the (i) emf and (ii) internal resistance of a single equivalent cell which can replace this combination.*

[2015]

SOLUTION

(a) Kirchhoff gave two rules to analyse complex electrical networks where simple series and parallel reduction does not work.

Junction Rule (Kirchhoff's Current Law): At any junction in a circuit, the algebraic sum of the currents meeting at that junction is zero.

$$\sum I = 0$$

In other words, the total current flowing into a junction equals the total current flowing out of it. This rule is justified by the law of conservation of charge. Since charge cannot pile up or be created at a junction in steady state, whatever charge enters per second must also leave per second.

Loop Rule (Kirchhoff's Voltage Law): Around any closed loop of a network, the algebraic sum of the changes in potential is zero.

$$\sum \Delta V = 0 \; i.e. \; \sum \varepsilon = \sum IR$$

This rule is justified by the law of conservation of energy. When a unit charge is carried around a closed loop and brought back to the starting point, the total

work done on it is zero because the electrostatic field is conservative, so the charge returns to the same potential.

(b) Consider two cells of emfs ε_1 and ε_2 with internal resistances r_1 and r_2, connected in parallel between two common points A (positive terminals joined) and B (negative terminals joined). Let V be the potential difference between A and B.

Let currents I_1 and I_2 flow out of the two cells, and let the total current drawn from the combination be I. By the junction rule,

$$I = I_1 + I_2$$

For cell 1, the terminal potential difference is

$$V = \varepsilon_1 - I_1 r_1 \Rightarrow I_1 = \frac{\varepsilon_1 - V}{r_1}$$

For cell 2,

$$V = \varepsilon_2 - I_2 r_2 \Rightarrow I_2 = \frac{\varepsilon_2 - V}{r_2}$$

Adding these,

$$I = I_1 + I_2 = \frac{\varepsilon_1 - V}{r_1} + \frac{\varepsilon_2 - V}{r_2}$$

$$I = \left(\frac{\varepsilon_1}{r_1} + \frac{\varepsilon_2}{r_2}\right) - V\left(\frac{1}{r_1} + \frac{1}{r_2}\right)$$

Solving for V,

$$V\left(\frac{1}{r_1} + \frac{1}{r_2}\right) = \left(\frac{\varepsilon_1}{r_1} + \frac{\varepsilon_2}{r_2}\right) - I$$

$$V = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2} - I\left(\frac{r_1 r_2}{r_1 + r_2}\right)$$

Comparing this with the single equivalent cell equation $V = \varepsilon_{eq} - I r_{eq}$, we identify

(i) $\varepsilon_{eq} = \dfrac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$

(ii) $r_{eq} = \dfrac{r_1 r_2}{r_1 + r_2}$ *i.e.* $\dfrac{1}{r_{eq}} = \dfrac{1}{r_1} + \dfrac{1}{r_2}$

Thus the parallel combination behaves as a **single cell of emf** $\varepsilon_{eq} = \dfrac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$ **and internal resistance** $r_{eq} = \dfrac{r_1 r_2}{r_1 + r_2}$.

TIP

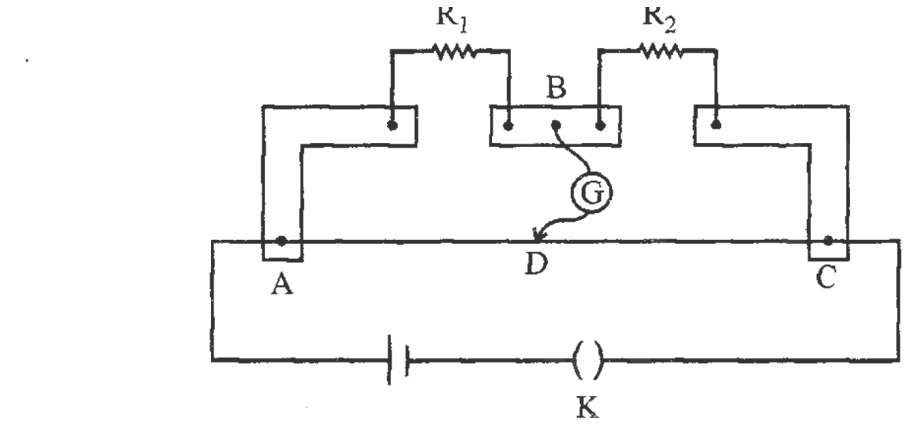
A neat check: the equivalent internal resistance is just the two internal resistances in parallel, exactly as you would expect for resistors. If you only remember that, you can verify your emf formula is dimensionally consistent.

Final answer. For two cells in parallel, $\varepsilon_{eq} = \dfrac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$ and $r_{eq} = \dfrac{r_1 r_2}{r_1 + r_2}$.

← *Back to Q321*

Q322

(a) State the working principle of a potentiometer. With the help of the circuit diagram, explain how a potentiometer is used to compare the emf's of two primary cells. Obtain the required expression used for comparing the emfs.
(b) Write two possible causes for one sided deflection in a potentiometer experiment.
OR
(a) State Kirchhoff's rules for an electric network. Using Kirchhoff's rules, obtain the balance condition in terms of the resistances of four arms of Wheatstone bridge.
(b) In the meterbridge experimental set up, shown in the figure, the null point 'D' is obtained at a distance of 40 cm *from end A of the meterbridge wire. If a resistance of* 10 Ω *is connected in series with* R_1*, null point is obtained at* **AD = 60 cm**. *Calculate the values of* R_1 *and* R_2.



[2013]

SOLUTION

(a) **Working principle of a potentiometer:** When a steady current flows through a wire of uniform cross-section and uniform composition, the potential drop across any portion of the wire is directly proportional to the length of that portion.

$$V \propto \Rightarrow V = k$$

where k is the potential gradient (potential drop per unit length) along the wire.

Circuit for comparing emfs: A long uniform wire AB is connected to a driver (auxiliary) battery, a rheostat and a key, so that a steady current flows through it from A to B. The positive terminals of both cells ε_1 and ε_2 are joined to end A. Their negative terminals are taken to a two-way key; through this key and a galvanometer G, the cell under test is connected to a jockey that can slide on the wire.

First, cell ε_1 is brought into the circuit using the two-way key. The jockey is moved along AB until the galvanometer shows zero deflection (null point) at a balancing length $_1$ from A. At balance, the potential drop across length $_1$ equals ε_1:

$$\varepsilon_1 = k\ _1$$

Next, cell ε_2 alone is connected and the new balancing length $_2$ is found, so that

$$\varepsilon_2 = k\ _2$$

Dividing the two equations,

$$\frac{\varepsilon_1}{\varepsilon_2} = \frac{1}{2}$$

Thus the **ratio of the emfs equals the ratio of their balancing lengths**, which is the required expression.

(b) Two possible causes for one-sided deflection in a potentiometer experiment:

(i) The emf of the driver (auxiliary) battery is less than the emf of the cell being measured, so the potential drop along the whole wire is smaller than the cell's emf and no balance (null point) can be obtained.

(ii) The positive terminals of the driver battery and the cell are not connected to the same end A of the wire (a connection or polarity error), or a connection is loose, so the galvanometer deflects to one side only.

OR

(a) **Kirchhoff's rules.** Junction rule: the algebraic sum of currents at any junction is zero, $\sum I = 0$ (conservation of charge). Loop rule: around any closed loop, $\sum \varepsilon = \sum IR$ (conservation of energy).

Balance condition of a Wheatstone bridge: Four resistances P, Q, R, S form the four arms of a quadrilateral ABCD. A galvanometer G is connected across one diagonal BD and a cell across the other diagonal AC.

At balance, the galvanometer current is zero ($\mathbf{I_g = 0}$). Hence the same current I_1 flows through P (arm AB) and Q (arm BC), and the same current I_2 flows through R (arm AD) and S (arm DC).

Applying the loop rule to loop ABDA (with $\mathbf{I_g = 0}$):

$$I_1 P - I_2 R = 0 \Rightarrow I_1 P = I_2 R$$

Applying the loop rule to loop BCDB (with $\mathbf{I_g = 0}$):

$$I_1 Q - I_2 S = 0 \Rightarrow I_1 Q = I_2 S$$

Dividing the first equation by the second,

$$\frac{I_1 P}{I_1 Q} = \frac{I_2 R}{I_2 S} \Rightarrow \frac{P}{Q} = \frac{R}{S}$$

This is the required balance condition of the Wheatstone bridge.

(b) In the meter bridge, R_1 is in the left gap and R_2 in the right gap. With null point at **AD = 40 cm**,

$$\frac{R_1}{R_2} = \frac{40}{100-40} = \frac{40}{60} = \frac{2}{3}$$

$$\Rightarrow R_1 = \frac{2}{3} R_2 \ldots(1)$$

When 10 Ω is added in series with R_1, the left-gap resistance becomes $(R_1 + 10)$ and the null point shifts to **AD = 60 cm**,

$$\frac{R_1 + 10}{R_2} = \frac{60}{100-60} = \frac{60}{40} = \frac{3}{2}$$

$$\Rightarrow R_1 + 10 = \frac{3}{2} R_2 \ldots(2)$$

Subtracting equation (1) from equation (2),

$$10 = \frac{3}{2} R_2 - \frac{2}{3} R_2 = \left(\frac{9-4}{6}\right) R_2 = \frac{5}{6} R_2$$

$$R_2 = \frac{10 \times 6}{5} = 12\ \Omega$$

From equation (1),

$$R_1 = \frac{2}{3} \times 12 = 8\ \Omega$$

Hence **$R_1 = 8\ \Omega$ and $R_2 = 12\ \Omega$.**

Final answer. OR (b): **$R_1 = 8\ \Omega$** and **$R_2 = 12\ \Omega$**; ratio of emfs $\frac{\varepsilon_1}{\varepsilon_2} = \frac{1}{2}$.

← *Back to Q322*

Q323

Deduce the condition for balance in a Wheatstone Bridge. Using the principle of Wheatstone Bridge, describe the method to determine the specific resistance of a wire

in the laboratory. Draw the circuit diagram and write the formula used. Write any two important precautions to observe while performing the experiment.

[2004]

SOLUTION

Balance condition of a Wheatstone bridge. Four resistances P, Q, R, S are connected to form the four arms of a quadrilateral ABCD. A galvanometer G is joined across the diagonal BD and a cell with a key is joined across the diagonal AC.

At balance, no current flows through the galvanometer, so $\mathbf{I_g = 0}$. Therefore the same current I_1 passes through arms P (AB) and Q (BC), and the same current I_2 passes through arms R (AD) and S (DC).

Applying Kirchhoff's loop rule to the loop ABDA (galvanometer arm carries no current),

$$I_1 P - I_2 R = 0 \Rightarrow I_1 P = I_2 R$$

Applying the loop rule to the loop BCDB,

$$I_1 Q - I_2 S = 0 \Rightarrow I_1 Q = I_2 S$$

Dividing the first relation by the second,

$$\frac{P}{Q} = \frac{R}{S}$$

This is the balance condition of the Wheatstone bridge.

Determination of specific resistance using a meter bridge. A meter bridge works on the principle of the Wheatstone bridge. A wire AC of length 100 cm and uniform area is stretched along a metre scale. The unknown wire (whose specific resistance is to be found) of resistance X is connected in the left gap, and a resistance box R is connected in the right gap. A cell with a key is connected across AC, and a galvanometer is connected between the central terminal and a jockey that slides over the wire.

The jockey is moved along the wire until the galvanometer shows zero deflection (null point) at a balancing length l from end A. The two parts of the bridge wire, of lengths l and $(100 - l)$, act as the other two arms. By the Wheatstone balance condition,

$$\frac{X}{R} = \frac{l}{100 - l}$$

$$X = R \cdot \frac{l}{100 - l}$$

Now the length L and radius r of the unknown wire are measured (radius with a screw gauge). The area of cross-section is $A = \pi r^2$. Using $X = \frac{\rho L}{A}$, the specific resistance (resistivity) is

$$\rho = \frac{X A}{L} = \frac{X \pi r^2}{L}$$

Formula used:

$$\rho = \frac{\pi r^2 X}{L}, \text{ where } X = R \cdot \frac{l}{100 - l}$$

Two important precautions:

(i) The jockey should be pressed gently and momentarily on the wire (not dragged or scraped along it), so that the wire is not heated or worn, keeping its resistance constant.

(ii) The value of the known resistance R should be so chosen that the null point lies near the middle of the bridge wire (around 50 cm), as this minimises the percentage error in the measured length l.

Final answer. $X = R \cdot \frac{l}{100 - l}$ and the specific resistance $\rho = \frac{\pi r^2 X}{L}$.

← *Back to Q323*